

## **Technical Reference Manual for Castalia**

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Build 601b3f5c

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# Part I System

# 1 About this manual

## 1.1 Scope and audience

This manual is the technical reference for the Castalia microcontroller. It is written for firmware developers and board designers, and it assumes familiarity with 32-bit microcontrollers, memory-mapped peripherals and the RISC-V instruction set. It describes the hardware as implemented: the address map, every peripheral and its registers, the interrupt fabric, the boot ROM and the processor core's configuration. The RISC-V specifications are normative for the instruction set and for the privileged architecture; where this manual restates them it does so only to say which parts are implemented and how. Where the implementation deviates from a specification, the deviation is stated where it applies.

Everything in this manual is generated from one chip configuration, the same one that produces the RTL, the C headers and the linker scripts (Section 2.3), so a value printed here is the value built.

## 1.2 Organisation of this manual

### **Part I, System**

What is on the chip, its package and pins, the address map, the multi-core architecture, and reset, clocks and power. Read this part first.

### **Part II, Processor**

The VestaRV core: its instruction set, control and status registers, privileged architecture, interrupts and exceptions, instruction timing, and the debug system.

### **Part III, Peripherals**

One chapter per peripheral, each with a functional description, a programming sequence, usage notes and the complete register description.

### **Part IV, Analog front end**

The mixed-signal front end and its characterisation.

### **Part V, Boot ROM and software**

The ROM monitor and the boot image format the boot ROM loads.

### **Appendices**

A software development quick start, the register index, errata, and the revision history of this document.

## 1.3 Related documents

Table 1.1 lists the documents this manual relies on and the generated artifacts that accompany it. This manual does not contain electrical characteristics, absolute maximum ratings, package mechanical drawings, or errata for individual silicon revisions; those belong to the datasheet and the errata sheet of a specific silicon revision.

Table 1.1: Documents and generated artifacts related to this manual.

| Document or artifact                                                                                                                                         | Role                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| RISC-V Instruction Set Manual, Volume I: Unprivileged ISA, version 20191213, with the ratified Bit-Manipulation extensions (Zba, Zbb, Zbc, Zbs, version 1.0) | Normative for the instruction set (Chapter 7).                                                                                                    |
| RISC-V Instruction Set Manual, Volume II: Privileged Architecture, version 1.12                                                                              | Normative for the machine-mode control and status registers and the trap architecture (Chapters 7 and 8).                                         |
| RISC-V External Debug Support, version 1.0                                                                                                                   | Normative for the debug transport and debug module (Chapter 11).                                                                                  |
| platform/common/out/software/include/MemoryMap.h, core_features.h, periph.S                                                                                  | Generated C and assembly headers: every base address, register offset, field mask and vector number in this manual under the header symbol names. |
| platform/common/out/linker-scripts/memory.x, periph.x                                                                                                        | Generated linker script fragments for the memory map of this build.                                                                               |
| docs/register_browser.html                                                                                                                                   | Interactive register browser built from the same memory map.                                                                                      |
| docs/chip_configurator.html                                                                                                                                  | Interactive editor for the chip configuration file.                                                                                               |

## 1.4 Conventions

### Numbers

Hexadecimal numbers carry the prefix `0x` (`0x4600`); binary numbers carry the prefix `0b` (`0b101`); decimal numbers have no prefix. A value of the form `0x8000 + 4n` is an address formula in which  $n$  is the array index.

### Units

Capacities are binary: 1 KiB is 1024 bytes. Rates and frequencies are decimal: 115,200 baud, 24 MHz. A word is 32 bits, a half-word 16 bits and a byte 8 bits.

### Registers and fields

A register is named by its peripheral template and register name, and a field by the dotted form `PERIPHERAL . REGISTER . FIELD`, for example `TIMx . TIMxCR . TEN`. In prose, register names are the template form (`PxIN`, `TIMxCR`), where  $x$  stands for the instance number; the instance form (`P0IN`, `TIM0CR`) appears only in the per-instance address tables and in the generated headers.

### Bit ranges

Bits are numbered from 0 at the least significant end. A range is written `[h:l]` with the high bit first; `[7:4]` is four bits. A single bit is written `[n]`.

### Set and clear

A bit is set when it holds 1 and cleared when it holds 0. Asserted means the active level of a signal, whatever its polarity; an active-low pin is named with a trailing N (`RESETN`).

### Reserved

A reserved bit reads as zero and must be written as zero. No exception is raised by an access to a reserved bit or to a reserved register offset inside a peripheral's window.

### Register width

Every register is described at its natural width, 8 or 32 bits, and every register occupies one 4-byte slot at a 4-byte aligned offset regardless of its width; the unused upper bytes of an 8-bit register are reserved.

### Narrow accesses

The bus carries four byte-lane write strobes, so a byte or half-word store reaches the peripheral with only the addressed lanes asserted. Reads always return the full word of the addressed slot. Whether a partial write is honoured is a property of the register: an 8-bit register lives in byte lane 0, and a register described as requiring a full-word write commits only stores that assert lane 0.

## 1.5 Register access types

Each field in a register description carries an access type from the list in Table 1.2. The list is the set the chip generator accepts, so it cannot drift from the register descriptions.

Table 1.2: Register access types

| Type | Meaning              | Read effect                 | Write effect                                                                                                                               |
|------|----------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| rw   | Read/write           | Returns the current value.  | Sets the field to the written value.                                                                                                       |
| r    | Read only            | Returns the current value.  | Ignored.                                                                                                                                   |
| r0   | Reserved, reads zero | Returns 0.                  | Ignored. Write 0.                                                                                                                          |
| r1   | Reserved, reads one  | Returns 1.                  | Ignored. Write 1.                                                                                                                          |
| rw0  | Read, write 0 to act | Returns the current value.  | A 0 in a bit position acts on that bit; a 1 leaves it unchanged.                                                                           |
| rw1  | Read, write 1 to act | Returns the current value.  | A 1 in a bit position acts on that bit (clears a flag, or applies the register's set, clear or toggle operation); a 0 leaves it unchanged. |
| w    | Write only           | Not defined by this manual. | Sets the field to the written value.                                                                                                       |
| w0   | Write 0 to trigger   | Not defined by this manual. | A 0 triggers the action; a 1 has no effect.                                                                                                |
| w1   | Write 1 to trigger   | Not defined by this manual. | A 1 triggers the action (a command strobe); a 0 has no effect.                                                                             |

## 1.6 The anatomy of a register description

Every register in Part III is described in the same layout. The example below is a fictitious register, DUMMY.DUMMYCR, chosen to show every element once; the callouts after it name each element.



## DUMMYxCR: Dummy control register

**Offset** 0x04 **Reset** 0x00000000 **Width** 32

Controls the dummy function. One or two sentences say what the register is for and point to the functional section that explains it.

| Bits   | Field | Type | Reset | Description                                                                                                                                                    |
|--------|-------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [31:8] | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                                              |
| [7:4]  | DIV   | rw   | 0x0   | Clock divider. The dummy clock is the source divided by $2^{\text{DIV}}$ .                                                                                     |
| [3:2]  | MODE  | rw   | 0b00  | Operating mode.<br>0b00 = DMODE_OFF: the function is off.<br>0b01 = DMODE_ONCE: one operation, then off.<br>0b10 = DMODE_CONT: continuous.<br>0b11 = reserved. |
| [1]    | DIF   | rw1  | 0     | Done interrupt flag. Set by hardware when an operation completes; write 1 to clear.                                                                            |
| [0]    | DEN   | rw   | 0     | Enable. 0 = disabled, 1 = enabled. Write this bit last.                                                                                                        |

### Heading

The register's template name and a short title. The heading appears in the chapter's own register list, not in the main table of contents. The x in the name is the instance number.

### Offset, Reset, Width

The byte offset from the instance base address (the base addresses are in the chapter's instance table), the reset value of the whole register, and its width in bits.

### Summary sentence

What the register is for, in one to three sentences, with a pointer to the functional description for anything longer.

### Field rows

One row per field, most significant first, every bit covered. **Bits** is the range, **Field** the name used in prose and in the generated headers (with the peripheral's field prefix), **Type** the access type of Table 1.2, **Reset** the field's reset value at the field's width.

### Enumerated values

Listed inside the description cell as value = SYMBOL: meaning. The symbol is the macro name in the generated header. Values are binary for fields of four bits or fewer and hexadecimal for wider fields.

### Reserved runs

Consecutive reserved bits are collapsed into one row whose field, type and reset are a dash.

## 1.7 Callouts

Two kinds of callout appear in this manual, and only two. A *Note* is advisory: it adds context or a recommendation, and nothing breaks if it is skipped. A *Caution* states a constraint the hardware enforces: ignoring it causes a wrong result, a hang, or lost data.

**Note:** A note looks like this. It is at most a few sentences long.

**Caution:** A caution looks like this. Software must satisfy the condition it states, because the hardware does not check it.

## 1.8 Glossary of terms and abbreviations

|                       |                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>ABI</b>            | Application binary interface: the register-usage and calling convention of Table 7.1.                                            |
| <b>AF</b>             | Alternate function: a peripheral signal a GPIO pin can carry instead of its plain I/O function; a pin has up to eight AF planes. |
| <b>AFE</b>            | Analog front end: the mixed-signal circuitry of Part IV.                                                                         |
| <b>CLINT</b>          | Core-local interruptor: the peripheral that provides each hart's software interrupt and timer interrupt.                         |
| <b>CPI</b>            | Cycles per instruction.                                                                                                          |
| <b>CRC</b>            | Cyclic redundancy check.                                                                                                         |
| <b>CSR</b>            | Control and status register: a register inside the processor core reached by the <code>csr*</code> instructions.                 |
| <b>DM</b>             | Debug module: the on-chip block that halts, resumes and inspects harts on behalf of an external debugger.                        |
| <b>DMI</b>            | Debug module interface: the register bus between the debug transport module and the debug module.                                |
| <b>DTM</b>            | Debug transport module: the JTAG-side block that converts scans into DMI accesses.                                               |
| <b>EIS</b>            | Electrochemical impedance spectroscopy: the swept-frequency impedance measurement the analog front end performs.                 |
| <b>GPIO</b>           | General-purpose input and output.                                                                                                |
| <b>hart</b>           | Hardware thread: one RISC-V execution context. Every hart on this chip is a separate core with its own TCM.                      |
| <b>I<sup>2</sup>C</b> | Inter-integrated circuit: the two-wire serial bus.                                                                               |
| <b>IPI</b>            | Inter-processor interrupt: a CLINT software interrupt sent from one hart to another.                                             |
| <b>IRQROUTER</b>      | Interrupt router: the peripheral that routes every interrupt source to any hart and provides the claim and complete registers.   |
| <b>ISA</b>            | Instruction set architecture.                                                                                                    |
| <b>ISR</b>            | Interrupt service routine.                                                                                                       |
| <b>IVT</b>            | Interrupt vector table: the table of jump instructions at the base of each hart's TCM.                                           |
| <b>JTAG</b>           | The IEEE 1149.1 test access port over which the debug transport operates.                                                        |
| <b>MCU</b>            | Microcontroller unit.                                                                                                            |
| <b>MTCMOS</b>         | Multi-threshold CMOS power gating: the header switches that cut the supply of a hart tile.                                       |

|                |                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>MUTEX</b>   | Hardware mutual-exclusion bank: the peripheral that provides atomically claimable locks.                                                |
| <b>NFC</b>     | Near-field communication: the peripheral that reads a 13.56 MHz tag interface and, on a field-powered board, supplies the chip.         |
| <b>NPU</b>     | Neural processing unit: the fixed-point matrix accelerator.                                                                             |
| <b>PGOOD</b>   | Power good: the supply-supervisor input that releases the boot gate on a field-powered board.                                           |
| <b>PMP</b>     | Physical memory protection: the RISC-V machine-mode memory protection unit, a configuration option.                                     |
| <b>POC</b>     | Power-on control: the input of the I/O pad ring that holds the pads in a safe state while the I/O supply rises. It is not a boot strap. |
| <b>POR</b>     | Power-on reset.                                                                                                                         |
| <b>PWRCTRL</b> | Power controller: the peripheral that gates hart tiles and holds the boot gate.                                                         |
| <b>QSPI</b>    | Quad serial peripheral interface: a four-data-line SPI variant, a configuration option.                                                 |
| <b>ROM</b>     | Read-only memory: the boot ROM at address 0x00000.                                                                                      |
| <b>SPI</b>     | Serial peripheral interface.                                                                                                            |
| <b>TCM</b>     | Tightly coupled memory: the private RAM of one hart, 8 KiB at 0x8000, reached without arbitration.                                      |
| <b>TRNG</b>    | True random number generator, a configuration option.                                                                                   |
| <b>UART</b>    | Universal asynchronous receiver and transmitter.                                                                                        |
| <b>W1C</b>     | Write one to clear: a flag bit that is cleared by writing 1 to it (access type rw1).                                                    |
| <b>WARL</b>    | Write any, read legal: a CSR field that accepts any written value and reads back a legal one.                                           |
| <b>XIP</b>     | Execute in place: fetching instructions directly from the external SPI flash through the extended-flash window.                         |

## 2 Overview

The Castalia microcontroller (MCU) is a five-hart multiprocessor built from five identical VestaRV cores, a 32-bit RISC-V processor, together with a boot ROM that contains the boot loader and a ROM monitor, a private tightly coupled memory (TCM) per hart, an arbitrated shared memory window for inter-hart communication, and the peripherals listed below. The multi-core architecture is described in Chapter 5, and Figure 2.1 shows the top-level organisation of this configuration.

### 2.1 Features

- Chip name: Castalia
- Compatible with RISC-V compiler RV32I[M][A][C][Zba][Zbb][Zbs][Zbc]
- 32-bit memory bus
- Contains 32 general purpose dual-port CPU registers
- Supports the RISC-V compressed instruction set and allows the use of the [C] compiler extension. Allowing the compiler to use the [C] extension reduces executable code size, but also takes the processor longer to fetch 32-bit long instructions that are aligned on a 2 but not 4 byte boundary from memory.
- Includes a multi-stage bit shifter in the ALU to save power and chip area for bit shift operations at the expense of speed
- Includes a fast 32-bit signed integer hardware multiplier in the ALU and allows the use of the [M] compiler extension
- Includes a medium speed 32-bit signed integer hardware divider and remainder calculator in the ALU
- Supports the RISC-V atomic instruction set (LR/SC and AMOs) and allows the use of the [A] compiler extension; on multi-hart configurations atomics are globally coherent across the shared window
- Supports the RISC-V bit-manipulation extensions Zba (address generation), Zbb (basic bit manipulation), Zbs (single-bit operations), and Zbc (carry-less multiplication)
- The read-only misa CSR (0x301) advertises the enabled instruction-set extensions at run time
- Supports maskable hardware interrupts generated by peripheral signals
- 5× VestaRV harts (cores), each with private RAM; hart ID readable via the `mhartid` CSR
- An arbitrated shared memory window with 80 KiB of shared RAM, a CLINT (inter-processor and per-hart timer interrupts), 16 hardware mutexes, and a per-hart peripheral interrupt router
- 6× 8-pin general purpose I/O (GPIO) ports with edge-triggered interrupts and per-pin multiplexed alternate functions (GPIO + up to 8 alternate functions per pin)
- 2× SPI interfaces (SPI0 provides memory-mapped access to external flash memory)
- 2× UART interfaces with hardware parity support
- 2× 32-bit timers with pulse-width modulation outputs and input capture units
- A CRC calculation engine (CRC16\_CDMA2000)

- 2× internal digitally controllable oscillators
- 2× external clock pins for clock generation and accurate timing
- A windowed watchdog timer
- A neural processing unit (NPU) co-processor for hardware acceleration of machine learning tasks
- Per-tile MTCMOS power gating with hardware gate/wake sequencing (cold-boot wake)
- 2× I<sup>2</sup>C interfaces (both master and slave mode)
- 1× NFC ISO 14443A tag / card-emulation engine (Miller/Manchester codec, CRC-A, anticollision, digital AFE boundary)
- Claim/complete peripheral interrupt delivery (PLIC-style) with any-vector-to-any-hart routing
- 4× three-electrode bipolar potentiostat channels, each with a programmable transimpedance stage (320 gain codes, 19.6 kΩ to 1.93 MΩ), a 10-bit SAR converter and a per-channel reference DAC; impedance spectroscopy is a firmware mode of the same channels

## 2.2 Block diagram

Figure 2.1 is the view to read first: the bus masters above the arbiter, the arbiter drawn as the single bus bar it behaves like, the memories and every peripheral family this configuration instantiates on one rank below it, and the package boundary in red with the pin groups that cross it. It is generated from the configuration, so the peripheral set and the instance counts in it are this build's. The memory system's own view of the chip, the one address column all five harts share, is Figure 4.1 in Chapter 4, and the per-pin bonding is Chapter 3.

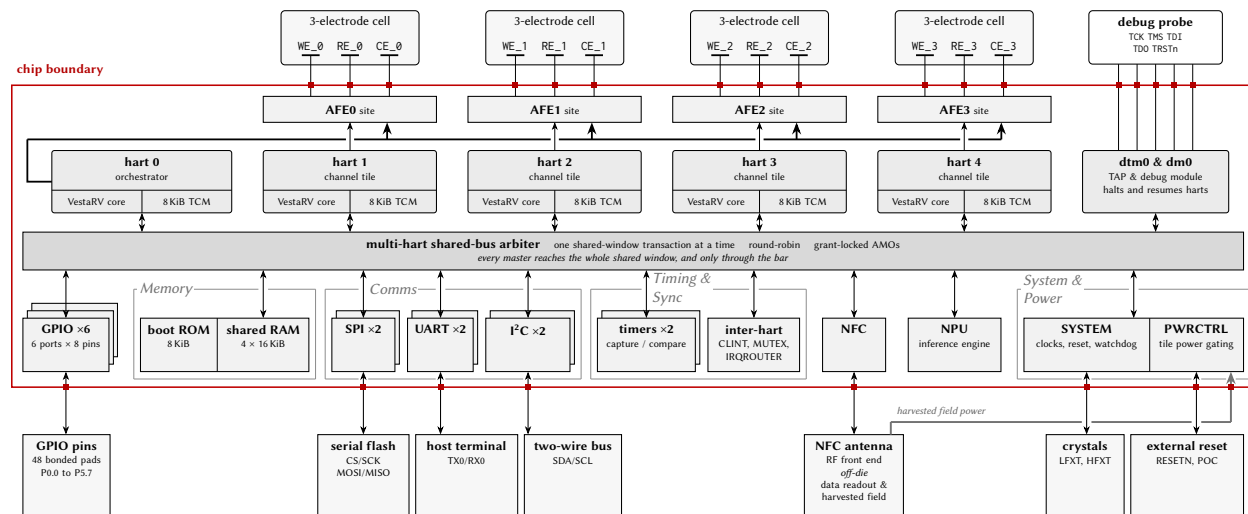


Figure 2.1: Castalia whole-chip diagram: the masters above one mp\_arbiter bus bar, every peripheral on one rank below it, and the package boundary in red.

## 2.3 Chip configuration

This document, the memory-map headers, the linker scripts, and the drop-in RTL are all generated from one chip configuration by `make chip`. The configuration is a small JSON document (every key optional; a

missing key keeps the Castalia default) passed as `make chip CONFIG=config.json`; an interactive front end for producing it ships with the generator (`docs/chip_configurator.html`). Table 2.1 lists every configuration knob together with the value used to build this document, Table 2.2 lists the geometry derived from those knobs, and Table 2.3 lists the ISA of each hart class this build emitted. The resolved configuration is also written to `config/ChipConfig.resolved.json` at every build, and `make verify [CONFIG=config.json]` proves a configuration by staging the generated RTL into a behavioural simulation environment and running the multi-core boot and ISA smoke suite against it. The core peripheral set and the pad ring are not configuration knobs; they are fixed template content, and the pad ring (Chapter 3) is derived from the generator’s package model. The second-instance peripherals (UART1, SPI1, TIMER1, I2C1) and the NPU are individually droppable through the `peripherals` section: a dropped instance’s register window reads zero, its interrupt vectors are reserved (the numbering is frozen), and its pins revert to plain GPIO.

Table 2.1: Chip configuration knobs (`make chip CONFIG=config.json`) and the values of this build

| Configuration key                 | This build | Meaning / valid values                                                                                       |
|-----------------------------------|------------|--------------------------------------------------------------------------------------------------------------|
| <code>chipName</code>             | Castalia   | non-empty string: renames the chip in the TRM and headers (CHIP_NAME env wins)                               |
| <code>numHarts</code>             | 5          | int 1..32: hart count; hart 0 is the always-on management hart and harts 1..N-1 are the power-gateable tiles |
| <code>orchestrator</code>         | true       | bool: hart 0 is the always-on orchestrator; harts 1..N-1 are gateable tiles                                  |
| <code>numMutexes</code>           | 16         | int 1..1024: hardware mutex bank size, the number of MUTEXn registers                                        |
| <code>registerFileDualPort</code> | true       | bool: docs-only, the register file is dual-port in the RTL either way                                        |
| <code>core.fetchAhead</code>      | true       | bool: C-extension fetch-ahead (one flip-flop; removes most of the straddling-fetch stall)                    |
| <code>isa.mul</code>              | true       | bool: M multiply                                                                                             |
| <code>isa.fastMul</code>          | true       | bool: docs-only, the multiplier is already single-cycle                                                      |
| <code>isa.div</code>              | true       | bool: M divide                                                                                               |
| <code>isa atomics</code>          | true       | bool: A extension (LR/SC + AMO)                                                                              |
| <code>isa.compressed</code>       | true       | bool: C extension                                                                                            |
| <code>isa.bitmanip</code>         | true       | bool: Zba/Zbb/Zbs/Zbc                                                                                        |
| <code>isa.minimalTiles</code>     | true       | bool: harts 1..N-1 drop M and B (rv32iac); hart 0 keeps the full ISA                                         |
| <code>isa.counters</code>         | false      | bool: Zicntr march suffix only (cycle/instrret always exist, time/timeh never do)                            |
| <code>isa.counters64</code>       | false      | bool: docs-only high counter halves (always present); needs isa.counters                                     |
| <code>isa.zicond</code>           | false      | bool: Zicond conditional-zero ops (czero.eqz/czero.nez)                                                      |
| <code>isa.zcb</code>              | false      | bool: Zcb extra compressed loads/stores and ALU ops; needs isa.compressed                                    |

*Table 2.1 continued on next page*

Table 2.1 continued from previous page

| Configuration key        | This build     | Meaning / valid values                                                                                                            |
|--------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| isa.zimop                | false          | bool: Zimop+Zcmop may-be-ops (mop.r/mop.rr rd<-0, c.mop.n nops)                                                                   |
| isa.zihint               | false          | bool: Zihintpause+Zihintntl (PAUSE = 16-cycle arbiter-yield window; ntl.* nops)                                                   |
| isa.zihpm                | false          | bool: Zihpm performance counters 3 and 4 (four chip events)                                                                       |
| isa.zawrs                | false          | bool: Zawrs wrs.nto/wrs.sto wait-on-reservation-set (needs isa.atomics)                                                           |
| isa.zabha                | false          | bool: Zabha byte/halfword AMOs; needs isa.atomics                                                                                 |
| isa.zacas                | false          | bool: Zacas compare-and-swap (amo-cas.w/.b/.h); needs isa.atomics                                                                 |
| isa.zicboz               | false          | bool: Zicboz cbo.zero cache-block zero (64-byte block)                                                                            |
| isa.zcmp                 | false          | bool: Zcmp compressed push/pop and register moves; needs isa.compressed                                                           |
| isa.zcmt                 | false          | bool: Zcmt compressed table jump and the jvt CSR; needs isa.compressed                                                            |
| isa.zbkb                 | false          | bool: Zbkb crypto bit-manipulation (pack/brev8/zip/unzip)                                                                         |
| isa.zbkc                 | false          | bool: Zbkc carry-less multiply (clmul/clmulh)                                                                                     |
| isa.zbkx                 | false          | bool: Zbkx crossbar permute (xperm8/xperm4)                                                                                       |
| isa.zkn                  | false          | bool: Zkn AES+SHA (Zknd+Zkne+Zknh)                                                                                                |
| isa.zfinx                | false          | bool: Zfinx single-precision floating point in the x registers                                                                    |
| priv.trapCsr             | true           | bool: standard M-mode trap CSRs and delivery; firmware opts in via mtrapctl                                                       |
| priv.umode               | false          | bool: user mode (privilege register, MPP stack, mcounteren); needs priv.trapCsr                                                   |
| priv.pmp                 | false          | bool: PMP (Smpmp) CSR bank and match unit; needs priv.umode                                                                       |
| priv.pmpEntries          | 16             | int, 8 or 16: PMP entry count when priv.pmp is true (default 16)                                                                  |
| debug.enable             | true           | bool: debug mode, the Debug Module and the JTAG DTM; needs priv.trapCsr                                                           |
| memory.romSize           | 8192 (8 KiB)   | int bytes, 1 KiB multiple <= 0x4000: boot ROM (0x0-0x1FFF at the shipped 8 KiB; the page runs to 0x3FFF and its tail is unmapped) |
| memory.tcmSizePerHart    | 8192 (8 KiB)   | int bytes, 1 KiB multiple <= 0x4000: per-hart TCM based at 0x8000 (top = 0x8000 + size - 1; 0x9FFF at the shipped 8 KiB)          |
| memory.sharedBulkRamSize | 65536 (64 KiB) | int bytes, multiple of 0x4000: shared bulk RAM from 0x10000, one bank per 16 KiB                                                  |

Table 2.1 continued on next page

Table 2.1 continued from previous page

| Configuration key        | This build       | Meaning / valid values                                                                                             |
|--------------------------|------------------|--------------------------------------------------------------------------------------------------------------------|
| memory.npuStagingRamSize | 16384 (16 KiB)   | int bytes, 1 KiB multiple <= 0x4000: NPU staging RAM (0xC000-0xFFFF)                                               |
| peripherals.npu          | true             | bool: false drops the NPU (slot 10 and the 0xC000 staging window read zero)                                        |
| peripherals.i2c1         | true             | bool: false drops I2C1 (slot 15 reads zero; SDA1/SCL1 revert to plain GPIO)                                        |
| peripherals.uart1        | true             | bool: false drops UART1 (slot 5 reads zero; TX1/RX1 revert to plain GPIO)                                          |
| peripherals.spi1         | true             | bool: false drops SPI1 (slot 3 reads zero; its four pins revert to plain GPIO)                                     |
| peripherals.timer1       | true             | bool: false drops TIMER1 (slot 7 reads zero; the T1 pins revert to plain GPIO)                                     |
| peripherals.cqAfeStubs   | true             | bool: the four AFE register stubs and the EIS engine stub (slot 12, 0x4C00)                                        |
| peripherals.qspi         | false            | bool: QSPI0 controller in slot 12 (0x4C00); needs cqAfeStubs false                                                 |
| peripherals.i3c          | false            | bool: I3C0 controller (MVP + DAA + IBI) at 0x6100                                                                  |
| peripherals.nfc          | true             | bool: NFC0 ISO 14443A tag engine at 0x6200 (the RF front end is off-die)                                           |
| peripherals.rtc          | false            | bool: RTC0 32.768 kHz wall clock with alarm and tick at 0x6500                                                     |
| peripherals.pwm          | false            | bool: PWM0 two-channel buffered PWM at 0x6600 (outputs on P2.2/P2.3)                                               |
| peripherals.onewire      | false            | bool: OW0 1-Wire master at 0x6700, open-drain DQ on P4.7                                                           |
| peripherals.fieldPower   | true             | bool: wires the field-power pins (PGOOD, harvest strap) into PWRCTRL                                               |
| peripherals.dma          | false            | bool: DMA0 multi-channel DMA at 0x6800; adds one more arbiter master                                               |
| peripherals.dmaChannels  | 4                | int, 2 or 4: DMA0 channel count when peripherals.dma is true (default 4)                                           |
| peripherals.i2ctarget    | false            | bool: I2CT0 hardware I2C target at 0x6A00, sharing the I2C0 pads                                                   |
| peripherals.trng         | false            | bool: TRNG0 ring-oscillator entropy source at 0x6900 (firmware must DRBG it)                                       |
| peripherals.trngRings    | 8                | int, 4 or 8: TRNG0 ring count when peripherals.trng is true (default 8)                                            |
| peripherals.eventFabric  | false            | bool: EVFAB0 event/trigger crossbar at 0x6B00 (8 channels, no IRQ vector)                                          |
| package.model            | castalia-lqfp100 | string: package pinout model, one of the pinout models defined in the generator (QFN-44, QFN-64 quad, or LQFP-100) |
| package.preliminary      | true             | bool: prints the Preliminary note in the TRM package section                                                       |



Table 2.2: Derived geometry of this configuration (computed, not configurable)

| Derived value               | This build                           | Notes                                                                                                       |
|-----------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------|
| ISA string (march)          | rv32imac_zba_zbb_zbs_zbc             | Advertised in the read-only misa CSR                                                                        |
| Shared-window address width | 16 bits (words)                      | Arbiter/tile word-address width; the window is $0x0$ to $2^{w+2} - 1$                                       |
| Shared RAM banks            | $4 \times 16$ KiB                    | One SRAM macro per bank behind the arbiter                                                                  |
| Extended flash base         | $0x40000$                            | First address decoded to the SPI-flash XIP path (hart 0 only); strictly the complement of the shared window |
| Interrupt vectors           | 121                                  | Vectors 83/84 are the CLINT software/timer interrupts                                                       |
| CLINT MTIME                 | $0x5020$                             | Layout is hart-count-derived: MSIPx at $0x5000 + 4h$ , MTIMECMPx from $0x5030$                              |
| Boot-loader mailbox rows    | $0x10500 + 0x10 \cdot \text{hartid}$ | Tile-loading rows {SRC, LEN, ENTRY} consumed by the boot ROM                                                |
| Stack pointer at reset      | $0xA000$                             | Top of each hart's private TCM, growing down                                                                |
| Peripheral count            | 21                                   | Instantiated peripherals (including CLINT/MUTEX/IRQROUTER)                                                  |

Table 2.3: Per-hart-class ISA of this build. Each row is a hart class the generator emitted; the ISA string is what that class advertises in its read-only misa CSR. Only M and Zb differ between classes (the generated TILE\_ENABLE\_MUL, TILE\_ENABLE\_DIV and TILE\_ENABLE\_BITMANIP constants gate them); A, C, the Z-series extensions and the whole privilege set reach every hart. Code that more than one class executes must be built for the narrowest row.

| Hart class                      | Harts | ISA string               | Privilege features       |
|---------------------------------|-------|--------------------------|--------------------------|
| Orchestrator (Soft core)        | 0     | rv32imac_zba_zbb_zbs_zbc | M-mode, trap CSRs, debug |
| Tile (Hardened hart_tile macro) | 1-4   | rv32iac                  | M-mode, trap CSRs, debug |

Every image more than one hart executes is built for the narrowest row, rv32iac: the boot ROM all 5 harts fetch at reset, the debug entry trampoline the Debug Module plants for whichever hart it halts, and any library linked into more than one image. A wider `-march` on one of those lets the assembler emit an instruction a tile traps as illegal, so the build flags are pinned to this string and graded on the linked ELF by `//software/testtools:tile_isa_test`. Images that run on hart 0 alone keep the full ISA.

### 3 Package and pins

**Note:** The package model for this configuration (LQFP-100) has not been finalised for Castalia; the bonding shown here is the planned assignment, not a confirmed bond-out. Pin assignments, package type, and pin count are subject to change.

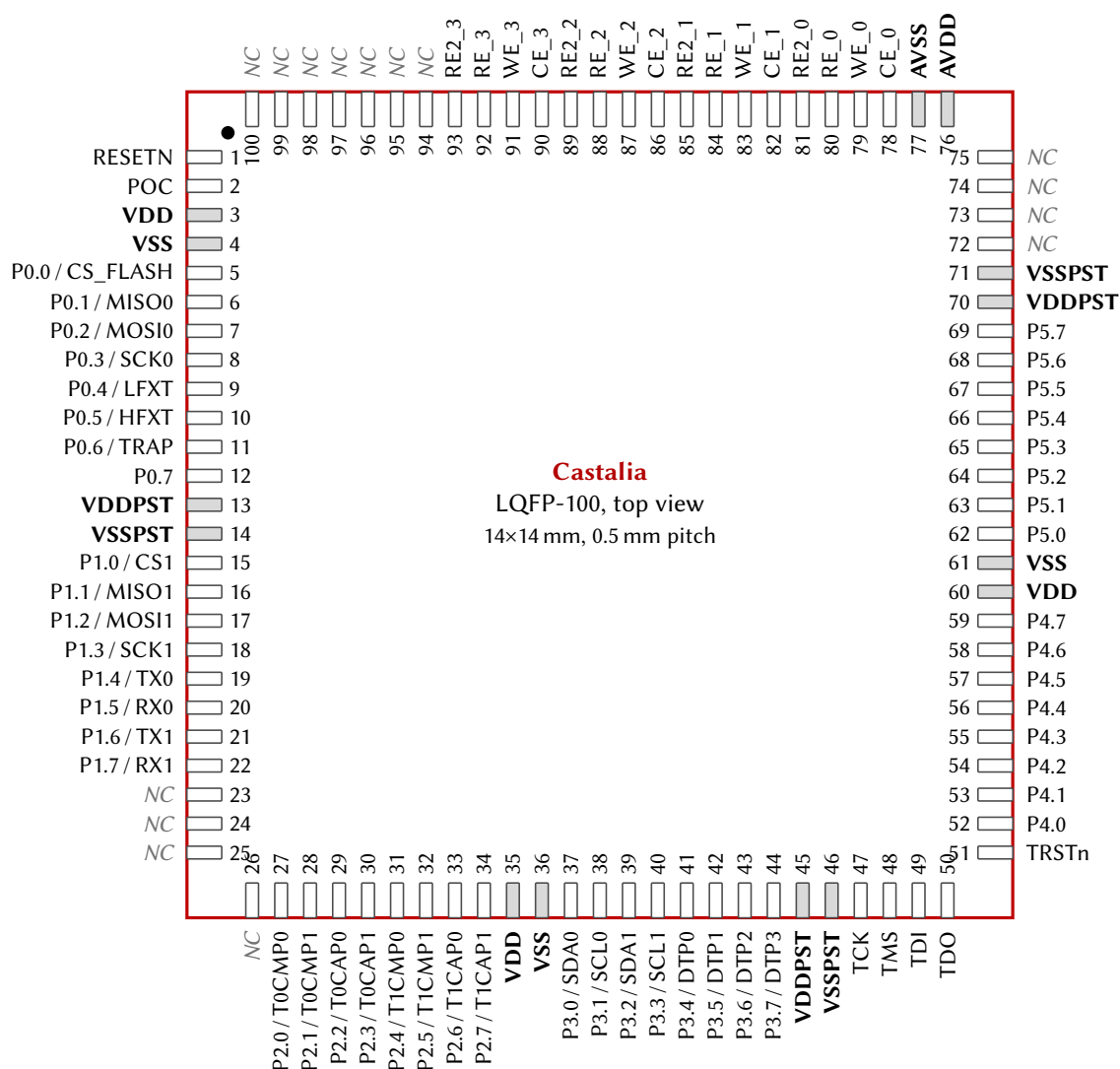


Figure 3.1: LQFP-100 pinout, top view (14 × 14 mm, 0.5 mm pitch); GPIO pins carry their AF0 function and power-rail pins are shaded.

The chip package is a LQFP-100 with dimensions 14 × 14 mm and pitch 0.5 mm. The package footprint, seen from the top, is shown in Figure 3.1, and the pins are listed in Table 3.1. The pin ordering, side assignment, and power domains in both the figure and the table are derived from the generator’s package model and are also exported machine-readably to config/PadRing.json at every make chip build.

Table 3.1: Package Pins

| #  | Pin    | I/O          | Power Domain         | Power Pins    |
|----|--------|--------------|----------------------|---------------|
| 1  | RESETN | Input        | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 2  | POC    | Input        | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 3  | VDD    | Power Input  | Digital Core (1.0 V) | VDD/VSS       |
| 4  | VSS    | Power Input  | Digital Core (1.0 V) | VDD/VSS       |
| 5  | P0.0   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 6  | P0.1   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 7  | P0.2   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 8  | P0.3   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 9  | P0.4   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 10 | P0.5   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 11 | P0.6   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 12 | P0.7   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 13 | VDDPST | Power Input  | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 14 | VSSPST | Power Input  | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 15 | P1.0   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 16 | P1.1   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 17 | P1.2   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 18 | P1.3   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 19 | P1.4   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 20 | P1.5   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 21 | P1.6   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 22 | P1.7   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 23 | NC     | -            | -                    | -             |
| 24 | NC     | -            | -                    | -             |
| 25 | NC     | -            | -                    | -             |
| 26 | NC     | -            | -                    | -             |
| 27 | P2.0   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 28 | P2.1   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 29 | P2.2   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 30 | P2.3   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 31 | P2.4   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 32 | P2.5   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 33 | P2.6   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 34 | P2.7   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 35 | VDD    | Power Input  | Digital Core (1.0 V) | VDD/VSS       |
| 36 | VSS    | Power Input  | Digital Core (1.0 V) | VDD/VSS       |
| 37 | P3.0   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 38 | P3.1   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 39 | P3.2   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 40 | P3.3   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 41 | P3.4   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 42 | P3.5   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 43 | P3.6   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 44 | P3.7   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |

*Table 3.1 continued on next page*

Table 3.1 continued from previous page

| #  | Pin    | I/O          | Power Domain         | Power Pins    |
|----|--------|--------------|----------------------|---------------|
| 45 | VDDPST | Power Input  | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 46 | VSSPST | Power Input  | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 47 | TCK    | Input        | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 48 | TMS    | Input        | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 49 | TDI    | Input        | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 50 | TDO    | Output       | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 51 | TRSTn  | Input        | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 52 | P4.0   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 53 | P4.1   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 54 | P4.2   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 55 | P4.3   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 56 | P4.4   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 57 | P4.5   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 58 | P4.6   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 59 | P4.7   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 60 | VDD    | Power Input  | Digital Core (1.0 V) | VDD/VSS       |
| 61 | VSS    | Power Input  | Digital Core (1.0 V) | VDD/VSS       |
| 62 | P5.0   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 63 | P5.1   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 64 | P5.2   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 65 | P5.3   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 66 | P5.4   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 67 | P5.5   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 68 | P5.6   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 69 | P5.7   | Input/Output | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 70 | VDDPST | Power Input  | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 71 | VSSPST | Power Input  | Digital I/O (3.3 V)  | VDDPST/VSSPST |
| 72 | NC     | -            | -                    | -             |
| 73 | NC     | -            | -                    | -             |
| 74 | NC     | -            | -                    | -             |
| 75 | NC     | -            | -                    | -             |
| 76 | AVDD   | Power Input  | Analog (2.5 V)       | AVDD/AVSS     |
| 77 | AVSS   | Power Input  | Analog (2.5 V)       | AVDD/AVSS     |
| 78 | CE_0   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 79 | WE_0   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 80 | RE_0   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 81 | RE2_0  | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 82 | CE_1   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 83 | WE_1   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 84 | RE_1   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 85 | RE2_1  | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 86 | CE_2   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 87 | WE_2   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 88 | RE_2   | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |
| 89 | RE2_2  | Input/Output | Analog (2.5 V)       | AVDD/AVSS     |

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| #   | Pin   | I/O          | Power Domain   | Power Pins |
|-----|-------|--------------|----------------|------------|
| 90  | CE_3  | Input/Output | Analog (2.5 V) | AVDD/AVSS  |
| 91  | WE_3  | Input/Output | Analog (2.5 V) | AVDD/AVSS  |
| 92  | RE_3  | Input/Output | Analog (2.5 V) | AVDD/AVSS  |
| 93  | RE2_3 | Input/Output | Analog (2.5 V) | AVDD/AVSS  |
| 94  | NC    | -            | -              | -          |
| 95  | NC    | -            | -              | -          |
| 96  | NC    | -            | -              | -          |
| 97  | NC    | -            | -              | -          |
| 98  | NC    | -            | -              | -          |
| 99  | NC    | -            | -              | -          |
| 100 | NC    | -            | -              | -          |

### 3.1 GPIO pins and alternate functions

Most of the digital I/O pins are GPIO pins. A GPIO pin is in either primary (GPIO) mode or alternate function (peripheral) mode; see the GPIOx chapter (Chapter 12). Table 3.2 lists every GPIO pin with its primary function, its reset-selected alternate function (AF0), and the reset state of its PxSEL, PxOUT, PxDIR, and PxREN bits. The remaining alternate functions each pin can select (AF1 to AF7) are in the alternate-function matrix, Table 3.3.

Table 3.2: GPIO Pin Functions and Reset Values

| Pin  | Primary Function | Alternate Function (AF0) | Reset Values  |          |            |              |
|------|------------------|--------------------------|---------------|----------|------------|--------------|
|      |                  |                          | SEL           | OUT      | DIR        | REN          |
| P0.0 | GPIO0            | CS_FLASH                 | 0 (Primary)   | 1 (High) | 1 (Output) | 0 (Disabled) |
| P0.1 | GPIO1            | MISO0                    | 1 (Alternate) | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P0.2 | GPIO2            | MOSI0                    | 1 (Alternate) | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P0.3 | GPIO3            | SCK0                     | 1 (Alternate) | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P0.4 | GPIO4            | LFXT                     | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P0.5 | GPIO5            | HFXT                     | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P0.6 | GPIO6            | TRAP                     | 1 (Alternate) | 0 (Low)  | 1 (Output) | 0 (Disabled) |
| P0.7 | BOOT             | -                        | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 1 (Enabled)  |
| P1.0 | GPIO8            | CS1                      | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P1.1 | GPIO9            | MISO1                    | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P1.2 | GPIO10           | MOSI1                    | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P1.3 | GPIO11           | SCK1                     | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P1.4 | GPIO12           | TX0                      | 1 (Alternate) | 0 (Low)  | 1 (Output) | 0 (Disabled) |
| P1.5 | GPIO13           | RX0                      | 1 (Alternate) | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P1.6 | GPIO14           | TX1                      | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P1.7 | GPIO15           | RX1                      | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P2.0 | GPIO16           | T0CMP0                   | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P2.1 | GPIO17           | T0CMP1                   | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P2.2 | GPIO18           | T0CAP0                   | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P2.3 | GPIO19           | T0CAP1                   | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |
| P2.4 | GPIO20           | T1CMP0                   | 0 (Primary)   | 0 (Low)  | 0 (Input)  | 0 (Disabled) |

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| Pin  | Primary Function | Alternate Function (AF0) | Reset Values |         |           |              |
|------|------------------|--------------------------|--------------|---------|-----------|--------------|
|      |                  |                          | SEL          | OUT     | DIR       | REN          |
| P2.5 | GPIO21           | T1CMP1                   | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P2.6 | GPIO22           | T1CAP0                   | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P2.7 | GPIO23           | T1CAP1                   | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.0 | GPIO24           | SDA0                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.1 | GPIO25           | SCL0                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.2 | GPIO26           | SDA1                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.3 | GPIO27           | SCL1                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.4 | GPIO28           | DTP0                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.5 | GPIO29           | DTP1                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.6 | GPIO30           | DTP2                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P3.7 | GPIO31           | DTP3                     | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.0 | GPIO32           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.1 | GPIO33           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.2 | GPIO34           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.3 | GPIO35           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.4 | GPIO36           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.5 | GPIO37           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.6 | GPIO38           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P4.7 | GPIO39           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P5.0 | GPIO40           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P5.1 | GPIO41           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P5.2 | GPIO42           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P5.3 | GPIO43           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P5.4 | GPIO44           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P5.5 | GPIO45           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 0 (Disabled) |
| P5.6 | GPIO46           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 1 (Enabled)  |
| P5.7 | GPIO47           | -                        | 0 (Primary)  | 0 (Low) | 0 (Input) | 1 (Enabled)  |

When a pin is in alternate function mode (its PxSEL bit is 1), its 3-bit PxAFS field selects which of up to eight alternate functions (AF0 to AF7) drives the pad. Table 3.3 lists, for every GPIO pin, the function selected by each PxAFS value. Plane 0 is the pin's AF0 function and is the reset selection; planes marked **Hi-Z** have no function assigned for that pin and configure the pad as a high-impedance input. The superscript on each function denotes its direction (<sup>in</sup> input, <sup>out</sup> output, <sup>io</sup> bidirectional), and a <sup>†</sup> marks each pin's reset plane (its PxAFS reset value). While a pin is in primary mode the selected plane has no effect on the pad, but it still routes relocatable peripheral inputs (Chapter 12).

Table 3.3: GPIO Alternate Function Selection (PxAFS)

| Pin  | PxAFS field value (selected AF plane) |                       |                       |                       |                       |                       |                       |                       |
|------|---------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|      | 0 (AF0)                               | 1                     | 2                     | 3                     | 4                     | 5                     | 6                     | 7                     |
| P0.0 | CS_FLASH <sup>out†</sup>              | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   |
| P0.1 | MISO0 <sup>io†</sup>                  | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  |
| P0.2 | MOSI0 <sup>out†</sup>                 | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    |
| P0.3 | SCK0 <sup>out†</sup>                  | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    |
| P0.4 | LFXT <sup>io†</sup>                   | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> |

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| Pin  | PxAFS field value (selected AF plane) |                                |                       |                       |                       |                       |                       |                       |
|------|---------------------------------------|--------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|      | 0 (AF0)                               | 1                              | 2                     | 3                     | 4                     | 5                     | 6                     | 7                     |
| P0.5 | HFXT <sup>io†</sup>                   | T1CMP1 <sup>out</sup>          | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> |
| P0.6 | TRAP <sup>out†</sup>                  | SCK1 <sup>out</sup>            | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> |
| P0.7 | Hi-Z <sup>†</sup>                     | MOSI1 <sup>out</sup>           | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> |
| P1.0 | CS1 <sup>in†</sup>                    | T0CMP0 <sup>out</sup>          | TX1 <sup>out</sup>    | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  |
| P1.1 | MISO1 <sup>io†</sup>                  | T0CMP1 <sup>out</sup>          | T0CMP0 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    |
| P1.2 | MOSI1 <sup>io†</sup>                  | T1CMP0 <sup>out</sup>          | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> |
| P1.3 | SCK1 <sup>io†</sup>                   | T1CMP1 <sup>out</sup>          | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> |
| P1.4 | TX0 <sup>out†</sup>                   | SDA1 <sup>io</sup>             | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> |
| P1.5 | RX0 <sup>io†</sup>                    | SCL1 <sup>io</sup>             | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> |
| P1.6 | TX1 <sup>out†</sup>                   | SDA0 <sup>io</sup>             | TX0 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   |
| P1.7 | RX1 <sup>io†</sup>                    | SCL0 <sup>io</sup>             | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> |
| P2.0 | T0CMP0 <sup>out†</sup>                | TX1 <sup>out</sup>             | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> |
| P2.1 | T0CMP1 <sup>out†</sup>                | RX1 <sup>io</sup>              | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    |
| P2.2 | T0CAP0 <sup>in†</sup>                 | SDA1 <sup>io</sup>             | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  |
| P2.3 | T0CAP1 <sup>in†</sup>                 | SCL1 <sup>io</sup>             | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    |
| P2.4 | T1CMP0 <sup>out†</sup>                | TX0 <sup>out</sup>             | T0CMP1 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> |
| P2.5 | T1CMP1 <sup>out†</sup>                | RX0 <sup>io</sup>              | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | SCK1 <sup>out</sup>   |
| P2.6 | T1CAP0 <sup>in†</sup>                 | SDA0 <sup>io</sup>             | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> |
| P2.7 | T1CAP1 <sup>in†</sup>                 | SCL0 <sup>io</sup>             | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> |
| P3.0 | SDA0 <sup>io†</sup>                   | T0CAP0 <sup>in</sup>           | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> |
| P3.1 | SCL0 <sup>io†</sup>                   | T0CAP1 <sup>in</sup>           | TX1 <sup>out</sup>    | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   |
| P3.2 | SDA1 <sup>io†</sup>                   | T1CAP0 <sup>in</sup>           | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  |
| P3.3 | SCL1 <sup>io†</sup>                   | T1CAP1 <sup>in</sup>           | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    |
| P3.4 | DTP0 <sup>io†</sup>                   | T0CMP0 <sup>out</sup>          | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   |
| P3.5 | DTP1 <sup>io†</sup>                   | T0CMP1 <sup>out</sup>          | RX0 <sup>io</sup>     | T0CMP0 <sup>out</sup> | T1CMP0 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  |
| P3.6 | DTP2 <sup>io†</sup>                   | T1CMP0 <sup>out</sup>          | T0CMP0 <sup>out</sup> | T0CMP1 <sup>out</sup> | T1CMP1 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | MISO1 <sup>io</sup>   |
| P3.7 | DTP3 <sup>io†</sup>                   | T1CMP1 <sup>out</sup>          | T0CMP1 <sup>out</sup> | T1CMP0 <sup>out</sup> | SCK1 <sup>out</sup>   | MOSI1 <sup>out</sup>  | TX0 <sup>out</sup>    | TX1 <sup>out</sup>    |
| P4.0 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P4.1 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P4.2 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P4.3 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P4.4 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P4.5 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P4.6 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P4.7 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.0 | Hi-Z                                  | NFC_RF_CLK <sup>in†</sup>      | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.1 | Hi-Z <sup>†</sup>                     | NFC_RF_RX <sup>in</sup>        | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.2 | Hi-Z <sup>†</sup>                     | NFC_FIELD_DETECT <sup>in</sup> | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.3 | Hi-Z <sup>†</sup>                     | NFC_RF_TXMOD <sup>out</sup>    | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.4 | Hi-Z <sup>†</sup>                     | NFC_RF_TX_EN <sup>out</sup>    | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.5 | Hi-Z <sup>†</sup>                     | NFC_AFE_EN <sup>out</sup>      | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.6 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |
| P5.7 | Hi-Z <sup>†</sup>                     | Hi-Z                           | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  | Hi-Z                  |

## 4 Memory map

|           |                                      |                                                                  |
|-----------|--------------------------------------|------------------------------------------------------------------|
| 0x00000   | Boot ROM, 8 KiB                      | Shared window<br>(all harts, arbitrated)                         |
| 0x02000   |                                      |                                                                  |
| 0x04000   | Peripheral registers, 4 KiB          |                                                                  |
| 0x05000   | CLINT, 4 KiB                         |                                                                  |
| 0x06000   | Mutex bank, 4 KiB                    |                                                                  |
| 0x07000   | IRQ router, 4 KiB                    | Private to each hart<br>(not arbitrated)                         |
| 0x08000   | Private TCM, 8 KiB per hart          |                                                                  |
| 0x0A000   | Private TCM (mirror), 8 KiB per hart | Shared window<br>(all harts, arbitrated)                         |
| 0x0C000   | NPU staging RAM, 16 KiB              |                                                                  |
| 0x10000   | Shared RAM, 64 KiB                   |                                                                  |
| 0x20000   | TCM apertures 0 to 4, 5 × 16 KiB     | Shared window<br>(hart 0's read-only view<br>of each hart's TCM) |
| 0x34000   |                                      | External SPI flash<br>(hart 0 only)                              |
| 0x40000   |                                      |                                                                  |
| 0xFFFFFFF | SPI flash (XIP)                      |                                                                  |

Figure 4.1: Castalia address space: one gapless 32-bit column, drawn once for all five harts; grey rows are unmapped and read zero.

The 32-bit memory bus is a single von Neumann address space: the ROM, the RAMs and the memory-mapped peripheral registers are all reached through one bus, and a pointer can address any of them. An access to an unmapped address reads zero, and a write to one is ignored; no bus error is raised.

The ROM holds the boot loader and the ROM monitor and is read-only. Each RAM can be read and written and may hold both code and data. The chip carries no on-die non-volatile memory: it boots from an external SPI flash device, and hart 0, and only hart 0, can additionally read that flash directly, and execute in place from it, through the extended-flash window at 0x40000 and above (the last region in Figure 4.1; see Section 13.4.6). At reset every hart fetches from the boot ROM at 0x00000 and dispatches on its hart ID: hart 0 copies the application image out of the flash into memory and enters it, and the remaining harts park until hart 0 loads and launches them (Section 5.3). Each RAM and ROM block can be individually powered off and on through the SYSTEM peripheral; a block that is powered off draws less static power, does not respond, and a RAM loses its contents.

Figure 4.1 is a single address column drawn for all five harts, and its braces name who reaches each region. Everything below 0x40000 other than the private band is the shared window: every hart reaches it through the arbiter, and all five harts see the same memory there (Chapter 5). The private band is the exception: at those addresses a hart reaches its own TCM instead of the shared bus, so the same address



names a different 8 KiB of RAM in each hart, and no hart can reach another's directly.

Table 4.1 is the same column as a table, one row per decoded region with its access attributes; the register-level index of every peripheral is Appendix B.

Table 4.1: Address map (R = readable, W = writable, X = executable; unmapped rows read zero)

| Start      | End        | Size        | Region                                                                                                                         | Attr. |
|------------|------------|-------------|--------------------------------------------------------------------------------------------------------------------------------|-------|
| 0x00000000 | 0x00001FFF | 8 KiB       | Boot ROM                                                                                                                       | R/X   |
| 0x00002000 | 0x00003FFF | 8 KiB       | Unmapped (reads zero)                                                                                                          | -     |
| 0x00004000 | 0x00004FFF | 4 KiB       | Peripheral registers                                                                                                           | R/W   |
| 0x00005000 | 0x00005FFF | 4 KiB       | CLINT                                                                                                                          | R/W   |
| 0x00006000 | 0x00006FFF | 4 KiB       | Mutex bank                                                                                                                     | R/W   |
| 0x00007000 | 0x00007FFF | 4 KiB       | IRQ router                                                                                                                     | R/W   |
| 0x00008000 | 0x00009FFF | 8 KiB       | Private TCM (private to each hart)                                                                                             | R/W/X |
| 0x0000A000 | 0x0000BFFF | 8 KiB       | Private TCM (mirror) (private to each hart; Alias: the band decodes 16 KiB and the array answers 8 KiB, so it repeats 2 times) | R/W/X |
| 0x0000C000 | 0x0000FFFF | 16 KiB      | NPU staging RAM                                                                                                                | R/W/X |
| 0x00010000 | 0x0001FFFF | 64 KiB      | Shared RAM                                                                                                                     | R/W/X |
| 0x00020000 | 0x00023FFF | 16 KiB      | TCM aperture 0 (hart 0 read-only view of a hart's TCM)                                                                         | R     |
| 0x00024000 | 0x00027FFF | 16 KiB      | TCM aperture 1 (hart 0 read-only view of a hart's TCM)                                                                         | R     |
| 0x00028000 | 0x0002BFFF | 16 KiB      | TCM aperture 2 (hart 0 read-only view of a hart's TCM)                                                                         | R     |
| 0x0002C000 | 0x0002FFFF | 16 KiB      | TCM aperture 3 (hart 0 read-only view of a hart's TCM)                                                                         | R     |
| 0x00030000 | 0x00033FFF | 16 KiB      | TCM aperture 4 (hart 0 read-only view of a hart's TCM)                                                                         | R     |
| 0x00034000 | 0x0003FFFF | 48 KiB      | Unmapped (reads zero)                                                                                                          | -     |
| 0x00040000 | 0xFFFFFFFF | 4194048 KiB | SPI flash (XIP) (hart 0 only; ; (addr + SPI0FOS) mod 2{24})                                                                    | R/X   |

Every register owns one word-aligned 4-byte slot, and registers are 32 bits wide except those of GPIOx (PxDIR, PxIE, PxIES, PxIF, PxIN, PxOUT, PxOUTC, PxOUTS, PxOUTT, PxREN, PxSEL, PxTASK); I2Cx (I2CxAMR, I2CxAR, I2CxFCR, I2CxMRX, I2CxMTX, I2CxSR, I2CxSRX, I2CxSTX); SPIx (SPIxSR); SYSTEM (BLOCKPWR, CLKDIVCR, CRCDATA, CRCSTATE, DCO0BIAS, DCO1BIAS, SYSCLKCR, WDTCSR, WDTSR); TIMERx (TIMxSR); UARTx (UARTxBR, UARTxCR, UARTxRX, UARTxSR, UARTxTX), which are narrower and occupy the low-order bits of their slot. The TCM apertures are the one sanctioned way across that line: hart 0, the orchestrator, reads hart  $h$ 's private TCM through the aperture at  $0x20000 + h \times 0x4000$ . The aperture stride is 16 KiB whatever the TCM size, so a 8 KiB TCM is mirrored inside its aperture. The apertures are read-only, and any other hart reading them gets zeros (Chapter 5). The extended-flash window is likewise hart 0's alone; the other harts' accesses in that range read zeros without stalling.

The interrupt vector table (IVT) begins at 0x8000, the base of each hart's TCM, and holds one 4-byte jump instruction per vector (Chapter 9).

Peripheral registers occupy one 4-byte slot each, at 4-byte aligned offsets from the instance base address, whatever their width. The GPIOx port registers are 8 bits wide, one bit per pin, and occupy byte lane 0 of their slot; a 32-bit register uses the whole slot. The register index in Appendix B gives the width of

every register.

## 5 Multi-core architecture

Castalia is a five-hart (five-core) multiprocessor built from five identical VestaRV cores. Each hart is a complete, single-issue, in-order RV32IMAC processor with its own private *tightly-coupled memory* (TCM), so the common case (instruction fetch and private data access) never contends with the other harts and runs at full core clock. Everything else in the low address space is *shared*: a single boot ROM, all peripherals, the inter-processor interrupt hardware (CLINT), a hardware mutex bank, an interrupt router, the NPU staging RAM, and 64 KiB of bulk RAM, all reached through one serializing round-robin arbiter. The main features of the multi-core architecture are listed below:

- 5 identical VestaRV harts, each identified by the read-only `mhartid` CSR (address `0xF14`)
- One shared boot ROM at `0x0`: all five harts reset to PC `0x0` and dispatch on `mhartid` (see Section 5.3)
- 8 KiB of private TCM per hart at `0x8000` to `0x9FFF` (the same local address in every hart), holding each hart's vector table, code, data, and stack
- All peripherals shared by all five harts at their legacy `0x4000`-page addresses, with the arbiter guaranteeing register-access atomicity
- 64 KiB of shared bulk RAM at `0x10000` to `0x1FFFF` for locks, mailboxes, staged tile programs, and inter-hart data
- CLINT with per-hart software interrupts (inter-processor interrupts) and a shared 64-bit `mtime` with per-hart compare registers
- 16 single-instruction hardware mutexes (MUTEX bank)
- Claim/complete peripheral interrupt delivery (IRQROUTER): any peripheral interrupt can be routed to any hart over one external-interrupt wire per hart, with exactly-once claiming
- LR/SC and AMO atomic instructions supported to shared RAM, including sub-word shared stores (byte lanes)
- Instruction fetch from shared memory is supported (the boot ROM is fetched this way by construction); parked harts sleep via the custom sleep instruction and are woken by a software interrupt; no busy-wait power burn

### 5.1 Harts and Roles

All five harts are instances of the same *hart tile*: an unmodified VestaRV core plus its own private TCM, replicating the proven single-core memory path, with every connection to the rest of the chip registered at the tile boundary. Since every hart sees the same address map (shared boot ROM, shared peripherals, private TCM at the same local address), the five harts are architecturally identical and code is hart-portable by construction.

Hart 0 is additionally the *management hart*. It is the only hart attached to the SPI-flash boot path and the extended flash region ( $\geq 0x40000$ ), and by convention it owns system management: the SYSTEM controller's clock configuration registers reprogram the master clock that all five harts run on, so only the

management hart may touch them, and only after quiescing the other harts (see the SYSTEM chapter). Peripheral interrupts are uniform across harts: every hart, hart 0 included, routes and receives them through its IRQROUTER row and the claim/complete mechanism (see the IRQROUTER chapter).

Hart 0 (mhartid zero, the management and boot hart) is additionally the *orchestrator*, and it is the one hart that is not a corner tile. The channel harts, mhartid 1 through 4, are hardened *hart tile* macros at the corners of the die; the orchestrator is soft logic placed in the centre band alongside the shared fabric, with its own 8 KiB TCM at the same private 0x8000 to 0x9FFF window. It is architecturally identical to the tiles (same ISA, same private memory map, same mhartid discovery) and differs in exactly three ways that software can observe:

- **It is always on, and every other hart is not.** The orchestrator sits outside the switched-rail power fabric entirely: there are no header switches for the centre band, so PWRCCR's bit 0 reads 0 and ignores writes, as it always has. What changes in this configuration is the other side of that sentence: *every* channel hart 1 to 4 now has a gate bit, including the corner tile that a four-hart chip would have called hart 0 and kept always-on. A blanket PWRCCR write therefore gates all 4 channel tiles and leaves the orchestrator running (see the PWRCTRL chapter).
- **It boots the chip and starts the others.** Boot mastership is the orchestrator's, because the orchestrator is hart 0: it is the hart attached to the SPI-flash boot path and the extended flash region, it stages each channel hart's image into that hart's loader mailbox row and ignites it with a CLINT software interrupt (Section 5.3), and it holds the management privilege from reset rather than receiving it in a hand-over.
- **It can read every hart's private TCM.** Each hart's TCM is also visible read-only in the shared window, hart  $h$  at  $0x20000 + 0x4000 \times h$  (the orchestrator's own included, at  $0x20000$ ), so the indexing is uniform. Each aperture is 16 KiB wide and the 8 KiB TCM repeats through it, because the aperture decode keeps the full window and the tile drops the top word-index bit. Only the management hart may read them; any other hart reads zero and has its write dropped, and writes are dropped for everyone (the windows are read-only by design: a write path into a running core's memory is a coherence hazard). A window belonging to a power-gated tile completes normally and returns zeros, so software checks the PWRCTRL sequencer state before trusting what it reads; zero is a legal TCM value.

The intended firmware model follows: the orchestrator boots the chip, starts and coordinates the channel harts, owns the front-end and its interrupts, inspects any channel hart's working memory through its TCM window, and stays up across any power-gating of the tiles.

Figure 5.1 follows one aperture read through the hardware, because the window is not a second port onto a shared memory: it is a request that leaves the shared fabric, borrows the target tile's own memory for four cycles, and comes back.

Software discovers which hart it is running on by reading the mhartid CSR:

```
csrr    t0, mhartid    # t0 = 0 .. (number of harts - 1)
```

## 5.2 The Shared Window

Everything below the extended flash region except each hart's private TCM is behind the arbiter. Every hart sees the same physical devices at the same addresses:

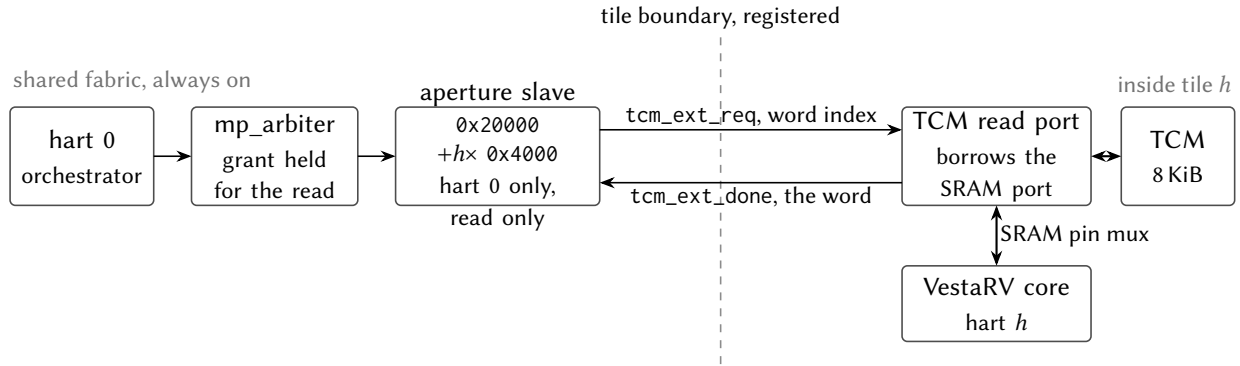


Figure 5.1: One read through a TCM aperture, from hart 0 to the target tile’s TCM and back.

| Address range      | Device                   | Notes                                               |
|--------------------|--------------------------|-----------------------------------------------------|
| 0x00000 to 0x01FFF | Boot ROM (8 KiB)         | Shared, read-only; all harts reset here             |
| 0x02000 to 0x03FFF | Unmapped                 | The ROM is smaller than its page; reads return zero |
| 0x04000 to 0x04FFF | Peripheral slots         | 16 slots of 256 bytes, legacy numbering             |
| 0x05000 to 0x05FFF | CLINT                    | IPIs (MSIPx), MTIME/MTIMECMPx                       |
| 0x06000 to 0x06FFF | MUTEX bank               | 16 hardware mutexes                                 |
| 0x07000 to 0x07FFF | IRQROUTER                | Per-hart routing rows + CLAIM/COMPLETE              |
| 0x0C000 to 0x0FFFF | NPU staging RAM (16 KiB) | Vector storage; NPU-owned during a run              |
| 0x10000 to 0x1FFFF | Shared bulk RAM (64 KiB) | Locks, mailboxes, inter-hart data                   |

(0x8000 to 0x9FFF is each hart’s **private** TCM and never reaches the arbiter.) The sixteen peripheral slots at 0x4000 keep the original single-core VestaRV slot numbering (GPIO0 at 0x4000, UART0 at 0x4400, SYSTEM at 0x4900, and so on), so single-core driver code runs unchanged; the difference is that every hart can now reach every peripheral.

Instruction fetch from the shared window is supported: the boot flow depends on it, and programs may execute directly from shared RAM. One caveat: compressed (RVC) instructions in shared-window code are not currently validated; code intended to execute from shared memory should be built without the C extension. Code in the private TCM has no such restriction.

### 5.2.1 Arbitration and Stalling

A round-robin arbiter serializes all shared-window transactions. The arbiter grants one hart at a time and holds the grant until that hart’s whole transaction completes, then advances the round-robin pointer, so no hart can be starved. While a hart waits for its grant, its clock is stalled transparently; software never sees a partial transaction and needs no special code for shared accesses. Because a stalled hart contributes nothing until its turn comes, frequently-pollled data structures (spin locks in particular) should use a backoff strategy (see Section 5.4).

An AMO (atomic read-modify-write) instruction holds the arbiter grant across both halves of its read-modify-write pair, so AMOs to shared RAM are atomic with respect to all other harts. A shared-window access costs a few master-clock cycles uncontended (versus a single cycle to the private TCM), which is why per-hart code and stacks belong in the TCM. Figure 5.2 shows one uncontended read at the arbiter’s own pins.

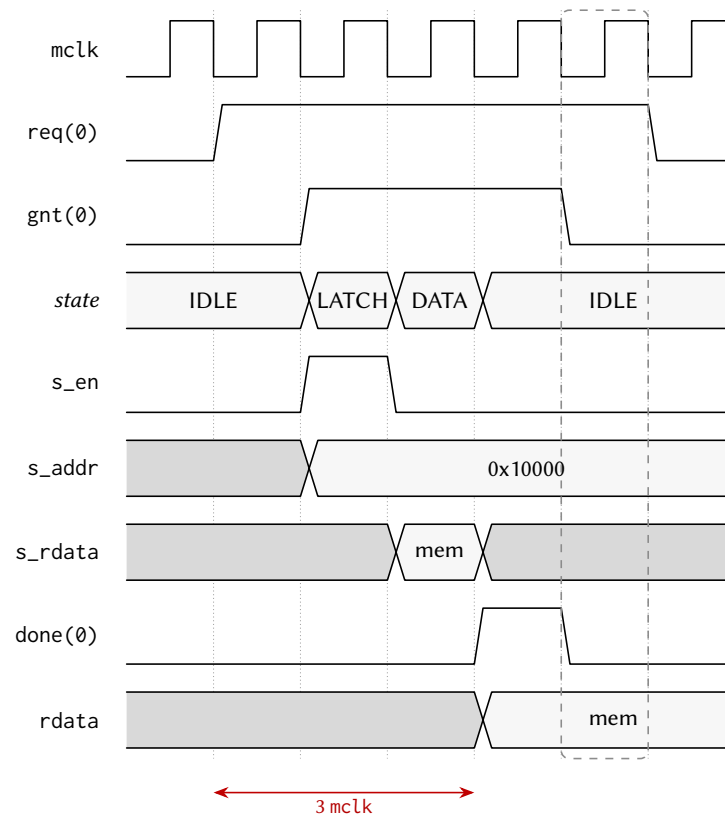


Figure 5.2: Timing diagram of one uncontended shared-window read at the mp\_arbiter pins; in the dashed ghost window req is stale-high for one cycle after done and is masked by need\_release.

One slave does not fit that walk. A read through a TCM aperture has to leave the shared fabric, borrow the target tile's own memory and come back, which takes six `mclk`, so the aperture slave holds the arbiter in `LATCH` with `s_stall` until the word is actually in its hand. It is the only slave in the design that does this, and the only one that can: nothing else could have been granted meanwhile, because the arbiter had already pinned the bus to this transaction. Figure 5.3 is the same set of pins as Figure 5.2, on that read.

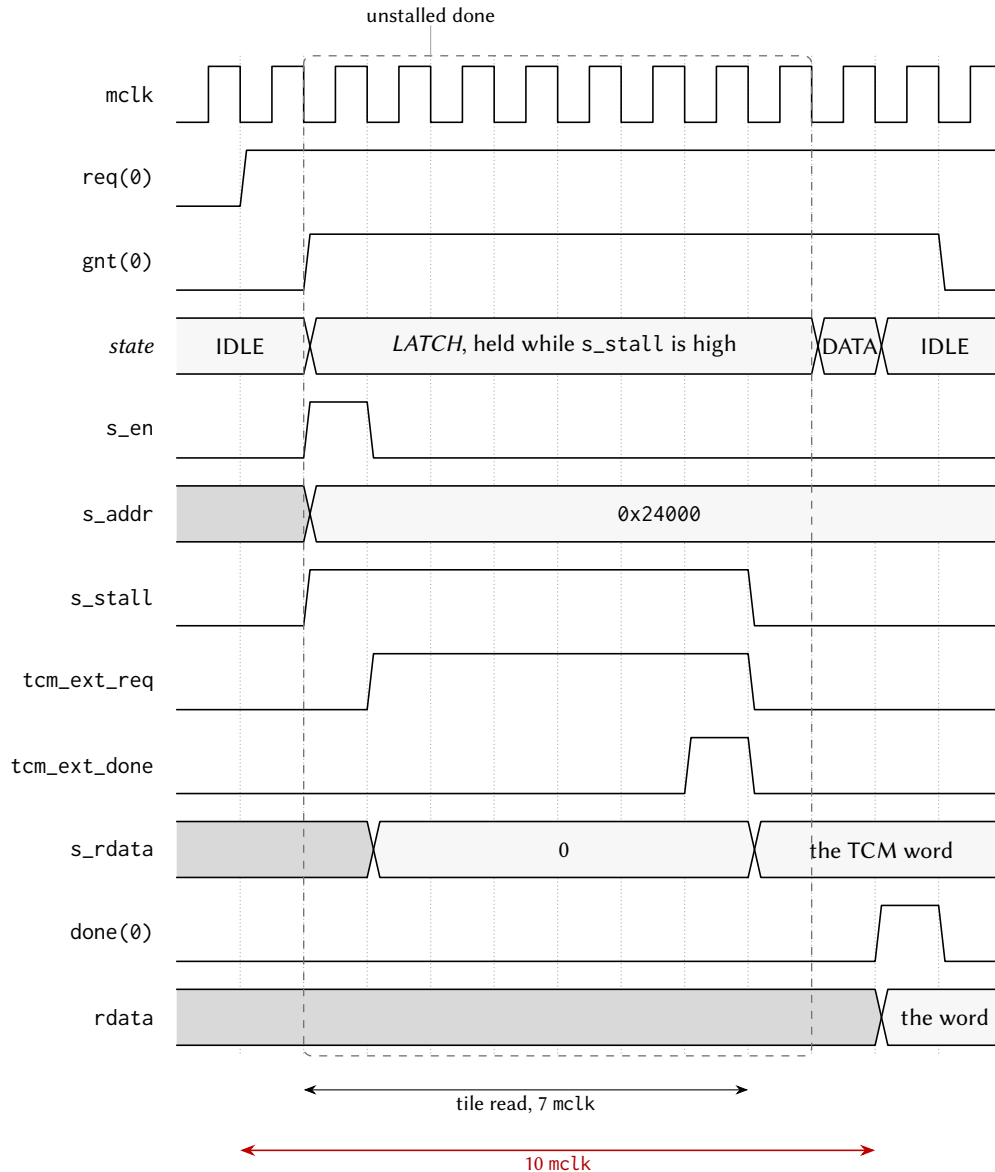


Figure 5.3: Timing diagram of one stalled shared-window read through a TCM aperture; the grey tick marks where an unstalled read would have completed, and the tile read (*SETTLE*, *READ*, *LATCH*, then one `mclk` of `tcm_ext_done`) runs while `s_stall` is high.

### 5.3 Boot and Tile Loading

All five harts come out of reset at PC `0x0` and fetch the shared boot ROM through the arbiter. The ROM immediately dispatches on `mhartid`; the whole flow is summarized in Figure 5.4:

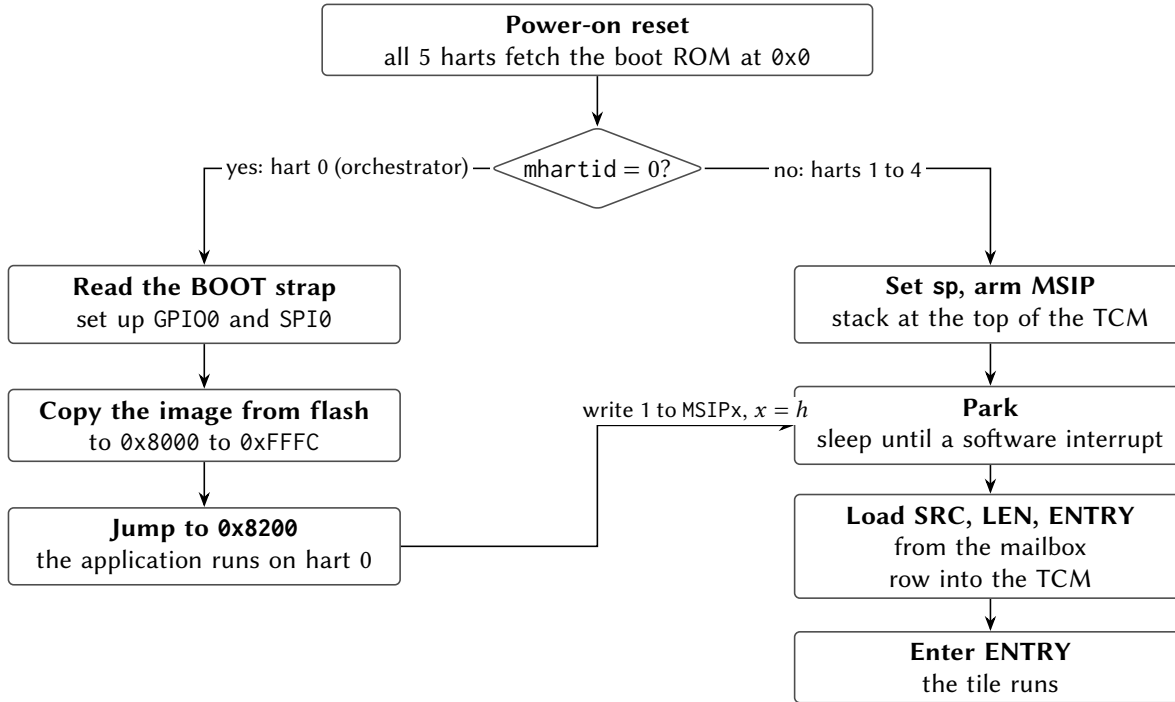


Figure 5.4: Flow chart of boot and tile loading across the harts.

**Hart 0** runs the full boot sequence, exactly as on the single-core chip: it configures GPIO0/SPI0, copies the program from SPI flash into the load window  $0x8000$  to  $0xFFFC$  (its TCM plus the NPU staging RAM), and jumps to the program at  $0x8200$ . Before any tile can be released, the boot ROM zeroes the shared-RAM mailbox region  $0x10000$  to  $0x107FF$ : shared RAM powers up with *unknown* contents on silicon, and this write-before-read guarantee is what makes “mailbox reads as zero” checks safe.

**Harts 1 to 4** set up a minimal interrupt context (a stack at the top of their TCM and a software-interrupt vector) and park in low-power sleep. A parked tile wakes only when hart 0 (or any running program) sends it a CLINT software interrupt. On wake, the ROM-resident loader reads the tile’s *loader mailbox row* in shared RAM:

| Mailbox  | Address             | Meaning                                                |
|----------|---------------------|--------------------------------------------------------|
| SRC[h]   | $0x10500 + 16h + 0$ | Word-aligned source address in shared space            |
| LEN[h]   | $0x10500 + 16h + 4$ | Words to copy to the tile’s TCM at $0x8000$ (0 = none) |
| ENTRY[h] | $0x10500 + 16h + 8$ | PC the tile enters with                                |

The loader clears the tile’s own MSIP level, copies LEN[h] words from SRC[h] into the tile’s private TCM at  $0x8000$  (a LEN of 2048 loads the full 8 KiB image: vector table, code, and interrupt handlers included), and enters ENTRY[h] with the stack pointer set to the top of the tile’s TCM and all other registers undefined.

The canonical release sequence for hart  $h$  is therefore: stage the tile’s program somewhere in shared RAM, write SRC[h] and LEN[h] and ENTRY[h], then write 1 to the hart’s MSIP register ( $0x5000 + 4h$ ). Because the mailbox row is consumed by the ROM only at wake time, the row must be complete *before* the MSIP write. Software interrupts sent to a tile *after* it has been loaded are handled by the loaded program’s own vector table, so the same CLINT mechanism serves both boot-time loading and run-time inter-processor interrupts.

**Stacks:** each hart needs its own stack in its own private TCM. When an interrupt is taken, the hardware pushes the return program counter at  $sp - 4$  (see Section 9), so every hart, including a freshly loaded tile



hart, must have a valid stack pointer before enabling interrupts or sleeping. The boot ROM guarantees this for parked and freshly-entered tiles; application code keeps the convention of placing each hart's stack at the top of its TCM, growing downward.

### 5.3.1 Harvested (field-powered) boot

In configurations wired for field power, hart 0's boot path has a second entry, selected by a hardware strap sampled at reset and reported by the PWRCTRL controller. Before any SPI or flash activity, hart 0 polls the strap-valid flag and, if the harvested-boot strap is set, branches to a ROM-resident service path instead of the SPI-flash download described above. This path is built for a chip running on harvested field energy, where the multi-kilobyte flash copy is the single most power-hungry cold-start step and is therefore skipped entirely.

The harvested path keeps both MCLK and SMCLK on HFXT (the SPI path's mid-boot SMCLK-to-LFXT switch does not run, so the clock configuration is written explicitly rather than inherited), then power-gates every tile hart through PWRCTRL (the tiles are already ROM-parked in WFI, the sanctioned quiesced state, so gating them is safe at any hart count). It configures the port-6 GPIO alternate-function plane for the six off-die NFC analog-front-end wires, provisions the NFC tag identity (a ROM-default UID, ATQA and SAK) and a short status payload, and arms the NFC block in LISTEN mode with auto-read enabled, so the hardware answers Type-2 read commands with no firmware in the loop. Finally it arms its own software-interrupt vector with a harmless clear-and-return handler (a stray MSIP must not disturb the parked hart), divides MCLK down (the parked service loop needs no speed, and dynamic power scales with the divide), and sleeps.

On a configuration built without the NFC block, the NFC register writes land in an unused alias of the MUTEX page and are harmless; the path is written to never *read* an NFC register, because a read of that alias would claim a mutex.

The exact sequence, in the order the ROM executes it (register addresses in the peripheral chapters):

1. **Common entry (shared with the SPI path):** interrupts off, all memory blocks powered, stack pointer set, and the shared-RAM mailbox region 0x10000 to 0x107FC zeroed (the write-before-read contract). At a 24 MHz MCLK this zeroing, 512 word writes through the shared-bus arbiter, dominates the harvested cold-start (about 625  $\mu$ s of the roughly 690  $\mu$ s from gate release to sleep).
2. **Strap branch:** poll PWRSTS until PWSTRAPVLD, then branch on PWSTRAP. (The sample completed a handful of cycles after reset release, long before this code runs; the poll is cheap insurance.)
3. **Clock story, explicitly:** write SYSCLOCKR = 0; both MCLK and SMCLK stay on HFXT. The SPI path's mid-boot switch of SMCLK to LFXT never runs on this path, so the choice is written rather than inherited.
4. **Gate the tiles:** write all-ones to PWRCTRL PWRCR. The register masks the write to the tile bits that exist, so the same ROM gates every tile hart at any hart count; the tiles are parked in their boot-ROM WFI, the sanctioned quiesced state.
5. **Pin mux:** select the alternate-function planes for the six off-die NFC AFE wires on port 6 pins 0 to 5 (PxSEL, then the AF-select fields). Pins 6 and 7 (the strap and PGOOD) are untouched and stay plain inputs.
6. **Provision the tag:** write the ROM-default identity (UID 0xC0FFEE01, ATQA 0x0044, SAK 0x00), point the payload index register at byte 0 with auto-increment, and write a three-byte status payload: 'H', 'V', then the live PWRSTS snapshot, the status record of the base chip. A fielded product replaces all of this by relaunching with real firmware.

7. **Arm LISTEN:** a *byte* store of NFCEN|LISTEN to the NFC control register's low lane. A full-word store would also write the second byte lane and clear the auto-read enable, whose reset default is set; auto-read is what lets the hardware answer Type-2 READs with the processor asleep.
8. **Stray-interrupt guard:** arm interrupt-vector slot 83 (the software interrupt) in hart 0's own TCM with a return stub, so an unexpected msip is cleared and returns the hart to sleep instead of vectoring into uninitialized memory.
9. **Slow down and sleep:** divide MCLK by eight (CLKDIVCR), then sleep. The clock reconfiguration is legal here because every other bus master is gated. From this point the NFC hardware serves readers autonomously; a field arrival can be taken as an interrupt only if firmware later opts in.

## 5.4 Synchronization

Castalia offers three synchronization mechanisms, from cheapest to most flexible. Figure 5.5 summarizes how to choose between them:

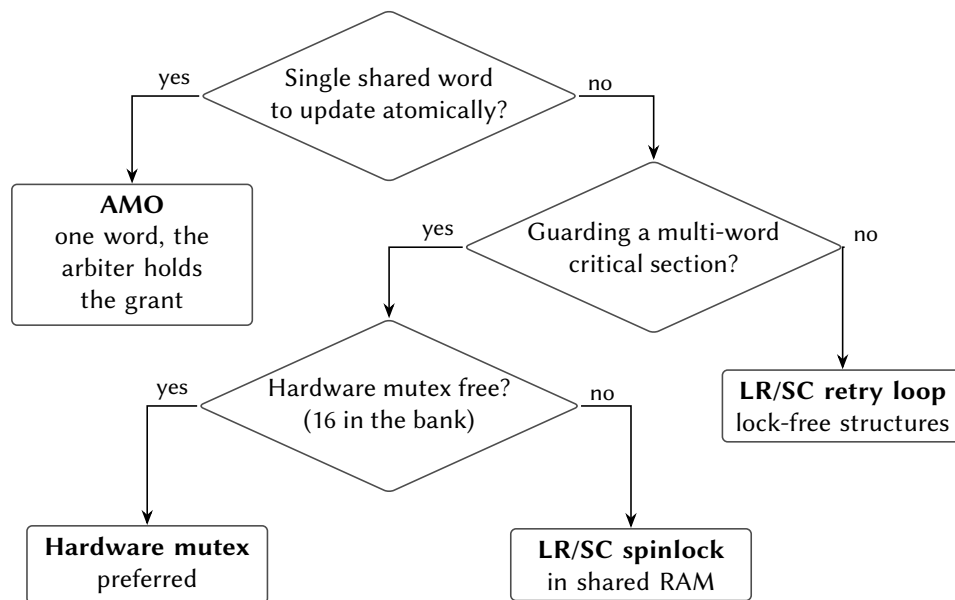


Figure 5.5: Decision chart for choosing a synchronization primitive.

1. **Hardware mutexes** (MUTEX bank): a single load claims a free mutex; a single store releases it. No retry loop hardware state, no reservation. This is the preferred lock for most uses. See the MUTEX chapter.
2. **LR/SC spinlocks in shared RAM:** standard RISC-V reservation-based locks. A failed SC has its write suppressed by the reservation unit, so it is safe under contention.
3. **AMO instructions to shared RAM:** atomic add, swap, and friends; the arbiter locks the grant across the read-modify-write pair.

Two rules apply to all of them:

- **Never** use LR/SC or AMO instructions on MUTEX bank addresses; hardware mutexes are claimed with plain loads and released with plain stores only.
- Under contention, spin with a *hart-scaled backoff*: delay between retries by an amount proportional to `mhartid` (for example  $\text{delay} = k \cdot (\text{mhartid} + 1)$ ) and bound the retry count. Multiple harts hammering the arbiter in lockstep can otherwise livelock a naive retry loop.
- Additionally, do not access the NPU staging RAM (0xC000 to 0xFFFF) while an NPU run may be active; poll the NPU's busy flag first (see the NPU chapter).

There are no data caches, so there is no cache-coherence protocol to consider: a shared-RAM write is globally visible as soon as the arbiter completes the transaction.

## 6 Reset, clocks and power

### 6.1 Reset

The chip is held in reset while the RESETN pin is low and leaves reset when it is released high. A power-on reset circuit asserts the internal reset while the core supply rises from zero and releases it once the supply has reached its operating level; the chip is in reset whenever either RESETN or the power-on reset is asserted. In reset, every memory-mapped register and every core state register is at its reset value. The RESETN input has an internal pull-up and may be left unconnected.

Every hart leaves reset fetching from the boot ROM at 0x00000. Hart 0 runs the boot loader; harts 1 to 4 park in the ROM until hart 0 releases them through the boot mailboxes in shared RAM (Section 5.3). The PWRCTRL boot gate can hold every hart in reset until a supply supervisor reports power good; see Section 6.4 and Chapter 18.

### 6.2 Boot mode selection

Hart 0's boot loader samples one strap after reset: GPIO pin P0.7, marked BOOT in Table 3.2. The pin is read once, through P0IN, after the boot loader has configured the SPI0 pins; it is an ordinary GPIO pin at every other time. Table 6.1 gives the two modes.

Table 6.1: Boot mode selected by the P0.7 strap.

| P0.7 at reset | Mode        | Behaviour                                                                                                        |
|---------------|-------------|------------------------------------------------------------------------------------------------------------------|
| Low           | ROM monitor | Hart 0 runs the interactive ROM monitor over UART0 at 115,200 baud (Chapter 25).                                 |
| High          | SPI flash   | Hart 0 reads the boot image from the SPI flash on SPI0, copies its segments into RAM and enters it (Chapter 26). |

On a field-powered board the harvested-boot strap sampled by PWRCTRL (PWRSTS) takes precedence: when it reads 1, hart 0 takes the harvested-boot path described in Chapter 18 before the P0.7 strap is consulted.

**Note:** The boot ROM reprograms the clock system during boot, so the SMCLK source at application entry is a boot-ROM implementation detail and may be the slow LFXT source. An application that uses an SMCLK-domain peripheral (a UART, SPI, I<sup>2</sup>C or a timer clock-source switch) selects its SMCLK source in the SYSTEM peripheral first (Section 16.3.1).

### 6.3 Clock tree

#### 6.3.1 Clock sources

The chip has four clock sources. HFXT and LFXT are external clocks supplied on the HFXT and LFXT pin functions (Section 3); DCO0 and DCO1 are on-chip digitally controlled oscillators whose frequency is set by the bias values in DCO0BIAS and DCO1BIAS, a higher bias giving a higher frequency. The design is characterised for a maximum clock frequency of 24 MHz, so HFXT must not exceed 24 MHz; LFXT is

intended to be 32.768 kHz so that time can be measured accurately. Operation above 24 MHz, from HFXT or from a DCO, is outside the characterised range and is not guaranteed.

HFXT and LFXT are enabled at reset and can be stopped with HFXTOFF and LFXTOFF; DCO0 and DCO1 are off at reset and are started with DC00ON and DC01ON. A source that is currently selected by either clock tree root stays enabled whatever its own bit says, so a mistaken write cannot stop the clock the chip is running on.

### 6.3.2 Roots and dividers

Two clocks form the roots of the tree in Figure 6.1: the main clock MCLK and the sub-main clock SMCLK. SMCLKSEL picks SMCLK's source from the four sources; MCLKSEL picks MCLK's from three of them and, on code 01, from SMCLK itself. That tap is taken after the SMCLK divider but before the SMCLKOFF gate, so a chip running MCLK out of SMCLK keeps its main clock when SMCLK is stopped for the peripherals. Each root may then be divided again by SYSMCLKDIV and SYSSMCLKDIV in CLKDIVCR. Both roots use glitch-free multiplexers, so a source can be changed while the chip runs.

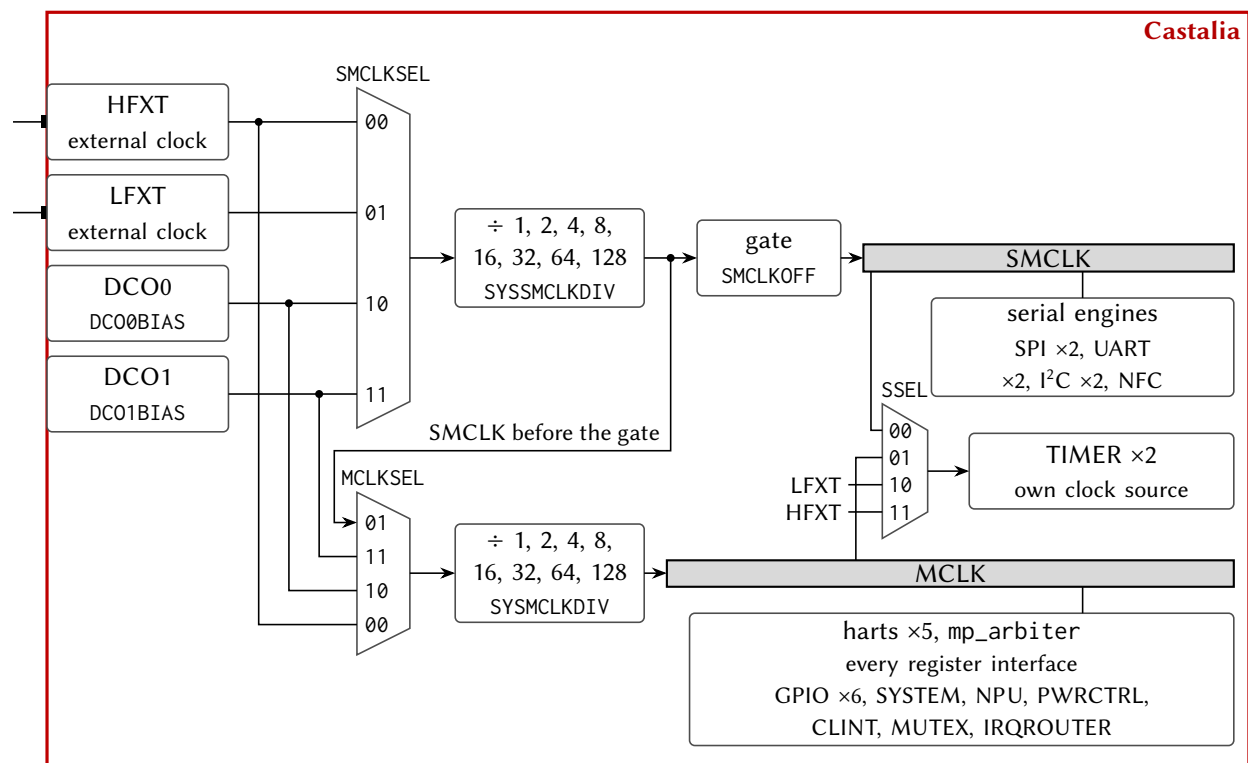


Figure 6.1: The clock tree: the four sources, the two selects and dividers, the SMCLK gate, and the blocks each root clocks.

MCLK drives all five harts, the multi-core arbiter and every always-on block, and it cannot be turned off. SMCLK drives the serial engines of the SPIx, UARTx, I2Cx and NFCx peripherals and is one of the TIMERx sources; it can be stopped with SMCLKOFF. The register interface of every peripheral is clocked from MCLK even where its engine runs on SMCLK, so a peripheral stays readable with SMCLK stopped. Two clocks come from outside this tree and are not reached by SYSCLOCKR: the NFC protocol core runs on the carrier recovered off die, and the debug TAP runs on the probe's TCK.

### 6.3.3 Clock state at reset and after boot

Both roots select HFXT at reset, so a valid clock on HFXT is required while the chip boots; the board must keep the oscillator driving HFXT running across reset rather than gating it from a pin the chip drives only after boot. After boot, software may move MCLK to another source and then stop HFXT.

The boot ROM reprograms the clock system during the SPI-flash boot and may leave SMCLK on LFXT when it enters the application. The SMCLK source at application entry is therefore a boot-ROM implementation detail, and every application that uses an SMCLK-domain peripheral, or a TIMERx clock source switch, selects its own SMCLK source in SYSCCLKCR first.

Because MCLK clocks every hart, a change to SYSCCLKCR or CLKDIVCR changes the clock of all five harts at once. By convention only hart 0 reconfigures the clocks, after the other harts have been quiesced.

## 6.4 Power domains

The chip has one always-on power domain and one switchable domain per channel tile. Everything the rest of the chip needs (the arbiter, the boot ROM, the shared RAM, the peripherals, the pad ring and hart 0) sits on the unswitched VDD rail. Each channel tile, harts 1 to 4, has a rail of its own behind PMOS header switches, with isolation cells clamping every signal that leaves the tile while it is unpowered, and a per-tile hardware sequencer in PWRCTRL that applies the controls in the legal order in both directions. Figure 6.2 draws the two domains and the four controls that gate one tile.

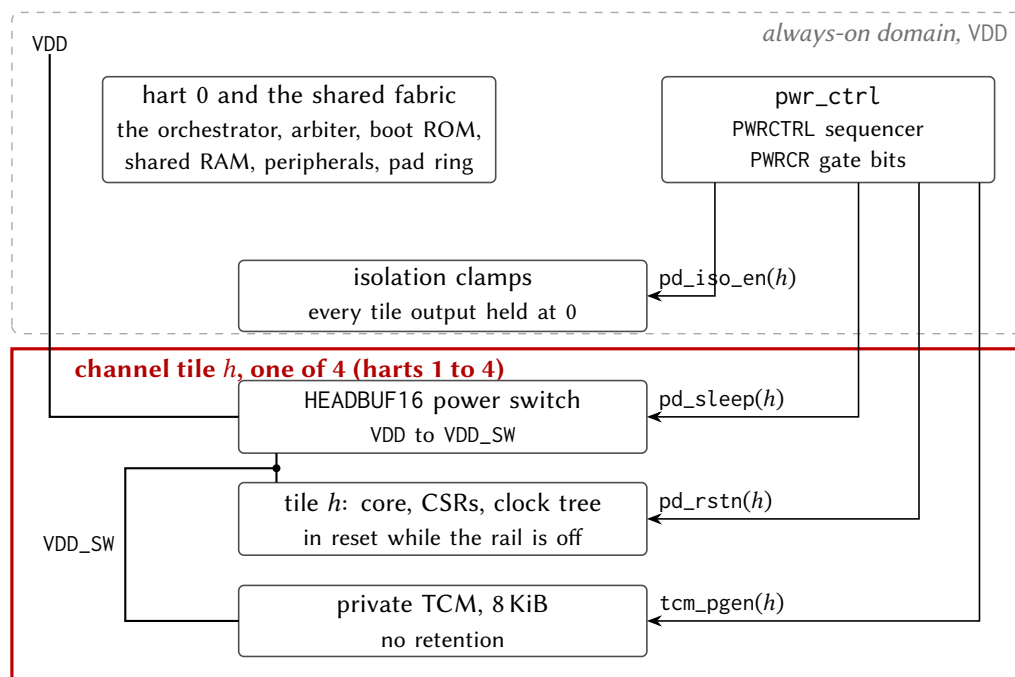


Figure 6.2: The always-on domain and one switched channel tile, with the four per-tile controls from pwr\_ctrl.

Hart 0, the orchestrator, is soft logic in the always-on centre band with no header switches of its own; it is permanently powered and cannot be gated. All domains power up on at reset, so chip boot does not depend on the controller, and PWRCTRL is inert until software gates a tile.

### 6.4.1 Sequencing

Four per-tile controls leave `pwr_ctrl`, and Figure 6.2 draws each at the gate it drives. `pd_iso_en(h)` clamps: it closes AND gates on the always-on side of the boundary, so every outbound signal of tile *h* reads 0 while it is asserted. `pd_sleep(h)` opens the switches: a HEADBUF16 chain carries VDD to the tile's VDD\_SW, and the chain is opened on the gate and closed again on the wake. `pd_rstn(h)` holds the tile in reset: `tile_rstn(h)` is `resetsn` AND `pd_rstn(h)` AND the field-power boot gate, asserted before the rail goes and released after it is back. `tcm_pgen` gates the TCM macro off the same `pd_sleep(h)`, so the TCM dies with the rail and a wake is a cold boot; its aperture then reads zeros, so software checks PWRSR first.

The sequencer applies them in the legal order in both directions. On the gate it clamps, then resets, then opens the switches; on the wake it closes the switches, waits for the rail to settle, unclamps, and then releases the reset. The PWRSR nibble of tile *h* reports the state the sequencer is in: 0 ON, 1 ISO, 2 RSTOFF, 3 OFF, 4 RAIL and 5 UNISO.

### 6.4.2 Cold gating

Gating is cold: a gated tile consumes no dynamic or switched-leakage power and retains nothing, including its register file, CSRs and private TCM. The isolation clamps drive every outbound tile signal to its inactive level, which is identical to the tile's reset state, so the rest of the chip observes a gated tile as a silent one; a read of its TCM aperture completes normally and returns zeros. When the tile is ungated it cold-boots through the shared boot ROM, parks, and is relaunched with the loader-row and software-interrupt sequence of Section 5.3. Any data a tile must keep across a power cycle is staged to shared RAM before gating.

### 6.4.3 Memory block power

The ROM, the two per-hart memory blocks and the first four shared RAM banks each have a power gate in BLOCKPWR (Section 16.3.2). A gated block halves its leakage and loses its contents; an access to a gated block returns undefined data. Every block powers up on at reset.

### 6.4.4 Hart sleep

Each hart can stop its own clock with the `sleep` instruction and is woken by an interrupt (Section 16.3.3). Sleep is a clock gate, not a power gate: the hart keeps its state and resumes where it stopped, so it is the right tool for a hart that waits for an event, while cold gating is the right tool for a tile that has no work for a long time.

### 6.4.5 Boot gate and field power

In configurations wired for field power (the harvested-energy NFC operating mode), PWRCTRL also decides when the chip may start executing, so that the harts are held quiescent until the harvested supply can carry a transaction. Two package pins take part: a power-good input (PGOOD, pin 69, P6.7) driven by an external supply supervisor, and a boot-mode strap (pin 68, P6.6) that selects harvested boot. Both pads reset to inputs with pull-downs, so an unconnected pin reads as power-not-good and normal boot, and a board that does not use the feature behaves as if it did not exist.

The boot gate is a hold-in-reset: an internal release signal is combined into every hart's outer reset, hart 0 included, so a held hart fetches nothing and issues no bus request, and the rest of the chip cannot distinguish it from a hart still in power-on reset. The strap is sampled once, eight MCLK cycles after reset release, and sets the hardware defaults of the gate: a strapped board arms the gate, makes PGOOD a release

condition and re-holds on brownout with no firmware at all, which resolves the problem that the gate holds the very harts that would otherwise program it. Software may add overrides through PWRWAKE: arm the gate, add PGOOD or the NFC field-detect level as release conditions, force a release, and choose between a one-shot release and re-holding whenever a release condition later drops. Field detect used this way is a wake path that consumes no interrupt vector; the same level remains available as an ordinary interrupt to firmware that is already running.

With re-hold enabled, a PGOOD drop re-asserts the hold on the next cycle and its return releases the harts into a fresh boot; since the chip keeps no retention, a full cold boot is the accurate model of losing and regaining the harvested rail. Two schemes are supported on the same silicon. In the batteryless scheme the gate is the mechanism: the chip holds until PGOOD, cold-boots the harvested-boot ROM path (Section 5.3.1), serves the reader, and re-holds when the field collapses. In the supercap duty-cycled scheme an external buffer stays charged, the gate is left disarmed after the first boot, and the service loop sleeps between transactions and resumes on the field-detect interrupt without re-entering reset. The register-level mechanism and its timing are in the PWRCTRL chapter.



# Part II Processor

## 7 VestaRV core

Every hart of the Castalia MCU is a VestaRV core, a 32-bit RISC-V processor with a single unified instruction and data bus. This chapter states what the core implements: its instruction set, its register conventions, its hart identities and its control and status registers. The RISC-V specifications named in Section 1.3 are normative for the instructions and registers themselves and are not restated here.

### 7.1 Instruction set

The ISA string of this build, the `-march` value the toolchain is given, is listed under *ISA string* in Table 2.2 and is `rv32imac_zba_zbb_zbs_zbc`. The same information is available at run time in the read-only `misa` register (Section 7.4). The extensions the core family implements are listed below with the configuration knob that selects each; which ones are present in this build is exactly what the ISA string says.

- **RV32I**, the base integer instruction set: 32 registers of 32 bits, integer arithmetic, logic, shifts, comparisons, loads and stores of bytes, half-words and words, branches and jumps. Always present.
- **M** (`isa.mul`, `isa.div`): hardware multiply, divide and remainder, signed and unsigned. Multiply is single-cycle; divide is iterative (Chapter 10).
- **A** (`isa.atomics`): load-reserved and store-conditional, and the atomic read-modify-write operations on 32-bit words. An atomic operation holds the shared-window arbiter grant across its whole read-modify-write pair, so it is atomic with respect to every hart (Section 5.4).
- **C** (`isa.compressed`): 16-bit encodings of the most common instructions. Code built with it is substantially smaller; its fetch-alignment cost is measured in Section 10.2.
- **Zba, Zbb, Zbc, Zbs** (`isa.bitmanip`): address generation (shift-and-add), basic bit manipulation (count, min and max, rotate, sign and zero extension, byte reversal), carry-less multiplication, and single-bit set, clear, invert and extract.
- **Zicsr**: the CSR access instructions. Always present.
- **Zicntr**: the cycle and instret counters, readable from the user views. Always present; time is not implemented (Section 7.4).

An instruction of an extension that is not configured into the build raises an illegal-instruction exception; what happens next depends on the trap delivery mode (Section 7.4.1).

Three instructions are specific to this core and are used by the runtime: `retirq` returns from a legacy-mode interrupt (Chapter 9), `sleep` gates the hart's clock until an interrupt arrives (Section 16.3.3), and `wake` is its counterpart. Their encodings are in the generated `periph.S`.

### 7.2 Registers and calling convention

The 32 general-purpose registers and their roles under the standard RISC-V calling convention (the `ilp32` ABI) are listed in Table 7.1. The compiler follows this convention, so hand-written assembly that follows it is drop-in compatible with compiled code. A register whose saver is the caller may be overwritten by any function call; a register whose saver is the callee is preserved across calls.

Table 7.1: General-purpose registers and their ABI roles.

| Register   | ABI name  | Role                                 | Saver  |
|------------|-----------|--------------------------------------|--------|
| x0         | zero      | Hard-wired zero (writes ignored)     | none   |
| x1         | ra        | Return address                       | Caller |
| x2         | sp        | Stack pointer                        | Callee |
| x3         | gp        | Global pointer                       | none   |
| x4         | tp        | Thread pointer                       | none   |
| x5         | t0        | Temporary, alternate link register   | Caller |
| x6, x7     | t1, t2    | Temporaries                          | Caller |
| x8         | s0/fp     | Saved register, frame pointer        | Callee |
| x9         | s1        | Saved register                       | Callee |
| x10, x11   | a0, a1    | Function arguments and return values | Caller |
| x12 to x17 | a2 to a7  | Function arguments                   | Caller |
| x18 to x27 | s2 to s11 | Saved registers                      | Callee |
| x28 to x31 | t3 to t6  | Temporaries                          | Caller |

### 7.3 Hart identities

Each hart reads its own identity in `mhartid`: 0 to 4. The boot ROM dispatches on it (Section 5.3), and it selects the hart's own CLINT and IRQROUTER rows. Hart 0 is the orchestrator: it is always powered, it runs the boot loader and the ROM monitor, and it alone reaches the TCM apertures and the extended-flash window (Chapter 5).

### 7.4 Control and status registers

Tables 7.2 to 7.3 list the CSRs the core implements, by the numbering of the RISC-V Privileged Architecture, version 1.12. The tables are partial in the sense the specification allows: a CSR not listed is not implemented, and Section 7.4.1 says what an access to it does. The trap CSRs of Table 7.3 are described field by field in Chapter 8, and the debug CSRs in Chapter 11.

Table 7.2: CSRs present in every configuration (partial list, Privileged Architecture 1.12 numbering).

| Number       | Name                | Access     | Implementation                                                                  |
|--------------|---------------------|------------|---------------------------------------------------------------------------------|
| 0x301        | misa                | read-only  | MXL = 1 (RV32) and one letter bit per configured extension. Writes are ignored. |
| 0xF11        | mvendorid           | read-only  | Reads zero (not a commercial vendor).                                           |
| 0xF12        | marchid             | read-only  | Reads zero.                                                                     |
| 0xF13        | mimpid              | read-only  | Reads zero.                                                                     |
| 0xF14        | mhartid             | read-only  | The hart's identity, 0 to 4.                                                    |
| 0xF15        | mconfigptr          | read-only  | Reads zero (no configuration structure).                                        |
| 0xB00, 0xB80 | mcycle, mcycleh     | read/write | 64-bit MCLK cycle counter, free-running; counts while the hart is stalled.      |
| 0xB02, 0xB82 | minstret, minstreth | read/write | 64-bit count of retired instructions.                                           |
| 0xC00, 0xC80 | cycle, cycleh       | read-only  | User view of mcycle.                                                            |
| 0xC02, 0xC82 | instret, instreth   | read-only  | User view of minstret.                                                          |

| Number                            | Name                                                        | Access    | Implementation                                                         |
|-----------------------------------|-------------------------------------------------------------|-----------|------------------------------------------------------------------------|
| 0x306                             | mcounteren                                                  | WARL      | Hard-wired zero unless user mode is configured.                        |
| 0x320                             | mcountinhibit                                               | WARL      | Hard-wired zero: the counters cannot be inhibited. Writes are ignored. |
| 0xB03 to 0xB1F,<br>0xB83 to 0xB9F | mhpmmcounter3 to<br>mhpmmcounter31 and<br>their high halves | WARL      | Hard-wired zero. Writes are ignored.                                   |
| 0x323 to 0x33F                    | mhpmevent3 to<br>mhpmevent31                                | WARL      | Hard-wired zero. Writes are ignored.                                   |
| 0xC03 to 0xC1F,<br>0xC83 to 0xC9F | hpmcounter3 to<br>hpmcounter31 and<br>their high halves     | read-only | Read zero.                                                             |

Table 7.3: CSRs present only in some configurations (partial list, Privileged Architecture 1.12 and Debug 1.0 numbering).

| Number                            | Name                                            | Configured by                  | Implementation                                                                                     |
|-----------------------------------|-------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------|
| 0x300                             | mstatus                                         | priv.trapCsr<br>(present here) | MIE, MPIE, MPP; MPRV and TW only with user mode. Chapter 8.                                        |
| 0x310                             | mstatush                                        | priv.trapCsr                   | Reads zero; writes are ignored.                                                                    |
| 0x305                             | mtvec                                           | priv.trapCsr                   | BASE read/write; MODE pinned to direct.                                                            |
| 0x304, 0x344                      | mie, mip                                        | priv.trapCsr                   | Bits MSIE/MSIP (3), MTIE/MTIP (7), MEIE/MEIP (11). mip is read-only.                               |
| 0x340                             | mscratch                                        | priv.trapCsr                   | Read/write scratch.                                                                                |
| 0x341, 0x342,<br>0x343            | mepc, mcause, mtval                             | priv.trapCsr                   | Written by hardware on a standard-mode trap entry (Table 8.2); mcause keeps bit 31 and bits 3:0.   |
| 0x7C0                             | mtrapctl                                        | priv.trapCsr                   | Custom: bit 0 LEGACY selects the trap delivery mode; resets to 1 (legacy).                         |
| 0x3A0 to 0x3A3,<br>0x3B0 to 0x3BF | pmpcfg0 to pmpcfg3,<br>pmpaddr0 to<br>pmpaddr15 | priv.pmp<br>(absent here)      | 16 entries; entries above the count read WARL zero. Chapter 8.                                     |
| 0x7B0 to 0x7B3                    | dcsr, dpc,<br>dscratch0,<br>dscratch1           | debug.enable<br>(present here) | Accessible in debug mode only; an access outside debug mode is an illegal instruction. Chapter 11. |

#### 7.4.1 Unimplemented CSRs and illegal accesses

A CSR instruction naming a register that is not in Tables 7.2 and 7.3 for this build is an illegal instruction (exception code 2). In particular time and timeh (0xC01, 0xC81) are not implemented, because no real-time source is wired to the core; use cycle and the known MCLK frequency instead. Two further rules apply in every configuration: a CSR instruction in a write form (any form other than a pure read with x0 as the source) aimed at the read-only quadrant (0xC00 to 0xFFFF) is illegal even when the register exists, and an access to the debug CSRs outside debug mode is illegal. A write to a WARL register listed as hard-wired (misa, mip, mstatush, mcountinhibit, the mhpmm\* ranges) is not illegal; it is accepted and ignored. An illegal CSR instruction commits nothing: no register, memory or CSR is changed.

What an illegal instruction does next depends on the trap delivery mode of Chapter 8. In the legacy mode, which every configuration selects at reset, there is no recoverable exception: the hart enters a terminal trap state and retires no further instruction until reset. In the standard mode, available where `priv.trapCsr` is configured and selected by clearing `mtrapctl.LEGACY`, the hart traps to `mtvec` with `mcause = 2` and `mtval` holding the faulting encoding.

**Caution:** The `unimp` pseudo-instruction assembles to a CSR write to `cycle` (`0xC0001073`), which is exactly the read-only-quadrant write described above. Reaching an `unimp` in legacy delivery mode halts the hart until reset.

## 8 Privileged Architecture

Every VestaRV hart executes at the RISC-V *machine* (M) privilege level, the level that owns the whole address space and every control register. What differs between configurations is how a hart *takes a trap* (an interrupt or an exception) and whether it can drop to the *user* (U) privilege level and be fenced off from memory it does not own.

Two trap architectures exist in this family. The **legacy** architecture is the one described in Section 9: a table of jump instructions at `0x8000`, a hardware push of the return address, and the `retirq` instruction to return. It is inherited from the single-core VestaRV chip, it is what the boot ROM, the interrupt dispatcher and every shipped example use, and it is active out of reset in *every* configuration. The **standard** architecture is the one written in the RISC-V privileged specification: a single trap entry point in `mtvec`, the cause and value registers `mcause/mtval`, a saved program counter in `mepc`, an interrupt-enable stack in `mstatus`, and the `MRET` instruction to return. It is a configuration option (`priv.trapCsr` in Table 2.1), and when it is built the two architectures *coexist* in the same hart, selected at run time by one bit.

This configuration includes the standard trap architecture; the sections below document it.

### 8.1 Machine Information Registers

Five read-only registers identify the hart and the implementation. They are present in *every* configuration (they are not part of the trap architecture and are not affected by `priv.trapCsr`), and reading any of them is always legal, at machine level, in either delivery mode.

| Address | Register                | Value                                                                                                                                  |
|---------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 0xF11   | <code>mvendorid</code>  | 0x00000000: no commercially registered JEDEC vendor.                                                                                   |
| 0xF12   | <code>marchid</code>    | 0x00000000: no architecture ID assigned by the RISC-V registry.                                                                        |
| 0xF13   | <code>mimpid</code>     | 0x00000000: no implementation revision encoded.                                                                                        |
| 0xF14   | <code>mhartid</code>    | The hart's own index, 0 to 4 (Section 5). The only one of the five that differs between harts, and the only one with a non-zero value. |
| 0xF15   | <code>mconfigptr</code> | 0x00000000: no configuration data structure in memory.                                                                                 |

**Zero is a defined answer, not a missing one.** The privileged specification requires all five registers to exist and gives zero a specific meaning in each: it says “*not implemented*” or “*not assigned*”, which is the truthful answer for a non-commercial teaching chip. Discovery software must therefore read them and interpret zero, rather than treat a zero as evidence that the register is absent; there is no way to tell the two apart by reading, which is why the specification requires the registers rather than the values.

**Writing one is an illegal instruction.** All five sit in the read-only quadrant of the register address space (`csr[11:10] = 11`), and any CSR instruction in a *write* form aimed at that quadrant traps, in every configuration, including the read-modify-write forms that would write back an unchanged value. Reading with `csrr` (or any form whose source operand is `x0`, which the hardware recognizes as a pure read) is the only legal access.

## 8.2 The Legacy Trap Mechanism

The legacy mechanism is a pure *interrupt* architecture: it delivers the hardware interrupt sources of Section 9 and nothing else.

- **Delivery** is vectored in hardware. Each of the 121 interrupt sources owns one 4-byte slot of the interrupt vector table at 0x8000, and the hardware jumps directly to the slot for the source it is taking. The slot holds a jump instruction to the service routine.
- **Entry** pushes the return program counter, and only that, to the stack at  $sp - 4$  and decrements  $sp$  by four. The stack pointer must therefore be valid before any interrupt can arrive.
- **Return** is the custom `retirq` instruction, which pops the saved program counter, increments  $sp$  by four and resumes. The custom `sleep` and `wake` instructions park and un-park the hart (the generated header spells the three of them `irq_return()`, `cpu_sleep()` and `cpu_wake()`; the bare-metal assembly environment spells them `iret`, `extinguish` and `ignite`).
- **Exceptions do not exist.** There is no recoverable exception in the legacy architecture. An illegal instruction, including an access to an unimplemented control register, puts the hart into a terminal trap state: the program counter stops advancing and the hart never retires another instruction. This is a deliberate, deterministic failure mode for a teaching chip, not an error state software can catch.
- **Handling is non-recursive** and the enable state of the three per-hart interrupt lines is fixed in hardware; there is no interrupt-enable control register in the legacy path.

Because the legacy mechanism pushes to the stack on entry, it assumes the stack pointer belongs to trusted code. That assumption is exactly what the standard architecture removes, which is why a chip that wants user mode needs the standard architecture underneath it.

## 8.3 Selecting a Delivery Mode: `mtrapctl`

The two architectures are selected by bit 0 (LEGACY) of the custom machine register `mtrapctl` at address 0x7C0, which lives in the architecturally reserved custom machine read/write range. All other bits read zero and ignore writes.

| <code>mtrapctl.LEGACY</code> | Mode     | Behavior                                                                                                                                                                                                                                           |
|------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (reset)                    | Legacy   | Exactly the mechanism of Section 8.2: vector table delivery, the stack push on entry, <code>retirq</code> to return, and a terminal halt on any illegal instruction.                                                                               |
| 0                            | Standard | Interrupts and exceptions vector through <code>mtvec</code> with the full <code>mepc/mcause/mtval/mstatus</code> sequence, and <code>MRET</code> returns. The vector table is inert, and <b>nothing is ever pushed to the stack</b> by trap entry. |

LEGACY resets to 1, so a chip built with the standard architecture still *boots* on the legacy path: the boot ROM, the tile loader, the interrupt dispatcher and existing application code run unmodified, and a program that never writes `mtrapctl` cannot tell the difference. Selecting the standard mode is an explicit software act.

**Software contract:** change `mtrapctl` only while no trap is in flight; that is, from ordinary program flow with interrupts quiesced, never from inside an interrupt service routine and never between a trap

entry and its matching return. The rule is the same class as the clock-reconfiguration contract in the SYSTEM chapter: the state of a trap that was taken under one delivery mode cannot be unwound by the other. Switching back to legacy is legal and is the recovery path.

The two custom instruction groups keep their encodings in both modes. `retirq` is only *defined* in legacy mode, where an interrupt sequence is in progress for it to acknowledge; executing it in standard mode is a software error whose only consequence is the stack-pointer damage the instruction itself does.

## 8.4 The Machine Trap Registers

Table 8.1 lists every control register the standard trap architecture adds, with the behavior of each field. Fields not listed read as zero and ignore writes; every field is WARL (write any value, read a legal value), so software must read back what it wrote rather than assume a value took.

Table 8.1: Machine trap control and status registers

| Address | Register | Fields and behavior                                                                                                                                                                                                                                                                   |
|---------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x300   | mstatus  | MIE (bit 3) global machine interrupt enable, and MPIE (bit 7) its saved copy. MPP (bits 12:11) is the saved previous privilege and is pinned to 11 (machine). MPRV (bit 17) makes <i>data</i> accesses use the privilege in MPP. TW (bit 21) traps WFI (no effect without user mode). |
| 0x310   | mstatush | Implemented as a legal register that reads zero and ignores writes (the high half of mstatus holds no field this chip implements).                                                                                                                                                    |
| 0x305   | mtvec    | BASE (bits 31:2) is the trap entry address, read/write. MODE (bits 1:0) is pinned to 00, <i>direct</i> mode: every trap enters at BASE, and the handler dispatches on mcause.                                                                                                         |
| 0x304   | mie      | Per-source interrupt enables: MSIE (bit 3) software, MTIE (bit 7) timer, MEIE (bit 11) external. <b>Resets to zero:</b> standard mode wakes up fully masked, unlike the legacy path whose three lines are hard-wired enabled.                                                         |
| 0x344   | mip      | Read-only mirror of the three live interrupt lines: MSIP (bit 3), MTIP (bit 7), MEIP (bit 11). Writes are ignored: every source is level-driven and is cleared at the source (a CLINT MSIPh write, an MTIMECMPx write, or the IRQROUTER claim/complete sequence).                     |
| 0x340   | mscratch | 32 bits of read/write scratch storage reserved for the trap handler. In standard mode the hardware saves nothing to memory, so this is where a handler keeps the pointer it needs before it owns a stack.                                                                             |
| 0x341   | mepc     | The trapped program counter. Bit 0 always reads zero; bit 1 is writable on a chip built with the compressed instruction set and reads zero otherwise.                                                                                                                                 |
| 0x342   | mcause   | The trap cause: bit 31 (the interrupt flag) distinguishes an interrupt from an exception, bits 3:0 carry the code. Bits 30:4 read zero. Hardware writes it on every trap entry per Table 8.2; software may write it.                                                                  |
| 0x343   | mtval    | The trap value, written by hardware per Table 8.2 and freely writable by software.                                                                                                                                                                                                    |



| Address | Register   | Fields and behavior                                                                                              |
|---------|------------|------------------------------------------------------------------------------------------------------------------|
| 0x306   | mcounteren | A legal register pinned to zero (its enables only have meaning with user mode built).                            |
| 0x7C0   | mtrapctl   | Custom. LEGACY (bit 0) selects the delivery mode and resets to 1; bits 31:1 are pinned to zero. See Section 8.3. |

## 8.5 Trap Entry, Trap Return, and Causes

In standard mode a trap entry performs **no memory transaction at all**. In one atomic step the hardware writes mepc, mcause and mtval per Table 8.2, copies MIE into MPIE and clears MIE (so the handler starts with interrupts disabled), records the interrupted privilege level in MPP, enters machine mode, and sets the program counter to mtvec’s BASE. The stack pointer is not read, not written and not adjusted, which is the entire point, because the interrupted stack pointer may belong to untrusted code.

MRET reverses it: the program counter is loaded from mepc, MIE is restored from MPIE, MPIE is set to 1, and the privilege level returns to MPP. ECALL and EBREAK are decoded as the environment call and breakpoint exceptions; both are illegal instructions on a chip without the standard trap architecture.

A minimal handler installation looks like this (the assembler needs the Zicsr extension enabled for the control-register instructions):

```
.option push
.option arch, +zicsr
la      t0, trap_handler
csrw    mtvec, t0      # direct-mode trap entry point
csrw    mie, zero      # start fully masked
csrwi   mtrapctl, 0    # leave legacy delivery: standard mode from here
.option pop
```

Table 8.2: Trap causes, values, and saved program counters

| Condition                                                                                                                          | mcause     | mtval                             | mepc                     |
|------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------|--------------------------|
| Instruction address misaligned                                                                                                     | 0          | The misaligned address            | The faulting instruction |
| Instruction access fault (memory protection)                                                                                       | 1          | The faulting fetch address        | The faulting instruction |
| Illegal instruction (including an access to an unimplemented or privileged control register, or a <i>write</i> to a read-only one) | 2          | The faulting instruction encoding | The faulting instruction |
| Breakpoint (EBREAK)                                                                                                                | 3          | 0                                 | The EBREAK               |
| Load access fault (memory protection)                                                                                              | 5          | The faulting data address         | The faulting instruction |
| Store or atomic access fault (memory protection)                                                                                   | 7          | The faulting data address         | The faulting instruction |
| Environment call from machine mode                                                                                                 | 11         | 0                                 | The ECALL                |
| Machine software interrupt (CLINT MSIPh)                                                                                           | 0x80000003 | 0                                 | The resume address       |

| Condition                                  | mcause     | mtval | mepc               |
|--------------------------------------------|------------|-------|--------------------|
| Machine timer interrupt (CLINT MTIMECMP0L) | 0x80000007 | 0     | The resume address |
| Machine external interrupt (IRQROUTER)     | 0x8000000B | 0     | The resume address |

Three details of Table 8.2 need stating explicitly.

- **The illegal-instruction value for a compressed instruction is the expanded encoding, not the raw halfword.** When a 16-bit compressed instruction traps, `mtval` carries the 32-bit instruction the decompressor produced from it, which is what the core actually rejected. The specification permits either the faulting encoding or zero here; reporting the expansion is more informative than zero and costs nothing, but a debugger that expects the raw halfword will not find it. This is a deliberate, documented deviation.
- **Interrupt priority is fixed:** external, then software, then timer. An interrupt is taken only when MIE is set and the same bit is set in both `mip` and `mie`.
- **Interrupts are sampled at instruction boundaries only**, and the multi-cycle instruction sequences (the compressed push/pop sequences, the cache-block zero instruction, and the table-jump sequence) run to completion uninterrupted, exactly as in the legacy mechanism.

## 8.6 Waiting for an Interrupt

Two instructions park a hart, and they are not interchangeable.

- The standard WFI instruction is available whenever the standard trap registers are built, in either delivery mode. It stops the hart until  $(mip \wedge mie) \neq 0$ , *regardless* of MIE. If MIE is set the hart then takes the interrupt; if it is clear the hart simply resumes at the following instruction, which is what makes the standard “sleep until an event, handle it inline” idiom work.
- The custom `sleep` instruction keeps its legacy meaning: it parks the hart until an interrupt is actually *taken*, and the service routine must execute `wake` before returning or the hart goes straight back to sleep. This is the instruction the boot ROM parks tile harts with.

Because WFI waits on  $mip \wedge mie$ , a hart that executes it with `mie` still zero waits forever. That is the specified behavior, not a defect; TW exists so that machine-mode software can stop less privileged code from doing it.

## 9 Interrupts and exceptions

The VestaRV CPU implements an interrupt system with 121 interrupt vectors, numbered 0 to 120, whose sources are enumerated in Table 9.1. The numbering is frozen: a vector belongs to the block that owns it whether or not this configuration instantiates that block, so vectors whose block is absent are reserved and counted with the rest rather than shifting the ones above them. Two of the 121 are the CLINT pair (83, the software or inter-processor interrupt; 84, the timer interrupt) added for multi-core operation; the other 119 sit on the IRQROUTER side of the fabric: the peripheral and system sources, the reserved slots frozen among them, and meip’s own slot. Each hart takes three interrupt lines: its own CLINT software and timer interrupts (vectors 83 and 84, hardwired enabled, delivered on dedicated per-hart wires) and the external interrupt meip (vector 85, a slot of its own that no source pends on), which delivers all peripheral and system sources through the IRQROUTER’s claim and complete mechanism: the vector-85 handler reads the router’s CLAIM register to discover and claim the pending source, dispatches on the returned vector number, and writes it back to complete (Chapter 23). Which sources reach which hart is programmed per hart in the router’s routing rows, identical for every hart, hart 0 included. All peripheral sources are masked at reset and must be routed in software before use; source priority is fixed (the lowest pending vector number is claimed first). A peripheral must additionally be configured to generate its interrupt at the peripheral level.

Table 9.1: Interrupt vectors

| Vector | Source | Description                                                   | Clear method                                 | Chapter |
|--------|--------|---------------------------------------------------------------|----------------------------------------------|---------|
| 0      | SYSTEM | Watchdog Timer Interrupt<br>IRQB_SYS_WDT                      | Write 1 to WDTSR.SYSWDTIF                    | 16      |
| 1      | GPIO0  | GPIO0 Bit 0 Interrupt<br>IRQB_GPIO0_B0                        | Write 1 to P0IF bit 0                        | 12      |
| 2      | GPIO0  | GPIO0 Bit 1 Interrupt<br>IRQB_GPIO0_B1                        | Write 1 to P0IF bit 1                        | 12      |
| 3      | GPIO0  | GPIO0 Bit 2 Interrupt<br>IRQB_GPIO0_B2                        | Write 1 to P0IF bit 2                        | 12      |
| 4      | GPIO0  | GPIO0 Bit 3 Interrupt<br>IRQB_GPIO0_B3                        | Write 1 to P0IF bit 3                        | 12      |
| 5      | GPIO0  | GPIO0 Bit 4 Interrupt<br>IRQB_GPIO0_B4                        | Write 1 to P0IF bit 4                        | 12      |
| 6      | GPIO0  | GPIO0 Bit 5 Interrupt<br>IRQB_GPIO0_B5                        | Write 1 to P0IF bit 5                        | 12      |
| 7      | GPIO0  | GPIO0 Bit 6 Interrupt<br>IRQB_GPIO0_B6                        | Write 1 to P0IF bit 6                        | 12      |
| 8      | GPIO0  | GPIO0 Bit 7 Interrupt<br>IRQB_GPIO0_B7                        | Write 1 to P0IF bit 7                        | 12      |
| 9      | SPI0   | SPI0 Transmission Complete Inter-<br>rupt<br>IRQB_SPI0_TC     | Write 1 to SPI0SR.SPITCIF, or read<br>SPI0RX | 13      |
| 10     | SPI0   | SPI0 Transmission Buffer Empty Inter-<br>rupt<br>IRQB_SPI0_TE | Write 1 to SPI0SR.SPITEIF                    | 13      |
| 11     | SPI1   | SPI1 Transmission Complete Inter-<br>rupt<br>IRQB_SPI1_TC     | Write 1 to SPI1SR.SPITCIF, or read<br>SPI1RX | 13      |

| Vector | Source | Description                                                | Clear method                             | Chapter |
|--------|--------|------------------------------------------------------------|------------------------------------------|---------|
| 12     | SPI1   | SPI1 Transmission Buffer Empty Interrupt<br>IRQB_SPI1_TE   | Write 1 to SPI1SR.SPITEIF                | 13      |
| 13     | UART0  | UART0 Receive Complete Interrupt<br>IRQB_UART0_RC          | Write 1 to UART0SR.RCIF, or read UART0RX | 14      |
| 14     | UART0  | UART0 Transmission Buffer Empty Interrupt<br>IRQB_UART0_TE | Write 1 to UART0SR.TEIF                  | 14      |
| 15     | UART0  | UART0 Transmission Complete Interrupt<br>IRQB_UART0_TC     | Write 1 to UART0SR.TCIF                  | 14      |
| 16     | TIMER0 | TIMER0 Capture 0 Interrupt<br>IRQB_TIMER0_CAP0             | Write 1 to TIM0SR.CAP0IF                 | 15      |
| 17     | TIMER0 | TIMER0 Capture 1 Interrupt<br>IRQB_TIMER0_CAP1             | Write 1 to TIM0SR.CAP1IF                 | 15      |
| 18     | TIMER0 | TIMER0 Overflow Interrupt<br>IRQB_TIMER0_OVF               | Write 1 to TIM0SR.OVIF                   | 15      |
| 19     | TIMER0 | TIMER0 Compare 0 Interrupt<br>IRQB_TIMER0_CMP0             | Write 1 to TIM0SR.CMP0IF                 | 15      |
| 20     | TIMER0 | TIMER0 Compare 1 Interrupt<br>IRQB_TIMER0_CMP1             | Write 1 to TIM0SR.CMP1IF                 | 15      |
| 21     | TIMER0 | TIMER0 Compare 2 Interrupt<br>IRQB_TIMER0_CMP2             | Write 1 to TIM0SR.CMP2IF                 | 15      |
| 22     | TIMER1 | TIMER1 Capture 0 Interrupt<br>IRQB_TIMER1_CAP0             | Write 1 to TIM1SR.CAP0IF                 | 15      |
| 23     | TIMER1 | TIMER1 Capture 1 Interrupt<br>IRQB_TIMER1_CAP1             | Write 1 to TIM1SR.CAP1IF                 | 15      |
| 24     | TIMER1 | TIMER1 Overflow Interrupt<br>IRQB_TIMER1_OVF               | Write 1 to TIM1SR.OVIF                   | 15      |
| 25     | TIMER1 | TIMER1 Compare 0 Interrupt<br>IRQB_TIMER1_CMP0             | Write 1 to TIM1SR.CMP0IF                 | 15      |
| 26     | TIMER1 | TIMER1 Compare 1 Interrupt<br>IRQB_TIMER1_CMP1             | Write 1 to TIM1SR.CMP1IF                 | 15      |
| 27     | TIMER1 | TIMER1 Compare 2 Interrupt<br>IRQB_TIMER1_CMP2             | Write 1 to TIM1SR.CMP2IF                 | 15      |
| 28     | GPIO1  | GPIO1 Bit 0 Interrupt<br>IRQB_GPIO1_B0                     | Write 1 to P1IF bit 0                    | 12      |
| 29     | GPIO1  | GPIO1 Bit 1 Interrupt<br>IRQB_GPIO1_B1                     | Write 1 to P1IF bit 1                    | 12      |
| 30     | GPIO1  | GPIO1 Bit 2 Interrupt<br>IRQB_GPIO1_B2                     | Write 1 to P1IF bit 2                    | 12      |
| 31     | GPIO1  | GPIO1 Bit 3 Interrupt<br>IRQB_GPIO1_B3                     | Write 1 to P1IF bit 3                    | 12      |
| 32     | GPIO1  | GPIO1 Bit 4 Interrupt<br>IRQB_GPIO1_B4                     | Write 1 to P1IF bit 4                    | 12      |
| 33     | GPIO1  | GPIO1 Bit 5 Interrupt<br>IRQB_GPIO1_B5                     | Write 1 to P1IF bit 5                    | 12      |
| 34     | GPIO1  | GPIO1 Bit 6 Interrupt<br>IRQB_GPIO1_B6                     | Write 1 to P1IF bit 6                    | 12      |
| 35     | GPIO1  | GPIO1 Bit 7 Interrupt<br>IRQB_GPIO1_B7                     | Write 1 to P1IF bit 7                    | 12      |
| 36     | GPIO2  | GPIO2 Bit 0 Interrupt<br>IRQB_GPIO2_B0                     | Write 1 to P2IF bit 0                    | 12      |

| Vector | Source   | Description                                                          | Clear method                                | Chapter |
|--------|----------|----------------------------------------------------------------------|---------------------------------------------|---------|
| 37     | GPIO2    | GPIO2 Bit 1 Interrupt<br>IRQB_GPIO2_B1                               | Write 1 to P2IF bit 1                       | 12      |
| 38     | GPIO2    | GPIO2 Bit 2 Interrupt<br>IRQB_GPIO2_B2                               | Write 1 to P2IF bit 2                       | 12      |
| 39     | GPIO2    | GPIO2 Bit 3 Interrupt<br>IRQB_GPIO2_B3                               | Write 1 to P2IF bit 3                       | 12      |
| 40     | GPIO2    | GPIO2 Bit 4 Interrupt<br>IRQB_GPIO2_B4                               | Write 1 to P2IF bit 4                       | 12      |
| 41     | GPIO2    | GPIO2 Bit 5 Interrupt<br>IRQB_GPIO2_B5                               | Write 1 to P2IF bit 5                       | 12      |
| 42     | GPIO2    | GPIO2 Bit 6 Interrupt<br>IRQB_GPIO2_B6                               | Write 1 to P2IF bit 6                       | 12      |
| 43     | GPIO2    | GPIO2 Bit 7 Interrupt<br>IRQB_GPIO2_B7                               | Write 1 to P2IF bit 7                       | 12      |
| 44     | GPIO3    | GPIO3 Bit 0 Interrupt<br>IRQB_GPIO3_B0                               | Write 1 to P3IF bit 0                       | 12      |
| 45     | GPIO3    | GPIO3 Bit 1 Interrupt<br>IRQB_GPIO3_B1                               | Write 1 to P3IF bit 1                       | 12      |
| 46     | GPIO3    | GPIO3 Bit 2 Interrupt<br>IRQB_GPIO3_B2                               | Write 1 to P3IF bit 2                       | 12      |
| 47     | GPIO3    | GPIO3 Bit 3 Interrupt<br>IRQB_GPIO3_B3                               | Write 1 to P3IF bit 3                       | 12      |
| 48     | GPIO3    | GPIO3 Bit 4 Interrupt<br>IRQB_GPIO3_B4                               | Write 1 to P3IF bit 4                       | 12      |
| 49     | GPIO3    | GPIO3 Bit 5 Interrupt<br>IRQB_GPIO3_B5                               | Write 1 to P3IF bit 5                       | 12      |
| 50     | GPIO3    | GPIO3 Bit 6 Interrupt<br>IRQB_GPIO3_B6                               | Write 1 to P3IF bit 6                       | 12      |
| 51     | GPIO3    | GPIO3 Bit 7 Interrupt<br>IRQB_GPIO3_B7                               | Write 1 to P3IF bit 7                       | 12      |
| 52     | UART1    | UART1 Receive Complete Interrupt<br>IRQB_UART1_RC                    | Write 1 to UART1SR.RCIF, or read<br>UART1RX | 14      |
| 53     | UART1    | UART1 Transmission Buffer Empty<br>Interrupt<br>IRQB_UART1_TE        | Write 1 to UART1SR.TEIF                     | 14      |
| 54     | UART1    | UART1 Transmission Complete Inter-<br>rupt<br>IRQB_UART1_TC          | Write 1 to UART1SR.TCIF                     | 14      |
| 55     | Reserved | Reserved (vector 55; formerly AFE0<br>Receive Complete)              | -                                           | -       |
| 56     | Reserved | Reserved (vector 56; formerly<br>SARADC0 Conversion Complete)        | -                                           | -       |
| 57     | I2C0     | I2C0 start received Interrupt<br>IRQB_I2C0_STR                       | Write 1 to I2C0SR.I2CSTR                    | 19      |
| 58     | I2C0     | I2C0 stop received Interrupt<br>IRQB_I2C0_spr                        | Write 1 to I2C0SR.I2CSPR                    | 19      |
| 59     | I2C0     | I2C0 master mode start condition sent<br>Interrupt<br>IRQB_I2C0_msts | Write 1 to I2C0SR.I2CMSTS                   | 19      |
| 60     | I2C0     | I2C0 master mode stop condition sent<br>Interrupt<br>IRQB_I2C0_msp   | Write 1 to I2C0SR.I2CMSPS                   | 19      |

| Vector | Source | Description                                                       | Clear method              | Chapter |
|--------|--------|-------------------------------------------------------------------|---------------------------|---------|
| 61     | I2C0   | I2C0 master mode arbitration lost Interrupt<br>IRQB_I2C0_marb     | Write 1 to I2C0SR.I2CMARB | 19      |
| 62     | I2C0   | I2C0 master mode transmit empty Interrupt<br>IRQB_I2C0_mtxe       | Write 1 to I2C0SR.I2CMTXE | 19      |
| 63     | I2C0   | I2C0 master mode NACK received Interrupt<br>IRQB_I2C0_mnr         | Write 1 to I2C0SR.I2CMNR  | 19      |
| 64     | I2C0   | I2C0 master mode transfer complete Interrupt<br>IRQB_I2C0_mxc     | Write 1 to I2C0SR.I2CMXC  | 19      |
| 65     | I2C0   | I2C0 slave address Interrupt<br>IRQB_I2C0_sa                      | Write 1 to I2C0SR.I2CSA   | 19      |
| 66     | I2C0   | I2C0 slave transmit empty Interrupt<br>IRQB_I2C0_stxe             | Write 1 to I2C0SR.I2CSTXE | 19      |
| 67     | I2C0   | I2C0 slave overflow Interrupt<br>IRQB_I2C0_sovf                   | Write 1 to I2C0SR.I2CSOVF | 19      |
| 68     | I2C0   | I2C0 slave mode NACK received Interrupt<br>IRQB_I2C0_snr          | Write 1 to I2C0SR.I2CSNR  | 19      |
| 69     | I2C0   | I2C0 slave mode transfer complete Interrupt<br>IRQB_I2C0_sxc      | Write 1 to I2C0SR.I2CSXC  | 19      |
| 70     | I2C1   | I2C1 start received Interrupt<br>IRQB_I2C1_STR                    | Write 1 to I2C1SR.I2CSTR  | 19      |
| 71     | I2C1   | I2C1 stop received Interrupt<br>IRQB_I2C1_spr                     | Write 1 to I2C1SR.I2CSPR  | 19      |
| 72     | I2C1   | I2C1 master mode start condition sent Interrupt<br>IRQB_I2C1_msts | Write 1 to I2C1SR.I2CMSTS | 19      |
| 73     | I2C1   | I2C1 master mode stop condition sent Interrupt<br>IRQB_I2C1_mspis | Write 1 to I2C1SR.I2CMSPS | 19      |
| 74     | I2C1   | I2C1 master mode arbitration lost Interrupt<br>IRQB_I2C1_marb     | Write 1 to I2C1SR.I2CMARB | 19      |
| 75     | I2C1   | I2C1 master mode transmit empty Interrupt<br>IRQB_I2C1_mtxe       | Write 1 to I2C1SR.I2CMTXE | 19      |
| 76     | I2C1   | I2C1 master mode NACK received Interrupt<br>IRQB_I2C1_mnr         | Write 1 to I2C1SR.I2CMNR  | 19      |
| 77     | I2C1   | I2C1 master mode transfer complete Interrupt<br>IRQB_I2C1_mxc     | Write 1 to I2C1SR.I2CMXC  | 19      |
| 78     | I2C1   | I2C1 slave address Interrupt<br>IRQB_I2C1_sa                      | Write 1 to I2C1SR.I2CSA   | 19      |
| 79     | I2C1   | I2C1 slave transmit empty Interrupt<br>IRQB_I2C1_stxe             | Write 1 to I2C1SR.I2CSTXE | 19      |
| 80     | I2C1   | I2C1 slave overflow Interrupt<br>IRQB_I2C1_sovf                   | Write 1 to I2C1SR.I2CSOVF | 19      |

| Vector | Source   | Description                                                                                       | Clear method                                       | Chapter |
|--------|----------|---------------------------------------------------------------------------------------------------|----------------------------------------------------|---------|
| 81     | I2C1     | I2C1 slave mode NACK received Interrupt<br>IRQB_I2C1_snr                                          | Write 1 to I2C1SR.I2CSNR                           | 19      |
| 82     | I2C1     | I2C1 slave mode transfer complete Interrupt<br>IRQB_I2C1_sxc                                      | Write 1 to I2C1SR.I2CSXC                           | 19      |
| 83     | CLINT    | CLINT software interrupt (IPI)<br>IRQB_CLINT_MSIP                                                 | Write 0 to the MSIPn register of the hart          | 20      |
| 84     | CLINT    | CLINT timer interrupt<br>IRQB_CLINT_MTIP                                                          | Write MTIMECMPn of the hart to a value above mtime | 20      |
| 85     | Reserved | Reserved (vector 85; coincides with the meip external-interrupt IVT slot, never a pending source) | -                                                  | -       |
| 86     | Reserved | Reserved (vector 86; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 87     | Reserved | Reserved (vector 87; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 88     | Reserved | Reserved (vector 88; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 89     | Reserved | Reserved (vector 89; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 90     | Reserved | Reserved (vector 90; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 91     | Reserved | Reserved (vector 91; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 92     | Reserved | Reserved (vector 92; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 93     | Reserved | Reserved (vector 93; I3C0 disabled by this configuration)                                         | -                                                  | -       |
| 94     | NFC0     | NFC0 RF Field-Detect Interrupt<br>IRQB_NFC0_FIELD                                                 | Write 1 to NFC0SR.NFCFIELDF                        | 22      |
| 95     | NFC0     | NFC0 Reader-Frame Received Interrupt<br>IRQB_NFC0_RXF                                             | Write 1 to NFC0SR.NFCRXFRAMEF                      | 22      |
| 96     | NFC0     | NFC0 Tag-Response Transmit-Done Interrupt<br>IRQB_NFC0_TXDONE                                     | Write 1 to NFC0SR.NFCTXDONEF                       | 22      |
| 97     | NFC0     | NFC0 RX CRC / Parity Error Interrupt<br>IRQB_NFC0_CRCERR                                          | Write 1 to NFC0SR.NFCCRCERRF and NFCPARERRF        | 22      |
| 98     | GPI04    | GPI04 Bit 0 Interrupt<br>IRQB_GPI04_B0                                                            | Write 1 to P4IF bit 0                              | 12      |
| 99     | GPI04    | GPI04 Bit 1 Interrupt<br>IRQB_GPI04_B1                                                            | Write 1 to P4IF bit 1                              | 12      |
| 100    | GPI04    | GPI04 Bit 2 Interrupt<br>IRQB_GPI04_B2                                                            | Write 1 to P4IF bit 2                              | 12      |
| 101    | GPI04    | GPI04 Bit 3 Interrupt<br>IRQB_GPI04_B3                                                            | Write 1 to P4IF bit 3                              | 12      |
| 102    | GPI04    | GPI04 Bit 4 Interrupt<br>IRQB_GPI04_B4                                                            | Write 1 to P4IF bit 4                              | 12      |
| 103    | GPI04    | GPI04 Bit 5 Interrupt<br>IRQB_GPI04_B5                                                            | Write 1 to P4IF bit 5                              | 12      |
| 104    | GPI04    | GPI04 Bit 6 Interrupt<br>IRQB_GPI04_B6                                                            | Write 1 to P4IF bit 6                              | 12      |

| Vector | Source   | Description                                                              | Clear method                  | Chapter |
|--------|----------|--------------------------------------------------------------------------|-------------------------------|---------|
| 105    | GPI04    | GPI04 Bit 7 Interrupt<br>IRQB_GPI04_B7                                   | Write 1 to P4IF bit 7         | 12      |
| 106    | GPI05    | GPI05 Bit 0 Interrupt<br>IRQB_GPI05_B0                                   | Write 1 to P5IF bit 0         | 12      |
| 107    | GPI05    | GPI05 Bit 1 Interrupt<br>IRQB_GPI05_B1                                   | Write 1 to P5IF bit 1         | 12      |
| 108    | GPI05    | GPI05 Bit 2 Interrupt<br>IRQB_GPI05_B2                                   | Write 1 to P5IF bit 2         | 12      |
| 109    | GPI05    | GPI05 Bit 3 Interrupt<br>IRQB_GPI05_B3                                   | Write 1 to P5IF bit 3         | 12      |
| 110    | GPI05    | GPI05 Bit 4 Interrupt<br>IRQB_GPI05_B4                                   | Write 1 to P5IF bit 4         | 12      |
| 111    | GPI05    | GPI05 Bit 5 Interrupt<br>IRQB_GPI05_B5                                   | Write 1 to P5IF bit 5         | 12      |
| 112    | GPI05    | GPI05 Bit 6 Interrupt<br>IRQB_GPI05_B6                                   | Write 1 to P5IF bit 6         | 12      |
| 113    | GPI05    | GPI05 Bit 7 Interrupt<br>IRQB_GPI05_B7                                   | Write 1 to P5IF bit 7         | 12      |
| 114    | Reserved | Reserved (vector 114; RTC0 source, disabled by this configuration)       | -                             | -       |
| 115    | Reserved | Reserved (vector 115; PWM0_FAULT source, disabled by this configuration) | -                             | -       |
| 116    | Reserved | Reserved (vector 116; PWM0_EVT source, disabled by this configuration)   | -                             | -       |
| 117    | Reserved | Reserved (vector 117; OW0 source, disabled by this configuration)        | -                             | -       |
| 118    | Reserved | Reserved (vector 118; DMA0_DONE source, disabled by this configuration)  | -                             | -       |
| 119    | Reserved | Reserved (vector 119; DMA0_ERR source, disabled by this configuration)   | -                             | -       |
| 120    | NPU      | NPU0 think-done Interrupt<br>IRQB_NPU0_TD                                | Write 1 to NPUSR.NPUTHINKDONE | 17      |

Figure 23.1, in the IRQROUTER chapter (Chapter 23), draws the whole fabric in one view: the source population of Table 9.1 collapsed by type, the IRQROUTER as the spine every one of those sources terminates in, and the five harts with the three lines each of them takes. It shows the one asymmetry: meip arrives from the router, and msip and mtip do not go near it.

## 9.1 Exceptions

In the legacy trap delivery mode selected at reset there is no recoverable exception: an illegal instruction, including an access to an unimplemented CSR, puts the hart into a terminal trap state (Section 7.4.1). Where the standard delivery mode is configured and selected, exceptions vector through mtvec with the mcause codes of Table 8.2 in Chapter 8. Table 9.2 lists the causes that originate inside the hart rather than on a numbered vector.

Table 9.2: CPU-internal interrupt causes

| Cause | Description        |
|-------|--------------------|
| Timer | CPU internal timer |



| Cause       | Description                           |
|-------------|---------------------------------------|
| Instruction | EBREAK, ECALL, or illegal instruction |
| Bus error   | Unaligned memory access               |

## 9.2 Interrupt vector table

The interrupt vector table (IVT) is located at the base of each hart's TCM, beginning at address `0x8000`. Unlike a pointer-based vector table, the VestaRV IVT contains direct jump instructions to the interrupt service routines (ISRs): each entry is a 4-byte RISC-V jump instruction (for example `j handler`) that vectors execution immediately to the corresponding ISR when an interrupt occurs. This eliminates the indirection of loading a function pointer and gives a deterministic, low-latency interrupt response.

The hardware does not mandate a specific IVT content, so software populates the IVT with jump instructions during initialisation. The generated linker scripts and startup code place the IVT at the beginning of the TCM, and the generated header files provide macros for ISR placement and IVT entry configuration.

## 9.3 Interrupt entry, exit and recursion

When an interrupt is taken, the hardware saves a minimal context: it pushes the return program counter (one 32-bit word) onto the stack at `sp-4`, decrements `sp` by 4, and vectors execution through the IVT entry for the interrupt. No other state is saved by hardware. Software is responsible for saving and restoring every general-purpose register the ISR uses. The generated runtime provides an interrupt entry wrapper (in the startup code) that saves the general-purpose registers on the stack (about 120 bytes per interrupt level), calls the C-level handler, restores the registers, and returns with the `ret irq` instruction, so C-coded ISRs need no special handling. Hand-written assembly ISRs must perform the equivalent save and restore themselves. Executing `ret irq` pops the hardware-saved program counter, increments `sp` by 4, and resumes execution at the interrupted location.

**Caution:** Because the first thing interrupt entry does is a store at `sp-4`, the stack pointer must point at valid, writable private memory before interrupts are enabled. This applies to every hart: a freshly released tile hart must load its stack pointer before unmasking interrupts or going to sleep (Chapter 5).

Interrupt handling is non-recursive: a running ISR is never preempted, and further pending interrupts (including a pending CLINT interrupt during a vector-85 service) are taken after the current ISR returns. Stack allocation must cover one interrupt level: the 4-byte hardware-saved return address plus the handler's register frame (about 120 bytes for the runtime wrapper) and the ISR's local variables. Source arbitration happens in the `IRQROUTER`; the core itself has no priority tiers.

## 9.4 Sleep mode and interrupt wake-up

If an interrupt occurs while the CPU is in sleep mode, the processor wakes from sleep and services the interrupt. On execution of the `ret irq` instruction, the CPU returns to sleep mode unless the ISR or the woken code reconfigures the clock and power settings. During sleep mode the CPU clock is gated to conserve power, and all register contents and the program counter are retained. If the clock source for MCLK is disabled during sleep, an enabled interrupt powers up and enables the source clock for the duration of interrupt servicing.

Clock and power management, including sleep mode configuration, are controlled through the SYSTEM peripheral (Section 16.3.3). If SMCLK is disabled, peripherals in the SMCLK domain cannot assert their interrupts and cannot wake the CPU.

## 10 Core performance and instruction timing

The VestaRV core is a multi-cycle machine on a single unified instruction and data bus, but it issues the fetch of the next instruction during the execute cycle of the current one. Most instructions therefore retire at one per MCLK cycle, and the core’s measured cycles-per-instruction (CPI) is much closer to 1 than the term “multi-cycle” on its own would suggest: over the benchmark suite of Section 10.3 it is 1.260. This chapter gives the cost of every instruction class and reports the CPI measured on that suite.

Every figure below is for one hart executing out of its own TCM with no shared-bus contention. An access that crosses the arbiter into the shared window, or that is made while another hart holds the grant, adds stall cycles on top of these numbers (Chapter 5). The core’s clock is gated while it waits, so a stalled cycle is still an MCLK cycle in which nothing retires and it is charged to CPI like any other.

### 10.1 Instruction timing

Table 10.1 gives the cost of each instruction class. A row for an optional extension applies only when that extension is configured for this build; a disabled extension’s instructions raise an illegal-instruction trap instead of executing. Atomic memory operations are the one class not re-measured here, because the bare-core bench has no second master; their 5-cycle cost is the design value.

Table 10.1: Measured instruction timing, in MCLK cycles.

| Instruction class                                                                              | MCLK cycles |
|------------------------------------------------------------------------------------------------|-------------|
| Arithmetic and logical (add, andi, lui, auipc)                                                 | 1           |
| Shift (sll, srl, sra and immediate forms)                                                      | 1           |
| Comparison (slt, sltu and immediate forms)                                                     | 1           |
| Bit manipulation (Zba, Zbb, Zbc, Zbs)                                                          | 1           |
| Multiply (M): mul, mulh, mulhu, mulhsu                                                         | 1           |
| Divide and remainder (M): div, divu, rem, remu                                                 | 36          |
| Load and store (lw, lh, lb, sw, sh, sb)                                                        | 2           |
| Conditional branch, whether taken or not taken                                                 | 1           |
| Jump and link (jal, jalr)                                                                      | 1           |
| CSR read (csrr)                                                                                | 1           |
| Atomic memory operations (A)                                                                   | 5           |
| <i>Additional</i> penalty: 32-bit instruction straddling a word boundary, reached sequentially | 0           |
| <i>Additional</i> penalty: the same, reached by a taken branch or jump                         | +1          |

Three entries are worth drawing out. Hardware multiply is single-cycle, so mul costs no more than add and there is no reason to hand-expand a multiplication into shifts and adds. Divide and remainder share one iterative divider and are, by a wide margin, the most expensive instructions the core executes; where the divisor is a constant, the compiler’s multiply-and-shift expansion is far faster than div and is worth checking for in the disassembly of any hot loop. And a conditional branch costs the same whether it is taken or not, because the core resolves the condition and issues the redirected fetch in the same cycle: there is no branch predictor, and no misprediction penalty to design around.

The two straddling rows are one instruction class split by how the core arrives at it, and Section 10.2 is about that split.

## 10.2 Compressed instructions and fetch alignment

The core fetches one 32-bit word per bus cycle. With the C extension enabled instructions are only guaranteed 2-byte aligned, so a 32-bit instruction may straddle a word boundary, and the core must have both halves before it can decode. Compressed (2-byte) instructions and word-aligned 32-bit instructions are never affected.

The core covers most of that case in hardware, with a fetch-ahead. When it is about to step sequentially onto a straddling 32-bit instruction, it can already see that instruction's lower half in the word it is holding, and the fetch its ordinary sequencing would issue is for the word it is holding already, a bus cycle that would carry nothing new. The fetch-ahead spends that cycle on the next word instead and latches the lower half at the same edge, so the straddling instruction dispatches in one cycle and costs nothing at all. The whole mechanism is one flip-flop, it issues no bus cycle the core would not have issued one cycle later, and it changes cycle counts only, never an architectural result. It is the core .fetchAhead knob of Table 2.1 (the ENABLE\_IF\_AHEAD generic on the core), it is on in this build, and turning it off is a performance choice and nothing else.

What the fetch-ahead cannot cover is a taken control transfer whose target is a half-word-aligned 32-bit instruction. It arms only on a sequential advance, because that is the only case in which the core can prove, from a half-word it can already see, what it is about to fetch; after a taken branch or jump the core arrives at the target holding nothing, and genuinely needs two fetches before it can decode. That case still costs one extra cycle, and it is the whole of the residual reported below. Both rows of Table 10.1 are measured directly: a straight-line block in which every 32-bit instruction straddles runs at 1.000 cycles per instruction, while a block of a taken branch, a straddling 32-bit target and a compressed instruction (three instructions that cost one cycle each on their own) takes 4 cycles.

Table 10.2 isolates what the C extension still costs by rebuilding the whole benchmark suite for RV32IMA instead of RV32IMAC. The compiler emits the identical instruction sequence in both builds (the retired-instruction counts match exactly, benchmark for benchmark) and only the encoding differs, so the entire difference in cycle count is the straddling-fetch penalty that survives the fetch-ahead. Across the suite it is 0.031 CPI, or 2.2% of all cycles.

Table 10.2: Straddling fetches that survive the fetch-ahead, measured with and without the C extension.

| <b>Benchmark</b> | <b>CPI (RV32IMAC)</b> | <b>CPI (RV32IMA)</b> | <b>Penalty, CPI</b> | <b>Cycles lost</b> |
|------------------|-----------------------|----------------------|---------------------|--------------------|
| dhrystone        | 1.504                 | 1.481                | 0.022               | 1.5%               |
| median           | 1.398                 | 1.397                | 0.000               | 0.0%               |
| memcpy           | 1.726                 | 1.726                | 0.000               | 0.0%               |
| multiply         | 1.163                 | 1.014                | 0.148               | 12.8%              |
| qsort            | 1.398                 | 1.330                | 0.068               | 4.8%               |
| rsort            | 1.412                 | 1.412                | 0.000               | 0.0%               |
| towers           | 1.682                 | 1.681                | 0.002               | 0.1%               |
| vvadd            | 1.374                 | 1.374                | 0.000               | 0.0%               |
| <b>Aggregate</b> | <b>1.443</b>          | <b>1.412</b>         | <b>0.031</b>        | <b>2.2%</b>        |

The aggregate row weights each benchmark by its retired-instruction count, and covers the eight benchmarks of Table 10.3 that were rebuilt both ways; spmv was not, which is why this aggregate differs from the one in that table.

Four of the eight benchmarks pay nothing measurable: their straddling instructions are all reached sequentially, and the hardware absorbs every one. What is left is concentrated in short, branch-dense loops, and multiply is the extreme case: its inner loop is seven instructions long and, in the measured

build, its head is a half-word-aligned 32-bit and, so the loop's own backward branch lands on a straddling instruction on every iteration, and one loop head accounts for the entire 12.8%.

That is the whole of the alignment advice. Aligning a hot loop's body buys nothing: the fetch-ahead already covers straight-line code, and a body of compressed instructions never straddled in the first place. What still pays is aligning branch targets: loop heads, and the entry of a hot function reached by a call. In C, `-falign-loops=4` does exactly that across a translation unit, and it complements hand alignment rather than replacing it: rebuilding `multiply` with it moves that one loop head onto a word boundary and recovers the entire residual, bringing the benchmark back to the RV32IMA figure in the table above, while an assembly hot spot or a computed jump target is aligned by hand with `.p2align 2` on the label. None of this is an argument against the C extension, which is configured on by default and earns its place by shrinking code substantially, decisive when a hart's private TCM is 8 KiB.

### 10.3 Measured CPI

Table 10.3 reports CPI over nine benchmarks derived from the `riscv-tests` benchmark suite, compiled `-O2` for RV32IMAC with the Zb\* extensions enabled. The measured region is each benchmark's own timed kernel, so it excludes program startup, the cache-warming pass, and result verification.

Table 10.3: Measured CPI by benchmark (timed kernel, single hart, no bus contention).

| Benchmark              | Instructions retired | CPI          |
|------------------------|----------------------|--------------|
| <code>spmv</code>      | 802,447              | 1.143        |
| <code>multiply</code>  | 20,892               | 1.163        |
| <code>vvadd</code>     | 2,408                | 1.374        |
| <code>median</code>    | 4,015                | 1.398        |
| <code>qsort</code>     | 123,205              | 1.398        |
| <code>rsort</code>     | 146,548              | 1.412        |
| <code>dhrystone</code> | 200,523              | 1.504        |
| <code>towers</code>    | 3,747                | 1.682        |
| <code>memcpy</code>    | 11,017               | 1.726        |
| <b>Aggregate</b>       | <b>1,314,802</b>     | <b>1.260</b> |

The aggregate is the total cycle count over the total retired-instruction count, so each benchmark contributes in proportion to the work it does. Read against Table 10.1, the spread has a simple explanation: the further a workload sits above 1.0, the more of its instruction mix is loads and stores, which cost two cycles each on the unified bus. Fetch alignment barely enters into it: `multiply` is register-bound and runs at 1.163, of which 0.148 is the one unaligned loop head of Section 10.2 and the rest is its own work, and `towers` and `memcpy` sit at the other end because they are dominated by memory traffic. Branches move none of it, since they are free.

### 10.4 Measuring application code

The core implements the `cycle/cycleh` and `instret/instreth` counters, and their machine-mode aliases `mcycle` and `minstret`, in every configuration. `cycle` counts MCLK cycles, free-running and independent of whether the core is stalled; `minstret` counts instructions actually retired. Bracketing a region with `rdcycle` and `rdinstret` and taking the differences therefore gives that region's exact cycle count and CPI on hardware, with no external instrumentation:

```

1 unsigned c0, i0, c1, i1;
2 __asm__ volatile ("rdcycle %0; rdinstret %1" : "=r"(c0), "=r"(i0));
3 hot_loop();
4 __asm__ volatile ("rdcycle %0; rdinstret %1" : "=r"(c1), "=r"(i1));
5 /* cycles = c1 - c0, instructions = i1 - i0 */

```

`time` and `timeh` are not implemented in any configuration, because no real-time source is wired to the core, and reading either raises an illegal-instruction trap. Use `cycle` for elapsed time, converting through the known MCLK frequency, or a timer peripheral for wall-clock measurement.

## 10.5 Measurement methodology

The numbers in this chapter are measured on the RTL, not estimated. Each benchmark is run to completion in a VHDL simulation of the bare core with a one-cycle-latency memory and no bus back-pressure, which is the timing a single hart sees executing from its own TCM. Cycles are counted on the free-running core clock, so any cycle in which the core's clock is gated is still counted. Instructions are counted from the core's internal instruction-retire strobe, the same signal that increments `minstret`, so the instruction count is architectural rather than an estimate derived from the simulation. The core under measurement is configured as this build ships it, fetch-ahead included.

The per-class costs in Table 10.1 come from paired micro-kernels: a loop containing 64 copies of the instruction under test is run against an otherwise identical empty loop, and the difference in cycles divided by the difference in retired instructions gives the marginal cost of that instruction alone, with the loop's own overhead subtracted out. Every micro-kernel body is word-aligned, so no class is charged a fetch cost that is not its own, with two deliberate exceptions: the two kernels that measure the straddling rows are built to straddle. One is a straight-line block of four instructions, two of them straddling 32-bit ones and every one of them reached sequentially, which measures 1.000 cycles per instruction; the other is a taken branch onto a straddling 32-bit target followed by a compressed instruction, three instructions whose measured classes cost one cycle each, which measures 4 cycles for the block.

# 11 Debug

## 11.1 Overview

Hardware debug reaches into the chip over dedicated wires that depend on no software: it stops a hart between instructions, reads and writes its registers and memory, and starts it again. It works on a hart stuck in a loop, on a hart that has taken an illegal instruction and stopped retiring, and, with halt-on-reset armed, on a hart before its first instruction. The Forth monitor over UART0 (Section 25) is the cooperative alternative and needs the chip to be executing correctly enough to answer.

- IEEE 1149.1 JTAG transport on five pins, with a 5-bit instruction register and a 41-bit DMI data register
- One Debug Module for the whole chip, reaching every hart, implementing RISC-V debug specification 1.0 (version = 3)
- Halt, resume, halt-on-reset and single-step per hart; halt groups across harts
- Abstract access-register commands with one data word and a two-word program buffer plus an implicit third word
- Debug mode in the hart with four CSRs (`dcsr`, `dpc`, `dscratch0`, `dscratch1`) and the DRET instruction
- Memory access through the halted hart's own load and store instructions; no system bus access, no hardware triggers

This configuration includes the debug system; the sections below document it.

## 11.2 Block diagram

Figure 11.1 shows the debug path as built in a configuration with the debug system: the probe and the transport on TCK, the two toggle signals that cross into the main clock domain, and the Debug Module as one more master on the shared bus with direct halt and resume wires to every hart.

## 11.3 Functional description

### 11.3.1 JTAG

JTAG is the IEEE 1149.1 serial protocol: four mandatory wires and an optional fifth, driving a sixteen-state Test Access Port (TAP) whose only input is TMS.

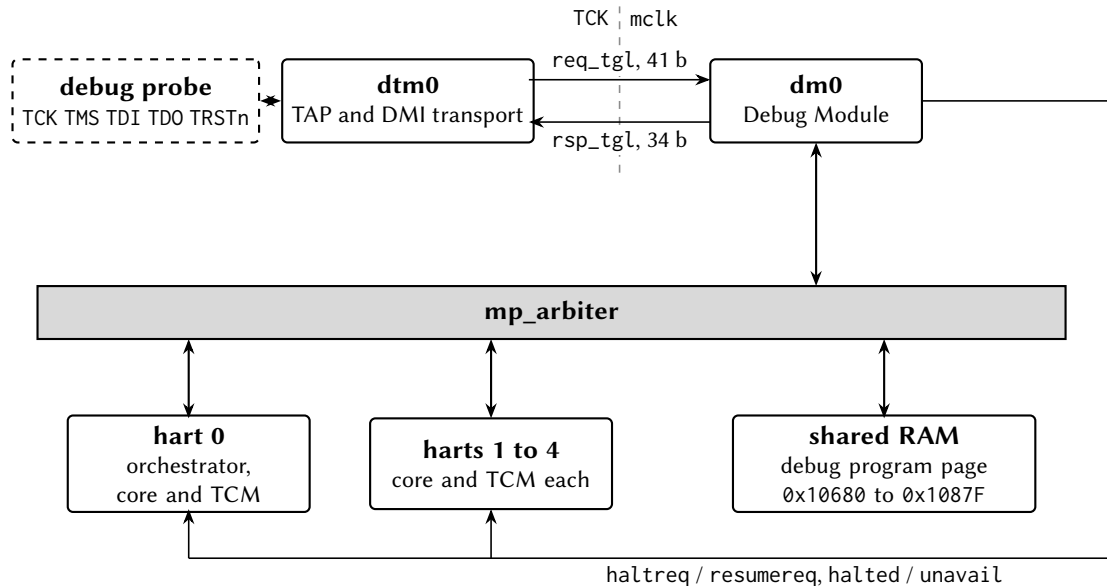


Figure 11.1: Block diagram of the debug path, from the probe to the harts.

| Wire  | Direction | Purpose                                                                                                                              |
|-------|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
| TCK   | in        | Test clock, supplied by the probe. Everything on the port is synchronous to it; it has no relationship to any clock inside the chip. |
| TMS   | in        | Test mode select, sampled on every rising TCK edge; this one wire drives the state machine.                                          |
| TDI   | in        | Serial input to the selected shift register.                                                                                         |
| TDO   | out       | Serial output of that register.                                                                                                      |
| TRSTn | in        | Optional asynchronous reset of the port; present on this chip.                                                                       |

The states form two lobes, one for scanning an instruction and one for scanning data, joined at a common idle state. Test-Logic-Reset is the reset state, and holding TMS high for five TCK clocks reaches it from any state, which is the standard's guaranteed recovery for a probe that has lost track of the machine. The IR lobe (Select-IR, Capture-IR, Shift-IR, Exit1-IR, Pause-IR, Exit2-IR, Update-IR) loads the instruction register; the DR lobe loads whichever data register the current instruction selects. Figure 11.2 is the machine with every transition this chip implements.

A scan has three phases: Capture loads the selected register in parallel from what it shadows, Shift moves one bit in from TDI and one out on TDO per TCK (least-significant bit first), and Update commits the shifted-in value. One scan therefore reads the old contents and writes the new ones at once, so a debugger's traffic is a stream of fixed-length shifts rather than read-then-write pairs. TMS and TDI are sampled on the rising TCK edge and TDO changes on the falling edge, which is why its transitions sit half a cycle after the others. Figure 11.3 follows one four-bit scan.

The port is passive with respect to the chip's own clocks: it answers while the harts run, while they are halted, and while they are held in reset. Because the standard's BYPASS instruction selects a single flip-flop, several devices can be chained (TDO of one to TDI of the next) and a debugger addresses one while the others contribute one bit of padding each.



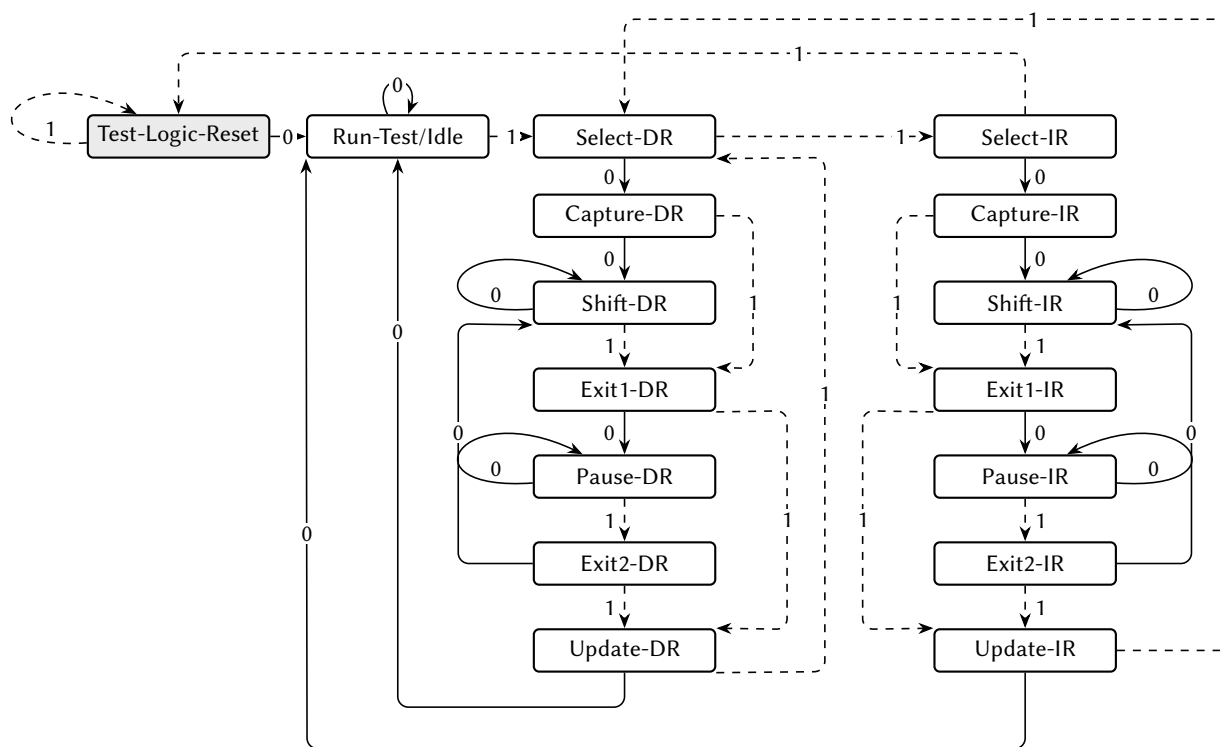


Figure 11.2: State diagram of the sixteen-state JTAG test access port.

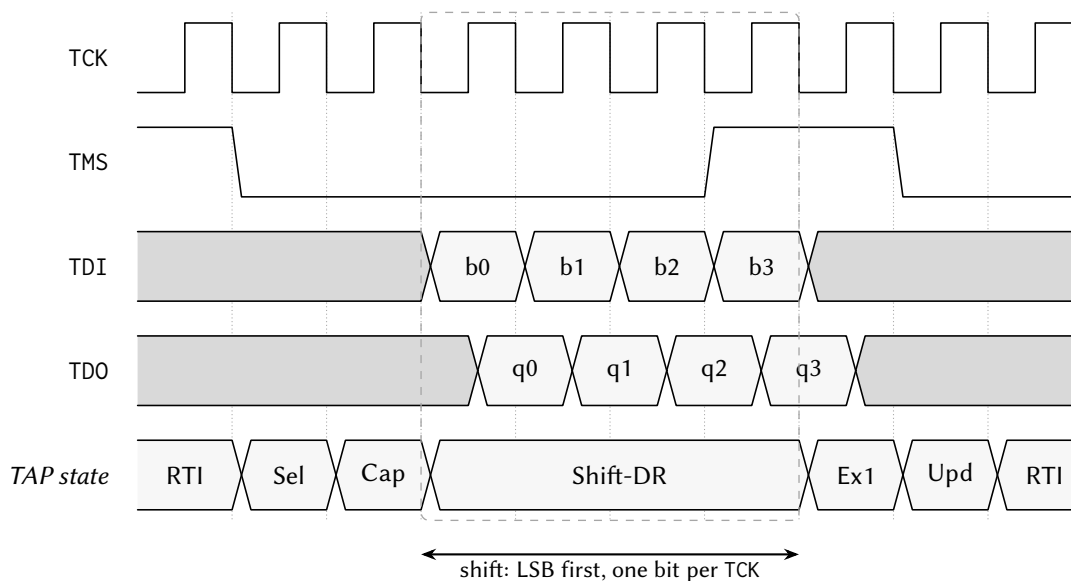


Figure 11.3: Timing diagram of one four-bit JTAG data-register scan.

### 11.3.2 The RISC-V debug stack

The RISC-V debug specification defines the meaning of the scanned bits in four layers, each independent of the internals of the one below it, so that one debugger drives different chips.

| Layer             | What it is                                                                                                                                                                                        |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DTM</b>        | The Debug Transport Module: the TAP plus the logic that turns one DR scan into one transaction on the interface below. It has no knowledge of harts.                                              |
| <b>DMI</b>        | The Debug Module Interface: a 7-bit address, 32-bit data bus with three operations (no-op, read, write). Anything that drives a DMI can debug the chip; JTAG is the transport this chip provides. |
| <b>DM</b>         | The Debug Module: registers on the DMI implementing run control (select a hart, halt it, query its state, run a short program on it, resume it). One DM serves every hart.                        |
| <b>Debug mode</b> | A state of the hart, effectively a privilege level above machine mode, in which the hart executes the debugger's code and has control registers that exist only there.                            |

A halted hart is not a frozen hart. Halting does not stop the clock; it diverts the hart, between instructions, to a fixed address where the debugger's code waits, and the hart executes there until told what to do. To read a register the debugger writes a two- or three-instruction program into memory, has the hart run it, and reads back where the program put the answer. Every property that follows derives from this: the debugger needs memory for its code, a register read can report an exception, the hart's `s0` and `s1` must be saved before anything else, and memory accesses on the debugger's behalf go through the ordinary bus with ordinary arbitration.

### 11.3.3 Debug port

The debug port occupies five pins, present only on the LQFP-100 package; a smaller package has the debug system in silicon with nowhere to bond it. The pins are ordinary 3.3 V digital I/O on the VDDPST/VSSPST domain, may be left unconnected, and are placed on the south and east edges away from the analog band.

| Pin | Signal | I/O    | Idle state when unconnected                                                                                       |
|-----|--------|--------|-------------------------------------------------------------------------------------------------------------------|
| 47  | TCK    | Input  | Pulled low.                                                                                                       |
| 48  | TMS    | Input  | Pulled high, as IEEE 1149.1 requires, so that five clocks from a disconnected probe still reach Test-Logic-Reset. |
| 49  | TDI    | Input  | Pulled high.                                                                                                      |
| 50  | TDO    | Output | Driven; held quiet between shift states so the pin is well behaved in a chain.                                    |
| 51  | TRSTn  | Input  | Pulled low; see below.                                                                                            |

The pull on TRSTn is a deliberate deviation from IEEE 1149.1, which wants the pin to idle high so that a port with a floating reset is available to a probe. This chip pulls it low, holding the TAP in reset, because the transport is optional: a board that does not use it leaves all five pins unconnected, and the desired behaviour is then a port that is inert rather than one that is attachable by anyone who touches the pins. A board that uses the port drives TRSTn high.

A DR scan of IDCODE returns `0x1CA57EEF`: version 1, part number `0xCA57`, manufacturer `0x777`, mandatory LSB 1. The manufacturer field is not a valid JEDEC JEP106 code and is not registered to anyone;

the value is deliberately invalid rather than borrowed, so discovery software identifies the part by its full IDCODE.

### 11.3.4 Debug Transport Module

The DTM implements the TAP and four data registers. The instruction register is five bits wide and resets to IDCODE (Table 11.1).

| IR value         | Instruction | DR width | Selects                                                                                                                    |
|------------------|-------------|----------|----------------------------------------------------------------------------------------------------------------------------|
| 0x01             | IDCODE      | 32       | The identification value. The reset instruction, so the first DR scan after a port reset returns the IDCODE.               |
| 0x10             | DTMCS       | 32       | The transport's control and status register.                                                                               |
| 0x11             | DMI         | 41       | One transaction on the Debug Module Interface.                                                                             |
| 0x1F             | BYPASS      | 1        | A single flip-flop, for a multi-device chain.                                                                              |
| <i>any other</i> | BYPASS      | 1        | Every unrecognised instruction selects the one-bit chain, so the selected length never depends on what was scanned before. |

Table 11.1: Debug Transport Module instruction register values.

Test-Logic-Reset sets the instruction register to IDCODE and nothing else: it does not clear the sticky status below and does not discard a waiting result, so a debugger that has recovered a lost TAP still clears the status with `dmireset`. `TRSTn` resets the entire transport.

DTMCS reports the transport's capabilities and carries the two recovery strobes:

| Bits    | Field        | Meaning                                                                                     |
|---------|--------------|---------------------------------------------------------------------------------------------|
| [3:0]   | version      | 1: satisfies the 0.13 and 1.0 debug specifications.                                         |
| [9:4]   | abits        | 7: the DMI address is seven bits, so the DMI register is $7 + 32 + 2 = 41$ bits.            |
| [11:10] | dmistat      | Sticky transaction status.                                                                  |
| [14:12] | idle         | 7: Run-Test/Idle clocks to insert between a DMI scan and the scan that collects its result. |
| [16]    | dmireset     | Write 1 to clear dmistat and abandon any outstanding transaction. Reads 0.                  |
| [17]    | dmihardreset | Write 1 to do the above and clear the stored result. Reads 0.                               |

The `idle` value is derived from the real latency of the slowest transaction class. A DMI access to one of the Debug Module's own registers completes in eight to ten cycles of the main clock; an access to `data0` or a program-buffer word is proxied into shared memory through the arbiter and takes tens of cycles. Against a TCK period of at least 140 ns (7.14 MHz or below) and a 24 MHz main clock, the worst class needs six idle cycles and 7 gives one cycle of margin. It is a recommendation, not a guarantee: at a faster TCK, or while the harts contend for the shared bus, a transaction can still be outstanding when the debugger returns, which is reported as busy and answered by clearing the status and reissuing.

The DMI data register is 41 bits (Figure 11.4): op [1:0], data [33:2], address [40:34]. On the way in, op is 00 for no operation, 01 for read and 10 for write; on the way out it is 00 for success, 10 for failed and 11 for busy. A no-operation scan starts nothing, which makes a capture-only scan (zeros shifted in to read the previous result) safe. The register is shifted least-significant bit first, so op goes in and comes out first; the field widths in the figure are drawn for legibility and the bit numbers are exact.

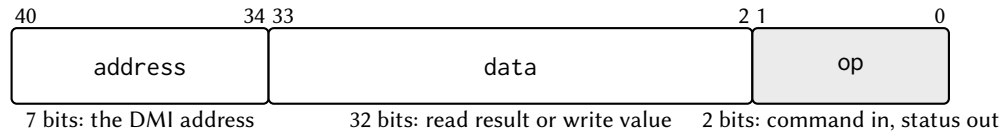


Figure 11.4: Field layout of the 41-bit DMI data register.

dmistat is sticky with two values: 3 (busy) if a scan is captured while a transaction is in flight or a new transaction is scanned in while one is outstanding, and 2 (failed) if a transaction fails, for instance on a DMI address the Debug Module does not implement. While dmistat is nonzero, further DMI scans are dropped, so that a debugger cannot issue a long sequence of writes after the first one failed. The status is cleared by dmireset or dmihardreset and by nothing else; neither Test-Logic-Reset nor a system reset clears it. Figure 11.5 follows one transaction across the clock boundary.

The Update-DR of a DMI scan writes the 41-bit request into a hold register and flips req\_tgl; the main-clock side synchronises the toggle through three flops and presents the held payload as dmi\_req\_valid for exactly two cycles, retiring it on the registered dmi\_req\_ready. The Debug Module's re-accept lockout is a timer, not a handshake: it reopens 9 mclk after a capture. A master that held valid until its response came back would still be asserting it then and would earn a second, duplicate accept: two responses for one request, sliding every later pair by one. The symptom is not a wrong answer; it is the previous answer. Two cycles leaves seven inside the window. The response latches op and data into a 34-bit hold and flips rsp\_tgl, which crosses back through three TCK flops before the shadow register updates.

The idle value of 7 follows from the same path. A debugger sees the result no earlier than 3 TCK (the response synchroniser) plus  $\lceil t_{DM}/T_{TCK} \rceil$ , where  $t_{DM}$  is 8 to 10 mclk for a Debug Module register and tens of cycles for data0 or a program-buffer word, which are proxied into shared RAM through the arbiter. At  $TCK \leq 7.14$  MHz against a 24 MHz mclk that is 6 cycles for the worst class; 7 adds one of margin and is the largest value the 3-bit field can hold. A faster TCK, or a contended bus, can still report busy.

### 11.3.5 Debug Module

There is one Debug Module for the chip. It runs on the always-running main clock, is reached by the DTM through the DMI, and, because it must place instructions where a hart will fetch them, it is also a master on the shared-memory arbiter alongside the harts (Section 5). Table 11.2 is its DMI address map.

Table 11.2: Debug Module registers at their DMI addresses.

| DMI  | Register  | Purpose                                                             |
|------|-----------|---------------------------------------------------------------------|
| 0x04 | data0     | The abstract-command data word, backed by shared memory.            |
| 0x10 | dmcontrol | Run control: enable the module, select a hart, halt it, resume it.  |
| 0x11 | dmstatus  | The selected hart's state and the module's capabilities. Read-only. |

*Table 11.2 continued on next page*

Table 11.2 continued from previous page

| DMI  | Register     | Purpose                                                                                           |
|------|--------------|---------------------------------------------------------------------------------------------------|
| 0x12 | hartinfo     | Reads zero: no scratch registers and no debugger-visible data0 address. Read-only.                |
| 0x13 | haltsum1     | Reads zero. Present because debuggers probe it during discovery.                                  |
| 0x16 | abstractcs   | Abstract-command status: busy, error code, sizes.                                                 |
| 0x17 | command      | Write to start an abstract command. Reads zero.                                                   |
| 0x18 | abstractauto | Reads zero; automatic re-execution is not implemented.                                            |
| 0x20 | progbuf0     | Program-buffer word 0, backed by shared memory.                                                   |
| 0x21 | progbuf1     | Program-buffer word 1, backed by shared memory.                                                   |
| 0x32 | dmcs2        | Halt-group membership of the selected hart.                                                       |
| 0x40 | haltsum0     | One bit per hart: halted and reachable. The only status register that ignores the hart selection. |

Any other DMI address returns failed, which because dmistat is sticky also stalls the transport until cleared. haltsum1 and abstractauto answer with a successful zero because debuggers routinely read them during discovery. hartinfo reads zero on purpose: with dataaccess = 0 a conforming debugger does not consult dataaddr and reaches memory through the program buffer, which is the only path this chip supports, and the real data0 address does not fit the twelve-bit field.

dmcontrol is where every session starts. dmactive (bit 0) must be written 1 before the module does anything; writing 0 resets the module's own state but not the harts. hartsel (bits [25:16]) selects the hart the status and command registers refer to; all ten bits are implemented, and an index that does not exist is reported as nonexistent rather than clamped, which is what makes hart-count discovery work. haltreq (bit 31) is a level that asks the selected hart to halt while set. resumereq (bit 30) is a write-one request that is queued rather than refused and ignored for a hart that is not halted or not reachable. setresethaltreq and clrresethaltreq (bits 3 and 2) arm and disarm halt-on-reset for the selected hart. ndmreset, hartreset and hasel are not implemented; they read zero and ignore writes.

dmstatus reports version = 3 (specification 1.0), authenticated = 1 (no authentication), hasresethaltreq = 1 and impebreak = 1. Hart state is reported through six pairs of bits, and because one hart is selected at a time both bits of each pair carry the same value:

- allhalted, anyhalted
- allrunning, anyrunning
- allunavail, anyunavail
- allnonexistent, anynonexistent
- allresumeack, anyresumeack
- allhavereset, anyhavereset

abstractcs reports progbufsize = 2 and datacount = 1. busy (bit 12) is set while a command runs, and the module stays readable so the debugger can poll. cmderr (bits [10:8]) is write-one-to-clear:

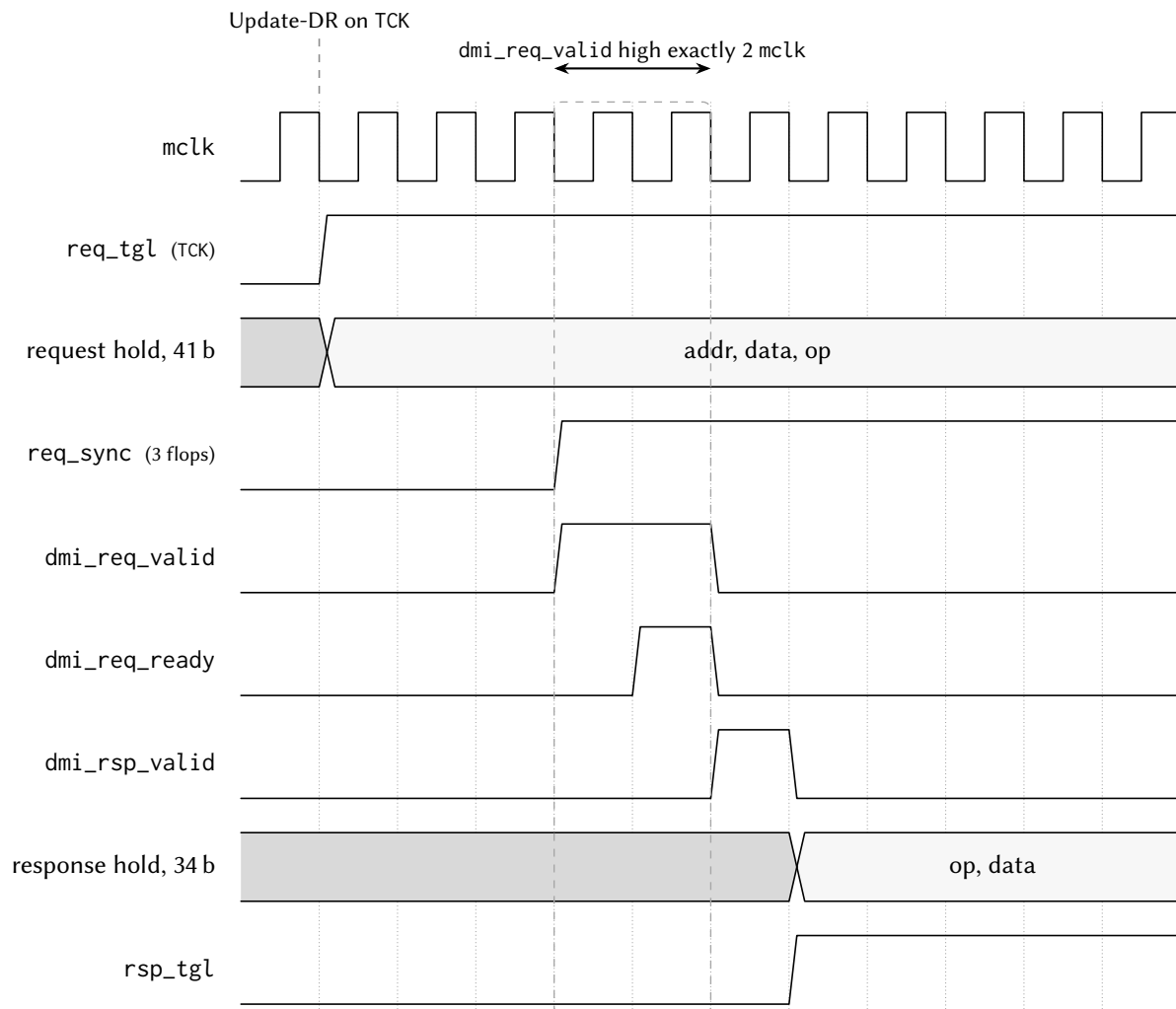


Figure 11.5: Timing diagram of one DMI transaction crossing between the two clock domains.

| <b>cmderr</b> | <b>Meaning</b> | <b>Raised when</b>                                                                                                         |
|---------------|----------------|----------------------------------------------------------------------------------------------------------------------------|
| 0             | none           | No error.                                                                                                                  |
| 1             | busy           | A command was written while one was running, or data0 or a program-buffer word was accessed while the bus engine was busy. |
| 2             | not supported  | The command type is not access-register, or the transfer size is not 32-bit.                                               |
| 3             | exception      | The hart took an exception while executing the command.                                                                    |
| 4             | halt/resume    | The selected hart was not halted when the command was written.                                                             |
| 7             | other          | The command did not complete within the module's internal timeout.                                                         |

A command written while `cmderr` is nonzero is ignored entirely: it does not run and does not clear the error.

`command` implements one command type, access register: it transfers one 32-bit value between `data0` and a register of the halted hart, optionally executing the program buffer afterwards. The fields are `aarsize` ([22:20], which must be 2), `postexec` (bit 18), `transfer` (bit 17), `write` (bit 16) and `regno` ([15:0]). Register numbers 0x1000 to 0x101F are x0 to x31; any other number is a CSR address. Access memory is not implemented and returns not supported; memory is reached by executing `lw` or `sw` from the program buffer.

`data0` and the program-buffer words are shared memory, not flip-flops: a DMI access to 0x04, 0x20 or 0x21 becomes a read or write of a word in shared memory issued by the Debug Module through the arbiter. That is why those addresses are the slow class behind `idle = 7`, and why an access to them while the bus engine is busy answers with zero data and sets `cmderr` to busy. The program buffer is two DMI-visible words plus an implicit third word supplied by hardware, which is why `impebreak` reads 1.

### 11.3.6 Debug mode in the hart

A hart enters debug mode for one of four reasons, recorded in `dcsr.cause`:

| <b>cause</b> | <b>Reason</b> | <b>Notes</b>                                                                                                         |
|--------------|---------------|----------------------------------------------------------------------------------------------------------------------|
| 1            | EBREAK        | A software breakpoint, taken only when <code>dcsr.ebreakm</code> is set (or when the hart is already in debug mode). |
| 3            | Halt request  | The Debug Module asserted <code>haltreq</code> .                                                                     |
| 4            | Single step   | The one instruction permitted by <code>dcsr.step</code> has retired.                                                 |
| 5            | Halt on reset | <code>setresethaltreq</code> was armed when this hart left reset.                                                    |

On entry the hart saves its program counter to `dpc`, records the cause and previous privilege level in `dcsr`, sets debug mode, and jumps to the fixed address 0x00010780 in the shared window. Four CSRs exist only in debug mode:

| Address | Register  | Contents                                                                                                                                                                                                                                           |
|---------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x7B0   | dcsr      | xdebugver [31:28] reads 4; ebreakm (15) makes EBREAK enter debug mode; ebreaku (12) does the same for user mode where it is configured; cause [8:6] is read-only; step (2) enables single-stepping; prv [1:0] is the privilege level to return to. |
| 0x7B1   | dpc       | The saved program counter. Bit 0 is hardwired zero; bit 1 is writable only where the compressed extension is configured.                                                                                                                           |
| 0x7B2   | dscratch0 | Scratch word for the debugger.                                                                                                                                                                                                                     |
| 0x7B3   | dscratch1 | Second scratch word.                                                                                                                                                                                                                               |

Every CSR instruction naming an address in 0x7B0 to 0x7BF is an illegal instruction outside debug mode, so a program cannot read the debugger's saved state or forge an entry. DRET (0x7B200073) leaves debug mode, restoring the program counter from dpc and the privilege level from dcsr.prv; outside debug mode it is an illegal instruction.

Five behaviours decide whether a session works:

- **The halt request is unmaskable.** It is recognised at the thirteen sequencer points at which an interrupt is recognised plus the terminal trap state, so a hart that has taken an illegal instruction and stopped retiring (Section 8.2) can be halted and inspected. Neither mstatus.MIE nor mtrapctl.LEGACY affects it.
- **An EBREAK records its own address in dpc,** not the address after it, so a debugger resuming past a software breakpoint adds the instruction's length itself. Every other cause is taken between instructions and dpc is the resume address.
- **No interrupt is taken in debug mode.** Sources stay pending as levels and are delivered after DRET, so a session neither loses interrupts nor runs the application's handlers underneath the debugger.
- **An exception in debug mode re-enters debug mode** at the same fixed address, leaving dpc, dcsr, mepc, mcause and mtval untouched, which is what makes a faulting abstract command report cmderr = 3 instead of ending the session.
- **Halt-on-reset is sampled once, at reset release.** Arming setresethaltreq affects the next time the hart leaves reset and does not halt a running hart.

Single-step diverts at the stepped instruction's own dispatch, so exactly one instruction retires per step; stepping is inactive inside debug mode. The hart therefore has exactly two states, running and debug mode; the four causes above enter debug mode at the debug entry address, and DRET returns to running at dpc.

### 11.3.7 Halt groups and unavailable harts

dmcs2 assigns the selected hart to one of eight halt groups; group 0, the reset value, is no group. When any member of a group halts for any reason, including its own EBREAK, the Debug Module requests a halt of every other member, so a breakpoint in one hart stops the cooperating set at very nearly the same instant. Halting is grouped; resuming is done per hart.

Unavailable is a third state, distinct from halted. A hart whose power domain is off cannot answer the Debug Module, and the signals that would report its state are clamped to zero at the domain boundary; a



clamped not-halted is indistinguishable from running, so the module takes a separate unavailable signal from the power controller and consults it before halted or running. The reported state resolves in the order nonexistent, unavailable, halted, running, and a gated hart reports allunavail/anyunavail with both halted and running clear. A halt request to a gated hart does nothing; the debugger powers the hart up through PWRCTRL first, and because a hart returning from power-down cold-boots the ROM and parks, halt-on-reset is armed before powering it up to catch it on its first instruction.

### 11.3.8 Debug program page

Because a halted hart executes code, the debug system owns one 512-byte region of the shared window:

| Address              | Contents                                                                                                                                         |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x10680              | data0, the abstract-command data word.                                                                                                           |
| 0x10684              | progbuf0                                                                                                                                         |
| 0x10688              | progbuf1                                                                                                                                         |
| 0x1068C              | The implicit third program-buffer word, written by the Debug Module.                                                                             |
| 0x10690 to 0x106EC   | Reserved for the Debug Module.                                                                                                                   |
| 0x106F0, 0x106F4     | The entry stub's repair copies of s0 and s1; not where a debugger reads those registers from.                                                    |
| 0x10700 + 4 <i>h</i> | The per-hart handshake word between the Debug Module and hart <i>h</i> .                                                                         |
| 0x10780 to 0x1087F   | The debug entry page, 64 words: the entry stub, the area into which the Debug Module writes each abstract command, and the common exit sequence. |

The boot ROM zeroes 0x10000 to 0x107FF at every boot, so every Debug Module data word below 0x10800 starts from a known value; above 0x10800 nothing is zero-filled, and every word there (the last eight words of the entry stub, the command body and the exit sequence) is code the Debug Module writes before anything reads it. The Debug Module writes the entry page itself, once when `dmactive` is raised and again before acting on any hart it has just seen halt, so nothing is staged in advance and the page cannot be read-only memory. The page is assembled without compressed instructions, because execution of compressed or misaligned instructions from the shared window is outside the tested envelope (Section 5).

While a hart is halted, the single home of the interrupted program's s0 and s1 is the CSR pair `dscratch0` and `dscratch1`: a debugger that reads or writes s0 reads or writes `dscratch0`, and a program-buffer change to s0 survives to the next read because the exit sequence adopts the architectural pair into the CSRs first. The two words at 0x106F0 and 0x106F4 are the stub's own repair copies for a hart that re-enters debug mode from inside the page; the Debug Module writes them only when it authored a new value, and nothing else reads them. A plain halt and resume with no register writes leaves s0 and s1 as it found them.

### 11.3.9 DMI register map, field by field

Table 11.2 says what each Debug Module register is for; the field-level tables below carry the bits. They are generated from `hdl/common/regs/rdl/debug_module.rdl`, the SystemRDL description that is the single source for every register description in this manual. The Debug Module has no row in the register index (Chapter B) because its registers answer on the DMI, not on the memory bus: no load or store reaches them.

Two conventions differ from Table 11.2. SystemRDL addresses in bytes, so each **Offset** below is the DMI address times four: `dmcontrol`, DMI `0x10`, prints `0x40`. Names carry the DM prefix the RTL uses, so `dmcontrol` appears as `DMCONTROL`.

Table 11.3: DEBUG register summary

| Offset       | Name           | Access | Reset      | Description                                |
|--------------|----------------|--------|------------|--------------------------------------------|
| 0x10         | DMDATA0        | rw     | 0x00000000 | Abstract data register 0, datacount = 1    |
| 0x14 to 0x3F | -              | -      | -          | Reserved                                   |
| 0x40         | DMCONTROL      | rw/w   | 0x00000000 | Debug Module control                       |
| 0x44         | DMSTATUS       | r      | 0x00000000 | Debug Module status, read only             |
| 0x48         | DMHARTINFO     | r0     | 0x00000000 | Hart information, read only                |
| 0x4C         | DMHALTSUM1     | r0     | 0x00000000 | Halt summary 1, read only                  |
| 0x50 to 0x57 | -              | -      | -          | Reserved                                   |
| 0x58         | DMABSTRACTCS   | r/rw1  | 0x00000000 | Abstract control and status                |
| 0x5C         | DMCOMMAND      | w      | 0x00000000 | Abstract command                           |
| 0x60         | DMABSTRACTAUTO | rw     | 0x00000000 | Abstract command autoexec                  |
| 0x64 to 0x7F | -              | -      | -          | Reserved                                   |
| 0x80         | DMPROGBUF0     | rw     | 0x00000000 | Program buffer word 0                      |
| 0x84         | DMPROGBUF1     | rw     | 0x00000000 | Program buffer word 1                      |
| 0x88 to 0xC7 | -              | -      | -          | Reserved                                   |
| 0xC8         | DMCS2          | rw/w   | 0x00000000 | Halt-group assignment of the selected hart |
| 0xCC to 0xDF | -              | -      | -          | Reserved                                   |
| 0xE0         | DMSBCS         | r0     | 0x00000000 | System Bus Access control and status       |
| 0xE4 to 0xFF | -              | -      | -          | Reserved                                   |
| 0x100        | DMHALTSUM0     | r      | 0x00000000 | Halt summary 0, read only                  |

### 11.3.10 DMDATA0: Abstract data register 0, datacount = 1

**Offset** 0x10 **Reset** 0x00000000 **Width** 32

Abstract data register 0, datacount = 1. Not a flop in this Debug Module: every access is PROXIED to the backing word at `DATA0_ADDR` through the DM's own shared-bus master, so a read returns whatever the halted hart's abstract command left there. An access while the master engine is busy is refused with `abstractcs.CMDERR = BUSY`. When `abstractauto.AUTOEXECDATA` is set, both a read and a write re-issue the last command at the retire of this access.

| Bits | Field   | Type | Reset      | Description                             |
|------|---------|------|------------|-----------------------------------------|
| 31:0 | DMDATA0 | rw   | 0x00000000 | Abstract command operand / result word. |

### 11.3.11 DMCONTROL: Debug Module control

**Offset** 0x40 **Reset** 0x00000000 **Width** 32

Debug Module control. `DMACTIVE` is the DM's own reset: writing 0 clears `haltreq`, the reset-halt requests, the halt-group pending set, the resume pending set, `busy`, `cmderr`, the plant-owed bits and `abstractauto`, and the RISING edge of `DMACTIVE` re-arms the entry-page plant. `HAVERESET` is deliberately NOT cleared by a `dmactive` toggle, because its value is a claim about the harts rather than about the DM. `NDMRESET` is not implemented and reads 0 (WARL).

| Bits  | Field             | Type | Reset | Description                                                                                                                                                                                                                                    |
|-------|-------------------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | DMHALTREQ         | rw   | 0     | Halt request for the selected hart, a level held until cleared.                                                                                                                                                                                |
| 30    | DMRESUMEREQ       | w    | 0     | Resume request for the selected hart. Queued, never refused and never dropped, but it means something only for a hart that is halted, exists and is available. Reads 0.                                                                        |
| 29    | -                 | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                              |
| 28    | DMACKHAVERESET    | w    | 0     | Acknowledge the selected hart's HAVERESET report. A reset arriving in the same cycle as the acknowledge survives it. Reads 0.                                                                                                                  |
| 27:26 | -                 | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                              |
| 25:16 | DMHARTSEL         | rw   | 0x000 | Hart select. An index at or above the hart count selects a nonexistent hart, which dmstatus reports through ANYNONEXISTENT / ALLNONEXISTENT; the all-ones probe a debugger uses to size the array is handled without indexing the hart arrays. |
| 15:4  | -                 | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                              |
| 3     | DMSETRESETHALTREQ | rw   | 0     | Set the selected hart's reset-halt request (halt it out of reset). Reads back the selected hart's current reset-halt state, so this one bit is a write-1 command in one direction and a status bit in the other.                               |
| 2     | DMCLRRESETHALTREQ | w    | 0     | Clear the selected hart's reset-halt request. SET (bit 3) is evaluated first, so a write with both bits set leaves the request set. Reads 0.                                                                                                   |
| 1     | -                 | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                              |
| 0     | DMACTIVE          | rw   | 0     | Debug Module active. 0 resets the DM's own state (not the harts); the 0-to-1 edge re-arms the entry-page plant. Every dmstatus poll writes this register with DMACTIVE set, which is why the plant is armed on the EDGE and not on the level.  |

### 11.3.12 DMSTATUS: Debug Module status, read only

**Offset** 0x44 **Reset** 0x00000000 **Width** 32

Debug Module status, read only. With hasel dropped, both halves of every any/all pair mirror the selected hart and BOTH are driven, because OpenOCD misreads them otherwise. VERSION = 3 (debug spec 1.0), IMPEBREAK = 1 (the implicit third program-buffer word is an ebreak), HASRESETHALTREQ = 1 and AUTHENTICATED = 1.

| Bits  | Field          | Type | Reset | Description                                               |
|-------|----------------|------|-------|-----------------------------------------------------------|
| 31:23 | -              | -    | -     | Reserved. Reads zero; write zero.                         |
| 22    | DMIMPEBREAK    | r    | 0     | Implicit ebreak after the program buffer. Constant 1.     |
| 21:20 | -              | -    | -     | Reserved. Reads zero; write zero.                         |
| 19    | DMALLHAVERESET | r    | 0     | Every selected hart has reset since the last acknowledge. |
| 18    | DMANYHAVERESET | r    | 0     | Any selected hart has reset since the last acknowledge.   |

| Bits | Field             | Type | Reset  | Description                                                                                                         |
|------|-------------------|------|--------|---------------------------------------------------------------------------------------------------------------------|
| 17   | DMALLRESUMEACK    | r    | 0      | Every selected hart has acknowledged its resume request.                                                            |
| 16   | DMANYRESUMEACK    | r    | 0      | Any selected hart has acknowledged its resume request.                                                              |
| 15   | DMALLNONEXISTENT  | r    | 0      | Every selected hart does not exist.                                                                                 |
| 14   | DMANYNONEXISTENT  | r    | 0      | Any selected hart does not exist.                                                                                   |
| 13   | DMALLUNAVAIL      | r    | 0      | Every selected hart is unavailable: it exists but the DM cannot reach it, because isolation or reset is holding it. |
| 12   | DMANYUNAVAIL      | r    | 0      | Any selected hart is unavailable.                                                                                   |
| 11   | DMALLRUNNING      | r    | 0      | Every selected hart is running.                                                                                     |
| 10   | DMANYRUNNING      | r    | 0      | Any selected hart is running.                                                                                       |
| 9    | DMALLHALTED       | r    | 0      | Every selected hart is halted.                                                                                      |
| 8    | DMANYHALTED       | r    | 0      | Any selected hart is halted.                                                                                        |
| 7    | DMAUTHENTICATED   | r    | 0      | Authenticated. Constant 1: this DM has no authentication.                                                           |
| 6    | -                 | -    | -      | Reserved. Reads zero; write zero.                                                                                   |
| 5    | DMHASRESETHALTREQ | r    | 0      | Reset-halt request supported. Constant 1: the wire exists.                                                          |
| 4    | -                 | -    | -      | Reserved. Reads zero; write zero.                                                                                   |
| 3:0  | DMVERSION         | r    | 0b0000 | Debug Module version. Constant 3 = debug specification 1.0.                                                         |

### 11.3.13 DMHARTINFO: Hart information, read only

**Offset** 0x48 **Reset** 0x00000000 **Width** 32

Hart information, read only. THE NULL CLAIM: all four fields (NSCRATCH, DATAACCESS, DATASIZE, DATAADDR) read 0, which says there are no scratch registers and no shadowed data region, so a debugger must use the program buffer. That is this design's only path. DATAADDR must NOT advertise DATA0\_ADDR: the field is 12 bits and sign-extended, so 0x10680 would publish as 0x680, a resolvable wrong address inside the read-only boot ROM. Clearing DATAACCESS is what makes that unreachable; the DM's own master engine still targets DATA0\_ADDR, it simply does not advertise it.

| Bits | Field | Type | Reset | Description                       |
|------|-------|------|-------|-----------------------------------|
| 31:0 | -     | -    | -     | Reserved. Reads zero; write zero. |

### 11.3.14 DMHALTSUM1: Halt summary 1, read only

**Offset** 0x4C **Reset** 0x00000000 **Width** 32

Halt summary 1, read only. Reads 0 with SUCCESS: a debugger probes this address and must not see a failed transaction. One 32-bit haltsum0 covers every hart count this chip supports.

| Bits | Field | Type | Reset | Description                       |
|------|-------|------|-------|-----------------------------------|
| 31:0 | -     | -    | -     | Reserved. Reads zero; write zero. |

### 11.3.15 DMABSTRACTCS: Abstract control and status

**Offset** 0x58 **Reset** 0x00000000 **Width** 32

Abstract control and status. PROGBUFSIZE = 2 and DATACOUNT = 1 are constants. CMDERR is sticky and write-1-to-clear ('cmderr\_r <= cmderr\_r and not d(10 downto 8)'), and NO ABSTRACT COMMAND STARTS WHILE IT IS SET: a command write arriving with a standing error is ignored outright, error and all, so the diagnosis is never overwritten with BUSY.

| Bits  | Field         | Type | Reset  | Description                                                                                                                   |
|-------|---------------|------|--------|-------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | -             | -    | -      | Reserved. Reads zero; write zero.                                                                                             |
| 28:24 | DMPROGBUFSIZE | r    | 0x00   | Program buffer size in words. Constant 2, with an implicit third word that is an ebreak.                                      |
| 23:13 | -             | -    | -      | Reserved. Reads zero; write zero.                                                                                             |
| 12    | DMBUSY        | r    | 0      | An abstract command is executing. The command WRITE itself completes immediately; this is what the debugger polls.            |
| 11    | -             | -    | -      | Reserved. Reads zero; write zero.                                                                                             |
| 10:8  | DMCMDERR      | rw1  | 0b000  | Sticky command error: 0 none, 1 busy, 2 not supported, 3 exception, 4 halt/resume state wrong. Write 1s to the bits to clear. |
| 7:4   | -             | -    | -      | Reserved. Reads zero; write zero.                                                                                             |
| 3:0   | DMDATACOUNT   | r    | 0b0000 | Number of abstract data registers. Constant 1.                                                                                |

### 11.3.16 DMCOMMAND: Abstract command

**Offset** 0x5C **Reset** 0x00000000 **Width** 32

Abstract command. Write-only: the read mux returns 0. Only cmdtype 0 (access register) with aarsize 2 (32-bit) is accepted; cmdtype 2 (access memory) answers CMDERR = NOT\_SUPPORTED so the debugger falls back to program-buffer lw/sw through the hart, which is this design's memory path. A write is refused, with the reason in abstractcs.CMDERR, when a command error is already standing (ignored outright, no new error), when a command is already running (BUSY), when the selected hart is not halted (HALTRES), or when the DM's master engine is busy (BUSY).

| Bits | Field     | Type | Reset      | Description                                                                                                   |
|------|-----------|------|------------|---------------------------------------------------------------------------------------------------------------|
| 31:0 | DMCOMMAND | w    | 0x00000000 | Abstract command word: [31:24] cmdtype, [22:20] aarsize, [17] postexec, [16] transfer, [15:0] regno. Reads 0. |

### 11.3.17 DMABSTRACTAUTO: Abstract command autoexec

**Offset** 0x60 **Reset** 0x00000000 **Width** 32

Abstract command autoexec. AUTOEXECDATA is bit 0 and AUTOEXECPROGBUF is bits 17:16, because datacount is 1 and progbufsize is 2; every other bit is hardwired 0, so the register is WARL by construction and an all-ones write reads back 0x00030001. A write arriving while a command executes DOES NOT LAND and reports CMDERR = BUSY: re-arming the trigger under a running command is exactly the race that rule exists to refuse. Cleared by a DMCONTROL.DMACTIVE toggle.

| Bits  | Field             | Type | Reset | Description                                                                                 |
|-------|-------------------|------|-------|---------------------------------------------------------------------------------------------|
| 31:18 | -                 | -    | -     | Reserved. Reads zero; write zero.                                                           |
| 17:16 | DMAUTOEXECPROGBUF | rw   | 0b00  | Re-issue the last abstract command on any access to progbuf0 (bit 16) or progbuf1 (bit 17). |
| 15:1  | -                 | -    | -     | Reserved. Reads zero; write zero.                                                           |

| Bits | Field          | Type | Reset | Description                                                                                                                                                                                   |
|------|----------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | DMAUTOEXECDATA | rw   | 0     | Re-issue the last abstract command on any access to data0, read or write: a read burst pulls results out and a write burst pushes operands in, and both need the command run once per access. |

### 11.3.18 DMPROGBUF0: Program buffer word 0

**Offset** 0x80 **Reset** 0x00000000 **Width** 32

Program buffer word 0. Proxied to the backing word like data0, and an access while the master engine is busy is refused with CMDERR = BUSY. A written word equal to the 32-bit ebreak encoding 0x00100073 is SUBSTITUTED with the position-correct jump to the epilogue, and a readback returns the substituted word because no shadow of the original is kept; that substitution does not touch the exception detector, so a program that genuinely traps still reports EXCEPTION.

| Bits | Field      | Type | Reset      | Description                        |
|------|------------|------|------------|------------------------------------|
| 31:0 | DMPROGBUF0 | rw   | 0x00000000 | Program buffer instruction word 0. |

### 11.3.19 DMPROGBUF1: Program buffer word 1

**Offset** 0x84 **Reset** 0x00000000 **Width** 32

Program buffer word 1. Same proxy, same busy refusal and same ebreak substitution as progbuf0. The implicit third word is an ebreak (dmstatus.IMPEBREAK = 1), so a two-word program needs no explicit terminator.

| Bits | Field      | Type | Reset      | Description                        |
|------|------------|------|------------|------------------------------------|
| 31:0 | DMPROGBUF1 | rw   | 0x00000000 | Program buffer instruction word 1. |

### 11.3.20 DMCS2: Halt-group assignment of the selected hart

**Offset** 0xC8 **Reset** 0x00000000 **Width** 32

Halt-group assignment of the selected hart. HGSELECT must be 0 (harts, not external triggers) and HGWRITE commits GROUP for the SELECTED hart; grouptype is 0. A read returns the selected hart's current group in bits 6:2.

| Bits | Field      | Type | Reset | Description                                                                                                                                                           |
|------|------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:7 | -          | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                     |
| 6:2  | DMGROUP    | rw   | 0x00  | Halt group of the selected hart. Read returns the group; a write commits it only when HGWRITE is 1 and HGSELECT is 0. The stored group is 3 bits, so bits 6:5 read 0. |
| 1    | DMHGWRITE  | w    | 0     | Commit GROUP to the selected hart. Reads 0.                                                                                                                           |
| 0    | DMHGSELECT | rw   | 0     | Halt-group selector: 0 = harts. Any other value makes the write a no-op.                                                                                              |

### 11.3.21 DMSBCS: System Bus Access control and status

**Offset** 0xE0 **Reset** 0x00000000 **Width** 32

System Bus Access control and status. READ-ZERO-SUCCESS and WRITES IGNORED-SUCCESS, both halves load-bearing. A debugger reads this address once per hart during examination and a FAILED read

aborts examination outright, while it writes sbcs to clear sberror and a failed write is one more gratuitous sticky transport event. Zeros are the honest answer, not a placation: SBVERSION 0, SBASIZE 0 and every access-width bit clear is the debug specification's own encoding for no system bus access. DMI addresses 0x39-0x3F remain unimplemented and answer 'failed'.

| Bits | Field | Type | Reset | Description                       |
|------|-------|------|-------|-----------------------------------|
| 31:0 | -     | -    | -     | Reserved. Reads zero; write zero. |

### 11.3.22 DMHALTSUM0: Halt summary 0, read only

**Offset** 0x100 **Reset** 0x00000000 **Width** 32

Halt summary 0, read only. One bit per hart, set when that hart is halted AND reachable (halted and not unavailable). One 32-bit word covers every hart count this chip supports, which is why haltsum1 reads 0.

| Bits | Field      | Type | Reset      | Description                                  |
|------|------------|------|------------|----------------------------------------------|
| 31:0 | DMHALTSUM0 | r    | 0x00000000 | Halted-and-reachable bitmap, bit h = hart h. |

## 11.4 Interrupts and events

Not applicable. The debug system raises no interrupt vector; a halted hart is observed through dmstatus, and interrupts pending on a hart in debug mode are delivered after DRET.

## 11.5 Programming

### 11.5.1 Halt, read a register, resume

The following session halts hart 2, reads its a1 register and resumes it. Each step is one or more JTAG scans; a DMI read is two scans (one to issue the read and one, after the idle clocks, to collect the result), because the transport returns the previous transaction's result on every capture. Figure 11.6 draws the same steps across the four agents that perform them.

1. Drive TRSTn high, then clock TMS high for five TCK cycles. The instruction register is now IDCODE; a 32-bit DR scan returns 0x1CA57EEF.
2. Scan 0x10 (DTMCS) into the instruction register and read it: version = 1, abits = 7, idle = 7. Insert seven Run-Test/Idle clocks after every DMI scan from here on.
3. Scan 0x11 (DMI) into the instruction register. Every scan below is a 41-bit DR scan of {address, data, op}.
4. Write 0x00000001 to dmcontrol (0x10): dmactive.
5. Write 0x80020001 to dmcontrol: haltreq, hartsel = 2, dmactive.
6. Read dmstatus (0x11) until allhalted (bit 9) is set. allunavail means hart 2 is powered down; allnonexistent means this configuration has fewer than three harts.
7. Write 0x00020001 to dmcontrol to drop the halt request. The hart stays halted, because halting is a state of the hart, not a level the module holds.

8. Write `0x00000700` to `abstractcs` (`0x16`) to clear any stale `cmderr`.
9. Write `0x0022100B` to `command` (`0x17`): `aarsize = 2`, `transfer`, `regno = 0x100B` (`x11`).
10. Read `abstractcs` until `busy` (bit 12) is clear, then check `cmderr`.
11. Read `data0` (`0x04`); this access is proxied through shared memory and is the one the seven idle clocks are sized for.
12. Write `0x40020001` to `dmcontrol`: `resumereq`, hart 2 selected. Read `dmstatus` until `allresumeack` (bit 17) is set.

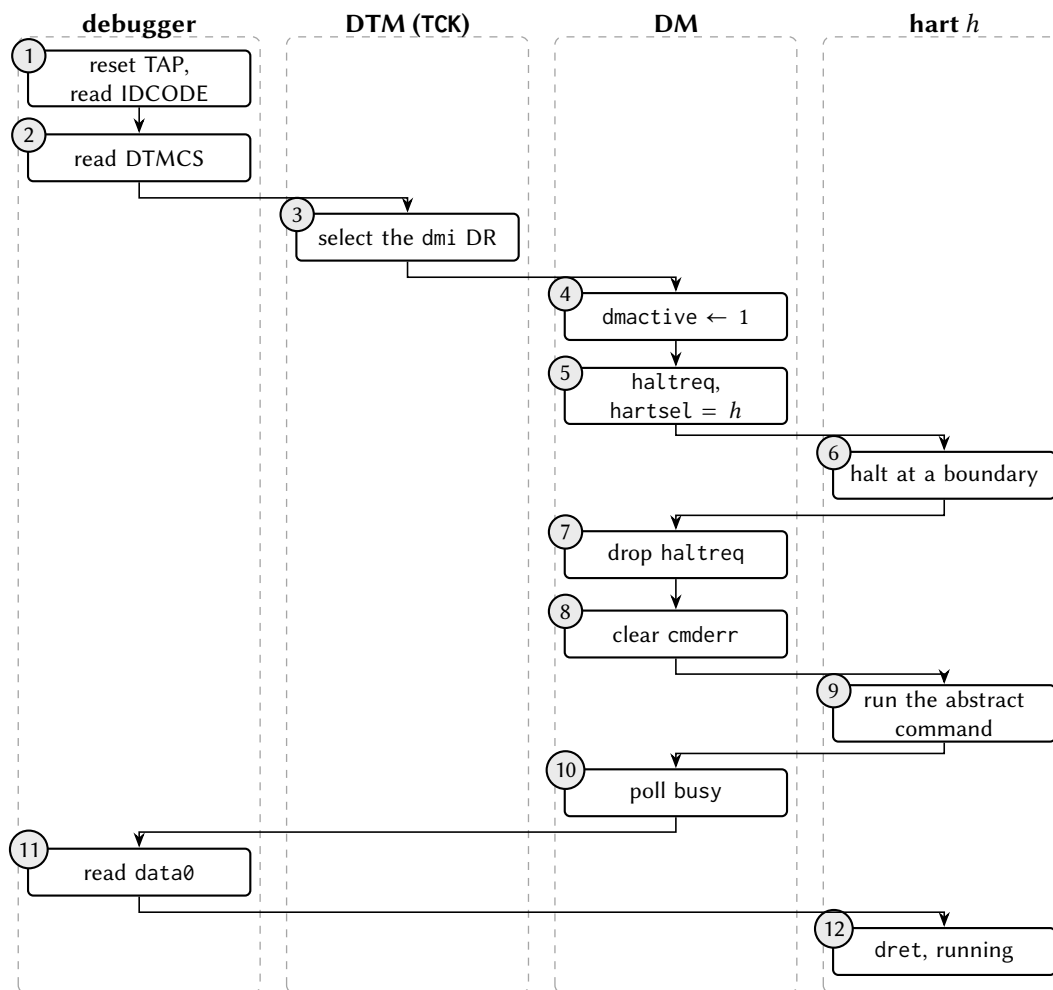


Figure 11.6: Swimlane diagram of the twelve steps across the four agents that perform them.

To write a register, put the value in `data0` first and set `write` (bit 16) in the command. To read or write memory, put an `lw` or `sw` in the program buffer, put the address in a register of the halted hart, and issue a command with `postexec` set; the hart executes the instruction with its own privilege and its own view of the address space. If a scan returns `op = 10` or `11`, the transport has latched a sticky status and is dropping scans: read `DTMCS`, write `dmireset`, and reissue.



### 11.5.2 Connecting a debugger

The part has been brought up end to end against unmodified OpenOCD 0.12.0 and riscv-none-elf-gdb 13.2, which attach, enumerate every hart, halt and resume harts individually, read and write registers and memory, set software breakpoints and single-step. The adapter clock bound and the other constraints of that combination are in Section 11.6. The bound follows from the requirement that three TCK periods cover one DMI transaction in the module's clock domain, because stock OpenOCD does not honour idle and learns the delay from failures:

| Traffic                                   | Minimum $T_{TCK}$ | Maximum $f_{TCK}$ |
|-------------------------------------------|-------------------|-------------------|
| Register class (dmcontrol, dmstatus, ...) | 139 ns            | 7.19 MHz          |
| Program-buffer class (all memory access)  | 278 ns            | 3.60 MHz          |

## 11.6 Usage notes

**Clock the adapter at 3 MHz or below.** Every memory access and software breakpoint goes through the program buffer, whose bound is 3.60 MHz and shrinks under shared-bus contention; the 7.14 MHz port rating applies only to a master that honours idle.

**Do not infer a reset from reset halt.** The module has no system reset (ndmreset is unimplemented), so the command halts every hart, re-reads the IDCODE and reports success while resetting nothing. For a genuine halt-on-reset, arm setresethaltreq and power-cycle the hart's tile through PWRCTRL.

**Experiment on harts 1 to 4, not hart 0.** The per-tile power controller is the lever behind every recovery procedure here; hart 0 has no gate bit, and the only thing that resets it is the resetn pin, which also resets the Debug Module and ends the session.

**Leave the program buffer through the exit sequence, not EBREAK.** A DMI write of exactly the 32-bit EBREAK encoding into progbuf0 or progbuf1 stores a jump to the common exit sequence instead, and reads back as that jump; only that encoding in those two registers is substituted. A program that genuinely traps is still reported as an exception.

**Write memory one word at a time.** abstractauto is unimplemented, and a debugger that assumes it, as OpenOCD's burst-write path does, writes only the first word of a block and drops the rest; loading a program image over the debug port does not work.

**Reach memory only through a halted, healthy hart.** There is no system bus access: memory is read and written by lw and sw executed from the program buffer, with the hart's own permissions and view of the address space, so a chip with no debuggable hart cannot have its memory inspected.

**Use software breakpoints only.** There are no hardware triggers: no watchpoints and no fetch breakpoints. A breakpoint replaces the instruction with EBREAK under dcsr.ebreakm, so breakpoints cannot be set in the boot ROM or any read-only region; data watchpoints by stepping are impractical.

**Keep application software out of 0x10680 to 0x1087F.** The region is ordinary shared RAM, so any master can overwrite it. Most damage repairs itself at the next halt, but a wrong first word at 0x10780 wedges the hart that halts into it: it takes an exception, re-enters at the same address, and is answered from the copy the tile already holds, so it never observes the repair. Recovery is a power cycle of that tile through PWRCTRL.

**Build with `priv.trapCsr` when enabling debug.** `debug.enable` requires the standard trap architecture (Section 8); the generator rejects the combination without it, because EBREAK is decoded on the standard privileged path.

**Resume group members one at a time.** Halt groups stop a set together; resume groups are not implemented.

**Expect specification behaviour, not simulator behaviour.** An unrecognised JTAG instruction selects BYPASS; `dmistat` is driven and reports sticky failure as well as sticky busy; idle reports the real requirement; an out-of-range `hartsel` reports nonexistent; `dmstatus.version` is 3. A debugger written against the specification finds the part conformant; one written against a particular simulator's quirks may not.

**The debug port is not yet in a signed-off layout.** The five pins exist in the chip-level netlist and the LQFP-100 package definition; the current signed-off layout predates them.

# Part III   Peripherals

## 12 General-purpose input and output (GPIOx)

### 12.1 Overview

Each GPIOx peripheral is one 8-pin general-purpose input/output port. A port is a parallel register set, so all eight pins are updated in one access, and every pin is also configurable on its own. Pin  $y$  of port  $x$  is written  $Px.y$  and is controlled by bit  $y$  of each port register. Every pin is either in GPIO mode, driven by the port registers, or in alternate-function mode, driven by one of up to eight peripheral functions selected per pin.

- 8 pins per port, individually configurable as input, output, or resistor-terminated input
- Atomic set, clear and toggle registers (PxOUTS, PxOUTC, PxOUTT), so no read-modify-write is needed
- Alternate-function mode per pin (PxSEL) with an 8-way function select per pin (PxAFS)
- Relocatable peripheral inputs, routed by PxAFS independently of PxSEL
- One interrupt per pin, 8 vectors per port, with programmable edge (PxIES)
- Event-fabric task select per pin (PxTASK) in configurations with the event fabric

### 12.2 Instances and signals

Table 12.1: GPIOx instances

| Instance | Base address | Interrupt vector(s) | Pins (AF)    |
|----------|--------------|---------------------|--------------|
| GPIO0    | 0x4000       | 1 to 8              | P0.0 to P0.7 |
| GPIO1    | 0x4100       | 28 to 35            | P1.0 to P1.7 |
| GPIO2    | 0x4800       | 36 to 43            | P2.0 to P2.7 |
| GPIO3    | 0x4D00       | 44 to 51            | P3.0 to P3.7 |
| GPIO4    | 0x6300       | 98 to 105           | P4.0 to P4.7 |
| GPIO5    | 0x6400       | 106 to 113          | P5.0 to P5.7 |

Table 12.2: GPIOx signals

| Signal | Direction     | Description                                                                                                         | Pin (AF) |
|--------|---------------|---------------------------------------------------------------------------------------------------------------------|----------|
| P0.0   | Bidirectional | General-purpose I/O GPIO0; AF0 CS_FLASH, AF1 TX0, AF2 TX1, AF3 T0CMP0, AF4 T0CMP1, AF5 T1CMP0, AF6 T1CMP1, AF7 SCK1 | P0.0     |
| P0.1   | Bidirectional | General-purpose I/O GPIO1; AF0 MIS00, AF1 TX1, AF2 T0CMP0, AF3 T0CMP1, AF4 T1CMP0, AF5 T1CMP1, AF6 SCK1, AF7 MOSI1  | P0.1     |

| Signal | Direction     | Description                                                                                                        | Pin (AF) |
|--------|---------------|--------------------------------------------------------------------------------------------------------------------|----------|
| P0.2   | Bidirectional | General-purpose I/O GPIO2; AF0 MOSI0, AF1 T0CMP0, AF2 T0CMP1, AF3 T1CMP0, AF4 T1CMP1, AF5 SCK1, AF6 MOSI1, AF7 TX0 | P0.2     |
| P0.3   | Bidirectional | General-purpose I/O GPIO3; AF0 SCK0, AF1 T0CMP1, AF2 T1CMP0, AF3 T1CMP1, AF4 SCK1, AF5 MOSI1, AF6 TX0, AF7 TX1     | P0.3     |
| P0.4   | Bidirectional | General-purpose I/O GPIO4; AF0 LFXT, AF1 T1CMP0, AF2 T1CMP1, AF3 SCK1, AF4 MOSI1, AF5 TX0, AF6 TX1, AF7 T0CMP0     | P0.4     |
| P0.5   | Bidirectional | General-purpose I/O GPIO5; AF0 HFXT, AF1 T1CMP1, AF2 SCK1, AF3 MOSI1, AF4 TX0, AF5 TX1, AF6 T0CMP0, AF7 T0CMP1     | P0.5     |
| P0.6   | Bidirectional | General-purpose I/O GPIO6; AF0 TRAP, AF1 SCK1, AF2 MOSI1, AF3 TX0, AF4 TX1, AF5 T0CMP0, AF6 T0CMP1, AF7 T1CMP0     | P0.6     |
| P0.7   | Bidirectional | General-purpose I/O BOOT; AF1 MOSI1, AF2 TX0, AF3 TX1, AF4 T0CMP0, AF5 T0CMP1, AF6 T1CMP0, AF7 T1CMP1              | P0.7     |
| P1.0   | Bidirectional | General-purpose I/O GPIO8; AF0 CS1, AF1 T0CMP0, AF2 TX1, AF3 T0CMP1, AF4 T1CMP0, AF5 T1CMP1, AF6 SCK1, AF7 MOSI1   | P1.0     |
| P1.1   | Bidirectional | General-purpose I/O GPIO9; AF0 MIS01, AF1 T0CMP1, AF2 T0CMP0, AF3 T1CMP0, AF4 T1CMP1, AF5 SCK1, AF6 MOSI1, AF7 TX0 | P1.1     |
| P1.2   | Bidirectional | General-purpose I/O GPIO10; AF0 MOSI1, AF1 T1CMP0, AF2 T1CMP1, AF3 SCK1, AF4 TX0, AF5 TX1, AF6 T0CMP0, AF7 T0CMP1  | P1.2     |
| P1.3   | Bidirectional | General-purpose I/O GPIO11; AF0 SCK1, AF1 T1CMP1, AF2 MOSI1, AF3 TX0, AF4 TX1, AF5 T0CMP0, AF6 T0CMP1, AF7 T1CMP0  | P1.3     |
| P1.4   | Bidirectional | General-purpose I/O GPIO12; AF0 TX0, AF1 SDA1, AF2 SCK1, AF3 MOSI1, AF4 TX1, AF5 T0CMP0, AF6 T0CMP1, AF7 T1CMP0    | P1.4     |

| Signal | Direction     | Description                                                                                                           | Pin (AF) |
|--------|---------------|-----------------------------------------------------------------------------------------------------------------------|----------|
| P1.5   | Bidirectional | General-purpose I/O GPIO13; AF0 RX0, AF1 SCL1, AF2 T1CMP1, AF3 SCK1, AF4 MOSI1, AF5 TX0, AF6 TX1, AF7 T0CMP0          | P1.5     |
| P1.6   | Bidirectional | General-purpose I/O GPIO14; AF0 TX1, AF1 SDA0, AF2 TX0, AF3 T0CMP0, AF4 T0CMP1, AF5 T1CMP0, AF6 T1CMP1, AF7 SCK1      | P1.6     |
| P1.7   | Bidirectional | General-purpose I/O GPIO15; AF0 RX1, AF1 SCL0, AF2 MOSI1, AF3 TX0, AF4 TX1, AF5 T0CMP0, AF6 T0CMP1, AF7 T1CMP0        | P1.7     |
| P2.0   | Bidirectional | General-purpose I/O GPIO16; AF0 T0CMP0, AF1 TX1, AF2 SCK1, AF3 MOSI1, AF4 TX0, AF5 T0CMP1, AF6 T1CMP0, AF7 T1CMP1     | P2.0     |
| P2.1   | Bidirectional | General-purpose I/O GPIO17; AF0 T0CMP1, AF1 RX1, AF2 T1CMP0, AF3 T1CMP1, AF4 SCK1, AF5 MOSI1, AF6 TX0, AF7 TX1        | P2.1     |
| P2.2   | Bidirectional | General-purpose I/O GPIO18; AF0 T0CAP0, AF1 SDA1, AF2 T0CMP0, AF3 T0CMP1, AF4 T1CMP0, AF5 T1CMP1, AF6 SCK1, AF7 MOSI1 | P2.2     |
| P2.3   | Bidirectional | General-purpose I/O GPIO19; AF0 T0CAP1, AF1 SCL1, AF2 T0CMP1, AF3 T1CMP0, AF4 T1CMP1, AF5 SCK1, AF6 MOSI1, AF7 TX0    | P2.3     |
| P2.4   | Bidirectional | General-purpose I/O GPIO20; AF0 T1CMP0, AF1 TX0, AF2 T0CMP1, AF3 T1CMP1, AF4 SCK1, AF5 MOSI1, AF6 TX1, AF7 T0CMP0     | P2.4     |
| P2.5   | Bidirectional | General-purpose I/O GPIO21; AF0 T1CMP1, AF1 RX0, AF2 TX0, AF3 TX1, AF4 T0CMP0, AF5 T0CMP1, AF6 T1CMP0, AF7 SCK1       | P2.5     |
| P2.6   | Bidirectional | General-purpose I/O GPIO22; AF0 T1CAP0, AF1 SDA0, AF2 SCK1, AF3 MOSI1, AF4 TX0, AF5 TX1, AF6 T0CMP0, AF7 T0CMP1       | P2.6     |
| P2.7   | Bidirectional | General-purpose I/O GPIO23; AF0 T1CAP1, AF1 SCL0, AF2 MOSI1, AF3 TX0, AF4 TX1, AF5 T0CMP0, AF6 T0CMP1, AF7 T1CMP0     | P2.7     |

| Signal | Direction     | Description                                                                                                           | Pin (AF) |
|--------|---------------|-----------------------------------------------------------------------------------------------------------------------|----------|
| P3.0   | Bidirectional | General-purpose I/O GPIO24; AF0 SDA0, AF1 T0CAP0, AF2 TX0, AF3 TX1, AF4 T0CMP0, AF5 T0CMP1, AF6 T1CMP0, AF7 T1CMP1    | P3.0     |
| P3.1   | Bidirectional | General-purpose I/O GPIO25; AF0 SCL0, AF1 T0CAP1, AF2 TX1, AF3 T0CMP0, AF4 T0CMP1, AF5 T1CMP0, AF6 T1CMP1, AF7 SCK1   | P3.1     |
| P3.2   | Bidirectional | General-purpose I/O GPIO26; AF0 SDA1, AF1 T1CAP0, AF2 T0CMP0, AF3 T0CMP1, AF4 T1CMP0, AF5 T1CMP1, AF6 SCK1, AF7 MOSI1 | P3.2     |
| P3.3   | Bidirectional | General-purpose I/O GPIO27; AF0 SCL1, AF1 T1CAP1, AF2 T0CMP1, AF3 T1CMP0, AF4 T1CMP1, AF5 SCK1, AF6 MOSI1, AF7 TX0    | P3.3     |
| P3.4   | Bidirectional | General-purpose I/O GPIO28; AF0 DTP0, AF1 T0CMP0, AF2 TX0, AF3 TX1, AF4 T0CMP1, AF5 T1CMP0, AF6 T1CMP1, AF7 SCK1      | P3.4     |
| P3.5   | Bidirectional | General-purpose I/O GPIO29; AF0 DTP1, AF1 T0CMP1, AF2 RX0, AF3 T0CMP0, AF4 T1CMP0, AF5 T1CMP1, AF6 SCK1, AF7 MOSI1    | P3.5     |
| P3.6   | Bidirectional | General-purpose I/O GPIO30; AF0 DTP2, AF1 T1CMP0, AF2 T0CMP0, AF3 T0CMP1, AF4 T1CMP1, AF5 SCK1, AF6 MOSI1, AF7 MIS01  | P3.6     |
| P3.7   | Bidirectional | General-purpose I/O GPIO31; AF0 DTP3, AF1 T1CMP1, AF2 T0CMP1, AF3 T1CMP0, AF4 SCK1, AF5 MOSI1, AF6 TX0, AF7 TX1       | P3.7     |
| P4.0   | Bidirectional | General-purpose I/O GPIO32                                                                                            | P4.0     |
| P4.1   | Bidirectional | General-purpose I/O GPIO33                                                                                            | P4.1     |
| P4.2   | Bidirectional | General-purpose I/O GPIO34                                                                                            | P4.2     |
| P4.3   | Bidirectional | General-purpose I/O GPIO35                                                                                            | P4.3     |
| P4.4   | Bidirectional | General-purpose I/O GPIO36                                                                                            | P4.4     |
| P4.5   | Bidirectional | General-purpose I/O GPIO37                                                                                            | P4.5     |
| P4.6   | Bidirectional | General-purpose I/O GPIO38                                                                                            | P4.6     |
| P4.7   | Bidirectional | General-purpose I/O GPIO39                                                                                            | P4.7     |
| P5.0   | Bidirectional | General-purpose I/O GPIO40; AF1 NFC_RF_CLK                                                                            | P5.0     |
| P5.1   | Bidirectional | General-purpose I/O GPIO41; AF1 NFC_RF_RX                                                                             | P5.1     |
| P5.2   | Bidirectional | General-purpose I/O GPIO42; AF1 NFC_FIELD_DETECT                                                                      | P5.2     |

| Signal | Direction     | Description                                     | Pin (AF) |
|--------|---------------|-------------------------------------------------|----------|
| P5.3   | Bidirectional | General-purpose I/O GPIO43; AF1<br>NFC_RF_TXMOD | P5.3     |
| P5.4   | Bidirectional | General-purpose I/O GPIO44; AF1<br>NFC_RF_TX_EN | P5.4     |
| P5.5   | Bidirectional | General-purpose I/O GPIO45; AF1<br>NFC_AFE_EN   | P5.5     |
| P5.6   | Bidirectional | General-purpose I/O GPIO46                      | P5.6     |
| P5.7   | Bidirectional | General-purpose I/O GPIO47                      | P5.7     |

Table 12.3: GPIOx interrupt vectors

| Vector | Instance | Description                        | Clear method          |
|--------|----------|------------------------------------|-----------------------|
| 1      | GPIO0    | GPIO0 Bit 0 InterruptIRQB_GPIO0_B0 | Write 1 to P0IF bit 0 |
| 2      | GPIO0    | GPIO0 Bit 1 InterruptIRQB_GPIO0_B1 | Write 1 to P0IF bit 1 |
| 3      | GPIO0    | GPIO0 Bit 2 InterruptIRQB_GPIO0_B2 | Write 1 to P0IF bit 2 |
| 4      | GPIO0    | GPIO0 Bit 3 InterruptIRQB_GPIO0_B3 | Write 1 to P0IF bit 3 |
| 5      | GPIO0    | GPIO0 Bit 4 InterruptIRQB_GPIO0_B4 | Write 1 to P0IF bit 4 |
| 6      | GPIO0    | GPIO0 Bit 5 InterruptIRQB_GPIO0_B5 | Write 1 to P0IF bit 5 |
| 7      | GPIO0    | GPIO0 Bit 6 InterruptIRQB_GPIO0_B6 | Write 1 to P0IF bit 6 |
| 8      | GPIO0    | GPIO0 Bit 7 InterruptIRQB_GPIO0_B7 | Write 1 to P0IF bit 7 |
| 28     | GPIO1    | GPIO1 Bit 0 InterruptIRQB_GPIO1_B0 | Write 1 to P1IF bit 0 |
| 29     | GPIO1    | GPIO1 Bit 1 InterruptIRQB_GPIO1_B1 | Write 1 to P1IF bit 1 |
| 30     | GPIO1    | GPIO1 Bit 2 InterruptIRQB_GPIO1_B2 | Write 1 to P1IF bit 2 |
| 31     | GPIO1    | GPIO1 Bit 3 InterruptIRQB_GPIO1_B3 | Write 1 to P1IF bit 3 |
| 32     | GPIO1    | GPIO1 Bit 4 InterruptIRQB_GPIO1_B4 | Write 1 to P1IF bit 4 |
| 33     | GPIO1    | GPIO1 Bit 5 InterruptIRQB_GPIO1_B5 | Write 1 to P1IF bit 5 |
| 34     | GPIO1    | GPIO1 Bit 6 InterruptIRQB_GPIO1_B6 | Write 1 to P1IF bit 6 |
| 35     | GPIO1    | GPIO1 Bit 7 InterruptIRQB_GPIO1_B7 | Write 1 to P1IF bit 7 |
| 36     | GPIO2    | GPIO2 Bit 0 InterruptIRQB_GPIO2_B0 | Write 1 to P2IF bit 0 |
| 37     | GPIO2    | GPIO2 Bit 1 InterruptIRQB_GPIO2_B1 | Write 1 to P2IF bit 1 |
| 38     | GPIO2    | GPIO2 Bit 2 InterruptIRQB_GPIO2_B2 | Write 1 to P2IF bit 2 |
| 39     | GPIO2    | GPIO2 Bit 3 InterruptIRQB_GPIO2_B3 | Write 1 to P2IF bit 3 |
| 40     | GPIO2    | GPIO2 Bit 4 InterruptIRQB_GPIO2_B4 | Write 1 to P2IF bit 4 |
| 41     | GPIO2    | GPIO2 Bit 5 InterruptIRQB_GPIO2_B5 | Write 1 to P2IF bit 5 |
| 42     | GPIO2    | GPIO2 Bit 6 InterruptIRQB_GPIO2_B6 | Write 1 to P2IF bit 6 |
| 43     | GPIO2    | GPIO2 Bit 7 InterruptIRQB_GPIO2_B7 | Write 1 to P2IF bit 7 |
| 44     | GPIO3    | GPIO3 Bit 0 InterruptIRQB_GPIO3_B0 | Write 1 to P3IF bit 0 |
| 45     | GPIO3    | GPIO3 Bit 1 InterruptIRQB_GPIO3_B1 | Write 1 to P3IF bit 1 |
| 46     | GPIO3    | GPIO3 Bit 2 InterruptIRQB_GPIO3_B2 | Write 1 to P3IF bit 2 |
| 47     | GPIO3    | GPIO3 Bit 3 InterruptIRQB_GPIO3_B3 | Write 1 to P3IF bit 3 |
| 48     | GPIO3    | GPIO3 Bit 4 InterruptIRQB_GPIO3_B4 | Write 1 to P3IF bit 4 |
| 49     | GPIO3    | GPIO3 Bit 5 InterruptIRQB_GPIO3_B5 | Write 1 to P3IF bit 5 |
| 50     | GPIO3    | GPIO3 Bit 6 InterruptIRQB_GPIO3_B6 | Write 1 to P3IF bit 6 |
| 51     | GPIO3    | GPIO3 Bit 7 InterruptIRQB_GPIO3_B7 | Write 1 to P3IF bit 7 |
| 98     | GPIO4    | GPIO4 Bit 0 InterruptIRQB_GPIO4_B0 | Write 1 to P4IF bit 0 |
| 99     | GPIO4    | GPIO4 Bit 1 InterruptIRQB_GPIO4_B1 | Write 1 to P4IF bit 1 |



| Vector | Instance | Description                        | Clear method          |
|--------|----------|------------------------------------|-----------------------|
| 100    | GPIO4    | GPIO4 Bit 2 InterruptIRQB_GPIO4_B2 | Write 1 to P4IF bit 2 |
| 101    | GPIO4    | GPIO4 Bit 3 InterruptIRQB_GPIO4_B3 | Write 1 to P4IF bit 3 |
| 102    | GPIO4    | GPIO4 Bit 4 InterruptIRQB_GPIO4_B4 | Write 1 to P4IF bit 4 |
| 103    | GPIO4    | GPIO4 Bit 5 InterruptIRQB_GPIO4_B5 | Write 1 to P4IF bit 5 |
| 104    | GPIO4    | GPIO4 Bit 6 InterruptIRQB_GPIO4_B6 | Write 1 to P4IF bit 6 |
| 105    | GPIO4    | GPIO4 Bit 7 InterruptIRQB_GPIO4_B7 | Write 1 to P4IF bit 7 |
| 106    | GPIO5    | GPIO5 Bit 0 InterruptIRQB_GPIO5_B0 | Write 1 to P5IF bit 0 |
| 107    | GPIO5    | GPIO5 Bit 1 InterruptIRQB_GPIO5_B1 | Write 1 to P5IF bit 1 |
| 108    | GPIO5    | GPIO5 Bit 2 InterruptIRQB_GPIO5_B2 | Write 1 to P5IF bit 2 |
| 109    | GPIO5    | GPIO5 Bit 3 InterruptIRQB_GPIO5_B3 | Write 1 to P5IF bit 3 |
| 110    | GPIO5    | GPIO5 Bit 4 InterruptIRQB_GPIO5_B4 | Write 1 to P5IF bit 4 |
| 111    | GPIO5    | GPIO5 Bit 5 InterruptIRQB_GPIO5_B5 | Write 1 to P5IF bit 5 |
| 112    | GPIO5    | GPIO5 Bit 6 InterruptIRQB_GPIO5_B6 | Write 1 to P5IF bit 6 |
| 113    | GPIO5    | GPIO5 Bit 7 InterruptIRQB_GPIO5_B7 | Write 1 to P5IF bit 7 |

## 12.3 Functional description

### 12.3.1 GPIO mode

In GPIO mode the pin is controlled by PxDIR, PxOUT and PxREN: PxDIR selects input or output, PxOUT selects the output level, and PxREN enables the internal pull resistor on an input. Table 12.4 lists the four states.

| PxDIR[y] | PxOUT[y] | PxREN[y] | Px.y state                              |
|----------|----------|----------|-----------------------------------------|
| 0        | -        | 0        | High-impedance input, resistor disabled |
| 0        | -        | 1        | Input with pull resistor enabled        |
| 1        | 0        | -        | Output driven low                       |
| 1        | 1        | -        | Output driven high                      |

Table 12.4: GPIO pin states.

PxIN always reads the digital level of the pad, in both directions and in both modes. The read value is latched on the falling edge of the memory enable signal, so a pin that changes during the access returns one stable sample.

### 12.3.2 Set, clear and toggle registers

Writing 1 to a bit of PxOUTS, PxOUTC or PxOUTT sets, clears or toggles the corresponding PxOUT bit; writing 0 has no effect. Each is a single write, so two harts, or a handler and the code it interrupted, can change different pins of one port without a read-modify-write race. PxOUTS and PxOUTT read back PxOUT; PxOUTC reads back its complement.

### 12.3.3 Alternate-function mode

Two registers together decide what drives a pin. PxSEL[y] = 1 puts Px.y in alternate-function mode, and the 4-bit field y of PxAFS (bits 4y + 3 to 4y, of which the low three are implemented) selects which function, AF0 to AF7, governs it. PxAFS resets to 0, so a pin in alternate-function mode drives its AF0 function, which is the pin's home peripheral function; software that only writes PxSEL therefore obtains the home function. Section 3 tabulates every pin's plane assignment.

In alternate-function mode the selected function owns the pin's direction, output and resistor controls, and PxDIR[y], PxOUT[y] and PxREN[y] have no effect until PxSEL[y] returns to 0. A field value that selects a plane on which the pin has no function leaves the pin as a high-impedance input. Some peripheral inputs, such as the timer capture pins, are inputs on every plane and are governed by PxDIR regardless of PxSEL, because the peripheral never drives them.

The planes above AF0 carry a shared pool of relocatable output functions (the UARTx transmitters, the TIMERx compare outputs, the SPI1 clock and master-out) fanned across the ports, so a peripheral output can be placed on whichever pin suits the board. Because there is one instance of each function, selecting it on several pins drives the same signal on all of them. A few plane slots carry bidirectional bus completions so that whole buses relocate as a set: both I2C pairs, the UART0 receiver alongside its transmitter, and the SPI1 master-in alongside its clock and master-out. A function that belongs to a peripheral this configuration omits leaves the pool, and its slots become high-impedance inputs.

### 12.3.4 Relocatable peripheral inputs

A peripheral input that appears at more than one pin (UART receivers, timer captures, I2C data and clock) is routed from the pin whose PxAFS field selects its plane, and from its home pin otherwise, regardless of PxSEL. Input routing ignores PxSEL so that the routing mirrors PxIN, which is always visible; a pin can therefore stay in GPIO output mode while its PxAFS field routes the pad into a peripheral input, which gives a software loopback for self-test. If the same input is selected on more than one pin, a fixed priority (newest alternate location first, then the older alternate, then home) picks one, so the outcome of that configuration error is deterministic.

### 12.3.5 Pin interrupts

Each pin has its own interrupt. With PxIE[y] set, the edge selected by PxIES[y] (0 = rising, 1 = falling) sets PxIF[y] and raises the pin's vector. Pin interrupts stay available in alternate-function mode alongside any interrupt the governing peripheral raises. PxIF is latched on the falling edge of the memory enable signal, so a handler reads one stable sample.

### 12.3.6 Event-fabric tasks

PxTASK selects which pins respond to the event fabric's OUT-SET and OUT-CLR tasks for this port; in a configuration without the event fabric the register is writable and has no effect. A task pulse is applied after any coincident register write in the same cycle, and CLR after SET, so a same-cycle conflict resolves to the cleared, safe direction.

## 12.4 Interrupts and events

Port x owns eight consecutive vectors, one per pin, in the order listed in Table 12.5. The interrupt is a level derived from PxIF[y] and PxIE[y].

## 12.5 Programming

### 12.5.1 Configuring a pin as an output

1. Write the initial level to PxOUT[y].
2. Clear PxSEL[y].

| Port  | Vectors    | Condition             | Set by  | Cleared by           |
|-------|------------|-----------------------|---------|----------------------|
| GPIO0 | 1 to 8     | Selected edge on P0.y | PxIF[y] | Writing 1 to PxIF[y] |
| GPIO1 | 28 to 35   | Selected edge on P1.y | PxIF[y] | Writing 1 to PxIF[y] |
| GPIO2 | 36 to 43   | Selected edge on P2.y | PxIF[y] | Writing 1 to PxIF[y] |
| GPIO3 | 44 to 51   | Selected edge on P3.y | PxIF[y] | Writing 1 to PxIF[y] |
| GPIO4 | 98 to 105  | Selected edge on P4.y | PxIF[y] | Writing 1 to PxIF[y] |
| GPIO5 | 106 to 113 | Selected edge on P5.y | PxIF[y] | Writing 1 to PxIF[y] |

Table 12.5: GPIOx interrupt vectors; pin y uses the port's base vector plus y.

3. Set PxDIR[y].

### 12.5.2 Routing a peripheral output to an alternate pin

1. Write the function's plane number to field y of PxAFS.
2. Set PxSEL[y].

### 12.5.3 Enabling a pin interrupt

1. Clear PxIE[y].
2. Write the edge to PxIES[y].
3. Write 1 to PxIF[y] to discard any stale flag.
4. Enable the pin's vector in the hart's IRQROUTER row.
5. Set PxIE[y].

## 12.6 Usage notes

**Disable the pin interrupt before changing its edge.** Writing PxIES[y] while PxIE[y] is set can raise a spurious interrupt, because the edge detector sees the selection change as an edge.

**Clear PxIF[y] with a 1 before returning.** The interrupt is a level; a handler that returns with the flag set is re-entered.

**Configure capture pins as inputs in PxDIR.** A timer capture input is governed by PxDIR on every plane, so setting PxSEL[y] alone does not make the pad an input.

**Select an input function on one pin only.** The priority among several selected pins is fixed and makes the error deterministic, not correct.

**Use the set, clear and toggle registers from shared code.** A read-modify-write of PxOUT from two harts loses one of the two updates.

## 12.7 Registers

Register offsets in this chapter are relative to the instance base address in Table 12.1 (GPIO0 0x4000, GPIO1 0x4100, GPIO2 0x4800, GPIO3 0x4D00, GPIO4 0x6300, GPIO5 0x6400); the generated header names each base GPIO0\_BASE and so on.

Table 12.6: GPIOx register summary

| Offset | Name   | Access | Reset      | Description                                |
|--------|--------|--------|------------|--------------------------------------------|
| 0x00   | PxIN   | r      | 0x00       | GPIO read pin register                     |
| 0x04   | PxOUT  | rw     | 0x00       | GPIO output drive register                 |
| 0x08   | PxOUTS | rw1    | 0x00       | GPIO output drive set register             |
| 0x0C   | PxOUTC | rw1    | 0x00       | GPIO output drive clear register           |
| 0x10   | PxOUTT | rw1    | 0x00       | GPIO output drive toggle register          |
| 0x14   | PxDIR  | rw     | 0x00       | GPIO pin direction register                |
| 0x18   | PxIF   | rw1    | 0x00       | GPIO interrupt flag register               |
| 0x1C   | PxIES  | rw     | 0x00       | GPIO interrupt edge select register        |
| 0x20   | PxIE   | rw     | 0x00       | GPIO interrupt enable register             |
| 0x24   | PxSEL  | rw     | 0x00       | GPIO peripheral select register            |
| 0x28   | PxREN  | rw     | 0x00       | GPIO resistor enable register              |
| 0x2C   | PxAFS  | rw     | 0x00000000 | GPIO alternate function select register    |
| 0x30   | PxTASK | rw     | 0x00       | GPIO event-fabric task pin-select register |

### 12.7.1 PxIN: GPIO read pin register

**Offset** 0x00 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO read pin register. Each bit corresponds to the input logic state of the GPIO pin of the same number. The register is latched on the falling edge of the memory enable signal. Reading a 0 in a bit indicates a logic low pin state; reading a 1 indicates a logic high state. This register always reflects the pin state regardless of pin direction or peripheral select settings.

| Bits | Field | Type | Reset | Description                                                                                                                                                           |
|------|-------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | IN    | r    | 0x00  | Pin input levels. Bit y reads the logic level of pin y as sampled at the start of the bus access, regardless of the pin direction or PxSEL setting. Bits 31:8 read 0. |

### 12.7.2 PxOUT: GPIO output drive register

**Offset** 0x04 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO output drive register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is set to GPIO (primary) mode in PxSEL. Write a 0 to the desired bit to make the corresponding pin output a logic low value; write a 1 to output a logic high value.

| Bits | Field | Type | Reset | Description                                                                                                                                                                           |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | OUT   | rw   | 0x00  | Output drive levels. Bit y is the level driven on pin y while the pin is an output (PxDIR bit y = 1) in GPIO mode (PxSEL bit y = 0). Resets to 0; bits 31:8 read 0 and ignore writes. |

### 12.7.3 PxOUTS: GPIO output drive set register

**Offset** 0x08 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO output drive set register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is in GPIO (primary) mode in PxSEL. Write a 1 to the desired bit to set the corresponding pin (make the pin output a logic high value). Writing a 0 has no effect. Reading this register is equivalent to reading the output drive register PxOUT.

| Bits | Field | Type | Reset | Description                                                                                                     |
|------|-------|------|-------|-----------------------------------------------------------------------------------------------------------------|
| 7:0  | OUTS  | rw1  | 0x00  | Output set. Writing 1 to bit y sets PxOUT bit y; writing 0 has no effect. Reads return the current PxOUT value. |

### 12.7.4 PxOUTC: GPIO output drive clear register

**Offset** 0x0C **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO output drive clear register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is in GPIO (primary) mode in PxSEL. Write a 1 to the desired bit to clear the corresponding pin (make the pin output a logic low value). Writing a 0 has no effect. Reading this register yields the inversion of the output drive register PxOUT.

| Bits | Field | Type | Reset | Description                                                                                                                        |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | OUTC  | rw1  | 0x00  | Output clear. Writing 1 to bit y clears PxOUT bit y; writing 0 has no effect. Reads return the inverse of the current PxOUT value. |

### 12.7.5 PxOUTT: GPIO output drive toggle register

**Offset** 0x10 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO output drive toggle register. Each bit corresponds to the output logic state of the GPIO pin of the same number. Only has an effect if the pin is configured as an output in PxDIR and is in GPIO (primary) mode in PxSEL. Write a 1 to the desired bit to toggle the corresponding pin state. Writing a 0 has no effect. Reading this register is equivalent to reading the output drive register PxOUT.

| Bits | Field | Type | Reset | Description                                                                                                           |
|------|-------|------|-------|-----------------------------------------------------------------------------------------------------------------------|
| 7:0  | OUTT  | rw1  | 0x00  | Output toggle. Writing 1 to bit y inverts PxOUT bit y; writing 0 has no effect. Reads return the current PxOUT value. |

### 12.7.6 PxDIR: GPIO pin direction register

**Offset** 0x14 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO pin direction register. Each bit corresponds to the GPIO pin of the same number. Only has an effect if the pin is configured in GPIO (primary) mode in PxSEL. Write a 0 to the desired bit to set the corresponding pin to input mode; write a 1 to set to output mode.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                           |
|------|-------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | DIR   | rw   | 0x00  | Pin direction. Bit y = 0 makes pin y an input and 1 makes it an output, effective only while the pin is in GPIO mode (PxSEL bit y = 0). Resets to 0 (all inputs); bits 31:8 read 0 and ignore writes. |

### 12.7.7 PxIF: GPIO interrupt flag register

**Offset** 0x18 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO interrupt flag register. Each bit corresponds to the GPIO pin of the same number. The register is latched on the falling edge of the memory enable signal. Reading a 0 in a bit indicates there is no pending interrupt for the corresponding pin; reading a 1 indicates there is a new interrupt pending for the corresponding pin. Write a 1 to each bit for which you wish to clear the interrupt flag. Writing 0 has no effect.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                            |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | IF    | rw1  | 0x00  | Interrupt flags. Hardware sets bit y when the edge selected by PxIES occurs on pin y while PxIE bit y is 1, and holds it until software writes 1 to the bit; writing 0 has no effect. Each flag drives the interrupt vector of its pin directly, and bits 31:8 read 0. |

### 12.7.8 PxIES: GPIO interrupt edge select register

**Offset** 0x1C **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO interrupt edge select register. Each bit corresponds to the GPIO pin of the same number. Write a 0 to the desired bit to set the corresponding pin interrupt to trigger on low-to-high (rising) edge; write a 1 to set to high-to-low (falling) edge triggering.

| Bits | Field | Type | Reset | Description                                                                                                          |
|------|-------|------|-------|----------------------------------------------------------------------------------------------------------------------|
| 7:0  | IES   | rw   | 0x00  | Interrupt edge select. Bit y = 0 makes a rising edge on pin y set PxIF bit y, 1 selects a falling edge. Resets to 0. |

### 12.7.9 PxIE: GPIO interrupt enable register

**Offset** 0x20 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO interrupt enable register. Each bit corresponds to the GPIO pin of the same number. Write a 0 to the desired bit to disable the pin interrupt; write a 1 to enable the pin interrupt. Each pin has an individual interrupt output that connects to the system interrupt vector table. Interrupts function in both GPIO (primary) and secondary function (peripheral) modes.

| Bits | Field | Type | Reset | Description                                                                                                                                      |
|------|-------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | IE    | rw   | 0x00  | Interrupt enable. Bit y = 1 allows the selected edge on pin y to set PxIF bit y; with the bit at 0 no flag is captured for the pin. Resets to 0. |

### 12.7.10 PxSEL: GPIO peripheral select register

**Offset** 0x24 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO peripheral select register. Each bit corresponds to the GPIO pin of the same number. Write a 0 to the desired bit to set the corresponding pin to GPIO (primary) mode; write a 1 to set the pin to alternate function

(peripheral) mode. When a pin is in alternate function (peripheral) mode, the governing peripheral takes control of the pin output, direction, and resistor enable states, and the PxOUT, PxDIR, and PxREN registers have no effect on the pin. WHICH alternate function governs the pin is selected by the pin's field in the PxAFS register: at reset all PxAFS fields are 0, selecting alternate function 0 (AF0). Pin interrupts remain available when in alternate function (peripheral) mode in addition to any interrupts the governing peripheral may generate. If a pin has no alternate function defined at the selected PxAFS plane, setting PxSEL to 1 will configure the pin as a high-impedance input.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                |
|------|-------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | SEL   | rw   | 0x00  | Pin function select. Bit y = 0 gives pin y to the GPIO registers, 1 hands its output level, direction and resistor enable to the alternate function chosen by the PxAFS field of the pin. Resets to 0; bits 31:8 read 0 and ignore writes. |

### 12.7.11 PxREN: GPIO resistor enable register

**Offset** 0x28 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO resistor enable register. Each bit corresponds to the GPIO pin of the same number. Only has an effect if the pin is configured in GPIO (primary) mode in PxSEL. Write a 0 to the desired bit to disable the pin pullup/pulldown resistor; write a 1 to enable the pin pullup/pulldown resistor.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                           |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | REN   | rw   | 0x00  | Pull resistor enable. Bit y = 1 enables the pull resistor of pin y while the pin is in GPIO mode (PxSEL bit y = 0); in alternate function mode the peripheral controls the resistor. Resets to 0; bits 31:8 read 0 and ignore writes. |

### 12.7.12 PxAFS: GPIO alternate function select register

**Offset** 0x2C **Reset** 0x00000000 **Width** 32 **Address** see Table 12.1

GPIO alternate function select register. One 4-bit field per pin (pin y occupies bits 4y+3 downto 4y; only the low 3 bits of each field are implemented, the top bit is reserved and reads 0). When a pin is in alternate function (peripheral) mode (PxSEL bit = 1), the value of its PxAFS field selects WHICH alternate function (AF0-AF7) controls the pin. Each pin's available alternate functions are listed in the pin configuration table in Section 3. The register resets to 0, so every pin comes out of reset selecting its AF0 (legacy) function. Selecting an AF plane with no function defined for the pin configures the pin as a high-impedance input. While a pin is in GPIO (primary) mode its PxAFS field has no effect on the pad, but it still routes relocatable peripheral INPUTS: a peripheral input function relocated to this pin (e.g. a UART receiver or timer capture) observes the pin whenever the PxAFS field selects it, regardless of PxSEL.

| Bits  | Field | Type | Reset | Description                                               |
|-------|-------|------|-------|-----------------------------------------------------------|
| 31    | -     | -    | -     | Reserved. Reads zero; write zero.                         |
| 30:28 | AFS7  | rw   | 0b000 | Alternate function select for pin 7 (0 = AF0 ... 7 = AF7) |
| 27    | -     | -    | -     | Reserved. Reads zero; write zero.                         |
| 26:24 | AFS6  | rw   | 0b000 | Alternate function select for pin 6 (0 = AF0 ... 7 = AF7) |
| 23    | -     | -    | -     | Reserved. Reads zero; write zero.                         |
| 22:20 | AFS5  | rw   | 0b000 | Alternate function select for pin 5 (0 = AF0 ... 7 = AF7) |
| 19    | -     | -    | -     | Reserved. Reads zero; write zero.                         |

| Bits  | Field | Type | Reset | Description                                               |
|-------|-------|------|-------|-----------------------------------------------------------|
| 18:16 | AFS4  | rw   | 0b000 | Alternate function select for pin 4 (0 = AF0 ... 7 = AF7) |
| 15    | -     | -    | -     | Reserved. Reads zero; write zero.                         |
| 14:12 | AFS3  | rw   | 0b000 | Alternate function select for pin 3 (0 = AF0 ... 7 = AF7) |
| 11    | -     | -    | -     | Reserved. Reads zero; write zero.                         |
| 10:8  | AFS2  | rw   | 0b000 | Alternate function select for pin 2 (0 = AF0 ... 7 = AF7) |
| 7     | -     | -    | -     | Reserved. Reads zero; write zero.                         |
| 6:4   | AFS1  | rw   | 0b000 | Alternate function select for pin 1 (0 = AF0 ... 7 = AF7) |
| 3     | -     | -    | -     | Reserved. Reads zero; write zero.                         |
| 2:0   | AFS0  | rw   | 0b000 | Alternate function select for pin 0 (0 = AF0 ... 7 = AF7) |

### 12.7.13 PxTASK: GPIO event-fabric task pin-select register

**Offset** 0x30 **Reset** 0x00 **Width** 8 **Address** see Table 12.1

GPIO event-fabric task pin-select register. Each bit corresponds to the GPIO pin of the same number and selects whether that pin participates in the EVFAB0 event-fabric output tasks: when the fabric fires the port's OUT-SET task, every pin whose PxTASK bit is 1 has its PxOUT bit set; when it fires the OUT-CLR task, every selected pin has its PxOUT bit cleared. The two task pulses are applied AFTER the CPU register write in the same cycle (a task wins its own pins against a coincident PxOUT write) and CLR is applied after SET (a same-cycle set+clear on an overlapping pin resolves to CLEAR, the safe direction). A toggle task is deliberately not offered. Resets to 0, so no pin is fabric-driven out of reset. In a configuration WITHOUT the event fabric (peripherals.eventFabric false) the register is still readable and writable but has no effect, because both task inputs are tied inactive. Only the port's implemented pins have bits; the rest read 0.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                              |
|------|-------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | TASK  | rw   | 0x00  | Event fabric task pin select. Bit y = 1 lets the EVFAB0 output set and output clear tasks of this port set or clear PxOUT bit y; with the bit at 0 the tasks leave pin y untouched. Resets to 0, so no pin responds to a task until software selects it. |



## 13 Serial peripheral interface (SPIx)

### 13.1 Overview

The Serial Peripheral Interface (SPIx) is a full-duplex synchronous serial interface. A bus has three shared wires (clock, master out, master in) and one chip select per slave. SPI0 is the boot-flash master and carries the memory-mapped flash window; SPI1, in configurations that include it, is a general-purpose port with both master and slave modes.

- Master mode on both ports; slave mode on SPI1
- 8-, 16- or 32-bit frames (SPIDL), LSB- or MSB-first (SPIMSB)
- SPI modes 0 to 3 as master (SPICPOL, SPICPHA); modes 1 and 3 as slave
- Optional byte swap on transmit and receive (SPITXSB, SPIRXSB)
- Baud divider from SMCLK (SPIBR)
- One-deep transmit queue for back-to-back frames without dead time
- Two interrupts per port: transfer complete, transmit register empty
- Flash extended memory on SPI0: memory-mapped reads and execute-in-place from the boot flash for hart 0 (SPIFEN, SPI0FOS)

### 13.2 Instances and signals

Table 13.1: SPIx instances

| Instance | Base address | Interrupt vector(s) | Pins (AF)                                                           |
|----------|--------------|---------------------|---------------------------------------------------------------------|
| SPI0     | 0x4200       | 9 to 10             | MISO0 P0.1 (AF0), MOSI0 P0.2 (AF0), SCK0 P0.3 (AF0)                 |
| SPI1     | 0x4300       | 11 to 12            | SCK1 P1.3 (AF0), MOSI1 P1.2 (AF0), CS1 P1.0 (AF0), MISO1 P1.1 (AF0) |

Table 13.2: SPIx signals

| Signal | Direction     | Description                                              | Pin (AF)   |
|--------|---------------|----------------------------------------------------------|------------|
| MISO0  | Bidirectional | SPI0 Master In Slave Out (connected to SPI flash memory) | P0.1 (AF0) |
| MOSI0  | Output        | SPI0 Master Out Slave In (connected to SPI flash memory) | P0.2 (AF0) |
| SCK0   | Output        | SPI0 serial clock (connected to SPI flash memory)        | P0.3 (AF0) |

| Signal | Direction     | Description              | Pin (AF)                                                                                                                                                                                                                                                                                                                                       |
|--------|---------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCK1   | Output        | SPI1 serial clock        | P0.0 (AF7), P0.1 (AF6), P0.2 (AF5), P0.3 (AF4), P0.4 (AF3), P0.5 (AF2), P0.6 (AF1), P1.0 (AF6), P1.1 (AF5), P1.2 (AF3), P1.3 (AF0), P1.4 (AF2), P1.5 (AF3), P1.6 (AF7), P2.0 (AF2), P2.1 (AF4), P2.2 (AF6), P2.3 (AF5), P2.4 (AF4), P2.5 (AF7), P2.6 (AF2), P3.1 (AF7), P3.2 (AF6), P3.3 (AF5), P3.4 (AF7), P3.5 (AF6), P3.6 (AF5), P3.7 (AF4) |
| MOSI1  | Output        | SPI1 Master Out Slave In | P0.1 (AF7), P0.2 (AF6), P0.3 (AF5), P0.4 (AF4), P0.5 (AF3), P0.6 (AF2), P0.7 (AF1), P1.0 (AF7), P1.1 (AF6), P1.2 (AF0), P1.3 (AF2), P1.4 (AF3), P1.5 (AF4), P1.7 (AF2), P2.0 (AF3), P2.1 (AF5), P2.2 (AF7), P2.3 (AF6), P2.4 (AF5), P2.6 (AF3), P2.7 (AF2), P3.2 (AF7), P3.3 (AF6), P3.5 (AF7), P3.6 (AF6), P3.7 (AF5)                         |
| CS1    | Input         | SPI1 chip select         | P1.0 (AF0)                                                                                                                                                                                                                                                                                                                                     |
| MISO1  | Bidirectional | SPI1 Master In Slave Out | P1.1 (AF0), P3.6 (AF7)                                                                                                                                                                                                                                                                                                                         |

Table 13.3: SPIx interrupt vectors

| Vector | Instance | Description                                              | Clear method                              |
|--------|----------|----------------------------------------------------------|-------------------------------------------|
| 9      | SPI0     | SPI0 Transmission Complete Interrupt<br>IRQB_SPI0_TC     | Write 1 to SPI0SR.SPITCIF, or read SPI0RX |
| 10     | SPI0     | SPI0 Transmission Buffer Empty Interrupt<br>IRQB_SPI0_TE | Write 1 to SPI0SR.SPITEIF                 |
| 11     | SPI1     | SPI1 Transmission Complete Interrupt<br>IRQB_SPI1_TC     | Write 1 to SPI1SR.SPITCIF, or read SPI1RX |
| 12     | SPI1     | SPI1 Transmission Buffer Empty Interrupt<br>IRQB_SPI1_TE | Write 1 to SPI1SR.SPITEIF                 |

### 13.3 Block diagram

Figure 13.1 shows how the two ports are wired in this configuration: SPI0 to the external boot flash on fixed pads, and SPI1 to the pool of relocatable pins. The flash chip select is a GPIO pad at reset, driven by the boot ROM while it loads the image, and bit 0 of P0SEL hands the pad driver to SPI0. Only hart 0 reaches the flash window, and it stalls while SPI0 reads on its behalf (Section 13.4.6).

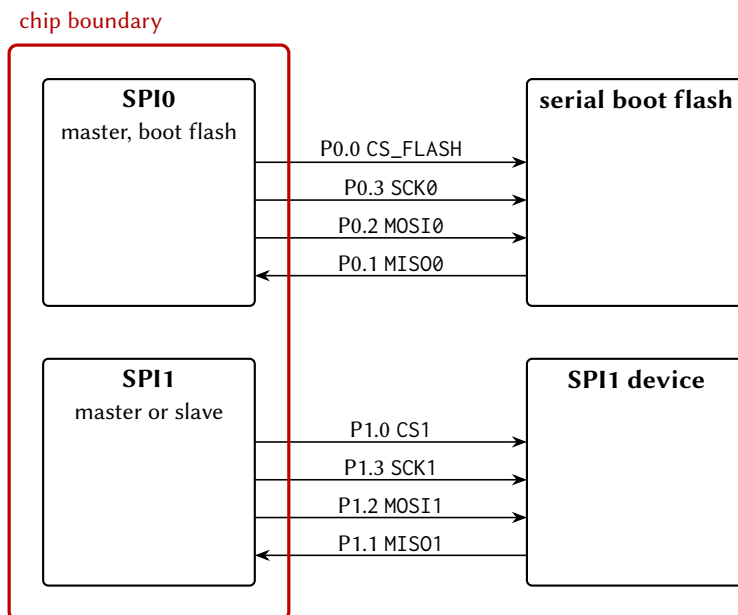


Figure 13.1: SPI connection diagram: SPI0 to the boot flash on fixed pads and SPI1, drawn as master, to an external device.

## 13.4 Functional description

### 13.4.1 Transfer

The bus is idle until the master asserts (drives low) the chip select of one slave. The master then drives the clock for one frame of 8, 16 or 32 bits; on each clock transition the master and the slave alternately update and latch the two data wires, so each frame moves one word in each direction at once. After a frame the master either sends another frame with the chip select still low, which makes the frames one packet, or deasserts the chip select. The slave must have a word ready for the first frame of a packet, because the master clocks it out whether or not the slave has data to send.

The order of update and latch, and the idle level of the clock, are set by clock polarity (CPOL) and clock phase (CPHA), together called the SPI mode (Table 13.4). Figure 13.2 draws all four modes for one 8-bit frame. Slave mode supports modes 1 and 3 only (CPHA = 1), because with CPHA = 0 the slave would have to present its first bit before the first clock edge, before it has seen the chip select settle.

| Mode | CPOL | CPHA | Clock idle | Data update edge   | Data latch edge    |
|------|------|------|------------|--------------------|--------------------|
| 0    | 0    | 0    | Low        | Trailing (falling) | Leading (rising)   |
| 1    | 0    | 1    | Low        | Leading (rising)   | Trailing (falling) |
| 2    | 1    | 0    | High       | Trailing (rising)  | Leading (falling)  |
| 3    | 1    | 1    | High       | Leading (falling)  | Trailing (rising)  |

Table 13.4: SPI mode key.

### 13.4.2 Bit and byte order

SPIMSB selects MSB-first (1) or LSB-first (0) within a frame. For 16- and 32-bit frames, SPITXSB and SPIRXSB swap the byte order of the transmitted and received words, so a word exchanged with a device of the other

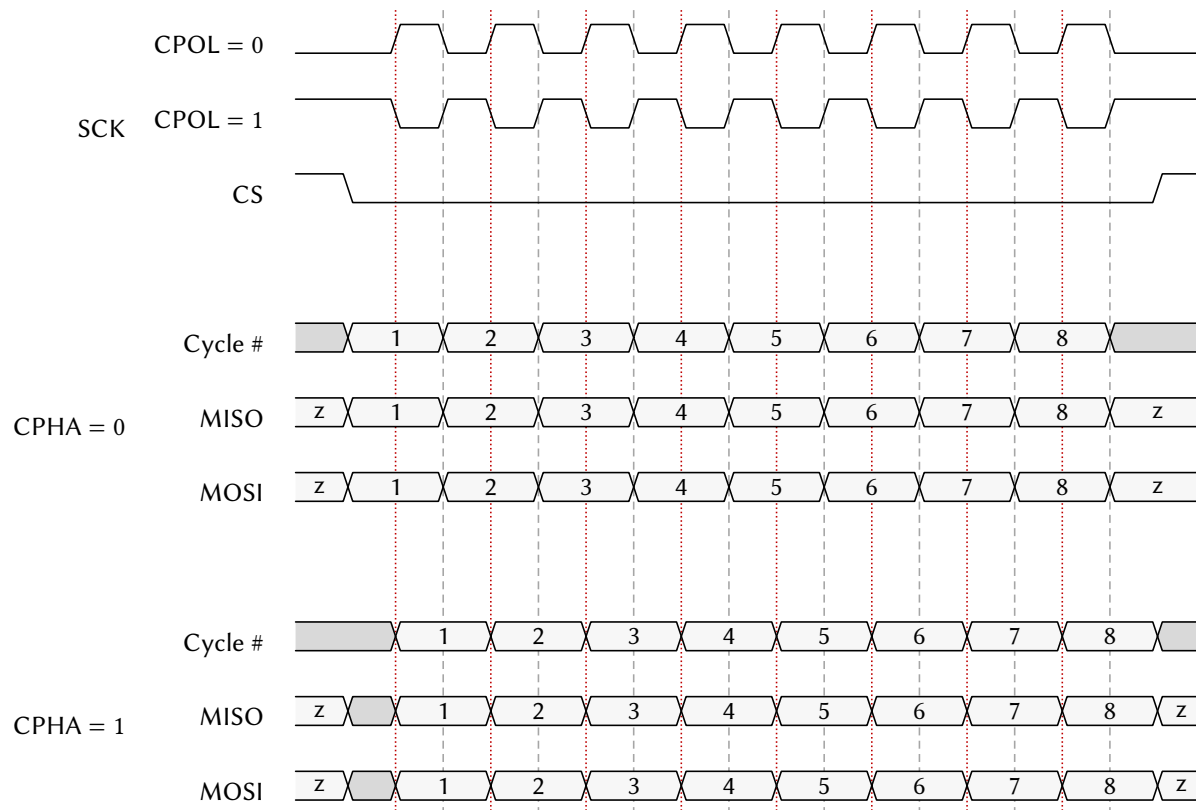


Figure 13.2: SPI timing for the four modes, one 8-bit frame.

endianness needs no software swap. The bit order is applied before the byte swap: a 16-bit MSB-first frame with SPITXSB set sends bits 7 to 0 and then bits 15 to 8. Figures 13.3 and 13.4 show both.

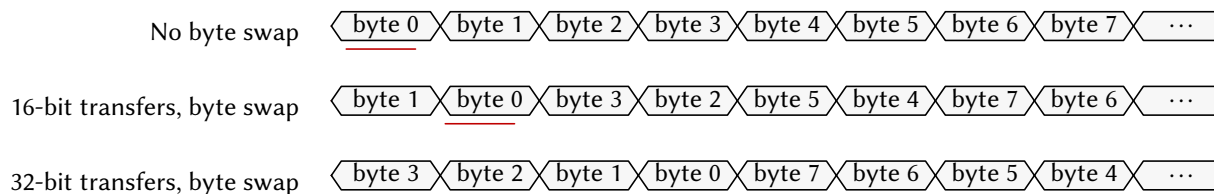


Figure 13.3: Byte order of a multi-byte frame with and without the swap.

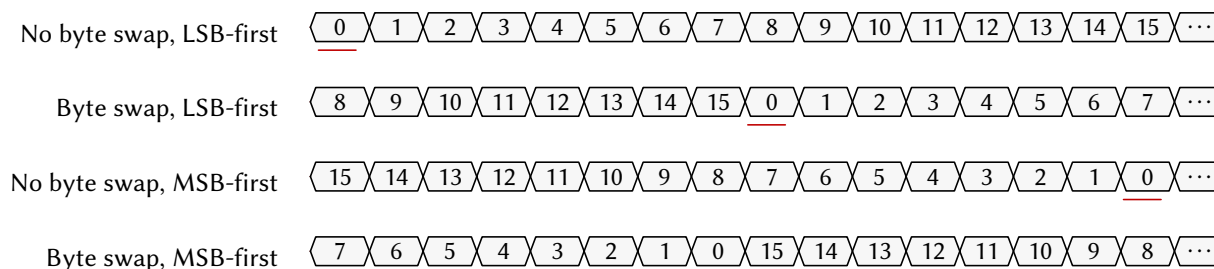


Figure 13.4: Bit order of a 16-bit frame.

### 13.4.3 Baud clock

In master mode the clock frequency is

$$f_{\text{SCK}} = \frac{f_{\text{SMCLK}}}{2 \times (1 + \text{SPIBR})}$$

so the fastest clock is half SMCLK. The port's serial engine runs on SMCLK and its register interface on MCLK, so the registers stay readable with SMCLK stopped.

### 13.4.4 Transmit queue

A write to SPIxTX queues one word. In master mode with no frame in progress the frame starts at once and SPITEIF is set immediately; with a frame in progress the queued word starts when that frame ends, and SPITEIF is set then. In slave mode the last word written is loaded when the master's first clock edge arrives, and SPITEIF is set at that edge. The queue holds one word, so a second write during one frame overwrites the first; software waits for SPITEIF = 1 or SPIBUSY = 0 before each write.

### 13.4.5 Status flags

SPITCIF is set when a frame completes and the received word is in SPIxRX; it is cleared by writing 1 or by reading SPIxRX. SPITEIF is set as described above and cleared by writing 1. SPIBUSY is 1 while the master is clocking a frame, and in slave mode while the chip select is asserted; it can read 0 between frames of one packet when no word was queued. Flash extended memory reads set none of the flags, because they are completed in hardware without software involvement.

### 13.4.6 Flash extended memory

With SPIFEN set, hart 0's loads and instruction fetches at or above  $0x40000$  become flash reads: SPI0 drives CS\_FLASH, sends the read command and a 24-bit address, and stalls hart 0 until the word returns. The window is decoded on hart 0's private bus, not through the arbiter, so the other harts have no path to the flash and read zeros there without stalling. The flash address is

$$\text{Flash address} = (\text{CPU address} + \text{SPI0FOS}) \bmod 2^{24}$$

so SPI0FOS relocates the flash contents within the window without moving data; to place flash address 0 at  $0x40000$  itself, SPI0FOS holds the two's complement of the window base modulo  $2^{24}$ . Each 32-bit read costs a command, an address and four data bytes on the serial clock, so the window suits read-only data, lookup tables and code that is not performance-critical.

### 13.4.7 Boot use of SPI0

The boot ROM uses SPI0 to load the application from the flash (Section 6). During that sequence the ROM drives CS\_FLASH as a GPIO output and the flash drives MISO0; at the end of the sequence the ROM puts the flash into deep power-down. After boot the port is free for ordinary transfers and for the extended memory window.

## 13.5 Interrupts and events

Each flag has an enable in SPIxCR (SPITCIE, SPITEIE); the interrupt is a level asserted while both are set. Table 13.5 lists the vectors.

| Vector | Port | Condition               | Set by  | Cleared by                   |
|--------|------|-------------------------|---------|------------------------------|
| 9      | SPI0 | Frame complete          | SPITCIF | Writing 1, or reading SPIxRX |
| 10     | SPI0 | Transmit register empty | SPITEIF | Writing 1                    |
| 11     | SPI1 | Frame complete          | SPITCIF | Writing 1, or reading SPIxRX |
| 12     | SPI1 | Transmit register empty | SPITEIF | Writing 1                    |

Table 13.5: SPIx interrupt vectors.

## 13.6 Programming

### 13.6.1 Initialisation as a master

1. Select the SMCLK source in SYSCLKCR (Section 16.3.1).
2. Set the PxSEL bits of the SCKx, MOSIx and MISOx pins, and their PxAFS fields if the port is relocated.
3. Configure each slave's chip-select pin as a GPIO output driven high.
4. Write SPIxCR.SPIBR for the clock rate.
5. Write SPIxCR.SPICPOL and SPIxCR.SPICPHA for the mode.

6. Write SPIxCR.SPIDL and SPIxCR.SPIMSB for the frame format.
7. Clear SPIxCR.SPISM.
8. Set SPIxCR.SPIEN.

### 13.6.2 Master transfer of one packet

1. Write 1 to every flag in SPIxSR.
2. Drive the slave's chip-select pin low.
3. Wait for SPITEIF = 1 or SPIBUSY = 0, then write the next word to SPIxTX.
4. Wait for SPITCIF = 1, then read SPIxRX; the read clears the flag.
5. Repeat from step 3 for each further frame.
6. Drive the chip-select pin high.

### 13.6.3 Initialisation as a slave (SPI1)

1. Select the SMCLK source in SYSCLKCR.
2. Set the PxSEL bits of the SCK1, MOSI1 and MISO1 pins.
3. Configure the CS1 pin as a high-impedance GPIO input; the port never drives it.
4. Write SPI1CR.SPICPHA = 1 and SPI1CR.SPICPOL for the mode.
5. Write SPI1CR.SPIDL and SPI1CR.SPIMSB.
6. Set SPI1CR.SPISM.
7. Set SPI1CR.SPIEN.
8. Write the first word to SPI1TX.

In slave mode SPITEIF at the start of each frame asks for the next word, and SPITCIF at the end delivers the received one in SPI1RX; the received word must be read before the next frame ends, because the next frame overwrites it.

### 13.6.4 Enabling the flash window

1. Initialise SPI0 as a master in mode 3 with 8-bit frames at a rate within the flash's specification.
2. Configure CS\_FLASH as a GPIO output driven high.
3. Send the flash's release-from-deep-power-down command (often 0xAB), because the boot ROM leaves the flash in deep power-down.
4. Send any vendor-specific configuration commands.
5. Write SPI0FOS.
6. Set the PxSEL bit of CS\_FLASH, which hands the pin to SPI0.
7. Set SPI0CR.SPIFEN.

## 13.7 Usage notes

**Select the SMCLK source before the first transfer.** The boot ROM may leave SMCLK on LFXT (32.768 kHz), where transfers appear to hang.

**Never read SPIxRX speculatively from a hart that does not own the transfer.** The read clears SPITCIF on any hart's access and consumes the owner's transfer-complete event; when ownership is unclear, clear flags by writing 1s to SPIxSR.

**Serialize whole transfers between harts.** The arbiter makes each register access atomic; frames from two harts interleave on the wire and race for SPIxRX (Section 5.4).

**Wait for the queue before each write.** A second write to SPIxTX within one frame overwrites the queued word.

**Keep other SPI0 slaves deselected across reset.** The boot ROM clocks the flash while every GPIO is a high-impedance input, so a slave whose chip select can float low drives MIS00 against the flash; pull the chip-select lines high externally or keep those slaves on SPI1.

**Use the flash window from hart 0 only.** Other harts read zeros there; a program that must run on a channel hart is staged into shared RAM or the hart's TCM first.

**Wake the flash before enabling the window.** The window issues read commands only; a flash left in deep power-down returns undefined data.

## 13.8 Registers

Register offsets in this chapter are relative to the instance base address in Table 13.1 (SPI0 0x4200, SPI1 0x4300); the generated header names each base SPI0\_BASE and so on.

Table 13.6: SPIx register summary

| Offset | Name    | Access | Reset      | Description                     |
|--------|---------|--------|------------|---------------------------------|
| 0x00   | SPIxCR  | rw     | 0x00000000 | SPI control register            |
| 0x04   | SPIxSR  | r/rw1  | 0x00       | SPI status register             |
| 0x08   | SPIxTX  | rw     | 0x00000000 | SPI transmit buffer register    |
| 0x0C   | SPIxRX  | r      | 0x00000000 | SPI receive buffer register     |
| 0x10   | SPIxFOS | rw     | 0x00000000 | SPI Flash memory address offset |

### 13.8.1 SPIxCR: SPI control register

**Offset** 0x00 **Reset** 0x00000000 **Width** 32 **Address** 0x4200 (SPI0), 0x4300 (SPI1)

| Bits  | Field | Type | Reset | Description                       |
|-------|-------|------|-------|-----------------------------------|
| 31:20 | -     | -    | -     | Reserved. Reads zero; write zero. |



| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                           |
|------|-------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19   | FEN   | rw   | 0     | Flash extended memory enable. When set, reads of the extended flash address range are served by the SPI peripheral, which issues the flash read command and returns the data over the system bus. Implemented on SPI0 only; reads 0 on SPI1.<br>0: Flash extended memory disabled<br>1: Flash extended memory enabled |
| 18   | SM    | rw   | 0     | Slave mode select. Effective on SPI1 only: SPI0 is master only and ignores this bit.<br>0: Master mode<br>1: Slave mode                                                                                                                                                                                               |
| 17   | TXSB  | rw   | 0     | Transmit byte swap. When set, 32-bit transfers swap bytes 3 with 0 and 2 with 1, and 16-bit transfers swap bytes 1 and 0; 8-bit transfers are unaffected.<br>0: Bytes not swapped<br>1: Bytes swapped                                                                                                                 |
| 16   | RXSB  | rw   | 0     | Receive byte swap. When set, 32-bit receptions swap bytes 3 with 0 and 2 with 1, and 16-bit receptions swap bytes 1 and 0; 8-bit receptions are unaffected.<br>0: Bytes not swapped<br>1: Bytes swapped                                                                                                               |
| 15:8 | BR    | rw   | 0x00  | SPI clock (SCK) baud rate control for master mode. Baud rate = $SMCLK / (2 * (1 + SPIBR))$ . For example, with SMCLK at 24 MHz: SPIBR=0 gives 12 MHz, SPIBR=1 gives 8 MHz, SPIBR=2 gives 6 MHz, SPIBR=5 gives 4 MHz, SPIBR=11 gives 2 MHz, SPIBR=23 gives 1 MHz.                                                      |
| 7    | EN    | rw   | 0     | SPI enable. When disabled, all SPI operations cease and the peripheral is held in reset state.<br>0: Disabled<br>1: Enabled                                                                                                                                                                                           |
| 6    | MSB   | rw   | 0     | Bit endianness select. Determines whether data is transmitted and received MSB-first or LSB-first.<br>0: LSB-first<br>1: MSB-first                                                                                                                                                                                    |
| 5    | TCIE  | rw   | 0     | SPI transmit complete interrupt enable. Interrupt triggers when a full SPI transfer (transmit and receive) completes.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                |
| 4    | TEIE  | rw   | 0     | SPI transmit register empty interrupt enable. Interrupt triggers when SPIxTX register is empty and ready to accept new data for the next transfer.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                   |
| 3:2  | DL    | rw   | 0b00  | SPI transmission data length select. Determines the number of bits transferred per SPI transaction.<br>0b00 SPIDL_8: 8-bit transfers<br>0b01 SPIDL_16: 16-bit transfers<br>0b10 SPIDL_32: 32-bit transfers<br>0b11 SPIDL_RES: Reserved (do not use)                                                                   |

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                          |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | CPOL  | rw   | 0     | SPI clock (SCK) polarity. Determines the idle state of the SCK line.<br>0: SCK idles low (SPIMODE0 or SPIMODE1)<br>1: SCK idles high (SPIMODE2 or SPIMODE3)                                                                                                                                          |
| 0    | CPHA  | rw   | 0     | SPI clock (SCK) phase. Determines when data is sampled relative to the SCK edge. In slave mode, only SPICPHA=1 is supported.<br>0: Data sampled on leading edge, shifted on trailing edge (SPIMODE0 or SPIMODE2)<br>1: Data shifted on leading edge, sampled on trailing edge (SPIMODE1 or SPIMODE3) |

### 13.8.2 SPIxSR: SPI status register

**Offset** 0x04 **Reset** 0x00 **Width** 8 **Address** 0x4204 (SPI0), 0x4304 (SPI1)

SPI status register. Provides real-time status of SPI transfer operations and interrupt flags.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                     |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:3  | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                               |
| 2    | BUSY  | r    | 0     | Indicates whether a SPI transfer is currently in progress. In master mode, set when a transfer starts and cleared when complete. In slave mode, set when chip select is asserted (driven low) and cleared when deasserted.<br>0: SPI is idle<br>1: SPI transfer in progress     |
| 1    | TCIF  | rw1  | 0     | SPI transfer complete interrupt flag. Set when a SPI transfer completes. Must be cleared by writing 1 to this bit or by reading SPIxRX register.<br>0: No transfer completed<br>1: Transfer completed                                                                           |
| 0    | TEIF  | rw1  | 0     | Transmit register empty interrupt flag, set when the transmitter has taken the SPIxTX contents so the register can accept the next word; a word written while a transfer is in progress is sent after it. Write 1 to clear.<br>0: SPIxTX not empty<br>1: SPIxTX empty and ready |

### 13.8.3 SPIxTX: SPI transmit buffer register

**Offset** 0x08 **Reset** 0x00000000 **Width** 32 **Address** 0x4208 (SPI0), 0x4308 (SPI1)

SPI transmit buffer register. In master mode, writing to this register initiates a new SPI transfer. In slave mode, writing to this register queues data to be transmitted during the next master-initiated transfer. Actual number of bits transmitted is determined by SPIDL setting.

| Bits | Field | Type | Reset      | Description                                                                                                                                                                                                                                            |
|------|-------|------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | TX    | rw   | 0x00000000 | Transmit data. In master mode a write starts a transfer of the low 8, 16 or 32 bits as selected by SPIDL; in slave mode the write loads the data shifted out during the next master-driven transfer. Reads return the last value written; resets to 0. |

### 13.8.4 SPIxRX: SPI receive buffer register

**Offset** 0x0C **Reset** 0x00000000 **Width** 32 **Address** 0x420C (SPI0), 0x430C (SPI1)

SPI receive buffer register. Contains the data received during the most recent SPI transfer. Reading this register also clears the SPITCIF flag. Valid data width depends on SPIDL setting; unused upper bits read as 0.

| Bits | Field | Type | Reset      | Description                                                                                                                                                           |
|------|-------|------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | RX    | r    | 0x00000000 | Received data from the most recent transfer, right aligned in the width selected by SPIDL with the unused upper bits reading 0. Reading this register clears SPITCIF. |

### 13.8.5 SPIxFOS: SPI Flash memory address offset

**Offset** 0x10 **Reset** 0x00000000 **Width** 32 **Address** 0x4210 (SPI0), 0x4310 (SPI1)

SPI Flash memory address offset. This 24-bit value is added to memory access addresses when flash extended memory mode is enabled (SPIFEN=1). Allows remapping of flash memory to different virtual addresses. The addition wraps around at 0x00FFFFFF. Available only on SPI0; reads as 0 on SPI1.

| Bits  | Field | Type | Reset    | Description                                                                                                                                                                                  |
|-------|-------|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | -     | -    | -        | Reserved. Reads zero; write zero.                                                                                                                                                            |
| 23:0  | FOS   | rw   | 0x000000 | Flash address offset, added to the bus address of every flash extended memory read while SPIFEN = 1; the 24-bit sum wraps. Resets to 0. Implemented on SPI0 only; on SPI1 the field reads 0. |

## 14 Universal asynchronous receiver and transmitter (UARTx)

### 14.1 Overview

The Universal Asynchronous Receiver/Transmitter (UARTx) is a two-wire, full-duplex asynchronous serial interface with hardware parity. UART0 is the console port used by the boot ROM's Forth interpreter; UART1, in configurations that include it, is identical.

- Independent transmitter and receiver, 8-bit frames, one stop bit
- Baud generator from SMCLK with 16 times oversampling (UARTxBR)
- Even or odd parity, generated and checked in hardware (UPEN, PSEL)
- Framing, parity and receive-overflow error flags (FEF, PEF, OVf)
- One-deep transmit queue for back-to-back frames
- Three interrupts per port: receive complete, transmit register empty, transmit complete
- Latched register reads for stable status during reception

### 14.2 Instances and signals

Table 14.1: UARTx instances

| Instance | Base address | Interrupt vector(s) | Pins (AF)                      |
|----------|--------------|---------------------|--------------------------------|
| UART0    | 0x4400       | 13 to 15            | TX0 P1.4 (AF0), RX0 P1.5 (AF0) |
| UART1    | 0x4500       | 52 to 54            | TX1 P1.6 (AF0), RX1 P1.7 (AF0) |

Table 14.2: UARTx signals

| Signal | Direction     | Description       | Pin (AF)                                                                                                                                                                                                                                                                                                   |
|--------|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX0    | Output        | UART0 transmitter | P0.0 (AF1), P0.2 (AF7), P0.3 (AF6), P0.4 (AF5), P0.5 (AF4), P0.6 (AF3), P0.7 (AF2), P1.1 (AF7), P1.2 (AF4), P1.3 (AF3), P1.4 (AF0), P1.5 (AF5), P1.6 (AF2), P1.7 (AF3), P2.0 (AF4), P2.1 (AF6), P2.3 (AF7), P2.4 (AF1), P2.5 (AF2), P2.6 (AF4), P2.7 (AF3), P3.0 (AF2), P3.3 (AF7), P3.4 (AF2), P3.7 (AF6) |
| RX0    | Bidirectional | UART0 receiver    | P1.5 (AF0), P2.5 (AF1), P3.5 (AF2)                                                                                                                                                                                                                                                                         |

| Signal | Direction     | Description       | Pin (AF)                                                                                                                                                                                                                                                                                       |
|--------|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TX1    | Output        | UART1 transmitter | P0.0 (AF2), P0.1 (AF1), P0.3 (AF7), P0.4 (AF6), P0.5 (AF5), P0.6 (AF4), P0.7 (AF3), P1.0 (AF2), P1.2 (AF5), P1.3 (AF4), P1.4 (AF4), P1.5 (AF6), P1.6 (AF0), P1.7 (AF4), P2.0 (AF1), P2.1 (AF7), P2.4 (AF6), P2.5 (AF3), P2.6 (AF5), P2.7 (AF4), P3.0 (AF3), P3.1 (AF2), P3.4 (AF3), P3.7 (AF7) |
| RX1    | Bidirectional | UART1 receiver    | P1.7 (AF0), P2.1 (AF1)                                                                                                                                                                                                                                                                         |

Table 14.3: UARTx interrupt vectors

| Vector | Instance | Description                                                | Clear method                             |
|--------|----------|------------------------------------------------------------|------------------------------------------|
| 13     | UART0    | UART0 Receive Complete Interrupt<br>IRQB_UART0_RC          | Write 1 to UART0SR.RCIF, or read UART0RX |
| 14     | UART0    | UART0 Transmission Buffer Empty Interrupt<br>IRQB_UART0_TE | Write 1 to UART0SR.TEIF                  |
| 15     | UART0    | UART0 Transmission Complete Interrupt<br>IRQB_UART0_TC     | Write 1 to UART0SR.TCIF                  |
| 52     | UART1    | UART1 Receive Complete Interrupt<br>IRQB_UART1_RC          | Write 1 to UART1SR.RCIF, or read UART1RX |
| 53     | UART1    | UART1 Transmission Buffer Empty Interrupt<br>IRQB_UART1_TE | Write 1 to UART1SR.TEIF                  |
| 54     | UART1    | UART1 Transmission Complete Interrupt<br>IRQB_UART1_TC     | Write 1 to UART1SR.TCIF                  |

### 14.3 Block diagram

Figure 14.1 shows the two-wire connection to an external device.

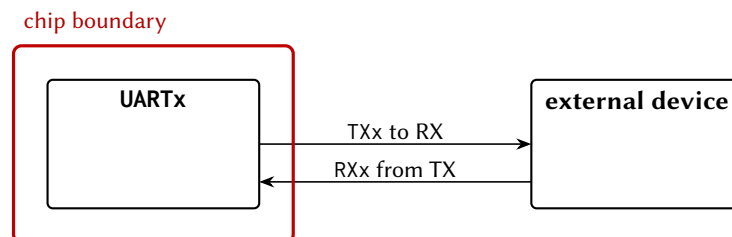


Figure 14.1: UART connection to an external device.

## 14.4 Functional description

### 14.4.1 Frame

Each wire is unidirectional and idles high. A frame is a start bit (low for one bit time  $T_B = 1/\text{baud}$ ), eight data bits LSB first, an optional parity bit, and a stop bit (high for  $T_B$ ). The transmitter may start the next frame immediately after the stop bit or leave the wire idle. Figure 14.2 shows one frame.

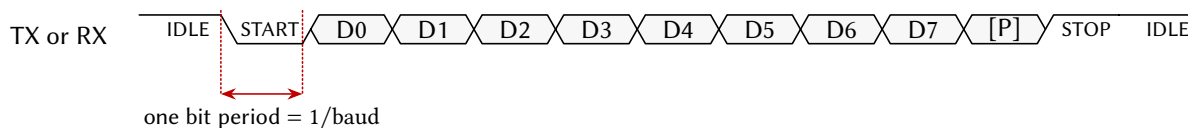


Figure 14.2: One UART frame with the optional parity bit.

### 14.4.2 Baud generation

Both ends must agree on the baud rate, because the receiver has no clock wire and samples each bit at the centre of its expected period. The receiver oversamples at 16 times the baud rate for noise immunity, and the rate is set by the 12-bit  $\text{UARTxBR}$ :

$$\text{baud} = \frac{f_{\text{SMCLK}}}{16 \times (\text{BR} + 1)}$$

With SMCLK at 24 MHz, BR = 12 gives 115,385 baud, BR = 25 gives 57,692 baud and BR = 51 gives 28,846 baud, each within 0.2 % of the standard rate. The serial engine runs on SMCLK and the register interface on MCLK, so the registers stay readable with SMCLK stopped.

### 14.4.3 Parity

With UPEN set the transmitter appends  $p = b_7 \oplus \dots \oplus b_0 \oplus \text{PSEL}$ , which is even parity for PSEL = 0 and odd for 1, and the receiver checks the same function and sets PEF on a mismatch. Parity detects any odd number of bit errors in a frame and misses any even number.

### 14.4.4 Transmit queue

A write to  $\text{UARTxTX}$  queues one byte. If no frame is in progress the frame starts at once and TEIF is set immediately; otherwise the byte starts when the current frame ends, and TEIF is set then. The queue holds one byte, so software waits for TEIF = 1 or TXBF = 0 before each write. TCIF is set when the last stop bit has been sent and the transmitter is idle.

### 14.4.5 Reception and status

RXBF is 1 while a frame is being received. When the stop bit has been sampled, the byte is placed in  $\text{UARTxRX}$ , RCIF is set, and FEF, PEF and OVF are updated: FEF when the stop bit was not high, which usually means the two ends disagree on rate or format; PEF on a parity mismatch; OVF when the previous byte had not been read, in which case it is lost. A byte received with FEF or PEF set is corrupt. Reading  $\text{UARTxRX}$  clears RCIF, FEF, PEF and OVF;  $\text{UARTxSR}$  and  $\text{UARTxRX}$  are latched on the falling edge of the memory enable signal, so a read during reception returns one stable sample.

## 14.5 Interrupts and events

The three interrupt flags in UARTxSR have enables in UARTxCR (CIE, TEIE, TCIE); each interrupt is a level asserted while both are set. Table 14.4 lists the vectors.

| Vector | Port  | Condition               | Set by | Cleared by                    |
|--------|-------|-------------------------|--------|-------------------------------|
| 13     | UART0 | Byte received           | RCIF   | Reading UARTxRX, or writing 1 |
| 14     | UART0 | Transmit register empty | TEIF   | Writing 1                     |
| 15     | UART0 | Transmission complete   | TCIF   | Writing 1                     |
| 52     | UART1 | Byte received           | RCIF   | Reading UARTxRX, or writing 1 |
| 53     | UART1 | Transmit register empty | TEIF   | Writing 1                     |
| 54     | UART1 | Transmission complete   | TCIF   | Writing 1                     |

Table 14.4: UARTx interrupt vectors.

## 14.6 Programming

### 14.6.1 Initialisation

1. Select the SMCLK source in SYSCLKCR (Section 16.3.1).
2. Set the PxSEL bits of the TXx and RXx pins, and their PxAFS fields if the port is relocated.
3. Write UARTxBR.BR for the baud rate.
4. Write UARTxCR.UPEN and UARTxCR.PSEL for parity.
5. Set the interrupt enables required in UARTxCR.
6. Set UARTxCR.UEN; while it is 0 the port is held in reset.

### 14.6.2 Transmitting a byte

1. Wait for TXBF = 0 or TEIF = 1.
2. Write the byte to UARTxTX.

### 14.6.3 Receiving a byte

1. Wait for RCIF = 1.
2. Read UARTxSR and reject the byte if FEF or PEF is set.
3. Read UARTxRX; the read clears the flags.

## 14.7 Usage notes

**Select the SMCLK source before the first frame.** With SMCLK on LFXT (32.768 kHz), as the boot ROM may leave it, each frame takes about 10 ms.

**Read the status before the data.** Reading UARTxRX clears the error flags, so a handler that reads the byte first cannot tell whether it was corrupt.

**Serialize whole messages between harts.** The arbiter makes each register access atomic; bytes from two harts interleave on the wire (Section 5.4).

**Clear TEIF and TCIF with a 1 before returning.** Both are levels into the interrupt controller.

**Leave UART0 to the console during boot.** The boot ROM's Forth interpreter owns UART0 at 115,200 baud until the application starts (Section 6).

## 14.8 Registers

Register offsets in this chapter are relative to the instance base address in Table 14.1 (UART0 0x4400, UART1 0x4500); the generated header names each base UART0\_BASE and so on.

Table 14.5: UARTx register summary

| Offset | Name    | Access | Reset  | Description                   |
|--------|---------|--------|--------|-------------------------------|
| 0x00   | UARTxCR | rw     | 0x00   | UART control register         |
| 0x04   | UARTxSR | r/rw1  | 0x00   | UART status register          |
| 0x08   | UARTxBR | rw     | 0x0000 | UART baud rate register       |
| 0x0C   | UARTxRX | r      | 0x00   | UART receive buffer register  |
| 0x10   | UARTxTX | rw     | 0x00   | UART transmit buffer register |

### 14.8.1 UARTxCR: UART control register

**Offset** 0x00 **Reset** 0x00 **Width** 8 **Address** 0x4400 (UART0), 0x4500 (UART1)

UART control register. Controls UART enable, parity configuration, and interrupt enable bits. When UCR.EN is disabled, the UART peripheral is held in reset state.

| Bits | Field | Type | Reset | Description                                                                                                                           |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| 7:6  | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                     |
| 5    | EN    | rw   | 0     | UART enable. When disabled, the UART peripheral is in reset state and the transmitter is idle.<br>0: UART disabled<br>1: UART enabled |
| 4    | PEN   | rw   | 0     | Parity enable. Enables parity generation on transmit and parity checking on receive.<br>0: Parity disabled<br>1: Parity enabled       |
| 3    | PSEL  | rw   | 0     | Parity select. Selects even or odd parity when parity is enabled.<br>0: Even parity<br>1: Odd parity                                  |



| Bits | Field | Type | Reset | Description                                                                                                                                                                                          |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | CIE   | rw   | 0     | Receive complete interrupt enable. Enables interrupt generation when a byte is successfully received.<br>0: RX complete interrupt disabled<br>1: RX complete interrupt enabled                       |
| 1    | TEIE  | rw   | 0     | Transmit empty interrupt enable. Enables interrupt generation when the transmit buffer becomes empty and ready for new data.<br>0: TX empty interrupt disabled<br>1: TX empty interrupt enabled      |
| 0    | TCIE  | rw   | 0     | Transmit complete interrupt enable. Enables interrupt generation when transmission is complete and the transmitter is idle.<br>0: TX complete interrupt disabled<br>1: TX complete interrupt enabled |

### 14.8.2 UARTxSR: UART status register

**Offset** 0x04 **Reset** 0x00 **Width** 8 **Address** 0x4404 (UART0), 0x4504 (UART1)

UART status register. Contains receiver and transmitter status flags and interrupt flags. Error flags (FEF, PEF, OVF, RCIF) are cleared by reading UARTxRX. Interrupt flags (RCIF, TEIF, TCIF) can also be cleared by writing a 1 to the respective bit.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                        |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | RXBF  | r    | 0     | Receiver busy flag. Set while the UART receiver is actively receiving a byte.<br>0: Receiver idle<br>1: Reception in progress                                                                                                      |
| 6    | TXBF  | r    | 0     | Transmitter busy flag. Set while the UART transmitter is actively transmitting a byte or has a pending transmission.<br>0: Transmitter idle<br>1: Transmission in progress                                                         |
| 5    | FEF   | r    | 0     | Framing error flag. Set when the stop bit is not detected (RX line not high at expected stop bit time). Cleared by reading UARTxRX.<br>0: No framing error<br>1: Framing error detected on last reception                          |
| 4    | PEF   | r    | 0     | Parity error flag. Set when received parity does not match expected parity (when parity is enabled). Cleared by reading UARTxRX.<br>0: No parity error<br>1: Parity error detected on last reception                               |
| 3    | OVF   | r    | 0     | Receive overflow flag. Set when a new byte is received before the previous byte in UARTxRX was read by the processor. Cleared by reading UARTxRX.<br>0: No receive data overrun<br>1: Receive data overflow detected               |
| 2    | RCIF  | rw1  | 0     | Receive complete interrupt flag. Set when a byte is successfully received and placed in UARTxRX. Cleared by reading UARTxRX or writing a 1 to this bit.<br>0: No pending RX complete interrupt<br>1: RX complete interrupt pending |

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                    |
|------|-------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | TEIF  | rw1  | 0     | Transmit empty interrupt flag. Set when the transmit buffer is empty and ready to accept new data. Write a 1 to this bit to clear it.<br>0: No pending TX empty interrupt<br>1: TX empty interrupt pending                     |
| 0    | TCIF  | rw1  | 0     | Transmit complete interrupt flag. Set when transmission is complete (all bits sent) and the transmitter is idle. Write a 1 to this bit to clear it.<br>0: No pending TX complete interrupt<br>1: TX complete interrupt pending |

### 14.8.3 UARTxBR: UART baud rate register

**Offset** 0x08 **Reset** 0x0000 **Width** 16 **Address** 0x4408 (UART0), 0x4508 (UART1)

UART baud rate register. Configures the baud rate divisor for the UART. The UART uses 16× oversampling.

| Bits  | Field | Type | Reset | Description                                                                                                                                                                                                  |
|-------|-------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:12 | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                            |
| 11:0  | BR    | rw   | 0x000 | Baud rate divisor. UART baud rate = $SMCLK \div (16 \times (BR + 1))$ . For example, with $SMCLK = 48\text{ MHz}$ : $BR = 25$ gives 115,200 baud; $BR = 51$ gives 57,600 baud; $BR = 103$ gives 28,800 baud. |

### 14.8.4 UARTxRX: UART receive buffer register

**Offset** 0x0C **Reset** 0x00 **Width** 8 **Address** 0x440C (UART0), 0x450C (UART1)

UART receive buffer register. Contains the last received byte. Reading this register clears the FEF, PEF, OVF, and RCIF flags in UARTxSR. The value is latched on the falling edge of the memory enable signal.

| Bits | Field | Type | Reset | Description        |
|------|-------|------|-------|--------------------|
| 7:0  | RX    | r    | 0x00  | Received data byte |

### 14.8.5 UARTxTX: UART transmit buffer register

**Offset** 0x10 **Reset** 0x00 **Width** 8 **Address** 0x4410 (UART0), 0x4510 (UART1)

UART transmit buffer register. Writing to this register loads the byte to transmit and initiates transmission. Do not write to this register while TXBF is set in UARTxSR.

| Bits | Field | Type | Reset | Description        |
|------|-------|------|-------|--------------------|
| 7:0  | TX    | rw   | 0x00  | Transmit data byte |

## 15 Timer (TIMERx)

### 15.1 Overview

The TIMERx peripheral is a 32-bit general-purpose timer/counter with a glitch-free clock source multiplexer, three output compare units and two input capture units.

- 32-bit counter (TIMxVAL), readable and writable at any time
- Three output compare units (TIMxCMP0 to TIMxCMP2); units 0 and 1 drive PWM outputs, unit 2 sets the period
- Two input capture units (TIMxCAP0, TIMxCAP1) with selectable edge
- Four clock sources (SMCLK, MCLK, LFXT, HFXT) and a divider from 1 to 32,768
- Glitch-free clock source switching while the timer runs
- Six interrupts: two capture, three compare, one overflow
- Latched register reads for stable values while the counter runs

### 15.2 Instances and signals

Table 15.1: TIMERx instances

| Instance | Base address | Interrupt vector(s) | Pins (AF)                                                                  |
|----------|--------------|---------------------|----------------------------------------------------------------------------|
| TIMER0   | 0x4600       | 16 to 21            | T0CMP0 P2.0 (AF0), T0CMP1 P2.1 (AF0), T0CAP0 P2.2 (AF0), T0CAP1 P2.3 (AF0) |
| TIMER1   | 0x4700       | 22 to 27            | T1CMP0 P2.4 (AF0), T1CMP1 P2.5 (AF0), T1CAP0 P2.6 (AF0), T1CAP1 P2.7 (AF0) |

Table 15.2: TIMERx signals

| Signal | Direction | Description      | Pin (AF)                                                                                                                                                                                                                                                                                                                           |
|--------|-----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T0CMP0 | Output    | TIMER0 Compare 0 | P0.0 (AF3), P0.1 (AF2), P0.2 (AF1), P0.4 (AF7), P0.5 (AF6), P0.6 (AF5), P0.7 (AF4), P1.0 (AF1), P1.1 (AF2), P1.2 (AF6), P1.3 (AF5), P1.4 (AF5), P1.5 (AF7), P1.6 (AF3), P1.7 (AF5), P2.0 (AF0), P2.2 (AF2), P2.4 (AF7), P2.5 (AF4), P2.6 (AF6), P2.7 (AF5), P3.0 (AF4), P3.1 (AF3), P3.2 (AF2), P3.4 (AF1), P3.5 (AF3), P3.6 (AF2) |

| Signal | Direction | Description      | Pin (AF)                                                                                                                                                                                                                                                                                                                                                               |
|--------|-----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T0CMP1 | Output    | TIMER0 Compare 1 | P0.0 (AF4), P0.1 (AF3), P0.2 (AF2), P0.3 (AF1), P0.5 (AF7), P0.6 (AF6), P0.7 (AF5), P1.0 (AF3), P1.1 (AF1), P1.2 (AF7), P1.3 (AF6), P1.4 (AF6), P1.6 (AF4), P1.7 (AF6), P2.0 (AF5), P2.1 (AF0), P2.2 (AF3), P2.3 (AF2), P2.4 (AF2), P2.5 (AF5), P2.6 (AF7), P2.7 (AF6), P3.0 (AF5), P3.1 (AF4), P3.2 (AF3), P3.3 (AF2), P3.4 (AF4), P3.5 (AF1), P3.6 (AF3), P3.7 (AF2) |
| T0CAP0 | Input     | TIMER0 Capture 0 | P2.2 (AF0), P3.0 (AF1)                                                                                                                                                                                                                                                                                                                                                 |
| T0CAP1 | Input     | TIMER0 Capture 1 | P2.3 (AF0), P3.1 (AF1)                                                                                                                                                                                                                                                                                                                                                 |
| T1CMP0 | Output    | TIMER1 Compare 0 | P0.0 (AF5), P0.1 (AF4), P0.2 (AF3), P0.3 (AF2), P0.4 (AF1), P0.6 (AF7), P0.7 (AF6), P1.0 (AF4), P1.1 (AF3), P1.2 (AF1), P1.3 (AF7), P1.4 (AF7), P1.6 (AF5), P1.7 (AF7), P2.0 (AF6), P2.1 (AF2), P2.2 (AF4), P2.3 (AF3), P2.4 (AF0), P2.5 (AF6), P2.7 (AF7), P3.0 (AF6), P3.1 (AF5), P3.2 (AF4), P3.3 (AF3), P3.4 (AF5), P3.5 (AF4), P3.6 (AF1), P3.7 (AF3)             |
| T1CMP1 | Output    | TIMER1 Compare 1 | P0.0 (AF6), P0.1 (AF5), P0.2 (AF4), P0.3 (AF3), P0.4 (AF2), P0.5 (AF1), P0.7 (AF7), P1.0 (AF5), P1.1 (AF4), P1.2 (AF2), P1.3 (AF1), P1.5 (AF2), P1.6 (AF6), P2.0 (AF7), P2.1 (AF3), P2.2 (AF5), P2.3 (AF4), P2.4 (AF3), P2.5 (AF0), P3.0 (AF7), P3.1 (AF6), P3.2 (AF5), P3.3 (AF4), P3.4 (AF6), P3.5 (AF5), P3.6 (AF4), P3.7 (AF1)                                     |
| T1CAP0 | Input     | TIMER1 Capture 0 | P2.6 (AF0), P3.2 (AF1)                                                                                                                                                                                                                                                                                                                                                 |
| T1CAP1 | Input     | TIMER1 Capture 1 | P2.7 (AF0), P3.3 (AF1)                                                                                                                                                                                                                                                                                                                                                 |

Table 15.3: TIMERx interrupt vectors

| Vector | Instance | Description                              | Clear method             |
|--------|----------|------------------------------------------|--------------------------|
| 16     | TIMER0   | TIMER0 Capture 0 InterruptIRQB_TIM0_CAP0 | Write 1 to TIM0SR.CAP0IF |
| 17     | TIMER0   | TIMER0 Capture 1 InterruptIRQB_TIM0_CAP1 | Write 1 to TIM0SR.CAP1IF |
| 18     | TIMER0   | TIMER0 Overflow InterruptIRQB_TIM0_OVF   | Write 1 to TIM0SR.OVIF   |
| 19     | TIMER0   | TIMER0 Compare 0 InterruptIRQB_TIM0_CMP0 | Write 1 to TIM0SR.CMP0IF |
| 20     | TIMER0   | TIMER0 Compare 1 InterruptIRQB_TIM0_CMP1 | Write 1 to TIM0SR.CMP1IF |
| 21     | TIMER0   | TIMER0 Compare 2 InterruptIRQB_TIM0_CMP2 | Write 1 to TIM0SR.CMP2IF |
| 22     | TIMER1   | TIMER1 Capture 0 InterruptIRQB_TIM1_CAP0 | Write 1 to TIM1SR.CAP0IF |
| 23     | TIMER1   | TIMER1 Capture 1 InterruptIRQB_TIM1_CAP1 | Write 1 to TIM1SR.CAP1IF |
| 24     | TIMER1   | TIMER1 Overflow InterruptIRQB_TIM1_OVF   | Write 1 to TIM1SR.OVIF   |

| Vector | Instance | Description                              | Clear method             |
|--------|----------|------------------------------------------|--------------------------|
| 25     | TIMER1   | TIMER1 Compare 0 InterruptIRQB_TIM1_CMP0 | Write 1 to TIM1SR.CMP0IF |
| 26     | TIMER1   | TIMER1 Compare 1 InterruptIRQB_TIM1_CMP1 | Write 1 to TIM1SR.CMP1IF |
| 27     | TIMER1   | TIMER1 Compare 2 InterruptIRQB_TIM1_CMP2 | Write 1 to TIM1SR.CMP2IF |

### 15.3 Block diagram

Figure 15.1 shows the clock path: the source multiplexer, the divider and the counter.

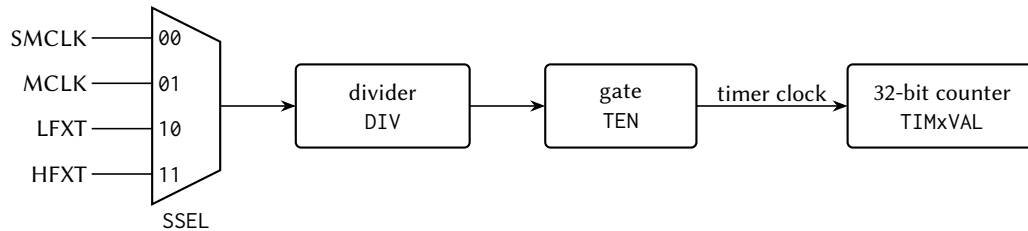


Figure 15.1: Clock path of the TIMERx peripheral.

## 15.4 Functional description

### 15.4.1 Clock source and divider

SSEL selects the timer clock from SMCLK, MCLK, LFXT or HFXT. The multiplexer is glitch-free, so the source can be changed while the timer is enabled without a runt pulse reaching the counter; the price is that the timer clock stops for a few cycles of the slower of the two clocks involved while the old source is released and the new one engages. DIV then divides the selected clock, and TIMxVAL increments on every rising edge of the divided clock while TEN is set.

The multiplexer powers up parked on the SMCLK slice and needs three edges of the clock being released before another source can engage. The first source selection after reset therefore waits three SMCLK edges at whatever rate SMCLK has at that moment, because the boot ROM may leave SMCLK on LFXT (32.768 kHz), and in that state the wait is about 92  $\mu$ s even when the selected source is MCLK or HFXT.

### 15.4.2 Counting

Setting TEN starts the counter from its current value; clearing it holds the value, which is kept until the timer is enabled again, written, or reset. TIMxVAL can be written at any time, enabled or not, and the new value takes effect immediately. Reads of TIMxVAL are latched on the falling edge of the memory enable signal, so a read during counting returns one stable sample.

### 15.4.3 Overflow and rollover

When the counter reaches  $2^{32} - 1$  it rolls over to zero on the next clock and sets OVIF. With CMP2RST set, the counter instead rolls over to zero on the clock after it equals TIMxCMP2, which sets the period of the PWM outputs, and CMP2IF is set at each such rollover while OVIF is not. The comparison is equality, not greater-or-equal: a counter that is above TIMxCMP2, because software wrote TIMxVAL or lowered TIMxCMP2 below the current value, counts on to  $2^{32} - 1$  before it wraps. Figure 15.2 shows the rollover.

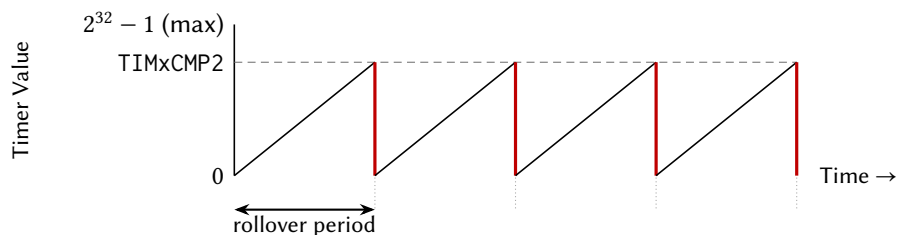


Figure 15.2: Rollover at TIMxCMP2 with CMP2RST set.

#### 15.4.4 Output compare and PWM

Compare units 0 and 1 each hold a compare value (TIMxCMP0, TIMxCMP1) that is compared with TIMxVAL continuously. The output TxCMPy toggles when the counter equals the compare value and again when the counter rolls over, so with CMP2RST set the pair TIMxCMP2 and TIMxCMPy sets the period and the duty cycle. CMP0IH and CMP1IH select the output level while the counter is below the compare value. CMP0OUT and CMP1OUT in TIMxSR report the current output levels whether or not the pins are in alternate-function mode. Figure 15.3 shows one PWM cycle.

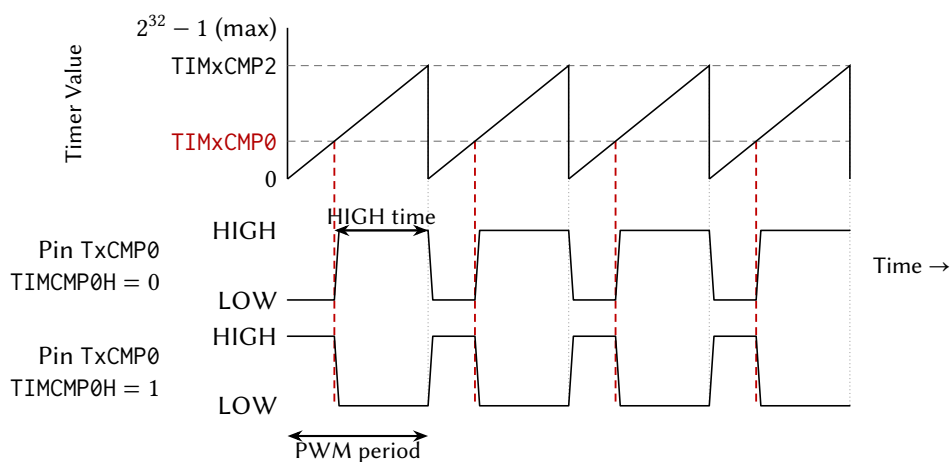


Figure 15.3: Output compare and PWM generation on TxCMP0.

#### 15.4.5 Input capture

Capture unit  $y$ , enabled by CAPyEN, waits for the edge selected by CAPyFE (0 = rising, 1 = falling) on TxCAPy, then copies TIMxVAL into TIMxCAPy and sets CAPyIF. A capture that arrives while CAPyIF is still set overwrites TIMxCAPy, so the flag is cleared, by writing 1 to it in TIMxSR, as soon as the value has been read. Figure 15.4 shows one capture: the selected edge on TxCAP0 latches TIMxVAL into TIMxCAP0 and sets CAP0IF on the same timer clock.

#### 15.4.6 Status register

TIMxSR holds the two output indicators and the six interrupt flags. It is latched on the falling edge of the memory enable signal, and every flag is write-1-to-clear.

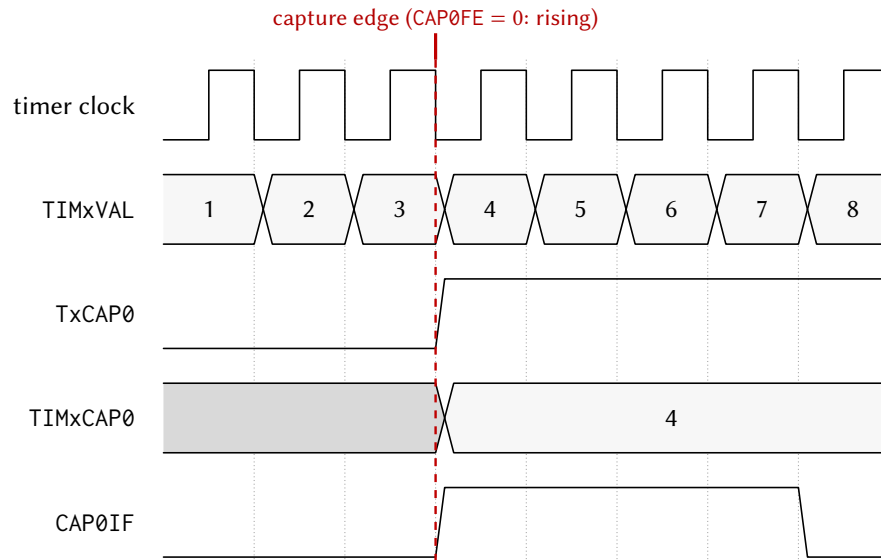


Figure 15.4: Input capture: an edge on TxCAP0 latches TIMxVAL into TIMxCAP0.

## 15.5 Interrupts and events

Each flag has an enable bit in TIMxCR; the interrupt is a level asserted while both are set. Table 15.4 lists the vectors of TIMER0; TIMER1, in configurations that include it, uses the same order from vector 22 (22 to 27).

| Vector | Condition                          | Set by | Cleared by          |
|--------|------------------------------------|--------|---------------------|
| 16     | Capture on T0CAP0                  | CAP0IF | Writing 1 to CAP0IF |
| 17     | Capture on T0CAP1                  | CAP1IF | Writing 1 to CAP1IF |
| 18     | Counter overflow from $2^{32} - 1$ | OVIF   | Writing 1 to OVIF   |
| 19     | Counter equals TIMxCMP0            | CMP0IF | Writing 1 to CMP0IF |
| 20     | Counter equals TIMxCMP1            | CMP1IF | Writing 1 to CMP1IF |
| 21     | Rollover at TIMxCMP2 (CMP2RST set) | CMP2IF | Writing 1 to CMP2IF |

Table 15.4: TIMER0 interrupt vectors; TIMER1 follows the same order from vector 22.

## 15.6 Programming

### 15.6.1 Initialisation as a PWM generator

1. Select a fast SMCLK source in SYSCCLKCR (Section 16.3.1).
2. Write the period minus one to TIMxCMP2.
3. Write the duty value to TIMxCMP0.
4. Write 0 to TIMxVAL.
5. Set the PxSEL bit of the TxCMP0 pin, and its PxAFS field if the output is relocated.

6. Write TIMxCR.SSEL and TIMxCR.DIV.
7. Set TIMxCR.CMP2RST and TIMxCR.CMP0IH as required.
8. Set TIMxCR.TEN.
9. Poll TIMxVAL until it has changed, because the clock handoff may hold the counter for a few cycles.

### 15.6.2 Initialisation for input capture

1. Clear the PxDIR bit of the TxCAPy pin, and write its PxAFS field if the input is relocated.
2. Write TIMxCR.CAPyFE for the edge.
3. Set TIMxCR.CAPyIE if an interrupt is wanted, and enable the vector in the hart's IRQROUTER row.
4. Set TIMxCR.CAPyEN.
5. Set TIMxCR.TEN.

## 15.7 Usage notes

**Select the SMCLK source before the first timer clock selection.** The multiplexer releases its reset slice only after three SMCLK edges, so with SMCLK on LFXT the first selection stalls the timer for about 92  $\mu$ s.

**Poll TIMxVAL after enabling.** Two reads immediately after setting TEN may return the same value while the clock handoff completes; the timer is running once the value has changed.

**Do not change CAPyFE while CAPyEN is set.** The edge detector can see the change as a capture event.

**Keep TIMxVAL below TIMxCMP2 when CMP2RST is set.** The rollover fires on equality only; a counter above the compare value runs to  $2^{32} - 1$ .

**Clear each flag with a 1 before returning.** Every flag in TIMxSR is a level into the interrupt controller.

**Own a timer from one hart at a time.** The arbiter serializes register accesses but not sequences; a second hart writing TIMxCR changes the configuration under the owner.

## 15.8 Registers

Register offsets in this chapter are relative to the instance base address in Table 15.1 (TIMER0 0x4600, TIMER1 0x4700); the generated header names each base TIMER0\_BASE and so on.

Table 15.5: TIMERx register summary

| Offset | Name   | Access | Reset      | Description                    |
|--------|--------|--------|------------|--------------------------------|
| 0x00   | TIMxCR | rw     | 0x00000000 | Timer/Counter control register |
| 0x04   | TIMxSR | r/rw1  | 0x00       | Timer/Counter status register  |



| Offset    | Name                | Access | Reset      | Description                                |
|-----------|---------------------|--------|------------|--------------------------------------------|
| 0x08      | TIMxVAL             | rw     | 0x00000000 | Timer/Counter value register               |
| 0x0C      | TIMxCMP0            | rw     | 0x00000000 | Timer/Counter Compare 0 threshold register |
| 0x10      | TIMxCMP1            | rw     | 0x00000000 | Timer/Counter Compare 1 threshold register |
| 0x14      | TIMxCMP2            | rw     | 0x00000000 | Timer/Counter Compare 2 threshold register |
| 0x18 + 4n | TIMxCAPn (n = 0..1) | r      | 0x00000000 | Timer/Counter Capture n value register     |

### 15.8.1 TIMxCR: Timer/Counter control register

**Offset** 0x00 **Reset** 0x00000000 **Width** 32 **Address** 0x4600 (TIMER0), 0x4700 (TIMER1)

Timer/Counter control register. Configures clock source, clock divider, capture/compare settings, and interrupt enables.

| Bits  | Field  | Type | Reset  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:20 | -      | -    | -      | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 19:16 | DIV    | rw   | 0b0000 | Timer/Counter clock divider. Divides the clock source selected by SSEL to generate the timer clock for TIMxVAL increments.<br>0b0000 DIV_1: /1 (no division)<br>0b0001 DIV_2: /2<br>0b0010 DIV_4: /4<br>0b0011 DIV_8: /8<br>0b0100 DIV_16: /16<br>0b0101 DIV_32: /32<br>0b0110 DIV_64: /64<br>0b0111 DIV_128: /128<br>0b1000 DIV_256: /256<br>0b1001 DIV_512: /512<br>0b1010 DIV_1024: /1,024<br>0b1011 DIV_2048: /2,048<br>0b1100 DIV_4096: /4,096<br>0b1101 DIV_8192: /8,192<br>0b1110 DIV_16384: /16,384<br>0b1111 DIV_32768: /32,768 |
| 15    | CMP1IH | rw   | 0      | Compare 1 initial PWM output level. Sets the TxCMP1 pin level when TIMxVAL < TIMxCMP1.<br>0: PWM output starts LOW<br>1: PWM output starts HIGH                                                                                                                                                                                                                                                                                                                                                                                          |
| 14    | CMP0IH | rw   | 0      | Compare 0 initial PWM output level. Sets the TxCMP0 pin level when TIMxVAL < TIMxCMP0.<br>0: PWM output starts LOW<br>1: PWM output starts HIGH                                                                                                                                                                                                                                                                                                                                                                                          |

| Bits | Field   | Type | Reset | Description                                                                                                                                                                                                                                                         |
|------|---------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13   | CAP1FE  | rw   | 0     | Capture 1 edge select. Selects which edge on TxCAP1 pin triggers a capture event.<br>0: Capture on rising edge<br>1: Capture on falling edge                                                                                                                        |
| 12   | CAP0FE  | rw   | 0     | Capture 0 edge select. Selects which edge on TxCAP0 pin triggers a capture event.<br>0: Capture on rising edge<br>1: Capture on falling edge                                                                                                                        |
| 11   | CAP1EN  | rw   | 0     | Capture 1 enable. Enables input capture on TxCAP1 pin.<br>0: Capture 1 disabled<br>1: Capture 1 enabled                                                                                                                                                             |
| 10   | CAP0EN  | rw   | 0     | Capture 0 enable. Enables input capture on TxCAP0 pin.<br>0: Capture 0 disabled<br>1: Capture 0 enabled                                                                                                                                                             |
| 9:8  | SSEL    | rw   | 0b00  | Timer/Counter clock source select. Glitch-free multiplexer prevents spurious transitions when switching sources.<br>0b00 SSEL_SMCLK: SMCLK<br>0b01 SSEL_MCLK: MCLK<br>0b10 SSEL_LFXT: LFXT (low frequency crystal)<br>0b11 SSEL_HFXT: HFXT (high frequency crystal) |
| 7    | CMP2RST | rw   | 0     | Timer/Counter reset on Compare 2 enable. Resets TIMxVAL to 0 when TIMxVAL equals TIMxCMP2.<br>0: Free-running mode (resets at $2^{32}-1$ )<br>1: Resets on TIMxVAL = TIMxCMP2                                                                                       |
| 6    | EN      | rw   | 0     | Timer/Counter enable. When disabled, timer clock is gated off and TIMxVAL holds its value.<br>0: Timer disabled<br>1: Timer enabled                                                                                                                                 |
| 5    | CAP1IE  | rw   | 0     | Capture 1 interrupt enable. Enables interrupt when CAP1IF flag is set.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                             |
| 4    | CAP0IE  | rw   | 0     | Capture 0 interrupt enable. Enables interrupt when CAP0IF flag is set.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                             |
| 3    | OVIE    | rw   | 0     | Timer/Counter overflow interrupt enable. Enables interrupt when OVIF flag is set.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                  |
| 2    | CMP2IE  | rw   | 0     | Compare 2 interrupt enable. Enables interrupt when CMP2IF flag is set.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                             |

| Bits | Field  | Type | Reset | Description                                                                                                             |
|------|--------|------|-------|-------------------------------------------------------------------------------------------------------------------------|
| 1    | CMP1IE | rw   | 0     | Compare 1 interrupt enable. Enables interrupt when CMP1IF flag is set.<br>0: Interrupt disabled<br>1: Interrupt enabled |
| 0    | CMP0IE | rw   | 0     | Compare 0 interrupt enable. Enables interrupt when CMP0IF flag is set.<br>0: Interrupt disabled<br>1: Interrupt enabled |

### 15.8.2 TIMxSR: Timer/Counter status register

**Offset** 0x04 **Reset** 0x00 **Width** 8 **Address** 0x4604 (TIMER0), 0x4704 (TIMER1)

Timer/Counter status register. Contains current compare output levels and interrupt flags. The register is latched on the falling edge of the memory enable signal for stable reads.

| Bits | Field   | Type | Reset | Description                                                                                                                                                                  |
|------|---------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | CMP1OUT | r    | 0     | Current value of the Compare 1 output pin TxCMP1. Reflects the actual PWM output level.<br>0: TxCMP1 output LOW<br>1: TxCMP1 output HIGH                                     |
| 6    | CMP0OUT | r    | 0     | Current value of the Compare 0 output pin TxCMP0. Reflects the actual PWM output level.<br>0: TxCMP0 output LOW<br>1: TxCMP0 output HIGH                                     |
| 5    | CAP1IF  | rw1  | 0     | Capture 1 interrupt flag. Set when TxCAP1 pin triggers a capture event. Write 1 to clear.<br>0: No pending capture 1 interrupt<br>1: Capture 1 interrupt pending             |
| 4    | CAP0IF  | rw1  | 0     | Capture 0 interrupt flag. Set when TxCAP0 pin triggers a capture event. Write 1 to clear.<br>0: No pending capture 0 interrupt<br>1: Capture 0 interrupt pending             |
| 3    | OVIF    | rw1  | 0     | Timer/Counter overflow interrupt flag. Set when timer overflows from $2^{32}-1$ to 0. Write 1 to clear.<br>0: No pending overflow interrupt<br>1: Overflow interrupt pending |
| 2    | CMP2IF  | rw1  | 0     | Compare 2 interrupt flag. Set when TIMxVAL equals TIMxCMP2. Write 1 to clear.<br>0: No pending compare 2 interrupt<br>1: Compare 2 interrupt pending                         |
| 1    | CMP1IF  | rw1  | 0     | Compare 1 interrupt flag. Set when TIMxVAL equals TIMxCMP1. Write 1 to clear.<br>0: No pending compare 1 interrupt<br>1: Compare 1 interrupt pending                         |

| Bits | Field  | Type | Reset | Description                                                                                                                                          |
|------|--------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | CMP0IF | rw1  | 0     | Compare 0 interrupt flag. Set when TIMxVAL equals TIMxCMP0. Write 1 to clear.<br>0: No pending compare 0 interrupt<br>1: Compare 0 interrupt pending |

### 15.8.3 TIMxVAL: Timer/Counter value register

**Offset** 0x08 **Reset** 0x00000000 **Width** 32 **Address** 0x4608 (TIMER0), 0x4708 (TIMER1)

Timer/Counter value register. Reads the current timer count. Writes immediately update the timer value regardless of enable state. The value is latched on the falling edge of the memory enable signal for stable reads.

| Bits | Field | Type | Reset      | Description               |
|------|-------|------|------------|---------------------------|
| 31:0 | VAL   | rw   | 0x00000000 | Current timer count value |

### 15.8.4 TIMxCMP0: Timer/Counter Compare 0 threshold register

**Offset** 0x0C **Reset** 0x00000000 **Width** 32 **Address** 0x460C (TIMER0), 0x470C (TIMER1)

Timer/Counter Compare 0 threshold register. When TIMxVAL equals this value, CMP0IF is set and the TxCMP0 PWM output toggles.

| Bits | Field | Type | Reset      | Description               |
|------|-------|------|------------|---------------------------|
| 31:0 | CMP0  | rw   | 0x00000000 | Compare 0 threshold value |

### 15.8.5 TIMxCMP1: Timer/Counter Compare 1 threshold register

**Offset** 0x10 **Reset** 0x00000000 **Width** 32 **Address** 0x4610 (TIMER0), 0x4710 (TIMER1)

Timer/Counter Compare 1 threshold register. When TIMxVAL equals this value, CMP1IF is set and the TxCMP1 PWM output toggles.

| Bits | Field | Type | Reset      | Description               |
|------|-------|------|------------|---------------------------|
| 31:0 | CMP1  | rw   | 0x00000000 | Compare 1 threshold value |

### 15.8.6 TIMxCMP2: Timer/Counter Compare 2 threshold register

**Offset** 0x14 **Reset** 0x00000000 **Width** 32 **Address** 0x4614 (TIMER0), 0x4714 (TIMER1)

Timer/Counter Compare 2 threshold register. When TIMxVAL equals this value, CMP2IF is set. If CMP2RST is enabled, timer resets to 0 (PWM period control).

| Bits | Field | Type | Reset      | Description                            |
|------|-------|------|------------|----------------------------------------|
| 31:0 | CMP2  | rw   | 0x00000000 | Compare 2 threshold value (PWM period) |

### 15.8.7 TIMxCAPn (n = 0..1): Timer/Counter Capture n value register

**Offset** 0x18 + 4n **Reset** 0x00000000 **Width** 32

Timer/Counter Capture n value register. Automatically latches TIMxVAL when TxCAPn pin edge triggers a capture event. The captured value is latched on the falling edge of the memory enable signal for stable reads.

---

| Bits | Field | Type | Reset      | Description                            |
|------|-------|------|------------|----------------------------------------|
| 31:0 | CAPn  | r    | 0x00000000 | Captured timer value from TxCAPn event |

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## 16 System control (SYSTEM)

### 16.1 Overview

The SYSTEM peripheral groups the chip-level control blocks: the clock tree roots and dividers, the memory block power gates, a CRC16 accumulator and the watchdog timer. Its effects are global (MCLK clocks all harts and SMCLK clocks the peripherals), so by convention only hart 0 reconfigures the clocks, after quiescing the other harts.

- Source selection and enables for MCLK and SMCLK (SYSCLKCR), dividers for both (CLKDIVCR), and bias control of the two on-chip oscillators (DC00BIAS, DC01BIAS)
- Power gates in BLOCKPWR for the ROM, the per-hart memory blocks and the first four shared RAM banks
- CRC16-CDMA2000 accumulator (CRCDATA, CRCSTATE)
- 24-bit watchdog with sixteen timeouts, password-protected control, optional interrupt and hardware reset (WDTCSR, WDTPASS, WDTSR, WDTVAL)
- One interrupt: watchdog timeout, vector 0

Interrupt routing is not managed here: every hart enables peripheral interrupts through its IRQROUTER row, and the CLINT vectors are hardwired into every hart. The register slots of the single-core chip's vectored interrupt controller (IRQENx, IRQPRIx, IRQCR) are reserved and read as zero.

### 16.2 Instances and signals

Table 16.1: SYSTEM instances

| Instance | Base address | Interrupt vector(s) | Pins (AF) |
|----------|--------------|---------------------|-----------|
| SYSTEM   | 0x4900       | 0                   | none      |

Table 16.2: SYSTEM interrupt vectors

| Vector | Instance | Description              | Clear method              |
|--------|----------|--------------------------|---------------------------|
| 0      | SYSTEM   | Watchdog Timer Interrupt | Write 1 to WDTSR.SYSWDTIF |

### 16.3 Functional description

#### 16.3.1 Clock control

The clock tree, its four sources and the clock state at reset and after boot are described in the Reset, clocks and power chapter (Section 6.3). SYSCLKCR holds the root selects and the source enables: MCLKSEL and SMCLKSEL pick the source of each root, SMCLKOFF stops SMCLK, HFXTOFF and LFXTOFF stop the external

sources, and DCO0ON and DCO1ON start the on-chip oscillators. A source selected by either root stays enabled regardless of its enable bit, so the running clock cannot be stopped by mistake. CLKDIVCR divides each root after its multiplexer, and DCO0BIAS and DCO1BIAS set the oscillator frequencies. Both root multiplexers are glitch-free, so a source change while running produces no runt pulse.

Because MCLK clocks every hart and SMCLK clocks the peripheral engines, writing SYSCLKCR or CLKDIVCR changes the timing of all five harts and every serial peripheral at once. Reducing the MCLK frequency, through the divider, a slower source or a lower DCO bias, is the main lever on dynamic power.

### 16.3.2 Memory block power

BLOCKPWR gates the supply of the ROM (SYSROMOFF), the two per-hart memory blocks (SYSRAM0OFF, SYSRAM1OFF) and the first four shared RAM banks individually (SYSSHB0OFF to SYSSHB3OFF); banks beyond the fourth are permanently powered. A gated block reduces leakage and loses its contents, because the gate disconnects the array's supply rail and there is no retention; an access to a gated block returns undefined data. All blocks power up on at reset, and the contents of a block that has been gated are undefined until written.

### 16.3.3 Hart sleep

Each hart can stop its own clock with the sleep instruction, which reduces that hart's dynamic power to zero while the peripherals and the other harts continue. A sleeping hart is woken only by an interrupt (a CLINT software or timer interrupt, or a peripheral interrupt routed to it) or by reset. On an interrupt the clock is re-enabled for the handler; when every pending interrupt has been serviced the hart returns to sleep, unless a handler executed the wake instruction, in which case the hart stays awake and continues after the sleep. The average power of a duty-cycled hart is its running power scaled by  $t_{\text{on}}/(t_{\text{on}} + t_{\text{off}})$ .

### 16.3.4 CRC accumulator

The CRC module computes CRC16-CDMA2000 over a byte sequence: width 16, polynomial 0xC857, initial value 0xFFFF, no input or output reflection, no output XOR. CRCSTATE holds the running value and CRCDATA accepts one byte per write. CRCSTATE is a true seed register: the value written to it is what the next CRCDATA byte folds against, so a partial state read out mid-message can be written back later and the chain resumes, and a protocol whose initial value is not 0xFFFF is computed by seeding that value instead. Reset behaves as a 0xFFFF seed write, so the first byte after reset needs no seed at all.

### 16.3.5 Watchdog

The watchdog is a 24-bit counter that increments on every MCLK cycle while SYSWDTEN is set. SYSWDTCDIV selects which counter bit, 16 to 31, raises the event on its 0-to-1 transition, so the timeout is  $2^{\text{SYSWDTCDIV}+16}$  MCLK cycles: at 24 MHz from about 2.73 ms to about 89.5 s. On the event, SYSWDTIF is set; if SYSWDTIE is set and vector 0 is routed to some hart, the interrupt is delivered through the IRQROUTER and, if SYSWDTHWRST is also set, the chip resets after that hart completes the claim. If the interrupt is disabled or unrouted, the chip resets immediately on the event when SYSWDTHWRST is set.

WDTCR is write-protected: a write of the unlock password 0x5F3759DF to WDTPASS opens a 64-MCLK-cycle window in which WDTCR accepts a write, so that a runaway program cannot disable the watchdog by chance. A write of the clear password 0xA0C8A620 to WDTPASS resets the counter to zero. SYSWDTRF in WDTSR records that the last reset was caused by the watchdog and survives reset. WDTVAL reads the counter, or zero while the watchdog is disabled.

## 16.4 Interrupts and events

Table 16.3 lists the one vector.

| Vector | Condition        | Set by   | Cleared by            |
|--------|------------------|----------|-----------------------|
| 0      | Watchdog timeout | SYSWDTIF | Writing 1 to SYSWDTIF |

Table 16.3: SYSTEM interrupt vector.

## 16.5 Programming

### 16.5.1 Selecting the clock sources

1. Quiesce the other harts, because the change reaches every hart.
2. Set SYSCLKCR.DC000N or SYSCLKCR.DC010N if a DCO is to be used, and write its bias register.
3. Write CLKDIVCR.SYSMCLKDIV and CLKDIVCR.SYSSMCLKDIV.
4. Write SYSCLKCR.SMCLKSEL.
5. Write SYSCLKCR.MCLKSEL.
6. Set SYSCLKCR.HFXTOFF or SYSCLKCR.LFXTOFF for any source no longer needed.

### 16.5.2 Computing a CRC

1. Write 0xFFFF to CRCSTATE, or the protocol's initial value, or a partial state saved earlier.
2. Write each byte, in order, to CRCDATA.
3. Read the result from CRCSTATE.

### 16.5.3 Starting the watchdog

1. Route vector 0 to the servicing hart in its IRQROUTER row if the interrupt is wanted.
2. Write 0x5F3759DF to WDPASS.
3. Write WDTCR with SYSWDTCDIV, SYSWDTIE, SYSWDTHWRST and SYSWDTEN set in the same write, within 64 MCLK cycles of the unlock.
4. Write 0xA0C8A620 to WDPASS periodically, within the timeout.

### 16.5.4 Putting a hart to sleep

1. Set a valid stack pointer, because interrupt entry stores to it.
2. Enable the wake source's vector in the hart's interrupt controller.
3. Execute sleep.
4. In the handler, execute wake before returning if the hart is to stay awake.



## 16.6 Usage notes

**Reconfigure the clocks from hart 0 only, with the other harts quiesced.** SYCLKCR and CLKDIVCR change the clock of every hart and every peripheral engine at once.

**Keep HFXT running across reset.** Both roots select HFXT at reset; a board that gates the oscillator from a chip-driven pin does not boot.

**Select the SMCLK source before using an SMCLK-domain peripheral.** The boot ROM may leave SMCLK on LFXT (32.768 kHz).

**Keep the bank in use powered.** A gated shared RAM bank or memory block loses its contents and returns undefined data; track which banks hold live stack or payload data before writing BLOCKPWR.

**Write WDTCR in one access within the unlock window.** The window closes 64 MCLK cycles after the unlock password; a read-modify-write that straddles it is discarded.

**Clear SYSWDTIF with a 1 in the handler.** The interrupt is a level; with SYSWDTHWRST set the chip resets after the claim is completed regardless.

## 16.7 Registers

Register offsets in this chapter are relative to the instance base address in Table 16.1 (SYSTEM 0x4900); the generated header names each base SYSTEM\_BASE.

Table 16.4: SYSTEM register summary

| Offset       | Name                | Access | Reset      | Description                                     |
|--------------|---------------------|--------|------------|-------------------------------------------------|
| 0x00         | SYCLKCR             | rw     | 0x0000     | System clock control register                   |
| 0x04         | CLKDIVCR            | rw     | 0x00       | MCLK and SMCLK clock divider control register   |
| 0x08         | BLOCKPWR            | rw     | 0x00       | Block power control register                    |
| 0x0C         | CRCDATA             | rw     | 0x00       | CRC input data register                         |
| 0x10         | CRCSTATE            | rw     | 0x0000     | CRC state register                              |
| 0x14 to 0x2F | -                   | -      | -          | Reserved                                        |
| 0x30         | WDTPASS             | w      | 0x00000000 | Watchdog timer password register                |
| 0x34         | WDTCR               | rw     | 0x00       | Watchdog timer control register                 |
| 0x38         | WDTSR               | rw1    | 0x00       | Watchdog timer status register                  |
| 0x3C         | WDTVAL              | r      | 0x00000000 | Watchdog timer value register                   |
| 0x40 + 4n    | DConBIAS (n = 0..1) | rw     | 0x0800     | Digitally controlled oscillator n bias register |

### 16.7.1 SYCLKCR: System clock control register

**Offset** 0x00    **Reset** 0x0000    **Width** 16    **Address** 0x4900

| Bits | Field    | Type | Reset | Description                                                                                                                                                                                                                                                     |
|------|----------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:9 | -        | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                               |
| 8    | DCO1ON   | rw   | 0     | Digitally controlled oscillator 1 (DCO1) power enable. Set to power on DCO1. DCO1 is automatically kept on if it is currently selected as the source for MCLK or SMCLK, regardless of this bit.<br>0: DCO1 powered off<br>1: DCO1 powered on                    |
| 7    | DCO0ON   | rw   | 0     | Digitally controlled oscillator 0 (DCO0) power enable. Set to power on DCO0. DCO0 is automatically kept on if it is currently selected as the source for MCLK or SMCLK, regardless of this bit.<br>0: DCO0 powered off<br>1: DCO0 powered on                    |
| 6    | HFXTOFF  | rw   | 0     | High frequency external crystal clock disable. Cannot be disabled if it is currently selected as the source for MCLK or SMCLK.<br>0: HFXT enabled<br>1: HFXT disabled                                                                                           |
| 5    | LFXTOFF  | rw   | 0     | Low frequency external crystal clock disable. Cannot be disabled if it is currently selected as the source for SMCLK.<br>0: LFXT enabled<br>1: LFXT disabled                                                                                                    |
| 4    | SMCLKOFF | rw   | 0     | Submain clock disable. Globally and unconditionally disables SMCLK, gating all peripherals clocked from SMCLK.<br>0: SMCLK enabled<br>1: SMCLK disabled                                                                                                         |
| 3:2  | SMCLKSEL | rw   | 0b00  | Submain clock source select<br>0b00 SMCLKSEL_HFXT: High Frequency Crystal Clock<br>0b01 SMCLKSEL_LFXT: Low Frequency Crystal Clock<br>0b10 SMCLKSEL_DCO0: Digitally Controlled Oscillator 0<br>0b11 SMCLKSEL_DCO1: Digitally Controlled Oscillator 1            |
| 1:0  | MCLKSEL  | rw   | 0b00  | Main clock source select (also CPU clock source select)<br>0b00 MCLKSEL_HFXT: High Frequency Crystal Clock<br>0b01 MCLKSEL_SMCLK: Submain Clock<br>0b10 MCLKSEL_DCO0: Digitally Controlled Oscillator 0<br>0b11 MCLKSEL_DCO1: Digitally Controlled Oscillator 1 |

### 16.7.2 CLKDIVCR: MCLK and SMCLK clock divider control register

**Offset** 0x04 **Reset** 0x00 **Width** 8 **Address** 0x4904

MCLK and SMCLK clock divider control register. Configures clock division for main and submain clocks using glitch-free multiplexers.

| Bits | Field | Type | Reset | Description                       |
|------|-------|------|-------|-----------------------------------|
| 7:6  | -     | -    | -     | Reserved. Reads zero; write zero. |

| Bits | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                               |
|------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5:3  | SMCLKDIV | rw   | 0b000 | SMCLK clock division selection. Division is applied after clock source selection through glitch-free divider multiplexer.<br>0b000 SYSSMCLKDIV_1: /1 (no division)<br>0b001 SYSSMCLKDIV_2: /2<br>0b010 SYSSMCLKDIV_4: /4<br>0b011 SYSSMCLKDIV_8: /8<br>0b100 SYSSMCLKDIV_16: /16<br>0b101 SYSSMCLKDIV_32: /32<br>0b110 SYSSMCLKDIV_64: /64<br>0b111 SYSSMCLKDIV_128: /128 |
| 2:0  | MCLKDIV  | rw   | 0b000 | MCLK clock division selection. Division is applied after clock source selection through glitch-free divider multiplexer.<br>0b000 SYSMCLKDIV_1: /1 (no division)<br>0b001 SYSMCLKDIV_2: /2<br>0b010 SYSMCLKDIV_4: /4<br>0b011 SYSMCLKDIV_8: /8<br>0b100 SYSMCLKDIV_16: /16<br>0b101 SYSMCLKDIV_32: /32<br>0b110 SYSMCLKDIV_64: /64<br>0b111 SYSMCLKDIV_128: /128          |

### 16.7.3 BLOCKPWR: Block power control register

**Offset** 0x08 **Reset** 0x00 **Width** 8 **Address** 0x4908

Block power control register. Controls power gating for the on-chip memory blocks; all bits reset to 0 (every block powered). Bits 6:3 gate the four low shared bulk-RAM banks individually: a gated bank loses its contents and stops responding, so software must keep the bank holding its stack, mailboxes or live data powered and must treat a re-powered bank as uninitialized. In configurations with more than four banks, banks 4 and up are always on.

| Bits | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                |
|------|---------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | -       | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                          |
| 6    | SHB3OFF | rw   | 0     | Shared bulk-RAM bank 3 (0x1C000-0x1FFFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage).<br>0: Bank 3 powered on<br>1: Bank 3 powered off (contents lost) |
| 5    | SHB2OFF | rw   | 0     | Shared bulk-RAM bank 2 (0x18000-0x1BFFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage).<br>0: Bank 2 powered on<br>1: Bank 2 powered off (contents lost) |

| Bits | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                |
|------|---------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | SHB10FF | rw   | 0     | Shared bulk-RAM bank 1 (0x14000-0x17FFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage).<br>0: Bank 1 powered on<br>1: Bank 1 powered off (contents lost) |
| 3    | SHB00FF | rw   | 0     | Shared bulk-RAM bank 0 (0x10000-0x13FFF) power control. When set, the bank is powered off: contents are LOST and accesses no longer respond. Reduces static power (a gated sram1p16k halves its leakage).<br>0: Bank 0 powered on<br>1: Bank 0 powered off (contents lost) |
| 2    | RAM10FF | rw   | 0     | RAM block 1 power control. When set, the block is powered off: its contents are lost and accesses to it no longer respond.<br>0: RAM block 1 powered on<br>1: RAM block 1 powered off                                                                                      |
| 1    | RAM00FF | rw   | 0     | RAM block 0 power control. When set, the block is powered off: its contents are lost and accesses to it no longer respond.<br>0: RAM block 0 powered on<br>1: RAM block 0 powered off                                                                                      |
| 0    | ROMOFF  | rw   | 0     | ROM power control. When set, the boot ROM is powered off and no longer responds to reads.<br>0: ROM powered on<br>1: ROM powered off                                                                                                                                       |

#### 16.7.4 CRCDATA: CRC input data register

**Offset** 0x0C **Reset** 0x00 **Width** 8 **Address** 0x490C

CRC input data register. Write the next byte of data to this register to update the CRC calculation. Uses CRC16-CDMA2000 polynomial 0xC857.

| Bits | Field   | Type | Reset | Description                                                                                                 |
|------|---------|------|-------|-------------------------------------------------------------------------------------------------------------|
| 7:0  | CRCDATA | rw   | 0x00  | CRC data input byte. Writing to this register feeds the byte into the CRC calculation and updates CRCSTATE. |

#### 16.7.5 CRCSTATE: CRC state register

**Offset** 0x10 **Reset** 0x0000 **Width** 16 **Address** 0x4910

CRC state register. Contains the current CRC16 calculation result. Write to this register to initialize or restart the CRC calculation. Read to obtain the computed CRC16 checksum.

| Bits | Field    | Type | Reset  | Description                                                                                                                              |
|------|----------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | CRCSTATE | rw   | 0x0000 | CRC state value. Initialize to 0xFFFF before starting CRC calculation. Read after processing all data bytes to get final CRC16 checksum. |

### 16.7.6 WDTPASS: Watchdog timer password register

**Offset** 0x30 **Reset** 0x00000000 **Width** 32 **Address** 0x4930

Watchdog timer password register. Write-only register for two security functions: (1) Write 0x5F3759DF to unlock WDTCR for 64 MCLK cycles, enabling writes to watchdog configuration. (2) Write 0xA0C8A620 to clear watchdog counter to 0, preventing timeout. Reading always returns 0.

| Bits | Field   | Type | Reset      | Description                                                                                                                                                        |
|------|---------|------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | WDTPASS | w    | 0x00000000 | Watchdog password. Write 0x5F3759DF (unlock password) to enable WDTCR writes for 64 MCLK cycles. Write 0xA0C8A620 (clear password) to reset watchdog counter to 0. |

### 16.7.7 WDTCR: Watchdog timer control register

**Offset** 0x34 **Reset** 0x00 **Width** 8 **Address** 0x4934

Watchdog timer control register. This register is protected and requires password unlock via WDTPASS before writing. Configures watchdog operation mode and timeout period.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                 |
|------|-------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | WDTEN | rw   | 0     | Watchdog enable. When set, the 24-bit watchdog counter counts MCLK cycles and a timeout occurs when the counter bit selected by SYSWDTCDIV rises. WDTCR is write protected: unlock it through WDTPASS first.<br>0: Watchdog disabled<br>1: Watchdog enabled |
| 6    | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                           |

| Bits | Field    | Type | Reset  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|----------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5:2  | WDTCDIV  | rw   | 0b0000 | <p>Watchdog timeout select. The timeout event occurs when counter bit SYSWDTCDIV + 16 rises, so the period is <math>2^{(\text{SYSWDTCDIV} + 16)}</math> MCLK cycles.</p> <p>0b0000 SYSWDTCDIV_65536: Bit 16: 65,536 MCLK cycles<br/> 0b0001 SYSWDTCDIV_131072: Bit 17: 131,072 MCLK cycles<br/> 0b0010 SYSWDTCDIV_262144: Bit 18: 262,144 MCLK cycles<br/> 0b0011 SYSWDTCDIV_524288: Bit 19: 524,288 MCLK cycles<br/> 0b0100 SYSWDTCDIV_1048576: Bit 20: 1,048,576 MCLK cycles<br/> 0b0101 SYSWDTCDIV_2097152: Bit 21: 2,097,152 MCLK cycles<br/> 0b0110 SYSWDTCDIV_4194304: Bit 22: 4,194,304 MCLK cycles<br/> 0b0111 SYSWDTCDIV_8388608: Bit 23: 8,388,608 MCLK cycles<br/> 0b1000 SYSWDTCDIV_16777216: Bit 24: 16,777,216 MCLK cycles<br/> 0b1001 SYSWDTCDIV_33554432: Bit 25: 33,554,432 MCLK cycles<br/> 0b1010 SYSWDTCDIV_67108864: Bit 26: 67,108,864 MCLK cycles<br/> 0b1011 SYSWDTCDIV_134217728: Bit 27: 134,217,728 MCLK cycles<br/> 0b1100 SYSWDTCDIV_268435456: Bit 28: 268,435,456 MCLK cycles<br/> 0b1101 SYSWDTCDIV_536870912: Bit 29: 536,870,912 MCLK cycles<br/> 0b1110 SYSWDTCDIV_1073741824: Bit 30: 1,073,741,824 MCLK cycles<br/> 0b1111 SYSWDTCDIV_2147483648: Bit 31: 2,147,483,648 MCLK cycles</p> |
| 1    | WDTIE    | rw   | 0      | <p>Watchdog interrupt enable. When set, a timeout sets SYSWDTIF and raises the watchdog interrupt (vector 0) to the harts the IRQROUTER routes it to; with SYSWDTHWRST also set, the system reset follows the servicing hart completing the interrupt. If the vector is routed to no hart, the reset (when enabled) occurs immediately on timeout.</p> <p>0: Watchdog interrupt disabled<br/> 1: Watchdog interrupt enabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0    | WDTHWRST | rw   | 0      | <p>Watchdog reset enable. When set, a timeout resets the system: immediately if the watchdog interrupt is disabled or routed to no hart, otherwise after the servicing hart completes the interrupt.</p> <p>0: Watchdog reset disabled<br/> 1: Watchdog reset enabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### 16.7.8 WDTSR: Watchdog timer status register

**Offset** 0x38 **Reset** 0x00 **Width** 8 **Address** 0x4938

Watchdog timer status register. Contains flags indicating watchdog reset and interrupt events. Flags are cleared by writing 1 to the respective bit.

| Bits | Field | Type | Reset | Description                       |
|------|-------|------|-------|-----------------------------------|
| 7:2  | -     | -    | -     | Reserved. Reads zero; write zero. |

| Bits | Field | Type | Reset | Description                                                                                                                                                                                            |
|------|-------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | WDTIF | rw1  | 0     | Watchdog interrupt flag, set on a timeout while SYSWDTIE = 1 and held until software writes 1 to it.<br>0: No watchdog interrupt pending<br>1: Watchdog interrupt occurred                             |
| 0    | WDTRF | rw1  | 0     | Watchdog reset flag, set when the last system reset was caused by the watchdog and held across resets until software writes 1 to it.<br>0: Reset not caused by watchdog<br>1: Reset caused by watchdog |

### 16.7.9 WDTVAL: Watchdog timer value register

**Offset** 0x3C **Reset** 0x00000000 **Width** 32 **Address** 0x493C

Watchdog timer value register. Read-only register containing current watchdog counter value. Counter increments on MCLK when watchdog is enabled. Returns 0 when watchdog is disabled.

| Bits  | Field  | Type | Reset    | Description                                                                                                                                           |
|-------|--------|------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | -      | -    | -        | Reserved. Reads zero; write zero.                                                                                                                     |
| 23:0  | WDTVAL | r    | 0x000000 | Watchdog counter value. 24-bit up-counter that increments on MCLK cycles. Watchdog event occurs when bit selected by WDTCDIV transitions from 0 to 1. |

### 16.7.10 DCONBIAS (n = 0..1): Digitally controlled oscillator n bias register

**Offset** 0x40 + 4n **Reset** 0x0800 **Width** 16

Digitally controlled oscillator n bias register. Controls DCON output frequency through bias voltage adjustment. Higher bias values generally produce higher frequencies.

| Bits  | Field    | Type | Reset | Description                                                                                                              |
|-------|----------|------|-------|--------------------------------------------------------------------------------------------------------------------------|
| 15:12 | -        | -    | -     | Reserved. Reads zero; write zero.                                                                                        |
| 11:0  | DCONBIAS | rw   | 0x800 | DCON bias adjustment value. 12-bit bias control for DCON frequency tuning. Default value loaded from constants on reset. |

## 17 Neural processing unit (NPU)

### 17.1 Overview

The NPU is a fixed-point neural network processing unit built around one multiply-accumulate datapath and a configurable sequencer. One engine provides four datapath modes, selected per computation by NPUMODE: a fully connected layer, a one-dimensional convolution, a binary XNOR-popcount layer and a general matrix multiply. Operands and results live in the 16 KiB staging RAM at 0xC000 to 0xFFFF, which is shared between the harts and the engine's compute port.

- Four datapath modes (NPUMODE): dense layer, 1-D convolution, XNOR-popcount, GEMM
- Q0.24 inputs, Q7.24 weights and accumulator, per-step rounding and saturation
- Five activation functions on one hardware sigmoid approximator (NPUACTF, NPUAEN): sigmoid, ReLU, tanh, clamp, exponential
- Optional bias term per neuron (NPUBEN); packed input and weight formats (NPUXPK, NPUWPK)
- Three word-index pointers into the staging RAM (NPUIVSAR, NPUWVSAR, NPUOVSAR) and two per-mode configuration words (NPUCFG1, NPUCFG2)
- One interrupt, think-done, vector 120

### 17.2 Instances and signals

Table 17.1: NPU instances

| Instance | Base address | Interrupt vector(s) | Pins (AF) |
|----------|--------------|---------------------|-----------|
| NPU      | 0x4A00       | 120                 | none      |

Table 17.2: NPU interrupt vectors

| Vector | Instance | Description                           | Clear method                  |
|--------|----------|---------------------------------------|-------------------------------|
| 120    | NPU      | NPU0 think-done InterruptIRQB_NPU0_TD | Write 1 to NPUSR.NPUTHINKDONE |

### 17.3 Block diagram

Figure 17.1 draws the whole arrangement: the register file and the staging RAM as the two paths a hart reaches the NPU on, the port multiplexer that decides which side drives the RAM's single port, and the engine behind it. The three pointer registers are word indices into the one staging array, so the input, weight and output regions in the figure are roles rather than fixed addresses. The mode and activation codes the engine implements are the value tables of NPUMODE and NPUACTF in NPUCR (Section 17.8.1); they are not repeated here.



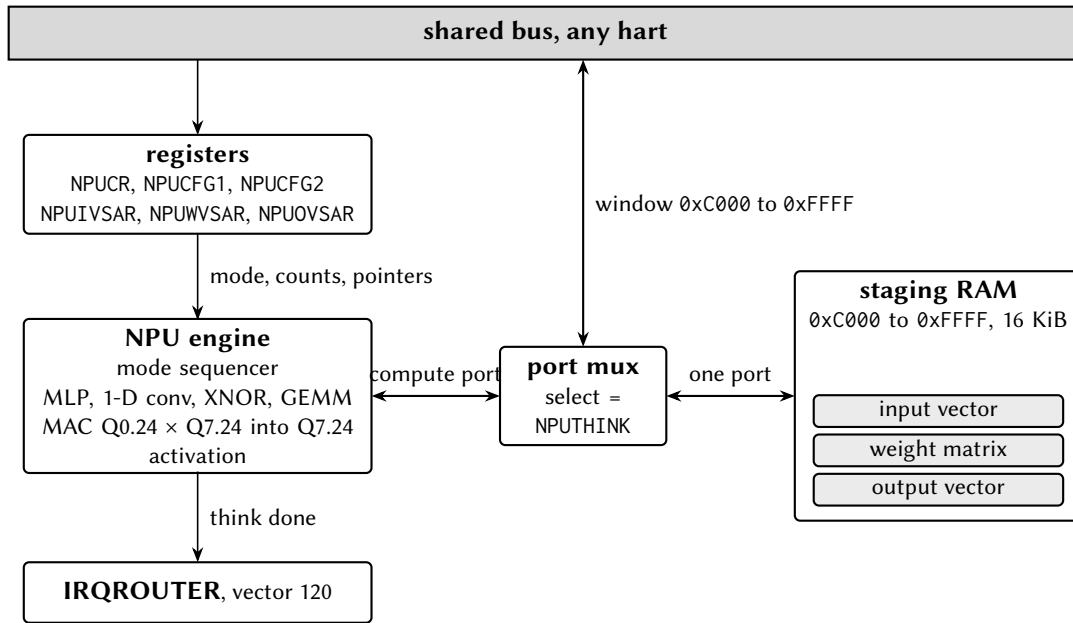


Figure 17.1: The NPU datapath and the staging RAM it borrows.

## 17.4 Functional description

### 17.4.1 Staging RAM ownership

The staging RAM is one single-port SRAM macro. While NPUTHINK is set the port belongs to the engine; the multiplexer select is NPUTHINK re-clocked on the free-running chip clock, so the port changes hands one cycle after a hart's start write lands and one cycle after the engine's completion pulse, and a hart's in-flight access is never cut in half. The hardware does not stop a hart from reaching into the window while the engine owns it: the access completes at the arbiter in the ordinary time and returns or overwrites whatever the RAM is holding for the engine. The rule that no hart touches 0xC000 to 0xFFFF while NPUTHINK reads 1 is therefore software's, and the poll on NPUTHINK is the whole of it.

The pointers NPUIVSAR, NPUWVSAR and NPUOVSAR are word indices into that array (byte offset from 0xC000 divided by 4), not addresses of separate memories. The sequencer trusts its configuration and bounds-checks nothing: a working set larger than the 4096 words, or count fields inconsistent with the staged data, wraps the 12-bit word index and corrupts neighbouring regions without the engine ever hanging.

### 17.4.2 Data formats

Inputs (dense and convolution inputs, GEMM A elements) are signed Q0.24 in bits 24:0 of a word, representing  $[-1, 1)$ ; bits 31:25 are ignored. Weights (and GEMM B elements) are signed Q7.24, representing  $[-128, 128)$ . Outputs are Q7.24 unless an activation narrows them. The accumulator is Q7.24 with round-to-nearest and saturation at  $\pm 128$  applied at every multiply-accumulate step in the fixed sequencer order, so a long dot product with large coefficients can clip before its end. XNOR mode packs 1-bit operands 32 per word and emits  $\pm 1.0$  Q7.24 words.

### 17.4.3 Packed operand formats

Two bits in NPUCR halve the staging footprint of a layer at the cost of resolution. With NPUPK set, the input vector holds two 16-bit Q0.15 elements per word, element  $2i$  in bits 15:0 and element  $2i + 1$  in bits 31:16, so element  $i$  is read from word NPUIVSAR +  $i/2$ ; the range stays  $[-1, 1)$  and firmware packs by arithmetic-shifting each Q0.24 input right by 9. The output vector is never packed, because outputs are full-width accumulator or activation words, so a layer whose outputs feed the next layer's inputs either leaves NPUPK clear or repacks between computations.

With NPUPK set, the weight matrix holds two 16-bit Q3.12 weights per word in the same element order as the unpacked layout; only the storage changes, and the element walk of every mode, including the convolution filter-block reload and the GEMM per-row reload, is unchanged. A maximal 256 by 256 layer then needs 8 re-stagings of the 4096-word staging RAM instead of 16. Q3.12 narrows the weight range from  $[-128, 128)$  to  $[-8, 8)$  and coarsens the resolution from  $2^{-24}$  to  $2^{-12}$ : firmware packs by arithmetic-shifting each Q7.24 weight right by 12 and must saturate first, because a weight outside  $[-8, 8)$  has no packed representation and the hardware only ever sees the packed half. The bias weight rides the same stream under the same range. Both bits are ignored in XNOR mode, which already packs 32 one-bit operands per word, and both are latched when NPUTHINK is set, so a write during a run takes effect at the next run.

### 17.4.4 Datapath modes

NPUNI and NPUNN hold a count minus one and are reinterpreted by each mode, as are NPUCFG1 and NPUCFG2.

- **Mode 0, dense layer** (reset default): NPUNI + 1 inputs at NPUIVSAR, NPUNN + 1 neurons. The weight matrix at NPUVVSAR holds one contiguous row per neuron; with NPUBEN set each row begins with a bias weight multiplied by an implicit input of 1.0. NPUCFG1 and NPUCFG2 are unused.
- **Mode 1, 1-D convolution**: NPUNI = taps - 1, NPUNN = filters - 1. NPUCFG1 packs stride, dilation, input length and channel count; NPUCFG2 holds the output length, which the sequencer trusts, so the padding convention is a firmware decision. Inputs are channel-major, weights filter-major (bias first per filter with NPUBEN), outputs filter-major.
- **Mode 2, XNOR-popcount**: activations and weights packed 32 per word LSB first. NPUNI = packed words per neuron - 1; the exact bit count  $K \leq 4096$  is in NPUCFG2 bits 12:0 and the unused tail bits of the last word are masked. A neuron emits +1.0 when  $2 \cdot \text{popcount}(\text{XNOR}(a, w)) - K$  is at least the signed threshold in NPUCFG1, else -1.0. NPUBEN and NPUAEN must be 0.
- **Mode 3, GEMM**:  $C = A \times B$  with  $A (M \times K, \text{row-major}, \text{Q0.24})$  at NPUIVSAR,  $B (K \times N, \text{column-major}, \text{Q7.24})$  at NPUVVSAR,  $C (M \times N, \text{row-major})$  at NPUOVSAR;  $K - 1$  in NPUNI,  $N - 1$  in NPUNN,  $M - 1$  in NPUCFG1 bits 7:0. There is no accumulation across runs, so  $K$  must fit one run; larger  $M$  or  $N$  tile by re-pointing the registers. Because  $C$  is row-major, a result chains directly as the next layer's  $A$ . NPUBEN must be 0.

Mode codes 4 to 7 run as mode 0. The mode, counts and activation selection are latched into run shadows when NPUTHINK is set, so a write during a run takes effect at the next run.

### 17.4.5 Activation functions

With NPUAEN set, each output passes through the function selected by NPUACTF before it is written; with NPUAEN clear the raw Q7.24 accumulator is written regardless. All five are shift-and-add wrappers around one piecewise-quadratic sigmoid approximator: 0 = sigmoid (Q0.24 in  $[0, 1)$ , saturating for  $|x| \geq 4$ ); 1 =

ReLU (Q7.24; an output at or above 1.0 is not a valid Q0.24 next-layer input); 2 = tanh as  $2\sigma(2x) - 1$  (in  $[-1, 1]$ , saturating for  $|x| \geq 2$ , a valid next-layer input); 3 = clamp to  $[-1, 1]$  (always chains); 4 = exponential as  $2\sigma(x)$  (Q1.24 in  $[0, 2]$ , the per-element scores of a softmax whose sum and normalisation firmware performs; subtracting the maximum logit first maps it to exactly 1.0). Codes 5 to 7 read as sigmoid. XNOR mode ignores the activation path.

### 17.4.6 Run time

Run time scales with the mode's loop product: a GEMM costs on the order of  $5 \cdot M \cdot N \cdot K$  cycles and a convolution  $5 \cdot C_{out} \cdot L_{out} \cdot C_{in} \cdot K$ , so a full-scope convolution can take hundreds of milliseconds at 24 MHz. XNOR mode is about an order of magnitude cheaper per output, because one fetched word pair carries 32 synapses.

## 17.5 Interrupts and events

The completion pulse that clears NPUTHINK also sets the sticky flag NPUTHINKDONE in NPUSR; the interrupt is the level of that flag gated by NPUTDIE, on the free-running clock. NPUTDIE resets to 0, so polling software is unaffected. Table 17.3 lists the vector.

| Vector | Condition            | Set by       | Cleared by                |
|--------|----------------------|--------------|---------------------------|
| 120    | Computation complete | NPUTHINKDONE | Writing 1 to NPUTHINKDONE |

Table 17.3: NPU interrupt vector.

## 17.6 Programming

### 17.6.1 Running one computation

1. Poll NPUCR.NPTHINK until it reads 0.
2. Stage the operands in the staging RAM in the selected mode's layout.
3. Write the input word index to NPUIVSAR.
4. Write the weight word index to NPUWVSAR.
5. Write the output word index to NPUOVSAR.
6. Write NPUCFG1 and NPUCFG2 if the mode uses them.
7. Write NPUCR with NPUMODE, NPUNI, NPUNN, NPUBEN, NPUAEN, NPUACTF and, for an interrupt, NPUTDIE.
8. Set NPUCR.NPTHINK.
9. Poll NPUCR.NPTHINK until it reads 0, or take vector 120 and write 1 to NPUSR.NPTHINKDONE in the handler.
10. Read the outputs from the staging RAM at NPUOVSAR.

## 17.7 Usage notes

**Do not touch the staging RAM while NPUTHINK reads 1.** The hardware neither blocks nor stalls the access; it returns or corrupts the engine's data.

**Check NPUTHINK from every hart, not only the one that took the interrupt.** The interrupt tells one hart the run has finished; any other hart consults the flag before reading results.

**Size the working set yourself.** Nothing in the sequencer checks bounds; an oversized working set wraps the 12-bit index and corrupts its neighbours silently.

**Saturate weights to  $[-8, 8)$  before packing them.** A Q7.24 weight outside that range has no Q3.12 representation; shifting without saturating wraps it.

**Chain layers through Q0.24-valid activations.** ReLU outputs at or above 1.0 are not valid inputs; use tanh, clamp or normalised weights when a result feeds the next layer.

**Clear NPUTHINKDONE with a 1 before completing the interrupt.** Reading NPUSR does not clear it, and the level re-enters the handler otherwise.

## 17.8 Registers

Register offsets in this chapter are relative to the instance base address in Table 17.1 (NPU 0x4A00); the generated header names each base NPU\_BASE.

Table 17.4: NPU register summary

| Offset | Name     | Access | Reset      | Description                   |
|--------|----------|--------|------------|-------------------------------|
| 0x00   | NPUCR    | rw     | 0x00000000 | NPU control register          |
| 0x04   | NPUIVSAR | rw     | 0x00000000 |                               |
| 0x08   | NPUWVSAR | rw     | 0x00000000 |                               |
| 0x0C   | NPUOVSAR | rw     | 0x00000000 |                               |
| 0x10   | NPUSR    | rw1    | 0x00000000 | NPU status register           |
| 0x14   | NPUCFG1  | rw     | 0x00000000 | NPU mode configuration word 1 |
| 0x18   | NPUCFG2  | rw     | 0x00000000 | NPU mode configuration word 2 |

### 17.8.1 NPUCR: NPU control register

Offset 0x00    Reset 0x00000000    Width 32    Address 0x4A00

| Bits  | Field | Type | Reset | Description                       |
|-------|-------|------|-------|-----------------------------------|
| 31:28 | -     | -    | -     | Reserved. Reads zero; write zero. |

| Bits  | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|-------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27    | XPk   | rw   | 0     | <p>Packed input format enable: when set, the input vector holds two 16-bit Q0.15 elements per staging word (element 2i in bits 15:0, element 2i+1 in bits 31:16), so element i is read from word NPUIVSAR + i/2; when clear each input occupies its own 32-bit word as Q0.24. The setting is latched when NPUTHINK is set and is ignored in XNOR-popcount mode. See the packed operand formats in the functional description.</p> <p>0: One input element per 32-bit word (reset default)<br/>1: Two 16-bit Q0.15 input elements per 32-bit word</p>                                                                                                           |
| 26    | WPK   | rw   | 0     | <p>Packed weight format enable: when set, the weight matrix holds two 16-bit Q3.12 weights per staging word (element 2i in bits 15:0, element 2i+1 in bits 31:16) in the same element order as the unpacked layout; when clear each weight occupies its own 32-bit word as Q7.24. The setting is latched when NPUTHINK is set and is ignored in XNOR-popcount mode. See the packed operand formats in the functional description for the range and saturation rules.</p> <p>0: One weight per 32-bit word (reset default)<br/>1: Two 16-bit Q3.12 weights per 32-bit word</p>                                                                                  |
| 25:23 | ACTF  | rw   | 0b000 | <p>Activation function applied to the accumulator output while NPUAEN = 1; with NPUAEN = 0 the raw accumulator passes through regardless of this field. The selection is latched when NPUTHINK is set; see the activation functions in the functional description for the exact formulas and output formats.</p> <p>0b000: logistic sigmoid approximation (reset default), output Q0.24 in [0, 1)<br/>0b001: ReLU, output Q7.24<br/>0b010: tanh approximation, output in [-1, 1)<br/>0b011: clamp to [-1, 1)<br/>0b100: exponential approximation <math>2^{\text{sigmoid}(x)}</math>, output Q1.24 in [0, 2)<br/>0b101 to 0b111: reserved, and behave as 0</p> |
| 22:20 | MODE  | rw   | 0b000 | <p>Datapath mode select. The mode is latched when NPUTHINK is set; see the datapath modes in the functional description for the operand layout of each mode.</p> <p>0b000: multilayer-perceptron mode (reset default)<br/>0b001: one-dimensional convolution mode<br/>0b010: XNOR-popcount binary mode (NPUBEN and NPUAEN must be 0)<br/>0b011: general matrix-multiply (GEMM) mode (NPUBEN must be 0)<br/>0b100 to 0b111: reserved, and behave as mode 0</p>                                                                                                                                                                                                  |
| 19    | TDIE  | rw   | 0     | <p>Think-done interrupt enable. When set, NPUSR.NPUTHINKDONE drives the NPU think-done interrupt (vector 120); when clear the flag is available for polling only.</p> <p>0: Interrupt disabled (polling only)<br/>1: Interrupt enabled</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                          |
|------|-------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18   | BEN   | rw   | 0     | Bias enable. When set, the first weight of each output neuron's row in the weight matrix is used as a bias term: it is multiplied by an implicit input of 1.0 and accumulated before the synaptic weights.<br>0: Disabled<br>1: Enabled                                              |
| 17   | AEN   | rw   | 0     | Activation function enable. When set, the logistic sigmoid approximation activation function is applied to the accumulator output. When cleared, the raw accumulator output is used (linear/identity).<br>0: Disabled (linear output)<br>1: Enabled (logistic sigmoid approximation) |
| 16   | THINK | rw   | 0     | Computation start and busy flag. Writing 1 starts the NPU; the bit reads 1 until the computation completes, when hardware clears it. While it reads 1 the staging RAM belongs to the NPU.<br>0: Idle (computation complete or not started)<br>1: Running (write 1 to start)          |
| 15:8 | NI    | rw   | 0x00  | Number of inputs in the input vector minus 1. The actual number of inputs is NPUNI + 1.                                                                                                                                                                                              |
| 7:0  | NN    | rw   | 0x00  | Number of output neurons minus 1. The actual number of outputs is NPUNN + 1.                                                                                                                                                                                                         |

### 17.8.2 NPUIVSAR

**Offset** 0x04 **Reset** 0x00000000 **Width** 32 **Address** 0x4A04

Input vector start word index within the shared NPU staging RAM: the byte offset from the start of the staging RAM (0xC000) divided by 4. For example, an input vector at byte address 0xC100 has word index 0x40. Each input is a signed Q0.24 value stored in bits 24:0 of its 32-bit SRAM word; bits 31:25 are ignored. Bit 24 is the sign bit. The input at index 0 is at word index NPUIVSAR. The input at index 1 is at word index NPUIVSAR + 1. The rest of the inputs follow in consecutive words. This one-input-per-word layout is what NPUCR.NPUXPK = 0 (the reset default) selects; with NPUXPK = 1 the vector is packed two 16-bit Q0.15 inputs per word and input index  $i$  is read from word index  $\text{NPUIVSAR} + i/2$ , low half first.

| Bits  | Field | Type | Reset | Description                                                                                     |
|-------|-------|------|-------|-------------------------------------------------------------------------------------------------|
| 31:12 | -     | -    | -     | Reserved. Reads zero; write zero.                                                               |
| 11:0  | IVSAR | rw   | 0x000 | Input vector start word index within the NPU staging RAM (byte offset from 0xC000 divided by 4) |

### 17.8.3 NPUWVSAR

**Offset** 0x08 **Reset** 0x00000000 **Width** 32 **Address** 0x4A08

Synaptic weight matrix start word index within the shared NPU staging RAM: the byte offset from the start of the staging RAM (0xC000) divided by 4. Each weight is a signed Q7.24 value occupying all 32 bits of its SRAM word; bit 31 is the sign bit. Weights are stored row-major, one per 32-bit word, in the following order: for each output neuron (0 through NPUNN), if bias is enabled (NPUBEN = 1), the first word in the row is the bias weight (multiplied by an implicit input of 1.0), followed by NPUNI + 1 synaptic weights for inputs 0 through NPUNI. If bias is disabled, each row contains NPUNI + 1 synaptic weights only. This one-weight-per-word layout is what NPUCR.NPUWPK = 0 (the reset default) selects; with NPUWPK = 1

the same weight sequence is packed two 16-bit Q3.12 weights per word, low half first, halving the word count of the block.

| Bits  | Field | Type | Reset | Description                                                                                               |
|-------|-------|------|-------|-----------------------------------------------------------------------------------------------------------|
| 31:12 | -     | -    | -     | Reserved. Reads zero; write zero.                                                                         |
| 11:0  | WVSAR | rw   | 0x000 | Synaptic weight matrix start word index within the NPU staging RAM (byte offset from 0xC000 divided by 4) |

#### 17.8.4 NPUOVSAR

**Offset** 0x0C **Reset** 0x00000000 **Width** 32 **Address** 0x4A0C

Output vector start word index within the shared NPU staging RAM: the byte offset from the start of the staging RAM (0xC000) divided by 4. Each output is a signed Q7.24 value occupying all 32 bits of its SRAM word; bit 31 is the sign bit. The output at index 0 is written to word index NPUOVSAR. The output at index 1 is at word index NPUOVSAR + 1, and so on.

| Bits  | Field | Type | Reset | Description                                                                                      |
|-------|-------|------|-------|--------------------------------------------------------------------------------------------------|
| 31:12 | -     | -    | -     | Reserved. Reads zero; write zero.                                                                |
| 11:0  | OVSAR | rw   | 0x000 | Output vector start word index within the NPU staging RAM (byte offset from 0xC000 divided by 4) |

#### 17.8.5 NPUSR: NPU status register

**Offset** 0x10 **Reset** 0x00000000 **Width** 32 **Address** 0x4A10

NPU status register. NPUTHINKDONE is write-1-to-clear and is never cleared by a read; the think-done interrupt (vector 120) is NPUTHINKDONE and NPUCR.NPUTDIE. Clearing NPUTHINKDONE does not affect NPUCR.NPUTHINK, and the staging RAM stays owned by the NPU until NPUCR.NPUTHINK reads 0.

| Bits | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                               |
|------|-----------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | -         | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                                                                                                         |
| 0    | THINKDONE | rw1  | 0     | Think-done flag, set by hardware when a computation finishes (the same event that clears NPUTHINK) and held until software writes 1 to it. It drives the think-done interrupt (vector 120) while NPUTDIE = 1. A completion that coincides with a write of 1 leaves the flag set.<br>0: No completed computation pending<br>1: A computation has completed |

#### 17.8.6 NPUCFG1: NPU mode configuration word 1

**Offset** 0x14 **Reset** 0x00000000 **Width** 32 **Address** 0x4A14

NPU mode configuration word 1. The interpretation depends on NPUCR.NPUMODE. Multilayer-perceptron mode (0): unused, no effect. One-dimensional convolution mode (1): bits 3:0 = stride S (1-15), bits 7:4 = dilation D (1-15), bits 23:8 = input length L per channel in samples, bits 31:24 = number of input channels minus 1 (the actual channel count  $C_{in}$  is this field + 1). XNOR-popcount mode (2): the full 32-bit signed firing threshold THRESH. A neuron fires (output +1.0) when  $2 * \text{popcount}(\text{XNOR}(\text{activations}, \text{weights})) - K$  is greater than or equal to THRESH, else outputs -1.0. GEMM mode (3): bits 7:0 = M - 1, the number of A rows minus 1 (the actual row count M is this field + 1); bits 31:8 are ignored in this mode.

| Bits | Field | Type | Reset      | Description                                                              |
|------|-------|------|------------|--------------------------------------------------------------------------|
| 31:0 | CFG1  | rw   | 0x00000000 | Per-mode configuration word 1 (interpretation selected by NPUCR.NPUMODE) |

### 17.8.7 NPUCFG2: NPU mode configuration word 2

**Offset** 0x18 **Reset** 0x00000000 **Width** 32 **Address** 0x4A18

NPU mode configuration word 2. The interpretation depends on NPUCR.NPUMODE. Multilayer-perceptron mode (0): unused, no effect. One-dimensional convolution mode (1): bits 15:0 = output length Lout per filter in samples. Lout is computed by the host (the sequencer trusts this value; the valid/same padding convention is a firmware decision) and every referenced input sample index  $j*S + k*D$  must be less than L. XNOR-popcount mode (2): bits 12:0 = K, the exact input bit count (1-4096); NPUNI must hold  $\text{ceil}(K/32) - 1$ , the packed words per neuron, and when  $K \bmod 32$  is nonzero the unused high bits of each last packed word are masked out of the popcount by hardware. GEMM mode (3): unused, no effect. Bits 31:16 are reserved and read 0.

| Bits  | Field | Type | Reset  | Description                                                              |
|-------|-------|------|--------|--------------------------------------------------------------------------|
| 31:16 | -     | -    | -      | Reserved. Reads zero; write zero.                                        |
| 15:0  | CFG2  | rw   | 0x0000 | Per-mode configuration word 2 (interpretation selected by NPUCR.NPUMODE) |



## 18 Power control (PWRCTRL)

### 18.1 Overview

The power controller (PWRCTRL) switches the power domains of the channel tiles, harts 1 to 4, and in configurations wired for field power it also owns the boot gate that holds every hart in reset until the harvested supply is good. The power domains themselves, and what cold gating means for software, are described in the Reset, clocks and power chapter (Section 6.4).

- One gate bit per channel tile in PWRCR (mask 0x1E); hart 0 has no gate bit
- A per-tile hardware sequencer that orders isolation, reset and the header switches in both directions
- Read-only sequencer state per tile in PWRSR, one nibble per hart
- Cold gating: a gated tile loses all state and cold-boots through the shared ROM on wake
- Boot gate configuration in PWRWAKE and live status in PWRSTS
- An event-fabric wake mask in TASKWKM: a task-wake pulse clears the PWRCR gate bits it selects
- All domains power up on at reset; the controller is inert until written

### 18.2 Instances and signals

Table 18.1: PWRCTRL instances

| Instance | Base address | Interrupt vector(s) | Pins (AF) |
|----------|--------------|---------------------|-----------|
| PWRCTRL  | 0x4B00       | none                | none      |

### 18.3 Functional description

#### 18.3.1 Gate sequencer

Setting bit  $h$  of PWRCR starts tile  $h$ 's power-down sequence: the isolation clamps assert on the always-on side of the boundary, the tile is held in reset, and only then do the header switches open, so that no floating signal ever reaches the shared fabric. Clearing the bit runs the sequence in the other order, with a rail-settle delay sized for the tile's header-switch chain before the reset is released. The sequencer never aborts a transition: a gate-bit change made mid-sequence takes effect when the current sequence completes. Bit 0 (PWRH0) is reserved, reads 0 and ignores writes, because hart 0 is always on; a write of 0x1E therefore gates every channel tile and leaves hart 0 running. The write is qualified by byte lane 0 only, so full-word stores are used.

PWRSR reports the sequencer state of each tile as a 4-bit code in nibble  $h$  (PWRST0 to PWRST4); 0 is on and 3 is off, with the intermediate codes marking the sequence in progress. Software polls the nibble after a write to PWRCR, because the transition takes several cycles and the tile must not be relaunched before its nibble reads 0.

### 18.3.2 Task wake

TASKWKM selects which tiles the event fabric may wake. A EVFAB task-wake pulse clears the PWRCR.PWRGATE bits that PWRTASKWKM selects, so those tiles run the same rail-up sequence a software gate-clear runs, with no hart awake to write PWRCR. The pulse is applied per bit after the bus writes of the same cycle, so a coincident PWRCR write merges and the task wins its own bits. The mask is qualified by byte lane 0 like PWRCR, bit 0 has no storage, and it resets to 0, so task wake is inert until software selects a tile. The tap never reaches the boot-gate release logic: a held boot gate stays held.

### 18.3.3 Boot gate

The boot gate is a hold-in-reset combined into every hart's outer reset. All of its state lives on the always-on MCLK domain and evaluates once per cycle. The three pad-side inputs (PGOOD, the strap and the NFC field-detect level) are double-flop synchronized, so they reach the gate two MCLK cycles late.

The strap is sampled once: a counter starts at reset release and, eight MCLK cycles later, latches the synchronized strap level into PWSTRAP and sets PWSTRAPVLD. A strapped board thereby arms the gate, adds PGOOD as a release condition and enables re-hold with no software. The gate is released during that sampling window so that a normal board's boot is never delayed; a strapped board's harts may therefore begin a first fetch and then be re-held, which is harmless because the hold is the same outer reset that cleared them at power-on and the eventual release runs a clean cold boot.

The effective policy combines the strap defaults with the PWRWAKE overrides:

- armed = PWGATEEN or strapped
- release on PGOOD = PWRLSPGOOD or strapped
- release on field detect = PWRLSFIELD
- re-hold = PWREHOLD or strapped
- release = PWSWRLS, or (PGOOD live and release on PGOOD), or (field live and release on field detect)
- hold = armed and not (release, when re-hold is set; the sticky latch of every past release, when it is clear)

The gate output is registered, so releases and re-holds are glitch-free and take effect one cycle after their condition resolves. With re-hold set the hold tracks the release condition: a PGOOD drop re-asserts the hold on the next cycle and the harts re-enter reset at once, because without retention there is nothing a grace period could save. With re-hold clear the first release is final until the next chip reset, for systems in which supervising software owns brownout policy. An armed gate with no release source holds until PWSWRLS, which production test uses to prove the gate is load-bearing.

PWRSTS reports the synchronized live levels (PWPGOODLIV, PWFIELDLIV), the sampled strap (PWSTRAP, PWSTRAPVLD), the gate state (PWB00THOLD) and the release latch (PWRRLSLATCH). Software can only observe the gate released or arm it for the next hold, because while the hold is active nothing executes.

## 18.4 Interrupts and events

Not applicable. The controller raises no vector; the NFC field-detect level used as a release source is available separately as NFCx vector 94, and tile state is polled in PWRSR.

## 18.5 Programming

### 18.5.1 Gating tile $h$

1. Confirm the tile is parked in its boot-ROM sleep or has finished its work, and that any data it must keep is in shared RAM.
2. Set bit  $h$  of PWRCCR.PWRGATE with a full-word store.
3. Poll nibble  $h$  of PWRSR until it reads 3.

### 18.5.2 Waking tile $h$

1. Clear bit  $h$  of PWRCCR.PWRGATE with a full-word store.
2. Poll nibble  $h$  of PWRSR until it reads 0.
3. Write the tile's loader mailbox row and send it a software interrupt (Section 5.3).

### 18.5.3 Arming the boot gate from software

1. Set PWRWAKE.PWRLSPGOOD, and PWRWAKE.PWRLSFIELD if the RF field is also a release source.
2. Set PWRWAKE.PWREHOLD if the hold is to track PGOOD.
3. Set PWRWAKE.PWGATEEN.

A harvested board needs none of this: with the strap high and the supervisor's output on PGOOD, the hardware defaults arm the gate, wait on PGOOD and re-hold on brownout, so the first instruction any hart executes runs after the supply is declared good.

### 18.5.4 Supercap duty-cycled service loop

1. Gate the unused tiles through PWRCCR.
2. Route NFCx vector 94 to the serving hart in its IRQROUTER row.
3. Provision and arm the NFC block in listen mode with auto-read.
4. Gate unused memories and banks in BLOCKPWR and divide MCLK in CLKDIVCR.
5. Execute sleep; a reader's field raises the interrupt and the hart resumes with its state intact.

## 18.6 Usage notes

**Gate only parked or quiesced tiles.** Gating is cold; a tile gated mid-transaction loses its work and any shared-bus transaction it had in flight is answered by the clamps.

**Use full-word stores to PWRCCR.** The write is qualified by byte lane 0 only.

**Poll PWRSR rather than adding delays.** The sequencer includes the rail-settle delay; the nibble reads 0 when the tile can be relaunched.

**Check PWRSR before trusting an aperture read.** A gated tile's TCM aperture returns zeros, and zero is a legal TCM value.

**Keep the boot gate disarmed on a normally powered board.** An armed gate with no release source holds every hart, hart 0 included, until PWSWRLS is written, which nothing can do while the hold is active.

## 18.7 Registers

Register offsets in this chapter are relative to the instance base address in Table 18.1 (PWRCTRL 0x4B00); the generated header names each base PWRCTRL\_BASE.

Table 18.2: PWRCTRL register summary

| Offset       | Name    | Access | Reset      | Description                                                         |
|--------------|---------|--------|------------|---------------------------------------------------------------------|
| 0x00         | PWRCR   | rw/r   | 0x00000000 | Power gate control                                                  |
| 0x04         | PWRSR   | r      | 0x00000000 | Power sequencer state, one read-only nibble per hart (bits 4h+3:4h) |
| 0x08 to 0x13 | -       | -      | -          | Reserved                                                            |
| 0x14         | PWRWAKE | rw     | 0x00000000 | Boot gate and wake source control                                   |
| 0x18         | PWRSTS  | r      | 0x00000000 |                                                                     |
| 0x1C         | TASKWKM | rw     | 0x00000000 | Event-fabric task-wake mask, one bit per gateable tile hart         |

### 18.7.1 PWRCR: Power gate control

**Offset** 0x00 **Reset** 0x00000000 **Width** 32 **Address** 0x4B00

Power gate control. Setting PWRGATE bit h powers tile hart h down (isolation, reset, rail off); clearing it powers the tile back up and cold-boots it. A request made while the sequencer is mid-sequence is honored when the sequence completes. Bit 0 (hart 0) is always on: it reads 0 and ignores writes.

| Bits | Field | Type | Reset  | Description                                                                                                                                                                                                                                                                                                            |
|------|-------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:5 | -     | -    | -      | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                                                                      |
| 4:1  | GATE  | rw   | 0b0000 | Gate request mask, one bit per tile hart: bit h = 1 powers tile hart h down and 0 powers it up, with a cold boot on the way up. Poll the PWRSR nibble of the hart for sequencer completion. Only the all-powered value is enumerated; any combination of bits is valid.<br>0b0000 PWRGATE_NONE: All tile harts powered |
| 0    | H0    | r    | 0      | Hart 0 is always-on: reads 0, writes ignored.                                                                                                                                                                                                                                                                          |

### 18.7.2 PWRSR: Power sequencer state, one read-only nibble per hart (bits 4h+3:4h)

**Offset** 0x04 **Reset** 0x00000000 **Width** 32 **Address** 0x4B04

Power sequencer state, one read-only nibble per hart (bits 4h+3:4h). 0 = ON, 1 = ISO (clamps asserting), 2 = RSTOFF (reset held, rail dying), 3 = OFF (gated), 4 = RAIL (waking, rail settling), 5 = UNISO (clamps releasing). Hart 0's nibble always reads 0. A tile is safely gated when its nibble reads 3, and fully awake (booting or parked in the ROM) when it returns to 0.

| Bits  | Field | Type | Reset  | Description                                    |
|-------|-------|------|--------|------------------------------------------------|
| 31:20 | -     | -    | -      | Reserved. Reads zero; write zero.              |
| 19:16 | ST4   | r    | 0b0000 | Tile hart 4 sequencer state.                   |
| 15:12 | ST3   | r    | 0b0000 | Tile hart 3 sequencer state.                   |
| 11:8  | ST2   | r    | 0b0000 | Tile hart 2 sequencer state.                   |
| 7:4   | ST1   | r    | 0b0000 | Tile hart 1 sequencer state.                   |
| 3:0   | ST0   | r    | 0b0000 | Hart 0 state: always 0 (ON, always-on domain). |

### 18.7.3 PWRWAKE: Boot gate and wake source control

**Offset** 0x14 **Reset** 0x00000000 **Width** 32 **Address** 0x4B14

Boot gate and wake source control. The harvested-boot strap drives the hardware defaults (a strapped board arms the gate, waits on PGOOD and re-holds on brownout with no software involvement); these bits OR software overrides on top. Resets to 0 (no override), so the gate is released on every normal boot. Write with full-word stores (byte lane 0 qualified, like PWRCCR).

| Bits | Field    | Type | Reset | Description                                                                                                                                                                                                                                                          |
|------|----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:5 | -        | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                    |
| 4    | EHOLD    | rw   | 0     | Re-hold policy: 1 re-asserts the boot hold whenever the release condition drops (a later PGOOD return then cold-boots the harts through the shared ROM); 0 makes the release one-shot, latched in PWRSTS.PWRRSLATCH. The strap ORs this bit in on a harvested board. |
| 3    | PWSWRLS  | rw   | 0     | Software release: 1 releases an armed boot gate unconditionally. With no other release source enabled, an armed gate holds until this bit is set.                                                                                                                    |
| 2    | LSFIELD  | rw   | 0     | Field release enable: 1 makes the synchronized NFC field level (PWRSTS.PWFIELDLIV) a release condition, so a reader arriving releases the boot gate. The field level is tied to 0 when NFC is absent or field power is disabled.                                     |
| 1    | LSPGOOD  | rw   | 0     | PGOOD release enable: 1 makes the synchronized PGOOD pad level (PWRSTS.PWPGOODLIV) a release condition. The strap ORs this bit in on a harvested board, which always waits on the supply supervisor.                                                                 |
| 0    | PWGATEEN | rw   | 0     | Boot gate arm: 1 holds every hart in reset until a release condition fires. The strap ORs this bit in, so a strapped board arms with no software.                                                                                                                    |

### 18.7.4 PWRSTS

**Offset** 0x18 **Reset** 0x00000000 **Width** 32 **Address** 0x4B18

Boot gate and wake source status, read only: the synchronized live pad levels, the one-shot strap sample that selects the harvested-boot branch of the boot ROM, and the gate state.

| Bits | Field    | Type | Reset | Description                                                                                                |
|------|----------|------|-------|------------------------------------------------------------------------------------------------------------|
| 31:6 | -        | -    | -     | Reserved. Reads zero; write zero.                                                                          |
| 5    | RLSLATCH | r    | 0     | One-shot release latched: set when an armed gate has been released with PWREHOLD = 0 and held until reset. |

| Bits | Field      | Type | Reset | Description                                                                                                                                                                                                     |
|------|------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4    | PWBOOTHOLD | r    | 0     | Boot gate state: 1 while the gate holds every hart in reset.                                                                                                                                                    |
| 3    | PWSTRAPVLD | r    | 0     | Strap sample valid: set a few MCLK cycles after reset release once the one-shot strap sample has been taken. Poll it before reading PWSTRAP.                                                                    |
| 2    | PWSTRAP    | r    | 0     | Harvested-boot strap sample: 1 = harvested boot (the boot ROM skips the SPI flash copy and runs the ROM-resident service loop), 0 = normal SPI boot. Sampled once after reset; later strap changes are ignored. |
| 1    | PWFIELDLIV | r    | 0     | Synchronized NFC field detect level; 0 when NFC is absent or field power is disabled.                                                                                                                           |
| 0    | PWPGOODLIV | r    | 0     | Synchronized PGOOD pad level (P6.7); reads 0 (power not good) when the pin is unconnected, because of the reset pull-down.                                                                                      |

### 18.7.5 TASKWKM: Event-fabric task-wake mask, one bit per gateable tile hart

**Offset** 0x1C **Reset** 0x00000000 **Width** 32 **Address** 0x4B1C

Event-fabric task-wake mask, one bit per gateable tile hart. A task\_wake pulse from the event fabric clears the PWRCR gate bits selected here, waking those tiles through the same MTCMOS rail-up sequence a register-cleared gate uses; the tap never touches the boot-gate release logic. Byte-lane-0 qualified like PWRCR, and bit 0 (hart 0, always on) has no storage. Resets to 0, so task-wake is inert out of reset.

| Bits | Field   | Type | Reset  | Description                                                                                                                          |
|------|---------|------|--------|--------------------------------------------------------------------------------------------------------------------------------------|
| 31:5 | -       | -    | -      | Reserved. Reads zero; write zero.                                                                                                    |
| 4:1  | TASKWKM | rw   | 0b0000 | Task-wake mask, one bit per tile hart: bit h = 1 lets a fabric task_wake pulse clear PWRCR.PWRGATE[h] and power tile hart h back up. |
| 0    | -       | -    | -      | Reserved. Reads zero; write zero.                                                                                                    |

## 19 Inter-integrated circuit interface (I2Cx)

### 19.1 Overview

The I2Cx peripherals are general-purpose I<sup>2</sup>C interfaces, each able to act as a bus master or as an addressed slave. Master and slave are driven through separate sets of control and status bits that share one pin pair, so a port can be configured for either role, or for both on a multi-master bus.

- Master transmitter and master receiver modes with START, repeated START and STOP generation (I2CMST, I2CMSP) and a programmable bus clock divider (I2CMDIV)
- Multi-master arbitration: losing the bus releases it and sets I2CMARB
- Slave transmitter and slave receiver modes with a 7-bit address (I2CxAR), a per-bit address mask (I2CxAMR) and optional general-call response (I2CGCE)
- Optional slave clock stretching (I2CSCS) through the ACK bit
- Separate transmit and receive data registers per role (I2CxMTX/I2CxMRX, I2CxSTX/I2CxSRX)
- Thirteen individually maskable interrupt sources, one vector each

### 19.2 Instances and signals

Table 19.1: I2Cx instances

| Instance | Base address | Interrupt vector(s) | Pins (AF)                        |
|----------|--------------|---------------------|----------------------------------|
| I2C0     | 0x4E00       | 57 to 69            | SDA0 P3.0 (AF0), SCL0 P3.1 (AF0) |
| I2C1     | 0x4F00       | 70 to 82            | SDA1 P3.2 (AF0), SCL1 P3.3 (AF0) |

Table 19.2: I2Cx signals

| Signal | Direction     | Description       | Pin (AF)                           |
|--------|---------------|-------------------|------------------------------------|
| SDA0   | Bidirectional | I2C0 serial data  | P1.6 (AF1), P2.6 (AF1), P3.0 (AF0) |
| SCL0   | Bidirectional | I2C0 serial clock | P1.7 (AF1), P2.7 (AF1), P3.1 (AF0) |
| SDA1   | Bidirectional | I2C1 serial data  | P1.4 (AF1), P2.2 (AF1), P3.2 (AF0) |
| SCL1   | Bidirectional | I2C1 serial clock | P1.5 (AF1), P2.3 (AF1), P3.3 (AF0) |

Table 19.3: I2Cx interrupt vectors

| Vector | Instance | Description                                    | Clear method             |
|--------|----------|------------------------------------------------|--------------------------|
| 57     | I2C0     | I2C0 start received Interrupt<br>IRQB_I2C0_STR | Write 1 to I2C0SR.I2CSTR |
| 58     | I2C0     | I2C0 stop received Interrupt<br>IRQB_I2C0_spr  | Write 1 to I2C0SR.I2CSPR |

| Vector | Instance | Description                                                       | Clear method              |
|--------|----------|-------------------------------------------------------------------|---------------------------|
| 59     | I2C0     | I2C0 master mode start condition sent Interrupt<br>IRQB_I2C0_msts | Write 1 to I2C0SR.I2CMSTS |
| 60     | I2C0     | I2C0 master mode stop condition sent Interrupt<br>IRQB_I2C0_msp   | Write 1 to I2C0SR.I2CMSPS |
| 61     | I2C0     | I2C0 master mode arbitration lost Interrupt<br>IRQB_I2C0_marb     | Write 1 to I2C0SR.I2CMARB |
| 62     | I2C0     | I2C0 master mode transmit empty Interrupt<br>IRQB_I2C0_mtxe       | Write 1 to I2C0SR.I2CMTXE |
| 63     | I2C0     | I2C0 master mode NACK received Interrupt<br>IRQB_I2C0_mnr         | Write 1 to I2C0SR.I2CMNR  |
| 64     | I2C0     | I2C0 master mode transfer complete Interrupt<br>IRQB_I2C0_mxc     | Write 1 to I2C0SR.I2CMXC  |
| 65     | I2C0     | I2C0 slave address Interrupt<br>IRQB_I2C0_sa                      | Write 1 to I2C0SR.I2CSA   |
| 66     | I2C0     | I2C0 slave transmit empty Interrupt<br>IRQB_I2C0_stxe             | Write 1 to I2C0SR.I2CSTXE |
| 67     | I2C0     | I2C0 slave overflow Interrupt<br>IRQB_I2C0_sovf                   | Write 1 to I2C0SR.I2CSOVF |
| 68     | I2C0     | I2C0 slave mode NACK received Interrupt<br>IRQB_I2C0_snr          | Write 1 to I2C0SR.I2CSNR  |
| 69     | I2C0     | I2C0 slave mode transfer complete Interrupt<br>IRQB_I2C0_sxc      | Write 1 to I2C0SR.I2CSXC  |
| 70     | I2C1     | I2C1 start received Interrupt<br>IRQB_I2C1_STR                    | Write 1 to I2C1SR.I2CSTR  |
| 71     | I2C1     | I2C1 stop received Interrupt<br>IRQB_I2C1_spr                     | Write 1 to I2C1SR.I2CSPR  |
| 72     | I2C1     | I2C1 master mode start condition sent Interrupt<br>IRQB_I2C1_msts | Write 1 to I2C1SR.I2CMSTS |
| 73     | I2C1     | I2C1 master mode stop condition sent Interrupt<br>IRQB_I2C1_msp   | Write 1 to I2C1SR.I2CMSPS |
| 74     | I2C1     | I2C1 master mode arbitration lost Interrupt<br>IRQB_I2C1_marb     | Write 1 to I2C1SR.I2CMARB |
| 75     | I2C1     | I2C1 master mode transmit empty Interrupt<br>IRQB_I2C1_mtxe       | Write 1 to I2C1SR.I2CMTXE |



| Vector | Instance | Description                                                   | Clear method              |
|--------|----------|---------------------------------------------------------------|---------------------------|
| 76     | I2C1     | I2C1 master mode NACK received Interrupt<br>IRQB_I2C1_mnr     | Write 1 to I2C1SR.I2CMNR  |
| 77     | I2C1     | I2C1 master mode transfer complete Interrupt<br>IRQB_I2C1_mxc | Write 1 to I2C1SR.I2CMXC  |
| 78     | I2C1     | I2C1 slave address Interrupt<br>IRQB_I2C1_sa                  | Write 1 to I2C1SR.I2CSA   |
| 79     | I2C1     | I2C1 slave transmit empty Interrupt<br>IRQB_I2C1_stxe         | Write 1 to I2C1SR.I2CSTXE |
| 80     | I2C1     | I2C1 slave overflow Interrupt<br>IRQB_I2C1_sovf               | Write 1 to I2C1SR.I2CSOVF |
| 81     | I2C1     | I2C1 slave mode NACK received Interrupt<br>IRQB_I2C1_snr      | Write 1 to I2C1SR.I2CSNR  |
| 82     | I2C1     | I2C1 slave mode transfer complete Interrupt<br>IRQB_I2C1_sxc  | Write 1 to I2C1SR.I2CSXC  |

## 19.3 Functional description

### 19.3.1 Wire protocol

Both roles use the same two-wire protocol on SDAx and SCLx. Both lines are open drain and require external pull-up resistors: a device drives a bit low by pulling the line down and releases it to let the pull-up present a one. Because every device can only pull down, two devices driving at once cannot damage each other, and that is what makes multi-master arbitration and slave clock stretching possible on the same pair of wires.

A transfer is framed by two conditions that are illegal during data: START is a falling edge on SDAx while SCLx is high, and STOP is a rising edge on SDAx while SCLx is high. Between them SDAx changes only while SCLx is low, so every data bit is stable across the whole SCLx high period and any receiver may sample it there. After the address byte and after each data byte, the transmitter releases SDAx for one clock and the receiver pulls it low for an ACK or leaves it high for a NACK. Figure 19.1 shows one byte write.

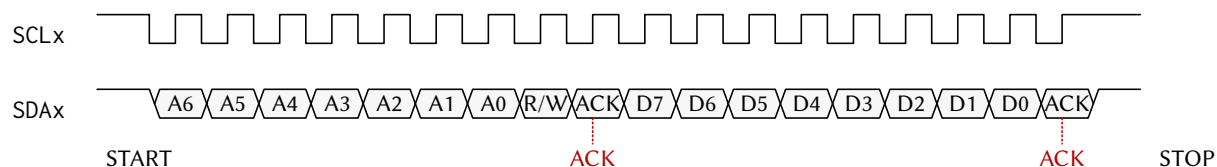


Figure 19.1: One I2C byte write on the wire: START, address, ACK, data byte, ACK, STOP.

The address byte carries the 7-bit slave address in its upper bits and the direction in its least significant bit: 0 selects a write (master transmitter), 1 a read (master receiver). A master that changes direction or address without giving up the bus issues a repeated START instead of a STOP.

### 19.3.2 Master operation

With I2CMEN set, the port drives SCLx at the rate set by I2CMDIV. Setting I2CMST in I2CxFCR sends a START (or a repeated START while the bus is held) followed by the byte in I2CxMTX; each subsequent write to I2CxMTX sends one byte, and each received byte lands in I2CxMRX. I2CMXC marks the end of each byte: in transmitter mode after the slave's ACK/NACK, in receiver mode before this master sends its own ACK/NACK, so software chooses the acknowledge before the next byte. I2CMNR reports a NACK from the slave. Setting I2CMSP sends a STOP. If another master wins the bus during arbitration, the port releases the lines and sets I2CMARB, because continuing to drive would corrupt the winner's transfer.

### 19.3.3 Slave operation

With I2CSEN set, the port compares every address byte against I2CxAR under the mask in I2CxAMR (a masked bit matches either value), and against the general-call address when I2CGCE is set. A match sets I2CSA, with I2CSTM reporting the direction. In slave receiver mode each byte lands in I2CxSRX and sets I2CSXC; a byte arriving before the previous one was read sets I2CSOVF. In slave transmitter mode each byte is taken from I2CxSTX, and I2CSTXE asks for the next one. With I2CSCS set, the port holds SCLx low through the ACK bit until firmware has serviced the flag, which trades bus latency for lossless flow control on a port whose hart may be busy elsewhere.

### 19.3.4 Clocking

The bus engine runs on SMCLK, divided by I2CMDIV to the bus rate; the register interface runs on MCLK, so the registers stay readable with SMCLK stopped. The boot ROM may leave SMCLK on LFXT (32.768 kHz), which would put the bus in the low kilohertz range.

## 19.4 Interrupts and events

Each flag in I2CxSR has its own enable in I2CxCR and its own vector. All thirteen flags are write-1-to-clear. Table 19.4 lists them for I2C0; I2C1, in configurations that include it, uses the same order from vector 70 (70 to 82).

## 19.5 Programming

### 19.5.1 Initialisation as a master

1. Select the SMCLK source in SYSCLKCR (Section 16.3.1).
2. Set the PxSEL bits of the SDAX and SCLx pins, and their PxAFS fields if the bus is relocated.
3. Write I2CxCR.I2CMDIV for the bus rate.
4. Set the interrupt enables required in I2CxCR.
5. Set I2CxCR.I2CMEN.

### 19.5.2 Initialisation as a slave

1. Select the SMCLK source in SYSCLKCR.
2. Set the PxSEL bits of the SDAX and SCLx pins.

| Vector | Condition                               | Set by  | Cleared by           |
|--------|-----------------------------------------|---------|----------------------|
| 57     | START or repeated START seen on the bus | I2CSTR  | Writing 1 to I2CSTR  |
| 58     | STOP seen on the bus                    | I2CSPR  | Writing 1 to I2CSPR  |
| 59     | Master START sent                       | I2CMSTS | Writing 1 to I2CMSTS |
| 60     | Master STOP sent                        | I2CMSPS | Writing 1 to I2CMSPS |
| 61     | Master arbitration lost                 | I2CMARB | Writing 1 to I2CMARB |
| 62     | Master transmit register empty          | I2CMTXE | Writing 1 to I2CMTXE |
| 63     | Master received NACK                    | I2CMNR  | Writing 1 to I2CMNR  |
| 64     | Master byte transfer complete           | I2CMXC  | Writing 1 to I2CMXC  |
| 65     | Slave addressed                         | I2CSA   | Writing 1 to I2CSA   |
| 66     | Slave transmit register empty           | I2CSTXE | Writing 1 to I2CSTXE |
| 67     | Slave receive overflow                  | I2CSOVF | Writing 1 to I2CSOVF |
| 68     | Slave received NACK                     | I2CSNR  | Writing 1 to I2CSNR  |
| 69     | Slave byte transfer complete            | I2CSXC  | Writing 1 to I2CSXC  |

Table 19.4: I2C0 interrupt vectors; I2C1 follows the same order from vector 70.

3. Write the slave address to I2CxAR and the mask to I2CxAMR.
4. Set I2CxCR.I2SCS if clock stretching is wanted, and I2CGCE for general-call response.
5. Set the interrupt enables required in I2CxCR.
6. Set I2CxCR.I2CSEN.

### 19.5.3 Master write of one byte

1. Write the address byte (address and direction 0) to I2CxMTX.
2. Set I2CxFCR.I2CMST.
3. Wait for I2CMXC; if I2CMNR is set, abort with a STOP.
4. Write the data byte to I2CxMTX and wait for I2CMXC.
5. Set I2CxFCR.I2CMSP.

The register descriptions in Section 19 give the remaining sequences (master read, slave transmit, slave receive) field by field.

## 19.6 Usage notes

**Set the SMCLK source before using the bus.** The bus clock is derived from SMCLK, and the boot-time SMCLK source is a boot-ROM implementation detail.

**Choose the acknowledge before the next byte in master receiver mode.** I2CMXC is raised before the master's ACK/NACK, because the last byte of a read must be answered with a NACK to end the transfer cleanly.

**Handle arbitration loss by retrying the whole transfer.** After I2CMARB the port has released the bus mid-transfer; the winner’s transaction must complete before a new START.

**Serialize whole transactions between harts.** The arbiter makes each register access atomic but not a transaction; a hart that shares a port claims a mutex around each transfer (Section 5.4).

**Clear every flag with a 1.** Writing 0 to I2CxSR has no effect; a handler that returns with its flag set is re-entered.

## 19.7 Registers

Register offsets in this chapter are relative to the instance base address in Table 19.1 (I2C0 0x4E00, I2C1 0x4F00); the generated header names each base I2C0\_BASE and so on.

Table 19.5: I2Cx register summary

| Offset | Name     | Access | Reset      | Description                          |
|--------|----------|--------|------------|--------------------------------------|
| 0x00   | I2xCxCR  | rw     | 0x00000000 | I2C control register                 |
| 0x04   | I2xCxFCR | w1     | 0x00       | I2C flow control register            |
| 0x08   | I2CxSR   | r/rw1  | 0x0000     | I2C status register                  |
| 0x0C   | I2CxMTX  | rw     | 0x00       | I2C master transmit register         |
| 0x10   | I2CxMRX  | r      | 0x00       | I2C master receive register          |
| 0x14   | I2CxSTX  | rw     | 0x00       | I2C slave transmit register          |
| 0x18   | I2CxSRX  | r      | 0x00       | I2C slave receive register           |
| 0x1C   | I2CxAR   | rw     | 0x00       | I2C this slave address register      |
| 0x20   | I2CxAMR  | rw     | 0x00       | I2C this slave address mask register |

### 19.7.1 I2CxCR: I2C control register

**Offset** 0x00    **Reset** 0x00000000    **Width** 32    **Address** 0x4E00 (I2C0), 0x4F00 (I2C1)

| Bits  | Field | Type | Reset | Description                                                                                                                                                                                                                                                            |
|-------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:22 | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                      |
| 21    | MEN   | rw   | 0     | Master enable. When set, the device waits for I2CMST, acts as a master from that start condition until the stop condition commanded by I2CMSP, and then resumes slave operation if I2CSEN is also set.<br>0: Disabled<br>1: Enabled                                    |
| 20    | SEN   | rw   | 0     | Slave enable. When set, the device listens for its address as an I2C slave. If master mode is also enabled, the device acts as a slave until I2CMST starts a master transfer and returns to slave operation when that transfer completes.<br>0: Disabled<br>1: Enabled |

| Bits  | Field  | Type | Reset  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|--------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19    | SN     | rw   | 0      | Slave NACK select. When set, the slave answers its address and every received byte with a NACK instead of an ACK. With clock stretching enabled, software can change this bit before releasing SCL to choose the reply for the current byte.<br>0: ACK<br>1: NACK                                                                                                                                                                                                                                                                                                                                                            |
| 18    | SCS    | rw   | 0      | I2C slave clock stretching enable. When enabled, this slave will hold the SCL line low during the ACK phase of the transmission to allow this slave more time. Note that the master will be left waiting for the ACK/NACK as long as this bit is set to 1.<br>0: Disabled<br>1: Enabled                                                                                                                                                                                                                                                                                                                                      |
| 17    | GCE    | rw   | 0      | I2C slave general call enable. When enabled, this slave will be addressed if a global call is issued on the bus.<br>0: Disabled<br>1: Enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 16:13 | MDIV   | rw   | 0b0000 | I2C master mode clock divider. The master mode finite state machine clock source is SMCLK, which is divided by a factor of $4 * 2^{I2CMDIV}$ .<br>0b0000 I2CMDIV_1: /1 (no division)<br>0b0001 I2CMDIV_2: /2<br>0b0010 I2CMDIV_4: /4<br>0b0011 I2CMDIV_8: /8<br>0b0100 I2CMDIV_16: /16<br>0b0101 I2CMDIV_32: /32<br>0b0110 I2CMDIV_64: /64<br>0b0111 I2CMDIV_128: /128<br>0b1000 I2CMDIV_256: /256<br>0b1001 I2CMDIV_512: /512<br>0b1010 I2CMDIV_1024: /1,024<br>0b1011 I2CMDIV_2048: /2,048<br>0b1100 I2CMDIV_4096: /4,096<br>0b1101 I2CMDIV_8192: /8,192<br>0b1110 I2CMDIV_16384: /16,384<br>0b1111 I2CMDIV_32768: /32,768 |
| 12    | SAIE   | rw   | 0      | I2C slave mode addressed interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 11    | STXEIE | rw   | 0      | I2C slave transmit register empty interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 10    | SOVFIE | rw   | 0      | I2C slave receive register overflow interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 9     | SNRIE  | rw   | 0      | I2C slave mode NACK received from master interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Bits | Field  | Type | Reset | Description                                                                                            |
|------|--------|------|-------|--------------------------------------------------------------------------------------------------------|
| 8    | SXCIE  | rw   | 0     | I2C slave mode transfer complete interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled     |
| 7    | MSTSIE | rw   | 0     | I2C master mode start condition sent interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled |
| 6    | MSPSIE | rw   | 0     | I2C master mode stop condition sent interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled  |
| 5    | MARBIE | rw   | 0     | I2C master mode arbitration error interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled    |
| 4    | MTXEIE | rw   | 0     | I2C master transmit register empty interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled   |
| 3    | MNRIE  | rw   | 0     | I2C master mode NACK received interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled        |
| 2    | MXCIE  | rw   | 0     | I2C master transfer complete interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled         |
| 1    | STRIE  | rw   | 0     | I2C start received interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled                   |
| 0    | SPRIE  | rw   | 0     | I2C stop received interrupt enable<br>0: Interrupt disabled<br>1: Interrupt enabled                    |

### 19.7.2 I2CxFCR: I2C flow control register

**Offset** 0x04 **Reset** 0x00 **Width** 8 **Address** 0x4E04 (I2C0), 0x4F04 (I2C1)

I2C flow control register. Writing a 1 to a bit in this register initiates or queues the associated command. Writing a 0 to a bit does nothing. Reading this register always returns the value 0.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                                                                                                                                           |
| 3    | SC    | w1   | 0     | I2C slave continue command. When clock stretching is enabled in slave mode, set this bit to tell the slave to continue with the ACK/NACK phase of the current byte by releasing SCL. This may only be set if clock stretching is enabled, slave mode is enabled, and the slave transfer complete flag has just been set.<br>0: No effect<br>1: Continue: release SCL for the ACK/NACK phase |

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2    | MST   | w1   | 0     | Master send start command. Writing 1 sends a start condition, or a repeated start if this master already controls the bus, after which at least one address frame must be sent before the command is used again. If the bus is busy the start is sent once it becomes idle, and if this master is mid-transaction the repeated start follows the transaction.<br>0: No effect<br>1: Send a start condition |
| 1    | MSP   | w1   | 0     | Master send stop command. Writing 1 sends a stop condition once this master has sent a start condition and at least one address frame. If this master is mid-transaction the stop follows the transaction.<br>0: No effect<br>1: Send a stop condition                                                                                                                                                     |
| 0    | MRB   | w1   | 0     | I2C master read byte command. Set this bit to 1 while in master receiver mode to read one byte from the slave. This master is required to have already sent the slave address and received an ACK from the slave before initiating this command.<br>0: No effect<br>1: Read next byte                                                                                                                      |

### 19.7.3 I2CxSR: I2C status register

Offset 0x08    Reset 0x0000    Width 16    Address 0x4E08 (I2C0), 0x4F08 (I2C1)

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                      |
|------|-------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | BS    | r    | 0     | I2C bus state indicator. This bit cannot be cleared by writing to the status register.<br>0: The I2C bus is idle<br>1: The I2C bus is active                                                                                     |
| 14   | MCB   | r    | 0     | I2C master controls bus indicator. This bit cannot be cleared by writing to the status register.<br>0: This master does not control the bus<br>1: This master controls the bus                                                   |
| 13   | STM   | r    | 0     | Slave transmitter mode indicator: the mode this slave was addressed for, valid only while I2CSA = 1. Read only; a status register write does not clear it.<br>0: Slave receiver mode<br>1: Slave transmitter mode                |
| 12   | SA    | rw1  | 0     | I2C slave mode addressed interrupt flag. Indicates that this slave has been addressed by another master. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt                                   |
| 11   | STXE  | rw1  | 0     | Slave transmit register empty interrupt flag, set when the slave latches the byte held in I2CxSTX for transmission, so the register can take the next byte. Write 1 to clear.<br>0: No pending interrupt<br>1: Pending interrupt |

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                    |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10   | SOVF  | rw1  | 0     | I2C slave receive register overflow interrupt flag. Indicates that this slave has failed to read one or more bytes from the I2CxSRX register before they were overwritten by another transmission. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt                                                       |
| 9    | SNR   | rw1  | 0     | Slave NACK received interrupt flag, set in slave transmitter mode when the master answers a transmitted byte with a NACK; an ACK does not set it. Write 1 to clear.<br>0: No pending interrupt<br>1: Pending interrupt                                                                                                                         |
| 8    | SXC   | rw1  | 0     | I2C slave mode transfer complete interrupt flag. This bit is set after this slave receives a byte of data from a master, but before this slave sends an ACK/NACK. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt                                                                                        |
| 7    | MSTS  | rw1  | 0     | I2C master mode start condition sent interrupt flag. This flag is set after this master sends a start condition or repeated start condition. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt                                                                                                             |
| 6    | MSPS  | rw1  | 0     | I2C master mode stop condition sent interrupt flag. This flag is set after this master sends a stop condition. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt                                                                                                                                           |
| 5    | MARB  | rw1  | 0     | Master arbitration loss interrupt flag, set when the value this master drives on SDA is overridden by another master; this master then releases the bus. Write 1 to clear.<br>0: No pending interrupt<br>1: Pending interrupt                                                                                                                  |
| 4    | MTXE  | rw1  | 0     | I2C master transmit register empty interrupt flag. This bit is set when this master latches the data stored in the master transmit register to indicate that the master transmit register is ready to accept another byte and queue it for transmission. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt |
| 3    | MNR   | rw1  | 0     | Master NACK received interrupt flag, set in master transmitter mode when the slave answers with a NACK; an ACK does not set it. Write 1 to clear.<br>0: No pending interrupt<br>1: Pending interrupt                                                                                                                                           |
| 2    | MXC   | rw1  | 0     | Master transfer complete interrupt flag, set in master transmitter mode after the data byte and the slave ACK/NACK, and in master receiver mode after the data byte and before this master sends its ACK/NACK. Write 1 to clear.<br>0: No pending interrupt<br>1: Pending interrupt                                                            |



| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                         |
|------|-------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | STR   | rw1  | 0     | I2C start condition received interrupt flag. This flag is set whenever a start condition or repeated start condition is detected on the bus, regardless of if this master or another sent it. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt |
| 0    | SPR   | rw1  | 0     | I2C stop condition received interrupt flag. This flag is set whenever a stop condition condition is detected on the bus, regardless of which device sent it. Write a 1 to this bit to clear it.<br>0: No pending interrupt<br>1: Pending interrupt                                  |

#### 19.7.4 I2CxMTX: I2C master transmit register

**Offset** 0x0C **Reset** 0x00 **Width** 8 **Address** 0x4E0C (I2C0), 0x4F0C (I2C1)

I2C master transmit register. Write the desired slave address and read/write bit to this register after sending a start condition to begin a transmission with a slave. Note that the desired slave address must occupy the upper seven bits and the read/write bit must occupy the least significant bit. If the read bit is 0, the master enters master transmitter mode. If the read bit is 1, the master enters master receiver mode. Write a byte of data to this register after sending the address frame or a data frame to send that byte of data to the slave. If this master is busy with a transmission when this register is written, it will send the byte after it finishes the transmission.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                                                                                     |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | xMTX  | rw   | 0x00  | Master transmit data. While I2CMEN = 1 a write loads the byte and starts sending it: the first byte after a start condition is the 7-bit slave address in bits 7:1 with the read/write bit in bit 0, and later bytes are data. Reads return the last byte written; resets to 0. |

#### 19.7.5 I2CxMRX: I2C master receive register

**Offset** 0x10 **Reset** 0x00 **Width** 8 **Address** 0x4E10 (I2C0), 0x4F10 (I2C1)

I2C master receive register. When in master receiver mode, read this register after the master transfer complete interrupt flag (I2CMXC) is set to get the received data byte.

| Bits | Field | Type | Reset | Description                                                                                                                   |
|------|-------|------|-------|-------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | xMRX  | r    | 0x00  | Master receive data. Hardware loads the byte received from the slave when I2CMXC is set in master receiver mode. Resets to 0. |

#### 19.7.6 I2CxSTX: I2C slave transmit register

**Offset** 0x14 **Reset** 0x00 **Width** 8 **Address** 0x4E14 (I2C0), 0x4F14 (I2C1)

I2C slave transmit register. When in slave transmitter mode, write to this register after the slave addressed flag (I2CSA) or the slave transaction complete flag (I2CSXC) has been set to queue the next byte to send to the master.

| Bits | Field | Type | Reset | Description                                                                                                                                             |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | xSTX  | rw   | 0x00  | Slave transmit data. The byte written here is sent to the master during the next slave transmitter data phase, after which I2CSTXE is set. Resets to 0. |

### 19.7.7 I2CxSRX: I2C slave receive register

**Offset** 0x18 **Reset** 0x00 **Width** 8 **Address** 0x4E18 (I2C0), 0x4F18 (I2C1)

I2C slave receive register. When in slave receiver mode, read this register after the slave transaction complete flag (I2CSXC) has been set to get the data byte sent from the master. Note that if this slave fails to clear the status register before another byte is received, the slave receive overflow flag (I2CSOVF) will be set.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                    |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | xSRX  | r    | 0x00  | Slave receive data. Hardware loads the byte received from the master when I2CSXC is set in slave receiver mode; a new byte arriving before the previous one is read sets I2CSOVF. Resets to 0. |

### 19.7.8 I2CxAR: I2C this slave address register

**Offset** 0x1C **Reset** 0x00 **Width** 8 **Address** 0x4E1C (I2C0), 0x4F1C (I2C1)

I2C this slave address register. When in slave mode, any master that sends an address frame containing this address will activate this slave.

| Bits | Field | Type | Reset | Description                                                                                                                                                                                                |
|------|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                          |
| 6:0  | xAR   | rw   | 0x00  | Own 7-bit slave address. A received address frame that matches this value under the I2CxAMR mask addresses this slave. Resets to the default slave address of the instance (0x79 for I2C0, 0x23 for I2C1). |

### 19.7.9 I2CxAMR: I2C this slave address mask register

**Offset** 0x20 **Reset** 0x00 **Width** 8 **Address** 0x4E20 (I2C0), 0x4F20 (I2C1)

I2C this slave address mask register. Any bit set to 1 in this register indicates that the corresponding bit in the slave address register is a wildcard. Only the slave address register bits that correspond to 0s in this register will be compared to the received slave address.

| Bits | Field | Type | Reset | Description                                                                                                                                     |
|------|-------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | -     | -    | -     | Reserved. Reads zero; write zero.                                                                                                               |
| 6:0  | xAMR  | rw   | 0x00  | Slave address mask. A 1 in bit i makes bit i of I2CxAR a wildcard for address matching; a 0 compares the bit. Resets to 0 (every bit compared). |

## 20 Core-local interruptor (CLINT)

### 20.1 Overview

The Core-Local Interruptor (CLINT) provides the two interrupt sources on which multi-core software is built: a per-hart software interrupt, used as the inter-processor interrupt (IPI), and a per-hart timer interrupt driven by one shared free-running 64-bit counter. The block sits in the shared window behind the multi-core arbiter, so any hart can raise, clear or program any hart's interrupts.

- One software-interrupt register per hart (MSIP0 to MSIP4), writable by any hart
- One shared free-running 64-bit counter (MTIMEL/MTIMEH), incrementing once per MCLK cycle
- One 64-bit compare register per hart (MTIMECMP0L/MTIMECMP0H to MTIMECMP4L/MTIMECMP4H)
- Level-sensitive delivery on two hardwired vectors per hart: 83 (software) and 84 (timer)
- Wakes a hart parked by the sleep instruction, so parked harts consume no dynamic power
- Compare registers reset to all ones, so no timer interrupt fires before one is programmed

### 20.2 Instances and signals

Table 20.1: CLINT instances

| Instance | Base address | Interrupt vector(s) | Pins (AF) |
|----------|--------------|---------------------|-----------|
| CLINT    | 0x5000       | 83 to 84            | none      |

Table 20.2: CLINT interrupt vectors

| Vector | Instance | Description                                       | Clear method                                       |
|--------|----------|---------------------------------------------------|----------------------------------------------------|
| 83     | CLINT    | CLINT software interrupt (IPI)<br>IRQB_CLINT_MSIP | Write 0 to the MSIPn register of the hart          |
| 84     | CLINT    | CLINT timer interrupt<br>IRQB_CLINT_MTIP          | Write MTIMECMPn of the hart to a value above mtime |

### 20.3 Functional description

#### 20.3.1 Software interrupts

Writing 1 to MSIPh asserts the software-interrupt level into hart  $h$ ; writing 0 deasserts it. The interrupt is a level, so the handler on hart  $h$  must write 0 to its own MSIPh before returning, because the level would otherwise re-enter the handler immediately.

The software interrupt is the wake mechanism for a parked hart: a hart that has enabled vector 83 and executed sleep resumes when any hart writes 1 to its MSIPh. The woken hart's handler clears the level and executes wake before returning, because a handler that returns without wake puts the hart back to sleep (Section 16.3.3). The boot ROM uses this same path to launch a freshly loaded tile (Section 5.3).

### 20.3.2 Timer interrupts

The counter is one 64-bit value shared by all five harts and clocked by the free-running MCLK. Each hart owns one 64-bit compare value; the timer level into hart  $h$  is asserted while the counter is greater than or equal to hart  $h$ 's compare value, and it is deasserted by moving the compare value into the future or back to all ones (the disarmed reset value).

The registers are 32 bits wide, so every 64-bit value is a low/high pair. The counter cannot be read atomically; Section 20.5 gives the read sequence that tolerates a carry between the two halves.

### 20.3.3 Clocking

The CLINT runs on MCLK and is never clock-gated, because a hart whose own clock is stopped by sleep can only be woken by logic that keeps running. The counter therefore advances through every sleep state, and its rate follows the MCLK divider in CLKDIVCR.

### 20.3.4 Address decoding

The block occupies shared-window page 1 from  $0x5000$ . MSIPh is at  $0x5000 + 4h$ , the counter pair starts at  $0x5020$ , and hart  $h$ 's compare pair is at  $0x5030 + 8h$ . Only the low address bits are decoded, so the registers alias every 128 bytes up to  $0x5FFF$ ; software uses the canonical addresses because the alias pattern is not part of the register interface and may change with the hart count.

## 20.4 Interrupts and events

Both vectors are hardwired into every hart's interrupt controller and do not pass through the IRQROUTER: each hart receives only its own MSIPh and its own compare match. Table 20.3 lists them.

| Vector | Condition                      | Set by                                     | Cleared by                                |
|--------|--------------------------------|--------------------------------------------|-------------------------------------------|
| 83     | Software interrupt to hart $h$ | Any hart writing 1 to MSIPh                | Writing 0 to MSIPh                        |
| 84     | Timer match for hart $h$       | Counter reaching hart $h$ 's compare value | Writing a compare value above the counter |

Table 20.3: CLINT interrupt vectors.

## 20.5 Programming

### 20.5.1 Reading the counter

1. Read MTIMEH.
2. Read MTIMEL.
3. Read MTIMEH again; if it differs from the first read, repeat from step 1, because the low half wrapped between the two reads.

### 20.5.2 Arming a timer interrupt for hart $h$

1. Write all ones to MTIMECMPnH, so that no intermediate compare value can be below the counter.
2. Write the low half of the target time to MTIMECMPnL.
3. Write the high half of the target time to MTIMECMPnH.
4. Enable vector 84 in hart  $h$ 's interrupt controller.

### 20.5.3 Sending a software interrupt to hart $h$

1. Write 1 to MSIPh.

## 20.6 Usage notes

**Clear the software-interrupt level before returning.** The level stays asserted until MSIPh is written 0; a handler that returns first is re-entered at once.

**Execute wake in the handler of a sleeping hart.** A hart woken by an interrupt returns to sleep after the handler unless the handler executed wake (Section 16.3.3).

**Program the compare pair high half first.** Writing the low half first can momentarily produce a compare value below the counter and raise a spurious timer interrupt.

**Do not send a software interrupt to a hart whose loader mailbox row is incomplete.** The boot ROM consumes the row when the parked hart wakes, so the row must be written before MSIPh (Section 5.3).

## 20.7 Registers

Register offsets in this chapter are relative to the instance base address in Table 20.1 (CLINT 0x5000); the generated header names each base CLINT\_BASE.

Table 20.4: CLINT register summary

| Offset       | Name                  | Access | Reset      | Description                                  |
|--------------|-----------------------|--------|------------|----------------------------------------------|
| 0x00 + 4n    | MSIPn (n = 0..4)      | rw     | 0x00000000 | Hart n software interrupt (IPI) register     |
| 0x14 to 0x1F | -                     | -      | -          | Reserved                                     |
| 0x20         | MTIMEL                | rw     | 0x00000000 | Machine time counter, lower 32 bits          |
| 0x24         | MTIMEH                | rw     | 0x00000000 | Machine time counter, upper 32 bits          |
| 0x28 to 0x2F | -                     | -      | -          | Reserved                                     |
| 0x30 + 8n    | MTIMECMPnL (n = 0..4) | rw     | 0xFFFFFFFF | Hart n timer compare register, lower 32 bits |
| 0x34 + 8n    | MTIMECMPnH (n = 0..4) | rw     | 0xFFFFFFFF | Hart n timer compare register, upper 32 bits |

### 20.7.1 MSIPn (n = 0..4): Hart n software interrupt (IPI) register

**Offset** 0x00 + 4n **Reset** 0x00000000 **Width** 32

Hart n software interrupt (IPI) register. Any hart may write it through the shared window: writing 1 raises the software interrupt level into hart n (interrupt vector 83); writing 0 clears it. The interrupt is level-sensitive, so hart n's service routine must write 0 here before returning.

| Bits | Field | Type | Reset | Description                                                                                                    |
|------|-------|------|-------|----------------------------------------------------------------------------------------------------------------|
| 31:1 | -     | -    | -     | Reserved. Reads zero; write zero.                                                                              |
| 0    | MSIPn | rw   | 0     | Software interrupt (IPI) level for hart n.<br>0: No software interrupt pending<br>1: Software interrupt raised |

### 20.7.2 MTIMEL: Machine time counter, lower 32 bits

**Offset** 0x20 **Reset** 0x00000000 **Width** 32 **Address** 0x5020

Machine time counter, lower 32 bits. mtime is a free-running 64-bit counter shared by all five harts that increments once per MCLK cycle. It is writable for initialization; a write merges only the addressed 32-bit half.

| Bits | Field  | Type | Reset      | Description      |
|------|--------|------|------------|------------------|
| 31:0 | MTIMEL | rw   | 0x00000000 | mtime bits 31:0. |

### 20.7.3 MTIMEH: Machine time counter, upper 32 bits

**Offset** 0x24 **Reset** 0x00000000 **Width** 32 **Address** 0x5024

Machine time counter, upper 32 bits. Read MTIMEH, then MTIMEL, then MTIMEH again (retry if the two MTIMEH reads differ) to obtain a coherent 64-bit time value.

| Bits | Field  | Type | Reset      | Description       |
|------|--------|------|------------|-------------------|
| 31:0 | MTIMEH | rw   | 0x00000000 | mtime bits 63:32. |

### 20.7.4 MTIMECMPnL (n = 0..4): Hart n timer compare register, lower 32 bits

**Offset** 0x30 + 8n **Reset** 0xFFFFFFFF **Width** 32

Hart n timer compare register, lower 32 bits. The timer interrupt level for hart n (interrupt vector 84) is raised while mtime >= mtimecmpn. Resets to all-ones (no interrupt). To set a new compare value without a spurious interrupt, first write all-ones to MTIMECMPnH, then the new lower half, then the real upper half.

| Bits | Field      | Type | Reset      | Description          |
|------|------------|------|------------|----------------------|
| 31:0 | MTIMECMPnL | rw   | 0xFFFFFFFF | mtimecmpn bits 31:0. |

### 20.7.5 MTIMECMPnH (n = 0..4): Hart n timer compare register, upper 32 bits

**Offset** 0x34 + 8n **Reset** 0xFFFFFFFF **Width** 32

| Bits | Field      | Type | Reset      | Description           |
|------|------------|------|------------|-----------------------|
| 31:0 | MTIMECMPnH | rw   | 0xFFFFFFFF | mtimecmpn bits 63:32. |

## 21 Hardware mutexes (MUTEX)

### 21.1 Overview

The hardware mutex bank (MUTEX) provides sixteen word-mapped advisory locks with single-instruction cross-hart mutual exclusion. A claim is one load and a release is one store; there is no reservation state and no retry loop in hardware. Atomicity comes from the multi-core arbiter, which serializes whole shared-window transactions, so two harts reading the same free mutex in the same cycle are still served one after the other.

- 16 independent mutexes, one word each (MUTEX0 to MUTEX15); the readback is a 4-bit owner marker in bits 3:0 and bits 31:4 read 0
- Claim with one load: reading a free mutex returns 0 and claims it in the same transaction
- Reading a held mutex returns the owner's marker ( $\text{mhartid} + 1$ ) and leaves it held
- Release with one store of 0; any hart may release, so a supervisor can recover a dead hart's lock
- Nonzero writes are ignored, so ownership cannot be forged
- All mutexes reset to free

### 21.2 Instances and signals

Table 21.1: MUTEX instances

| Instance | Base address | Interrupt vector(s) | Pins (AF) |
|----------|--------------|---------------------|-----------|
| MUTEX    | 0x6000       | none                | none      |

### 21.3 Functional description

#### 21.3.1 Claim and release

Each mutex word holds either 0 (free) or the marker of the hart that holds it,  $\text{mhartid} + 1$ . The marker is four bits wide (the arbiter's granted-master width plus one), so it occupies bits 3:0 and bits 31:4 always read 0; the release comparison is against the full 32-bit written word, so only a store of exactly 0 frees a mutex. A read of a free mutex returns 0 and, in the same arbiter transaction, writes the reading hart's marker into the word; the arbiter attributes the read to the requesting hart, which is how the bank knows whose marker to store. A read of a held mutex returns the holder's marker and changes nothing, so a hart that reads its own mutex twice sees its own marker and is not disturbed. Figure 21.1 shows two harts racing for one mutex.

A store of 0 frees the mutex. The store is deliberately not qualified by owner, so that a supervising hart can release the mutex of a hart that has died while holding it; correct software releases only mutexes it holds. A store of any nonzero value is ignored, because the only legitimate nonzero content is a marker the hardware writes itself.

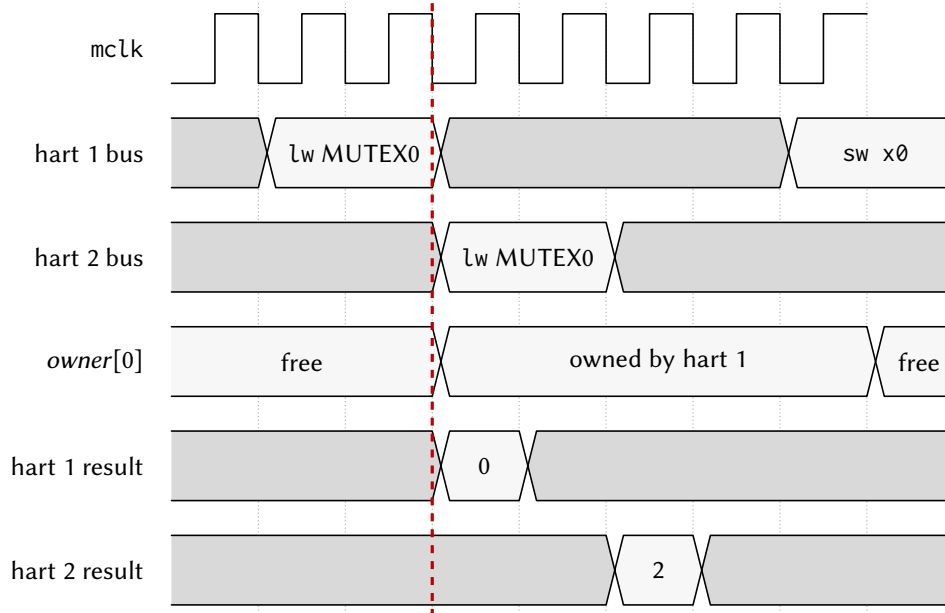


Figure 21.1: Two harts racing for the same mutex: the first read returns 0 and takes the word, the second returns the holder's marker (`mhartid + 1`) and must retry, and the holder releases with `sw x0`.

### 21.3.2 Address decoding

The bank occupies shared-window page 2 from `0x6000`, with mutex  $n$  at `0x6000 + 4n`. Only the low address bits are decoded, so the words alias throughout the page; software uses the canonical addresses.

## 21.4 Interrupts and events

Not applicable. The bank raises no interrupt; a hart that fails a claim retries under software control.

## 21.5 Programming

### 21.5.1 Claiming mutex $n$

1. Load `MUTEXn` with a plain `lw`.
2. If the value is 0, the mutex is held by this hart; otherwise the value is the holder's marker and the claim failed.
3. On failure, delay by an amount proportional to `mhartid + 1` and repeat from step 1, with a bounded retry count.

### 21.5.2 Releasing mutex $n$

1. Store 0 to `MUTEXn` with a plain `sw`.



## 21.6 Usage notes

**Treat the locks as advisory.** The hardware does not block a hart that accesses a protected resource without claiming its mutex; the protocol is enforced by software only.

**Never use `lr.w`, `sc.w` or an AMO on a mutex address.** A suppressed store-conditional reaches the bank as an ordinary read and claims the mutex, and an AMO performs a read and a write whose values the bank does not expect. Only `lw` and `sw` are legal (Section 5.4).

**Back off between retries.** five harts re-reading a contested mutex in lockstep waste arbiter bandwidth and can starve useful work; scale the delay with the hart index and bound the retry count.

**Release only what this hart holds.** The bank does not check the releasing hart; a stray store of 0 frees another hart's lock silently.

## 21.7 Registers

Register offsets in this chapter are relative to the instance base address in Table 21.1 (MUTEX 0x6000); the generated header names each base MUTEX\_BASE.

Table 21.2: MUTEX register summary

| Offset    | Name               | Access | Reset      | Description      |
|-----------|--------------------|--------|------------|------------------|
| 0x00 + 4n | MUTEXn (n = 0..15) | rw     | 0x00000000 | Hardware mutex n |

### 21.7.1 MUTEXn (n = 0..15): Hardware mutex n

**Offset** 0x00 + 4n **Reset** 0x00000000 **Width** 32

Hardware mutex n. Read to claim: a returned value of 0 means the mutex was free and the reading hart now holds it; a nonzero value is the current owner's marker (hartid+1). Write 0 to release.

| Bits | Field | Type | Reset  | Description                                                                                                                                                                                                                                                                                                                                                           |
|------|-------|------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:4 | -     | -    | -      | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                                                                                                                     |
| 3:0  | OWNn  | rw   | 0b0000 | Owner marker: 0 = free (and a read that returns 0 claims the mutex), h+1 = held by hart h.<br>0b0000 MTXOWNn_FREE: Mutex free (a read returning this value claims the mutex)<br>0b0001 MTXOWNn_H0: Held by hart 0<br>0b0010 MTXOWNn_H1: Held by hart 1<br>0b0011 MTXOWNn_H2: Held by hart 2<br>0b0100 MTXOWNn_H3: Held by hart 3<br>0b0101 MTXOWNn_H4: Held by hart 4 |

## 22 Near-field communication interface (NFCx)

### 22.1 Overview

The NFC controller (NFCx) is an ISO/IEC 14443 Type A tag emulation engine. It presents a passive contactless tag to an external reader: it decodes the reader-to-tag frames, runs the tag transaction state machine, and load-modulates the tag's response. The 13.56 MHz analog front end is off-die; the block is digital only and connects to an external front end through a small demodulated interface.

- ISO 14443A tag emulation: REQA/WUPA to ATQA, bit-frame anticollision with a 4-byte UID to SAK, then Type-2 READ answered from a firmware-filled payload window (NFC\_AUTOREAD)
- Miller decoding of reader frames; Manchester subcarrier ( $f_c/16$ ) load modulation of tag responses
- Byte framing with odd parity and CRC\_A (reflected 0x8408, initial value 0x6363)
- Provisioned 4-byte UID (NFCxUID), programmable ATQA and SAK (NFCxCFG), hardware BCC
- Two 64-byte indexed windows behind one index/data pair (NFCxIDX, NFCxDATA): the payload window and the received-frame buffer
- Programmable protocol timing (NFCxTIM): bit period, subcarrier divisor, frame delay time
- Registered read path: no register read has a side effect
- Four interrupt sources, vectors 94 to 97

### 22.2 Instances and signals

Table 22.1: NFCx instances

| Instance | Base address | Interrupt vector(s) | Pins (AF)                                                                                                                                         |
|----------|--------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| NFC0     | 0x6200       | 94 to 97            | NFC_RF_CLK P5.0 (AF1), NFC_RF_RX P5.1 (AF1), NFC_FIELD_DETECT P5.2 (AF1), NFC_RF_TXMOD P5.3 (AF1), NFC_RF_TX_EN P5.4 (AF1), NFC_AFE_EN P5.5 (AF1) |

Table 22.2: NFCx signals

| Signal           | Direction | Description                     | Pin (AF)   |
|------------------|-----------|---------------------------------|------------|
| NFC_RF_CLK       | Input     | NFC0 off-die RF carrier clock   | P5.0 (AF1) |
| NFC_RF_RX        | Input     | NFC0 off-die RX Miller envelope | P5.1 (AF1) |
| NFC_FIELD_DETECT | Input     | NFC0 off-die RF field detect    | P5.2 (AF1) |
| NFC_RF_TXMOD     | Output    | NFC0 off-die TX load modulation | P5.3 (AF1) |
| NFC_RF_TX_EN     | Output    | NFC0 off-die TX enable          | P5.4 (AF1) |
| NFC_AFE_EN       | Output    | NFC0 off-die AFE enable         | P5.5 (AF1) |

Table 22.3: NFCx interrupt vectors

| Vector | Instance | Description                                                   | Clear method                                   |
|--------|----------|---------------------------------------------------------------|------------------------------------------------|
| 94     | NFC0     | NFC0 RF Field-Detect Interrupt<br>IRQB_NFC0_FIELD             | Write 1 to NFC0SR.NFCFIELDF                    |
| 95     | NFC0     | NFC0 Reader-Frame Received Interrupt<br>IRQB_NFC0_RXF         | Write 1 to NFC0SR.NFCRXFRAMEF                  |
| 96     | NFC0     | NFC0 Tag-Response Transmit-Done Interrupt<br>IRQB_NFC0_TXDONE | Write 1 to NFC0SR.NFCTXDONEF                   |
| 97     | NFC0     | NFC0 RX CRC / Parity Error Interrupt<br>IRQB_NFC0_CRCERR      | Write 1 to NFC0SR.NFCCRCERRF and<br>NFCPARERRF |

## 22.3 Functional description

### 22.3.1 Analog front-end boundary

The block never touches the carrier. Its interface to the external front end is a demodulated receive envelope (1 = field present, 0 = a reader pause), an asynchronous field-present level, the carrier-derived clock `rf_clk` that clocks the protocol core, a load-modulation drive output, a transmit-active window, and a listen-power enable. These wires are alternate functions on port 6 (Section 3).

The block spans three clock domains: the bus clock for the register interface, a free-running SMCLK reference that hosts the synchronizers and the write-1-to-clear retirement, and `rf_clk`. Quasi-static configuration (UID, ATQA, SAK, timing) is latched per transaction so that a multi-byte value is never sampled while software is updating it, and received-frame bytes cross by a flag-gated handshake: the buffer is written before `NFCRXFRAMEF` is raised, so a handler that sees the flag can read the frame.

### 22.3.2 Transaction engine

With `NFCEN` set the protocol core runs; with `NFCLISTEN` also set it answers readers. The engine walks the 14443A sequence on its own: `REQA` or `WUPA` is answered with the `ATQA` from `NFCxCFG`, the anticollision cascade with the `UID` from `NFCxUID` and its hardware-derived `BCC`, and `SELECT` with the `SAK`. With `NFCAUTOREAD` set (its reset value), a Type-2 `READ` is answered directly from the payload window without firmware. Any other command, or every command when `NFCAUTOREAD` is clear, is surfaced to firmware: `NFCRXFRAMEF` is set, `NFCxRXST` reports the command byte, length and CRC/parity result, and the raw frame is readable through the frame buffer. Firmware answers with `NFCxTXCTL`: it fills the payload window, writes the length to `NFCTXLEN`, selects CRC appending with `NFCTXAPPCRC`, and sets `NFCTXG0`.

### 22.3.3 Indexed windows

Both 64-byte windows are reached through one index/data pair: `NFCIDXSEL` in `NFCxIDX` selects the payload window (0) or the received-frame buffer (1), `NFCIDX` selects the byte, and each access to `NFCxDATA` reads or writes it. With `NFCIDXAINC` set the index advances after every access, so a record is streamed with consecutive data accesses.

### 22.3.4 Protocol timing

`NFCxTIM` carries the bit period (`NFCETU`), the subcarrier half period (`NFCSUBCDIV`) and the frame delay time (`NFCFDT`), all counted in `rf_clk` ticks. The reset values follow the 13.56 MHz grid; a simulation bench

writes them small to compress protocol time.

## 22.4 Interrupts and events

The four flags in NFCxSR are write-1-to-clear and each has an enable in NFCxCR. Table 22.4 lists them. The field-detect level is also available to the boot gate in PWRCTRL as a wake source, which does not use a vector.

| Vector | Condition                          | Set by                 | Cleared by               |
|--------|------------------------------------|------------------------|--------------------------|
| 94     | RF field detected                  | NFCFIELD               | Writing 1 to NFCFIELD    |
| 95     | Reader frame received for firmware | NFCRXFRAMEF            | Writing 1 to NFCRXFRAMEF |
| 96     | Tag response transmit done         | NFCTXDONEF             | Writing 1 to NFCTXDONEF  |
| 97     | Receive CRC or parity error        | NFCCRCERRF, NFCPARERRF | Writing 1 to the flag    |

Table 22.4: NFC0 interrupt vectors.

## 22.5 Programming

### 22.5.1 Provisioning and arming the tag

1. Write SYSCLKCR so that SMCLK runs from HFXT (Section 16.3.1).
2. Set the PxSEL bits and PxAFS fields of the six front-end pins on port 6.
3. Write the UID to NFCxUID.
4. Write NFCxCFG.NFCATQA and NFCxCFG.NFCSAK.
5. Write NFCxIDX with NFCIDXSEL = 0, NFCIDX = 0 and NFCIDXAINC = 1.
6. Write each payload byte to NFCxDATA.
7. Set the interrupt enables required in NFCxCR.
8. Set NFCxCR.NFCEN and NFCxCR.NFCLISTEN.

### 22.5.2 Servicing a frame delivered to firmware

1. Read NFCxRXST for the command, length and CRC/parity result.
2. Write NFCxIDX with NFCIDXSEL = 1 and NFCIDXAINC = 1, then read the frame bytes from NFCxDATA.
3. Fill the payload window with the response.
4. Write NFCxTXCTL with NFCTXLEN, NFCTXAPPCRC and NFCTXGO.
5. Write 1 to NFCxSR.NFCRXFRAMEF.

## 22.6 Usage notes

**Keep SMCLK on HFXT while the tag is armed.** The synchronizers and flag retirement run on SMCLK; a stopped or slow SMCLK stalls them.

**Use a byte store to arm LISTEN when the payload is auto-read.** A full-word write of NFCxCR also writes the byte lane holding NFCAUTOREAD; writing 0 there disables the hardware answer to READ.

**Provision before enabling.** The UID, ATQA, SAK and timing are latched per transaction, so a value written while a reader is mid-transaction takes effect at the next one.

**Do not read the block through an alias when it is absent.** In a configuration without the NFC block its address range aliases the MUTEX page, and a read there claims a mutex.

## 22.7 Registers

Register offsets in this chapter are relative to the instance base address in Table 22.1 (NFC0 0x6200); the generated header names each base NFC0\_BASE.

Table 22.5: NFCx register summary

| Offset | Name      | Access | Reset      | Description                                                             |
|--------|-----------|--------|------------|-------------------------------------------------------------------------|
| 0x00   | NFCxCR    | rw/w1  | 0x00001000 | NFC control register                                                    |
| 0x04   | NFCxSR    | r/rw1  | 0x00000000 | NFC status register                                                     |
| 0x08   | NFCxUID   | rw     | 0x00000000 | Provisioned single-size 4-byte tag UID, used by bit-frame anticollision |
| 0x0C   | NFCxCFG   | rw     | 0x00000044 |                                                                         |
| 0x10   | NFCxTIM   | rw     | 0x088004D4 |                                                                         |
| 0x14   | NFCxRXST  | r      | 0x00000000 |                                                                         |
| 0x18   | NFCxIDX   | rw     | 0x00000000 |                                                                         |
| 0x1C   | NFCxDATA  | rw     | 0x00000000 | The byte at NFCIDX in the window selected by NFCIDXSEL                  |
| 0x20   | NFCxTXCTL | rw     | 0x00000000 |                                                                         |
| 0x24   | NFCxDBG   | r      | 0x00000000 | Debug telemetry / bench cross-checks: frame counters                    |

### 22.7.1 NFCxCR: NFC control register

**Offset** 0x00 **Reset** 0x00001000 **Width** 32 **Address** 0x6200

NFC control register. Enables the protocol core, arms the tag to respond, selects the auto-answer path, and holds the interrupt enables and the carrier-division note. Resets to 0 except NFCAUTOREAD, which resets to 1. Program the identity and timing registers before setting NFCEN.

| Bits  | Field | Type | Reset  | Description                                                                                                                                  |
|-------|-------|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 31:20 | -     | -    | -      | Reserved. Reads zero; write zero.                                                                                                            |
| 19:16 | RFDIV | rw   | 0b0000 | Carrier-division note. Documents the assumed division from the 13.56 MHz carrier to rf_clk (informational; the off-die AFE supplies rf_clk). |
| 15:13 | -     | -    | -      | Reserved. Reads zero; write zero.                                                                                                            |

| Bits | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                             |
|------|----------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12   | AUTOREAD | rw   | 1     | Auto-answer READ path (resets to 1). When set, hardware answers a Type-2 READ command directly from the payload window without firmware intervention. When clear, standard frames are surfaced to firmware through NFCxRXST and the RX buffer for a firmware-composed response.<br>0: Firmware-handled reads<br>1: Hardware auto-answer |
| 11   | CRCIE    | rw   | 0     | CRC / parity-error interrupt enable. When set, an RX CRC or parity error (NFCCRCERRF or NFCPARERRF) drives interrupt vector 97.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                        |
| 10   | TXIE     | rw   | 0     | Transmit-done interrupt enable. When set, the NFCTXDONEF flag drives interrupt vector 96.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                                                              |
| 9    | RXFIE    | rw   | 0     | Frame-received interrupt enable. When set, the NFCRXFRAMEF flag drives interrupt vector 95.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                                                            |
| 8    | FIELDIE  | rw   | 0     | Field-detect interrupt enable. When set, the NFCFIELDF flag drives interrupt vector 94.<br>0: Interrupt disabled<br>1: Interrupt enabled                                                                                                                                                                                                |
| 7:3  | -        | -    | -     | Reserved. Reads zero; write zero.                                                                                                                                                                                                                                                                                                       |
| 2    | HALTCLR  | w1   | 0     | Halt-clear pulse. Writing 1 forces the transaction FSM from the HALT state back to IDLE (a self-clearing, write-1-style event pulse); it reads 0.<br>0: No action<br>1: Force HALT to IDLE                                                                                                                                              |
| 1    | LISTEN   | rw   | 0     | Listen arm. When set, the tag responds to reader commands once a field is present; when clear, the tag stays deaf even in a field.<br>0: Deaf (no response)<br>1: Armed to respond                                                                                                                                                      |
| 0    | EN       | rw   | 0     | NFC enable. When 0 the rf_clk protocol core is held in reset; it does NOT wipe the UID, CFG, payload window, or status flags (disable-preserves-data rule). Set to 1 to run the tag engine.<br>0: Disabled (protocol core in reset)<br>1: Enabled                                                                                       |

### 22.7.2 NFCxSR: NFC status register

**Offset** 0x04 **Reset** 0x00000000 **Width** 32 **Address** 0x6204

NFC status register. NFCBUSY, NFCFIELDLIVE, NFCHALTED, and NFCSTATE are read-only live status; the five event flags (bits 1-5) are write-1-to-clear (write a 1 to a bit to clear it; writing 0 leaves it unchanged). The volatile bits are captured by the registered pre-latch read.

| Bits  | Field     | Type | Reset  | Description                                                                                                                                                                                                                          |
|-------|-----------|------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | -         | -    | -      | Reserved. Reads zero; write zero.                                                                                                                                                                                                    |
| 11:8  | STATE     | r    | 0b0000 | Transaction FSM state: 0 = POWER_OFF, 1 = IDLE, 2 = READY, 3 = ACTIVE, 4 = HALT.                                                                                                                                                     |
| 7     | HALTED    | r    | 0      | Halted. Reads 1 while the transaction FSM is in the HALT state (the tag has been HLTA / halted by the reader).<br>0: Not halted<br>1: FSM halted                                                                                     |
| 6     | FIELDLIVE | r    | 0      | Field-live level. The synchronized RF field-present level (1 while a reader field is on the tag).<br>0: No field<br>1: Field present                                                                                                 |
| 5     | PARERRF   | rw1  | 0      | RX parity-error flag. Set when a received frame fails an odd-parity check; drives vector 97 (folded with NFCCRCERRF) when NFCCRCIE is set. Write 1 to clear.<br>0: No event<br>1: RX parity error                                    |
| 4     | CRCERRF   | rw1  | 0      | RX CRC-error flag. Set when a received frame fails the CRC_A check; drives vector 97 when NFCCRCIE is set. Write 1 to clear.<br>0: No event<br>1: RX CRC error                                                                       |
| 3     | TXDONEF   | rw1  | 0      | Transmit-done flag. Set when the tag response end-of-frame has been sent; drives vector 96 when NFCTXIE is set. Write 1 to clear.<br>0: No event<br>1: Tag response sent                                                             |
| 2     | RXFRAMEF  | rw1  | 0      | Reader-frame-received flag. Set when a reader frame has landed for firmware (its CRC / parity result is summarized in NFCxRXST); drives vector 95 when NFCRXFIE is set. Write 1 to clear.<br>0: No event<br>1: Reader frame received |
| 1     | FIELDF    | rw1  | 0      | Field-detect flag. Set on a synchronized rising edge of the RF field-present input; drives vector 94 when NFCFIELDIE is set. Write 1 to clear.<br>0: No event<br>1: RF field detected                                                |
| 0     | BUSY      | r    | 0      | Busy. Reads 1 while a reader frame is being received or a tag response is in flight.<br>0: Idle<br>1: RX or TX in progress                                                                                                           |

### 22.7.3 NFCxUID: Provisioned single-size 4-byte tag UID, used by bit-frame anticollision

**Offset** 0x08 **Reset** 0x00000000 **Width** 32 **Address** 0x6208

Provisioned single-size 4-byte tag UID, used by bit-frame anticollision. Byte order on air is UID byte 0 first: UID[7:0] is the first byte, UID[31:24] the last. Hardware derives the BCC check byte. Latched transaction-

locally at the start of each transaction.

| Bits | Field | Type | Reset      | Description                                  |
|------|-------|------|------------|----------------------------------------------|
| 31:0 | UID   | rw   | 0x00000000 | 4-byte UID (UID0 in bits 7:0, first on air). |

#### 22.7.4 NFCxCFG

**Offset** 0x0C **Reset** 0x00000044 **Width** 32 **Address** 0x620C

Tag identity response configuration: the ATQA answer-to-request word and the SAK select-acknowledge byte. Latched transaction-locally.

| Bits  | Field | Type | Reset  | Description                                                                            |
|-------|-------|------|--------|----------------------------------------------------------------------------------------|
| 31:24 | -     | -    | -      | Reserved. Reads zero; write zero.                                                      |
| 23:16 | SAK   | rw   | 0x00   | Select Acknowledge byte (resets to 0x00) returned after the final anticollision level. |
| 15:0  | ATQA  | rw   | 0x0044 | Answer To Request, Type A (resets to 0x0044). Bits 7:0 are the first byte on air.      |

#### 22.7.5 NFCxTIM

**Offset** 0x10 **Reset** 0x088004D4 **Width** 32 **Address** 0x6210

Protocol timing divisors, all in rf\_clk ticks, latched transaction-locally so a mid-count reload never glitches. The reset values are the real 13.56 MHz grid; a bench compresses all three for fast simulation.

| Bits  | Field   | Type | Reset  | Description                                                                                                                                                                   |
|-------|---------|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | SUBCDIV | rw   | 0x08   | Subcarrier half-period, in rf_clk ticks (resets to 8, giving the fc/16 load-modulation subcarrier).                                                                           |
| 23:16 | ETU     | rw   | 0x80   | Elementary Time Unit (bit period), in rf_clk ticks (resets to 128).                                                                                                           |
| 15:0  | FDT     | rw   | 0x04D4 | Frame Delay Time, in rf_clk ticks (resets to approximately 1236, the ISO 14443A tag response grid). The composed response is released when the FDT down-counter reaches zero. |

#### 22.7.6 NFCxRXST

**Offset** 0x14 **Reset** 0x00000000 **Width** 32 **Address** 0x6214

Received-frame inspection, for firmware-handled frames (NFC\_AUTOREAD = 0 or an unrecognized command). Read side-effect-free; the frame byte payload is read separately through the indexed RX buffer (NFCxIDX / NFCxDATA with NFCIDXSEL = 1). Pre-latched.

| Bits  | Field   | Type | Reset | Description                                                       |
|-------|---------|------|-------|-------------------------------------------------------------------|
| 31:18 | -       | -    | -     | Reserved. Reads zero; write zero.                                 |
| 17    | RXPOROK | r    | 0     | Parity OK for the last received frame.0: Parity bad1: Parity good |
| 16    | RXCRCOK | r    | 0     | CRC OK for the last received frame.0: CRC bad1: CRC good          |
| 15:8  | RXLEN   | r    | 0x00  | Received length in bytes, including the CRC.                      |
| 7:0   | CMD     | r    | 0x00  | Command byte: the first byte of the last received standard frame. |



### 22.7.7 NFCxIDX

**Offset** 0x18 **Reset** 0x00000000 **Width** 32 **Address** 0x6218

Byte-index pointer into one of the two 64-byte windows accessed through NFCxDATA. The index persists across accesses; NFCIDXSEL selects which window. Mirrors the indexed-window idiom used by I3C's Device Address Table.

| Bits | Field   | Type | Reset | Description                                                                                                                                  |
|------|---------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 31:9 | -       | -    | -     | Reserved. Reads zero; write zero.                                                                                                            |
| 8    | IDXSEL  | rw   | 0     | Window select for NFCxDATA.<br>0: Payload TX window (64 B, firmware-filled, HW-read)<br>1: RX frame buffer (64 B, HW-filled, firmware-read)  |
| 7    | -       | -    | -     | Reserved. Reads zero; write zero.                                                                                                            |
| 6    | IDXAINC | rw   | 0     | Auto-increment. When set, NFCIDX increments after each NFCx-DATA access (streaming).<br>0: Index held<br>1: Auto-increment after each access |
| 5:0  | IDX     | rw   | 0x00  | Byte index (0 to 63) into the selected window. Hardware advances it by one modulo 64 after every NFCxDATA access while NFCIDX-AINC = 1.      |

### 22.7.8 NFCxDATA: The byte at NFCIDX in the window selected by NFCIDXSEL

**Offset** 0x1C **Reset** 0x00000000 **Width** 32 **Address** 0x621C

The byte at NFCIDX in the window selected by NFCIDXSEL. The payload TX window is read/write (firmware fills the record the reader collects; hardware reads it for the auto-answer READ); the RX frame buffer is read-only (hardware fills it from the reader frame; writes are ignored). Auto-increments per NFCIDXAINC. Pre-latched on read (no side effects).

| Bits | Field | Type | Reset | Description                            |
|------|-------|------|-------|----------------------------------------|
| 31:8 | -     | -    | -     | Reserved. Reads zero; write zero.      |
| 7:0  | DATA  | rw   | 0x00  | Window data byte at the current index. |

### 22.7.9 NFCxTXCTL

**Offset** 0x20 **Reset** 0x00000000 **Width** 32 **Address** 0x6220

Firmware-composed response control (used when NFCAUTOREAD = 0 or for a vendor command). INERT in this MVP (auto-answer is the default path); present in the frozen register map so the firmware-WRITE stage bolts on without a map change.

| Bits  | Field    | Type | Reset | Description                                                                              |
|-------|----------|------|-------|------------------------------------------------------------------------------------------|
| 31:10 | -        | -    | -     | Reserved. Reads zero; write zero.                                                        |
| 9     | TXAPPCRC | rw   | 0     | Append CRC_A to the firmware-composed response.<br>0: No CRC appended<br>1: Append CRC_A |

| Bits | Field | Type | Reset | Description                                                                                                                                                                            |
|------|-------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8    | TXGO  | rw   | 0     | Launch a firmware-composed response (a lane-0 write is a held-level launch, suppressed unless the FSM is in ACTIVE and awaiting a firmware reply).<br>0: No launch<br>1: Send response |
| 7:0  | TXLEN | rw   | 0x00  | Number of payload bytes to send from the payload TX window.                                                                                                                            |

### 22.7.10 NFCxDBG: Debug telemetry / bench cross-checks: frame counters

**Offset** 0x24   **Reset** 0x00000000   **Width** 32   **Address** 0x6224

Debug telemetry / bench cross-checks: frame counters. Read-only; resets to 0.

| Bits  | Field      | Type | Reset  | Description              |
|-------|------------|------|--------|--------------------------|
| 31:16 | TXFRAMECNT | r    | 0x0000 | Transmitted-frame count. |
| 15:0  | RXFRAMECNT | r    | 0x0000 | Received-frame count.    |

## 23 Interrupt router (IRQROUTER)

### 23.1 Overview

The interrupt router (IRQROUTER) is the chip's peripheral interrupt controller. Every deglitched peripheral interrupt level terminates here, and delivery follows the claim/complete model of a RISC-V platform-level interrupt controller: the router raises one external-interrupt wire (meip, vector 85) to each hart, and the servicing hart reads CLAIM to discover and atomically claim the pending source. Routing is per hart and per vector, so a peripheral's interrupt can be steered to whichever hart currently owns that peripheral.

- One row of four 32-bit enable words per hart (HxENL, HxENM, HxENU, HxENX), one bit per vector, covering vectors 0 to 120
- One meip wire per hart (vector 85); CLAIM read claims the lowest pending enabled vector, CLAIM write completes it
- Exactly-once delivery: a claimed vector is hidden from every hart until completed
- Fixed priority: the lowest pending vector number wins
- Read-only raw-level words (PENDL, PENDM, PENDU, PENDX) and under-service words (INSVCL, INSVCM, INSVCU, INSVCX)
- Resets fully masked, so the router is inert until programmed
- Runs on the free-running MCLK, so a routed interrupt wakes a hart whose clock is stopped by sleep

The CLINT vectors (83 and 84) are never delivered through meip: each hart receives its own software and timer interrupts on dedicated wires, so the corresponding row bits are writable but have no effect.

### 23.2 Instances and signals

Table 23.1: IRQROUTER instances

| Instance  | Base address | Interrupt vector(s) | Pins (AF) |
|-----------|--------------|---------------------|-----------|
| IRQROUTER | 0x7000       | none                | none      |

### 23.3 Block diagram

Figure 23.1 draws the router in its place in the chip: every peripheral interrupt source on one side, the per-hart enable rows and the claim/complete stage in the middle, and the harts on the other. The two CLINT levels reach the harts on their own wires and never pass through the router.

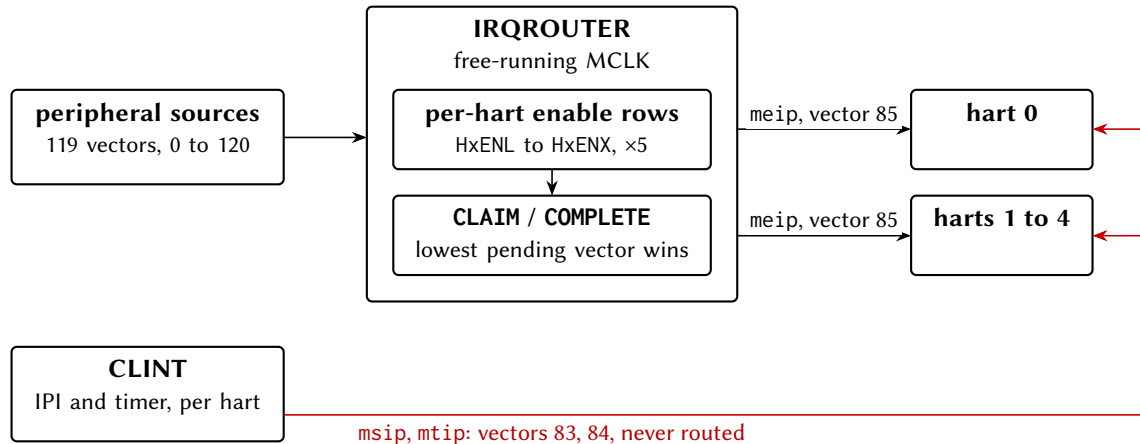


Figure 23.1: The interrupt fabric: peripheral sources, the router’s enable rows and claim/complete stage, the harts, and the CLINT pair in red because it bypasses the router.

## 23.4 Functional description

### 23.4.1 Routing rows

Hart  $h$ ’s row is the four words HxENL (vectors 0 to 31), HxENM (32 to 63), HxENU (64 to 95) and HxENX (96 to 120), with  $x = h$ . Bit  $v \bmod 32$  of the word selected by  $v \div 32$  enables vector  $v$  for that hart. The packing is contiguous, so bit  $b$  of HxENU is vector  $64 + b$ . Any hart may write any row through the shared window, which is what allows a supervising hart to hand a peripheral to another hart. The rows start at the block base, 16 bytes per hart; CLAIM and the status words sit at the fixed offsets  $0x800$ ,  $0x810$  and  $0x820$ , independent of the hart count.

### 23.4.2 Claim and complete

Hart  $h$ ’s meip wire is asserted while some vector  $v$  is pending (deglitched level high), enabled in row  $h$ , and not under service. A read of CLAIM by hart  $h$  returns the lowest such  $v$  and marks it under service, which hides it from every hart’s meip until it is completed; the arbiter attributes the read to the requesting hart, so the router evaluates the reader’s own row. A read that finds nothing pending for the reader returns  $0xFFFFFFFF$ , which happens when another hart claimed the only pending source first. A write of  $v$  to CLAIM completes it: the vector becomes deliverable again, and if its level is still asserted it pends again immediately. Figure 23.2 shows one delivery.

The handler cell in the figure stands for many mclk of software. The handler clears the level at the peripheral before it completes, because a level still high at completion pends the source again at once. Completion is not qualified by owner, so that a supervising hart can complete a vector on behalf of a hart that died while servicing it; the under-service words show which vectors are stuck. Every hart, hart 0 included, uses this path; row 0 is hart 0’s live routing row.

### 23.4.3 Status words

The four PENDx words carry the raw deglitched level of every vector, whether enabled or not, and the four INSVcx words carry the under-service state. Both sets are read-only and exist for diagnosis and recovery.

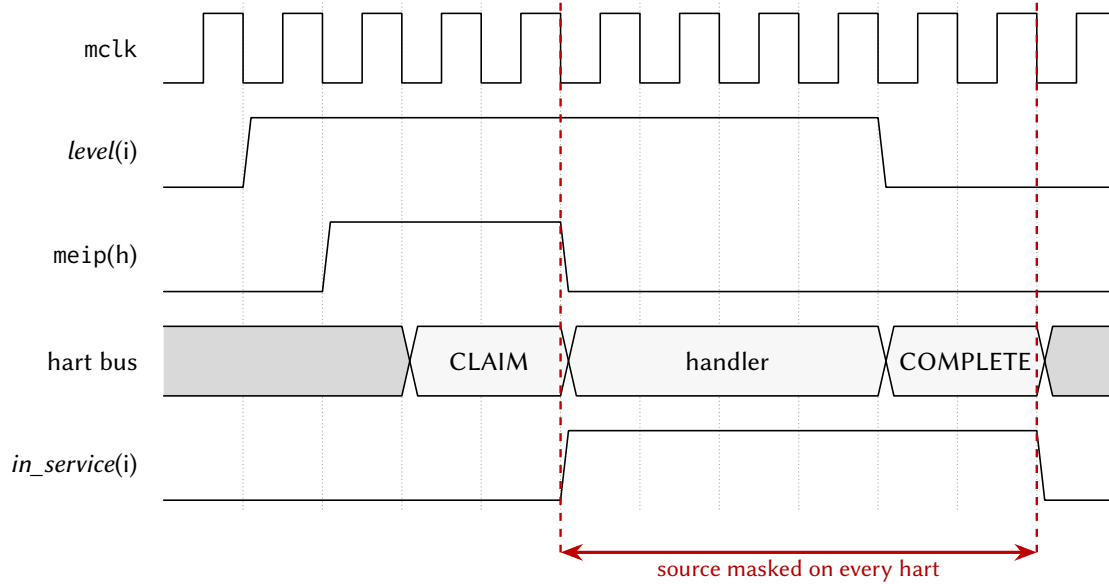


Figure 23.2: Claim/complete delivery of one peripheral interrupt.

## 23.5 Interrupts and events

The router does not raise a vector of its own; it drives vector 85 (meip) into each hart. Table 23.2 states the condition.

| Vector | Condition                                                          | Set by                                       | Cleared by                                       |
|--------|--------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------|
| 85     | A vector pending, enabled in this hart's row and not under service | The peripheral asserting its interrupt level | Reading CLAIM (hides the vector until completed) |

Table 23.2: IRQROUTER delivery into the harts.

## 23.6 Programming

### 23.6.1 Routing a peripheral to hart *h*

1. Clear the peripheral's vector bits in the previous owner's row (HxENL to HxENX with *x* the previous owner).
2. Set the same bits in hart *h*'s row.
3. Enable vector 85 in hart *h*'s interrupt controller.

### 23.6.2 Servicing a routed interrupt

1. Read CLAIM; if the value is 0xFFFFFFFF, return, because the interrupt was spurious.
2. Service the source and clear the interrupt flag at the peripheral.
3. Write the claimed vector number to CLAIM.

## 23.7 Usage notes

**Clear the level at the peripheral before completing.** A completed vector whose level is still asserted pends again at once, so a handler that completes first re-enters immediately.

**Treat 0xFFFFFFFF from CLAIM as spurious.** It is the normal outcome when several harts enable the same vector and another hart claimed it first.

**Route a vector to one hart at a time unless the sources are idempotent.** Exactly-once delivery guarantees that one hart services each event, not which hart.

**Recover a stuck vector from a supervisor.** Read the INSVCx words to find a vector left under service by a dead hart and write its number to CLAIM from any hart.

## 23.8 Registers

Register offsets in this chapter are relative to the instance base address in Table 23.1 (IRQROUTER 0x7000); the generated header names each base IRQROUTER\_BASE.

Table 23.3: IRQROUTER register summary

| Offset         | Name             | Access | Reset      | Description                                             |
|----------------|------------------|--------|------------|---------------------------------------------------------|
| 0x00 + 16n     | HnENL (n = 0..4) | rw     | 0x00000000 | Hart n interrupt routing register, vectors 31:0         |
| 0x04 + 16n     | HnENM (n = 0..4) | rw     | 0x00000000 | Hart n interrupt routing register, vectors 63:32        |
| 0x08 + 16n     | HnENU (n = 0..4) | rw     | 0x00000000 | Hart n interrupt routing register, vectors 95:64        |
| 0x0C + 16n     | HnENX (n = 0..4) | rw     | 0x00000000 |                                                         |
| 0x50 to 0x7FF  | -                | -      | -          | Reserved                                                |
| 0x800          | CLAIM            | rw     | 0x00000000 | Interrupt claim/complete register (M19)                 |
| 0x804 to 0x80F | -                | -      | -          | Reserved                                                |
| 0x810          | PENDL            | r      | 0x00000000 |                                                         |
| 0x814          | PENDM            | r      | 0x00000000 | Raw pending interrupt levels, vectors 63:32 (read-only) |
| 0x818          | PENDU            | r      | 0x00000000 | Raw pending interrupt levels, vectors 95:64 (read-only) |
| 0x81C          | PENDX            | r      | 0x00000000 |                                                         |
| 0x820          | INSVCL           | r      | 0x00000000 |                                                         |
| 0x824          | INSVCM           | r      | 0x00000000 | Under-service flags, vectors 63:32 (read-only)          |
| 0x828          | INSVCU           | r      | 0x00000000 | Under-service flags, vectors 95:64 (read-only)          |
| 0x82C          | INSVCX           | r      | 0x00000000 |                                                         |

**23.8.1 HnENL (n = 0..4): Hart n interrupt routing register, vectors 31:0****Offset** 0x00 + 16n **Reset** 0x00000000 **Width** 32

Hart n interrupt routing register, vectors 31:0. Each bit enables delivery of the corresponding interrupt vector to hart n via its meip wire (vector 85) and the CLAIM/COMPLETE mechanism.

| Bits | Field | Type | Reset      | Description                                               |
|------|-------|------|------------|-----------------------------------------------------------|
| 31:0 | HnENL | rw   | 0x00000000 | Interrupt enable bits for vectors 31:0, routed to hart n. |

**23.8.2 HnENM (n = 0..4): Hart n interrupt routing register, vectors 63:32****Offset** 0x04 + 16n **Reset** 0x00000000 **Width** 32

| Bits | Field | Type | Reset      | Description                                                |
|------|-------|------|------------|------------------------------------------------------------|
| 31:0 | HnENM | rw   | 0x00000000 | Interrupt enable bits for vectors 63:32, routed to hart n. |

**23.8.3 HnENU (n = 0..4): Hart n interrupt routing register, vectors 95:64****Offset** 0x08 + 16n **Reset** 0x00000000 **Width** 32

Hart n interrupt routing register, vectors 95:64. Bits 19 and 20 correspond to the CLINT vectors 83 and 84, which are delivered on dedicated hardwired wires and never through meip: these two bits are writable but have no effect.

| Bits | Field | Type | Reset      | Description                                                |
|------|-------|------|------------|------------------------------------------------------------|
| 31:0 | HnENU | rw   | 0x00000000 | Interrupt enable bits for vectors 95:64, routed to hart n. |

**23.8.4 HnENX (n = 0..4)****Offset** 0x0C + 16n **Reset** 0x00000000 **Width** 32

Hart n interrupt routing register, vectors 120:96 (bits 24:0; upper bits read as 0).

| Bits  | Field | Type | Reset      | Description                                                 |
|-------|-------|------|------------|-------------------------------------------------------------|
| 31:25 | -     | -    | -          | Reserved. Reads zero; write zero.                           |
| 24:0  | HnENX | rw   | 0x00000000 | Interrupt enable bits for vectors 120:96, routed to hart n. |

**23.8.5 CLAIM: Interrupt claim/complete register (M19)****Offset** 0x800 **Reset** 0x00000000 **Width** 32 **Address** 0x7800

Interrupt claim/complete register (M19). READ = claim: atomically returns the lowest pending interrupt vector that is enabled in the READING hart's routing row and not already under service, and marks it under service (masking it from every hart's meip). Returns 0xFFFFFFFF if nothing is pending for the reader; a handler seeing that value treats the interrupt as spurious and simply returns. WRITE = complete: write the claimed vector number to end its service; the vector becomes deliverable again (and pends immediately if its level is still asserted, so clear the level at the peripheral BEFORE completing). Completion is deliberately not qualified by owner, so a supervisor hart can complete on behalf of a hung hart (recovery); written values that are not valid vector numbers, including a stored 0xFFFFFFFF, are ignored.

| Bits | Field | Type | Reset      | Description                                                                                              |
|------|-------|------|------------|----------------------------------------------------------------------------------------------------------|
| 31:0 | CLAIM | rw   | 0x00000000 | Read: claimed vector number (0xFFFFFFFF = none pending for this hart). Write: vector number to complete. |

### 23.8.6 PENDL

**Offset** 0x810 **Reset** 0x00000000 **Width** 32 **Address** 0x7810

Raw pending interrupt levels, vectors 31:0 (read-only; deglitched peripheral levels before enable/claim masking). Debug and polling aid.

| Bits | Field | Type | Reset      | Description                                   |
|------|-------|------|------------|-----------------------------------------------|
| 31:0 | PENDL | r    | 0x00000000 | Deglitched interrupt levels for vectors 31:0. |

### 23.8.7 PENDM: Raw pending interrupt levels, vectors 63:32 (read-only)

**Offset** 0x814 **Reset** 0x00000000 **Width** 32 **Address** 0x7814

| Bits | Field | Type | Reset      | Description                                    |
|------|-------|------|------------|------------------------------------------------|
| 31:0 | PENDM | r    | 0x00000000 | Deglitched interrupt levels for vectors 63:32. |

### 23.8.8 PENDU: Raw pending interrupt levels, vectors 95:64 (read-only)

**Offset** 0x818 **Reset** 0x00000000 **Width** 32 **Address** 0x7818

| Bits | Field | Type | Reset      | Description                                    |
|------|-------|------|------------|------------------------------------------------|
| 31:0 | PENDU | r    | 0x00000000 | Deglitched interrupt levels for vectors 95:64. |

### 23.8.9 PENDX

**Offset** 0x81C **Reset** 0x00000000 **Width** 32 **Address** 0x781C

Raw pending interrupt levels, vectors 120:96 (bits 24:0, read-only; upper bits read as 0). Completes the raw-level readback for the fourth enable word (added with the peripheral-library program; before it, sources above 95 were routable and claimable but absent from the debug readback).

| Bits  | Field | Type | Reset      | Description                                     |
|-------|-------|------|------------|-------------------------------------------------|
| 31:25 | -     | -    | -          | Reserved. Reads zero; write zero.               |
| 24:0  | PENDX | r    | 0x00000000 | Deglitched interrupt levels for vectors 120:96. |

### 23.8.10 INSVCL

**Offset** 0x820 **Reset** 0x00000000 **Width** 32 **Address** 0x7820

Under-service (claimed, not yet completed) flags, vectors 31:0 (read-only). Debug and recovery visibility: a stuck bit here means a hart claimed the vector and never completed it; any hart can recover by writing the vector number to CLAIM.

| Bits | Field  | Type | Reset      | Description                           |
|------|--------|------|------------|---------------------------------------|
| 31:0 | INSVCL | r    | 0x00000000 | Under-service flags for vectors 31:0. |



**23.8.11 INSVCN: Under-service flags, vectors 63:32 (read-only)****Offset** 0x824   **Reset** 0x00000000   **Width** 32   **Address** 0x7824

| Bits | Field  | Type | Reset      | Description                            |
|------|--------|------|------------|----------------------------------------|
| 31:0 | INSVCN | r    | 0x00000000 | Under-service flags for vectors 63:32. |

**23.8.12 INSVCU: Under-service flags, vectors 95:64 (read-only)****Offset** 0x828   **Reset** 0x00000000   **Width** 32   **Address** 0x7828

| Bits | Field  | Type | Reset      | Description                            |
|------|--------|------|------------|----------------------------------------|
| 31:0 | INSVCU | r    | 0x00000000 | Under-service flags for vectors 95:64. |

**23.8.13 INSVCX****Offset** 0x82C   **Reset** 0x00000000   **Width** 32   **Address** 0x782C

Under-service flags, vectors 120:96 (bits 24:0, read-only; upper bits read as 0). Completes the under-service readback for the fourth enable word.

| Bits  | Field  | Type | Reset      | Description                             |
|-------|--------|------|------------|-----------------------------------------|
| 31:25 | -      | -    | -          | Reserved. Reads zero; write zero.       |
| 24:0  | INSVCX | r    | 0x00000000 | Under-service flags for vectors 120:96. |

# **Part IV   Analog front end**

## 23.9 Analog Front-End Subsystem

This chip carries a four-channel bipolar-potentiostat analog front end (AFE) for amperometric and voltammetric electrochemical sensing, organised as four per-quadrant measurement *sites*. Impedance spectroscopy is a firmware mode of those same channels and not a separate block: the reference DAC drives a square-wave excitation and firmware demodulates the converter record in quadrature, so no impedance hardware exists. Each site drives its own electrode group (counter/working/reference/RE2) brought out on the package (Section 3). The analog blocks themselves (the potentiostat and the transimpedance ADC path) are analog IP that is not yet integrated; what *is* present is the complete *digital access path*: a register stub for each site and for the EIS sweep controller, each a fully-functional shared-window arbiter slave with the ownership gate and interrupt path described below. The 0x7C00 site remains allocated for that sweep controller and its I/Q accumulator; no analog stage sits behind it. Software (and the verification suite) programs and reads these exactly as it will the final analog blocks; the stubs hold placeholder storage and drive their interrupt from a software-settable flag until the analog IP replaces them.

Figure 23.3 is the arrangement: the five analog register sites in one rank under the shared-window arbiter of Figure 2.1, the electrode pads that leave the die under each AFE site, and the analog stages that are not yet integrated. Each site bonds four electrode pads; the figure draws three and omits RE2.

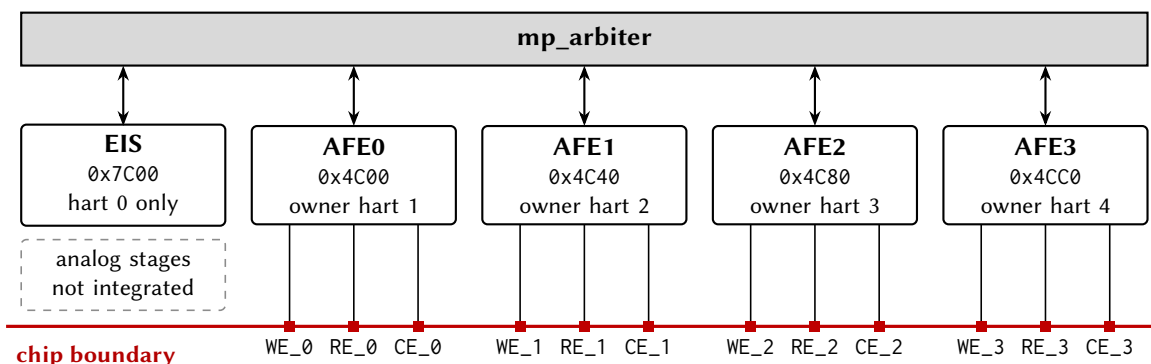


Figure 23.3: Analog front-end connectivity: each AFE site answers its owning hart and hart 0, and the EIS site answers hart 0 only. Each site bonds four electrode pads; the figure draws three and omits RE2.

### 23.9.1 Blocks, addresses, and ownership

The five blocks live in otherwise-reserved shared-window space, so they do not disturb the peripheral memory map: the four AFE sites occupy the four 64-byte sub-slots of the reserved page-0 slot 12 at 0x4C00, and the EIS sweep controller occupies the top quarter of the IRQ-router page at 0x7C00. Every block is a 16-word (64-byte) register file reached through the multi-core arbiter like any other shared slave; access is qualified by the *ownership gate* (Section 23.9.2).

| Block | Base   | IRQ source | Access allowed when          |
|-------|--------|------------|------------------------------|
| AFE0  | 0x4C00 | 55         | s_master = 1 or s_master = 0 |
| AFE1  | 0x4C40 | 55         | s_master = 2 or s_master = 0 |
| AFE2  | 0x4C80 | 55         | s_master = 3 or s_master = 0 |
| AFE3  | 0x4CC0 | 55         | s_master = 4 or s_master = 0 |
| EIS   | 0x7C00 | 56         | s_master = 0                 |

Here `s_master` is the granted-master (hart) index that the multi-core arbiter attributes to the in-flight transaction, the same attribution the hardware mutex bank and the IRQ-router CLAIM use. Because each block decodes only its low four address bits, its 16 words fill its 64-byte sub-slot exactly; the EIS block additionally aliases across the `0x7C00` to `0x7FFF` quarter it owns.

### 23.9.2 Ownership gating

Each AFE site is owned by one hart (AFE0 to hart 1, AFE1 to hart 2, AFE2 to hart 3, AFE3 to hart 4), and it answers only when `s_master` is that hart *or* `s_master` = 0. Hart 0 is the management hart, so it reaches every site (this is what lets it demultiplex the shared interrupt, below); every other hart sees only its own site. The EIS sweep controller is instantiated hart-0-only (`s_master` = 0), so other harts request a sweep through a software mailbox convention rather than touching it directly. The gate is hardware-enforced inside the slave and keys off `s_master` alone: there is no way to forge ownership, and no cross-hart data leaks. A *denied* read returns 0 and a *denied* write is silently dropped: no bus error, no stall, no change to the arbiter contract. All registers reset to 0, so a block is a provable no-op (interrupt low, reads 0) until its owner writes it.

### 23.9.3 Register map (per block)

All five blocks share one 16-word register layout (they are the same hardware entity, differing only in the ownership gate). Offsets are byte offsets from the block base; each word is 32 bits. Detailed bit fields for the placeholder registers are owned by the analog IP specification and will be filled in when that IP is integrated.

| Offset       | Name            | Access | Description                                                                                                                                                                                                                                                        |
|--------------|-----------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x00         | CTRL            | RW     | Control. Placeholder for the analog IP. As a bring-up test hook (until the analog IP drives real events), a write whose data bit 0 is 1 soft-sets IF bit 0, which exercises the block's level-interrupt path end to end.                                           |
| 0x04         | DACPAT          | RW     | DAC excitation-pattern control. Placeholder for the analog IP.                                                                                                                                                                                                     |
| 0x08         | TIA             | RW     | Transimpedance-amplifier gain-range select. Placeholder for the analog IP.                                                                                                                                                                                         |
| 0x0C         | SWM             | RW     | Switch-matrix / analog-multiplexer configuration. Placeholder for the analog IP.                                                                                                                                                                                   |
| 0x10         | ADCC            | RW     | ADC control. Placeholder for the analog IP.                                                                                                                                                                                                                        |
| 0x14         | ADCD            | RW     | ADC data. Placeholder for the analog IP.                                                                                                                                                                                                                           |
| 0x18         | STAT            | RW     | Status. Placeholder for the analog IP.                                                                                                                                                                                                                             |
| 0x1C         | IF              | W1C    | Interrupt-flag word. While any bit is set the block drives its level interrupt high; write a 1 to a bit to clear that bit (write-1-to-clear). Reads are side-effect-free. This is the word hart 0 reads to demultiplex which site raised the shared AFE interrupt. |
| 0x20 to 0x3C | <i>reserved</i> | RW     | Scratch / reserved (plain read-write storage).                                                                                                                                                                                                                     |

### 23.9.4 Interrupt relay

The AFE/EIS interrupts reuse two vector numbers that the digital-only respin left reserved, so nothing is renumbered: the four AFE sites are OR-combined onto a single shared interrupt at source 55 (formerly

the AFE0 vector), and the EIS sweep controller drives source 56 (formerly the SARADC0 vector). Both are delivered through the IRQ router (Section 23) like any other peripheral source, and both are routed, by software convention in the routing rows, to *hart 0 only*.

Routing source 55 to hart 0 is forced, not merely chosen: because the ownership gate lets only hart 0 read all four AFE flag registers, hart 0 is the only hart that can tell which site fired. The relay is: hart 0 takes the external interrupt, CLAIMs source 55 at the router, reads the four IF words to demultiplex the originating site(s), clears the level at each firing site (write-1-to-clear on IF), COMPLETEs source 55, and then notifies the owning hart through the usual CLINT software-interrupt (msip) mailbox. Sensor-rate latency through this relay is immaterial. Source 56 (EIS) is intrinsically hart-0-only and needs no demultiplex.

Giving each AFE site its own interrupt identity (growing the source map, so a site can interrupt its quadrant hart directly without the hart-0 relay) is a deliberately deferred option: it would renumber the CLINT and external-interrupt vectors and disturb the boot-ROM park contract, so it is only worth doing if a future workload needs hart-local AFE interrupt latency.

## 24 Analog Circuitry

Castalia carries a mixed-signal front end alongside the digital subsystem documented in the preceding chapters. This chapter describes the analog blocks themselves and reports their simulated performance; the memory-mapped registers through which software reaches them are documented with the peripherals in Section 23.9.3, a placeholder map until the analog IP is integrated.

Unless stated otherwise, every figure and table in this chapter is taken from schematic-level simulation of the block’s own testbench (pre-layout, with no extracted parasitics), and is therefore a design-intent characterisation rather than silicon data. Two places run on an extracted netlist and say so where they do: the SAR ADC section and the code-level probe of the impedance mode. Each block section names the testbench it came from and the process, temperature and supply points that were simulated, so the coverage behind each number is explicit. Numbers are regenerated directly from the simulator’s results database rather than transcribed by hand.

### Blocks documented in this revision

The analog front end is being brought up incrementally. The electrochemical models the benches use are defined once in Section 24.1, and the nonlinear electrode model that cyclic voltammetry needs in Section 24.2. This revision documents the local bias generator, the class-AB amplifier that serves as both the channel’s transimpedance amplifier (A2) and its potentiostat control amplifier (A1), the complete potentiostat pixel system built from a pair of those amplifiers around a three-electrode cell, the 10-bit SAR analog-to-digital converter that closes the channel, the programmable feedback resistor that sets the channel’s transimpedance gain, and the 12-bit R-2R reference DAC with the supply block that feeds it. Section 24.6 documents not a block but a second measurement mode of the channel: impedance spectroscopy performed by the potentiostat itself, with no impedance hardware. Each block section closes with the register settings that observe that block on a real die, and Section 24.11 lists the per-chip constants those measurements produce. Both are design intent: the analog test bus and the electrode isolation switches they use are not yet in the schematics.

### Characterisation coverage

| Block                     | Section | Benches | Corners        | Monte Carlo |
|---------------------------|---------|---------|----------------|-------------|
| Bias generator            | 24.3    | 3       | ✓ <sup>†</sup> | n/a         |
| Class-AB amplifier        | 24.4    | 6       | ✓ <sup>†</sup> | ✓           |
| Potentiostat pixel system | 24.5    | 6       | ✓ <sup>†</sup> | ✓           |
| Impedance spectroscopy    | 24.6    | 1       | ✓ <sup>†</sup> | ✓           |
| SAR ADC                   | 24.7    | 2       | ✓              | ✓           |
| Feedback resistor         | 24.8    | 5       | ✓              | ✓           |
| Reference DAC             | 24.9    | 6       | ✓ <sup>†</sup> | ✓           |
| Four-channel macro        | 24.10   | 1       |                |             |

<sup>†</sup> Corner run with the passive model rows pinned at typical: a lower bound. See below.

Every block now carries a corner run, but a corner run is not the same claim in every row: the five rows marked <sup>†</sup> are cornered over the transistors alone, for the reason set out immediately below. The impedance

row is not a block but a mode that adds no devices, and its corner and Monte Carlo runs cover the small-signal measurement only. The four-channel row is nominal only: the macro adds no device the block rows do not already corner, and its own claim is that the four channels are independent, which a corner shifts equally on all four. *n/a* marks a block whose section reports no Monte Carlo because the devices at issue carry no per-instance mismatch terms: the bias generator's start-up branch sits on model cards that give a mismatch run a spread of zero from exactly the devices under test. The converter's Monte Carlo has been run and is reported as a negative result for a related reason (Section 24.7.3); nothing in this chapter quotes a converter gain or offset  $\sigma$ . Corner runs cover process, temperature and supply at the five points listed in each block section. Monte Carlo runs cover process and mismatch together unless a table states mismatch-only; a corner figure and a Monte Carlo  $\sigma$  are different quantities and are reported separately throughout. Design targets, where quoted alongside a measurement, are stated in the block section that reports the measurement.

**The pinned-passive caveat.** The corner set that serves this chapter moves the sixteen transistor model rows only. The resistor, capacitor and metal-capacitor rows hold their typical sections in every corner, so poly sheet resistance, a  $\pm 35\%$  quantity, does not move at all. It sets the programmable feedback resistor of Section 24.8 directly, and it sets the reference resistor of the local bias generator, and through that the tail current, transconductance, bandwidth and supply current of every amplifier in this chapter. Every corner column marked  $\dagger$  above is therefore the spread of the transistors alone, and is a lower bound on the true spread.

Measured on the same benches with the passive rows freed: the control amplifier's unity-gain frequency spreads 15.89 MHz to 36.09 MHz against the published 22.36 MHz to 22.91 MHz, a box  $36\times$  wider, and its supply current 81.5  $\mu\text{A}$  to 193.8  $\mu\text{A}$  against 110.0  $\mu\text{A}$  to 136.7  $\mu\text{A}$ ,  $4.2\times$ ; worst-case gain margin falls from 8.10 dB to 7.60 dB. On the transimpedance loop at gain code 63 the feedback resistance spreads  $-33.4\%$  and  $+35.5\%$  against the published  $+0.44\%$ , a box  $157\times$  wider; its  $-3$  dB frequency spreads 8.53 kHz to 16.14 kHz against 10.90 kHz to 11.32 kHz,  $18\times$ ; and worst-case loop phase margin falls from  $71.69^\circ$  to  $61.52^\circ$ .

The two mixed corners cannot be repaired. The foundry ships the passive families with fast, slow and typical sections only (there is no mixed-corner section for a resistor or a capacitor), so fs and sf had to pin those rows, and the pinned table was then cloned into ff and ss, where a real section does exist. That is why the tables look partly right: the bias generator's fs and sf columns agree with the freed matrix to three digits while its ff and ss columns diverge by  $47\%$  and  $25\%$ , because the two columns that were always going to be pinned are the two that are still correct. Only ff and ss can carry the true spread.

**The Monte Carlo has been the only honest process bound in this chapter all along.** Its statistical model section varies the resistors, and the  $11.45\%$  process  $\sigma$  it measures on the feedback resistance is  $3\sigma \approx 34\%$ : the box the corner set was missing. Every Monte Carlo table and figure in this chapter therefore stands as published. So does the converter of Section 24.7, whose extracted array carries no capacitor model card and whose poly strips are fill with no dc path; the programmable feedback resistor of Section 24.8, whose corner axis is the resistor family by construction and which is the one section that already shows the  $\pm 35\%$  band; the offset and tracking tables, which are matching-limited and move *less* when the passives are freed than the published corner box already allows; and the reference DAC's zero-scale, full-scale and LSB columns, which are ratiometric and identical to six digits pinned or freed.

**The corrected corner set.** Define it once as a file rather than as a table typed into each setup, so that a corrected table cannot be cloned with its defects: one section per corner, each a complete model-row list. Five sections, named so that the pin is visible from a corner label: `castalia_tt`, `castalia_ff_all` and `castalia_ss_all` with every row moved, plus `castalia_fs_mos` and `castalia_sf_mos`, which cannot

move the passives and never will. Verify from the netlist that executed, never from the setup:

```
grep -E 'cor_(res|mim|rtmom)\.scs' <run>/input.scs
```

must show the corner letter on the passive rows. One gap survives the fix: the statistical model section omits the MIM statistics, so MIM capacitor variation is bounded by neither the corner set nor the Monte Carlo today; the 22 pF compensation capacitor of Section 24.4 is the device that makes that matter.

## 24.1 Electrochemical Models

Every electrode measurement in this chapter uses one of the two lumped models of Figures 24.1 and 24.2, valued in Table 24.1. The electrode impedance sets the feedback factor of both the potentiostat and the transimpedance loop, so those blocks' stability figures depend on it directly.

The three-electrode cell is what the potentiostat drives: into the working electrode,  $R_{ct} \parallel C_{dl}$  in series with  $R_u$ , about 31 k $\Omega$  at DC and falling toward  $R_u = 1$  k $\Omega$  above roughly 5 Hz. No bench reported here uses the single-interface model.

Both models are linear and frequency-independent, with no Warburg term and no potential dependence, so they give small-signal behaviour about a bias point rather than a full electrochemical response. Both are also nominal: no element carries a spread, so nothing in this chapter accounts for electrode-to-electrode variation, which on a real wound interface is likely to exceed the process spread reported beside it.

| Model                | Element  | Value         | Represents                                 |
|----------------------|----------|---------------|--------------------------------------------|
| three-electrode cell | $R_{ce}$ | 1 k $\Omega$  | solution resistance, CE to RE              |
|                      | $R_u$    | 1 k $\Omega$  | uncompensated resistance, RE to WE surface |
|                      | $R_{ct}$ | 30 k $\Omega$ | charge transfer at the WE interface        |
|                      | $C_{dl}$ | 1 $\mu$ F     | double layer at the WE interface           |
| single interface     | $R_s$    | 10 k $\Omega$ | series solution resistance                 |
|                      | $R_{ct}$ | 1 M $\Omega$  | charge transfer                            |
|                      | $C_{dl}$ | 10 nF         | double layer                               |

Table 24.1: Component values of the electrochemical models. The three-electrode values are the subcircuit defaults; the transimpedance benches override them, and the values used are stated with each measurement.

The closed-loop measurements of Section 24.4 override the defaults with  $R_u = 100 \Omega$  or 1 k $\Omega$ ,  $R_{ct} = 10$  k $\Omega$  or 1 M $\Omega$ , and  $C_{dl} = 100$  nF or 1  $\mu$ F; the value used is stated with each measurement.



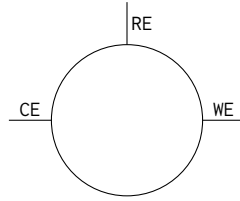


Figure 24.1: Three-electrode cell model, the load the potentiostat actually drives, shown as the symbol used throughout this chapter. It is a lumped linear network between the counter, reference and working electrodes; its elements and their values are given in Table 24.1. The model is frequency-independent (no Warburg diffusion term and no potential dependence), so it represents small-signal behaviour about a bias point rather than a full electrochemical response.

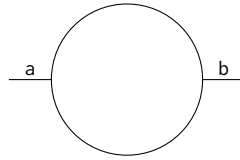


Figure 24.2: Single electrode-electrolyte interface, presented as a two-terminal element and shown as the same abstracted symbol. Its elements and their values are given in Table 24.1; they describe a much smaller double layer and a much larger charge-transfer resistance than the three-electrode cell. None of the benches reported in this chapter uses it.

## 24.2 Voltammetric Electrode Model

This section documents a *simulation model*, not a circuit block and not silicon: a Verilog-A element standing in for the electrode-electrolyte interface when a bench sweeps potential and expects a voltammetric peak.

A linear network cannot produce one: its extrema stay pinned at the switching potentials whatever the element values (Figure 24.3, dashed). A peak needs Butler-Volmer kinetics, exponential in overpotential about  $E^{0'}$ , together with diffusional depletion, which turns the exponential rise into a  $t^{-1/2}$  decay; kinetics alone give a sigmoid.

The faradaic current is Butler-Volmer in heterogeneous-rate-constant form, emitted anodic-positive to match the transimpedance amplifier:

$$i_f = nFAk^0 \left[ C_R(0, t) e^{(1-\alpha)nf(E-E^{0'})} - C_O(0, t) e^{-\alpha nf(E-E^{0'})} \right], \quad f = \frac{F}{RT}, \quad (24.1)$$

both exponents clamped at  $\pm \text{expmax}$  through `limexp`. Surface concentrations follow diffusion into a finite Nernst layer,

$$\frac{\partial C_s}{\partial t} = D \frac{\partial^2 C_s}{\partial x^2}, \quad s \in \{O, R\}, \quad (24.2)$$

with flux set by  $i_f$  at the surface and bulk pinned at the layer edge,

$$D \frac{\partial C_O}{\partial x} \Big|_{x=0} = +\frac{i_f}{nFA}, \quad D \frac{\partial C_R}{\partial x} \Big|_{x=0} = -\frac{i_f}{nFA}, \quad C_s(\delta, t) = C_s^*. \quad (24.3)$$

The simulator solves the finite-volume form of (24.2) on an expanding grid of 28 slabs per species,

$$h_j \frac{dC_j}{dt} = \frac{D}{d_{j-1}} (C_{j-1} - C_j) - \frac{D}{d_j} (C_j - C_{j+1}), \quad h_j = h_0 \beta^j, \quad (24.4)$$

with  $h_0 = 0.5 \mu\text{m}$ ,  $\beta = 1.280$  and  $\delta = 1.79 \text{ mm}$ . Each slab is one electrical node carrying its concentration scaled by  $1 \times 10^{-6} \text{ mol cm}^{-3}$ , since the raw value solves to noise below the simulator's voltage tolerance. Parameters are in Table 24.2, in the CGS units of the source literature.

Benches instantiate it as the cell of Figure 24.1 with the charge-transfer branch replaced by this element. Only voltammetry needs it; the linear cell remains correct for small-signal impedance work.

| Parameter | Symbol   | Default             | Unit                         | Quantity                             |
|-----------|----------|---------------------|------------------------------|--------------------------------------|
| E0        | $E^{0'}$ | 0.0                 | V                            | formal potential of the couple       |
| nel       | $n$      | 1                   | —                            | electrons transferred                |
| alpha     | $\alpha$ | 0.5                 | —                            | transfer coefficient                 |
| k0        | $k^0$    | 0.1                 | $\text{cm s}^{-1}$           | standard heterogeneous rate constant |
| Dif       | $D$      | $5 \times 10^{-6}$  | $\text{cm}^2 \text{ s}^{-1}$ | diffusion coefficient, both species  |
| Area      | $A$      | 0.01                | $\text{cm}^2$                | electrode area                       |
| Cox       | $C_O^*$  | $1 \times 10^{-6}$  | $\text{mol cm}^{-3}$         | bulk concentration, oxidised species |
| Cred      | $C_R^*$  | 0                   | $\text{mol cm}^{-3}$         | bulk concentration, reduced species  |
| Temp      | $T$      | 298.15              | K                            | interface temperature                |
| expmax    | —        | 40                  | —                            | exponent clamp on the kinetic terms  |
| gpar      | —        | $1 \times 10^{-15}$ | S                            | DC shunt across the terminals        |

Table 24.2: Parameters of the Verilog-A electrode model, with the defaults the standalone validation runs of Table 24.3 use. Units are CGS, as in the source literature: a bulk concentration of  $1 \text{ mmol L}^{-1}$  is  $1 \times 10^{-6} \text{ mol cm}^{-3}$  and an area of  $1 \text{ mm}^2$  is  $0.01 \text{ cm}^2$ . The last two rows are numerical guards, not physical quantities; the exponent clamp is never reached over the  $\pm 0.8 \text{ V}$  window of this front end. The grid constants (28 slabs per species,  $h_0$ ,  $\beta$ ,  $\delta$ ) are compile-time constants of the module, not parameters, and are given in Section 24.2.

| Scan rate<br>$\text{mV s}^{-1}$ | $i_p$ Randles–Ševčík<br>$\mu\text{A}$ | $i_p$ model<br>$\mu\text{A}$ | Error<br>% | Error, $C_{dl} \rightarrow 0$<br>% | $\Delta E_p$<br>mV |
|---------------------------------|---------------------------------------|------------------------------|------------|------------------------------------|--------------------|
| 25                              | 0.951                                 | 0.952                        | +0.08      | −0.45                              | 58.78              |
| 50                              | 1.345                                 | 1.349                        | +0.32      | −0.42                              | 59.85              |
| 100                             | 1.902                                 | 1.915                        | +0.67      | −0.38                              | 60.41              |
| 200                             | 2.690                                 | 2.721                        | +1.16      | −0.33                              | 62.33              |
| 400                             | 3.804                                 | 3.875                        | +1.86      | −0.25                              | 64.74              |

Table 24.3: Peak metrics of the electrode model against the Randles–Ševčík law, first cathodic scan, conditions as Figure 24.3. The last-but-one column repeats the measurement with  $C_{dl}$  removed: the remaining  $-0.45\%$  to  $-0.25\%$  is a scan-rate-independent discretisation bias of the 28-slab grid, so the whole scan-rate dependence of the error is double-layer charging, which the ablation reproduces as  $C_{dl}v/i_p$  to two decimals. Peak separation is inflated by  $iR$  drop across  $R_s = 500 \Omega$ , which grows with scan rate; with  $R_s$  and  $C_{dl}$  removed it is  $57.04 \text{ mV}$  at every rate, against Nicholson's reversible  $2.218RT/F = 56.9 \text{ mV}$ .

**Validation.** Peak current scales  $4.07\times$  between  $25 \text{ mV s}^{-1}$  and  $400 \text{ mV s}^{-1}$  against the  $4.00\times$  of the  $\sqrt{v}$  law, running high against Randles–Ševčík by  $0.08\%$  to  $1.86\%$  (Table 24.3). Kinetics walk the Matsuda–Ayabe progression correctly: at  $50 \text{ mV s}^{-1}$ ,  $k^0 = 0.1 \text{ cm s}^{-1}$  gives  $\Lambda = 18.1$ , reversible ( $\Delta E_p = 60.0 \text{ mV}$ ,  $i_p$  within  $0.11\%$ ), and  $k^0 = 1 \times 10^{-3} \text{ cm s}^{-1}$  gives  $\Lambda = 0.181$ ,  $\Delta E_p = 167 \text{ mV}$ ,  $i_p$   $15.5\%$  low.

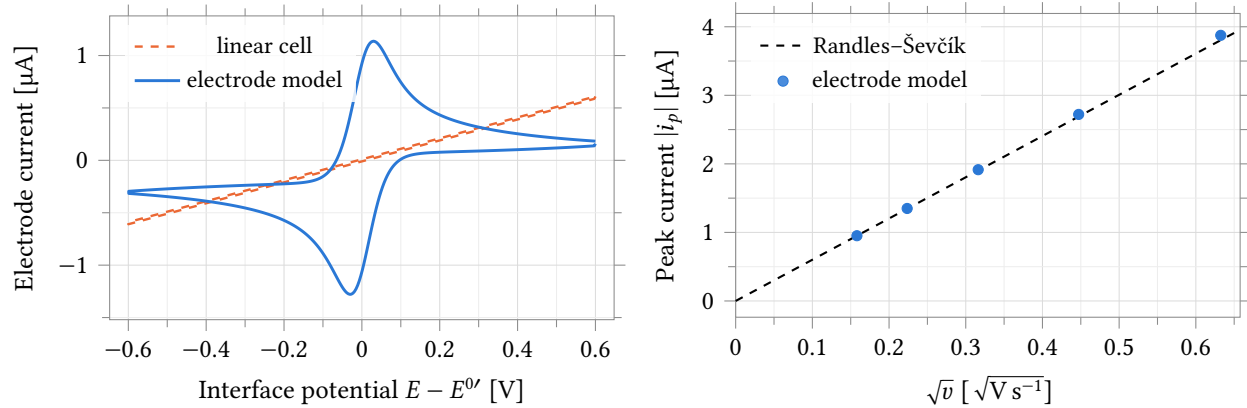


Figure 24.3: Standalone validation of the electrode model against closed-form voltammetry, ideal potential source, model defaults of Table 24.2 with  $R_s = 500 \Omega$  and  $C_{dl} = 0.2 \mu\text{F}$  in series and in parallel respectively,  $25^\circ\text{C}$ . Left: current against interface potential over a  $\pm 0.6 \text{ V}$  triangular sweep at  $50 \text{ mV s}^{-1}$ , second cycle, for the nonlinear model (solid) and for the linear cell of Section 24.1 driven identically (dashed,  $R_{ct} = 1 \text{ M}\Omega$ ); the linear response is a ramp with its extrema pinned at the switching potentials (its  $C_{dl}$  box is only  $20 \text{ nA}$  wide at this scan rate and invisible here), while the nonlinear one peaks  $30 \text{ mV}$  either side of  $E^{0'}$  and decays afterwards. Right: peak cathodic current of the *first* scan against  $\sqrt{v}$  over  $25 \text{ mV s}^{-1}$  to  $400 \text{ mV s}^{-1}$ , against the Randles-Ševčík line, of slope  $6.015 \mu\text{A}$  per  $\sqrt{\text{V s}^{-1}}$ ; the two panels use different cycles because Randles-Ševčík describes the first scan into an undepleted bulk, while the second cycle is the repeatable steady response and sits 3 % to 5 % below it. Numbers in Table 24.3.

Discretisation, not the solver, is the accuracy limit: eight slabs per species is unusable ( $i_p$  high by 232 %), 16 and above agree within 0.35 %, and 28 ships. Dropping the maximum time step from 20 ms to 0.5 ms moves  $i_p$  below the fourth significant figure. Roughly 60 sweeps ran without a convergence failure, and the exponentials survive  $\pm 12 \text{ V}$  of overpotential with the clamp removed, so the clamp and shunt conductance are insurance.

One design limit follows. At  $50 \text{ mV s}^{-1}$  on  $1 \text{ mm}^2$ , double-layer charging is 0.74 % of the peak at  $1 \text{ mmol L}^{-1}$  but 37 % of a  $27 \text{ nA}$  peak at  $0.02 \text{ mmol L}^{-1}$ : below roughly  $0.1 \text{ mmol L}^{-1}$  the faradaic peak is buried in charging current rather than in the front end's noise, and the remedy is a slower scan, not more gain.

**First measurement with the rev-2 channel in the loop.** The cyclic-voltammetry test is the potentiostat bench of Section 24.5 with the linear cell swapped for the nonlinear one and nothing else changed: ideal  $R_f = 35 \text{ k}\Omega$ ,  $C_{dl} = 0.2 \mu\text{F}$ ,  $R_u = R_{ce} = 1 \text{ k}\Omega$ , electrode at  $E^{0'} = 0.3 \text{ V}$ ,  $1 \text{ mmol L}^{-1}$ ,  $k^0 = 0.1 \text{ cm s}^{-1}$ , commanded over a  $0.75 \text{ V}$  to  $1.75 \text{ V}$  triangle at  $2 \text{ V s}^{-1}$ , typical process,  $37^\circ\text{C}$  for silicon and electrode alike.

The peak measures  $8.666 \mu\text{A}$  against  $8.74 \mu\text{A}$  predicted ( $8.34 \mu\text{A}$  Randles-Ševčík plus  $0.40 \mu\text{A}$  of  $C_{dl} dv/dt$ ),  $-0.8 \%$ : the nonlinearity survives the loop rather than being reshaped by it. Read as  $i_{we} = (v_{out} - V_{CM})/R_f$ , the peak moves the output  $0.30 \text{ V}$  from  $V_{CM} = 1.25 \text{ V}$ , 38 % of the  $\pm 0.80 \text{ V}$  window, so the sweep is unclipped. The four wound-panel analytes run as corners on the same test (Figure 24.4).

#### What these numbers do not cover.

- Nominal transients only: one process point, one temperature, no corner set and no Monte Carlo. The analyte corners vary electrode parameters alone.
- A Verilog-A element carries no mismatch statistics, so every Monte Carlo  $\sigma$  in this chapter holds the electrode nominal and reports circuit variation only.

- Table 24.3 runs at 25 °C; the in-loop run is the only 37 °C result. Temp is a plain model parameter, so a new bench must set it explicitly or the kinetics silently revert to 25 °C.  $D$  is held at its 25 °C value throughout, so the  $\sim 30\%$  rise in a real aqueous  $D$  over 25 °C to 37 °C is not modelled.
- Randles–Ševčík assumes semi-infinite diffusion; the model pins bulk at  $\delta = 1.79$  mm. That holds by  $16\times$  at  $50\text{ mV s}^{-1}$  ( $\sqrt{Dt} = 0.11$  mm) but not for a thin unstirred film, for which a  $93\text{ }\mu\text{m}$  variant attenuates the reverse peak to 0.65 of the forward. Quote  $\delta$  with any peak from this model.
- $k^0$  shapes the peak but neither identifies nor quantifies an analyte, and it is the least-supported parameter: no rate constant on carbon is published for these four markers, so the values in use are fitted to a measured  $\Delta E_p$ . A gap in the literature, not the model.

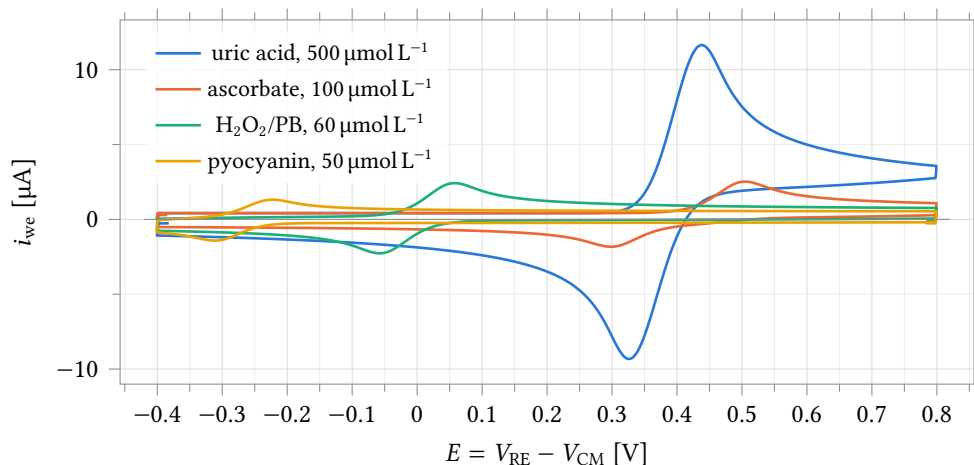


Figure 24.4: Voltammograms of the four-analyte wound panel through the rev-2 channel, one electrode parameter set per channel, from the cyclic-voltammetry Maestro corner run of Section 24.2 ( $2\text{ V s}^{-1}$ ,  $R_f = 35\text{ k}\Omega$ , conditions of the in-loop run above). Peak potential identifies the analyte and peak height its concentration; each trace is one full cycle started at that analyte’s non-faradaic vertex.  $E^{0'}$  and concentrations are literature-anchored;  $k^0$  is fitted (no measured value exists on carbon for any of the four), and reversal peaks appear for the chemically irreversible couples because following chemistry is not modelled. The two cathodic markers span  $-1.4\text{ }\mu\text{A}$  to  $-2.4\text{ }\mu\text{A}$ , unmeasurable by the unidirectional rev-1 front end.

**Differential-pulse voltammetry through the channel.** The same bench runs a staircase-plus-pulse program at the published assay envelope (Figure 24.5), every  $4\text{ mV}$  tread quantised to  $152.6\text{ }\mu\text{V}$  and current sampled  $5\text{ ms}$  before each edge. Four reporters give four resolved peaks, apexes within  $2\text{ mV}$  and  $1\%$  of the assay targets. The quantisation is not the shipped one: the reference DAC of Section 24.9 is 12-bit with a  $610.35\text{ }\mu\text{V}$  LSB, four times coarser, so a  $4\text{ mV}$  tread is 6 or 7 codes rather than 26 and the  $2\text{ mV}$  apex figure is not a channel measurement. Three findings bound how the numbers read:

- *No residual charging baseline.* A blank gives under  $1\text{ pA}$  at every step,  $45\text{ ms}$  of a  $50\text{ ms}$  pulse being far beyond the  $200\text{ nF}/1\text{ k}\Omega$  time constant, so  $\Delta I = I(S_2) - I(S_1)$  cancels charging entirely.
- *The  $560\text{ k}\Omega$  tap slew-limits on every edge.* A  $50\text{ mV}$  step into  $200\text{ nF}$  is  $10\text{ nC}$  against a  $\sim 2.2\text{ }\mu\text{A}$  output stage, leaving the applied potential wrong for  $\sim 20\text{ ms}$ . Settled samples are unaffected, but the transfer turns superlinear above  $\sim 350\text{ nA}$  (up to  $30\%$  at the top of the tap; proportional within  $3\%$  below it), which is why Figure 24.6 uses a *measured* transfer.
- *Full scale is consumed by  $I(S_2)$ , not  $\Delta I$ .* The largest sampled current is  $1.02\text{--}1.04\times$  the differential peak and reaches  $84\%$  of the  $\pm 2.06\text{ }\mu\text{A}$  tap; do headroom accounting on  $I(S_2)$ .

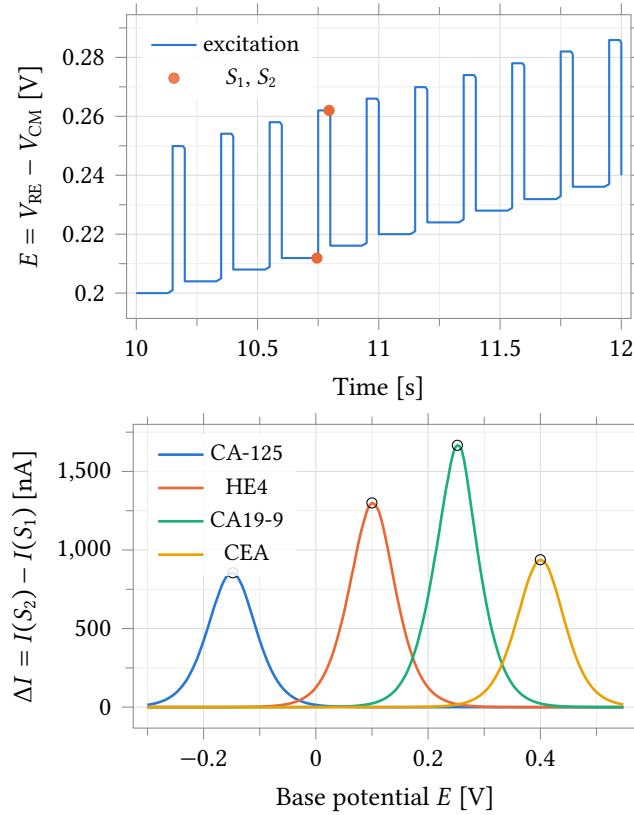


Figure 24.5: Differential-pulse voltammetry through the rev-2 channel, four-reporter immunoassay panel, one electrode parameter set per reporter ( $R_f = 560 \text{ k}\Omega$ ,  $C_{dl} = 200 \text{ nF}$ ,  $37^\circ\text{C}$ ). Top: a 2 s excerpt of the excitation (4 mV treads, 50 mV / 50 ms pulses at a 200 ms period, every level quantised to  $152.6 \mu\text{V}$ ), with the two current-sampling instants 5 ms before each edge. Bottom: the differential current over the full  $-0.300 \text{ V}$  to  $0.548 \text{ V}$  scan (213 steps, 42.6 s); peak potential separates the reporters, peak height carries concentration (circles mark the apexes: 855 nA, 1300 nA, 1666 nA and 938 nA). Reporter kinetics as in the cyclic-voltammetry runs; the panel differs from Figure 24.4 in readout, gain tap and reporter set. The  $152.6 \mu\text{V}$  step is finer than the  $610.35 \mu\text{V}$  LSB of the shipped 12-bit reference DAC (Section 24.9), so the apex accuracy read off this panel is a lower bound on the channel's.

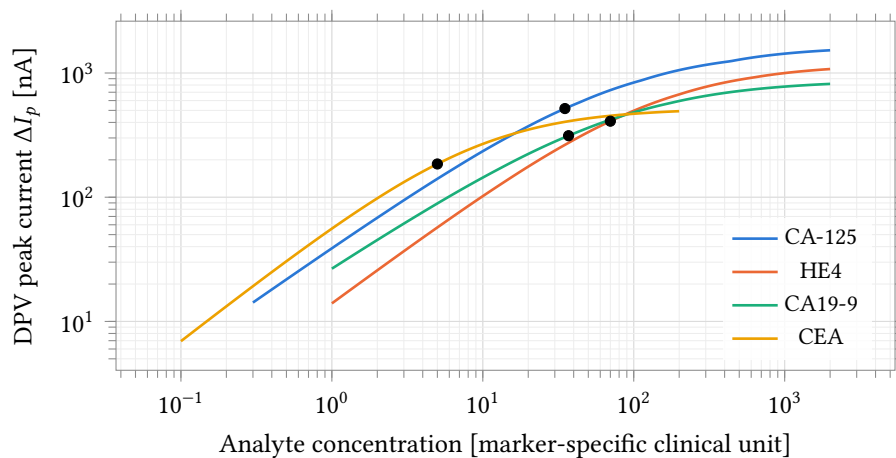


Figure 24.6: Dose response of the four-reporter panel: published-assay binding curves composed with the current transfer *measured* from the DPV runs of Figure 24.5 (11 simulated concentrations per reporter, 560 kΩ tap). Filled circles are the clinical decision points through the same chain: CA-125 35 U/mL → 517 nA, HE4 70 pmol L<sup>-1</sup> → 409 nA, CA19-9 37 U/mL → 313 nA, CEA 5 ng mL<sup>-1</sup> → 185 nA. The concentration axis carries each marker's own clinical unit; only the current axis is common. The binding model is analytic; the electrode and channel behind the current axis are simulated.

| Element                  | Form                                                                       | Source                   |
|--------------------------|----------------------------------------------------------------------------|--------------------------|
| <i>Implemented</i>       |                                                                            |                          |
| Equivalent circuit       | $R_s + C_{dl} \parallel Z_f$ ; the model is $Z_f$                          | Randles 1947             |
| Faradaic current         | $i = nFAk^0 [C_R(0, t)e^{(1-\alpha)nf\eta} - C_O(0, t)e^{-\alpha nf\eta}]$ | B&F Eq. 3.3.11           |
| Mass transport           | $\partial C / \partial t = D \partial^2 C / \partial x^2$                  | B&F Eq. 4.4.2            |
| Surface boundary         | $i / (nFA) = D \partial C_O / \partial x _{x=0}$                           | B&F Eq. 4.4.29           |
| Outer boundary           | $C(\delta, t) = C^*$ , Nernst diffusion layer                              | B&F Sec. 1.4.2           |
| Discretisation           | box (finite-volume) method                                                 | Feldberg 1969; B&S Ch. 3 |
| Space grid               | $h_j = h_0 \beta^j$                                                        | B&S Ch. 7                |
| <i>Validated against</i> |                                                                            |                          |
| Peak current             | $i_p = (2.69 \times 10^5) n^{3/2} AD^{1/2} C^* v^{1/2}$                    | B&F Eq. 6.2.19           |
| Peak separation          | $\Delta E_p = 2.218 RT / nF$                                               | Nicholson 1965           |
| Reversibility            | $\Lambda = k^0 / \sqrt{\pi D n f v}$                                       | Matsuda & Ayabe 1955     |

Table 24.4: Every relation the electrode model implements, in the form it is implemented, with the source it is taken from; the same map is carried in the model's own header. The lower group is not implemented: those are the closed forms the model is validated against in Table 24.3. Here  $\eta = E - E^{0'}$  and  $f = F/RT$ . Equation numbers are those of the second edition of Bard and Faulkner; the third edition rennumbers.

**Sources.** The abbreviations of Table 24.4 are: B&F, A. J. Bard and L. R. Faulkner, *Electrochemical Methods: Fundamentals and Applications*, 2nd ed., Wiley, 2001; B&S, D. Britz and J. Strutwolf, *Digital Simulation in Electrochemistry*, 4th ed., Springer, 2016; Randles 1947, J. E. B. Randles, *Discuss. Faraday Soc.* 1:11, 1947; Feldberg 1969, S. W. Feldberg, *Electroanal. Chem.* 3:199, 1969; Nicholson 1965, R. S. Nicholson, *Anal. Chem.* 37:1351, 1965; Matsuda & Ayabe 1955, H. Matsuda and Y. Ayabe, *Z. Elektrochem.* 59:494, 1955. The peak-current law is due independently to Randles (*Trans. Faraday Soc.* 44:327, 1948) and Ševčík (*Coll. Czech. Chem. Commun.* 13:349, 1948); the numerical form quoted in Table 24.4 is the 25 °C one of B&F Eq. 6.2.19.

### 24.3 Local Wide-Swing Cascode Bias Generator

The local bias generator (schematic cell `anatop_biasgen_local`, drawn from `BiasGenCascWideSwing`) is the cell each analog block instantiates to develop its own cascode bias rails, as distinct from the single chip-level `BiasGenCascWideSwingGlobal` that drives the global `bn! / bp!` nets. It presents four outputs, the NMOS mirror and cascode gates ( $V_{\text{BNM}}$ ,  $V_{\text{BNC}}$ ) and the PMOS mirror and cascode gates ( $V_{\text{BPM}}$ ,  $V_{\text{BPC}}$ ), generated from a resistor-degenerated reference core in the wide-swing cascode arrangement, so the cascode gates sit only one saturation voltage away from the mirror gates and the mirrors keep their output devices in saturation over a wide compliance range. An `En` input gates the cell, and a start-up branch guarantees it leaves the degenerate zero-current state at power-up. A 6-bit trim word, `BiasAdj<5:0>`, moves the reference current so the operating point can be corrected after silicon.

The characterisation below is from schematic-level simulation (no extracted parasitics) of the `BiasGenCascWideSwing_tb` testbench, over the process, temperature and supply points of Table 24.5. All rail voltages are referred to `vss!`, and the cell is enabled throughout.

| Point | Process | Temperature | Supply |
|-------|---------|-------------|--------|
| 1     | tt      | 27 °C       | 2.5 V  |
| 2     | ff      | 25 °C       | 2.5 V  |
| 3     | fs      | 25 °C       | 2.5 V  |
| 4     | sf      | 25 °C       | 2.5 V  |
| 5     | ss      | 25 °C       | 2.5 V  |

Table 24.5: Process, temperature and supply points simulated for the local bias generator. These points move the sixteen transistor model rows only (the resistor and capacitor rows stay at their typical sections), so every corner spread in this section is a lower bound (pinned-passive caveat, page 207).

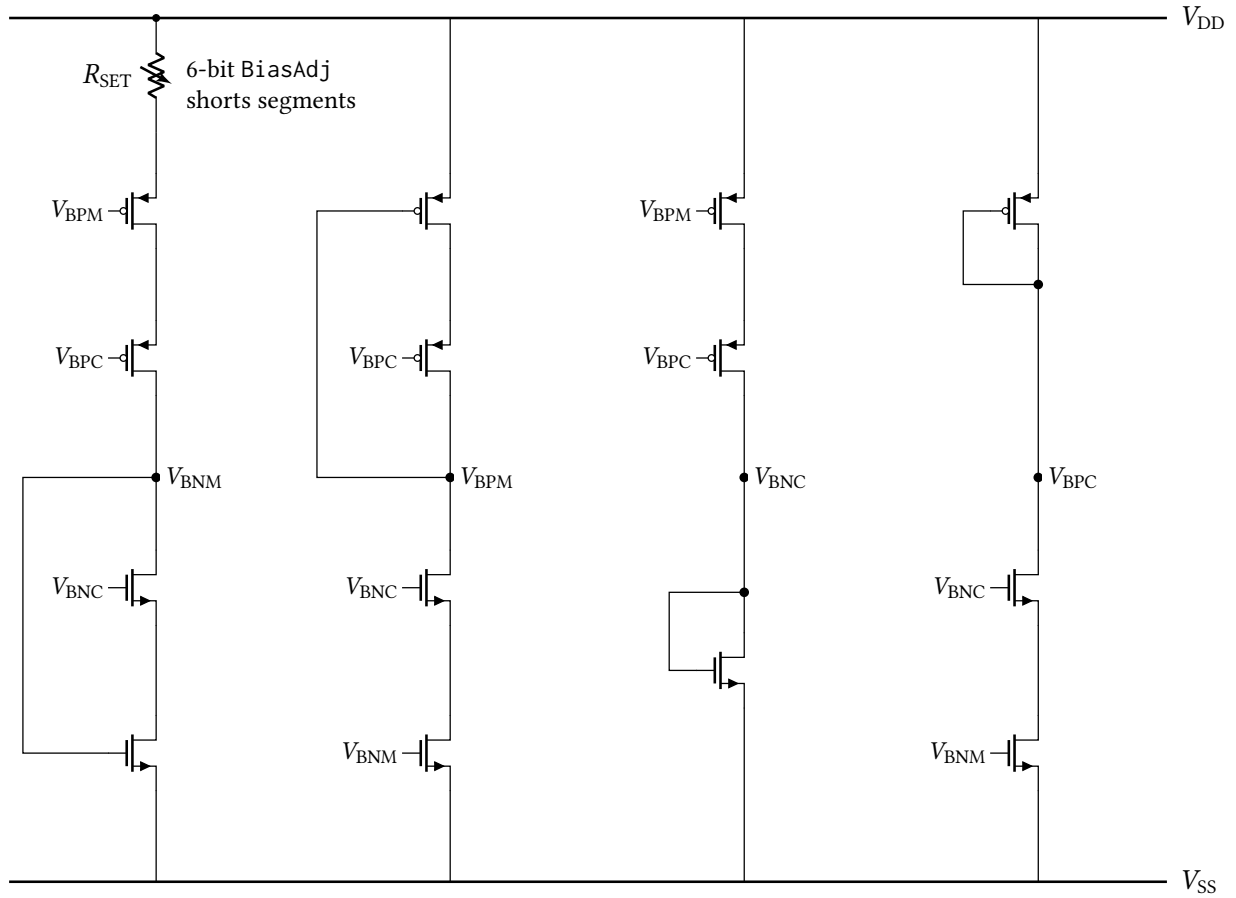


Figure 24.7: Simplified core of the local wide-swing cascode bias generator, drawn from the `anatop_biasgen_local` schematic. Four self-biased branches develop the four rails: the reference branch (left) passes the current set by the trimmable resistor  $R_{SET}$ , and the remaining branches mirror it to produce  $V_{BPM}$ ,  $V_{BNC}$  and  $V_{BPC}$ . Each branch's own self-bias (diode) connection is drawn, from the rail it generates back to the gate it drives in the same branch; the cross-branch gate connections are identified by net name rather than drawn as buses. The enable switches, the decoupling capacitors on each rail and the individual segments of the resistor string are omitted for clarity.

| Parameter | Unit    | tt     | ff     | fs     | sf     | ss     | Spread |
|-----------|---------|--------|--------|--------|--------|--------|--------|
| $V_{BNM}$ | mV      | 614.3  | 578.9  | 584.6  | 644.0  | 657.5  | 78.6   |
| $V_{BNC}$ | mV      | 776.2  | 765.4  | 745.4  | 807.1  | 805.7  | 61.7   |
| $V_{BPM}$ | mV      | 1815.4 | 1841.7 | 1804.2 | 1826.6 | 1777.6 | 64.1   |
| $V_{BPC}$ | mV      | 1589.3 | 1571.3 | 1578.4 | 1600.2 | 1575.2 | 28.9   |
| $I_{DD}$  | $\mu A$ | 24.38  | 35.97  | 24.23  | 24.52  | 18.38  | 17.59  |

Table 24.6: DC operating point of the local bias generator at the nominal trim code ( $BiasAdj = 33$ ),  $25^\circ C$ , by process corner; the last column is the spread across corners. The degeneration resistor is moved with the process corner here. An earlier corner set held it at typical in every corner and reported a supply-current spread of  $0.30 \mu A$ : that resistor sets the reference current, so pinning it removed almost all of the block's corner spread, and the  $fs$  and  $sf$  columns still agree with it because the resistor family has no skew section.



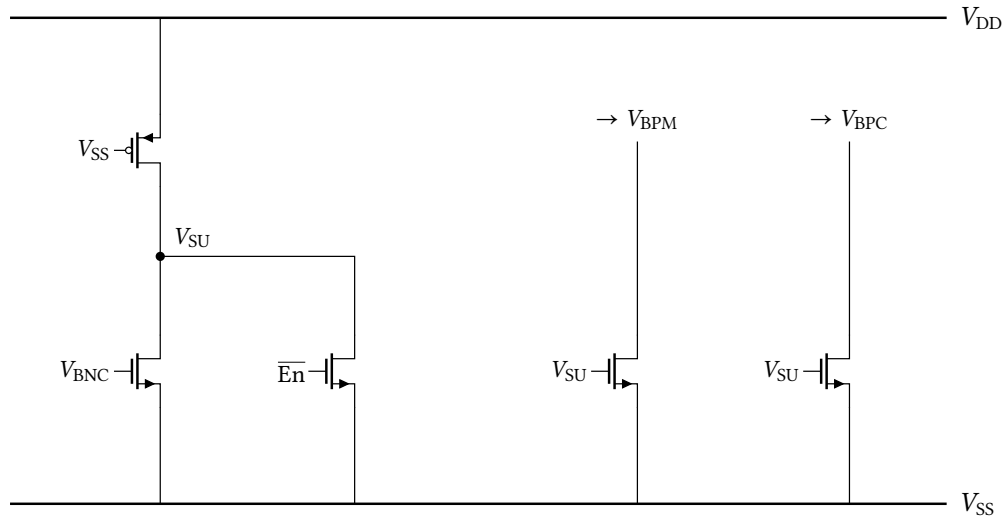


Figure 24.8: Start-up branch. A gate-grounded, permanently-on weak PMOS pulls  $V_{SU}$  up at power-on; while  $V_{SU}$  is high it drags  $V_{BPM}$  and  $V_{BPC}$  down, forcing current into the core and breaking the degenerate zero-current state. Once the core is running, the rising  $V_{BNC}$  turns on the left NMOS and collapses  $V_{SU}$ , so the branch draws no static current in normal operation, the behaviour visible in the power-up transient of Figure 24.16. A second device holds  $V_{SU}$  low whenever the cell is disabled.

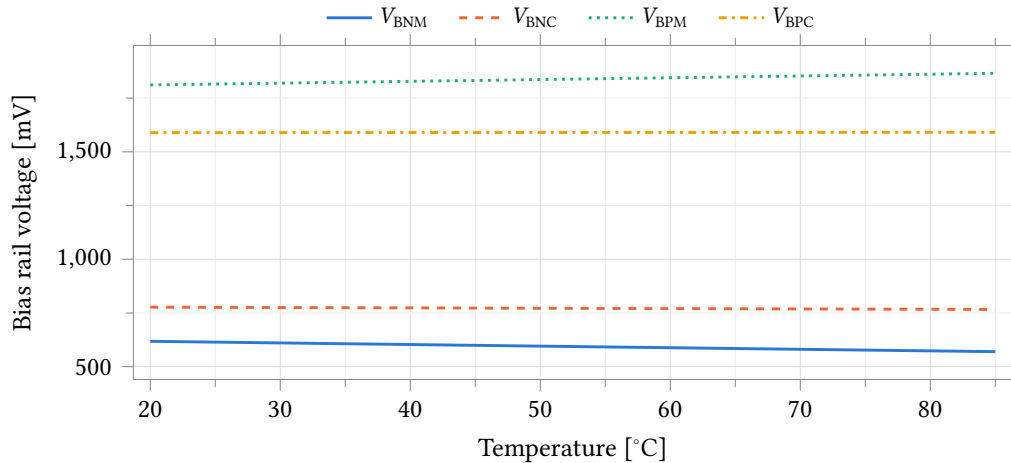


Figure 24.9: The four local bias rails against temperature at the typical corner.  $V_{BNM}$  and  $V_{BNC}$  are referred to  $vss!$ ;  $V_{BPM}$  and  $V_{BPC}$  sit near the supply.

| Parameter        | Unit          | Min    | Mean   | Max    | Spread |
|------------------|---------------|--------|--------|--------|--------|
| $V_{\text{BNM}}$ | mV            | 569.9  | 593.8  | 618.1  | 48.2   |
| $V_{\text{BNC}}$ | mV            | 766.3  | 771.5  | 777.1  | 10.8   |
| $V_{\text{BPM}}$ | mV            | 1811.2 | 1838.4 | 1865.4 | 54.2   |
| $V_{\text{BPC}}$ | mV            | 1589.2 | 1590.0 | 1590.9 | 1.7    |
| $I_{\text{DD}}$  | $\mu\text{A}$ | 24.16  | 25.50  | 26.81  | 2.65   |

Table 24.7: Extremes of the four bias rails and of the quiescent supply current over the simulated temperature sweep: typical corner, nominal supply, 23 points from 20 °C to 85 °C. Spread is the maximum minus the minimum of that sweep and is neither a corner nor a mismatch figure; the corner temperature coefficients are Table 24.8. Nothing here bounds behaviour below 20 °C or above 85 °C.

| Parameter        | Unit   | tt   | ff   | fs   | sf   | ss   |
|------------------|--------|------|------|------|------|------|
| $V_{\text{BNM}}$ | ppm/°C | 1248 | 1432 | 1324 | 1179 | 1097 |
| $V_{\text{BNC}}$ | ppm/°C | 215  | 308  | 232  | 198  | 134  |
| $V_{\text{BPM}}$ | ppm/°C | 453  | 449  | 455  | 451  | 459  |
| $V_{\text{BPC}}$ | ppm/°C | 17   | 53   | 14   | 19   | 24   |
| $I_{\text{DD}}$  | ppm/°C | 1598 | 1518 | 1624 | 1573 | 1673 |

Table 24.8: Temperature coefficient of each bias rail and of the supply current, taken as the total variation over the simulated temperature range normalised to the mean. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

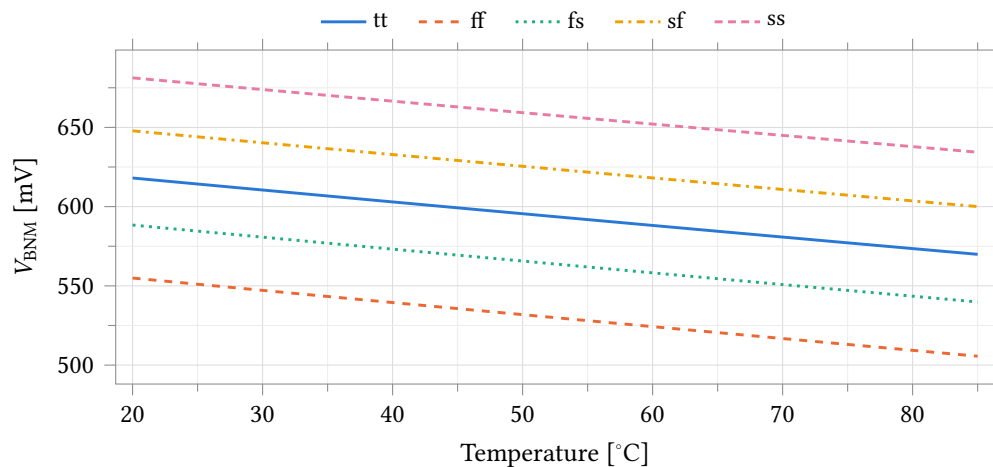


Figure 24.10: NMOS mirror bias  $V_{\text{BNM}}$  against temperature over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

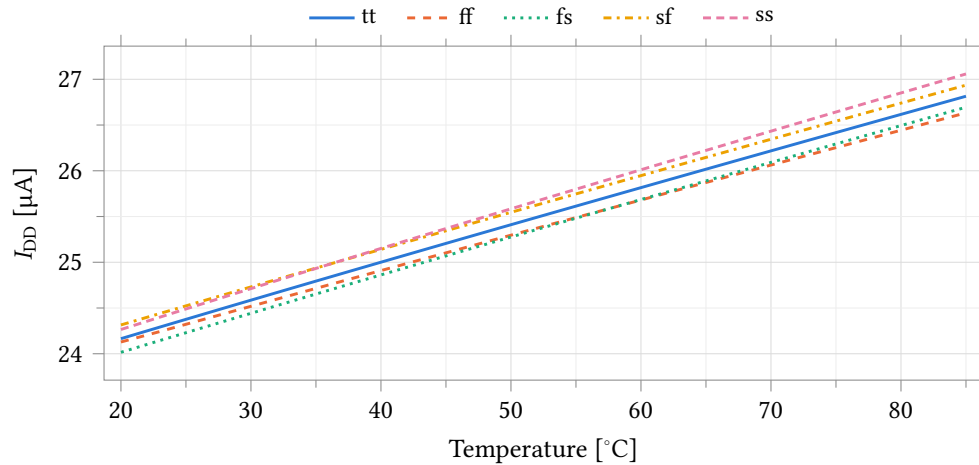


Figure 24.11: Quiescent supply current of the bias generator against temperature over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter                        | Unit       | tt    | ff    | fs    | sf    | ss    |
|----------------------------------|------------|-------|-------|-------|-------|-------|
| $V_{\text{BNM}}$                 | % $V^{-1}$ | 0.30  | 0.32  | 0.31  | 0.28  | 0.31  |
| $V_{\text{BNC}}$                 | % $V^{-1}$ | 0.53  | 0.55  | 0.54  | 0.50  | 0.55  |
| $V_{\text{DD}} - V_{\text{BPM}}$ | % $V^{-1}$ | 0.31  | 0.33  | 0.30  | 0.31  | 0.32  |
| $V_{\text{DD}} - V_{\text{BPC}}$ | % $V^{-1}$ | 0.61  | 0.62  | 0.60  | 0.61  | 0.65  |
| $I_{\text{DD}}$                  | % $V^{-1}$ | 18.74 | 20.21 | 18.55 | 19.23 | 17.98 |

Table 24.9: Line sensitivity: total variation over the simulated supply range, normalised to the mean and to the swept supply span. The PMOS rails are supply-referred (they track  $V_{\text{DD}}$  by design), so it is the supply-referred drops  $V_{\text{DD}} - V_{\text{BPM}}$  and  $V_{\text{DD}} - V_{\text{BPC}}$  that are quoted here, not the raw rail voltages, whose sensitivity against  $v_{\text{ss}}$ ! would simply restate that tracking. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

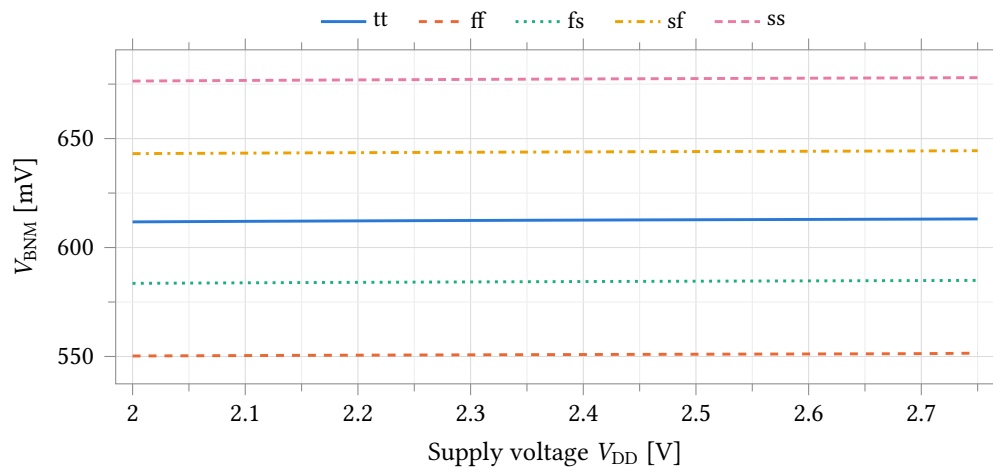


Figure 24.12: Supply rejection of the NMOS mirror bias:  $V_{\text{BNM}}$  against  $V_{\text{DD}}$  over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

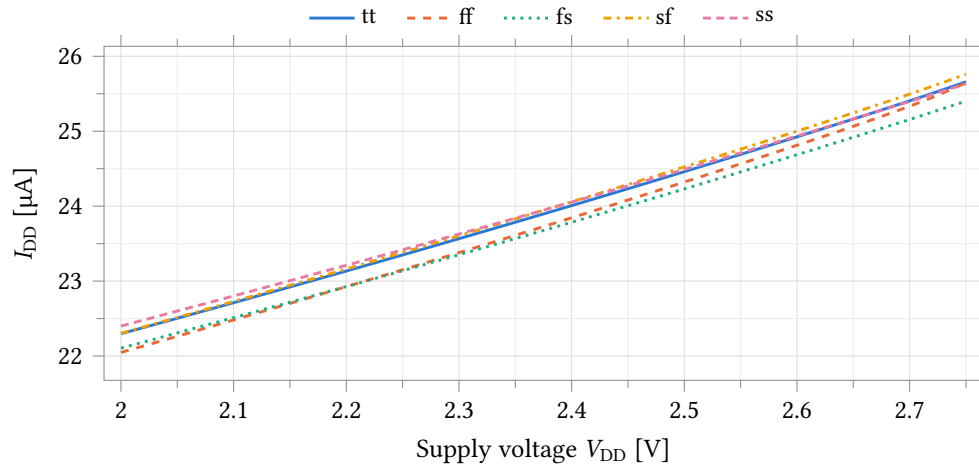


Figure 24.13: Quiescent supply current against supply voltage over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter        | Unit          | tt    | ff    | fs    | sf    | ss    |
|------------------|---------------|-------|-------|-------|-------|-------|
| $V_{\text{BNM}}$ | mV            | 89.6  | 87.5  | 88.7  | 89.7  | 91.0  |
| $V_{\text{BNC}}$ | mV            | 198.2 | 190.2 | 194.8 | 198.8 | 203.8 |
| $V_{\text{BPM}}$ | mV            | 98.9  | 96.4  | 98.3  | 98.6  | 100.7 |
| $V_{\text{BPC}}$ | mV            | 283.2 | 271.6 | 281.2 | 281.4 | 291.4 |
| $I_{\text{DD}}$  | $\mu\text{A}$ | 39.87 | 39.32 | 39.55 | 39.81 | 40.07 |

Table 24.10: Trim range: total swing of each quantity between BiasAdj codes 0 and 63. Those are the extremes of the six codes the bench samples. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

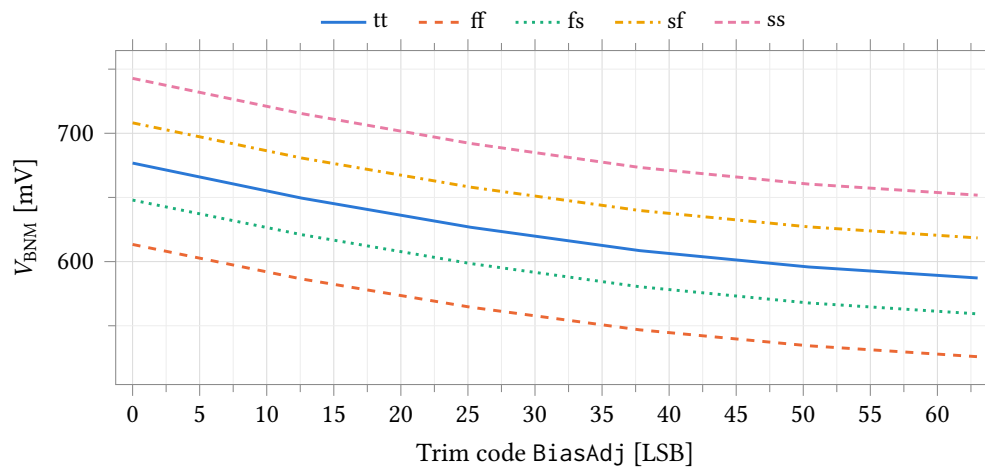


Figure 24.14: Trim characteristic:  $V_{\text{BNM}}$  against the BiasAdj word over all process corners. The bench sweeps the trim word as a continuous variable, so only the six codes 0, 12.6, 25.2, 37.8, 50.4 and 63 are sampled: this is the shape of the trim law, not a per-code characterisation. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

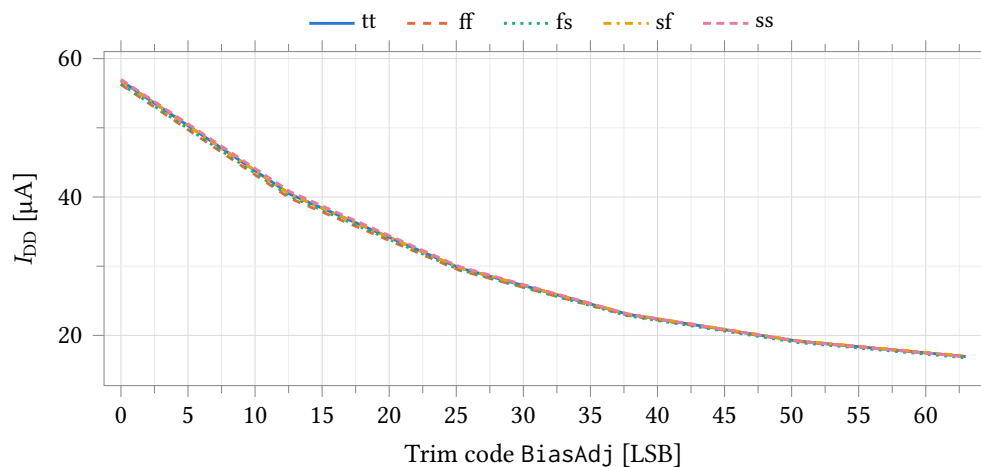


Figure 24.15: Trim characteristic: quiescent supply current against the BiasAdj word over all process corners, at the same six sampled codes. The current is strongly monotonic in the trim word, falling by a factor of about 3.4 from code 0 to code 63. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter        | Unit          | tt   | ff   | fs   | sf   | ss   |
|------------------|---------------|------|------|------|------|------|
| $V_{\text{BNM}}$ | $\mu\text{s}$ | 0.85 | 0.77 | 0.83 | 0.87 | 0.91 |
| $V_{\text{BNC}}$ | $\mu\text{s}$ | 0.95 | 0.87 | 0.93 | 0.97 | 1.01 |
| $V_{\text{BPM}}$ | $\mu\text{s}$ | 1.10 | 1.10 | 1.10 | 1.10 | 1.10 |
| $V_{\text{BPC}}$ | $\mu\text{s}$ | 1.10 | 1.10 | 1.10 | 1.10 | 1.10 |

Table 24.11: Power-up settling time of each bias rail, defined as the last instant at which the rail is still outside a 1 % band around its final value. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

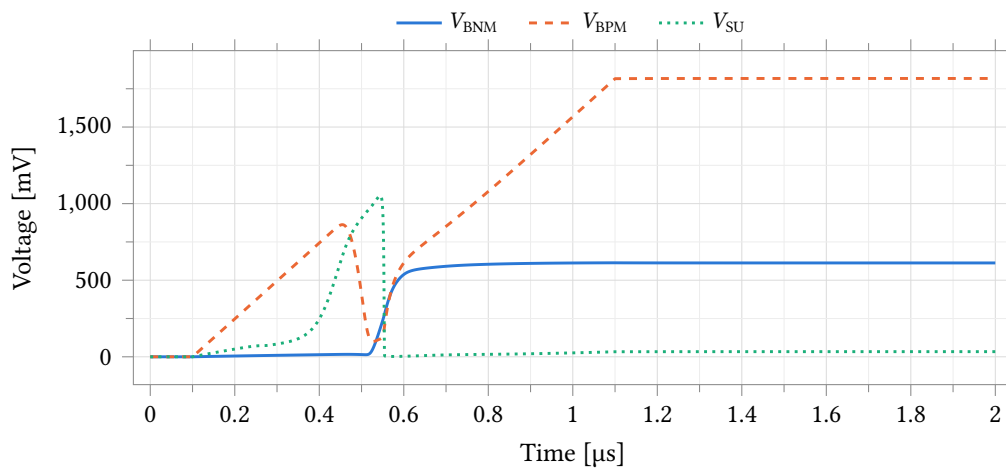


Figure 24.16: Power-up transient at the typical corner. The start-up node  $V_{\text{SU}}$  pulls the core out of its zero-current state, after which both mirror rails settle to their steady-state values.

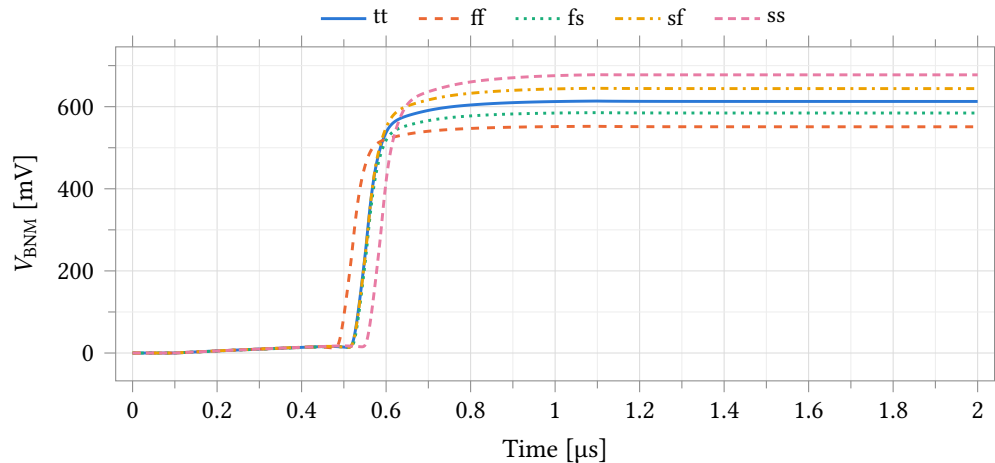


Figure 24.17: Power-up settling of  $V_{\text{BNM}}$  over all process corners. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

### 24.3.1 Start-up

Start-up is verified across 101 corner points (supply ramp rate over five decades, five process corners, four temperatures and three trim codes, plus a brown-out depth sweep and an enable cycle) and all 101 start, none into the degenerate state. The settled operating point is bit-identical across all five decades of ramp rate: the spread of  $V_{\text{BNM}}$  and of  $I_{\text{DD}}$  over the six ramp rates is zero to six digits at every process corner. Ramp rate changes when the generator arrives, not where.

| Supply ramp       | $t_{\text{set}}(V_{\text{BNM}})$ | ratio | $t_{\text{set}}(V_{\text{BPC}})$ | ratio |
|-------------------|----------------------------------|-------|----------------------------------|-------|
| 100 ns            | 211 ns                           | 2.11  | 232 ns                           | 2.32  |
| 1 $\mu\text{s}$   | 741 ns                           | 0.741 | 994 ns                           | 0.994 |
| 10 $\mu\text{s}$  | 4.40 $\mu\text{s}$               | 0.440 | 9.94 $\mu\text{s}$               | 0.994 |
| 100 $\mu\text{s}$ | 43.5 $\mu\text{s}$               | 0.435 | 99.4 $\mu\text{s}$               | 0.994 |
| 1 ms              | 436 $\mu\text{s}$                | 0.436 | 994 $\mu\text{s}$                | 0.994 |
| 10 ms             | 4.36 ms                          | 0.436 | 9.94 ms                          | 0.994 |

Table 24.12: Settling time of the two extreme bias rails against supply ramp rate, at tt and 25 °C, defined as the last instant outside a 1 % band around the value reached; the ratio is settling time over ramp time. Above 10  $\mu\text{s}$  the block is ramp-limited and the rails settle at a fixed fraction of the ramp; below 1  $\mu\text{s}$  it is circuit-limited, and the floor is 211 ns and 232 ns here, 487 ns and 641 ns at the slow corner.

**No brown-out floor exists.** The supply is ramped up, held, dipped and returned, at thirteen depths from 2.0 V down to 0 V and at three hold times. Every dip recovers exactly: the ratio of  $V_{\text{BNM}}$  after to before is 1.000 000 to six digits at every depth and every corner, and the supply current returns to its pre-dip value. How far the rails actually fall is the honest caveat: at 0 V the only discharge path for the rail decoupling is subthreshold leakage, so a short dip is not a power-down and  $V_{\text{BNM}}$  sags only 72 %. Held 100 ms at 0 V the rails do collapse, to 0.27 mV at the slow hot corner, and recovery is still exact. A recovery edge slower than 1 ms is untested.

The enable input is the stronger test, and it passes. De-asserted, it pulls both NMOS rails to ground

and holds the start-up node down, so the cell is forced into the true zero-current state:  $V_{\text{BNM}} = 0.000\,00\text{ V}$ , against the 96.8 % collapse a 100 ms power-down manages. On re-assert both the rails and the supply current return to 1.000 000 of their previous values at all six cells tested.

One limit on the coverage. The devices that do the starting sit on model cards that carry no per-instance mismatch terms, so a mismatch Monte Carlo of start-up as the cell stands would report zero contribution from exactly the mechanism under test, and none has been run. The results above are corner-based and do not depend on it: the process corner reaches those devices through the model-card parameters they do consume.

### 24.3.2 Measuring this block on chip

1. Set AFE\_DBGOWNSSEL = 5; step AFE\_DBGGSEL over slots 2–5 for the four global rails. A channel's own rails are per-channel slots 10 and 11 under AFE\_DBGSEL.
2. Read on the buffered pad atb\_o with AFE\_DBGBUFEN = 1, never on the direct pad: these are gate nets with picoamperes of drive, and the direct path moves what it measures.
3. Compare against the tt column of Table 24.6: 614.3 mV and 776.2 mV on the NMOS pair, 1815.4 mV and 1589.3 mV on the PMOS pair, corner spread 29 mV to 79 mV. A rail or near-zero on any of the four means the start-up branch did not fire; no measurement elsewhere on the die is meaningful until it does.
4. Trim from the supply pin, not through the multiplexer: step the 6-bit code and keep the one whose total analog supply current is nearest design, 2.7 % per code over a 39.9  $\mu\text{A}$  span (Table 24.10), against a 12.5 % Monte Carlo spread the corner runs do not show. The code is per die;  $\pm\frac{1}{2}$  code leaves  $\pm 1.4\%$ .
5. Recovery: chip-level slot 8 drives the reference-current node from atb\_d with AFE\_DBGDIREN = 1, the last path to a working amplifier on a die whose generator did not start.

This test fabric is design intent; it is not yet in the schematics.

## 24.4 Class-AB Amplifier

One cell, anatop\_tia\_amp\_ab (34 transistors; schematic, symbol, layout and extracted views), serves as both amplifiers of a potentiostat channel: the transimpedance amplifier A2 (Section 24.4.2) and the control amplifier A1 (Section 24.4.3). The cell name records its first application; the device is role-independent and the two roles differ only in the network around it. All data in this section is schematic-level, with no extracted parasitics.

The topology is drawn in Figure 24.18. A complementary rail-to-rail input stage places an NMOS and a PMOS pair in parallel, both active at the 1.25 V common mode used throughout this section; the two pairs fold into a cascode summing stage, which drives a class-AB output stage whose two output devices are both actively driven. Cascode (Ahuja) compensation is taken from the common source node of the PMOS cascodes, C1, and of the NMOS cascodes, C2. Both nodes are brought out as pins and returned to the output by one 500 fF capacitor each, outside the cell, so compensation is set per application without editing the cell.

Bias comes from the local bias generator of Section 24.3 over the global cascode rails, at trim code BiasAdj = 37 throughout this section.

The device-level characterisation below is taken at a 2.5 V supply into a 5 pF load on three benches: the unity-feedback bench of Figure 24.19, an open-loop bench with the input offset nulled before measurement, and a swept-load drive bench. Coverage is five corner points (the typical statistical point at nominal and 37 °C, and the ff, fs, sf and ss skew corners at 25 °C, uniform across every test of this bench) and a 200-sample Monte Carlo run varying process and mismatch together at 37 °C.

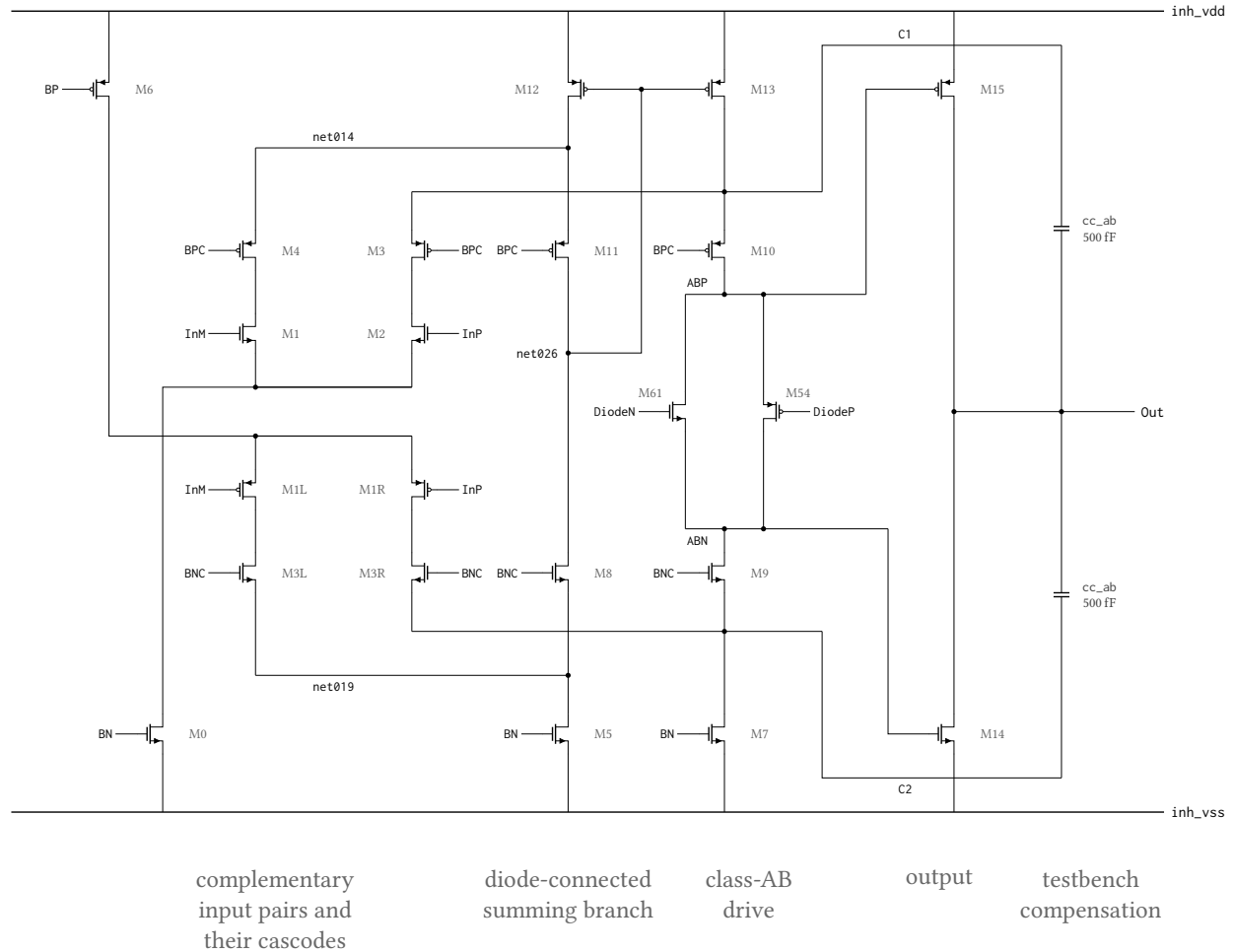


Figure 24.18: Signal-path topology of the class-AB amplifier `anatop_tia_amp_ab`, drawn from its netlist. The complementary pairs M1/M2 (NMOS, tail M0) and M1L/M1R (PMOS, tail M6) fold into two cascode branches; the InM side terminates in the diode-connected branch M12–M11–M8–M5, whose summing node `net026` sets M13, which restates that current into the InP branch, and the floating pair M54/M61 holds ABP above ABN so both output devices M15 and M14 are actively driven. C1 and C2 are the common source nodes of the PMOS cascodes M10/M3 and of the NMOS cascodes M9/M3R, brought out as pins; the two `cc_ab` capacitors that return them to `Out` are testbench components outside the cell, 500 fF per side. Simplified: the enable switch and its four power-down devices, and the bias generator that develops DiodeN/DiodeP, are omitted, so 22 of the cell’s 34 transistors are drawn; BP/BPC/BN/BNC are the enabled forms of the `bp!/bpc!/bn!/bnc!` globals and device dimensions are not shown. Identically labelled nets are connected; wires crossing without a filled dot are not.



| Point | Process | Temperature | Supply |
|-------|---------|-------------|--------|
| 1     | tt      | 37 °C       | 2.5 V  |
| 2     | ff      | 25 °C       | 2.5 V  |
| 3     | fs      | 25 °C       | 2.5 V  |
| 4     | sf      | 25 °C       | 2.5 V  |
| 5     | ss      | 25 °C       | 2.5 V  |

Table 24.13: Process, temperature and supply points simulated for the class-AB amplifier in its control-amplifier role. Point 1 is the typical statistical process point, evaluated with zero statistical offset; points 2–5 are the imported foundry skew corners. Temperature is uniform across the seven benches of this run at every point: 37 °C at point 1 and 25 °C at points 2–5, so the typical point carries the body-temperature condition and the skew corners do not. Every figure and table in this section is schematic-level. These points move the sixteen transistor model rows only (the resistor and capacitor rows stay at their typical sections), so every corner spread in this section is a lower bound (pinned-passive caveat, page 207).

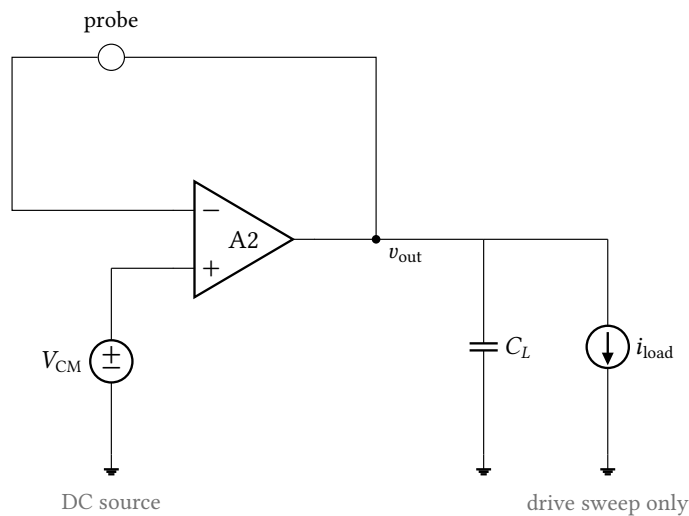


Figure 24.19: Unity-feedback bench. The amplifier’s output is returned to its inverting input through a current probe and loaded by  $C_L = 5$  pF; the loop-gain analysis through that probe gives the amplifier’s own gain and margins, and DC and noise analyses on the same wiring give the offset and output noise. The drive measurement adds the swept load current  $i_{load}$  on the same schematic. Supplies, the bias rails and the enable pin are omitted here and in Figure 24.22.

| Parameter                  | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|----------------------------|------|--------|--------|--------|--------|--------|--------|
| Open-loop gain $A_{OL}$    | dB   | 82.74  | 80.28  | 83.30  | 83.31  | 86.75  | 6.47   |
| Unity-gain frequency $f_u$ | MHz  | 22.474 | 22.942 | 22.621 | 22.587 | 22.358 | 0.584  |
| Phase margin               | °    | 73.43  | 74.26  | 73.81  | 73.43  | 72.76  | 1.49   |
| Gain margin                | dB   | 8.81   | 9.75   | 8.73   | 8.95   | 8.12   | 1.63   |

Table 24.14: Amplifier stability in unity-gain feedback by process corner. Unity-feedback bench (Figure 24.19) at  $v_{cm} = 1.25$  V,  $C_L = 5$  pF,  $C_c = 500$  fF, 2.5 V supply; the typical point is at 37 °C and the four skew corners at 25 °C. Measured through a probe in the feedback wire, so the quantity is the loop gain of the unity-gain configuration and not the amplifier loaded by an application network. Gain margin is read at the first crossing of zero loop phase above the unity-gain frequency. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter                       | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|---------------------------------|------|--------|--------|--------|--------|--------|--------|
| Open-loop gain $A_{OL}$         | dB   | 82.77  | 79.88  | 83.14  | 83.37  | 84.94  | 5.06   |
| Phase margin                    | °    | 73.52  | 74.70  | 74.04  | 72.86  | 73.34  | 1.84   |
| Dominant pole $f_{-3\text{dB}}$ | kHz  | 1.592  | 2.286  | 1.538  | 1.494  | 1.225  | 1.061  |
| Gain-bandwidth product          | MHz  | 22.781 | 23.187 | 22.894 | 22.939 | 22.587 | 0.600  |

Table 24.15: Open-loop small-signal response by process corner. The amplifier is driven differentially about mid-supply with no feedback,  $C_L = 5$  pF,  $C_c = 500$  fF, 2.5 V supply, 37 °C at the typical point and 25 °C at the four skew corners. The input offset is nulled before the sweep and the DC operating point is linear at every point (output 0.60 V to 1.78 V across corners), so the open-loop numbers are valid; they agree with the unity-feedback loop gain of Table 24.14 to within 1.8 dB and the phase margins to within 0.6°. The  $-3$  dB frequency varies inversely with the DC gain at a unity-gain frequency that is nearly corner-independent. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter            | Unit | tt    | ff    | fs    | sf    | ss    | Spread |
|----------------------|------|-------|-------|-------|-------|-------|--------|
| Input offset voltage | mV   | 0.622 | 0.649 | 0.624 | 0.569 | 0.456 | 0.193  |

Table 24.16: Systematic input offset by process corner. The differential input is swept about mid-supply into a 5 pF load, 2.5 V supply, 37 °C at the typical point and 25 °C at the four skew corners; the entry is the differential input at which the output crosses mid-supply. These are corner figures with zero statistical offset, so they carry only the topology's systematic term: the random offset a manufactured channel carries is the Monte Carlo result of Table 24.19 and is an order of magnitude larger.

| Parameter                             | Unit     | tt     | ff     | fs     | sf     | ss     | Spread |
|---------------------------------------|----------|--------|--------|--------|--------|--------|--------|
| Load current at 50 mV error, sourcing | mA       | 2.224  | 2.564  | 2.239  | 2.331  | 2.051  | 0.513  |
| Load current at 50 mV error, sinking  | mA       | −2.159 | −2.551 | −2.334 | −2.099 | −1.908 | 0.643  |
| Closed-loop output resistance         | $\Omega$ | 1.811  | 1.646  | 1.862  | 1.484  | 1.612  | 0.378  |

Table 24.17: Output drive by process corner. The amplifier is in unity-gain feedback at  $v_{cm} = 1.25$  V with  $C_L = 5$  pF,  $C_c = 500$  fF, 2.5 V supply and the load current swept  $-10$  mA to  $10$  mA; the typical point is at  $37^\circ\text{C}$  and the four skew corners at  $25^\circ\text{C}$ . The first two entries are the load current at which the output departs from  $v_{cm}$  by 50 mV, not a hard current limit; load current is signed positive out of the amplifier, so the sinking entry is negative. Output resistance is the slope of the same curve at zero load and is the quantity that sets load regulation in Table 24.35. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter      | Unit          | tt     | ff     | fs     | sf     | ss     | Spread |
|----------------|---------------|--------|--------|--------|--------|--------|--------|
| Supply current | $\mu\text{A}$ | 121.03 | 141.34 | 114.20 | 122.81 | 111.14 | 30.20  |
| Dissipation    | $\mu\text{W}$ | 302.57 | 353.34 | 285.50 | 307.03 | 277.85 | 75.49  |

Table 24.18: Quiescent supply current and dissipation by process corner, at the DC operating point of the open-loop bench with both inputs at mid-supply and no load. 2.5 V supply,  $37^\circ\text{C}$  at the typical point and  $25^\circ\text{C}$  at the four skew corners. The measurement is taken at the supply terminal of the bench and therefore covers the amplifier together with the local bias generator that feeds it; at the typical point the amplifier draws  $97\ \mu\text{A}$  of the  $121\ \mu\text{A}$  total and the bias generator the remainder. One local bias generator serves one channel, so this is the per-channel budget. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

### 24.4.1 Monte Carlo

200 samples, process and mismatch together, 37 °C, under the conditions of Section 24.4. Per-channel yield is the fraction of samples inside the specification entered on the test; the four-channel figure is its fourth power, the four channels being nominally identical and assumed independent.

Input offset has a mean of 0.583 mV and  $\sigma = 5.201$  mV (Figure 24.20, Table 24.19); against the  $\pm 1$  mV limit entered on the test the yield is 12.0 % per channel and 0.0 % over four. In unity-gain feedback the phase margin averages 73.40° with a worst sample of 69.21° and the gain margin 8.89 dB with a worst sample of 6.98 dB, so all 200 samples clear the 60° and 6 dB limits; open-loop gain averages 82.64 dB in that configuration (Table 24.20) and 82.16 dB on the open-loop bench (Table 24.21). Output noise integrated 0.1 Hz to 100 kHz is 30.44  $\mu$ V (Table 24.20). Drive at 50 mV of output error is 2.224 mA sourcing and  $-2.158$  mA sinking, mean  $\pm 3\sigma$  of  $\pm 0.252$  mA and  $\pm 0.313$  mA (Table 24.23). Supply current, which covers the amplifier together with the local bias generator that feeds it, has a median of 122.81  $\mu$ A over a 1st-to-99th percentile range of 91.58  $\mu$ A to 172.09  $\mu$ A, for a median dissipation of 307.0  $\mu$ W (Figure 24.21, Table 24.22).

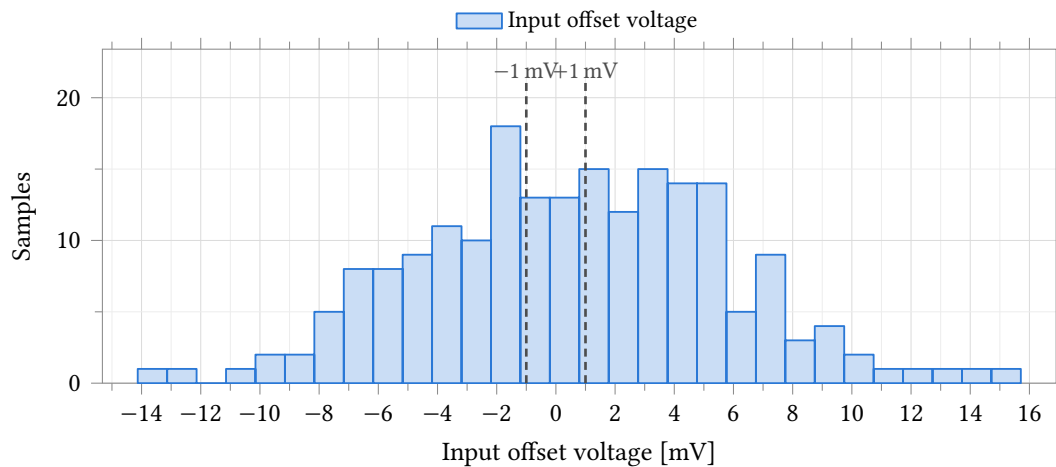


Figure 24.20: Input offset voltage of the amplifier. **Process and mismatch**, 37 °C, 200 samples, 2.5 V supply,  $C_L = 5$  pF,  $C_c = 500$  fF, a 5 pF load, differential input swept about mid-supply. The distribution is symmetric about 0.58 mV, so the systematic term of Table 24.16 is small against the random one, and the  $\pm 1$  mV rules are the limits entered on the test; 24 of 200 samples fall inside them. Statistics are in Table 24.19.

| Parameter            | Unit | N   | Mean  | $\sigma$ | Mean $\pm 3\sigma$ | Min     | Max    | Spec        | Yield | 4 ch |
|----------------------|------|-----|-------|----------|--------------------|---------|--------|-------------|-------|------|
| Input offset voltage | mV   | 200 | 0.583 | 5.201    | 0.583 $\pm$ 15.604 | -14.130 | 15.709 | $\pm 1.000$ | 12.0  | 0.0  |

Table 24.19: Monte Carlo input offset voltage: **process and mismatch**, 37 °C, 200 samples, 2.5 V supply,  $C_L = 5$  pF,  $C_c = 500$  fF, a 5 pF load. The distribution is symmetric (Figure 24.20), so mean  $\pm 3\sigma$  is quoted here and not for the skewed quantities in the tables that follow. Spec and yield are the  $\pm 1$  mV limits entered on the test; the four-channel column assumes independent channels. This run varies process as well as mismatch, so  $\sigma$  is die-to-die and is an upper bound on the within-die channel-to-channel term. The same offset appears at the reference electrode as the zero-current tracking error of Table 24.38.

| Parameter                       | Unit          | $N$ | Mean  | $\sigma$ | $p_1$ | $p_{99}$ | Worst | Spec           | Yield | 4 ch  |
|---------------------------------|---------------|-----|-------|----------|-------|----------|-------|----------------|-------|-------|
| Open-loop gain $A_{OL}$         | dB            | 200 | 82.64 | 1.20     | 79.98 | 85.52    | 79.44 | $\geq 60.00$   | 100.0 | 100.0 |
| Phase margin                    | $^\circ$      | 200 | 73.40 | 1.37     | 69.60 | 75.73    | 69.21 | $\geq 60.00$   | 100.0 | 100.0 |
| Gain margin                     | dB            | 200 | 8.89  | 0.71     | 7.19  | 10.35    | 6.98  | $\geq 6.00$    | 100.0 | 100.0 |
| Output noise, 0.1 Hz to 100 kHz | $\mu\text{V}$ | 200 | 30.44 | 1.04     | 28.32 | 33.11    | 34.02 | $\leq 1220.00$ | 100.0 | 100.0 |

Table 24.20: Monte Carlo amplifier stability and noise in unity-gain feedback: **process and mismatch**,  $37^\circ\text{C}$ , 200 samples,  $v_{\text{cm}} = 1.25\text{ V}$ ,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ ,  $2.5\text{ V}$  supply. This is the amplifier alone, with no application network in the loop, and is not comparable with Table 24.37. Every sample clears all three stability limits entered on the test, and percentiles and the extreme sample are quoted rather than  $\pm 3\sigma$  because the margins are not normally distributed. The noise row is the output noise of this configuration integrated 0.1 Hz to 100 kHz, within the 0.1 Hz to 100 MHz span of the analysis.

| Parameter                        | Unit     | $N$ | Mean   | $\sigma$ | $p_1$  | $p_{99}$ | Worst | Spec         | Yield | 4 ch  |
|----------------------------------|----------|-----|--------|----------|--------|----------|-------|--------------|-------|-------|
| Open-loop gain $A_{OL}$          | dB       | 200 | 82.16  | 1.06     | 80.04  | 84.60    | 79.50 | $\geq 40.00$ | 100.0 | 100.0 |
| Phase margin                     | $^\circ$ | 200 | 73.62  | 1.42     | 69.80  | 76.21    | 68.93 | $\geq 40.00$ | 100.0 | 100.0 |
| Dominant pole $f_{-3\text{ dB}}$ | kHz      | 200 | 1.723  | 0.276    | 1.158  | 2.453    | –     | –            | –     | –     |
| Gain-bandwidth product           | MHz      | 200 | 22.815 | 3.566    | 16.191 | 31.741   | –     | –            | –     | –     |

Table 24.21: Monte Carlo open-loop response: **process and mismatch**,  $37^\circ\text{C}$ , 200 samples, amplifier driven differentially about mid-supply with no feedback,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ ,  $2.5\text{ V}$  supply. The input offset is nulled per sample from the offset measurement of the same run, leaving the DC operating point linear on every sample; all four quantities are measurable on all 200. Only gain and phase margin carry limits; the two bandwidth rows were not graded and their spec, yield and worst-case columns are therefore empty. Gain agrees with the unity-feedback loop gain of Table 24.20 to within 0.5 dB of mean. Percentiles are quoted rather than  $\pm 3\sigma$ : the two bandwidth distributions are right-skewed, both being inversely proportional to a transconductance.

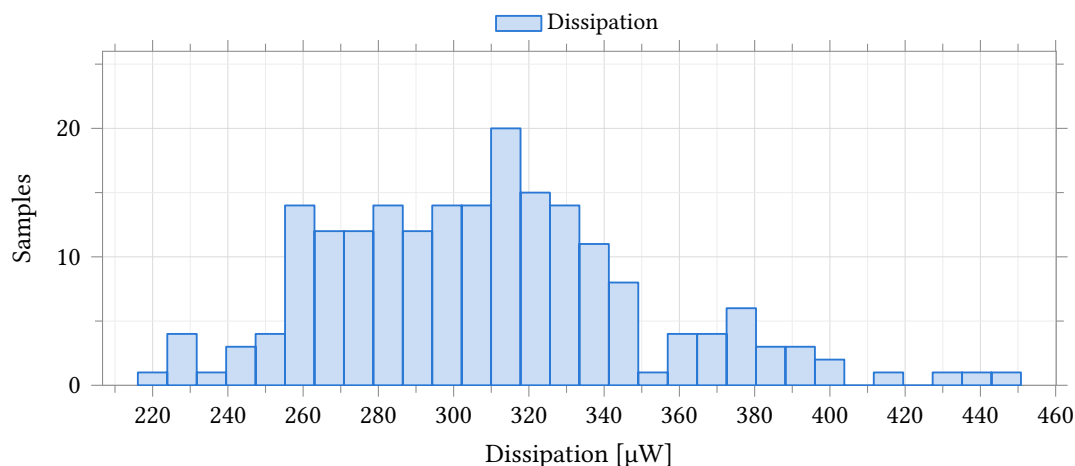


Figure 24.21: Quiescent dissipation of one channel's control amplifier and its local bias generator together. **Process and mismatch**,  $37^\circ\text{C}$ , 200 samples,  $2.5\text{ V}$  supply, both inputs at mid-supply, no load. The distribution is unimodal and right-skewed, the tail belonging to the fast-process samples in which every bias branch runs harder; no limit was entered on this output, so none is drawn. The measurement is taken at the supply terminal of the bench, so it is the per-channel budget and not the amplifier alone.

| Parameter      | Unit          | $N$ | Mean   | $\sigma$ | Median | $p_1$ | $p_{99}$ | Min   | Max    |
|----------------|---------------|-----|--------|----------|--------|-------|----------|-------|--------|
| Supply current | $\mu\text{A}$ | 200 | 123.42 | 16.80    | 122.81 | 91.58 | 172.09   | 86.43 | 180.31 |
| Dissipation    | $\mu\text{W}$ | 200 | 308.6  | 42.0     | 307.0  | 228.9 | 430.2    | 216.1 | 450.8  |

Table 24.22: Monte Carlo quiescent supply current and dissipation. **Process and mismatch**, 37 °C, 200 samples, 2.5 V supply, both inputs at mid-supply, no load. The figure covers one channel’s control amplifier together with the local bias generator that feeds it, measured at the supply terminal of the bench; at the typical point the amplifier draws 97  $\mu\text{A}$  of the 121  $\mu\text{A}$  total and the bias generator the remainder. Both distributions are the same measurement and are right-skewed (Figure 24.21), so the median and percentiles are quoted and mean  $\pm 3\sigma$  is not. No limit was entered on either output, so the spec and yield columns are omitted rather than left blank.

| Parameter              | Unit     | $N$ | Mean   | $\sigma$ | Mean $\pm 3\sigma$ | Min    | Max    | Spec         | Yield | 4 ch  |
|------------------------|----------|-----|--------|----------|--------------------|--------|--------|--------------|-------|-------|
| Load current, sourcing | mA       | 200 | 2.224  | 0.084    | 2.224 $\pm$ 0.252  | 2.041  | 2.443  | $\geq 0.033$ | 100.0 | 100.0 |
| Load current, sinking  | mA       | 200 | -2.158 | 0.104    | -2.158 $\pm$ 0.313 | -2.511 | -1.860 | –            | –     | –     |
| Output resistance      | $\Omega$ | 200 | 1.850  | 0.252    | 1.850 $\pm$ 0.757  | 1.218  | 2.908  | –            | –     | –     |

Table 24.23: Monte Carlo output drive. **Process and mismatch**, 37 °C, 200 samples, amplifier in unity-gain feedback at  $v_{\text{cm}} = 1.25\text{ V}$ ,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ , 2.5 V supply, load current swept  $-10\text{ mA}$  to  $10\text{ mA}$ . The first two entries are the load current at which the output departs from  $v_{\text{cm}}$  by 50 mV, not a hard current limit; load current is signed positive out of the amplifier, so the sinking entry is negative. All three distributions are symmetric, so mean  $\pm 3\sigma$  applies. Every sample clears the sourcing limit entered on the test; the other two rows were not graded.

#### 24.4.2 As the Transimpedance Amplifier (A2)

As A2 the amplifier holds the working electrode at  $v_{\text{cm}}$  on its inverting input and converts the electrode current across the feedback resistor,  $V_{\text{OUT}} = v_{\text{cm}} - I_{\text{WE}}R_f$ . Both current polarities pass through  $R_f$ , so the ADC sees offset binary about the  $v_{\text{cm}}$  code.  $R_f$  is a 16-tap string spanning 35 k $\Omega$  to 560 k $\Omega$ .

The bench is Figure 24.22:  $R_f = 560\text{ k}\Omega$ , the only tap characterised here and the only one to which the figures of this part apply, with  $C_f = 22\text{ pF}$ , a 5 pF load and the Randles electrode of Figure 24.1 at  $R_s = 100\text{ }\Omega$ ,  $R_{\text{ct}} = 1\text{ M}\Omega$ ,  $C_{\text{dl}} = 100\text{ nF}$ . The typical point and the Monte Carlo run are at 37 °C, the four skew corners at 25 °C.

**Monte Carlo.** 200 samples, process and mismatch together, 37 °C, at the conditions above. Output offset at zero electrode current has a mean of  $-1.08\text{ mV}$  and  $\sigma = 8.91\text{ mV}$  (Figure 24.25, Table 24.29); against the  $\pm 5\text{ mV}$  limit entered on the test the yield is 48.0 % per channel and 5.3 % over four. Loop gain at DC averages 78.65 dB, and every sample clears the 45° phase-margin limit, the worst at 64.7° (Figure 24.26, Table 24.31). Signal bandwidth averages 8.84 kHz against a 1 kHz floor met on all 200 samples (Table 24.32). Input current noise integrated 0.1 Hz to 100 Hz is 341.5 pA against the 2 nA channel budget (Figure 24.27, Table 24.33); it is referred to the 560 k $\Omega$  tap and scales inversely with the tap selected.

#### 24.4.3 As the Potentiostat Control Amplifier (A1)

As A1 the amplifier drives the counter electrode and closes its loop on the reference electrode, holding that electrode at the commanded potential. The commanded potential is swept over the full 0 V to 2.5 V supply range.

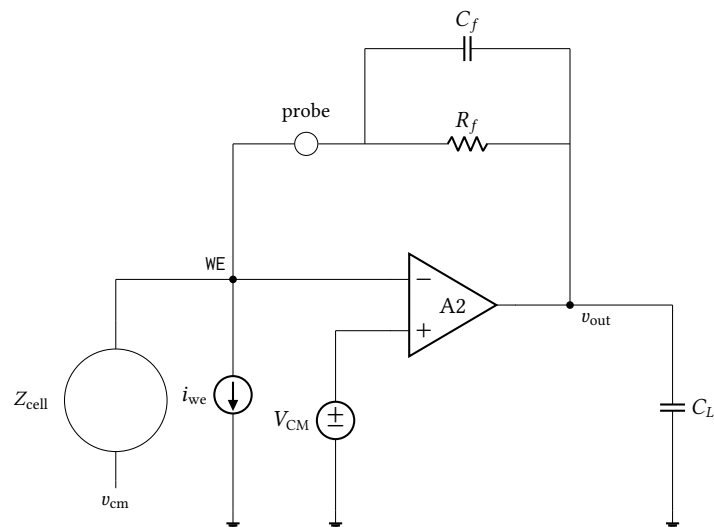


Figure 24.22: The closed transimpedance loop bench, the only one in this section that closes feedback around the electrode.  $R_f \parallel C_f$  returns the output to the working electrode through a current probe,  $i_{we}$  is the test source, and  $Z_{cell}$  is the Randles cell of Figure 24.1, returned to  $v_{cm}$ . The loop gain, phase margin and crossover are measured through the probe; the DC accuracy, closed-loop transfer and noise figures come from the same wiring.

| Point | Process | Temperature | Supply |
|-------|---------|-------------|--------|
| 1     | tt      | 37 °C       | 2.5 V  |
| 2     | ff      | 25 °C       | 2.5 V  |
| 3     | fs      | 25 °C       | 2.5 V  |
| 4     | sf      | 25 °C       | 2.5 V  |
| 5     | ss      | 25 °C       | 2.5 V  |

Table 24.24: Process, temperature and supply points simulated for the transimpedance application. Point 1 is the typical statistical process point, evaluated with zero statistical offset, at the 37 °C on-body temperature; points 2–5 are the imported foundry skew corners at 25 °C. Every figure and table in this part is schematic-level and covers only the  $R_f = 560 \text{ k}\Omega$  feedback tap; nothing here characterises the other fifteen taps of the gain string. These points move the sixteen transistor model rows only (the resistor and capacitor rows stay at their typical sections), so every corner spread in this section is a lower bound (pinned-passive caveat, page 207).

| Parameter                     | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|-------------------------------|------|--------|--------|--------|--------|--------|--------|
| Output offset at zero current | mV   | -0.967 | -1.056 | -0.987 | -0.832 | -0.762 | 0.294  |
| Gain error                    | ppm  | -167.3 | -221.9 | -150.6 | -161.0 | -106.3 | 115.6  |
| Integral non-linearity        | LSB  | 0.021  | 0.033  | 0.016  | 0.023  | 0.011  | 0.022  |

Table 24.25: DC accuracy by process corner, closed transimpedance loop bench (Figure 24.22):  $R_f = 560 \text{ k}\Omega$ ,  $C_f = 22 \text{ pF}$ ,  $C_L = 5 \text{ pF}$ ,  $C_c = 500 \text{ fF}$ ,  $\text{BiasAdj} = 37$ ,  $v_{\text{cm}} = 1.25 \text{ V}$ ,  $2.5 \text{ V}$  supply, Randles electrode with  $R_s = 100 \Omega$ ,  $R_{\text{ct}} = 1 \text{ M}\Omega$ ,  $C_{\text{dl}} = 100 \text{ nF}$ . Point tt is the typical statistical point at  $37^\circ \text{C}$ ; the four skew corners are at  $25^\circ \text{C}$ . Gain error and INL are evaluated over the central  $\pm 0.9$  of the full-scale input-current range and are referred to the nominal  $2.441 \text{ mV}$  ADC LSB; the converter’s measured step is  $1.591 \text{ mV}$  (Table 24.58), so multiply the LSB row by  $1.535$  to read it as converter codes. These are corner figures with zero statistical offset: the offset a manufactured channel carries is the Monte Carlo result of Table 24.29. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

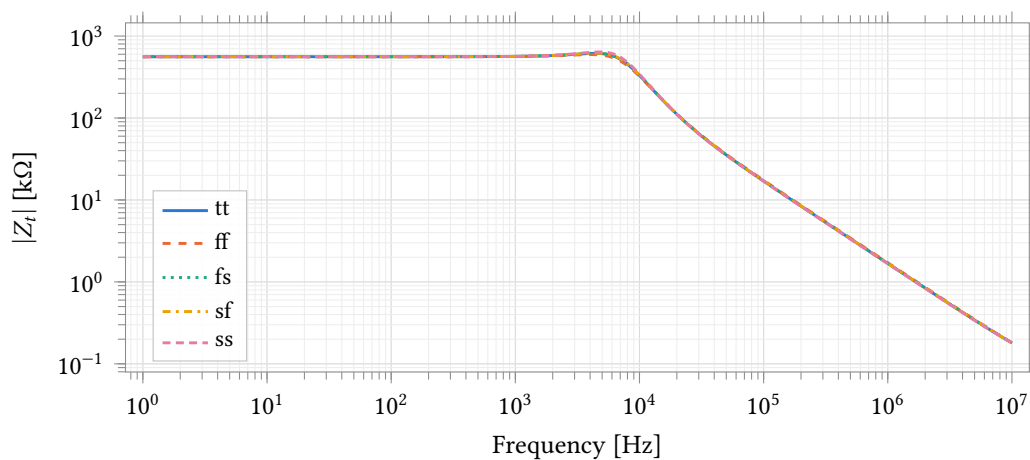


Figure 24.23: Closed-loop transimpedance magnitude over five process corners, conditions as Figure 24.24. The AC stimulus is a  $1 \text{ A}$  current into the working electrode, so the ordinate is transimpedance directly. The feedback resistor is an ideal element here rather than the 16-tap string, which is why the corners separate only near the band edge; the numbers are in Table 24.26. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter                          | Unit             | tt      | ff      | fs      | sf      | ss      | Spread |
|------------------------------------|------------------|---------|---------|---------|---------|---------|--------|
| Transimpedance at $1 \text{ Hz}$   | $\text{k}\Omega$ | 559.933 | 559.913 | 559.937 | 559.938 | 559.957 | 0.044  |
| Deviation from $R_f$               | ppm              | -118.8  | -155.7  | -111.9  | -110.7  | -76.2   | 79.4   |
| Signal bandwidth $f_{-3\text{dB}}$ | $\text{kHz}$     | 8.868   | 8.651   | 8.927   | 8.935   | 9.149   | 0.498  |
| Gain peaking                       | $\text{dB}$      | 0.812   | 0.523   | 0.850   | 0.860   | 1.156   | 0.633  |

Table 24.26: Small-signal transimpedance by process corner, bench and conditions as Table 24.25. The feedback resistor is ideal in this measurement, so the deviation of  $Z_t(1 \text{ Hz})$  from  $R_f$  is set by finite loop gain alone and carries no resistor tolerance. Peaking is referred to  $Z_t(1 \text{ Hz})$  and the bandwidth is the  $-3 \text{ dB}$  point of the same curve. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).



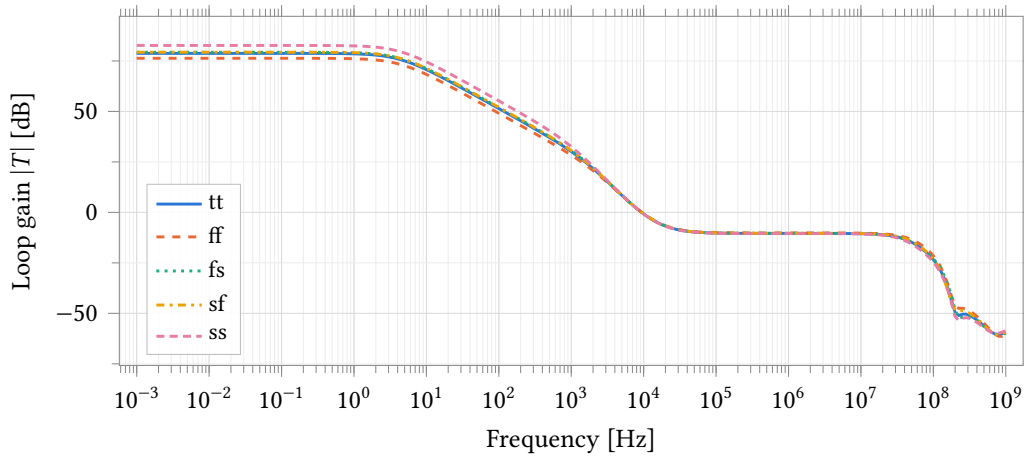


Figure 24.24: Transimpedance loop gain over five process corners, measured through a probe in series with the feedback network (Figure 24.22).  $R_f = 560\text{ k}\Omega$ ,  $C_f = 22\text{ pF}$ ,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ ,  $\text{BiasAdj} = 37$ ,  $2.5\text{ V}$ , Randles electrode with  $R_s = 100\text{ }\Omega$ ,  $R_{ct} = 1\text{ M}\Omega$ ,  $C_{dl} = 100\text{ nF}$ .  $tt$  is at  $37^\circ\text{C}$ , the four skew corners at  $25^\circ\text{C}$ . This is the loop around the amplifier, the feedback network and the electrode together, not the amplifier's own gain; the two are tabulated separately in Table 24.27 and Table 24.14. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter       | Unit     | tt    | ff    | fs    | sf    | ss    | Spread |
|-----------------|----------|-------|-------|-------|-------|-------|--------|
| Loop gain at DC | dB       | 78.74 | 76.32 | 79.28 | 79.32 | 82.69 | 6.37   |
| Loop crossover  | kHz      | 9.375 | 9.475 | 9.428 | 9.428 | 9.360 | 0.115  |
| Phase margin    | $^\circ$ | 76.50 | 80.30 | 76.23 | 76.13 | 72.78 | 7.52   |
| Gain margin     | dB       | 21.81 | 20.67 | 21.90 | 21.24 | 22.49 | 1.82   |

Table 24.27: Transimpedance loop stability by process corner, conditions as Table 24.25. Measured through the feedback probe of Figure 24.22, so the loop includes the amplifier, the feedback network and the electrode; the amplifier's own margins are in Table 24.14. Gain margin is read at the first crossing of zero loop phase above the unity-gain frequency. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter                              | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|----------------------------------------|------|--------|--------|--------|--------|--------|--------|
| Output noise, 0.1 Hz to 1000 Hz        | mV   | 1.690  | 1.543  | 1.656  | 1.647  | 1.753  | 0.210  |
| Input current noise, 0.1 Hz to 10 Hz   | pA   | 68.23  | 62.35  | 66.73  | 66.54  | 70.61  | 8.27   |
| Input current noise, 0.1 Hz to 100 Hz  | pA   | 340.24 | 310.83 | 333.13 | 331.71 | 352.48 | 41.65  |
| Input current noise, 0.1 Hz to 1000 Hz | nA   | 3.017  | 2.755  | 2.957  | 2.941  | 3.130  | 0.375  |

Table 24.28: Noise by process corner, conditions as Table 24.25. The noise analysis runs 0.1 Hz to 10 kHz at 20 points per decade; the three input-referred rows are the output noise integrated to 10 Hz, 100 Hz and 1 kHz and divided by  $R_f$ , and no figure is integrated beyond the analysis stop frequency. The input-referred rows therefore scale inversely with the selected gain tap. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

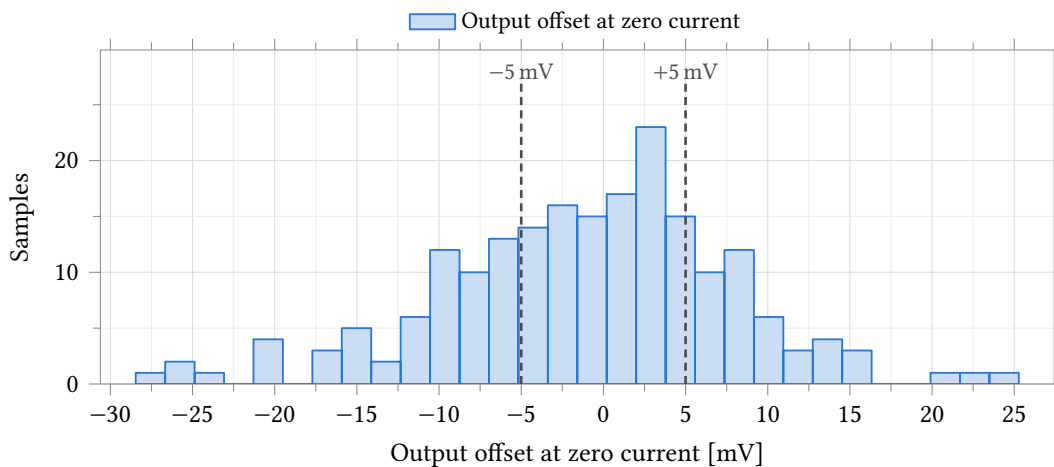


Figure 24.25: Output offset at zero electrode current, referred to  $v_{cm}$ . **Process and mismatch**, 37 °C, 200 samples,  $R_f = 560\text{ k}\Omega$ ,  $C_f = 22\text{ pF}$ ,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ , BiasAdj = 37, 2.5 V. The rules are the  $\pm 5\text{ mV}$  limits entered on the test.  $\sigma = 8.91\text{ mV}$  (sample deviation, as in every table of this section); 96 of 200 samples lie within the limits, 5.3 % across four independent channels. Statistics are in Table 24.29.

| Parameter                     | Unit | $N$ | Mean  | $\sigma$ | Mean $\pm 3\sigma$ | Min    | Max   | Spec       | Yield | 4 ch |
|-------------------------------|------|-----|-------|----------|--------------------|--------|-------|------------|-------|------|
| Output offset at zero current | mV   | 200 | -1.08 | 8.91     | $-1.08 \pm 26.72$  | -28.46 | 25.28 | $\pm 5.00$ | 48.0  | 5.3  |

Table 24.29: Monte Carlo output offset at zero electrode current. **Process and mismatch**, 37 °C, 200 samples,  $R_f = 560\text{ k}\Omega$ ,  $C_f = 22\text{ pF}$ ,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ , BiasAdj = 37, 2.5 V, Randles electrode with  $R_s = 100\text{ }\Omega$ ,  $R_{ct} = 1\text{ M}\Omega$ ,  $C_{dl} = 100\text{ nF}$ . The distribution is approximately Gaussian, so mean  $\pm 3\sigma$  is quoted here and not for the skewed quantities in the tables that follow; the worst sample,  $-28.5\text{ mV}$ , lies just outside it. Spec and yield are the  $\pm 5\text{ mV}$  limits entered on the test; the four-channel column assumes independent channels. This run varies process as well as mismatch, so  $\sigma$  is die-to-die and is an upper bound on the within-die channel-to-channel term.

| Parameter              | Unit       | $N$ | Mean   | $\sigma$ | $p_1$  | $p_{99}$ | Worst | Spec         | Yield | 4 ch  |
|------------------------|------------|-----|--------|----------|--------|----------|-------|--------------|-------|-------|
| Gain error             | ppm        | 200 | -171.7 | 23.1     | -235.5 | -128.0   | 249.3 | $\pm 2000.0$ | 100.0 | 100.0 |
| Transimpedance at 1 Hz | k $\Omega$ | 200 | 559.93 | 0.01     | 559.91 | 559.95   | —     | —            | —     | —     |
| Deviation from $R_f$   | ppm        | 200 | -121.3 | 17.4     | -165.2 | -89.2    | —     | —            | —     | —     |
| Integral non-linearity | LSB        | 200 | 0.022  | 0.006    | 0.012  | 0.040    | 0.051 | $\leq 0.500$ | 100.0 | 100.0 |
| Total unadjusted error | LSB        | 200 | 2.90   | 2.34     | 0.14   | 10.62    | 11.72 | $\leq 4.00$  | 75.0  | 31.6  |

Table 24.30: Monte Carlo transfer accuracy, conditions as Table 24.29. Gain error, INL and total unadjusted error are evaluated over the central  $\pm 0.9$  of the full-scale input-current range and the two LSB rows are referred to the nominal 2.441 mV ADC LSB; the converter’s measured step is 1.591 mV (Table 24.58), so multiply those two rows by 1.535 to read them as converter codes. Total unadjusted error correlates with  $|V_{OS}|$  at  $r = +1.000$  across the 200 samples: at this gain tap it is the offset of Table 24.29 expressed in LSB, not an independent result, and its 4 LSB limit corresponds to an offset of about 9.8 mV, which is why its yield is not the offset yield. Gain error and INL are set by the amplifier and pass on every sample; the feedback resistor is an ideal element here and contributes no tolerance. Percentiles and worst case are quoted rather than  $\pm 3\sigma$  because INL and total unadjusted error are right-skewed.

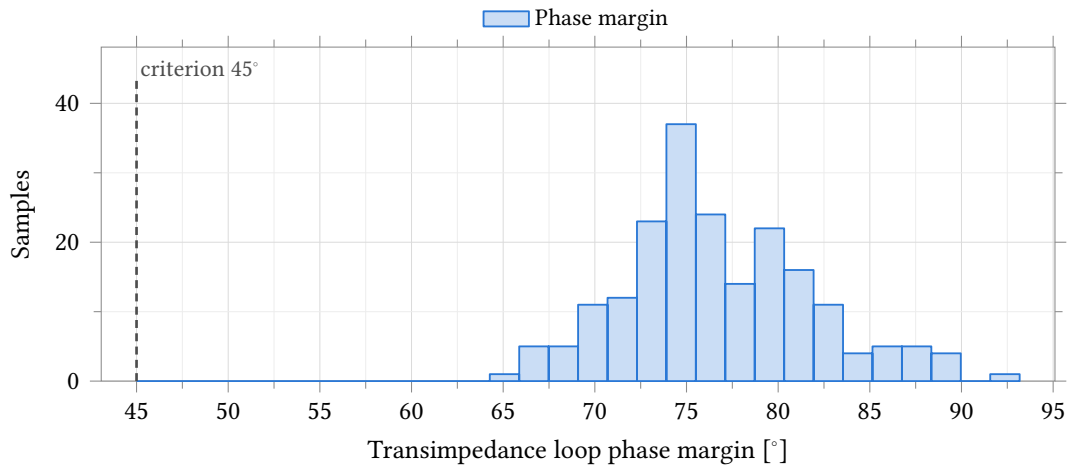


Figure 24.26: Phase margin of the transimpedance loop, conditions as Figure 24.25. The loop is measured through the feedback probe of Figure 24.22 and includes the amplifier, the feedback network and the Randles electrode ( $R_s = 100 \Omega$ ,  $R_{ct} = 1 \text{ M}\Omega$ ,  $C_{dl} = 100 \text{ nF}$ ). All 200 samples clear the  $45^\circ$  criterion, minimum  $64.7^\circ$ . The distribution is right-skewed, so percentiles and the extreme sample are quoted in Table 24.31 rather than  $\pm 3\sigma$ .

| Parameter       | Unit     | N         | Mean  | $\sigma$ | $p_1$ | $p_{99}$ | Worst | Spec         | Yield | 4 ch  |
|-----------------|----------|-----------|-------|----------|-------|----------|-------|--------------|-------|-------|
| Loop gain at DC | dB       | 200       | 78.65 | 1.23     | 75.89 | 81.26    | 74.89 | $\geq 60.00$ | 100.0 | 100.0 |
| Phase margin    | $^\circ$ | 200       | 76.7  | 5.2      | 66.4  | 89.6     | 64.7  | $\geq 45.0$  | 100.0 | 100.0 |
| Gain margin     | dB       | 167 / 200 | 22.06 | 0.64     | 20.79 | 23.70    | –     | –            | –     | –     |
| Loop crossover  | kHz      | 200       | 9.41  | 0.98     | 7.56  | 11.85    | –     | –            | –     | –     |

Table 24.31: Monte Carlo transimpedance loop stability, conditions as Table 24.29. Measured through the feedback probe of Figure 24.22, so the loop includes the amplifier, the feedback network and the electrode. Gain margin reports  $N = 167/200$  because the loop phase never reaches the critical value on the other 33 samples and no gain margin exists for them; those samples are held out of the statistics and are not failures. Percentiles and worst case are quoted rather than  $\pm 3\sigma$  for the margins, which are not normally distributed.

| Parameter                          | Unit | N   | Mean | $\sigma$ | Mean $\pm 3\sigma$ | Min  | Max  | Spec        | Yield | 4 ch  |
|------------------------------------|------|-----|------|----------|--------------------|------|------|-------------|-------|-------|
| Signal bandwidth $f_{-3\text{dB}}$ | kHz  | 200 | 8.84 | 0.34     | $8.84 \pm 1.03$    | 7.90 | 9.73 | $\geq 1.00$ | 100.0 | 100.0 |
| Gain peaking                       | dB   | 200 | 0.82 | 0.23     | $0.82 \pm 0.68$    | 0.30 | 1.52 | $\leq 2.00$ | 100.0 | 100.0 |

Table 24.32: Monte Carlo closed-loop bandwidth and gain peaking, conditions as Table 24.29. Both are read from the transimpedance magnitude with a 1 A AC current into the working electrode; peaking is referred to  $Z_t(1 \text{ Hz})$ . Both distributions are symmetric, so mean  $\pm 3\sigma$  applies. Every sample meets the 1 kHz bandwidth floor and the 2 dB peaking ceiling entered on the test.

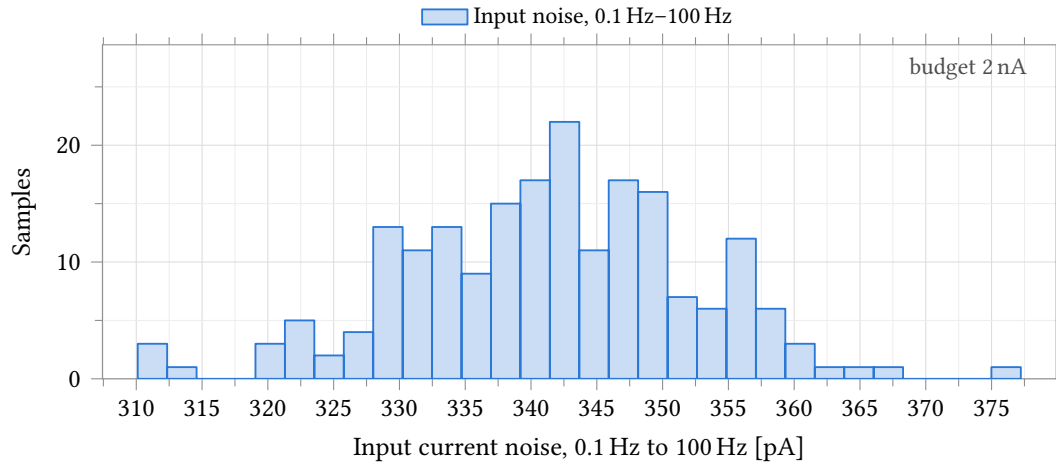


Figure 24.27: Input-referred current noise integrated 0.1 Hz to 100 Hz, conditions as Figure 24.25. Output noise divided by  $R_f$ , from a noise analysis that runs 0.1 Hz to 10 kHz at 20 points per decade. The 2 nA budget entered on the test is about six times the measured value and is annotated at the axis edge rather than drawn, so that the 310 pA to 377 pA spread stays legible. The figure scales inversely with the selected gain tap.

| Parameter                  | Unit | $N$ | Mean  | $\sigma$ | Mean $\pm 3\sigma$ | Min   | Max   | Spec          | Yield | 4 ch  |
|----------------------------|------|-----|-------|----------|--------------------|-------|-------|---------------|-------|-------|
| Output noise, 0.1 Hz–1 kHz | mV   | 200 | 1.696 | 0.055    | $1.696 \pm 0.165$  | 1.539 | 1.874 | –             | –     | –     |
| Input noise, 0.1 Hz–10 Hz  | pA   | 200 | 68.5  | 2.2      | $68.5 \pm 6.7$     | 62.2  | 75.8  | –             | –     | –     |
| Input noise, 0.1 Hz–100 Hz | pA   | 200 | 341.5 | 11.0     | $341.5 \pm 33.0$   | 310.1 | 377.2 | $\leq 2000.0$ | 100.0 | 100.0 |
| Input noise, 0.1 Hz–1 kHz  | nA   | 200 | 3.029 | 0.098    | $3.029 \pm 0.294$  | 2.749 | 3.346 | –             | –     | –     |

Table 24.33: Monte Carlo noise of the transimpedance stage, conditions as Table 24.29. The noise analysis runs 0.1 Hz to 10 kHz at 20 points per decade; the three input-referred rows are the output noise integrated to 10 Hz, 100 Hz and 1 kHz and divided by  $R_f$ , and no figure is integrated beyond the analysis stop frequency. All four distributions are symmetric, so mean  $\pm 3\sigma$  applies. The input-referred rows scale inversely with the selected gain tap; the 2 nA entry on the 100 Hz row is the channel budget.

The bench is Figure 24.28: a Randles cell with  $R_{CE} = 1\text{ k}\Omega$ ,  $R_u = 1\text{ k}\Omega$ ,  $R_{ct} = 10\text{ k}\Omega$  and  $C_{dl} = 1\text{ }\mu\text{F}$ , the only electrode model simulated, and the working electrode held at  $1.25\text{ V}$ . No load current is injected: the cell current follows from the sweep against the cell impedance, and the regulation figures at  $\pm 10\text{ }\mu\text{A}$  are read off the sweep where the cell current crosses those values. Temperature is split as for the device tables,  $37^\circ\text{C}$  at the typical point and  $25^\circ\text{C}$  at the four skew corners.

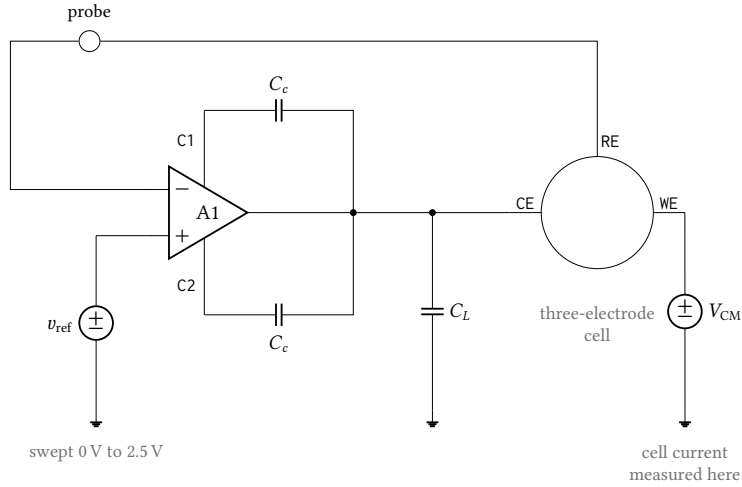


Figure 24.28: Cell-drive bench. The amplifier drives the counter electrode of a three-electrode cell and closes its loop on the reference electrode, with the commanded potential  $v_{ref}$  swept over the full  $0\text{ V}$  to  $2.5\text{ V}$  supply range. The working electrode is held at  $V_{CM}$  by an ideal source: no load current is injected, so the cell current is set by the sweep against the cell impedance and is measured in that source, and the regulation errors quoted at  $\pm 10\text{ }\mu\text{A}$  are read off the sweep at the points where the cell current reaches those values. The probe in the reference-electrode return measures the loop gain with the cell inside the loop; it is not the amplifier's own gain, and the two are tabulated separately. Cell elements are  $R_{ce} = R_u = 1\text{ k}\Omega$  and  $R_{ct} = 10\text{ k}\Omega$  in parallel with  $C_{dl} = 1\text{ }\mu\text{F}$  (Figure 24.1), the load is  $C_L = 5\text{ pF}$ , and each external compensation capacitor  $C_c$  is  $500\text{ fF}$ .

**Monte Carlo.** 200 samples, process and mismatch together,  $37^\circ\text{C}$ , at the conditions above. Cell-loop gain at DC averages  $72.13\text{ dB}$  with a 1st percentile of  $68.67\text{ dB}$ , and every sample clears the  $60^\circ$  phase-margin limit, the worst at  $80.13^\circ$  (Figure 24.30, Table 24.37). Tracking error at zero cell current, the amplifier's input offset carried to the reference electrode, has a mean of  $-0.694\text{ mV}$ , a median of  $-0.267\text{ mV}$  and  $\sigma = 5.708\text{ mV}$ ; over the fixed  $0.4\text{ V}$  to  $2.1\text{ V}$  command window the worst magnitude averages  $5.494\text{ mV}$  and reaches  $23.082\text{ mV}$ , meeting the  $10\text{ mV}$  limit on  $87.0\%$  of channels and  $57.3\%$  of four-channel chips (Table 24.38). With that offset removed, load regulation is  $-22.17\text{ }\mu\text{V}$  at  $10\text{ }\mu\text{A}$  and  $23.97\text{ }\mu\text{V}$  at  $-10\text{ }\mu\text{A}$ ; all 200 samples lie inside  $\pm 1\text{ mV}$ , for  $100\%$  per channel and over four (Table 24.39). Counter-electrode compliance is  $2.4621\text{ V}$  at the ceiling and  $25.69\text{ mV}$  at the floor, spreading  $\pm 3.9\text{ mV}$  and  $\pm 3.27\text{ mV}$  at  $3\sigma$  (Table 24.40).

| Parameter       | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|-----------------|------|--------|--------|--------|--------|--------|--------|
| Loop gain at DC | dB   | 72.24  | 71.85  | 72.24  | 73.57  | 74.11  | 2.26   |
| Loop crossover  | MHz  | 10.440 | 10.835 | 10.508 | 10.533 | 10.223 | 0.612  |
| Phase margin    | °    | 81.88  | 82.14  | 82.05  | 81.83  | 81.68  | 0.45   |

Table 24.34: Control-loop stability driving an electrochemical cell, by process corner. Cell-drive bench (Figure 24.28): Randles cell with  $R_{CE} = 1\text{ k}\Omega$ ,  $R_u = 1\text{ k}\Omega$ ,  $R_{ct} = 10\text{ k}\Omega$ ,  $C_{dl} = 1\text{ }\mu\text{F}$ , working electrode at 1.25 V,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ , 2.5 V supply, 37 °C at the typical point and 25 °C at the four skew corners. The probe sits in the reference-electrode sense wire, so the loop measured here includes the cell; it is a different loop from Table 24.14 and the two are not comparable. Loop gain is 10 dB lower than the unity-feedback figure because the cell’s 12 k $\Omega$  loads an open-loop output resistance of about 25 k $\Omega$ ; crossover halves with it and the phase margin improves. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

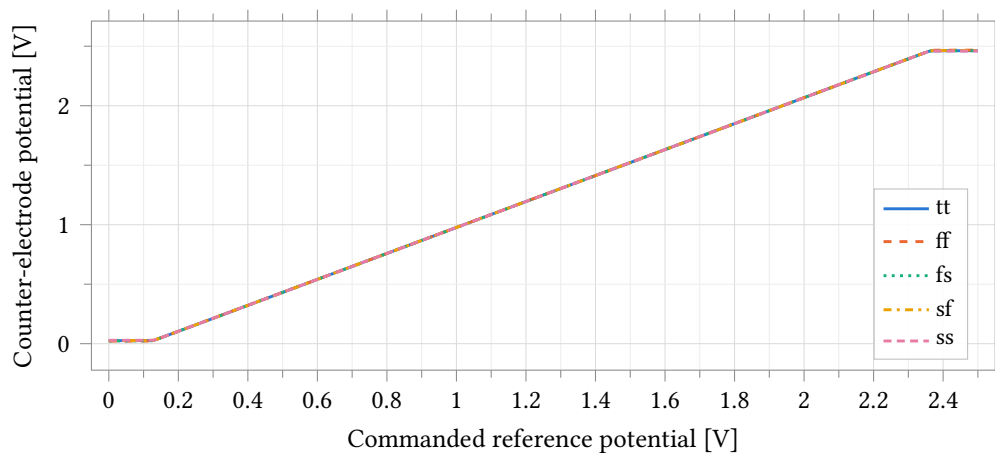


Figure 24.29: Counter-electrode potential against commanded reference potential over five process corners, cell-drive bench (Figure 24.28). The amplifier drives a Randles cell with  $R_{CE} = 1\text{ k}\Omega$ ,  $R_u = 1\text{ k}\Omega$ ,  $R_{ct} = 10\text{ k}\Omega$ ,  $C_{dl} = 1\text{ }\mu\text{F}$  with the working electrode held at 1.25 V,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ , 2.5 V supply; the typical point is at 37 °C and the four skew corners at 25 °C. The central segment rises with slope 12/11, the divider between the 1 k $\Omega$  counter-electrode branch and the 11 k $\Omega$  remainder of the cell; the flat ends are the amplifier against its rails, and their heights are the compliance limits tabulated in Table 24.36. Over the whole central segment the cell current is set by the cell resistance and the commanded potential, not by the amplifier.

| Parameter                              | Unit          | tt     | ff     | fs     | sf     | ss     | Spread |
|----------------------------------------|---------------|--------|--------|--------|--------|--------|--------|
| Tracking error at zero cell current    | $\mu\text{V}$ | -619.5 | -676.9 | -632.9 | -533.2 | -488.1 | 188.7  |
| Worst tracking error over the window   | $\mu\text{V}$ | 771.5  | 838.8  | 794.8  | 667.8  | 715.9  | 171.0  |
| Regulation error at $10\ \mu\text{A}$  | $\mu\text{V}$ | -23.00 | -24.25 | -23.01 | -19.60 | -18.22 | 6.04   |
| Regulation error at $-10\ \mu\text{A}$ | $\mu\text{V}$ | 22.63  | 23.52  | 22.81  | 19.08  | 18.31  | 5.20   |

Table 24.35: Reference-potential tracking by process corner, conditions as Table 24.34. Tracking error is the reference-electrode potential minus the commanded value. The first row is that error at zero cell current and is the loop's DC offset; the second is its worst magnitude over the fixed 0.4 V to 2.1 V command window, offset-inclusive. The last two rows are the load-regulation error at  $\pm 10\ \mu\text{A}$  of cell current with the zero-current error subtracted at the same corner, which is the quantity a differential measurement sees; they are two orders of magnitude smaller than the offset that dominates the rows above. The window is fixed for every corner and every sample, so a weak corner cannot report better accuracy over a narrower range.

| Parameter                        | Unit          | tt      | ff      | fs      | sf      | ss      | Spread |
|----------------------------------|---------------|---------|---------|---------|---------|---------|--------|
| Counter-electrode ceiling        | V             | 2.4622  | 2.4671  | 2.4627  | 2.4641  | 2.4596  | 0.0075 |
| Counter-electrode floor          | mV            | 25.67   | 21.88   | 23.43   | 26.11   | 28.28   | 6.40   |
| Cell current delivered, anodic   | $\mu\text{A}$ | 101.013 | 101.422 | 101.055 | 101.176 | 100.796 | 0.626  |
| Cell current delivered, cathodic | $\mu\text{A}$ | 102.028 | 102.343 | 102.214 | 101.991 | 101.810 | 0.534  |

Table 24.36: Counter-electrode compliance and delivered cell current by process corner, conditions as Table 24.34. The first two rows are the extremes of the counter-electrode potential over the full 0 V to 2.5 V command sweep (Figure 24.29); they are measured directly at the electrode and are independent of the loop offset, which is why compliance is published from them. Headroom to each rail is under 40 mV. The current rows are the largest anodic and cathodic currents reached in the same sweep; with  $12\ \text{k}\Omega$  between the counter and working electrodes they are the compliance limits divided by that resistance, so they are a property of this load and do not transfer to another electrode. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter       | Unit     | N   | Mean   | $\sigma$ | $p_1$ | $p_{99}$ | Worst | Spec         | Yield | 4 ch  |
|-----------------|----------|-----|--------|----------|-------|----------|-------|--------------|-------|-------|
| Loop gain at DC | dB       | 200 | 72.13  | 1.27     | 68.67 | 74.79    | —     | —            | —     | —     |
| Phase margin    | $^\circ$ | 200 | 81.88  | 0.57     | 80.55 | 83.02    | 80.13 | $\geq 60.00$ | 100.0 | 100.0 |
| Loop crossover  | MHz      | 200 | 10.497 | 1.523    | 7.649 | 14.203   | —     | —            | —     | —     |

Table 24.37: Monte Carlo control-loop stability driving an electrochemical cell: **process and mismatch**,  $37\ ^\circ\text{C}$ , 200 samples, cell-drive bench (Figure 24.28) with a Randles cell of  $R_{\text{CE}} = 1\ \text{k}\Omega$ ,  $R_u = 1\ \text{k}\Omega$ ,  $R_{\text{ct}} = 10\ \text{k}\Omega$ ,  $C_{\text{dl}} = 1\ \mu\text{F}$ , working electrode at 1.25 V,  $C_L = 5\ \text{pF}$ ,  $C_c = 500\ \text{fF}$ , 2.5 V supply. The probe sits in the reference-electrode sense wire, so this loop includes the cell and is not comparable with Table 24.20. All 200 samples clear the phase-margin limit entered on the test with better than  $20^\circ$  to spare, and margin is better with the cell attached than without it: the resistive load costs 10 dB of gain, crossover falls to half the unloaded value. Loop gain and crossover were not graded, and percentiles and the extreme sample are quoted rather than  $\pm 3\sigma$  because the crossover distribution is right-skewed.

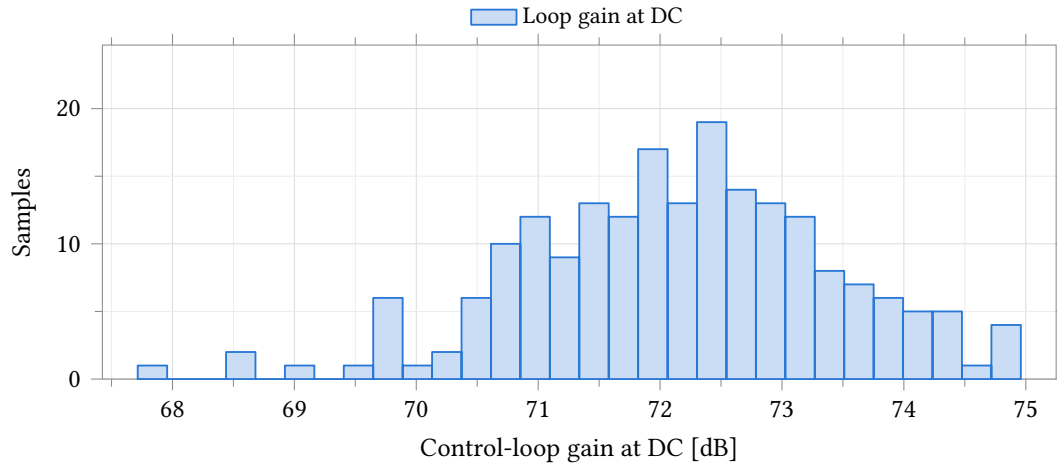


Figure 24.30: DC gain of the control loop closed around an electrochemical cell. **Process and mismatch**, 37 °C, 200 samples, cell-drive bench (Figure 24.28) with a Randles cell of  $R_{CE} = 1\text{ k}\Omega$ ,  $R_u = 1\text{ k}\Omega$ ,  $R_{ct} = 10\text{ k}\Omega$ ,  $C_{dl} = 1\text{ }\mu\text{F}$ , working electrode at 1.25 V,  $C_L = 5\text{ pF}$ ,  $C_c = 500\text{ fF}$ , 2.5 V supply. The loop is measured through the reference-electrode sense wire and therefore includes the cell, whose 12 k $\Omega$  loads an open-loop output resistance of about 25 k $\Omega$  and is 10 dB below the unity-feedback gain of Table 24.20. The spread is 7 dB peak to peak and no sample falls below 67 dB. No limit was entered on this output.

| Parameter             | Unit | <i>N</i> | Mean   | $\sigma$ | Median | $p_1$   | $p_{99}$ | Worst  | Spec          | Yield | 4 ch |
|-----------------------|------|----------|--------|----------|--------|---------|----------|--------|---------------|-------|------|
| Zero-current error    | mV   | 200      | -0.694 | 5.708    | -0.267 | -16.532 | 13.086   | -      | -             | -     | -    |
| Worst error in window | mV   | 200      | 5.494  | 4.463    | 4.423  | 0.358   | 21.869   | 23.082 | $\leq 10.000$ | 87.0  | 57.3 |

Table 24.38: Monte Carlo reference-potential tracking, conditions as Table 24.37. Tracking error is the reference-electrode potential minus the commanded value; the first row is that error at zero cell current, the second its worst magnitude over the fixed 0.4 V to 2.1 V command window. Both are offset-inclusive, and the worst-error row tracks the magnitude of the zero-current row at  $r = 0.986$  across the 200 samples: what is measured here is the amplifier’s input offset carried to the electrode, not a gain error, and calibrating the offset removes almost all of it (load regulation with that offset taken out is Table 24.39). The window is frozen at the same two voltages for every sample, so a weak sample cannot report better accuracy over a narrower range. Both distributions are right-skewed, so the median, percentiles and worst sample are quoted and mean  $\pm 3\sigma$  is not.

| Parameter                       | Unit          | <i>N</i> | Mean   | $\sigma$ | Mean $\pm 3\sigma$ | Min    | Max    | Spec          | Yield | 4 ch  |
|---------------------------------|---------------|----------|--------|----------|--------------------|--------|--------|---------------|-------|-------|
| Regulation at 10 $\mu\text{A}$  | $\mu\text{V}$ | 200      | -22.17 | 27.35    | -22.17 $\pm$ 82.04 | -99.30 | 74.00  | $\pm 1000.00$ | 100.0 | 100.0 |
| Regulation at -10 $\mu\text{A}$ | $\mu\text{V}$ | 200      | 23.97  | 34.41    | 23.97 $\pm$ 103.22 | -76.00 | 167.80 | $\pm 1000.00$ | 100.0 | 100.0 |

Table 24.39: Monte Carlo load regulation of the reference potential, conditions as Table 24.37. Each entry is the tracking error at  $\pm 10\text{ }\mu\text{A}$  of cell current with the same sample’s zero-current error subtracted, so it is the change the loop allows when the cell starts to draw current and carries no offset term. Both distributions are symmetric, so mean  $\pm 3\sigma$  applies, and their means are of opposite sign: the reference potential droops under anodic current and rises under cathodic current, by about 2.3  $\mu\text{V}$  per microamp of cell current. All 200 samples lie inside  $\pm 1\text{ mV}$ , the worst by a factor of six; that limit is stated here for the derived quantity because the run graded only the offset-inclusive error. Regulation is more than two hundred times smaller than the raw tracking error of Table 24.38.



| Parameter                        | Unit          | $N$ | Mean    | $\sigma$ | Mean $\pm 3\sigma$  | Min     | Max     |
|----------------------------------|---------------|-----|---------|----------|---------------------|---------|---------|
| Counter-electrode ceiling        | V             | 200 | 2.4621  | 0.0013   | $2.4621 \pm 0.0039$ | 2.4591  | 2.4654  |
| Counter-electrode floor          | mV            | 200 | 25.69   | 1.09     | $25.69 \pm 3.27$    | 22.38   | 28.94   |
| Cell current delivered, anodic   | $\mu\text{A}$ | 200 | 101.011 | 0.109    | $101.011 \pm 0.327$ | 100.757 | 101.282 |
| Cell current delivered, cathodic | $\mu\text{A}$ | 200 | 102.026 | 0.091    | $102.026 \pm 0.272$ | 101.755 | 102.302 |

Table 24.40: Monte Carlo counter-electrode compliance and delivered cell current, conditions as Table 24.37. The first two rows are the extremes of the counter-electrode potential over the full 0 V to 2.5 V command sweep, measured at the electrode itself and therefore independent of the loop offset; this is the published compliance figure. Headroom is under 35 mV to the upper rail and under 29 mV to the lower one, and each spreads under 7 mV across the 200 samples: compliance is set by the output stage’s saturation and is almost insensitive to matching. The current rows are the largest anodic and cathodic currents reached in the same sweep; with 12 k $\Omega$  between the counter and working electrodes they are the compliance limits divided by that resistance and do not transfer to another electrode. All four distributions are symmetric, and no limit was entered on any of them, so the spec and yield columns are omitted.

#### 24.4.4 Measuring this block on chip

Input offset is the term that decides yield; a loopback is the only way to see it.

1. Open the reference-electrode isolation switch; close the unity-gain loopback around A1.
2. Select per-channel slot 5 (the amplifier output, inboard of the counter-electrode isolation switch). Read it on atb\_d with AFE\_DBGDIEN = 1, or on-chip with AFE\_ADCSRSEL = 4 and averaging; unbuffered either way: a test buffer built from the same devices carries an offset of the same class.
3. Repeat at three reference codes across the working window. The intercept is the input offset,  $\sigma = 5.20$  mV (Table 24.19); the slope is the finite-loop-gain tracking error.
4. For the transimpedance role: isolate the working electrode and read slot 4 the same way. That reading is output-referred,  $\sigma = 8.91$  mV (Table 24.29); divided by the active gain it is the channel’s zero-current error.

This test fabric is design intent; it is not yet in the schematics.

### 24.5 Potentiostat Pixel System

The pixel system is the rev-2 grounded-WE two-amp potentiostat: one complete analog channel built from two instances of the class-AB amplifier of Section 24.4 around a three-electrode electrochemical cell (Figure 24.31). Control amplifier A1 drives the counter electrode (CE) and closes its loop on the reference electrode (RE), holding RE at the commanded potential  $v_{\text{ref}}$  against whatever counter-electrode drive that requires. Transimpedance amplifier A2 holds the working electrode (WE) at a fixed common-mode potential  $V_{\text{CM}}$  through the programmable feedback resistor  $R_f$  and returns the cell current as a single-ended voltage,  $v_{\text{out}} = V_{\text{CM}} + i_{\text{we}} \cdot R_f$ , ready for the single-ended 10-bit SAR converter. The electrode is randles\_cell (Section 24.1):  $R_u = R_{\text{ce}} = 1$  k $\Omega$  fixed,  $R_{\text{ct}}$  and  $C_{\text{dl}}$  swept over the ranges stated with each measurement below.

**Sign convention (measured, firmware-critical).** Positive  $i_{\text{we}}$  flows from the working electrode to ground, and  $v_{\text{out}}$  rises with it:  $dv_{\text{out}}/di_{\text{we}} = +R_f$ . This is the orientation this bench actually measures (accuracy slope +1 against injected truth current, Table 24.42); the TB-07 formula of the system verification plan assumes the opposite source orientation and inverts every sign derived from it. Firmware reading the ADC code must use  $i_{\text{we}} = (v_{\text{out}} - V_{\text{CM}})/R_f$ , not its negation.

The characterisation below covers, in order: bipolar accuracy against a truth current injected at the working electrode (the rev-2 headline result); the potentiostat control loop; stability across the electrode grid; system noise; CV and step transients; and the Monte Carlo zero-current offset. All data is schematic-level, with no extracted parasitics.

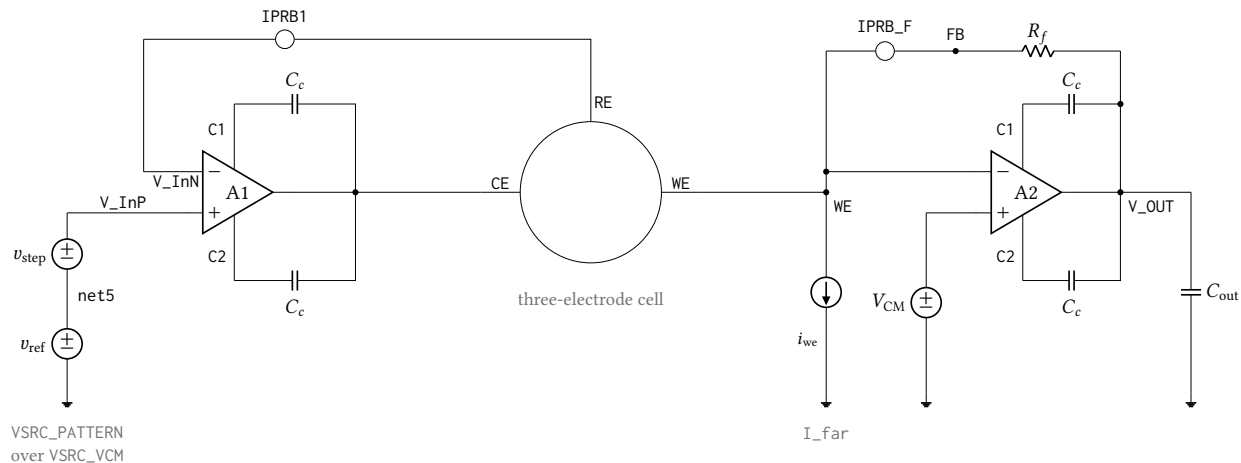


Figure 24.31: The complete rev-2 potentiostat pixel as simulated, `tb_anatop_pixel` view `PotFbTB`: control amplifier A1 drives the counter electrode and closes its loop on the reference electrode, while transimpedance amplifier A2 holds the working electrode at  $V_{CM}$  and returns the cell current through  $R_f$  to  $V_{OUT}$ . Both amplifiers are the same `anatop_tia_amp_ab` class-AB cell, each with its own external compensation capacitors  $C_c = 500$  fF from pins C1/C2 onto its own output; supplies,  $E_n$  and the shared local bias generator are omitted. `IPRB1` and `IPRB_F` are 0 V current probes inserted only so that `stb` can break the control loop and the transimpedance loop respectively – they add no impedance and no offset. `I_far` injects the accuracy-truth current with positive  $i_{we}$  flowing from the working electrode to ground, so  $v_{out} = V_{CM} + i_{we} \cdot R_f$  and the measured current is the probe current in `IPRB_F`; the opposite source orientation inverts the sign of every accuracy number. Cell elements are  $R_{ce} = R_u = 1$  k $\Omega$  fixed with  $R_{ct}$  and  $C_{dl}$  swept over the electrode grid,  $R_f$  is the programmable feedback resistor (35 k $\Omega$  to 560 k $\Omega$ ), and  $C_{out} = 5$  pF.

### 24.5.1 Bipolar Accuracy

Figure 24.32 sweeps the truth current  $i_{we}$  through zero at the operational  $R_f = 35$  k $\Omega$  tap. The error crosses zero smoothly, with no kink between the sourcing and sinking halves of the sweep: the same TIA feedback path carries current in both directions, so there is no crossover distortion of the kind a complementary NMOS/PMOS mirror front end would show at zero current. The crossing slope is  $-1/T$  exactly (finite loop gain, not a nonlinearity), and the crossing offset is  $-1.1$  nA at the operational 1 M $\Omega$  electrode.

Table 24.42 gives the operational accuracy at  $R_f = 35$  k $\Omega$  into the 10 k $\Omega$  electrode: 0.41–0.61 LSB worst-case error across the five process corners. Table 24.43 gives the operational accuracy at the maximum gain tap,  $R_f = 560$  k $\Omega$  into the 1 M $\Omega$  electrode operational at that gain: 0.253–0.376 LSB. These are the two gain taps with a matched operational electrode simulated; the 140 and 560 k $\Omega$  rows measured against the 10 k $\Omega$  electrode (Table 24.44) are a finite-loop-gain demonstration only and are not comparable accuracy figures – see that table’s caption.

At 35 k $\Omega$  the error is range-dependent: 0.41–0.61 LSB over the full  $\pm 33$   $\mu$ A span, improving to 0.21–0.30 LSB over the  $\pm 2.3$   $\mu$ A subrange. The growth toward full scale is the  $1/T$  gain error of the transimpedance loop; the 560 k $\Omega$  figure is already a full-scale measurement at its gain.

| Point | Process | Temperature | Supply |
|-------|---------|-------------|--------|
| 81    | tt      | 37 °C       | 2.5 V  |
| 82    | ff      | 25 °C       | 2.5 V  |
| 83    | fs      | 25 °C       | 2.5 V  |
| 84    | sf      | 25 °C       | 2.5 V  |
| 85    | ss      | 25 °C       | 2.5 V  |

Table 24.41: Process, temperature and supply points simulated for the potentiostat pixel system (design point 17 of the archived Interactive.20 run:  $R_f = 35\text{ k}\Omega$ ,  $R_{ct} = 10\text{ k}\Omega$ ,  $C_{dl} = 100\text{ nF}$ ). Point 1 is the typical statistical process point at 37 °C; points 2–5 are the imported foundry skew corners at 25 °C. Every figure and table in this section is schematic-level. These points move the sixteen transistor model rows only (the resistor and capacitor rows stay at their typical sections), so every corner spread in this section is a lower bound (pinned-passive caveat, page 207).

Every LSB in this subsection is referred to the nominal 2.441 mV converter step; the measured step is 1.591 mV (Table 24.58), so multiply by 1.535 to read these errors as converter codes.

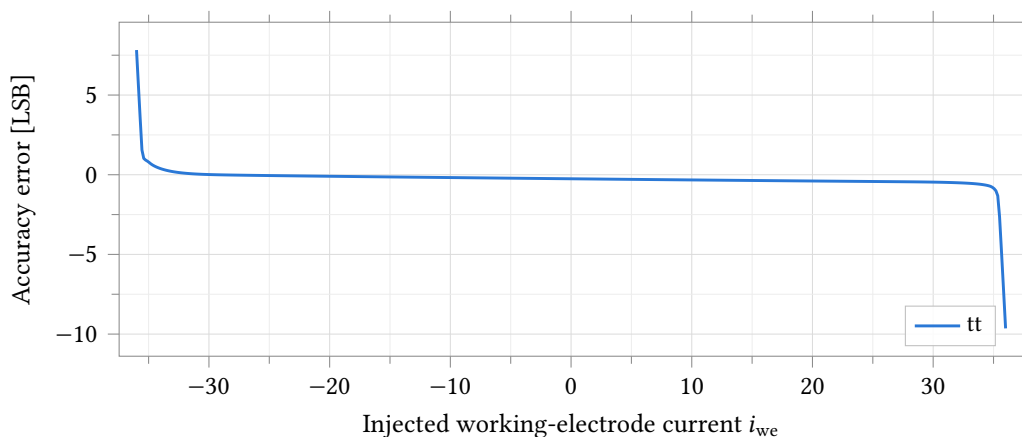


Figure 24.32: Bipolar accuracy error against injected working-electrode current,  $R_f = 35\text{ k}\Omega$ ,  $R_{ct} = 10\text{ k}\Omega$  electrode, typical process point, 37 °C, full  $\pm 36\text{ }\mu\text{A}$  sweep (full scale  $\pm 33\text{ }\mu\text{A}$ ). The zero crossing is smooth with no kink or discontinuity between the sourcing and sinking halves of the sweep: the error’s slope through zero is  $-1/T$  exactly (Table 24.42) and the same single anatop\_tia\_amp\_ab TIA feedback path serves both current directions, unlike the rev-1 PMOS-only mirror it replaces. The error axis is referred to the nominal 2.441 mV step; the converter’s measured step is 1.591 mV (Table 24.58), so multiply by 1.535 to read it as converter codes.

**Finite-loop-gain error away from the operational electrode.** Table 24.44 sweeps a small band around zero at all four gain taps with the electrode held at the stiff 10 k $\Omega$  cell used for the control-loop bench. The 140, 315 and 560 k $\Omega$  rows are not the accuracy of those taps – 10 k $\Omega$  is not the electrode those taps are designed against – but they isolate the mechanism: the zero-crossing slope error tracks  $-1/T$  within 4 % at every tap, so the growth in worst-case error with  $R_f$  is finite closed-loop gain, not a distortion that appears only at high gain.

| Parameter                 | Unit | tt       | ff       | fs       | sf       | ss       | Spread  |
|---------------------------|------|----------|----------|----------|----------|----------|---------|
| Worst full-scale error    | LSB  | 0.539    | 0.606    | 0.521    | 0.483    | 0.409    | 0.197   |
| Zero-crossing error       | nA   | −17.704  | −19.342  | −18.087  | −15.235  | −13.949  | 5.393   |
| Zero-crossing slope error |      | −0.00052 | −0.00060 | −0.00051 | −0.00046 | −0.00038 | 0.00022 |

Table 24.42: Bipolar accuracy at the  $R_f = 35\text{ k}\Omega$  gain tap, the operational point at the  $R_{ct} = 10\text{ k}\Omega$  electrode, by process corner. Truth current  $i_{we}$  swept  $-36\text{ }\mu\text{A}$  to  $36\text{ }\mu\text{A}$  (full scale is  $33\text{ }\mu\text{A}$ ), error referred to the nominal  $2.441\text{ mV}$  ADC LSB and clipped to  $\pm$  full scale before taking the worst case. Over the full range the worst-case error is  $0.41\text{--}0.61$  LSB across corners; over the  $\pm 2.3\text{ }\mu\text{A}$  subrange it improves to  $0.21\text{--}0.30$  LSB, the error growing toward full scale as the finite transimpedance loop gain ( $\varepsilon \propto 1/T$ ). The converter's measured step is  $1.591\text{ mV}$ , not the nominal  $2.441\text{ mV}$  (Table 24.58); multiply every LSB figure here by  $1.535$  to read it as converter codes. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter                 | Unit | tt       | ff       | fs       | sf       | ss       | Spread  |
|---------------------------|------|----------|----------|----------|----------|----------|---------|
| Worst full-scale error    | LSB  | 0.333    | 0.376    | 0.331    | 0.293    | 0.253    | 0.123   |
| Zero-crossing error       | nA   | −1.107   | −1.209   | −1.130   | −0.952   | −0.872   | 0.337   |
| Zero-crossing slope error |      | −0.00012 | −0.00016 | −0.00011 | −0.00011 | −0.00008 | 0.00008 |

Table 24.43: Bipolar accuracy at the  $R_f = 560\text{ k}\Omega$  gain tap, the operational point at the  $R_{ct} = 1\text{ M}\Omega$  electrode, by process corner – **the headline rev-2 accuracy figure**. Truth current swept  $\pm 2.3\text{ }\mu\text{A}$  (full scale is  $2.06\text{ }\mu\text{A}$ ), error referred to the nominal  $2.441\text{ mV}$  ADC LSB and clipped to  $\pm$  full scale before taking the worst case. The converter's measured step is  $1.591\text{ mV}$ , not the nominal  $2.441\text{ mV}$  (Table 24.58); multiply every LSB figure here by  $1.535$  to read it as converter codes. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

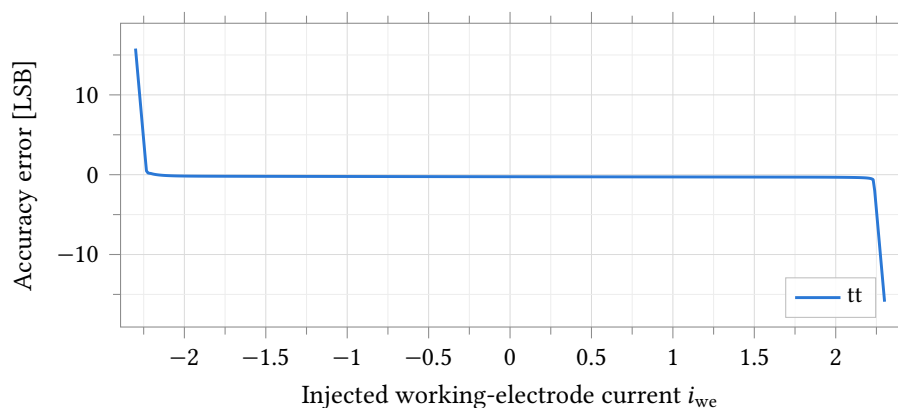


Figure 24.33: Accuracy error at the operational max-gain point,  $R_f = 560\text{ k}\Omega$  into  $R_{ct} = 1\text{ M}\Omega$ , typical process point,  $37\text{ }^\circ\text{C}$ . The truth current sweep is  $\pm 2.3\text{ }\mu\text{A}$ , just beyond the  $2.06\text{ }\mu\text{A}$  full-scale current at this gain tap; the crossing at zero is smooth, with no kink, over the full swept range shown. The error axis is referred to the nominal  $2.441\text{ mV}$  step; the converter's measured step is  $1.591\text{ mV}$  (Table 24.58), so multiply by  $1.535$  to read it as converter codes.

| $R_f$  | Worst error over the sweep [LSB] | Zero-crossing error [nA] | Zero-crossing slope error ( $= -1/T$ ) |
|--------|----------------------------------|--------------------------|----------------------------------------|
| 35000  | 0.271                            | -17.704                  | -0.00052                               |
| 140000 | 0.407                            | -4.426                   | -0.00118                               |
| 315000 | 0.935                            | -1.967                   | -0.00233                               |
| 560000 | 3.101                            | -1.107                   | -0.00394                               |

Table 24.44: Finite-loop-gain error against feedback resistance at the typical statistical process point, 37 °C, truth current swept  $-2.3\text{ }\mu\text{A}$  to  $2.3\text{ }\mu\text{A}$  with  $R_{ct} = 10\text{ k}\Omega$  held fixed across the whole sweep. **The 315 and 560 k $\Omega$  rows are NOT the accuracy of those gain taps** –  $R_{ct} = 10\text{ k}\Omega$  is a stiff, non-operational electrode at those gains (the operational electrode is  $1\text{ M}\Omega$ , Table 24.43). They demonstrate that the residual error is finite closed-loop gain alone: the zero-crossing slope error is  $-1/T$  to within 4 % at every tap, and the worst-case error scales with  $R_f$  as  $T$  falls. The LSB column is referred to the nominal  $2.441\text{ mV}$  step; the converter’s measured step is  $1.591\text{ mV}$  (Table 24.58), so multiply it by 1.535 to read it as converter codes.

### 24.5.2 Control Loop

The commanded reference potential  $v_{\text{ref}}$  is swept  $0\text{ V}$  to  $2.5\text{ V}$ ; Table 24.45 gives the resulting loop stability and Table 24.46 the tracking error. Loop gain exceeds  $71.8\text{ dB}$  and crossover exceeds  $21.2\text{ MHz}$  at every corner, both well past the  $60\text{ dB}/60^\circ$  limits entered on the bench; the worst tracking error stays at or below  $0.814\text{ mV}$  over the fixed  $0.4\text{ V}$  to  $2.1\text{ V}$  evaluation window at every corner. Table 24.47 and Table 24.48 give the compliance: the reference electrode tracks over  $0.054\text{ V}$  to  $2.44\text{ V}$  (a  $10\text{ mV}$  error criterion, offset-inclusive) and the loop delivers  $\pm 52.5\text{ }\mu\text{A}$  into the cell at every corner, read through the TIA feedback probe since no separate truth-current source is active on this bench.

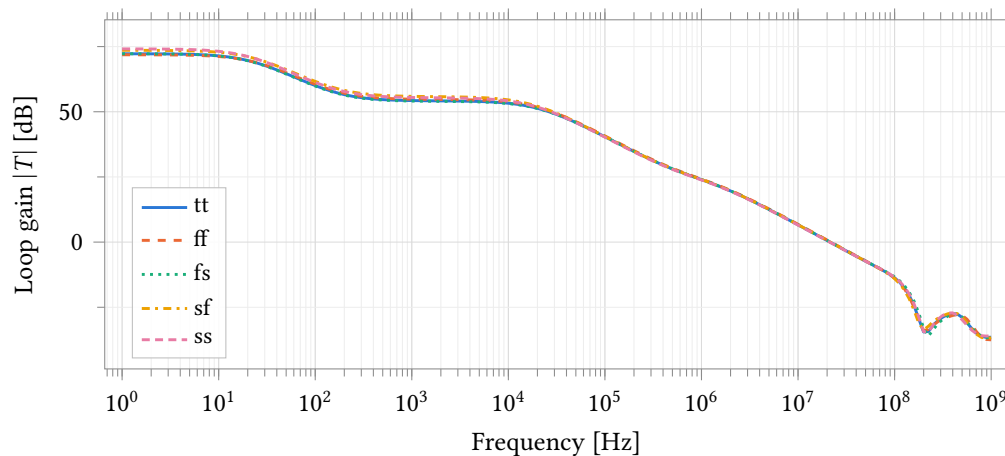


Figure 24.34: Potentiostat control-loop gain over five process corners, measured through the  $0\text{ V}$  probe IPRB1 in the RE feedback path (Figure 24.31).  $R_f = 35\text{ k}\Omega$ , cell  $R_{ce} = R_u = 1\text{ k}\Omega$ ,  $R_{ct} = 10\text{ k}\Omega$ ,  $C_{dl} = 100\text{ nF}$ ,  $2.5\text{ V}$  supply. This is the full loop – A1, A2, both feedback networks and the electrode together – and is not the open-loop amplifier gain of Section 24.4. tt is at 37 °C, the four skew corners at 25 °C. The corner separation drawn here is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter       | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|-----------------|------|--------|--------|--------|--------|--------|--------|
| Loop gain at DC | dB   | 72.24  | 71.85  | 72.25  | 73.57  | 74.11  | 2.26   |
| Loop crossover  | MHz  | 21.475 | 21.959 | 21.595 | 21.578 | 21.276 | 0.683  |
| Phase margin    | °    | 79.51  | 79.18  | 79.98  | 79.09  | 79.81  | 0.89   |

Table 24.45: Potentiostat control-loop stability by process corner, conditions as Figure 24.34. Every corner exceeds the 60° phase-margin and 60 dB DC-gain limits entered on the bench by a wide margin. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter                            | Unit               | tt     | ff     | fs     | sf     | ss     | Spread |
|--------------------------------------|--------------------|--------|--------|--------|--------|--------|--------|
| Zero-current tracking error          | mV                 | −0.620 | −0.677 | −0.633 | −0.533 | −0.488 | 0.189  |
| Worst tracking error over the window | mV                 | 0.748  | 0.814  | 0.773  | 0.646  | 0.692  | 0.168  |
| Worst tracking-error slope           | mV V <sup>−1</sup> | 1.735  | 0.541  | 1.534  | 2.022  | 2.442  | 1.901  |

Table 24.46: Reference-potential tracking by process corner. Cell-drive bench (Figure 24.31),  $v_{\text{ref}}$  swept 0 V to 2.5 V,  $R_f = 35 \text{ k}\Omega$ , cell  $R_{\text{ce}} = R_u = 1 \text{ k}\Omega$ ,  $R_{\text{ct}} = 10 \text{ k}\Omega$ ,  $C_{\text{dl}} = 100 \text{ nF}$ , 2.5 V supply. Tracking error is the reference-electrode potential minus the commanded value; the worst-error row is evaluated over the fixed 0.4 V to 2.1 V design-variable window ( $v_{\text{ref\_lo}}/v_{\text{ref\_hi}}$ ), the same window for every corner, and is offset-inclusive. All five corners clear the 1 mV  $\text{re\_err\_max}$  and 50 mV V<sup>−1</sup>  $\text{re\_err\_slope}$  limits entered on the bench.

| Parameter                          | Unit | tt    | ff    | fs    | sf    | ss    | Spread |
|------------------------------------|------|-------|-------|-------|-------|-------|--------|
| RE tracking range (10 mV crit.)    | V    | 2.382 | 2.386 | 2.383 | 2.382 | 2.379 | 0.007  |
| TIA compliance range (10 mV crit.) | V    | 0.805 | 0.806 | 0.805 | 0.805 | 0.804 | 0.002  |

Table 24.47: RE tracking range and TIA compliance range by process corner, conditions as Table 24.46. Both are the width of the region over which the respective error stays inside a 10 mV threshold; the threshold is on the total error including the loop’s static offset, so a corner with more offset reports a narrower range for the same underlying loop gain – read these next to Table 24.46 rather than as an independent compliance figure. The TIA range is  $\pm 35 \mu\text{A} \times 35 \text{ k}\Omega$  at the 10 k $\Omega$  electrode used here.

| Parameter                        | Unit          | tt     | ff     | fs     | sf     | ss     | Spread |
|----------------------------------|---------------|--------|--------|--------|--------|--------|--------|
| Counter-electrode ceiling        | V             | 2.4804 | 2.4829 | 2.4806 | 2.4814 | 2.4790 | 0.0039 |
| Counter-electrode floor          | mV            | 13.15  | 11.20  | 11.99  | 13.38  | 14.49  | 3.29   |
| Cell current delivered, anodic   | $\mu\text{A}$ | 52.501 | 52.595 | 52.530 | 52.517 | 52.444 | 0.151  |
| Cell current delivered, cathodic | $\mu\text{A}$ | 52.496 | 52.591 | 52.526 | 52.513 | 52.439 | 0.153  |
| Supply current                   | $\mu\text{A}$ | 218.71 | 250.19 | 204.17 | 230.66 | 196.65 | 53.54  |

Table 24.48: Counter-electrode compliance, delivered cell current and supply current by process corner, conditions as Table 24.46. Cell current is measured through the TIA feedback probe IPRB\_F, which carries the entire working-electrode current by KCL since no truth current is injected on this bench; both directions clear the 33  $\mu\text{A}$  limits entered on the test by more than 50 %. Supply current covers both amplifiers, both local bias generators and the shared global bias generator. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

### 24.5.3 Stability Across the Electrode Grid

Table 24.49 gives the transimpedance-loop phase margin at the typical process point over all twelve  $R_f \times R_{ct} \times C_{dl}$  cells simulated. Phase margin stays inside  $89.3^\circ$  to  $93.3^\circ$  over the entire twelve-cell grid and all five process corners – every combination simulated, not only the typical point shown here. No cell needs an explicit compensation capacitor  $C_F$  across the electrode range simulated: the loop is unconditionally stable at both gain taps, both electrode resistances and all three double-layer capacitances tested.

| $R_f$  | $C_{dl}$ [ $\mu$ F] | $R_{ct}$ [ $k\Omega$ ] | TIA loop gain at 1 Hz [dB] | TIA loop phase margin [ $^\circ$ ] |
|--------|---------------------|------------------------|----------------------------|------------------------------------|
| 35000  | 0.10                | 10                     | 66.57                      | 91.36                              |
| 35000  | 1.00                | 10                     | 66.56                      | 91.50                              |
| 35000  | 1.00                | 1000                   | 81.71                      | 91.50                              |
| 35000  | 10.00               | 10                     | 65.47                      | 91.51                              |
| 35000  | 10.00               | 1000                   | 70.80                      | 91.51                              |
| 35000  | 0.10                | 1000                   | 82.23                      | 91.36                              |
| 560000 | 0.10                | 10                     | 48.07                      | 90.21                              |
| 560000 | 0.10                | 1000                   | 78.52                      | 90.21                              |
| 560000 | 1.00                | 10                     | 48.05                      | 92.30                              |
| 560000 | 1.00                | 1000                   | 70.70                      | 92.30                              |
| 560000 | 10.00               | 10                     | 46.68                      | 92.51                              |
| 560000 | 10.00               | 1000                   | 51.44                      | 92.51                              |

Table 24.49: Transimpedance-loop stability across the electrode grid at the typical statistical process point ( $37^\circ\text{C}$ , zero statistical offset). Randles electrode,  $R_u = R_{ce} = 1\text{ k}\Omega$  fixed,  $C_{dl}$  and  $R_{ct}$  swept over the twelve  $R_f \times C_{dl} \times R_{ct}$  combinations simulated (rows grouped by  $R_f$ ). Phase margin stays inside  $89.3^\circ$  to  $93.3^\circ$  over all twelve cells and all five process corners of this grid (verified from the same extracted data, not tabulated in full here for width) – the transimpedance loop needs no explicit compensation capacitor  $C_F$  across the electrode range simulated.

### 24.5.4 System Noise

Table 24.50 gives output and input-referred noise over the same electrode grid. Output noise spans 0.129 mV to 11.97 mV across the full twelve-cell grid; at  $R_f = 35\text{ k}\Omega$  it stays under 0.75 mV everywhere, but at  $R_f = 560\text{ k}\Omega$  it grows to 2.0 mV to 12 mV – 1.3 to 7.5 ADC LSB on the converter’s measured 1.591 mV step (Table 24.58). The growth tracks  $C_{dl}$ , not  $R_{ct}$ : the closed-loop noise gain is  $e_n \cdot (1 + R_f/Z_{\text{electrode}})$ , and  $|Z_{\text{electrode}}|$  falls with  $C_{dl}$  over the integration band, so a larger double-layer capacitance raises the noise gain even though it does not change  $R_{ct}$ . Input-referred current noise at  $R_f = 560\text{ k}\Omega$  reaches 3.6 nA to 21 nA across the grid against a 2 nA design budget – exceeded by a factor of two to ten at the maximum gain tap. This mechanism is invisible to the standalone TIA bench of Section 24.4.2, which uses an ideal feedback resistor with no electrode network in the loop; publishing it here, with the mechanism, is why the pixel-system noise grid exists.

### 24.5.5 Transients

Figure 24.35 is a cyclic-voltammetry sweep at the  $1\text{ M}\Omega$  electrode: the working-electrode current stays within the loop’s linear region for the entire sweep, unclipped (Table 24.51). Figure 24.37 is the current response to a 400 mV step in the commanded reference potential at the same electrode. **The edges rail**

| $R_f$  | $C_{dl}$ [ $\mu$ F] | $R_{ct}$ [ $k\Omega$ ] | Output noise, 0.1 Hz to 1000 Hz [mV] | Input-referred current noise [nA] |
|--------|---------------------|------------------------|--------------------------------------|-----------------------------------|
| 35000  | 0.10                | 10                     | 0.186                                | 5.309                             |
| 35000  | 1.00                | 10                     | 0.477                                | 13.640                            |
| 35000  | 1.00                | 1000                   | 0.474                                | 13.534                            |
| 35000  | 10.00               | 10                     | 0.743                                | 21.242                            |
| 35000  | 10.00               | 1000                   | 0.746                                | 21.308                            |
| 35000  | 0.10                | 1000                   | 0.141                                | 4.031                             |
| 560000 | 0.10                | 10                     | 2.739                                | 4.891                             |
| 560000 | 0.10                | 1000                   | 2.187                                | 3.905                             |
| 560000 | 1.00                | 10                     | 7.299                                | 13.033                            |
| 560000 | 1.00                | 1000                   | 7.287                                | 13.013                            |
| 560000 | 10.00               | 10                     | 11.343                               | 20.255                            |
| 560000 | 10.00               | 1000                   | 11.402                               | 20.360                            |

Table 24.50: System output and input-referred noise across the electrode grid, conditions and process point as Table 24.49. Output noise integrated 0.1 Hz to 1000 Hz; input-referred current noise is the same integral divided by  $R_f$  and is therefore not comparable across the two  $R_f$  blocks by itself. At  $R_f = 560\text{ k}\Omega$  noise grows with  $C_{dl}$ , not  $R_{ct}$ : the closed-loop noise gain is  $e_n \cdot (1 + R_f/Z_{\text{electrode}})$  and  $|Z_{\text{electrode}}|$  falls with  $C_{dl}$  over the 0.1 kHz to 1 kHz integration band, which the standalone TIA bench of Section 24.4.2 (ideal  $R_f$ , no electrode network) cannot show.

**transiently by design:** the bench exercises the step edge itself, and the current briefly saturates against the loop’s slew-limited transient (Table 24.52) before settling to the value the 1 M $\Omega$  cell impedance sets. Read the transient extremes as edge behaviour, not a sustained current rating.

| Parameter                 | Unit    | Min     | Max    | Spread |
|---------------------------|---------|---------|--------|--------|
| Commanded RE potential    | V       | 0.749   | 1.749  | 1.000  |
| Working-electrode current | $\mu$ A | −20.473 | 20.440 | 40.913 |

Table 24.51: Extremes of the CV sweep and its current response, typical process point, conditions as Figure 24.36. The current stays inside the loop’s linear region for the whole sweep – unclipped, unlike the 10 k $\Omega$  electrode used for the accuracy and control-loop measurements elsewhere in this section.

### 24.5.6 Monte Carlo: Zero-Current Offset

200 samples, **mismatch only**, nominal process at 25 °C, the 10 k $\Omega$  electrode, one 200-sample sub-run at each of the four gain taps. The reference-electrode offset  $\sigma(v_{\text{re},\text{zero}}) = 5.72\text{ mV}$  (Table 24.55) is the clean, gain-independent statistic: it is the control loop’s own tracking offset carried to the electrode, identical to five significant figures at all four gain taps. The output-referred current offset  $\sigma(i_{\text{zero}})$  is nearly gain-independent in its own right, 0.729  $\mu$ A to 0.843  $\mu$ A across the four taps (Table 24.54) – consistent with  $\sigma(i_{\text{zero}}) \approx \sigma(v_{\text{re},\text{zero}})/R_{\text{cell}}$ . Referred to the nominal 2.441 mV ADC LSB the same offset current scales with  $R_f$ :  $\sigma \approx 12/43/96/167\text{ LSB}$  at 35/140/315/560 k $\Omega$  (Figure 24.38, Figure 24.39). At 560 k $\Omega$  some samples reach the output rails (Table 24.53) – this bounds the range a vcm-mux offset calibration must cover. Applying the same 5.72 mV loop offset to the operational 1 M $\Omega$  electrode of Table 24.43 predicts  $\approx 1.3\text{ LSB}$  at 560 k $\Omega$  – a channel-to-channel spread on top of, not instead of, the corner-level finite-gain error of that table. Both



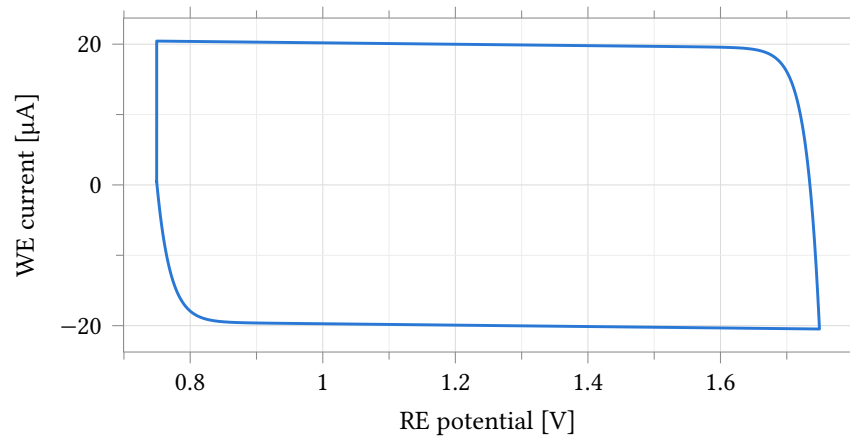


Figure 24.35: Working-electrode current against commanded reference-electrode potential over one triangular voltammetry sweep, typical process point,  $37^{\circ}\text{C}$ ,  $R_f = 35\text{ k}\Omega$ ,  $R_{ct} = 1\text{ M}\Omega$  electrode. The trace is an ellipse-like loop rather than a single-valued curve because the double-layer capacitance  $C_{dl}$  carries a current proportional to  $dv_{re}/dt$  in addition to the resistive charge-transfer current; the two straight sweep segments (RE potential linear in time, Figure 24.36) are why the loop closes into two nearly-parallel branches rather than a smooth curve. The current stays unclipped over the whole sweep (Table 24.51).

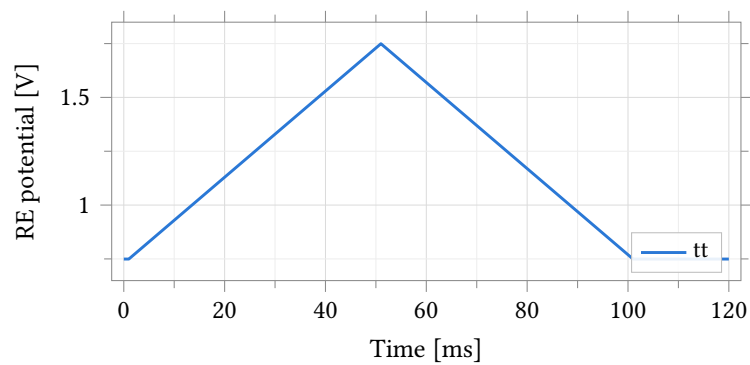


Figure 24.36: Commanded reference-electrode potential against time for the triangular voltammetry sweep, typical process point,  $37^{\circ}\text{C}$ ,  $R_f = 35\text{ k}\Omega$ ,  $R_{ct} = 1\text{ M}\Omega$  electrode. Paired with the current response of Figure 24.35.

| Parameter                   | Unit          | Min     | Max    | Spread | Settling |
|-----------------------------|---------------|---------|--------|--------|----------|
| Working-electrode current   | $\mu\text{A}$ | -35.456 | 35.315 | 70.771 | 49.001   |
| Counter-electrode potential | V             | 1.204   | 1.695  | 0.491  | 49.002   |

Table 24.52: Extremes and settling of the step response, conditions as Figure 24.37. `settle` is the last exit from a  $\pm 1\%$  band around the final value; the extremes are transient peaks reached during the rail-limited edge and are not a sustained operating rating.

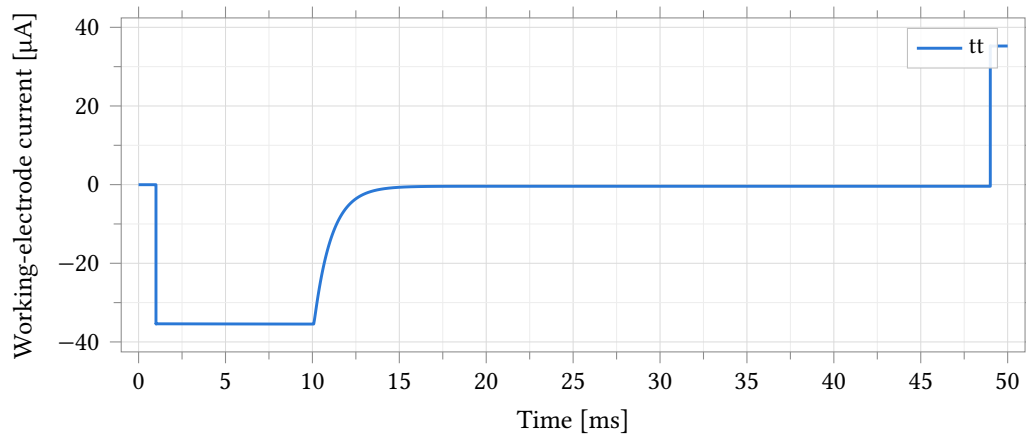


Figure 24.37: Step response of the working-electrode current to a 400 mV step in the commanded reference potential, typical process point, 37 °C,  $R_f = 35 \text{ k}\Omega$ ,  $R_{ct} = 1 \text{ M}\Omega$  electrode, 1  $\mu\text{s}$  edge. The transient overshoots and briefly rails against the loop’s slew-limited edges immediately after the step – by design, the bench exercises the edge rather than a settled operating condition – before settling to the value set by the cell impedance.

LSB figures are on the nominal step; the converter’s measured step is 1.591 mV (Table 24.58), so multiply by 1.535 to read them as converter codes.

| Parameter                                                   | Unit          | $N$ | Mean   | $\sigma$ | Min     | Max    | Spec       | Yield |
|-------------------------------------------------------------|---------------|-----|--------|----------|---------|--------|------------|-------|
| Zero-current offset, $R_f = 560 \text{ k}\Omega$            | $\mu\text{A}$ | 200 | -0.002 | 0.729    | -1.856  | 2.231  | -          | -     |
| Zero-current offset in LSB, $R_f = 560 \text{ k}\Omega$     | LSB           | 200 | -0.49  | 167.33   | -425.83 | 511.74 | $\pm 1.00$ | 0.5   |
| Output voltage at zero current, $R_f = 560 \text{ k}\Omega$ | V             | 200 | 1.249  | 0.408    | 0.211   | 2.499  | -          | -     |

Table 24.53: Monte Carlo zero-current offset at  $R_f = 560 \text{ k}\Omega$ ,  $R_{ct} = 10 \text{ k}\Omega$  electrode. **Mismatch only**, 25 °C nominal process, 200 samples.  $\sigma(i_{\text{zero}}) = 0.729 \mu\text{A}$  here against  $0.729 \mu\text{A}$  to  $0.843 \mu\text{A}$  across all four gain taps (Table 24.54) – nearly gain-independent, as expected for an amplifier-referred current offset.  $v_{\text{out\_zero}}$  ranges 0.21 V to 2.50 V, i.e. some samples reach the output rails at this gain – this bounds the range a  $v_{\text{cm-mux}}$  offset calibration must cover.  $i_{\text{zero\_lsb}}$  and the  $\pm 1$  LSB limit entered on the test are referred to the nominal 2.441 mV step, 1.535 times the converter’s measured 1.591 mV step (Table 24.58); only the yield against that literal limit is quoted, not a four-channel figure, because the limit itself is not gain-scaled (see Figure 24.39).

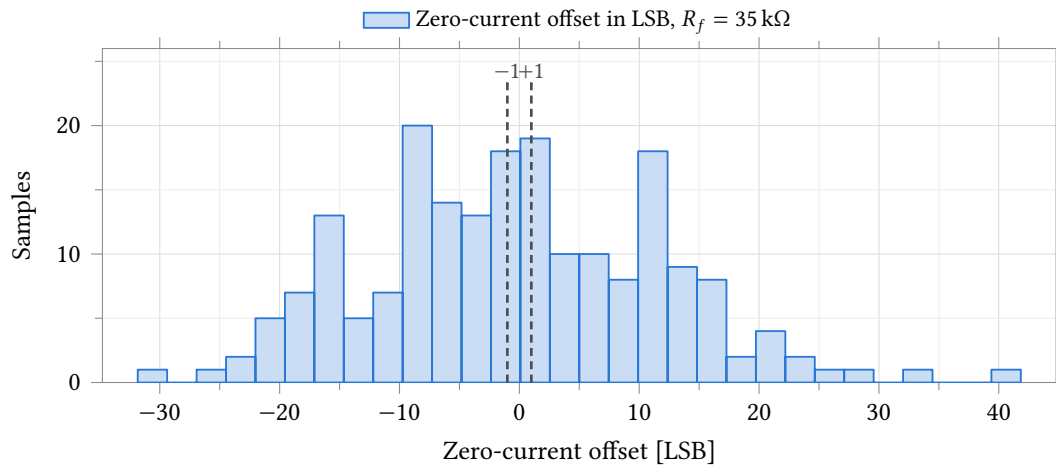


Figure 24.38: Zero-current output offset at  $R_f = 35\text{ k}\Omega$ , in nominal 2.441 mV ADC LSB. **Mismatch only**, 25 °C nominal process, 200 samples,  $R_{ct} = 10\text{ k}\Omega$  electrode. The limit shown is the  $\pm 1$  LSB entered on the test in Assembler – comparable in kind across the four gain taps but mechanically harder to meet as  $R_f$  rises, since the underlying offset is closer to gain-independent in current (Table 24.53). The converter’s measured step is 1.591 mV, not the nominal 2.441 mV (Table 24.58); multiply the axis and the limit by 1.535 to read them as converter codes. Statistics are in Table 24.53.

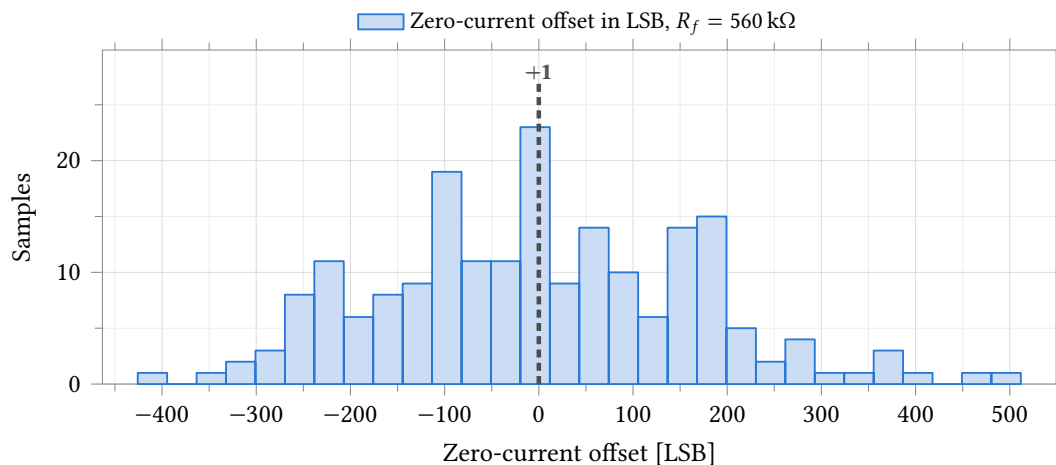


Figure 24.39: Zero-current output offset at  $R_f = 560\text{ k}\Omega$ , conditions as Figure 24.38 but at  $R_{ct} = 10\text{ k}\Omega$  (the same stiff electrode as the finite-loop-gain demonstration of Table 24.44, NOT the 1 M $\Omega$  operational electrode of Table 24.43). Some samples reach the sweep’s output rails (Table 24.53); the LSB spread here is 14 $\times$  wider than at 35 k $\Omega$  for the same underlying current offset, since the same offset current is multiplied through a 16 $\times$  larger  $R_f$ .

| Parameter                                      | Unit          | N   | Mean   | $\sigma$ | Min    | Max   |
|------------------------------------------------|---------------|-----|--------|----------|--------|-------|
| Zero-current offset, $R_f = 35\text{ k}\Omega$ | $\mu\text{A}$ | 200 | -0.019 | 0.843    | -2.221 | 2.918 |

Table 24.54: Monte Carlo zero-current offset at  $R_f = 35\text{ k}\Omega$ , conditions as Table 24.53.  $\sigma = 0.843\text{ }\mu\text{A}$ ; the four-gain range 0.729  $\mu\text{A}$  to 0.843  $\mu\text{A}$  (35k highest, 560k lowest) is the full spread of this same statistic recomputed independently at each of the four staged gain blocks (140k: 0.758  $\mu\text{A}$ , 315k: 0.742  $\mu\text{A}$ ) – all four traceable to `skill/px_mc_slice.py` and the corresponding `px_mc_g*_mc.csv` files.

| Parameter                                         | Unit | $N$ | Mean   | $\sigma$ | Min     | Max    |
|---------------------------------------------------|------|-----|--------|----------|---------|--------|
| RE zero-current offset, $R_f = 35\text{ k}\Omega$ | mV   | 200 | -0.686 | 5.718    | -18.623 | 16.457 |

Table 24.55: Monte Carlo reference-electrode zero-current offset, the gain-independent statistic. **Mismatch only**, 25 °C nominal process, 200 samples.  $\sigma = 5.7176\text{ mV}$ , identical to five significant figures at all four gain taps (verified: 5.71757, 5.71757, 5.71758, 5.71759 mV at 35k/140k/315k/560k) – this is the control loop’s own reference-tracking offset (Table 24.46), which does not depend on which TIA gain tap is selected. At the operational  $1\text{ M}\Omega$  electrode this offset predicts  $\sigma(i_{\text{zero}}) \approx \sigma(v_{\text{os}})/R_{\text{cell}}$ ; using the measured 5.72 mV against the operational electrode’s own  $R_{\text{ct}} + R_u \approx 1.001\text{ M}\Omega$  gives  $\approx 5.7\text{ nA} \approx 1.3\text{ LSB}$  at 560 k $\Omega$  – consistent with, and about 3.5 $\times$  larger than, the corner-level finite-gain error of Table 24.43.

### 24.5.7 Measuring this block on chip

Two nodes separate the four faults that all read as a stuck converter code.

1. Select slot 5: the counter-electrode drive must sit mid-rail and follow the reference code.
2. Select slot 6, the reference-electrode buffer output: it must equal the commanded reference within 16 mV ( $3\sigma$  of A1’s offset).
3. Triage on that pair, not on the converter (it reports a rail for both faults): counter electrode railed with the summing node at common mode = control-loop or electrode fault; counter electrode unrailed with the summing node off common mode = transimpedance fault.
4. Read the summing node itself at slot 7 only through its series guard and with the mid node clamped: it is a virtual ground at 0.824 nA per LSB at the top gain code, and an ungranted site must stay doubly isolated from it.
5. Zero-current constant: open the working-electrode isolation switch (cell current is zero by construction), read the transimpedance output at every gain code the channel will use, subtract the converter’s zero code. Uncalibrated spread:  $\sigma = 12.1\text{--}167.3\text{ LSB}$  over the four taps (Section 24.5.6), on the nominal 2.441 mV step; multiply by 1.535 for the converter’s measured 1.591 mV step (Table 24.58).

This test fabric is design intent; it is not yet in the schematics.

## 24.6 Impedance Spectroscopy

The channel measures impedance with no impedance hardware. A small excitation rides on the reference-DAC setpoint of Section 24.9, A1 applies it at the reference electrode, A2 and the converter read the resulting current through the same programmable feedback resistor a dc measurement uses (Section 24.8), and firmware demodulates the two records in quadrature, an I/Q multiply-accumulate over whole excitation periods. The excitation is a toggle between two DAC codes, so the mode adds a clock divider and an accumulate to the channel and nothing else.

Everything below runs on the netlist of that complete channel at a 2.5 V analog supply, and is schematic-level except where the converter appears, which it does as its extracted view. The nominal point is the typical process corner at 37 °C; only the small-signal measurement was repeated over the chapter’s four process corners at 25 °C, over 0 °C to 85 °C, and over Monte Carlo. The electrodes are the linear cell model of Section 24.1,  $R_u = 1\text{ k}\Omega$  throughout, with  $R_{\text{ct}} \parallel C_{\text{dl}}$  of  $1\text{ M}\Omega \parallel 100\text{ nF}$ ,  $100\text{ k}\Omega \parallel 1\text{ }\mu\text{F}$  and  $10\text{ k}\Omega \parallel 10\text{ }\mu\text{F}$ . Sweeps carry 20 points per decade. **The small-signal measurements were taken on a replica that carries no feedback capacitor**, so their phase numbers and the band edge that follows from them are optimistic; the transient campaign runs through the shipping channel with the 22 pF present, and the accuracy note states both.

**The pinned-passive caveat of page 207 governs every corner number in this section.** Neither the ladder that is  $R_f$  nor the capacitor that sets the feedback pole corners here. The Monte Carlo does vary the poly, and measures  $\sigma = 11.1\%$  on  $R_f$  with process and mismatch together at the nominal corner. Across the five-corner set the current the channel reports through the nominal feedback law spans  $-33.3\%$  to  $35.7\%$  while the current that actually flows is constant to  $0.12\%$  and the impedance derived from it to  $0.025\%$ , all of it the poly sheet: the ratiometric calibration is not a refinement of this mode but a condition of it.

**The error is set by the noise gain  $1 + R_f/|Z|$  at the frequency being measured, not by the electrode's chemistry.** Gain-ranged to the lowest code in compliance and with no calibration applied, the channel holds  $0.44\%$  of magnitude and  $1.04^\circ$  of phase from  $0.01\text{ Hz}$  to  $20\text{ kHz}$  on all five loads, and  $0.33\%$  and  $0.29^\circ$  at and below  $5\text{ kHz}$ ; ranging badly costs an order of magnitude. The 320-tap ladder covers  $236\ \Omega$  to  $14.79\text{ M}\Omega$ , 4.80 decades, each gain code spanning 2.81 of them between the  $\pm 0.80\text{ V}$  output compliance and one converter code of output fundamental; adjacent codes overlap by more than a decade, so there are no impedance seams. The usable band is  $0.1\text{ Hz}$  to  $5.3\text{ kHz}$ , set by a  $\pm 0.573^\circ$  phase gate at the feedback pole and not by magnitude; with the poly and the capacitor cornered as well it falls to  $2.5\text{ kHz}$  on a worst-case die. **Every figure in this paragraph is a small-signal measurement on a replica that carries no feedback capacitor.** Adding the  $22\text{ pF}$  the shipping stage has, and changing nothing else, leaves the magnitude envelope alone ( $0.311$  to  $0.326\%$  at  $5\text{ kHz}$ ) and multiplies the phase error by four,  $0.265^\circ$  to  $1.049^\circ$ : the uncorrected band edge is therefore well below  $5.3\text{ kHz}$ , not at it. The same capacitor moves the transimpedance loop crossing from  $1.06\text{ MHz}$  to the amplifier's own  $22.2\text{ MHz}$  and the gain margin from  $28.0\text{ dB}$  to  $9.76\text{ dB}$ , against a  $10\text{ dB}$  specification; phase margin  $84.0^\circ$  still passes.

**Measured through the converter.** An  $11.000\text{ k}\Omega$  load at gain code 20, excited by the real reference ladder and read through the input multiplexer and an extracted converter, demodulates from 256 converter codes to  $+0.343\%$  of magnitude and  $-2.94^\circ$  of phase against the small-signal analysis of the same load and gain code (Table 24.56). Three demodulations of that one record attribute the magnitude error completely, and **the converter's whole cost is one number**: a  $+0.636\%$  inter-channel gain mismatch, which two points of gain calibration remove. The phase term is the sampling grid and not the converter: one converter serving two channels samples them half a conversion apart, costing  $180 f/f_s$  on a two-level excitation: deterministic, so it needs no calibration; uncorrected, the  $\pm 0.573^\circ$  gate caps an interleaved read at  $2.77\text{ kHz}$ .

Table 24.56: Impedance demodulated from the ten-bit words of the extracted converter: 256 conversions at  $869\,565\,\text{s}^{-1}$ , the multiplexer interleaving the transimpedance output and the reference electrode, excitation  $13.587\,\text{kHz}$  into  $11.000\,\text{k}\Omega$  at gain code 20. Magnitude rows accumulate to the total; phase rows are independent, and  $-180\,f/f_s$  is the analytic half-conversion grid offset the two interleaved ones sit on. Both are referred to the small-signal analysis of the same load and gain code, itself  $+0.295\,\%$  and  $0.405^\circ$  from the analytic load. **One record, one die, one frequency into a resistive load**, no noise sources in the deck: a deterministic attribution with no distribution behind it.

| Quantity                                               | Unit     | Value  | $\pm 0.573^\circ$ gate | Reading          |
|--------------------------------------------------------|----------|--------|------------------------|------------------|
| <i>Magnitude, against the small-signal analysis</i>    |          |        |                        |                  |
| Transient path                                         | points   | +0.126 |                        | measured         |
| Sampled system                                         | points   | +0.855 |                        | measured         |
| Gain mismatch                                          | points   | -0.638 |                        | measured         |
| Total, from codes                                      | points   | +0.343 |                        | measured         |
| <i>Phase, departure from the small-signal analysis</i> |          |        |                        |                  |
| From codes, interleaved                                | $^\circ$ | -2.938 | —                      | measured         |
| Sampled, interleaved                                   | $^\circ$ | -2.895 | —                      | measured         |
| Predicted, $-180\,f/f_s$                               | $^\circ$ | -2.813 | —                      | analytic         |
| Analog, no sampling                                    | $^\circ$ | -0.013 | ✓                      | measured         |
| Deskewed by $180\,f/f_s$                               | $^\circ$ | -0.082 | ✓                      | corrected        |
| Two records, one grid                                  | $^\circ$ | +0.083 | ✓                      | measured         |
| Pure sine, same grids                                  | $^\circ$ | 0.000  | ✓                      | analytic control |

**Measured through the complete channel.** A square-wave record taken through the shipping channel, with the excitation entering at the reference ladder and the true electrode current read in series with the working electrode, separates the demodulator from the instrument. **The demodulator, the two-code square and the boxcar contribute  $0.01\,\%$  of magnitude and  $0.01^\circ$** ; the settled impedance is within  $0.015\,\%$  of the model on the typical load, inside  $0.15\,\%$  at every gain code, and inside  $0.025\,\%$  across the corner set. The converter costs  $+0.160\,\%$  and  $0.063^\circ$ , reconstructing its analog input to  $0.305\,\text{LSB rms}$ . What remains is the instrument, and it is two terms that do not overlap (Table 24.57): the feedback pole, which is  $\arctan(2\pi f R_f C_F)$  and which the demodulator removes exactly by dividing by  $R_f/(1 + j\omega R_f C_F)$  rather than by  $R_f$ , and the control amplifier tracking the commanded potential,  $0.15\,\%$  and  $0.09^\circ$  at  $1\,\text{kHz}$  rising to  $2.1\,\%$  and  $8.4^\circ$  at  $50\,\text{kHz}$ . With the pole restored and the record taken ratiometrically against a known impedance, charge-transfer resistance and double-layer capacitance come out within  $0.15\,\%$  at  $1\,\text{kHz}$  and  $0.40\,\%$  at  $5\,\text{kHz}$ . **That last claim assumes the electrolyte resistance is supplied:** taken as the real part at the top sweep frequency, as the firmware note describes, it carries a  $+9.9\,\%$  bias on this load before any measurement error and degrades both extracted parameters by an order of magnitude.

Table 24.57: Error of one square-wave record per frequency, demodulated four ways so the sources separate: at the cell, at the commanded potential, through the nominal feedback-resistance law, and through that law with the feedback pole restored. Conditions are the linear cell model at  $R_u = 1 \text{ k}\Omega$ ,  $R_{ct} = 10 \text{ k}\Omega$ ,  $C_{dl} = 3.18 \text{ nF}$ , gain code 20, typical corner,  $37^\circ\text{C}$ , one recorded excitation period, converter idle. The nominal-law column is  $\arctan(2\pi f R_f C_F)$  to within 6 % at every frequency, so it is the feedback pole and nothing else; what survives the correction is the control amplifier tracking the commanded potential, which is why the band edge exists at all.

|                        | True $ Z $        | At the cell | + tracking | + nominal law | + law, pole restored |
|------------------------|-------------------|-------------|------------|---------------|----------------------|
| <i>Magnitude error</i> |                   |             |            |               |                      |
| 1 kHz                  | 10 788.6 $\Omega$ | +0.034 %    | +0.179 %   | −0.238 %      | −0.256 %             |
| 2 kHz                  | 10 221.4 $\Omega$ | +0.049 %    | +0.171 %   | −0.235 %      | −0.307 %             |
| 5 kHz                  | 7814.0 $\Omega$   | +0.031 %    | −0.011 %   | −0.108 %      | −0.559 %             |
| 10 kHz                 | 5003.7 $\Omega$   | −0.007 %    | −0.568 %   | +0.647 %      | −1.135 %             |
| 20 kHz                 | 2841.1 $\Omega$   | −0.011 %    | −1.821 %   | +4.488 %      | −2.376 %             |
| 50 kHz                 | 1480.0 $\Omega$   | +0.003 %    | +2.123 %   | +40.61 %      | +1.751 %             |
| <i>Phase error</i>     |                   |             |            |               |                      |
| 1 kHz                  | −10.259°          | +0.078      | +0.170     | +1.159        | +0.066               |
| 2 kHz                  | −19.702°          | +0.061      | +0.195     | +2.274        | +0.089               |
| 5 kHz                  | −39.783°          | +0.015      | +0.294     | +5.692        | +0.244               |
| 10 kHz                 | −53.118°          | −0.007      | +0.641     | +11.472       | +0.672               |
| 20 kHz                 | −55.985°          | −0.005      | +2.078     | +23.088       | +2.206               |
| 50 kHz                 | −42.038°          | +0.002      | +8.385     | +52.348       | +8.703               |

Phase errors in degrees. Gain code 20 puts the feedback pole at 52.4 kHz.

### 24.6.1 From Codes to Resistance and Capacitance

Firmware holds two code sequences per frequency, the potential channel and the transimpedance output, taken on a grid of  $N$  samples per excitation period over  $M$  whole periods. One multiply and accumulate per channel reduces each to its fundamental phasor,

$$\hat{X} = \frac{2}{MN} \sum_{n=0}^{MN-1} c_n e^{-j2\pi n/N}, \quad j^2 = -1, \quad (24.5)$$

giving  $\hat{V}_P$  and  $\hat{V}_{OUT}$ . The converter LSB is never applied: it cancels in every ratio below, as does the  $4/\pi$  fundamental of the two-level excitation.

Since A2 holds the working electrode at the common mode across  $R_f$  (Section 24.4.2), the electrode current in that bin is  $\hat{I}_{WE} = -\hat{V}_{OUT}/R_f$ , so with a per-chip dc calibration of the feedback resistance

$$\hat{Z} = \frac{\hat{V}_P}{\hat{I}_{WE}} = -R_f \frac{\hat{V}_P}{\hat{V}_{OUT}}, \quad (24.6)$$

the minus sign being the transimpedance inversion. Measuring a known  $Z_{cal}$  at the same frequency and gain code removes  $R_f$  altogether, and with it the feedback pole and the quantisation bias:

$$\hat{Z} = Z_{cal} \frac{\hat{V}_P \hat{V}_{OUT}^{cal}}{\hat{V}_{OUT} \hat{V}_P^{cal}}. \quad (24.7)$$

Against the cell model of Section 24.1,

$$Z(\omega) = R_u + \frac{R_{ct}}{1 + j\omega R_{ct} C_{dl}}, \quad (24.8)$$

the extraction is two divisions and no curve fit. Take  $R_u$  as  $\Re \hat{Z}$  at the top of the band, where the parallel branch has collapsed, then invert one mid-band point:

$$Y = \frac{1}{\hat{Z} - R_u} = \frac{1}{R_{ct}} + j\omega C_{dl}, \quad R_{ct} = \frac{1}{\Re Y}, \quad C_{dl} = \frac{\Im Y}{\omega}. \quad (24.9)$$

Both readings come from the same inversion, so a single frequency inside 0.1 Hz to 5.3 kHz yields both, and the accuracy of each is the accuracy of  $\hat{Z}$  at that frequency, which the calibration paragraphs above bound.

The potential phasor  $\hat{V}_p$  is whatever the converter multiplexer presents as that channel. On this cell that is the commanded potential, not a reference-electrode tap, which no multiplexer slot carries (the converter readout note records the slot map); the difference is worth 0.02 % to 0.2 % of magnitude here and up to 3.5 % on the excitation (the excitation and averaging note carries that figure).

## 24.7 SAR Analog-to-Digital Converter

`anatop_pixel_adc` is a 10-bit single-ended charge-redistribution SAR converter (Figure 24.40). One instance per channel digitises the transimpedance output of Section 24.5:  $v_{out} = V_{CM} + i_{we} \cdot R_f$  enters the converter and leaves as a 10-bit code, offset-binary about the code that corresponds to  $V_{CM}$ . The capacitor array is both the sampling network and the feedback DAC; its layout is Figure 24.41.

Conversion is externally clocked from a 20 MHz trigger clock, and the macro advances one bit trial per falling edge of that clock. The bench steps the input from 0 V to 2.5 V in 0.25 V steps and reads the code at the settled point of the first completed conversion, where the converter's ready output is high at every step.

**The delivered stimulus converts four times too slowly.** The clock pattern the converter peripheral emits carries three pulse groups per 1150 ns (clear, sample, convert) but holds the convert phase statically high, so it presents a single falling edge where the ten bit trials need ten. The ten trials therefore consume the falling edges of the following three groups, and ready asserts every 4.600  $\mu$ s: 217.4 kSa/s, not the 869.6 kSa/s the 1150 ns group period implies. Replacing the held level with ten 20 MHz pulses in the same 500 ns window and changing nothing else closes each conversion in 1150 ns and returns *identical codes* at every input level. The defect is in the stimulus, not the converter: the fix is one commented-out line of the peripheral's register-transfer source, and both protocols are reported separately throughout Section 24.7.2.

**Coverage.** All data in this section comes from a *post-extraction* netlist (Quantus, C-only coupled parasitic extraction of the ADC macro) and is, with the code-level impedance record of Section 24.6 that splices this same macro into the channel, the only extracted data in this chapter; every other section is schematic-level. The distinction is not incidental: the capacitor array exists only in layout, so a schematic-level simulation of this block would not be a simulation of this converter. The transfer, timing and power figures are measured at five process corners (Table 24.59) and the nominal point is 37 °C, 2.5 V analog and 1.0 V digital supplies. A 100-sample process-and-mismatch Monte Carlo also exists, and its result is that this netlist cannot produce a matching distribution at all (Section 24.7.3); no gain or offset  $\sigma$  is quoted anywhere below. The 0.25 V sweep step is 157 codes. A second sweep of the unclipped window in 0.0625 V steps, 39 codes apart, carries the linearity figures, and the corner runs use a 20-point staircase at 0.1 V, 63 codes apart. No grid resolves individual codes, so nothing here is a DNL or INL measurement.



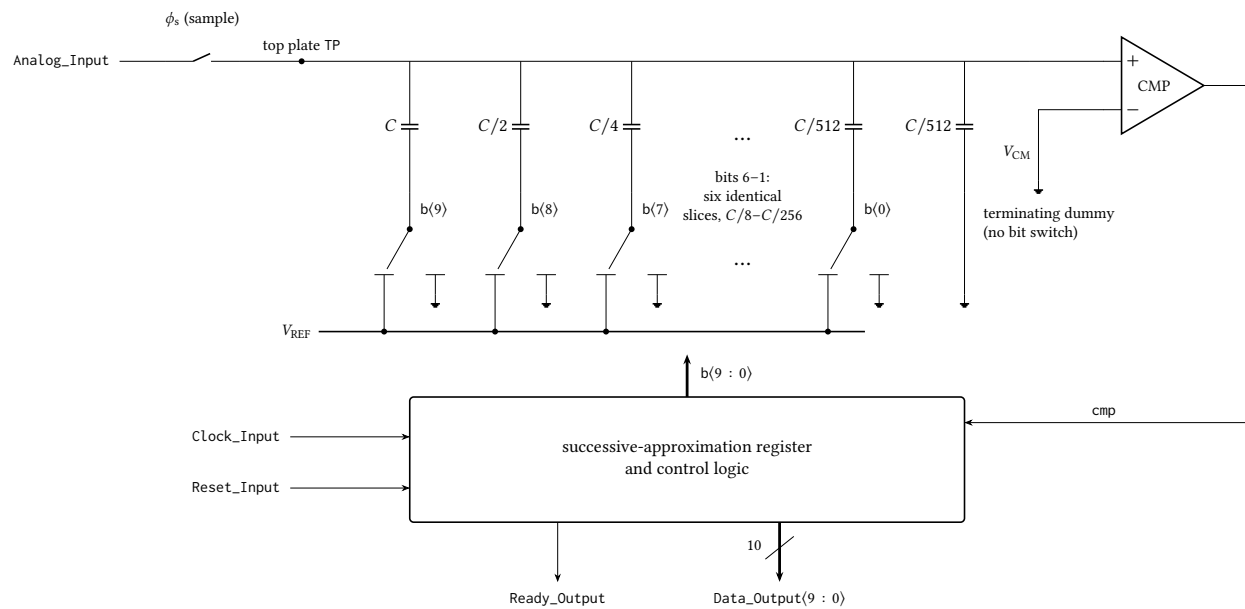


Figure 24.40: Generic single-ended charge-redistribution SAR ADC, drawn to explain the architecture of the 10-bit converter rather than to reproduce its implemented transistor-level schematic. The input is sampled through  $\phi_s$  onto the top plate of a binary-weighted array; the logic then switches each bottom plate from  $V_{REF}$  to ground one bit at a time, most significant first, and keeps or discards the trial from the comparator decision, so the conversion takes ten comparator cycles and the top plate converges to  $V_{CM}$ . Bits 6–1 are elided: six further slices identical to those drawn sit between  $C/4$  and  $C/512$ , and the terminating dummy  $C/512$  carries no bit switch, which makes the total array  $2C$  and the LSB weight  $C/512$ . Only the external port names are those of the implemented block; the sampling network, comparator topology, clock generation and reset behaviour are representative, and no device dimensions are stated.

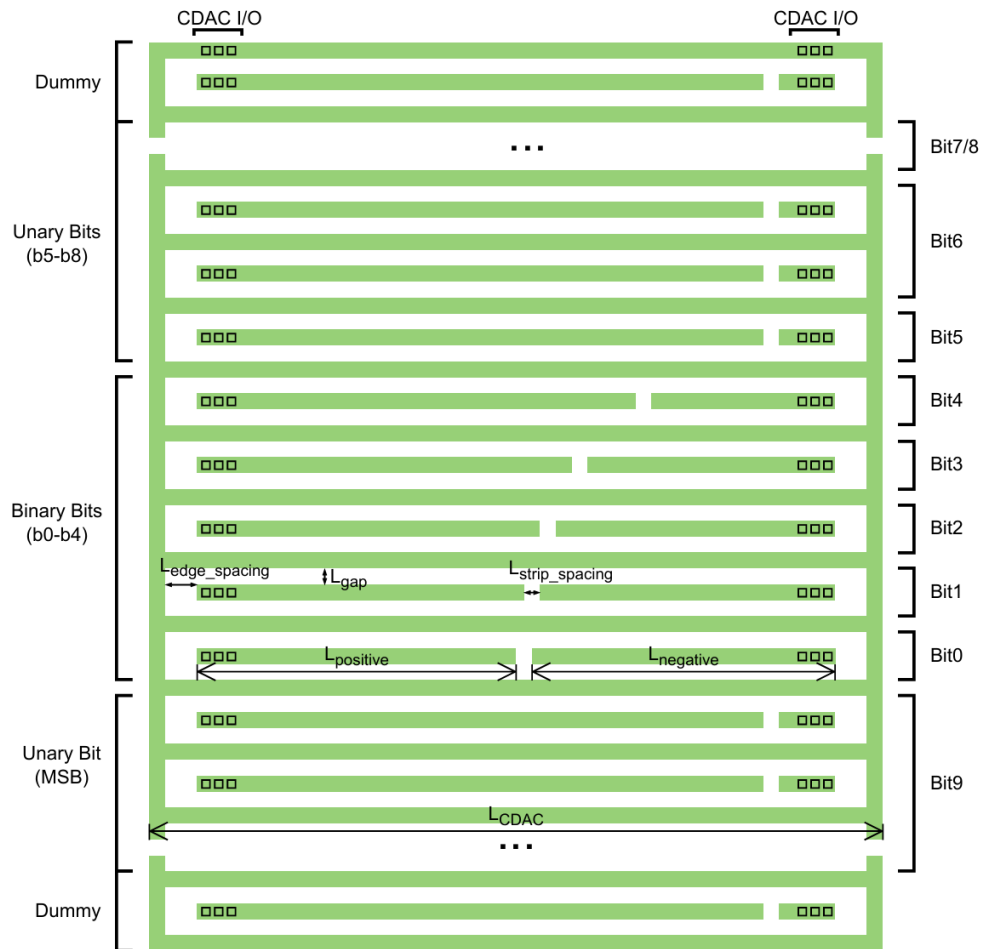


Figure 24.41: Floor plan of the 10-bit segmented capacitor array of `anatop_pixel_adc`. The five lower bits b0–b4 are binary-weighted within single strips, each strip split into a positive and a negative section whose boundary sets the weight; the upper bits b5–b8 and the MSB b9 are unary, built from repeated identical strips, and every row terminates in CDAC I/O at both ends. Unary segmentation of the large weights makes their ratio a count of identical unit strips rather than the dimension of one large device, so it is set by matching; the dummy rows at the top and bottom of the array give the outermost active strips the same local surroundings as interior ones. The array exists only in layout, which is why this section’s data is taken from an extracted netlist.

### 24.7.1 Transfer Function

The converter is monotonic across the whole sweep, reading code 0 at 0 V and code 1023 at 2.5 V (Figure 24.42). Codes climb 1.535 times faster than ideal: a least-squares fit to the 24 unclipped points of a 0.0625 V sweep over 0.5 V to 2.0 V gives 628.64 codes per volt against the ideal 409.60, so the effective full-scale input span is 1.627 V rather than the 2.5 V supply span, and the effective LSB is 1.591 mV rather than 2.441 mV (Table 24.58). Extrapolating that fit puts code 0 at 0.441 V and code 1023 at 2.068 V, a window centred on 1.255 V, the 1.25 V channel common mode. The converter clips outside that window, losing 34.9 % of the intended input range, split between the two ends rather than concentrated at one: a full-scale transimpedance swing about  $V_{CM}$  is truncated symmetrically.

**Mechanism.** The 1.535 factor is a product of two, and the smaller of them is the parasitic. The array is a *differencing* capacitor DAC: each bit drives two plates in antiphase, so the charge it injects is proportional to the difference of the two, and it is that difference which is binary, to four figures over the 512:1 weight range, while the physical plates are not; the positive and negative segments of each bit row are visible in Figure 24.41. Building the sub-LSB weights as small differences of much larger plates is what makes them matchable, and the price is that all of that plate loads the top node: the summed physical plate is 1.235 times the summed weight. The extracted top-plate parasitic and the comparator input add a further 1.243, and  $1.235 \times 1.243 = 1.535$ . The parasitic alone is 0.226 of the physical array, not the 0.535 of the array a single-mechanism reading infers. The slope then closes in one expression,

$$m = \frac{C_{\text{top,total}}}{V_{\text{ref}} C_{\text{unit}}},$$

giving 627.8 codes per volt against the 628.59 the corner staircase measures (Table 24.59), 0.13 %.

Seven perturbation runs separate the terms. Two known capacitances added to the top plate predict the perturbed slope from one fitted  $C_{\text{top}}$  to better than 0.03 %, with nothing else in the converter changed. Lowering the analog supply scales the slope by exactly the reference ratio and moves the window centre with it, while forcing the comparator's reference alone moves the centre and leaves the slope unchanged to 0.003 %, so the span is set by the reference and the centre by the comparator's own supply-halving resistor divider, and the 1.25 V centre is that divider rather than an offset. Halving and then removing the antiphase plates moves the slope the predicted way and destroys the binary weighting at the same time, taking the worst residual about the fitted line from 0.50 LSB to 7.02 and 10.36 LSB. Nothing here is a layout defect that a parasitic-free layout would remove: the larger factor is the topology.

Departure from the fitted line is at most 0.52 LSB over the fitted range (Figure 24.43). That is a residual of a 24-point fit at 39 codes per step, not a linearity figure.

Referred to its own range instead of the nominal one the picture separates (Table 24.58). The usable window is 0.441 V to 2.068 V, 1.627 V wide against the 2.5 V span and centred on the common mode (Figure 24.44); 34.9 % of the nominal range is unreachable, split 0.441 V low and 0.432 V high. Across that window the effective LSB is 1.591 mV rather than 2.441 mV, and each code sits within 0.523 LSB of the measured line, 0.294 LSB rms (Figure 24.45). The two defects are therefore separable: the range is short by a third, and within what remains the transfer is straight to better than half a code at the sampled inputs.

### 24.7.2 Supply Current and Conversion Energy

Power depends on which clock protocol the converter is driven with, so both are reported (Table 24.60). On the stimulus as delivered a conversion takes 4.600  $\mu\text{s}$  and costs 12.95  $\mu\text{W}$ , 59.6 pJ per conversion; at the design rate the same converter draws 18.09  $\mu\text{W}$  and 20.8 pJ. Four times the throughput for 40 % more

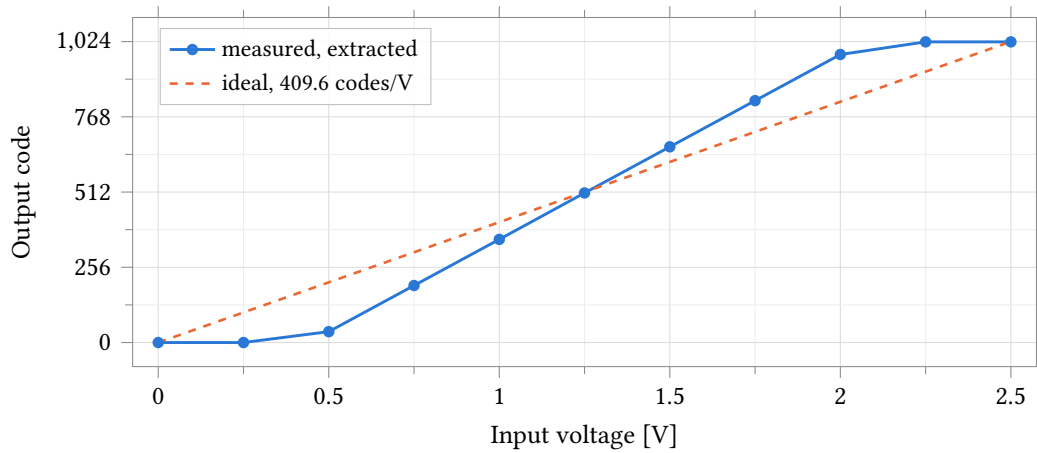


Figure 24.42: Output code against input voltage for the extracted anatop\_pixel\_adc, nominal corner, 37°C, 2.5 V analog and 1.0 V digital supplies, with the ideal 409.6 codes/V line overlaid. The transfer is monotonic and reaches both code limits, but its slope is 628.64 codes/V, 1.535 times ideal, so the flat segments below 0.44 V and above 2.07 V are the code limits, not the ends of the sweep. Markers are the eleven simulated points; the 0.25 V step is 157 codes, so the curve between them is interpolation and carries no linearity information.

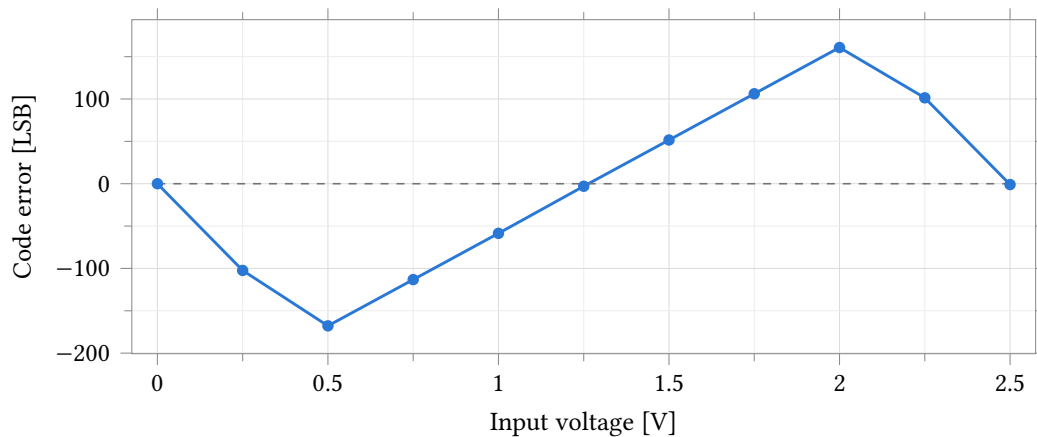


Figure 24.43: Code error against input voltage, measured code minus the ideal 409.6 codes/V line, from the same run as Figure 24.42. The error is the 1.535 slope ratio, not nonlinearity: it reaches  $-167.8$  LSB at 0.5 V and 160.8 LSB at 2.0 V, crossing zero near the 1.25 V common mode where the measured and ideal lines intersect. It also returns near zero at 0 V and 2.5 V, where the clipped output coincides with the ideal endpoints; those two points are not accuracy. Referred to the fitted line instead, the residual over 0.5 V to 2.0 V is at most 0.52 LSB.

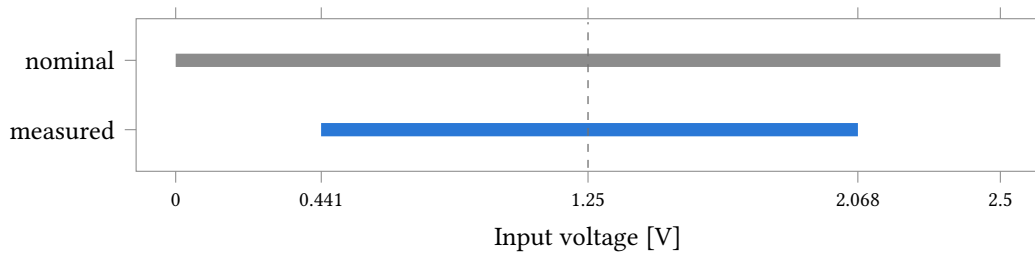


Figure 24.44: Usable input window of the converter against the nominal 0 V to 2.5 V supply span. The measured window runs 0.441 V to 2.068 V, 1.627 V wide, so 34.9 % of the nominal span is unreachable: 0.441 V at the bottom and 0.432 V at the top, near enough symmetric about the 1.25 V common mode (dashed). Outside that window the output sits at a code rail, so a transimpedance swing wider than  $\pm 0.81$  V about  $V_{CM}$  is truncated rather than digitised.

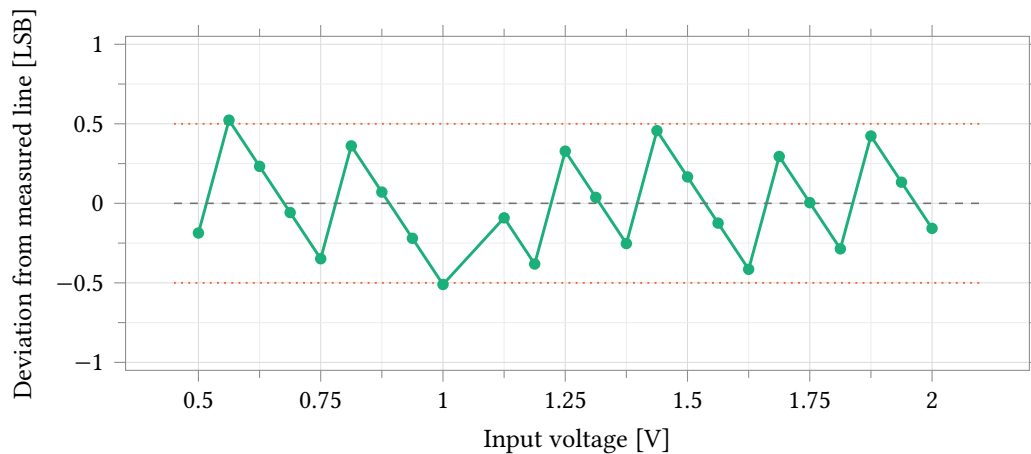


Figure 24.45: Deviation of each measured code from the converter's own best-fit line, 628.64 codes/V, over a 0.0625 V sweep of the unclipped window. Referred to its own scale rather than the nominal one the converter is straight to 0.523 LSB worst case and 0.294 LSB rms, so the 168 LSB of Figure 24.43 is gain error alone and carries no linearity information. The worst point sits just outside the  $\pm 0.5$  LSB dotted guides. Twenty-four points 39 codes apart bound the deviation at the sampled inputs; a DNL or INL measurement needs every code and this grid does not provide one. A twenty-fifth point at 1.0625 V is absent because its simulation returned an error.

| Parameter                        | Unit    | Value  |
|----------------------------------|---------|--------|
| Fitted slope, 0.5 V to 2.0 V     | codes/V | 628.64 |
| Ideal slope, 1024/2.5 V          | codes/V | 409.60 |
| Slope ratio, measured / ideal    |         | 1.535  |
| Effective LSB                    | mV      | 1.591  |
| Nominal LSB, 2.5 V/1024          | mV      | 2.441  |
| Effective full-scale input span  | V       | 1.627  |
| Input at code 0, extrapolated    | V       | 0.441  |
| Input at code 1023, extrapolated | V       | 2.068  |
| Centre of usable window          | V       | 1.255  |
| Input range lost to clipping     | %       | 34.9   |
| Range utilisation                | %       | 65.1   |
| Worst residual about the fit     | LSB     | 0.52   |
| RMS residual about the fit       | LSB     | 0.29   |

Table 24.58: Transfer parameters of the extracted `anatop_pixel_adc` from test `adc_dc`, nominal corner, 37 °C, 2.5 V analog and 1.0 V digital supplies. The slope is a least-squares fit to the 24 unclipped points of a 0.0625 V sweep over 0.5 V to 2.0 V; the code-0, code-1023 and centre voltages are extrapolations of that fit and are not measured points. The worst and rms residuals are the departure of those 24 points from the fit, not an INL: the step is 39 codes. Every entry is reproduced to 0.10 % at all five process corners (Table 24.59); no mismatch spread attaches to any entry, and none can be measured from this netlist (Section 24.7.3).

| Parameter             | Unit    | tt 37 °C | ff     | fs     | sf     | ss     |
|-----------------------|---------|----------|--------|--------|--------|--------|
| Fitted slope          | codes/V | 628.59   | 628.59 | 628.40 | 628.73 | 628.13 |
| Effective LSB         | mV      | 1.5909   | 1.5909 | 1.5914 | 1.5905 | 1.5920 |
| Input at code 0       | V       | 0.4408   | 0.4408 | 0.4407 | 0.4408 | 0.4399 |
| Usable span           | V       | 1.6274   | 1.6274 | 1.6280 | 1.6271 | 1.6286 |
| Worst residual        | LSB     | 0.495    | 0.495  | 0.490  | 0.477  | 0.484  |
| RMS residual          | LSB     | 0.294    | 0.294  | 0.277  | 0.288  | 0.302  |
| Monotonic points      |         | 20/20    | 20/20  | 20/20  | 20/20  | 20/20  |
| Conversion period     | μs      | 1.150    | 1.150  | 1.150  | 1.150  | 1.150  |
| Analog supply current | μA      | 6.360    | 4.787  | 5.453  | 8.437  | 15.429 |
| Total power           | μW      | 18.09    | 16.77  | 15.77  | 23.74  | 43.06  |
| Energy per conversion | pJ      | 20.8     | 19.3   | 18.1   | 27.3   | 49.5   |

Table 24.59: Transfer, timing and power of the extracted converter at five process corners: typical at 37 °C and the four skew corners at 25 °C, on a 20-point staircase at 0.1 V (63 codes apart) fitted over 0.55 V to 1.95 V. The transfer is invariant to 0.10 % because it is a ratio of extracted fixed capacitances, which the corner rows do not move; the residuals are departures from that fit at 63 codes per step, not an INL. Supply current is averaged over one whole conversion at the design-rate protocol of Section 24.7.2, and depends on the sweep step because the switching term dominates.

power is the expected trade (the static term is carried four times longer in the slow case), and the energy per conversion is the figure that changes, by  $2.9\times$ .

Power corners by  $2.7\times$  where the transfer does not (Table 24.59):  $15.8\mu\text{W}$  to  $43.1\mu\text{W}$  across the five process points, the slow corner alone a factor of two above the other four. That signature is leakage and bias rather than switching, consistent with the  $1.0\text{ V}$  core logic slowing down.

Currents are averaged over one whole conversion, and they depend on the sweep step: the nominal point reads  $6.36\mu\text{A}$  on a 63-code step and  $12.56\mu\text{A}$  on a 220-code step, because the switching term dominates. A converter current quoted without its code step is not a comparable number.

| Parameter                     | Unit          | As delivered | Design rate | Corner range  |
|-------------------------------|---------------|--------------|-------------|---------------|
| Trigger clock                 | MHz           | 20           | 20          | 20            |
| Falling edges per conversion  |               | 10           | 10          | 10            |
| Conversion period             | $\mu\text{s}$ | 4.600        | 1.150       | 1.150         |
| Conversion rate               | kSa/s         | 217.4        | 869.6       | 869.6         |
| Analog supply current, 2.5 V  | $\mu\text{A}$ | 4.737        | 6.360       | 4.79–15.43    |
| Digital supply current, 2.5 V | nA            | 2.0          | —           | —             |
| Digital supply current, 1.0 V | $\mu\text{A}$ | 1.09         | —           | —             |
| Total power                   | $\mu\text{W}$ | 12.95        | 18.09       | 15.77–43.06   |
| Energy per conversion         | pJ            | 59.6         | 20.8        | 18.1–49.5     |
| Fitted slope                  | codes/V       | 628.85       | 628.59      | 628.13–628.73 |
| Input at code 0               | V             | 0.4410       | 0.4408      | 0.4399–0.4408 |

Table 24.60: Conversion timing, supply current and conversion energy of the extracted converter on the two clock protocols of Section 24.7, nominal corner,  $37^\circ\text{C}$ . Currents are averaged over one whole conversion rather than sampled, and on a 63-code sweep step: the switching term dominates, so the same point reads  $12.56\mu\text{A}$  on a 220-code step, and any quoted current must name its step. The corner range is the design-rate protocol across the five corners of Table 24.59. Over a whole conversion the digital 2.5 V rail carries leakage only.

### 24.7.3 Monte Carlo: What This Netlist Cannot Measure

A 100-sample process-and-mismatch Monte Carlo of the extracted converter runs clean and measures nothing about matching. Two of the three input levels return the same code in all 100 samples; the third dithers between the two codes adjacent to the boundary it already sits on. A mismatch-only control returns  $\sigma = 0$  on every level and on the supply current, and the full run reproduces the process-only run to every printed digit.

Three structures account for it, and none is fixable in post-processing. Extraction flattens away the inline model wrappers in which the kit’s per-instance mismatch parameters are declared, so the converter’s transistors sit on raw model cards that the statistical block has no path to: the simulator reports five of its six mismatch parameters as non-referenced. The capacitor array is extracted linear capacitors with no statistical model of any kind, so the matching that actually sets a charge-redistribution converter’s linearity is not represented at all. The one remaining matched structure, the resistor divider that sets the comparator’s reference, ships with its mismatch flag cleared, which multiplies the model’s mismatch terms to zero.

**The  $\sigma$  on this converter’s gain and offset are therefore unmeasured, not small.** Nothing below should be read as a matching result. What the run does establish is the analog bias spread, which is

process and is real: 11.4 % on the analog supply current and 10.5 % on total power, 10.5  $\mu$ W to 16.4  $\mu$ W, with conversion timing identical in every sample.

Forcing the reference divider's mismatch on bounds the one contribution the netlist can express: 0.821 mV, 0.52 LSB of  $1\sigma$  on the converter zero,  $\pm 1$  code, and 0.012 % on gain: the divider moves the comparison threshold, not the array's charge balance. That is small against the three-code offset the nominal converter already carries and smaller still than the channel's own loop offset. Measuring the array itself needs a different bench: a schematic capacitor DAC built from the kit's statistical metal-fringe capacitor device rather than from extracted linear elements. A precedent bench for exactly that exists in the legacy library and has not been run against this array.

#### 24.7.4 Measuring this block on chip

The converter is characterised end to end with no analog pad measurement, and its gain has never been checked in system.

1. Gain and intercept: set AFE\_ADCSRSEL = 4, drive an external ramp onto atb\_d (AFE\_DBGDIEN = 1, AFE\_DBGBUFEN = 0), fit slope and intercept per die. The 1.535 $\times$  slope, 0.441 V intercept and 1.591 mV effective LSB of Table 24.58 hold to 0.10 % across process corners in simulation; their die-to-die spread is neither measured nor simulable (Section 24.7.3), which is what makes this step per-die rather than per-lot.
2. Zero code: set AFE\_ADCSRSEL = 1 and average the common-mode rail. That code is the origin of the current axis and carries the converter's offset and the common-mode value together, the combination a current reading needs.
3. Dither before averaging: with the electrode isolated the amplifier output noise is tens of microvolts, below one code, so step the common-mode DAC over three of its 529  $\mu$ V codes (together they span one converter code); only then does averaging reach the quantisation floor.
4. Observe the sampling node itself at per-channel slot 9 between conversions only.

This test fabric is design intent; it is not yet in the schematics.

### 24.8 Programmable feedback resistor

anatop\_tia\_rprog is the transimpedance gain element of the rev-2 channel: a digitally programmable resistor that sets  $R_F$  between the TIA output and the working-electrode virtual ground. Its law is

$$R = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N, \quad N = \text{bin}(\text{ResEn} < 5 : 0 >) + 64 \cdot \text{popcount}(\text{ThEn} < 3 : 0 >),$$

$N = 0 \dots 319$ , so every 6 k $\Omega$  value from 18 k $\Omega$  to 1.932 M $\Omega$  is reachable and the map is monotonic by construction. The ladder is a series chain of identical poly units (a three-unit fixed base, six binary-weighted segments of 1 to 32 units, and four thermometer segments of 64 units), each switched segment shunted by a CMOS bypass transmission gate (the thermometer gates at twice the binary gates' width; Figure 24.46). A control bit at 1 opens its gate and inserts the segment; terminals are A, B and the ten control bits, with the gate supplies arriving by inherited connection.

The endpoints follow from the converter, not from the ladder. The SAR's effective input window is 1.63 V,  $\pm 0.815$  V about  $V_{CM}$  (§24.7), essentially coincident with the TIA's measured  $\pm 0.80$  V compliance. At  $N = 0$  (19.6 k $\Omega$  typical) that window carries  $\pm 40.7$   $\mu$ A, covering the  $\pm 33$   $\mu$ A application target; at  $N = 319$  (1.931 M $\Omega$ ) the range is  $\pm 0.41$   $\mu$ A with a current quantum of 0.82 nA, an order above the 49 pA the loop actually delivers in the amperometric band (Section 24.8.1), so at maximum gain the resolution floor is quantisation rather than noise. Larger  $R_F$  would buy nothing: the noise floor does not shrink with  $R_F$ , and headroom does: at  $V_{CM} = 1.25$  V on 2.5 V the top code can carry at most  $\approx 0.65$   $\mu$ A before the terminal voltage rails.



All data is schematic-level; the cell has no layout view yet. The bench forces 500 nA through A–B with B held at  $V_{CM}$  and reads  $R$  from the terminal voltage; one swept variable decodes into all ten control bits, and the decode was verified against an independent per-code deck to  $0.18\ \Omega$  over all 320 codes.

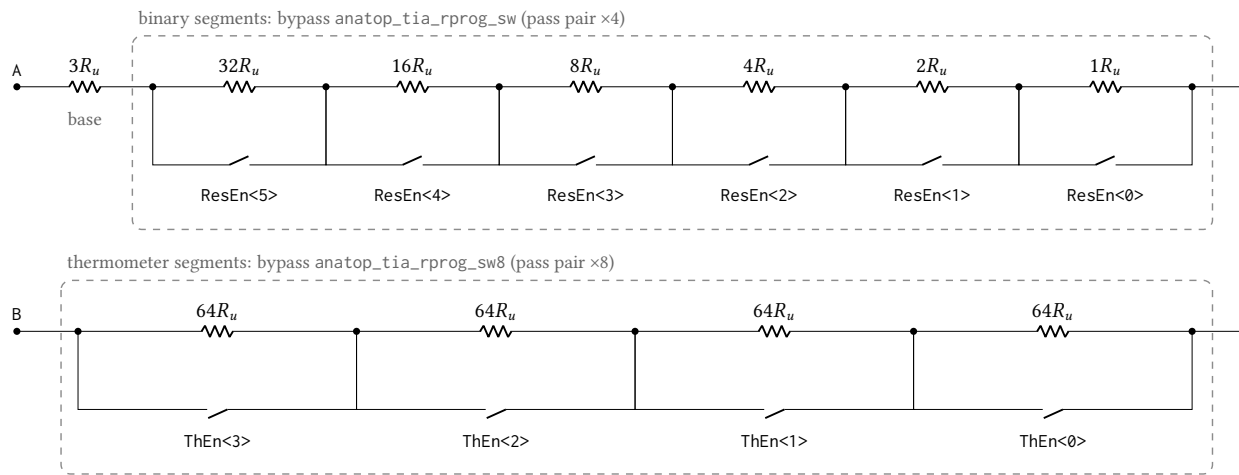


Figure 24.46: Segment and bypass-switch structure of `anatop_tia_rprog`, transcribed from the executed spec-run netlist. The unit  $R_u$  is one series poly resistor; from terminal A the chain runs through the never-switched three-unit base, the six binary-weighted segments of 32 down to 1 unit, and the four identical 64-unit thermometer segments to terminal B. Each switched segment is shunted by a CMOS transmission gate with a local inverter: the control pin `EnBar` is the `ResEn/ThEn` bit itself and drives the PMOS pass gate directly and the NMOS pass gate through the inverter, so a bit at 1 turns both pass devices off and inserts the segment. The pass pair has multiplicity 4 in the binary switch (`_sw`) and 8 in the thermometer switch (`_sw8`), which halves the thermometer gates' on-resistance where up to six binary gates close at a seam. Three grounded dummy units accompany the 322 active units for matching.

Corners move every code by the same sheet-resistance factor, so matching, the property that decides whether a 64-unit thermometer segment can invert a step against 63 binary units, is invisible to them. The Monte Carlo run therefore grades the seam directly, per sample, alongside the absolute spread at four codes.

Every graded code carries a  $\pm 5\%$  absolute window about its typical value, and the seam a monotonicity limit at zero; all five yield 100.0% over the 200 samples (Table 24.65). The margins are not close: the nearest limit sits  $14.2\sigma$  from the mean at  $N=0$ ,  $54\sigma$  at  $N=63$ ,  $55\sigma$  at  $N=64$  and  $134\sigma$  at  $N=319$ , and the worst seam sample is  $14.7\sigma$  above zero. Mismatch is therefore nowhere near yield-limiting for this block; the items that need engineering attention are the deterministic ones graded in Table 24.66: bottom-code linearity and slow-corner top-code compliance.

**All four seams, all 320 codes.** A second 200-sample run sweeps the whole code space in every sample, so each sample is graded at all four thermometer seams and none can contribute to one seam and not another (Table 24.64). **No sample is non-monotonic anywhere:** zero codes with a negative step, in 200 samples across 320 codes. The nearest approach to zero is not a seam but the smallest ordinary step,  $5470\ \Omega$  in the worst sample,  $137\sigma$  from zero on its own distribution; the four seam steps clear zero by 14.7 to 17.7 standard deviations. The worst of them is the first, and predictably so: it is the only seam at which a thermometer segment switches in with no thermometer segment already in circuit, so its step is the smallest and its relative spread the largest.

The ratio the seams rest on measures well. Sixty-four thermometer units against sixty-three binary

| Point | Process | Temperature | Supply |
|-------|---------|-------------|--------|
| 1     | typ     | 27 °C       | 2.5 V  |
| 2     | ff-res  | 27 °C       | 2.5 V  |
| 3     | ss-res  | 27 °C       | 2.5 V  |
| 4     | ff -40C | -40 °C      | 2.5 V  |
| 5     | ss 125C | 125 °C      | 2.5 V  |

Table 24.61: Process and temperature points simulated for the programmable feedback resistor. Point 1 is the typical statistical process point (castalia\_typ\_25) at 27 °C. Points 2–3 skew the poly resistor family alone (ff\_res/ss\_res with typical MOS, 27 °C) and points 4–5 skew MOS and resistor together at the temperature extremes (ff\_25+ff\_res at -40 °C, ss\_25+ss\_res at 125 °C), so the ladder term and the switch-resistance term of each code can be told apart. The \_res family has no sf/fs sections, so no cross corner exists to run. This is a corner run and carries no device mismatch.

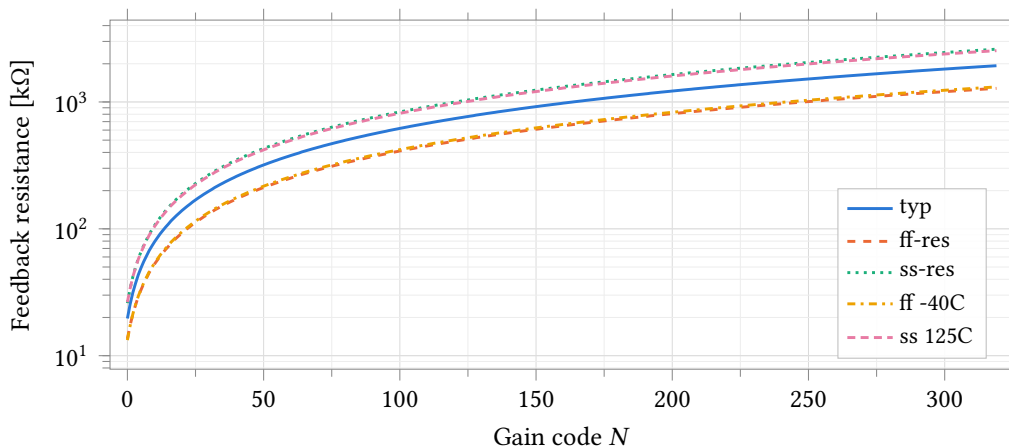


Figure 24.47: Measured resistance against gain code at all five corners, 500 nA forced through the A–B terminals with B held at  $V_{CM} = 1.25$  V. The law is  $R = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$  with  $N$  the binary value of ResEn<5:0> plus 64 per set ThEn<3:0> thermometer bit; every corner is strictly monotonic over all 320 codes, including the four thermometer seams. The corner families move the absolute value by -33 to +35 %, which is the poly sheet-resistance tolerance and must be absorbed by gain calibration, not by code choice.

| Parameter       | Unit       | typ      | ff-res   | ss-res   | ff -40C  | ss 125C  | Spread   |
|-----------------|------------|----------|----------|----------|----------|----------|----------|
| $R(N=0)$        | k $\Omega$ | 19.648   | 13.579   | 25.947   | 13.263   | 26.407   | 13.143   |
| $R(N=63)$       | k $\Omega$ | 396.1817 | 262.8186 | 534.6769 | 269.5412 | 520.7830 | 271.8583 |
| $R(N=64)$       | k $\Omega$ | 403.6009 | 268.0920 | 544.3539 | 274.4177 | 531.2663 | 276.2619 |
| $R(N=319)$      | M $\Omega$ | 1.9308   | 1.2802   | 2.6064   | 1.3137   | 2.5369   | 1.3262   |
| Seam step 63→64 | k $\Omega$ | 7.419    | 5.273    | 9.677    | 4.876    | 10.483   | 5.607    |
| Step, min       | $\Omega$   | 5576     | 3574     | 7668     | 3812     | 7250     | 4094     |
| Step, max       | $\Omega$   | 8509     | 6445     | 10618    | 5687     | 11659    | 5972     |
| Step, mean      | $\Omega$   | 5991.0   | 3970.5   | 8089.1   | 4076.7   | 7870.0   | 4118.6   |

Table 24.62: Code-space summary by corner, 500 nA test current. step\_min/step\_max are taken from the point-to-point derivative of the 320-code sweep and step\_mean from the endpoints; step\_min is positive at every corner, which is the monotonicity statement. The seam step (code 63 to 64, one 64-unit thermometer segment in, six binary segments out) exceeds a normal step everywhere. At ss 125C the top code reaches 2.553 V at the forced 500 nA, 53 mV over the rail; a 250 nA control run bounds the cost of that at 0.0017 % on the top code and  $\leq 0.036$  % anywhere, so the values are reported as measured.

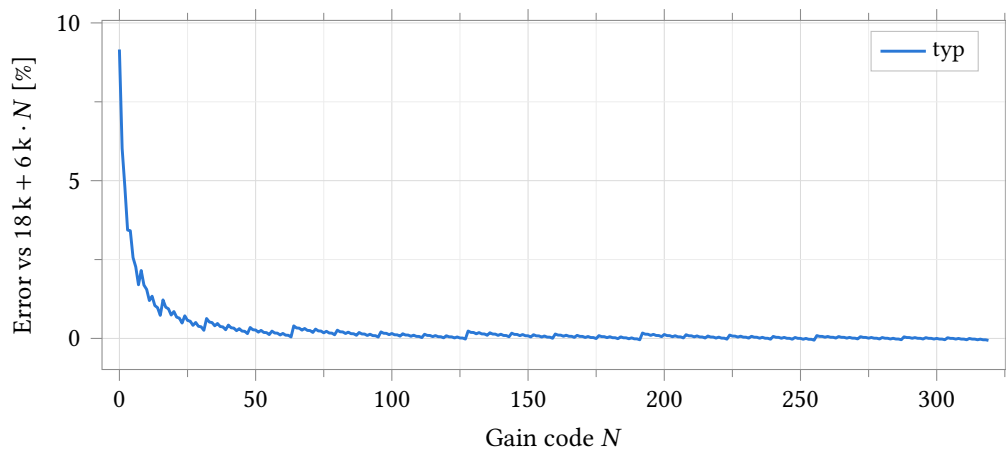


Figure 24.48: Typical-corner deviation from the ideal code law. The +9.2 % at  $N = 0$  is the series resistance of the ten closed bypass switches (1.65 k $\Omega$ , about 165  $\Omega$  each), which decays as  $1/R$  once real ladder segments dominate; by  $N = 319$  the error is  $-0.063$  %. The ripple is the carry structure: at each binary carry and at each thermometer seam the set of closed switches changes, so the  $R_{on}$  term steps while the ladder term is exact.

| Parameter              | Unit       | typ    |
|------------------------|------------|--------|
| Chord resistance       | k $\Omega$ | 19.639 |
| Worst deviation        | mV         | 1.998  |
| Worst deviation        | ADC LSB    | 1.255  |
| Secant- $R$ modulation | %          | 0.504  |

Table 24.63: Drive linearity at the bottom code ( $N = 0$ ), typical corner, sweep  $\pm 90$  % of the  $\pm 0.8$  V-window full scale. The chord is the straight line through the sweep endpoints; the worst deviation of 1.998 mV (1.26 ADC LSB) sits at mid-range, where the closed bypass gates run at lower on-resistance than at the endpoints that pin the chord. The companion runs at  $N = 63$  and  $N = 319$  measure 5.03  $\mu$ V (0.0032 LSB) and 4.87  $\mu$ V (0.0031 LSB) respectively (run directories simruns/tb\_anatop\_tia\_rprog\_lin\_n63, \_n319).

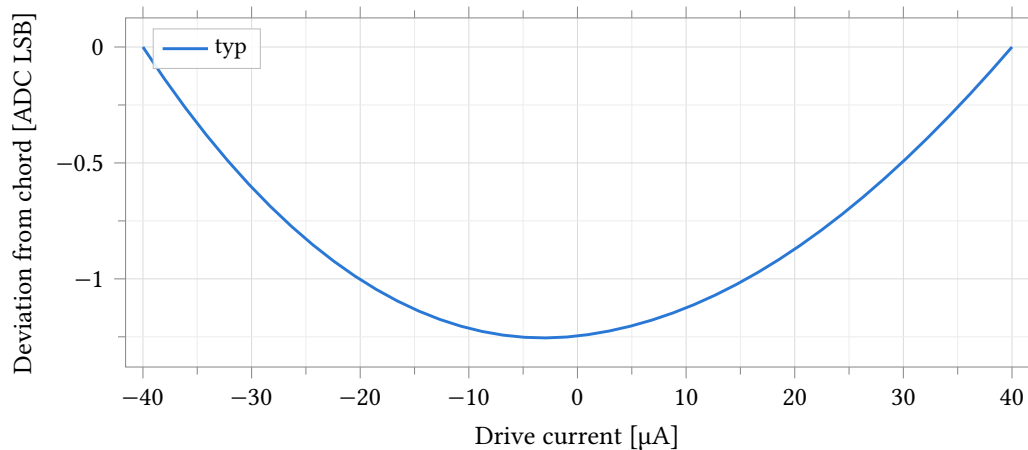


Figure 24.49: Drive nonlinearity of the bottom code ( $N = 0$ ,  $R \approx 19.6 \text{ k}\Omega$ ): voltage deviation from the endpoint chord over  $\pm 90\%$  of full scale ( $\pm 40 \mu\text{A}$ ), typical corner, expressed in  $1.592 \text{ mV ADC LSB}$ . All ten bypass gates are closed at this code and carry the full signal current; their on-resistance rises with the terminal's excursion from  $V_{\text{CM}}$ , so the chord, pinned at the sweep endpoints where that excursion is largest, overestimates the mid-range voltage by up to  $1.25 \text{ LSB}$ , and the deviation is bowl-shaped with a slight skew from the gates' voltage asymmetry. The same measurement at  $N = 63$  and  $N = 319$ , where the ladder dominates and current is  $20\text{--}100\times$  smaller, stays below  $0.004 \text{ LSB}$  and is tabulated, not plotted.

units read  $1.0197$  against an ideal  $1.015873$  ( $0.38\%$  high, which is switch resistance and does not cancel between the two differences), with a spread of  $0.132\%$ , i.e.  $0.13\%$  on one thermometer segment. It is the spread and not the offset that decides monotonicity, and the offset calibrates out.

| Seam                      | Mean<br>k $\Omega$ | $\sigma$<br>$\Omega$ | $\sigma/\mu$<br>% | Worst sample<br>k $\Omega$ | Margin<br>$\sigma$ |
|---------------------------|--------------------|----------------------|-------------------|----------------------------|--------------------|
| $N = 63 \rightarrow 64$   | 7.424              | 504.0                | 6.79              | 5.878                      | 14.7               |
| $N = 127 \rightarrow 128$ | 8.003              | 458.6                | 5.73              | 6.590                      | 17.5               |
| $N = 191 \rightarrow 192$ | 8.494              | 481.3                | 5.67              | 7.214                      | 17.7               |
| $N = 255 \rightarrow 256$ | 8.250              | 468.5                | 5.68              | 6.782                      | 17.6               |

Table 24.64: Monte Carlo of the four thermometer seam steps, 200 mismatch samples, seed 12345, low-discrepancy sampling, nominal process,  $25^\circ\text{C}$ ,  $500 \text{ nA}$  test current. Each sample sweeps all 320 codes, so every sample is graded at every seam. The last column is the margin from a negative step, in standard deviations of that seam's own distribution; the minimum step anywhere in the code space is  $137\sigma$  from zero.

The bench carries 14 pass/fail specifications, graded by the simulator against every corner and every Monte Carlo sample (Table 24.66). The limits are screening bounds chosen from the measured data: the endpoint windows allow the  $\pm 35\%$  poly sheet-resistance span plus margin, so a violation flags a broken device or model, not process spread. Twelve of the fourteen pass at every corner and every sample.

The two exceptions are honest and bounded.  $v_{\text{top}}$  exceeds its  $2.45 \text{ V}$  limit at the two slow-resistor corners ( $2.553 \text{ V}$  at ss-res,  $2.518 \text{ V}$  at ss  $125^\circ\text{C}$ ): at  $+35\%$  resistance the fixed  $500 \text{ nA}$  test current rails the  $2.5 \text{ V}$  supply at the top codes. That is the measurement drive running out of compliance at the slow corner, not a device defect (the same compliance ceiling already documented as the  $\sim 0.65 \mu\text{A}$  top-code operating limit), and the  $250 \text{ nA}$  control run bounds its effect on the reported resistances at  $\leq 0.036\%$  (Table 24.62).

| Parameter       | Unit       | $N$ | Mean     | $\sigma$ | Yield |
|-----------------|------------|-----|----------|----------|-------|
| $R(N=0)$        | k $\Omega$ | 200 | 19.650   | 0.071    | 100.0 |
| $R(N=63)$       | k $\Omega$ | 200 | 396.4515 | 0.3608   | 100.0 |
| $R(N=64)$       | k $\Omega$ | 200 | 403.8759 | 0.3644   | 100.0 |
| $R(N=319)$      | M $\Omega$ | 200 | 1.93212  | 0.00071  | 100.0 |
| Seam step 63→64 | k $\Omega$ | 200 | 7.424    | 0.504    | 100.0 |

Table 24.65: Monte Carlo of the programmable resistor at four codes plus the per-sample seam step. **Mismatch only**, model section `castalia_pm_25`, nominal process, 200 samples, seed 12345, low-discrepancy sampling, 25 °C, 500 nA test current. Relative spread falls with code as units accumulate:  $\sigma/\mu = 0.359\%$  at  $N=0$  (three ladder units plus ten switch Rons), 0.091 % at  $N=63$ , 0.037 % at  $N=319$ ; the gain ratio  $R(319)/R(0)$  spreads 0.354 %. Yield grades each code against a  $\pm 5\%$  absolute window about its typical value and the seam against zero (limits in Table 24.66): all five yield 100.0 % over the 200 samples, with the nearest limit  $14.2\sigma$  from the mean ( $N=0$ ).

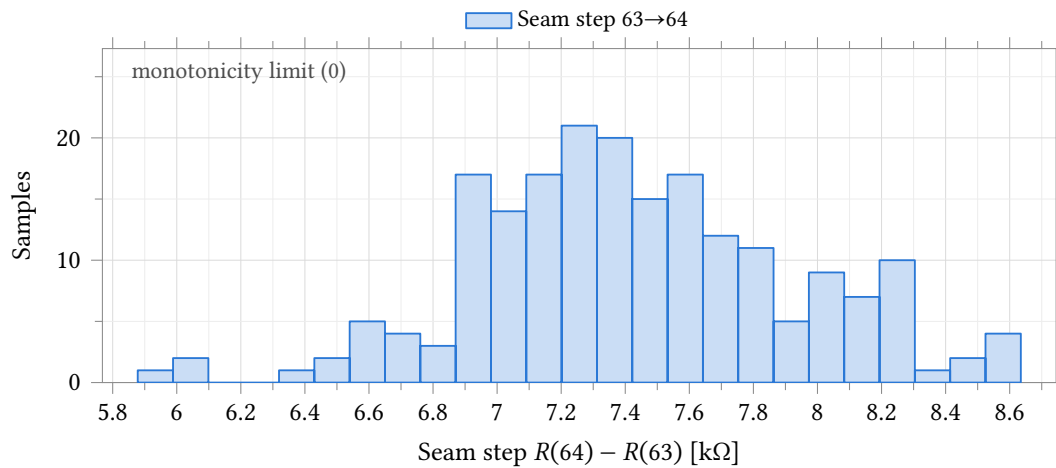


Figure 24.50: Per-sample step across the first thermometer seam (code 63 to 64: one 64-unit segment switches in while all six binary segments switch out), the code transition most exposed to mismatch. **Mismatch only**, model section `castalia_pm_25`, nominal process, 200 samples, seed 12345, low-discrepancy sampling, 25 °C. Mean 7.42 k $\Omega$ ,  $\sigma = 504 \Omega$ , worst sample 5.88 k $\Omega$ ,  $14.7\sigma$  above the monotonicity limit at zero, so no realistic mismatch closes the seam.

lsb\_err fails its 0.5-LSB budget at four of five corners (1.24 typical, 3.60 worst at ss 125C, passing only at ff -40C with 0.41): this is the bottom-code switch-Ron bowl of Figure 24.49, largest where the gates are slowest and hottest. It is accepted as documented: the chord-resistance error  $r_{lin\_err\_pct}$  stays below 1 % everywhere (0.12–0.53 %), and the fix, if bottom-code linearity ever matters, is wider bypass gates. One bench note: the linearity sweep’s drive source includes the  $i_{test}$  term ( $i = i_{test} + f \cdot 0.8 / R_{ideal}$ , mirroring the netlist source exactly), so the chord is taken about the same operating point the code sweep uses; against that corrected drive the typical worst deviation reads 1.24 LSB where Table 24.63 reported 1.26.

| Specification                                            | Unit       | Limit     | Measured    | Verdict               |
|----------------------------------------------------------|------------|-----------|-------------|-----------------------|
| <i>Code sweep, 320 codes, five corners</i>               |            |           |             |                       |
| $R$ at $N=0$                                             | k $\Omega$ | 12–28     | 13.26–26.41 | pass                  |
| $R$ at $N=319$                                           | M $\Omega$ | 1.25–2.65 | 1.280–2.606 | pass                  |
| Step, min                                                | k $\Omega$ | > 1       | 3.57–7.67   | pass                  |
| Step, mean                                               | k $\Omega$ | 3.8–8.2   | 3.97–8.09   | pass                  |
| Non-monotonic steps                                      | count      | 0         | 0           | pass                  |
| Error vs $18k + 6k \cdot N$ , max                        | %          | < 55      | 9.2–46.7    | pass                  |
| $V(A)$ at top code                                       | V          | < 2.45    | 1.89–2.55   | marginal <sup>a</sup> |
| <i>Drive linearity at <math>N=0</math>, five corners</i> |            |           |             |                       |
| Chord deviation, max                                     | ADC LSB    | < 0.5     | 0.41–3.60   | fail <sup>b</sup>     |
| Secant- $R$ error                                        | %          | < 1       | 0.12–0.53   | pass                  |
| <i>Monte Carlo, 200 mismatch samples, typical</i>        |            | Nominal   | Std. dev.   |                       |
| $R(N=0)$                                                 | k $\Omega$ | 19.650    | 0.071       | pass, 200/200         |
| $R(N=63)$                                                | k $\Omega$ | 396.452   | 0.361       | pass, 200/200         |
| $R(N=64)$                                                | k $\Omega$ | 403.876   | 0.364       | pass, 200/200         |
| $R(N=319)$                                               | M $\Omega$ | 1.93212   | 0.00071     | pass, 200/200         |
| Seam step 63→64                                          | k $\Omega$ | 7.424     | 0.504       | pass, 200/200         |

<sup>a</sup> Exceeded at ss-res (2.553 V) and ss 125C (2.518 V): test-drive compliance, not a device defect (see text).

<sup>b</sup> Fails at four of five corners; passes at ff -40C (0.41). Bottom-code switch-Ron bowl, accepted (see text).

Table 24.66: Specification grading: the 14 limits carried by the bench and the verdict. Corner rows list the limit and the measured extremes across all five corners of Table 24.61; the endpoint and step windows are screening bounds ( $\pm 35$  % corner span plus margin). Monte Carlo rows list the nominal (mean) and standard deviation over the 200 mismatch samples (typical process, 25 °C), each graded against a  $\pm 5$  % absolute-accuracy window about its typical value and the seam against zero.

### 24.8.1 In-loop noise

Inside the transimpedance loop the ladder is not a noise contributor. Its own Johnson noise is 0.0008 % to 0.0065 % of the input-referred noise power in the dc-amperometric band, four to five orders below the amplifier (Table 24.67), and replacing the whole switched ladder by a single ideal resistor of the measured value changes the answer by 0.02 % at the worst code. The roughly three hundred open bypass switches contribute nothing measurable. Loop stability is unchanged by the noise run (phase margin 74.9° to 92.9°, gain margin 21.65 dB to 21.79 dB), so these are the operating points the stability grid already reports.

**No noise figure for this block is meaningful without its band.** Over 0.1 Hz to 10 Hz the input-referred current noise is 48.9 pA to 1060 pA depending on gain code. Over 0.1 Hz–1 kHz it is 3.00 nA at every code, and over 0.1 Hz–10 kHz 26.7 nA at every code. That is not a ladder defect: above the frequency at which the feedback impedance exceeds the electrode impedance the noise gain is set by the electrode capacitance, so the input-referred noise becomes independent of  $R_f$ , which is what the identical 132 fA/ $\sqrt{\text{Hz}}$

spot noise at 1 kHz across six decades of  $R_f$  shows. The same design is 49 pA or 27 nA according to where the integral stops, so the band edges are fixed design variables rather than tracked to each code's own bandwidth: letting them follow would make the slow, high-gain codes report less noise merely for being slower.

Against the 2 nA input-referred budget in the amperometric band, codes  $N \geq 63$  sit 25× to 40× under it and the bottom code 1.9× under. At the top code the measured 48.9 pA is 17× below the 0.824 nA current quantum, so the resolution floor at maximum gain is quantisation, not noise.

| $N$ | $R_f$            | Input-referred noise, rms |              | Ladder share<br>of noise power |
|-----|------------------|---------------------------|--------------|--------------------------------|
|     |                  | 0.1 Hz to 10 Hz           | 0.1 Hz–1 kHz |                                |
| 0   | 19.63 k $\Omega$ | 1060.5 pA                 | 3.304 nA     | 0.0008 %                       |
| 63  | 394.8 k $\Omega$ | 81.4 pA                   | 3.005 nA     | 0.0065 %                       |
| 127 | 777.1 k $\Omega$ | 60.0 pA                   | 3.004 nA     | 0.0061 %                       |
| 191 | 1.159 M $\Omega$ | 53.6 pA                   | 3.004 nA     | 0.0051 %                       |
| 255 | 1.542 M $\Omega$ | 50.6 pA                   | 3.003 nA     | 0.0043 %                       |
| 319 | 1.924 M $\Omega$ | 48.9 pA                   | 3.003 nA     | 0.0037 %                       |

Table 24.67: Input-referred current noise of the assembled transimpedance loop against gain code, 37 °C, nominal process, at a 100 nF double-layer capacitance and the 22 pF compensation capacitor. Noise is referred through the measured transimpedance rather than by an input-referral option, which does not reach the netlist from this setup. The last column is the feedback resistor's own Johnson noise as a fraction of the noise power in the 0.1 Hz to 10 Hz band. The two bands differ because the noise gain above the feedback/electrode impedance crossover is set by the electrode, not by  $R_f$ .

Three limits on what this section claims. First, the resistance, linearity and mismatch data are two-terminal measurements of the ladder alone; only noise and loop stability have been measured with the ladder inside the transimpedance loop (Section 24.8.1), at one electrode and one process corner, and settling with the real switches is still open. Second, the Monte Carlo is mismatch-only at 25 °C; no process-plus-mismatch statistics exist. Third, the bottom code is the block's weak point, and all three of its symptoms share one root cause: the 1.65 k $\Omega$  closed-switch stack (+9.2 % at  $N = 0$ ), the 1.26-LSB drive nonlinearity, and the 4× larger relative mismatch spread are all the ten bypass gates in series with a three-unit ladder. Absolute gain at any code calibrates out against the corner and mismatch spread; if the bottom code ever needs uncalibrated accuracy, the fix is wider bypass gates, not more ladder. Integration into `anatop_tia` (whose legacy 16-bit `tia_gain` bus becomes these ten control bits) is pending, as is layout.

### 24.8.2 Measuring this block on chip

1. Force  $\pm$  full-scale current into the working-electrode pin with no cell present; read the transimpedance output at per-channel slot 4 on `atb_d`, one gain code at a time. Scale the drive per gain code or the high-gain codes rail; skip the bottom code: its drive nonlinearity is 1.26 LSB typical and no two-point slope sees it.
2. Read at the pad, not through the converter: the converter window truncates samples that rail, and the two-point slope wanted here is the end-to-end constant in amperes per code, absorbing the resistor's  $\pm 35$  % sheet tolerance (Table 24.62), the 1.65 k $\Omega$  closed-switch stack and the converter's own gain in one number.
3. Switch resistance: slots 7 and 8 straddle the working-electrode isolation switch; their difference under the same forced current is that switch's  $i \cdot R_{\text{on}}$ , which calibration stores rather than minimises.

4. Gain refresh without equipment: switch the on-chip reference resistor between the counter-electrode node and the summing node, take the same two-point slope, and scale the stored constant by the change in that ratio. Sheet resistance cancels; matching alone remains, 0.04 % to 0.36 % (Table 24.65). This test fabric is design intent; it is not yet in the schematics.

## 24.9 Reference DAC

anatop\_pixel\_dacR2R12, a 12-bit voltage-mode R-2R DAC, produces a channel's reference potential. It was fed by anatop\_pixel\_dac\_psu, a supply block deleted from anatop\_pixel on 2026-08-24; the ladder now runs from the 2.5 V analog rail directly, and the deleted block is kept on record in Section 24.9.2. All data in this section is schematic-level, with no extracted parasitics; layout and extracted views exist for both cells and have not been simulated.

The ladder is drawn in Figure 24.51. Its terminals are A<11:0>, En and Out; the supply arrives by inherited connection and netlists as inh\_vdd/inh\_vss, so there is no supply pin on the symbol. Each code bit is gated through a two-input gate whose output high level is the ladder reference, so whatever drives inh\_vdd sets full scale directly. That gate was a 1.0 V standard cell until 2026-09-05 and is now a custom 2.5 V cell; every corner and Monte Carlo number below is measured on the current one, and the change is recorded once, after the nominal linearity table. Every element of the ladder is the same unit resistor ( $R$  is two units in series,  $2R$  four), which makes the binary weighting a ratio of identical devices and the converter's linearity a matching property rather than a design margin. A nodal solve of the transcribed network gives  $V_{\text{out}}/V_{\text{ref}} = D/4096$  exactly at every code, so none of the nonlinearity reported below is the ratio design.

Three testbench views drive the two cells. DacTB puts the DAC on an ideal 2.5 V rail and measures the ladder alone, which is the tapeout arrangement. DacPSUTB exercises the supply block by itself against a current load, and DacWPSUTB runs both together with the DAC's inherited supply resolved to VddDac by a netSet property; both of those measure the removed block.

Coverage is five corner points (the typical statistical point and the ff, fs, sf and ss skew corners) and a 200-sample Monte Carlo run varying process and mismatch together, both on the current cell. Temperature is 25 °C on both DAC benches throughout; the four supply-block benches run at 27 °C at the typical point and 25 °C at the skew corners. The two DAC benches carry tightened solver tolerances ( $\text{reltol} = 1 \times 10^{-6}$ ,  $\text{vabstol} = 1 \times 10^{-9}$ ,  $\text{iabstol} = 1 \times 10^{-15}$ ): the LSB through the supply block is 529  $\mu\text{V}$  and the ADE defaults permit several LSB of error per DC solve, of which DNL is a difference.

One of this section's two defects survives into the tapeout channel. The ladder is matching-limited and fails its own linearity limits on an ideal rail (Section 24.9.1), and no corner point sees it, or can, because a process corner moves every ladder resistor together. The other, the supply block modulating the reference by 25.8 mV peak to peak across the code range against a quarter-LSB budget of 133  $\mu\text{V}$ , went with the block it belonged to (Section 24.9.2): it is no longer modelled, not smaller.

**The bit driver is a 2.5 V cell, and the tables above are measured on it.** The gate on each code bit was a 1.0 V standard cell sitting on the 2.5 V analog rail, which is a reliability defect independent of linearity; it was replaced on 2026-09-05 by a custom cell on the 2.5 V device family. **The driver's on-resistance is a linearity term, not a timing term:** its pull-up and pull-down differ, and at the most significant carry that difference appears divided by a  $2R$  leg and multiplied by 2048. For the cell removed, that product is 0.495 LSB against the 0.511 LSB of differential nonlinearity it actually produced, which is the whole of the carry error.

**The replacement is better than the cell it replaces on both sides.** Over the same five corners, the same bench and the same tolerance, integral nonlinearity at the worst corner falls from 0.425 LSB to



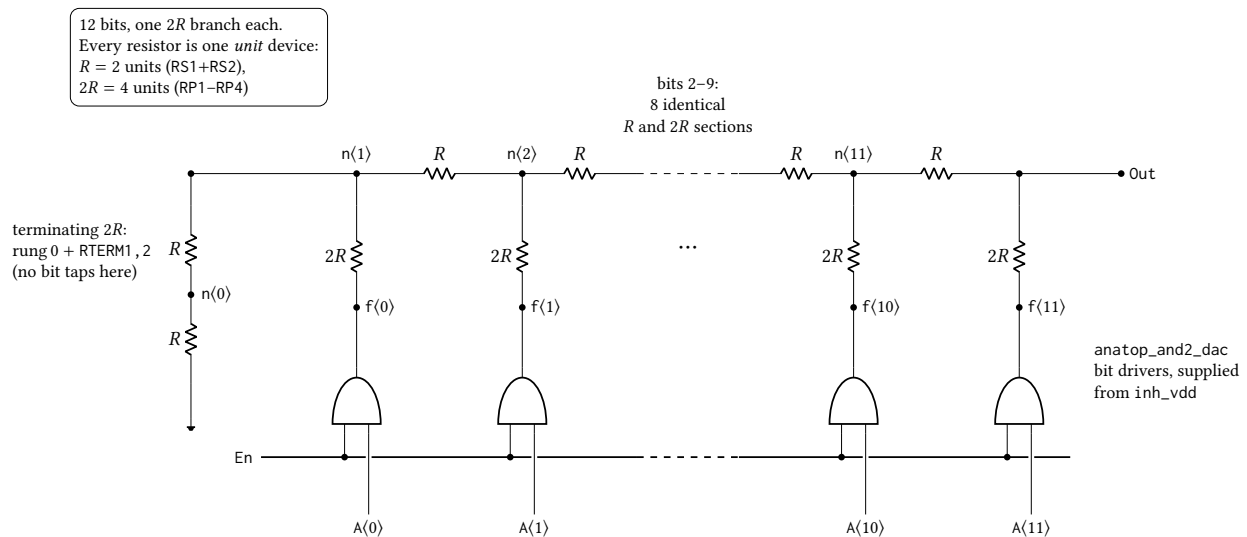


Figure 24.51: The 12-bit voltage-mode  $R$ – $2R$  ladder of `anatotop_pixel_dacR2R12`, transcribed from the `DacNonLinTB` netlist. Bit  $k$  drives tap  $n(k+1)$  through its own  $2R$  branch, so the twelve taps are  $n(1)$ – $n(11)$  plus Out, and the MSB branch lands directly on Out at the top of the ladder;  $n(0)$  carries no bit, its rung and  $R_{TERM1,2}$  together forming the terminating  $2R$ . Bits 2–9 are elided: eight further sections identical to those drawn sit between  $n(2)$  and  $n(11)$ . Every element is the same unit resistor, so the network itself is exact and gives  $\text{Out} = (\text{code}/4096) V_{\text{ref}}$ ; the reference is the *output high level* of the `anatotop_and2_dac` drivers, which sit in series with each  $2R$  branch and bound the ladder in place of the rails: measured zero scale is 20 nV above  $V_{\text{SS}}$  and full scale 1.2  $\mu\text{V}$  below the ideal  $(4095/4096) V_{\text{ref}}$  on a 2.5 V rail. Device dimensions are proprietary and are not stated.

| Point | Process | Temperature | Supply |
|-------|---------|-------------|--------|
| 1     | tt      | 25 °C       | 2.5 V  |
| 2     | ff      | 25 °C       | 2.5 V  |
| 3     | fs      | 25 °C       | 2.5 V  |
| 4     | sf      | 25 °C       | 2.5 V  |
| 5     | ss      | 25 °C       | 2.5 V  |

Table 24.68: Process, temperature and supply points simulated for the 12-bit  $R$ – $2R$  DAC and its supply block: point 1 is the typical statistical process point evaluated at nominal, points 2–5 the imported foundry skew corners. Temperature and solver tolerance are not uniform across this run: the two DAC benches sit at 25 °C at all five points and run  $\text{reltol} = 1 \times 10^{-6}$ ,  $\text{vabstol} = 1 \times 10^{-9}$ ,  $\text{iabstol} = 1 \times 10^{-15}$ , while the four supply-block benches (load regulation, line regulation, output impedance, startup) sit at 27 °C at point 1 and 25 °C at points 2–5 and run the ADE defaults  $1 \times 10^{-3}$ ,  $1 \times 10^{-6}$ ,  $1 \times 10^{-12}$ ; the column below reports the DAC benches. This is a corner run and carries no device mismatch;  $R$ – $2R$  linearity is a matching property, so every nonlinearity figure in this section is a systematic term and none of it is a yield statement. These points move the sixteen transistor model rows only (the resistor and capacitor rows stay at their typical sections), so every corner spread in this section is a lower bound (pinned-passive caveat, page 207).

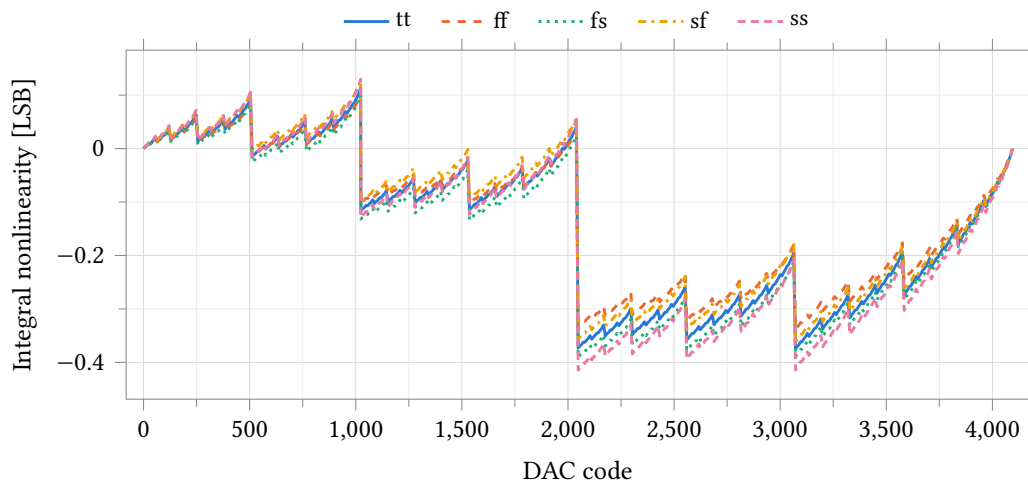


Figure 24.52: Integral nonlinearity against code over five process corners, ladder on an ideal 2.5 V rail, 25 °C,  $\text{reltol} = 1 \times 10^{-6}$ . INL is taken from the endpoint line through each corner's own zero- and full-scale outputs, so it is zero at codes 0 and 4095 by construction; the run is the 2026-09-06 five-corner sweep on the tapeout cell, with the custom 2.5 V bit driver `anatop_and2_dac`. The positive extreme is code 1023 at every corner, +0.10 to +0.13 LSB; the negative extreme is the major carry at code 2048 or 3072, which sit within 0.006 LSB of each other at -0.34 to -0.41 LSB, so which of the two wins flips with the corner while the shape does not. Corner run, no device mismatch: R-2R linearity is a matching property, so this envelope is the systematic term alone.

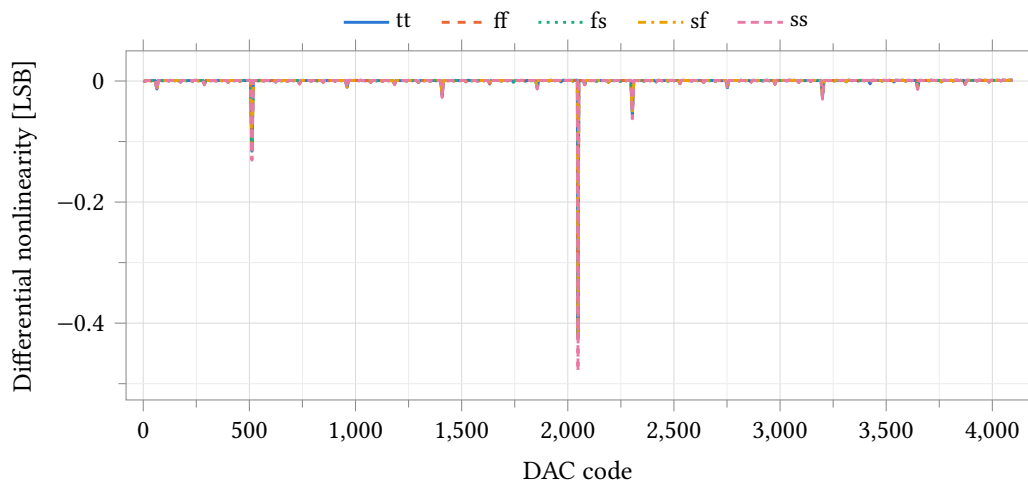


Figure 24.53: Differential nonlinearity against code over five process corners, conditions as Figure 24.52. DNL is the first difference of the transfer divided by the endpoint LSB, minus one. The worst step is the mid-scale major carry at code 2048, -0.37 to -0.48 LSB; at tt the sub-carries measure -0.23 at code 1024, -0.19 at 3072, -0.12 at 512 and -0.11 LSB at 2560. The 4095-point curve is stride-decimated for the plot, so the comb drawn is a sample of the code grid and not every tooth of it; the extreme is preserved exactly. The converter is monotonic at every corner, with 0.52 LSB of margin to the -1 LSB limit at worst.

| Parameter            | Unit          | tt       | ff       | fs       | sf       | ss       | Spread   |
|----------------------|---------------|----------|----------|----------|----------|----------|----------|
| Zero-scale output    | $\mu\text{V}$ | 0.02     | 0.25     | 0.01     | 0.06     | 0.00     | 0.25     |
| Full-scale output    | V             | 2.499388 | 2.499388 | 2.499388 | 2.499388 | 2.499388 | 0.000000 |
| LSB size             | $\mu\text{V}$ | 610.351  | 610.351  | 610.351  | 610.351  | 610.351  | 0.000    |
| INL, most positive   | LSB           | 0.113    | 0.101    | 0.100    | 0.127    | 0.130    | 0.030    |
| INL, most negative   | LSB           | -0.376   | -0.338   | -0.390   | -0.362   | -0.414   | 0.076    |
| INL, worst magnitude | LSB           | 0.376    | 0.338    | 0.390    | 0.362    | 0.414    | 0.076    |
| DNL, most positive   | LSB           | 0.002    | 0.001    | 0.002    | 0.002    | 0.002    | 0.000    |
| DNL, most negative   | LSB           | -0.423   | -0.374   | -0.424   | -0.424   | -0.479   | 0.105    |
| DNL, worst magnitude | LSB           | 0.423    | 0.374    | 0.424    | 0.424    | 0.479    | 0.105    |

Table 24.69: Static accuracy of the ladder on an ideal 2.5 V rail by process corner, DacTB, 25 °C, all 4096 codes,  $\text{reltol} = 1 \times 10^{-6}$ , on the tapeout cell with the custom 2.5 V bit driver `anatop_and2_dac`; the ideal endpoints for this bench are 2.499 390 V full scale and 610.35  $\mu\text{V}$  per LSB ( $2.5 \times 4095/4096$ ), and the measured LSB is within 0.35 nV of that at every corner. INL and DNL are referred to the endpoint line through each corner's own zero- and full-scale outputs, so gain and offset are removed and only shape is reported. Corner run, no device mismatch: these are systematic terms, and the bench saves the output node only, so no supply current is measured on it (the supply block's quiescent current is Table 24.78). The scale columns stand: zero scale, full scale and the LSB are ratiometric and identical to seven digits whether the passive model rows are pinned at typical or freed. The INL and DNL columns are a lower bound, because the pass-gate on-resistance is a transistor quantity against a poly ladder; freed, worst DNL is -0.557 and worst INL -0.488 LSB, both at ff (pinned-passive caveat, page 207).

0.414 LSB and the most negative differential step from -0.535 LSB to -0.479 LSB, still at code 2048 and still monotonic at every point. The Monte Carlo of Section 24.9.1 moves the same way, to 42.0 % monotonic and 20.5 % within one LSB of integral nonlinearity against 40.0 % and 21.0 %. **The conclusion of Section 24.9.1 is unchanged: what limits this converter is ladder matching, and no driver fixes it.**

### 24.9.1 Matching

R-2R linearity is a matching property, so it is the Monte Carlo run and not the corner set that measures it. On an ideal 2.5 V rail, over 200 samples varying process and mismatch together with the custom 2.5 V bit driver `anatop_and2_dac` the tapeout cell carries, INL is 1.49 LSB mean against a 1 LSB limit and the worst sample reaches 3.39 LSB; DNL reaches -6.50 LSB (Table 24.70). 20.5 % of samples meet  $\text{INL} < 1 \text{ LSB}$  and 42.0 % are monotonic ( $\text{DNL} > -1 \text{ LSB}$ ); 18.5 % meet both, which over four independent channels is a 0.12 % chip yield.

The failure is matching and nothing else. Full scale, LSB and gain error are set by the rail and by ladder attenuation, not by device ratios, and they barely move across the same 200 samples:  $\sigma$  is 1.2  $\mu\text{V}$  on a 2.4994 V full scale and 0.50 ppm about a -0.51 ppm gain error (Table 24.71). All of the spread in Table 24.70 is therefore resistor mismatch in the ladder.

Referencing the ladder to  $\text{VddDac}$  instead of an ideal rail did not change this either. The ratiometric figures of Table 24.74 (1.51 LSB mean INL, -1.75 LSB mean DNL, 40.0 % monotonic) reproduce the ideal-rail figures to within 5 %, because dividing the reference out removes the supply block and leaves the same ladder. That table and the raw columns beside it, 36.8 LSB and -10.8 LSB, are the earlier campaign: the 1.0 V bit driver, and the supply block of Section 24.9.2 that the tapeout channel no longer has. They are kept as the corroboration of the matching result, not as a specification.

The corner set sees none of it. Ratiometric INL spans 0.432 LSB to 0.450 LSB across ff, fs, sf and ss, a 4 %

spread, while the Monte Carlo samples of the same quantity scatter from 0.35 LSB to 3.49 LSB. A process corner moves every ladder resistor in the same direction, which a ratio cancels; the corner table is context and not a linearity result.

INL and DNL are both skewed (Fisher skew 0.74 and  $-1.14$ ), so the tables quote the median, the first and ninety-ninth percentiles and the worst sample rather than  $\text{mean} \pm 3\sigma$ .

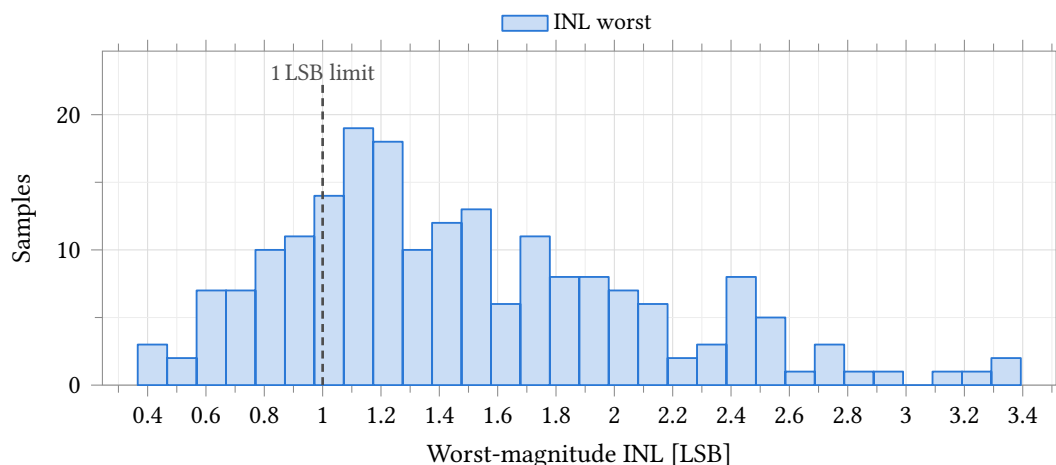


Figure 24.54: Integral nonlinearity of the  $R$ - $2R$  ladder driven from an ideal 2.5 V reference, with the custom 2.5 V bit driver `anatop_and2_dac` the tapeout cell carries. **Process and mismatch**, model section `castalia_pm_25`, 200 samples, seed 12345, low-discrepancy sampling, 25 °C, all 4096 codes. The dashed rule is the 1 LSB limit entered on the test; 41 of 200 samples fall inside it and the worst reaches 3.393 LSB. The distribution is right-skewed (Fisher skew  $+0.74$ ), so Table 24.70 quotes the median and percentiles rather than  $\text{mean} \pm 3\sigma$ . LSB size, full scale and gain error are fixed to a part in  $10^6$  on this bench (Table 24.71), so the whole of this spread is resistor mismatch in the ladder and none of it is the reference.

### 24.9.2 Removed block: the DAC supply regulator

**The two per-pixel DAC supply regulators were deleted from `anatop_pixel` on 2026-08-24, returning 806.8  $\mu\text{W}$ , and both ladders now run from the 2.5 V analog rail directly.** Everything in this subsection was measured through that block and describes a path the tapeout channel does not have, so the 25.8 mV reference movement below is no longer modelled rather than smaller, and the 528.6  $\mu\text{V}$  LSB it implies does not apply to the shipping cell.

Divide the reference out and the ladder's own transfer is well behaved; leave it in and the same sweep is  $-37$  LSB of INL and  $-10.8$  LSB of DNL (Table 24.73). The difference is entirely reference movement, and the mechanism is not the one a reader expects.

In an  $R$ - $2R$  the binary weighting comes from ladder attenuation, not from branch current: every  $2R$  branch switched to the reference draws a comparable current. The current drawn from  $V_{\text{dd}}\text{DAC}$  therefore tracks the bit *pattern*, not the code value. An ideal-ladder model gives a reference current of zero at code 0,  $0.25 V_{\text{ref}}/R$  at code 2048 and  $0.333 V_{\text{ref}}/R$  at code 4095, with a maximum of  $1.444 V_{\text{ref}}/R$  at code 1365 = 0b010101010101, the alternating-bit pattern, and exactly where the measured  $V_{\text{dd}}\text{DAC}$  minimum sits, in every corner but ff and in all 200 Monte Carlo samples. A one-parameter fit  $V_{\text{DDDAC}} = V_0 - R_{\text{out}}I_{\text{ref}}$  accounts for 94 % of the variance of the measured curve, at a correlation of  $-0.97$  and a residual of 931  $\mu\text{V}$  rms against a 25.70 mV spread. Figure 24.57 is that curve. It closes on the measured nonlinearity: droop times code weight accounts for  $-36.9$  LSB of the  $-37.1$  LSB of raw INL, so 99.5 % of the delivered block's static error is reference modulation and not a ladder error.

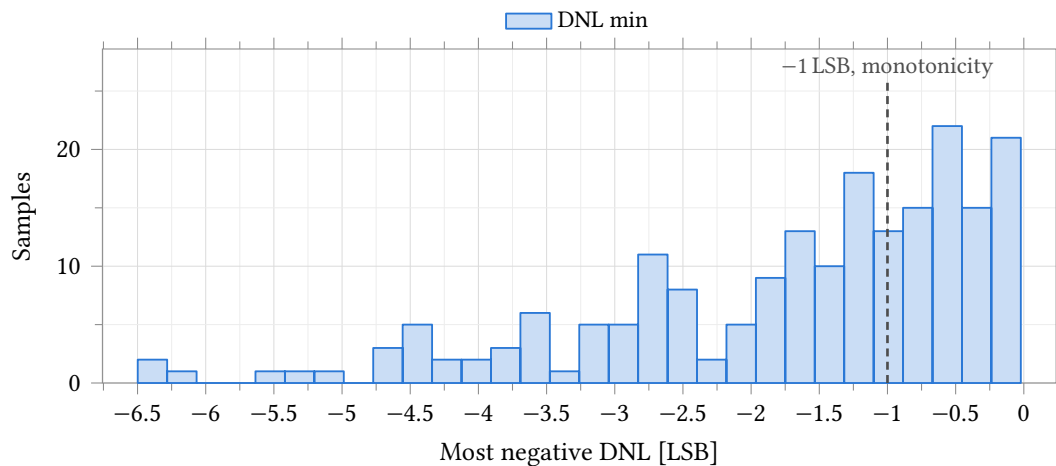


Figure 24.55: Worst negative differential nonlinearity, same bench, samples and bit driver as Figure 24.54. The dashed rule at  $-1$  LSB is the monotonicity limit entered on the test: 84 of 200 samples clear it, so 42.0 % of channels are monotonic and  $0.420^4 = 3.1$  % of four-channel chips are monotonic on all four. Graded on integral nonlinearity as well, 37 of 200 pass and the four-channel figure is 0.12 %. The distribution is left-skewed (skew  $-1.14$ ) with a tail to  $-6.499$  LSB, so the median  $-1.287$  LSB and the percentiles of Table 24.70 describe it and  $\text{mean} \pm 3\sigma$  does not.

| Parameter | Unit | N   | Mean     | $\sigma$ | Median   | $p_1$    | $p_{99}$ | Worst    | Spec          | Yield | 4 ch |
|-----------|------|-----|----------|----------|----------|----------|----------|----------|---------------|-------|------|
| INL worst | LSB  | 200 | 1.485    | 0.620    | 1.364    | 0.426    | 3.287    | 3.393    | $\leq 1.000$  | 20.5  | 0.2  |
| INL max   | LSB  | 200 | 1.154    | 0.610    | 1.031    | 0.194    | 2.886    | –        | –             | –     | –    |
| INL min   | LSB  | 200 | $-1.485$ | 0.620    | $-1.364$ | $-3.287$ | $-0.426$ | –        | –             | –     | –    |
| DNL min   | LSB  | 200 | $-1.682$ | 1.415    | $-1.287$ | $-6.161$ | $-0.058$ | $-6.499$ | $\geq -1.000$ | 42.0  | 3.1  |
| DNL max   | LSB  | 200 | 1.123    | 1.128    | 0.713    | 0.007    | 4.760    | –        | –             | –     | –    |
| DNL worst | LSB  | 200 | 2.249    | 1.314    | 1.847    | 0.441    | 6.161    | 6.499    | $\leq 0.750$  | 9.0   | 0.0  |

Table 24.70: Monte Carlo static linearity of the  $R$ - $2R$  ladder on an ideal 2.5 V reference, with the custom 2.5 V bit driver anatop\_and2\_dac: **process and mismatch**, model section castalia\_pm\_25, 200 samples, seed 12345, low-discrepancy sampling, 25 °C, all 4096 codes,  $\text{reltol } 1 \times 10^{-6}$ , yield columns as per cent of samples and of four independent channels. The ladder fails on its own: 41 of 200 samples hold INL below 1 LSB, 84 of 200 are monotonic, and both together on 37 of 200, 18.5 % of channels and 0.12 % of four-channel chips. Only the three worst-case rows carry limits entered on the test; `inl_max`, `inl_min` and `dnl_max` were not graded and their spec, yield and worst-case cells are therefore empty. Worst-magnitude INL is the negative extreme on all 200 samples, so the first and third rows are one measurement with opposite sign, while `inl_max` is the independent positive extreme. All three graded rows are skewed (Figures 24.54 and 24.55), so the median and percentiles are quoted and  $\text{mean} \pm 3\sigma$  is not.

| Parameter         | Unit          | $N$ | Mean     | $\sigma$ | Min      | Max      |
|-------------------|---------------|-----|----------|----------|----------|----------|
| LSB size          | $\mu\text{V}$ | 200 | 610.3513 | 0.0003   | 610.3505 | 610.3521 |
| Full-scale output | V             | 200 | 2.499388 | 0.000001 | 2.499385 | 2.499392 |
| Zero-scale offset | LSB           | 200 | 0.0000   | 0.0000   | 0.0000   | 0.0003   |
| Gain error        | ppm           | 200 | -0.51    | 0.50     | -1.74    | 0.93     |

Table 24.71: Monte Carlo scale factors, same bench and samples as Table 24.70. **Process and mismatch**, castalia\_pm\_25, 200 samples, 25 °C. On an ideal reference these four are set by the rail and by the ladder’s attenuation ratio, not by matching: LSB size spreads 0.31 nV on 610.35  $\mu\text{V}$ , full scale 1.2  $\mu\text{V}$  on 2.5 V, and gain error 0.50 ppm about a -0.51 ppm mean. All of the linearity spread of Table 24.70 is therefore ladder mismatch and none of it is a gain or offset term. No limit was entered on any of these outputs, so the spec, yield and worst-case columns are omitted rather than left blank; mean  $\pm 3\sigma$  is not quoted either, because at a part in  $10^6$  the distribution shapes are solver residue and not device statistics.

Two consequences follow that are easy to get wrong. First, the -10.86 LSB DNL is real non-monotonicity of the block as delivered and not a numerical artifact: tightening `reltol` from  $1 \times 10^{-3}$  to  $1 \times 10^{-6}$ , `vabstol` from  $1 \times 10^{-6}$  to  $1 \times 10^{-9}$  and `iabstol` from  $1 \times 10^{-12}$  to  $1 \times 10^{-15}$  moves `VddDac` by at most 193  $\mu\text{V}$  anywhere and leaves the structure of the curve intact. Second, endpoint-anchored INL makes the block look worse still, because codes 0 and 4095 both sit near the top of the droop curve and the whole bow therefore lands in INL at maximum leverage.

The requirement on the supply block follows from the DAC rather than from an assumed target. A quarter-LSB reference budget is  $0.25 \times 528.96 \mu\text{V} = 133 \mu\text{V}$ , against the 25.8 mV the ladder’s own code-dependent current actually produces: the supply’s output impedance has to fall by a factor of 193, to about  $5 \Omega$ . Measured over the 200 Monte Carlo samples it is 894  $\Omega$  small-signal at zero load (Table 24.81) and 1133  $\Omega$  as a chord across the full 0  $\mu\text{A}$  to 100  $\mu\text{A}$  load sweep (Table 24.77); the typical corner gives 902  $\Omega$  and 1137  $\Omega$ . That is what the topology of Figure 24.56 gives: the block is a shunt clamp, not a regulated loop, so its output impedance is the  $1/g_m$  of the shunt device at a quiescent current of 160  $\mu\text{A}$  and is flat from DC to 100 kHz. Closing the factor of 193 is a topology change, not a trim. `VddDac` is the ladder reference today, not merely a gated supply for the digital side: `Out/VddDac` is 0.499 890 at code 2048 and 0.999 733 at code 4095, against an ideal  $4095/4096 = 0.999\,756$ .

The supply-side entries that follow are one defect measured five ways, not five problems: `vddac_pp`, the raw `dnl_min`, `load_reg`, `dvddac_load` and `zout_max` are all the same output impedance. `load_reg` and `zout_lf` correlate at  $r = 0.97$  across the 200 samples, and `vddac_pp` against the raw `dnl_min` at  $r = -0.94$ .

### 24.9.3 Scope of these measurements

All data in this section is schematic-level. Layout and `av_extracted` views exist for both cells and have not been simulated, so no number here carries interconnect parasitics, which matter for an R-2R ladder, whose accuracy is a ratio of resistances that layout can degrade.

**Settling time and glitch impulse were not measured.** The code bus is driven by a `busset12` model that holds a static code, so every measurement in this section is a sequence of DC solves. Code-step transients, and with them settling time, glitch impulse and the major-carry glitch that an R-2R is prone to, require a different stimulus and are not characterised.

**No settling or overshoot figure of merit is quoted for the supply block.** The corner run’s startup transient is valid and is plotted in Figure 24.65: `VddDac` reaches its zero-load DC value to within 140  $\mu\text{V}$  at every corner and holds it. What cannot be published from it is a number, because the bench’s `t_win`

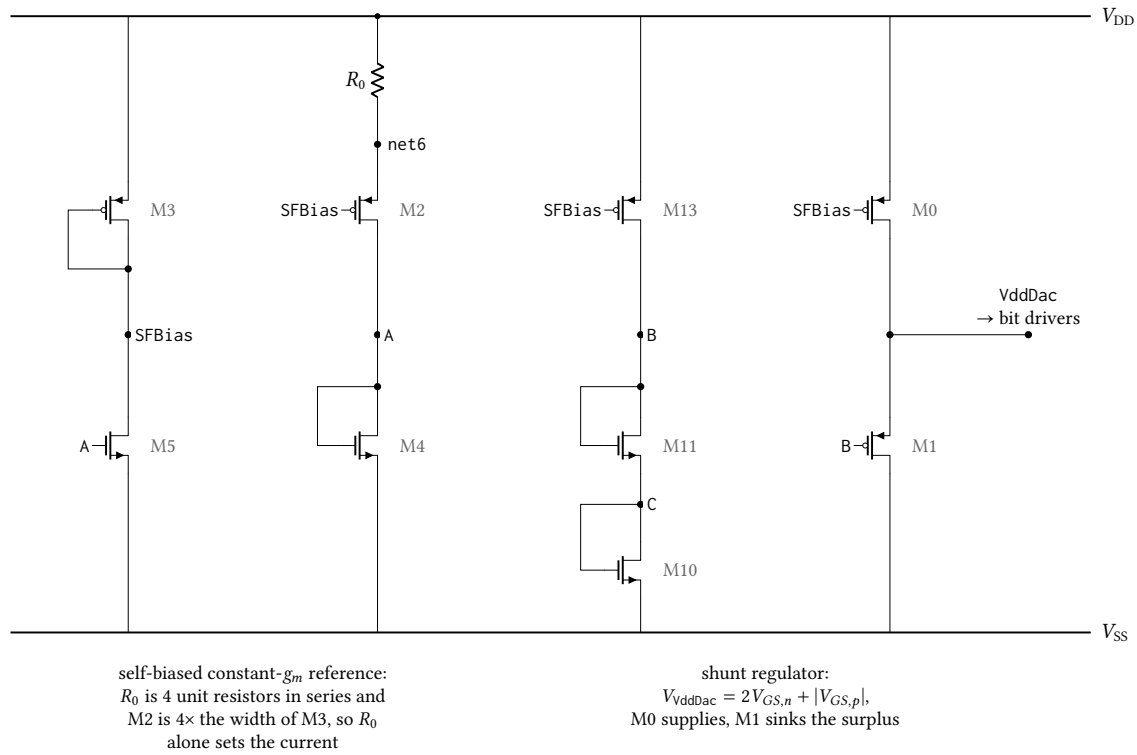


Figure 24.56: Core of anatop\_pixel\_dac\_psu, the shunt regulator that generates the VddDac rail the ladder uses as its reference, transcribed from the LoadRegTB netlist. The self-biased constant- $g_m$  reference on the left sets SFBias; M13 drives that current through the stacked diodes M11 and M10 to hold B two gate-source voltages above  $V_{SS}$ , and M1 sinks whatever of M0's current the ladder leaves, clamping VddDac one PMOS gate-source voltage above B: 2.177 V unloaded from a 2.5 V rail. There is no error amplifier, so load regulation is set by M1's  $g_m$  alone. The five start-up and enable devices (the weak pull-up and its two kick devices on SU, the shutdown pull-up on SFBias and the En inverter) are omitted; none carries current once the loop is running. Device dimensions are proprietary and are not stated.

| Parameter                  | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|----------------------------|------|--------|--------|--------|--------|--------|--------|
| Reference, maximum         | V    | 2.1655 | 1.9938 | 2.1056 | 2.2232 | 2.3511 | 0.3573 |
| Reference, minimum         | V    | 2.1398 | 1.9642 | 2.0788 | 2.1954 | 2.3221 | 0.3578 |
| Reference excursion, pk-pk | mV   | 25.700 | 29.572 | 26.783 | 27.842 | 29.053 | 3.872  |
| Code at the minimum        |      | 1365   | 1317   | 1365   | 1365   | 1365   | 48     |

Table 24.72: Ladder reference excursion over the full code sweep by process corner, DacWPSUTB at 25 °C with zero external load. The ladder is the only load on the supply block, so this is reference modulation the DAC imposes on itself; against the 133  $\mu$ V limit entered on the test it is 190 to 220 times over. The minimum lands on code 1365 = 0b0101010101 at four corners and 1317 at ff. The maximum is at code 0 at tt, fs and ss but at code 4095 at ff and sf, which is why the excursion is quoted peak-to-peak and not as a droop from zero scale.

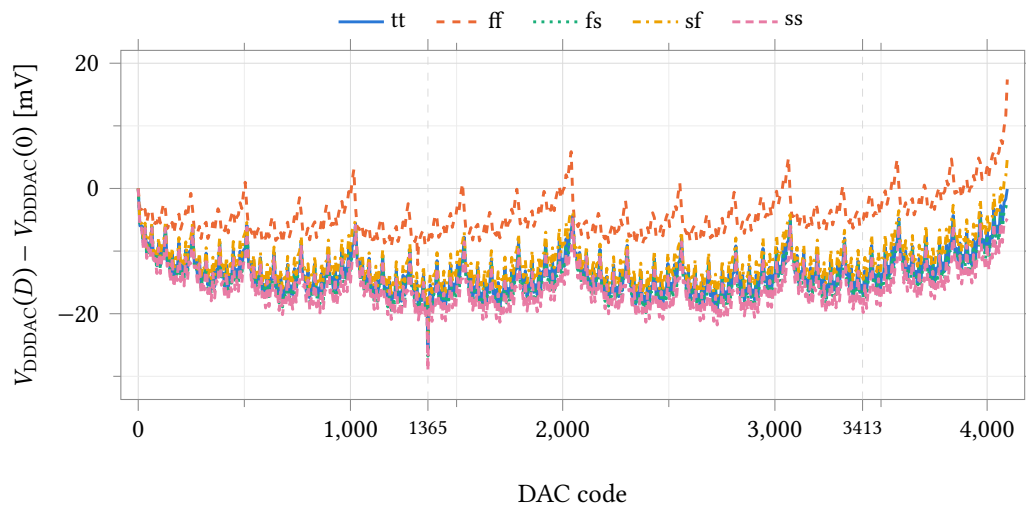


Figure 24.57: Ladder reference  $V_{DDDAC}$  referred to its value at code 0, against DAC code, over five process corners, DacWPSUTB at 25 °C with the external load current set to zero. The ladder is the only load on the supply block and its current is code-dependent, so the reference the ladder divides moves 25.7 mV to 29.6 mV peak-to-peak with no external load at all; the absolute levels, which differ by 357 mV across corners, are Table 24.72. The two marked codes are alternating-bit patterns: 1365 = 0b0101010101 is the global minimum at four of the five corners (1317 at ff), and 3413 = 1365 + 2048 droops 23.4 mV at 2.5× the code weight and therefore sets the worst raw INL of Figure 24.58. Reference droop times code accounts for −36.9 of the −37.1 LSB measured there, so the raw nonlinearity of the delivered block is reference modulation and not a ladder error.

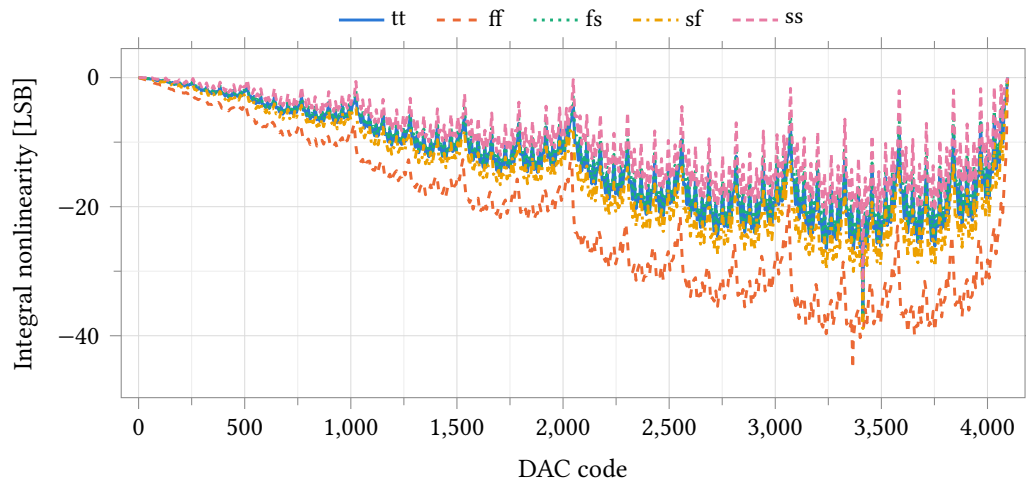


Figure 24.58: Integral nonlinearity against code over five process corners with the ladder referenced to  $V_{ddDac}$  from the supply block, zero external load, 25 °C. Endpoint-line INL as in Figure 24.52; the vertical scale is tens of LSB, not tenths. The bow reaches −37.1 LSB at tt (code 3413) and −45.1 LSB at ff (code 3365) and tracks the reference droop of Figure 24.57 scaled by code. Dividing each code’s output by the reference measured at that code removes it and leaves 0.43 to 0.45 LSB (Table 24.73), so the ladder is unchanged from the ideal-rail case.



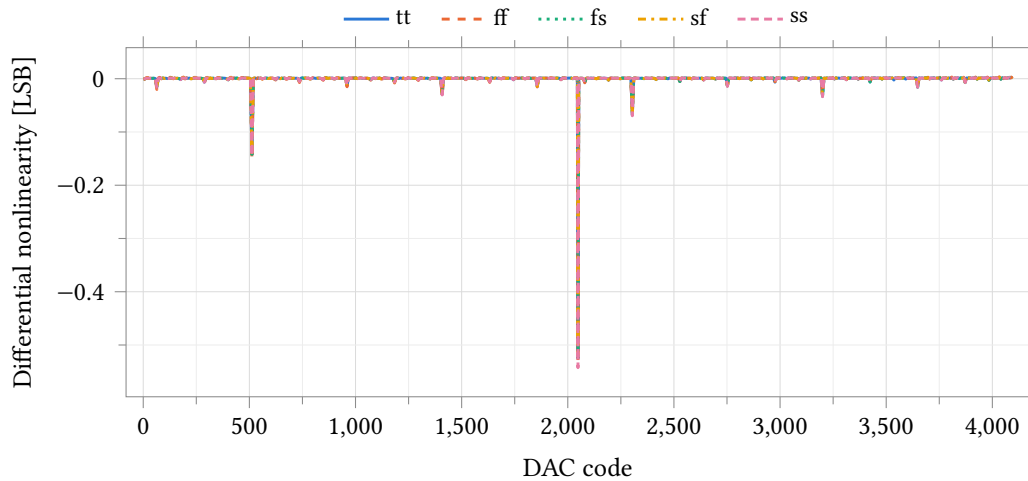


Figure 24.59: Ratiometric differential nonlinearity against code over five process corners: each code's output is divided by the reference measured at that code before the LSB and the first difference are taken, so the reference droop divides out and the ladder alone is measured. Conditions as Figure 24.58; the 4095-point curve is stride-decimated for the plot as in Figure 24.53. The major-carry hierarchy is the ideal-rail one ( $-0.53$  at code 2048, then  $-0.28$  at 1024,  $-0.24$  at 3072,  $-0.14$  at 512 and  $-0.13$  LSB at 2560), and the worst step matches Figure 24.53 to within 0.03 LSB. Raw DNL on the same bench is  $-9.8$  to  $-18.4$  LSB (Table 24.73); the difference is the reference step between adjacent codes and nothing else.

| Parameter                  | Unit          | tt        | ff        | fs        | sf       | ss        | Spread  |
|----------------------------|---------------|-----------|-----------|-----------|----------|-----------|---------|
| Full-scale output          | V             | 2.16484   | 1.99313   | 2.10312   | 2.22263  | 2.34569   | 0.35256 |
| LSB size                   | $\mu\text{V}$ | 528.646   | 486.684   | 513.576   | 542.747  | 572.810   | 86.125  |
| INL, worst (raw)           | LSB           | 37.064    | 45.087    | 37.145    | 38.890   | 32.546    | 12.541  |
| DNL, most negative (raw)   | LSB           | $-10.864$ | $-18.356$ | $-11.542$ | $-9.765$ | $-10.836$ | 8.592   |
| INL, worst (ratio)         | LSB           | 0.442     | 0.450     | 0.438     | 0.443    | 0.432     | 0.018   |
| DNL, most negative (ratio) | LSB           | $-0.525$  | $-0.530$  | $-0.525$  | $-0.518$ | $-0.543$  | 0.024   |
| DNL, worst (ratio)         | LSB           | 0.525     | 0.530     | 0.525     | 0.518    | 0.543     | 0.024   |
| Zero-scale offset (ratio)  | LSB           | 0.077     | 0.325     | 0.058     | 0.139    | 0.051     | 0.273   |
| Gain error (ratio)         | ppm           | $-41.3$   | $-170.9$  | $-46.0$   | $-56.0$  | $-30.6$   | 140.2   |

Table 24.73: Static accuracy with the ladder referenced to  $V_{\text{ddDac}}$  from the supply block,  $\text{DacWPSUTB}$ , zero external load,  $25^\circ\text{C}$ , all 4096 codes,  $\text{reltol} = 1 \times 10^{-6}$ . The first four rows are the block as delivered, reference included; the last five divide the output by the measured  $V_{\text{DDDAC}}$  at each code and so measure the ladder alone. The pair attributes the error: raw INL of 32.5 to 45.1 LSB collapses to 0.43 to 0.45 LSB ratiometric and raw DNL of  $-9.8$  to  $-18.4$  LSB to  $-0.52$  to  $-0.54$  LSB, so 99% of the raw nonlinearity is the reference modulation of Figure 24.57. Gain error is referred to the ideal ratiometric full scale  $4095/4096 = 0.999756$ ; it is not an absolute-voltage gain error, which the line regulation of Table 24.78 dominates. Corner run, no device mismatch.

| Parameter      | Unit | N   | Mean    | $\sigma$ | Median  | $p_1$   | $p_{99}$ | Worst   | Spec          | Yield | 4 ch |
|----------------|------|-----|---------|----------|---------|---------|----------|---------|---------------|-------|------|
| INL worst      | LSB  | 200 | 1.514   | 0.627    | 1.388   | 0.418   | 3.326    | 3.491   | $\leq 1.000$  | 20.5  | 0.2  |
| DNL min        | LSB  | 200 | -1.747  | 1.441    | -1.324  | -6.252  | -0.063   | -6.629  | $\geq -1.000$ | 40.0  | 2.6  |
| DNL worst      | LSB  | 200 | 2.271   | 1.328    | 1.913   | 0.412   | 6.252    | 6.629   | $\leq 0.750$  | 7.0   | 0.0  |
| INL worst, raw | LSB  | 200 | 36.776  | 2.308    | 36.911  | 30.562  | 41.460   | —       | —             | —     | —    |
| DNL min, raw   | LSB  | 200 | -10.816 | 0.833    | -10.837 | -12.522 | -8.906   | -12.700 | $\geq -1.000$ | 0.0   | 0.0  |

Table 24.74: Monte Carlo linearity with the ladder referenced to VddDac from the on-chip supply block, ratiometric rows first and raw rows below: **process and mismatch**, castalia\_pm\_25, 200 samples, 25 °C, all 4096 codes, no external load on VddDac, yields as per cent of samples and of four channels. The ratiometric rows normalise each code to the reference at that code and so measure the ladder alone; they reproduce this campaign’s own ideal-rail run sample for sample, at  $r = 1.000$  on both INL and DNL, and their 20.5 % and 40.0 % yields are that run’s 21.0 % and 40.0 %, not the 20.5 % and 42.0 % Table 24.70 now reports from the replacement bit driver. The raw rows include the reference and are 24 and 6 times worse; the run’s own `inl_from_ref`, which is the INL predicted from the reference error alone, averages 36.770 LSB against the raw 36.776 LSB, so the reference accounts for essentially all of the difference and no second matching mechanism is needed to explain it. Raw `inl_lsb` carries no entered limit and its spec, yield and worst-case cells are empty; raw `dnl_min` was graded and no sample meets it. The three ratiometric rows are skewed (+0.74, -1.10, +0.89), so the median and percentiles are quoted throughout.

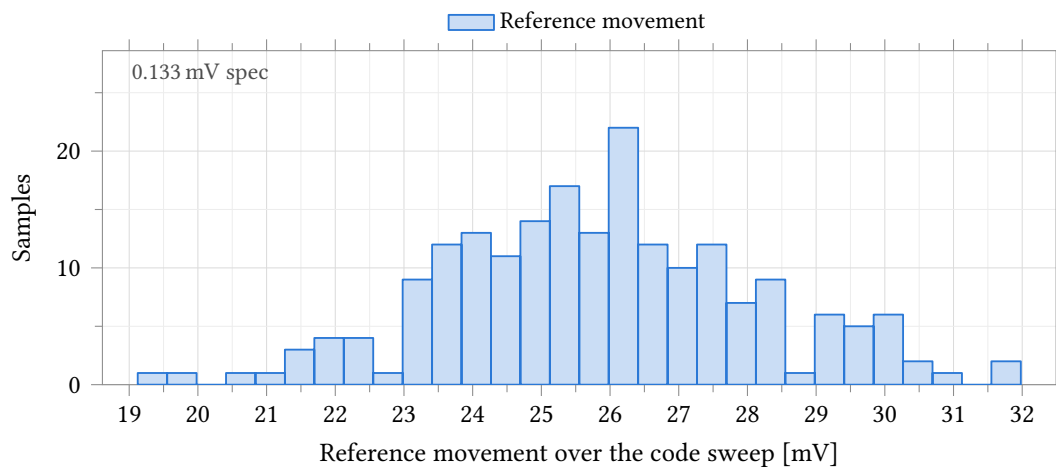


Figure 24.60: Peak-to-peak movement of the DAC reference VddDac across the 4096-code sweep, with the ladder referenced to the on-chip supply block instead of an ideal rail. **Process and mismatch**, castalia\_pm\_25, 200 samples, 25 °C, no external load on VddDac. The quarter-LSB budget of 133  $\mu$ V entered on the test sits about 200 times below the data, so it is annotated at the left axis edge rather than drawn and the distribution is not compressed into a single bar; the closest sample misses it by a factor of 144. Meeting it needs the supply’s output impedance down by a factor of 193, to about 5  $\Omega$ , against 894  $\Omega$  small-signal (Table 24.81) and 1133  $\Omega$  as a 0  $\mu$ A to 100  $\mu$ A chord (Table 24.77).

| Parameter          | Unit          | N   | Mean   | $\sigma$ | Mean $\pm 3\sigma$  | Min    | Max    | Spec         | Yield | 4 ch |
|--------------------|---------------|-----|--------|----------|---------------------|--------|--------|--------------|-------|------|
| Reference movement | mV            | 200 | 25.835 | 2.302    | $25.835 \pm 6.905$  | 19.124 | 31.980 | $\leq 0.133$ | 0.0   | 0.0  |
| Reference min      | V             | 200 | 2.1412 | 0.0554   | $2.1412 \pm 0.1663$ | 1.9821 | 2.3066 | –            | –     | –    |
| Reference max      | V             | 200 | 2.1671 | 0.0571   | $2.1671 \pm 0.1712$ | 2.0030 | 2.3348 | –            | –     | –    |
| LSB size           | $\mu\text{V}$ | 200 | 528.96 | 13.88    | $528.96 \pm 41.64$  | 488.91 | 569.53 | –            | –     | –    |
| Full scale         | V             | 200 | 2.1661 | 0.0568   | $2.1661 \pm 0.1705$ | 2.0021 | 2.3323 | –            | –     | –    |

Table 24.75: Monte Carlo behaviour of the VddDac reference itself, same bench and samples as Table 24.74; only vddac\_pp carries an entered limit, so the other four rows have empty spec, yield and worst-case cells.

**Process and mismatch**, castalia\_pm\_25, 200 samples, 25 °C; yields are per cent of samples and of four channels. Full scale is 2.166 V and the LSB 529.0  $\mu\text{V}$  rather than 2.5 V and 610.3  $\mu\text{V}$ , because the supply block delivers about 2.15 V; the reference then moves 25.8 mV peak to peak across the code sweep against the 133  $\mu\text{V}$  quarter-LSB budget, which no sample meets. The excursion is not monotonic in code: in an  $R-2R$  ladder the binary weighting comes from attenuation, so every  $2R$  branch switched to the reference draws a comparable current and the reference current follows the bit pattern rather than the code value: a nodal solve of this topology gives  $I_{\text{ref}} = 0$  at code 0,  $0.250 V_{\text{ref}}/R$  at 2048 and 0.333 at 4095, with a maximum of 1.444 at code 1365 (0b010101010101), which is exactly where vddac\_min lands on all 200 samples, and a one-parameter fit  $V = V_0 - R_{\text{out}}I_{\text{ref}}$  accounts for 94 % of the excursion. All five distributions are near-symmetric (skew  $\leq 0.13$ ), so mean  $\pm 3\sigma$  applies.

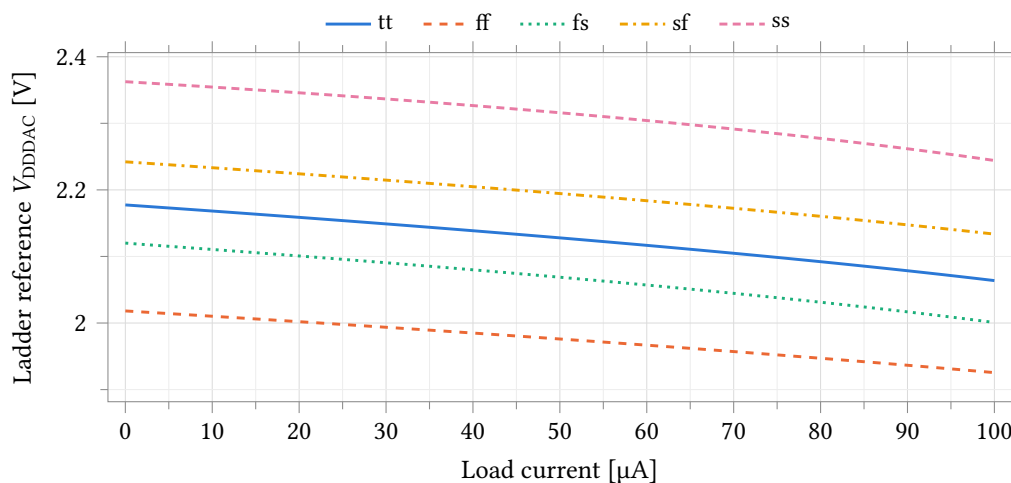


Figure 24.61:  $V_{\text{DDDAC}}$  against external load current over five process corners, supply block alone, 27 °C at point 1 and 25 °C at points 2–5, ADE default tolerances. There is no feedback loop around the output, so the slope is the block’s own output resistance and the reference falls 92.7 mV to 119.3 mV across the 0  $\mu\text{A}$  to 100  $\mu\text{A}$  sweep. The curve steepens with load, which is why the chord in Table 24.76 exceeds the small-signal  $|Z_{\text{out}}|$  of Table 24.80. This sweep is an external load only: the ladder is not connected on this bench, and its own code-dependent draw is Figure 24.57.

| Parameter                | Unit     | tt      | ff     | fs      | sf      | ss      | Spread |
|--------------------------|----------|---------|--------|---------|---------|---------|--------|
| Reference at zero load   | V        | 2.1774  | 2.0182 | 2.1201  | 2.2422  | 2.3625  | 0.3444 |
| Reference at 100 $\mu$ A | V        | 2.0638  | 1.9255 | 2.0008  | 2.1336  | 2.2442  | 0.3187 |
| Load-induced drop        | mV       | 113.667 | 92.664 | 119.287 | 108.585 | 118.361 | 26.623 |
| Load regulation          | $\Omega$ | 1136.7  | 926.6  | 1192.9  | 1085.9  | 1183.6  | 266.2  |

Table 24.76: Supply-block load regulation by process corner:  $V_{DDDAC}$  against an external load current swept 0  $\mu$ A to 100  $\mu$ A, DacPSUTB, 27 °C at point 1 and 25 °C at points 2–5, ADE default tolerances. Load regulation is the chord across the whole sweep, not the slope at the origin, and runs 18 % to 51 % above the small-signal  $|Z_{out}|$  of Table 24.80. The ladder is not connected on this bench; its own code-dependent current is Figure 24.57. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

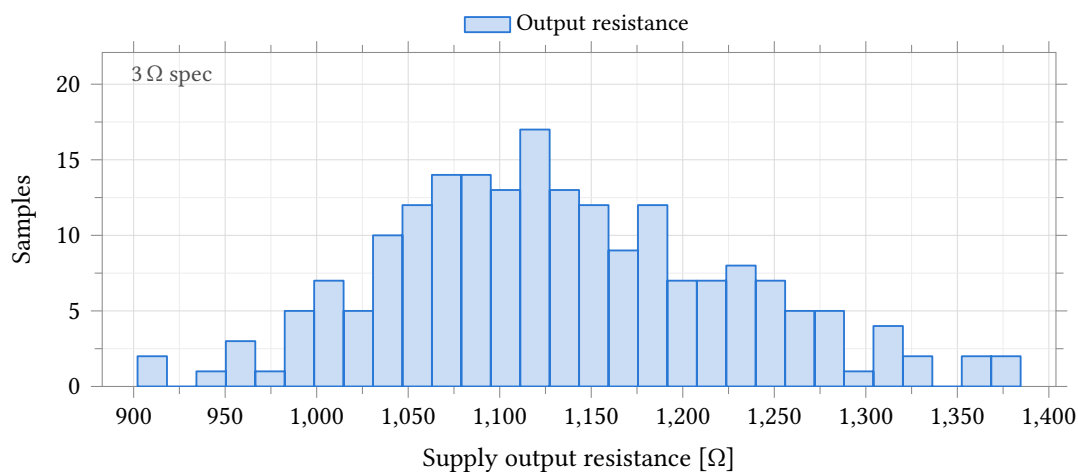


Figure 24.62: Output resistance of the DAC supply block, taken as the chord  $\Delta V/\Delta I$  over a 0  $\mu$ A to 100  $\mu$ A load sweep on VddDac. **Process and mismatch**, castalia\_pm\_25, 200 samples, 27 °C. The 3  $\Omega$  limit entered on the test is two orders of magnitude below the data and is annotated at the left axis edge rather than drawn; no sample meets it. This is not a second defect: the reference movement of Figure 24.60 is this impedance multiplied by the ladder’s own code-dependent reference current.

| Parameter            | Unit     | $N$ | Mean    | $\sigma$ | Mean $\pm 3\sigma$   | Min    | Max     | Spec         | Yield | 4 ch |
|----------------------|----------|-----|---------|----------|----------------------|--------|---------|--------------|-------|------|
| Reference, no load   | V        | 200 | 2.1785  | 0.0595   | 2.1785 $\pm$ 0.1784  | 2.0053 | 2.3495  | –            | –     | –    |
| Reference, full load | V        | 200 | 2.0653  | 0.0593   | 2.0653 $\pm$ 0.1778  | 1.9003 | 2.2462  | –            | –     | –    |
| Load droop           | mV       | 200 | 113.281 | 9.335    | 113.281 $\pm$ 28.006 | 90.214 | 138.458 | $\leq 0.133$ | 0.0   | 0.0  |
| Output resistance    | $\Omega$ | 200 | 1132.8  | 93.4     | 1132.8 $\pm$ 280.1   | 902.1  | 1384.6  | $\leq 3.0$   | 0.0   | 0.0  |

Table 24.77: Monte Carlo load regulation of the DAC supply block. **Process and mismatch**, castalia\_pm\_25, 200 samples, 27 °C, VddDac loaded 0  $\mu$ A to 100  $\mu$ A, ADE default tolerances, looser than the two DAC benches, which run 25 °C and reltol  $1 \times 10^{-6}$ ; yields are per cent of samples and of four channels. VddDac droops 113.3 mV over that sweep, a chord output resistance of 1132.8  $\Omega$  against the 3  $\Omega$  limit entered on the test; no sample meets either graded limit, and the two reference-voltage rows were not graded. The last two rows are one measurement scaled by the load current, and both are near-symmetric (skew +0.29), so mean  $\pm 3\sigma$  applies. Together with the reference movement of Table 24.75 and the small-signal impedance of Table 24.81 this is one defect measured three ways, not three problems.

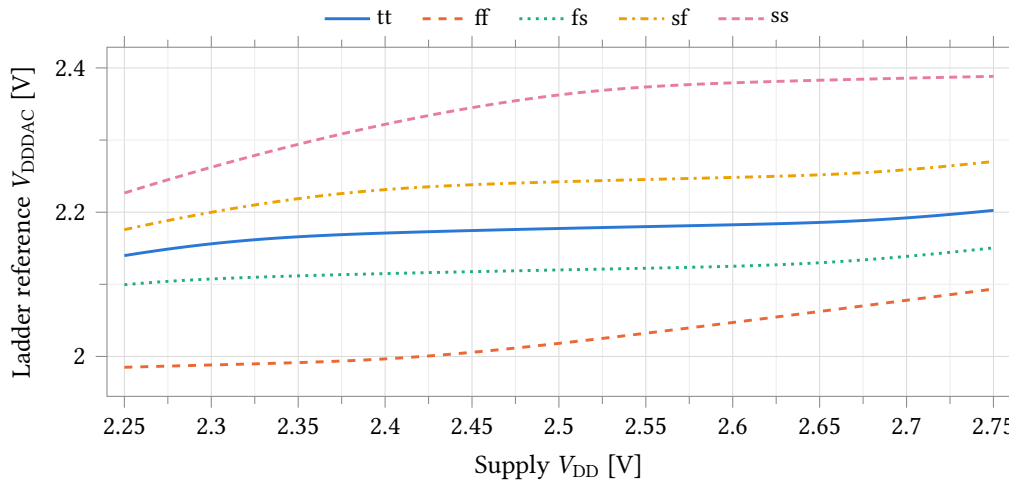


Figure 24.63:  $V_{DDDAC}$  against the supply, swept 2.25 V to 2.75 V, over five process corners at zero external load, 27 °C at point 1 and 25 °C at points 2–5. The slope is the line regulation tabulated in Table 24.78; it is not constant across the sweep, so the tabulated value is the chord between the two endpoints. The corner spread of the reference itself, 344 mV at nominal supply, is larger than anything the supply sweep produces.

| Parameter                          | Unit       | tt     | ff     | fs     | sf     | ss     | Spread |
|------------------------------------|------------|--------|--------|--------|--------|--------|--------|
| Reference at 2.5 V                 | V          | 2.1774 | 2.0182 | 2.1200 | 2.2422 | 2.3625 | 0.3444 |
| Reference change, 2.25 V to 2.75 V | mV         | 62.67  | 108.46 | 51.11  | 94.46  | 161.70 | 110.59 |
| Line regulation                    | $V V^{-1}$ | 0.1253 | 0.2169 | 0.1022 | 0.1889 | 0.3234 | 0.2212 |
| Quiescent supply current           | $\mu A$    | 158.05 | 194.10 | 152.07 | 161.96 | 129.18 | 64.91  |

Table 24.78: Supply-block line regulation and quiescent current by process corner:  $V_{DDDAC}$  with the supply swept 2.25 V to 2.75 V at zero external load, DacPSUTB, 27 °C at point 1 and 25 °C at points 2–5. With no feedback loop, 0.102  $V V^{-1}$  to 0.323  $V V^{-1}$  of any supply movement reaches the reference, and every absolute output voltage of the DAC moves with it; the ratiometric rows of Table 24.73 are what survives. The current is read at the bench supply terminal and covers the supply block only; the ladder is not connected on this bench. The quiescent-current column is a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter         | Unit       | N   | Mean   | $\sigma$ | Median | $p_1$  | $p_{99}$ | Worst   | Spec         | Yield | 4 ch |
|-------------------|------------|-----|--------|----------|--------|--------|----------|---------|--------------|-------|------|
| Line shift        | mV         | 200 | 78.750 | 23.770   | 75.740 | 38.686 | 144.474  | 157.026 | $\leq 5.000$ | 0.0   | 0.0  |
| Line regulation   | $V V^{-1}$ | 200 | 0.1575 | 0.0475   | 0.1515 | 0.0774 | 0.2889   | –       | –            | –     | –    |
| Quiescent current | $\mu A$    | 200 | 159.56 | 12.97    | 159.41 | 134.34 | 193.35   | –       | –            | –     | –    |
| Dissipation       | $\mu W$    | 200 | 398.89 | 32.42    | 398.52 | 335.85 | 483.38   | –       | –            | –     | –    |

Table 24.79: Monte Carlo line regulation and quiescent power of the DAC supply block. **Process and mismatch**, castalia\_pm\_25, 200 samples, 27 °C, supply swept 2.25 V to 2.75 V, VddDac unloaded, ADE default tolerances; yields are per cent of samples and of four channels. VddDac follows the supply by 78.8 mV over that 0.5 V span, a rejection of only 0.158  $V V^{-1}$ , against the 5 mV limit entered on the test that no sample meets. Only dvddac\_line was graded; the other three rows have empty spec, yield and worst-case cells, the last two being the same measurement scaled by the 2.5 V supply. All four are right-skewed (+0.58, +0.58, +0.45, +0.45), so the median and percentiles are quoted and mean  $\pm 3\sigma$  is not.

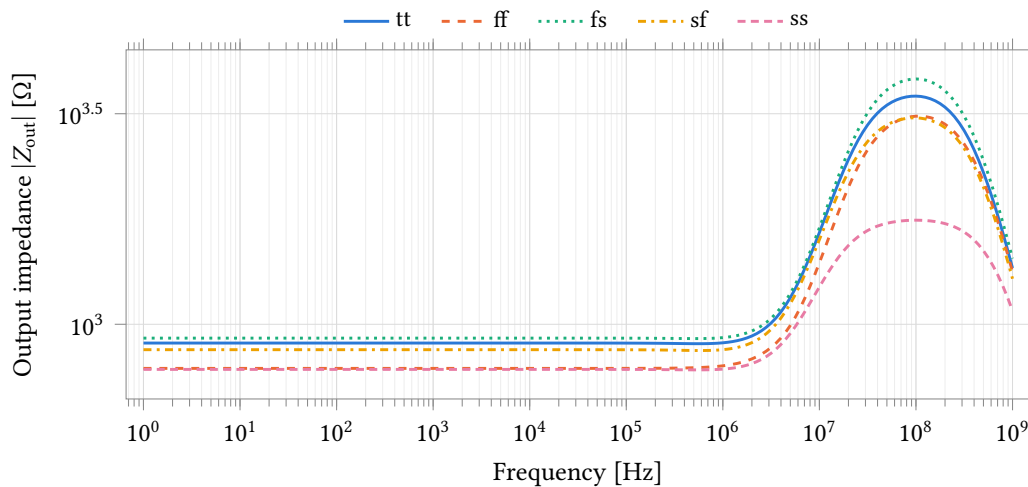


Figure 24.64: Magnitude of the supply block’s output impedance against frequency over five process corners, 27 °C at point 1 and 25 °C at points 2–5. The AC load source carries unit magnitude, so  $|V_{\text{DDDAC}}|$  is the impedance in ohms. It is flat at 781 Ω to 927 Ω from 1 Hz to 100 kHz and peaks at 1.8 kΩ to 3.8 kΩ near 100 MHz. This is the small-signal impedance at zero load; the large-signal chord over the 0 μA to 100 μA sweep is 18 % to 51 % higher (Table 24.76).

| Parameter                   | Unit | tt     | ff     | fs     | sf     | ss     | Spread |
|-----------------------------|------|--------|--------|--------|--------|--------|--------|
| Output impedance at 1 Hz    | Ω    | 902.5  | 785.9  | 927.1  | 870.3  | 781.5  | 145.6  |
| Output impedance at 1 kHz   | Ω    | 902.5  | 785.9  | 927.1  | 870.3  | 781.5  | 145.6  |
| Output impedance at 100 kHz | Ω    | 902.2  | 786.0  | 926.9  | 870.0  | 781.3  | 145.6  |
| Output impedance, peak      | Ω    | 3482.1 | 3120.4 | 3825.2 | 3092.0 | 1766.9 | 2058.3 |
| Frequency of the peak       | MHz  | 100.0  | 112.2  | 100.0  | 100.0  | 100.0  | 12.2   |

Table 24.80: Supply-block small-signal output impedance by process corner, DacPSUTB AC sweep from 1 Hz to 1 GHz at 20 points per decade, 27 °C at point 1 and 25 °C at points 2–5. The AC load source carries unit magnitude, so  $|V_{\text{DDDAC}}|$  reads directly in ohms. The impedance is flat to 100 kHz, within 0.3 Ω of the 1 Hz value at every corner, and peaks three decades higher in frequency at two to four times the flat value. Grid points are 12 % apart, so the peak frequency is located only to that resolution and the ff entry is one grid step from the other four. These corner spreads are a lower bound: the corner set holds the passive model rows at typical (pinned-passive caveat, page 207).

| Parameter              | Unit     | <i>N</i> | Mean   | $\sigma$ | Median | $p_1$  | $p_{99}$ | Worst  | Spec        | Yield | 4 ch |
|------------------------|----------|----------|--------|----------|--------|--------|----------|--------|-------------|-------|------|
| $Z_{\text{out, LF}}$   | $\Omega$ | 200      | 894.3  | 46.6     | 893.8  | 779.4  | 990.8    | –      | –           | –     | –    |
| $Z_{\text{out, peak}}$ | $\Omega$ | 200      | 3402.1 | 299.5    | 3442.8 | 2369.2 | 3853.9   | 3872.2 | $\leq 10.0$ | 0.0   | 0.0  |
| Peak frequency         | MHz      | 200      | 98.94  | 6.23     | 100.00 | 89.13  | 112.20   | –      | –           | –     | –    |

Table 24.81: Monte Carlo small-signal output impedance of the DAC supply block, AC swept 1 Hz to 1 GHz. **Process and mismatch**, castalia\_pm\_25, 200 samples, 27 °C, VddDac unloaded, ADE default tolerances; yields are per cent of samples and of four channels. The impedance is flat at 894.3  $\Omega$  to within 0.4  $\Omega$  from 1 Hz to 100 kHz, and this is the value that sets the reference movement of Table 24.75, the ladder’s code-dependent current being a DC excursion.  $z_{\text{out\_max}}$  is the peak over the whole band and lands at 98.9 MHz, loop peaking well above anything the ladder excites; it is the only graded row here and no sample meets its 10  $\Omega$  limit.  $z_{\text{out\_max}}$  is strongly left-skewed (skew  $-1.82$ ), so the median and percentiles are quoted;  $z_{\text{out\_lf}}$  and the peak frequency carry no entered limit and their spec, yield and worst-case cells are empty.

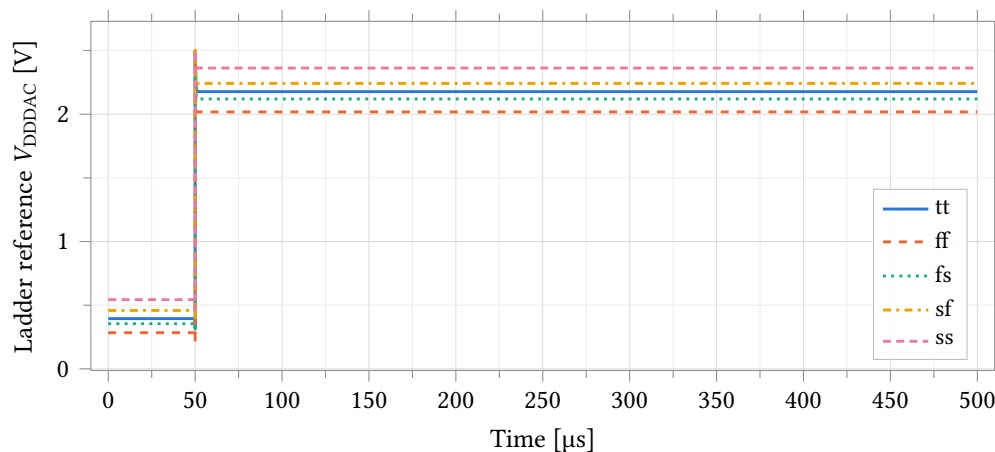


Figure 24.65:  $V_{\text{DDDAC}}$  against time over five process corners, PSUStartupTB: En rises at 50  $\mu\text{s}$  into a 500  $\mu\text{s}$  window at zero external load, 27 °C at point 1 and 25 °C at points 2–5. With En low the node sits at 0.28 V to 0.54 V; at the 1 ns enable edge the pass device is driven hard on and the node is pulled to the 2.5 V supply rail for 64 ns to 100 ns before falling back. It reaches the zero-load DC value of Table 24.76 to within 140  $\mu\text{V}$  at every corner and holds it to the end of the window. No settling-time or overshoot figure of merit is published from this bench: the design variable  $t_{\text{win}}$  sets both the En pulse width and the settlingTime evaluation window, so the two cannot be specified independently.

variable sets both the En pulse width and the settlingTime evaluation window, so the two cannot be specified independently. The Monte Carlo run of the same test is separately unusable: its enable pulse is delayed 1 s against a 500  $\mu$ s window, so En never asserts and every startup output it reports grades the disabled state. Note from the figure that the reference is pulled to the full 2.5 V supply rail for 64 ns to 100 ns at the enable edge; any code held through an enable event sees that excursion.

**Disabled-state supply current was not measured in this run.** The enable bench is not part of the corner or Monte Carlo runs reported here. The last measurement of it, 59.45  $\mu$ A against a 1  $\mu$ A target, comes from a superseded setup and is not quoted as a result; it is a separate finding from the reference problem of Section 24.9.2 and needs re-measuring.

Which supply the ladder was referenced to is stated on every table and figure: the DacTB results are against an ideal 2.5 V rail and characterise the ladder, which is the tapeout arrangement, and the DacWPSUTB results are against VddDac and characterise the block as it stood before the supply regulator was deleted. The corner run carries no mismatch information and the Monte Carlo run varies process and mismatch together; the two are different quantities and are never totalled.

#### 24.9.4 Measuring this block on chip

1. Select per-channel slot 2 with AFE\_DBGISOEN = 1 and read the reference output on atb\_d. The isolation switch is required, not optional: leakage on an R-2R output is a code-dependent error.
2. Absolute anchor: take three codes with a voltmeter to fix the DAC's offset and step in volts. Nothing on the die can supply that scale; it anchors the channel's whole potential axis.
3. Sweep all 4096 codes into the converter with AFE\_ADCSRSEL = 2 (75 ms per channel at sixteen averages and the design conversion rate, four times that on the clock pattern as delivered, Section 24.7.2). The supply regulators were deleted from the cell (Section 24.9.2), so the ladder reference is the 2.5 V analog rail and the one-parameter droop fit applies only if a regulator is reinstated. Do not store the per-code residual table; through this converter each entry carries as much quantisation as the error it corrects (0.07 mV rms for 4 KiB per channel), but run the sweep regardless: it is where a broken ladder shows.
4. Chip-level slot 7 reads the DAC supply node directly, which is the only place a reinstated regulator's 25.8 mV code-dependent excursion could be confirmed or refuted on silicon. Chip-level slot 6 reads the shared common-mode DAC, forceable through AFE\_VCMISOEN.

This test fabric is design intent; it is not yet in the schematics.

### 24.10 Four-Channel Macro

The four channels are independent by construction: each carries its own reference and common-mode converters, its own feedback ladder, its own ten-bit converter and its own local bias generator, and shares only the analog supplies and the two test pads. Running all four at once is therefore a check that the assembly is right, not a test of a shared resource.

**Site-to-site spread is zero codes.** With 10  $\mu$ A drawn from every working electrode at the bottom gain code, all four sites return code 633, their transimpedance outputs agreeing to 42  $\mu$ V, and the control and transimpedance pedestals reproduce the single-channel figures (−0.56 mV of applied-potential error, −0.59 mV at the working electrode) rather than adding to them. Driven with a different current, a different polarity and a different gain code per site, simultaneously, the four codes land symmetrically about the common-mode code at 257, −247, 244 and −245 for 20  $\mu$ A, −20  $\mu$ A, 1  $\mu$ A and −0.2  $\mu$ A across 107-fold of gain.

**The macro draws 4.55 mW, 1.14 mW per channel, of which the digital rail is 14.2  $\mu$ A.** The neighbour-disturbance column of Table 24.82 is **not a crosstalk measurement**: this bench has ideal supplies and



per-site bias, so it carries no coupling path, and the same 1.1 LSB of ripple appears with no neighbour stepping at all. It is each channel's own converter sampling its own transimpedance output. A crosstalk figure needs the post-layout supply and substrate network. Corners and Monte Carlo were not run on the macro; every block below it carries its own.

Table 24.82: The four channels driven together at the nominal corner, 37 °C, with the converter as its extracted view and a 20 MHz trigger clock (1.150  $\mu$ s per conversion). The mixed-condition rows carry a different current, polarity and gain code per site at the same instant. The disturbance column is measured with only site 0 stepping, but the same figure appears when no site steps at all, so it is each channel's own converter sampling its own output and not coupling: this bench has ideal supplies and per-site bias and therefore no coupling path. Nominal only; corners and Monte Carlo live in the block sections.

|                                                                    | Site 0   | Site 1                       | Site 2   | Site 3       |
|--------------------------------------------------------------------|----------|------------------------------|----------|--------------|
| <i>Identical drive, 10 <math>\mu</math>A at gain code 0</i>        |          |                              |          |              |
| Code returned                                                      | 633      | 633                          | 633      | 633          |
| Transimpedance output (V)                                          | 1.449192 | 1.449195                     | 1.449234 | 1.449195     |
| Applied-potential error (mV)                                       | −0.56    | −0.56                        | −0.56    | −0.56        |
| Working-electrode error (mV)                                       | −0.59    | −0.59                        | −0.59    | −0.59        |
| <i>Mixed drive, simultaneous</i>                                   |          |                              |          |              |
| Current forced ( $\mu$ A)                                          | +20      | −20                          | +1       | −0.2         |
| Gain code                                                          | 0        | 0                            | 63       | 319          |
| Code, referred to common mode                                      | +257     | −247                         | +244     | −245         |
| <i>Neighbour disturbance, site 0 stepping 40 <math>\mu</math>A</i> |          |                              |          |              |
| Quiet-site output (mV pp)                                          | —        | 1.751                        | 1.750    | 1.751        |
| <i>Power, all four channels</i>                                    |          |                              |          |              |
| 2.5 V supply                                                       |          |                              |          | 1.815 mA     |
| 1.0 V supply                                                       |          |                              |          | 14.2 $\mu$ A |
| Total                                                              |          | 4.55 mW, 1.14 mW per channel |          |              |

## 24.11 Per-Chip Calibration

Every correction is a stored digital constant except one, the bias generator's 6-bit trim code. Table 24.83 is the complete set of knobs: what each corrects, how it is measured, and when it is applied. The measurement itself is described in each block's section, under *Measuring this block on chip*.

Only two constants are absolute (one voltage at the reference DAC output, one current forced at the working electrode) and both need the production tester. The other eight are zeros and ratios the channel measures against itself through its own converter, which is what lets the power-up refresh run with no equipment.

Order matters twice. At test, apply the bias trim before anything else is measured: it moves every transconductance behind every constant that follows. In a reading, subtract the zero code, scale by the gain constant for the active gain code, then subtract the drop across the working-electrode switch and, where an impedance sweep has supplied it, the uncompensated electrolyte resistance. Potential-axis corrections apply where the waveform is tabulated, not per sample.

**Residual.** Each zero is the difference of two conversions, so its floor is 0.65 mV rms, two quantisations of the converter's 1.591 mV step; the common-mode DAC has to dither the input to reach it. On the potential

| Constant                           | Scope   | Corrects                                                                                            | Measured by                                                                                  | Applied                                                     |
|------------------------------------|---------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Bias trim code, 6 bit              | chip    | Bias-current spread,<br>$\sigma/\mu = 12.5\%$ in MC                                                 | Step the code; keep the one whose total analog supply current is nearest design (Tab. 24.10) | Test; reloaded at power-up before anything else is measured |
| Reference DAC offset and step      | channel | Absolute potential scale; full scale 2.166 V, $\sigma = 56.8$ mV (Tab. 24.75)                       | Tester voltmeter at three codes, direct test pad                                             | Test only                                                   |
| Reference droop parameter          | channel | 25.8 mV pp movement of the reference with code                                                      | 4096-code sweep into the converter; one-parameter fit (§24.9.2)                              | Test only; applied where the waveform is built              |
| Converter zero code                | channel | Converter offset and the common-mode value together                                                 | Convert the common-mode rail and average                                                     | Test, power-up, and between potential steps                 |
| A1 offset and tracking slope       | channel | Applied-potential error, $\sigma = 5.20$ mV (Tab. 24.19)                                            | A1 in unity-gain loopback, three reference codes (§24.4)                                     | Test and power-up; applied where the waveform is built      |
| Current zero, per gain code        | channel | Zero-current offset, $\sigma = 12.1$ – $167.3$ LSB (§24.5.6), that row on the nominal 2.441 mV step | Working electrode isolated; read at each gain code                                           | Test and power-up                                           |
| Transimpedance gain, per gain code | channel | $\pm 35\%$ resistor tolerance, switch stack and converter gain, in one number (Tab. 24.62)          | Tester forces $\pm$ full-scale current at the working-electrode pad                          | Test; rescaled at power-up by the ratio below               |
| Internal gain ratio, per gain code | channel | Drift of the gain constant                                                                          | Same slope against an on-chip reference resistor (§24.8)                                     | Power-up                                                    |
| Working-electrode switch $R_{on}$  | channel | $i \cdot R_{on}$ : 3.3 mV at 33 $\mu$ A through 100 $\Omega$                                        | Force current; read both sides of the switch                                                 | Test; applied per sample                                    |

Table 24.83: Per-chip calibration constants, the complete set of digital knobs. *Test* constants need external equipment and are written once; *power-up* constants are re-measured on-die at every start and overwrite the stored copy. At eight gain codes per channel the whole block is under 128 bytes per channel, so it is staged from flash with a channel hart’s image rather than read back during a measurement. LSB is the converter’s measured 1.591 mV step (Table 24.58) unless a row states otherwise. No calibration has been executed, in simulation or on silicon, and the isolation switches and test multiplexer these measurements need are design intent, not schematics.

axis the residual is 1.38 mV rms (that floor, the 0.93 mV droop-fit residual and 0.79 mV of uncorrected ladder INL in quadrature) against 5.71 mV of uncalibrated  $\sigma$ . That holds over the specified 0.4 V to 2.1 V window only: outside it the tracking error is no longer a constant and no stored constant removes it. On the current axis it is 0.71 mV rms, 0.45 LSB on the converter's measured 1.591 mV step (Table 24.58), plus 0.1 % to 0.4 % of reading, against 12.1 LSB to 167.3 LSB uncalibrated, which are on the nominal 2.441 mV step. A constant removes the current zero only while that zero stays inside the converter window, which requires  $R_f/R_{\text{cell}} < 47.5$ ; the 560 k $\Omega$  tap into a 10 k $\Omega$  electrode (Section 24.5.6) is outside it and those samples are recoverable by nothing.

**Out of reach.** The reference DAC's  $-6.6$  LSB of DNL is lost command resolution, up to 3.5 mV at the worst code, and no constant creates a code the ladder does not produce. A reference-electrode junction potential is indistinguishable from A1's offset, so only a zero taken *wet* against that electrode absorbs it. The temperature dependence of every constant here is unquantified: no block in this chapter has a temperature-resolved Monte Carlo, so how far a power-up re-measurement tracks a 37 °C operating point is not known.

# **Part V   Boot ROM and software**

## 25 ROM monitor

When the chip leaves reset with the P0.7 strap low (Section 6.2), hart 0 runs the ROM monitor: an interactive Forth interpreter stored in the boot ROM that communicates over UART0 at 115,200 baud, 8 data bits, no parity, one stop bit. The monitor is a bring-up console and the flash programming path of Chapter 26; it is not a development environment.

On start-up the monitor prints a one-line banner ending in `rv4th-rom!`, an empty line, and the prompt `>`, and is then ready to receive characters. The leading word of the banner is the `ASIC_NAME` string compiled into the ROM source and does not name this chip.

### 25.1 Interpreter model

Forth is a stack language. Every value is a 32-bit signed integer on one data stack, the top of which is called TOS. Typing a number pushes it: a plain decimal integer, or a hexadecimal value with the prefix `0x` in either case and without leading zeros (`0xA4`). Typing a word (a function name) executes it; a word pops its arguments from the stack and pushes its results. Nothing is executed until a line feed is received, at which point the whole line is executed left to right. An input the interpreter does not recognise as a number or a word prints `?`. Stack underflow is not an error: a pop from an empty stack yields 0.

By default the monitor echoes every received character back to the host, which suits a terminal emulator; `0 echo` switches the echo off for a programmatic host.

### 25.2 Word list

The words below are listed with their stack effect in the usual Forth notation: the items left of the double dash are the arguments as they are typed (the rightmost is the TOS when the word runs), and the items right of it are the results, rightmost on top. Arguments are popped before the word executes and results are pushed after it finishes.

#### 25.2.1 Memory Access Functions

- `@ ( addr -- )`

Read directly from the memory address and push the value to the TOS

Arguments: (in order from TOS down, reverse text order):

1. `addr:` (TOS) The address to read from (must be aligned to a 4-byte (32-bit) boundary)

Return values: none

- `! ( x addr -- )`

Write directly to memory address

Arguments: (in order from TOS down, reverse text order):

1. `addr:` (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. `x:` The value to write to the memory address

Return values: none

- `+`! ( x addr -- )

Add to value, direct memory access (`*addr += x`)

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. x: The value to add to the value at the memory address

Return values: none

- `sbi` ( bitnum addr -- )

Set bit, direct memory access (`*addr |= (1 « bitnum)`)

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. bitnum: The bit number

Return values: none

- `cbi` ( bitnum addr -- )

Clear bit, direct memory access (`*addr &= ~(1 « bitnum)`)

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. bitnum: The bit number

Return values: none

- `mask` ( bitmask x addr -- )

Change only masked bits, direct memory access (`*addr = (*addr & ~bitmask) | (x & bitmask)`)

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address to write to (must be aligned to a 4-byte (32-bit) boundary)
2. x: The value to write (only masked bits will be written)
3. bitmask: The bit mask. Only bits with 1s can be changed

Return values: none

### 25.2.2 Print and Input Functions

- `.` ( x -- )

Prints the TOS as an integer

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) The value to print

Return values: none

- **h.** ( x -- )

Prints the TOS as a hex string

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) The value to print

Return values: none

- **echo** ( bool -- )

Change the echo state, which controls whether the Forth interpreter repeats all of its received characters

Arguments: (in order from TOS down, reverse text order):

1. bool: (TOS) If 0, does not echo. Otherwise, it does

Return values: none

- **emit** ( x -- )

Prints the char at the TOS (integers are mapped to their ASCII counterparts)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) The ASCII value of the char to print

Return values: none

- **key** ( -- char )

Waits for a char over UART and pushes the ASCII value of the char to the TOS

Arguments: none

Return values: (in order from TOS down, reverse text order):

1. char: (TOS) The char that was received over UART

- **list** ( -- )

Prints all currently available functions

Arguments: none

Return values: none

### 25.2.3 Arithmetic Functions

- **+** ( y x -- ret )

Adds two numbers (y + x)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **-** ( y x -- ret )

Subtracts two numbers ( $y - x$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **\*** ( y x -- retprodhi retprodlo )

Multiplies two numbers ( $y * x$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. retprodlo: (TOS) Lower 32 bits of the return product
2. retprodhi: Upper 32 bits of the return product

- **/%** ( y x -- retdiv retrem )

Divides two numbers with remainder ( $y / x$  and  $y \% x$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. retrem: (TOS) The remainder (or modulo) result
2. retdiv: The integer division result

- **neg** ( x -- ret )

Returns the negative (twos complement) of x

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **abs** ( x -- ret )

Absolute value ( $|x|$ )

Arguments: (in order from TOS down, reverse text order):



1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

#### 25.2.4 Bitwise Logic Functions

- `~ ( x -- ret )`

Bitwise complement/inversion ( $\sim x$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `| ( y x -- ret )`

Bitwise OR ( $y | x$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `& ( y x -- ret )`

Bitwise AND ( $y \& x$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `^ ( y x -- ret )`

Bitwise XOR ( $y \wedge x$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `*2 ( x -- ret )`

Shift left 1 bit ( $x \ll 1$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `/2 ( x -- ret )`

Shift right 1 bit ( $x \gg 1$ )

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `sll ( x bits -- ret )`

Shift left logical ( $x \ll \text{bits}$ )

Arguments: (in order from TOS down, reverse text order):

1. bits: (TOS) Signed integer
2. x: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- `srl ( x bits -- ret )`

Shift right logical ( $x \gg \text{bits}$ ), no sign extension

Arguments: (in order from TOS down, reverse text order):

1. bits: (TOS) Signed integer
2. x: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

### 25.2.5 Comparison Functions

- `> ( y x -- ret )`

Greater than. Returns 1 if  $y > x$ , returns 0 otherwise.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) 1 if true, 0 if false

- **<** ( y x -- ret )

Less than. Returns 1 if  $y < x$ , returns 0 otherwise.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) 1 if true, 0 if false

- **==** ( y x -- ret )

Equals. Returns 1 if  $y == x$ , returns 0 otherwise.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) 1 if true, 0 if false

### 25.2.6 Boolean Functions

- **not** ( x -- ret )

Logical not (!x). Returns 1 if x is equal to 0, returns 0 otherwise

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **and** ( y x -- ret )

Logical and ( $y \&\& x$ ). Returns 1 if y and x are both true (nonzero), returns 0 otherwise

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **or** ( y x -- ret )

Logical or ( $y \parallel x$ ). Returns 1 if y or x are true (nonzero), returns 0 otherwise

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer
2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

### 25.2.7 Stack Manipulation Functions

- **dup** ( x -- x x )

Duplicates the TOS (pops the TOS and then pushes the value to the TOS twice)

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Value to duplicate

Return values: (in order from TOS down, reverse text order):

1. x: (TOS) The duplicated value
2. x: The original TOS

- **drop** ( x -- )

Pops the TOS and sets it as the internal Forth i value. Usually used to delete the TOS.

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) The value that is dropped

Return values: none

- **swap** ( y x -- x y )

Swaps the TOS with the value below the TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Old TOS
2. y: Old value below the TOS

Return values: (in order from TOS down, reverse text order):

1. y: (TOS) Old value below the TOS
2. x: Old TOS

- **over** ( y x -- y x y )

Copies the value below the TOS and pushes it to the TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed Integer

2. y: Signed Integer

Return values: (in order from TOS down, reverse text order):

1. y: (TOS) Signed Integer
2. x: Signed Integer
3. y: Signed Integer

- **tuck** ( y x -- x y x )

Copies the TOS and inserts it two after the TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed Integer
2. y: Signed Integer

Return values: (in order from TOS down, reverse text order):

1. x: (TOS) Signed Integer
2. y: Signed Integer
3. x: Signed Integer

- **roll** ( n -- )

Removes value at index n of the stack and pushes it to the TOS (zero indexed)

Arguments: (in order from TOS down, reverse text order):

1. n: (TOS) Stack index (zero indexed)

Return values: none

- **pick** ( n -- )

Copies value at index n of the stack and pushes it to the TOS (zero indexed)

Arguments: (in order from TOS down, reverse text order):

1. n: (TOS) Stack index (zero indexed)

Return values: none

- **depth** ( -- ret )

Gets size of stack and pushes it to the top of the stack (making the new stack size the value of TOS + 1)

Arguments: none

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

### 25.2.8 SPI Flash R/W and Programming Functions

- **fr** ( bin length addr -- )

Reads data from the SPI flash.

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address in the SPI flash to begin reading from
2. length: The number of bytes to read
3. bin: If true (nonzero), transmits binary data, otherwise transmits ASCII hex string

Return values: none

- **fe** ( conf addr -- eraseSuccessful )

Erases a 256-byte page from the SPI flash memory. The start address must be 256-byte aligned.

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The start address of the page to erase in the SPI flash. Must be 256-byte aligned.
2. conf: If equal to 123, erases the page and returns 1, otherwise does not erase and returns 0

Return values: (in order from TOS down, reverse text order):

1. eraseSuccessful: (TOS) Nonzero if successful, zero if failed

- **fw** ( bin addr -- )

Writes data to the SPI flash. Must write exactly 256 bytes at a time. The start address must be 256-byte aligned.

Arguments: (in order from TOS down, reverse text order):

1. addr: (TOS) The address in the SPI flash to begin writing to. Must be 256-byte aligned.
2. bin: If true (nonzero), expects to receive binary data, otherwise expects to receive an ASCII hex string

Return values: none

### 25.2.9 Miscellaneous Functions

- **rst** ( -- )

Performs a soft reset (turns on all memory cells and jumps to address 0 in the ROM, does not reset clocks or hardware)

Arguments: none

Return values: none

- **clk** ( meastime clkselel -- freq )

Measures selected clock frequency

Arguments: (in order from TOS down, reverse text order):

1. clkselel: (TOS) Clock select. 0 = SMCLK; 1 = MCLK; 2 = LFXT; 3 = HFXT

2. meastime: Measurement time. 0 = 1 s; 1 = 0.5 s; 2 = 0.25 s; 3 = 0.125 s

Return values: (in order from TOS down, reverse text order):

1. freq: (TOS) Selected clock frequency in Hz

- **min** ( y x -- ret )

Returns minimum value of y and x

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **max** ( y x -- ret )

Returns maximum value of y and x

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

2. y: Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **swpb** ( x -- ret )

Swap bits 15 down to 8 with bits 7 down to 0 of TOS, also causes bits 31 down to 16 to all be 0

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

- **swphw** ( x -- ret )

Swap upper 16 bits with lower 16 bits of TOS

Arguments: (in order from TOS down, reverse text order):

1. x: (TOS) Signed integer

Return values: (in order from TOS down, reverse text order):

1. ret: (TOS) Return value

## 25.3 Interpreter limits

The interpreter keeps all of its mutable state in hart 0's TCM, of which the interrupt vector table takes the first 121 words and the top 1 KiB is reserved for the C stack; the interpreter's arrays are sized to what is left. The capacities are fixed by the ROM and are not configurable at run time:

- Data (math) stack: 128 cells of 32 bits.
- Return (address) stack: 64 cells of 32 bits. Each active `do` loop uses three of them, so loop nesting is bounded well below that.
- Dictionary: 64 user-defined words, 512 bytes total of word names including the separating space after each, and 1024 opcodes of compiled body shared by all of them.
- Input line buffer: 128 characters.
- Word buffer: 64 characters, so a single word name is at most 63 characters.

Only three of these are checked at run time: math stack underflow, input line overflow and word buffer overflow. Exceeding the dictionary capacities or underflowing the return stack is not detected and corrupts interpreter state. A program that outgrows these limits belongs in the SPI flash boot mode of Chapter 26.

**Caution:** The monitor's receive path is a polled loop with no flow control, and it echoes each character as it takes it. A host that delivers a line back to back at 115,200 baud can lose characters; a host that leaves one character time of idle line between frames does not.

## 25.4 Defining words

A new word is defined with `: name body ;`, where `body` is any sequence of numbers and words. The new word may be used as soon as the semicolon has been read, even later on the same line. A user-defined word can only push and pop the stack and call other words. For example, a word that increments the TOS:

```
: inc 1 + ;
```

## 25.5 Conditionals

`if` and `then` form a conditional and may only be used inside a word definition. `if` pops the TOS and executes the words up to the matching `then` only when the popped value is nonzero. An `else` clause is optional:

```
cond if wordsIfTrue else wordsIfFalse then
```

For example, a word that prints `cool` when the TOS equals 5:

```
: cool? 5 == if 108 111 111 99 emit emit emit emit then ;
```

## 25.6 Loops

Two loop forms exist, and both may only be used inside a word definition. `begin ... until` executes its body, then pops the TOS at `until` and repeats the body while the popped value is zero. `stop start do ... loop` is a counted loop: it runs its body with the index taking the values `start` to `stop - 1`. Inside a counted loop the word `i` pushes the index of the innermost loop, and `j` and `k` the indices of the two enclosing loops. For example, a word that prints the numbers 0 to 9:

```
: looptest 10 0 do i . loop ;
```



## 26 Boot image format

When the P0.7 strap is high at reset (Section 6.2), hart 0's boot loader reads a boot image from the SPI flash on SPI0, copies its segments into RAM and enters it. The image is a stream of 32-bit words beginning at flash offset 0x000000, read with one continuous high-frequency read command. Table 26.1 is the authoritative description of the stream.

Table 26.1: Boot image stream, as read by the boot ROM from flash offset 0x000000.

| Command word | Following words                             | Effect                                                                                                                               |
|--------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 0x10ADBEEF   | START, END, then (END - START)/4 data words | Load segment: the data words are stored to RAM at START, START + 4, ..., END - 4. END is exclusive.                                  |
| 0xDEADBEEF   | START, END, VALUE                           | Fill segment: every word from START to END - 4 is set to VALUE.                                                                      |
| 0xCAFEBAFE   | none                                        | Execute: the loader ends the flash read, restores the SPI0 pins, selects the boot clocks, clears the registers and enters the image. |

Every segment must lie inside the load window:  $0x8000 \leq \text{START} < \text{END} \leq 0x1FFFF + 1$ , which spans the private TCM, the NPU staging RAM and the shared RAM. A segment outside the window, or any other command word, traps the boot loader. The entry address is the word at 0x10640 in shared RAM if the image has loaded a nonzero value there, and 0x8200 otherwise; the loader zeroes the shared-RAM mailbox region before reading the flash, so an image that publishes nothing enters at 0x8200.

Words are transferred most-significant byte first: the flash byte at offset  $4k$  is bits 31:24 of word  $k$ . A tool that writes the flash from a little-endian memory image must therefore reverse the byte order within every word.

A standard image, as produced by the build scripts, consists of a fill or load segment that zeroes the vector table region 0x8000 to 0x8200, one load segment per contiguous run of nonzero program words (a run of eight or more zero words separates segments, and each segment is padded by two trailing zero words), and the execute word. The image's first instruction is at 0x8200 in the TCM unless it publishes an entry address; an image larger than the TCM executes from the shared RAM and publishes its entry there. The vector table itself is written by the image's startup code.

### 26.1 Programming the flash from the ROM monitor

The ROM monitor (Chapter 25) writes, erases and reads the flash in 256-byte pages, and the provided `forthprog.py` script (Appendix A) drives it to program a whole image from an Intel hex file. The image occupies flash pages from offset 0x000000 upward, as many as its length requires.

#### **bin addr fw**

Writes one page at flash address `addr`, which must be a multiple of 256. If `bin` is nonzero the monitor expects 256 bytes of binary data; if zero, a 512-character ASCII hex string. After the line feed the monitor prints `$` when it is ready for the data. When the data has arrived it prints the CRC of the page as four hex characters (CRC16/CDMA2000: polynomial 0xC857, initial value 0xFFFF, no final XOR, no reflection). The host answers with a single `Y` to commit the write; any other character aborts it and leaves the flash unchanged.

**conf addr fe**

Erases the page at `addr` to 0xFF when `conf` equals 123 and pushes 1; otherwise does nothing and pushes 0. Erasing is faster than writing and is the right way to handle a blank page.

**fr**

Reads the flash back; see Section 25.2.8.

**Note:** The flash device specifies a page program time of up to 25 ms (typically 10 ms) and a page erase time of up to 25 ms (typically 6 ms). The handshake of `fw` is long enough that no extra delay is needed between writes; consecutive erases at a high baud rate may need one.

After the image is written, the chip runs it after a reset with the `P0.7` strap high. The `rst` word restarts hart 0 at the boot ROM without a pin reset, so a host that controls the strap can switch modes without toggling `RESETN`.

## A Software development quick start

This appendix lists the commands needed to build a program for the Castalia MCU and load it into the SPI flash. The memory map header, the linker script fragments and the startup code are generated by make chip (Section 1.3).

### A.1 Toolchain

The RISC-V GNU toolchain is required. The prebuilt xPack distribution supports every extension the core implements and is the quickest route on Linux; Windows users run the same steps under the Windows Subsystem for Linux.

1. Download the current release archive (xpack-riscv-none-elf-gcc\*-linux-x64.tar.gz) from <https://github.com/xpack-dev-tools/riscv-none-elf-gcc-xpack/releases>.
2. Extract it and export its location; the version number in the path is the one downloaded:

```
mkdir -p ~/riscv-toolchain
tar -xf xpack-riscv-none-elf-gcc-*.tar.gz -C ~/riscv-toolchain/
export RISCV_TOOLCHAIN_DIR=~/riscv-toolchain/xpack-riscv-none-elf-
gcc-13.2.0-2
export PATH=$RISCV_TOOLCHAIN_DIR/bin:$PATH
```

The two export lines belong in the shell profile (~/.bashrc or ~/.profile). The build system requires RISCV\_TOOLCHAIN\_DIR.

3. Check the installation with riscv-none-elf-gcc --version.

Building the toolchain from source is an alternative when a specific version is needed. On Ubuntu or Linux Mint:

```
sudo apt install autoconf automake autotools-dev curl libmpc-dev
libmpfr-dev libgmp-dev gawk build-essential bison flex texinfo gperf
libtool patchutils bc zlib1g-dev libexpat-dev
git clone --recursive https://github.com/riscv/riscv-gnu-toolchain
cd riscv-gnu-toolchain && mkdir build && cd build
../configure --with-arch=rv32imac_zba_zbb_zbc_zbs --prefix=/opt/riscv
sudo make
```

The tools are then installed under /opt/riscv/bin with the prefix riscv32-unknown-elf-; the commands below use the xPack prefix riscv-none-elf-, and the two are interchangeable.

### A.2 Compiling

Each source file is compiled to an object file:

```
riscv-none-elf-gcc -c \
-march=rv32imac_zba_zbb_zbc_zbs \
-mabi=ilp32 -O2 -Wall -Wextra \
-I./include -g \
main.c -o main.o
```

- `-march=...` names the extensions the compiler may use and must match this build's ISA string (Table 2.2); the value above is the full VestaRV set. An extension can be omitted from the string for compatibility testing or size comparison.
- `-mabi=ilp32` selects the integer calling convention of Table 7.1.
- `-O2` is the recommended optimisation level; `-Os` minimises size, `-Og` favours debugging.
- `-I` adds an include directory, such as the generated `MemoryMap.h` location.
- `-g` keeps debugging symbols, which the disassembly listing and the debugger use.

The provided `start.S` must be the first object given to the linker: its first instruction is placed at `0x8200`, the address the boot loader enters (Chapter 26), and it initialises the vector table and calls `main`. A C++ project declares `extern "C" int main()` so the entry symbol is not mangled, and may pass `-fno-exceptions -fno-unwind-tables -fno-asynchronous-unwind-tables` to shrink or remove the `.eh_frame` section.

### A.3 Linking

The objects are linked with the provided linker script, which defines the ROM and RAM regions, the vector table, the entry address and the interrupt handler symbol from the generated fragments:

```
riscv-none-elf-gcc -march=rv32imac_zba_zbb_zbc_zbs -mabi=ilp32 \
  -T MCU.ld -L linkerfiles_dir -Wl,-Map=program.map -g \
  start.o main.o -o program.elf
```

- `-T MCU.ld` names the linker script and `-L` the directory of the fragments it includes (`memory.x`, `periph.x`).
- `-Wl,-Map=` writes the map file with every symbol's address.
- `-flto` enables link-time optimisation.
- `-nostdlib` drops the C library; it also drops its software integer and floating-point routines and its initialisation code, so it is used only when size demands it.

### A.4 Loading the program

An Intel hex file is produced from the ELF and written to the SPI flash through the ROM monitor by the provided script, which also switches the strap and resets the chip on boards that support it:

```
riscv-none-elf-objcopy -O ihex program.elf program.hex
python3 forthprog.py program.hex --port /dev/ttyUSB0 --verify
```

The script needs the IntelHex Python package. Its `--skipBlank` option skips erasing pages the image does not use, and `--verify` reads the flash back after writing. The byte order and page handshake it implements are those of Chapter 26.

## B Register index

This appendix lists every peripheral instance of this build with its base address and every register with its absolute address, offset and size. Each register name links to its description in Part III.

Table B.1: Register index

| Register                                 | Address | Offset | Size | Access | Reset      |
|------------------------------------------|---------|--------|------|--------|------------|
| <b>GPIO0</b> base 0x4000 (shared window) |         |        |      |        |            |
| P0IN                                     | 0x4000  | 0x00   | 1    | r      | 0x00       |
| P0OUT                                    | 0x4004  | 0x04   | 1    | rw     | 0x01       |
| P0OUTS                                   | 0x4008  | 0x08   | 1    | rw1    | 0x00       |
| P0OUTC                                   | 0x400C  | 0x0C   | 1    | rw1    | 0x00       |
| P0OUTT                                   | 0x4010  | 0x10   | 1    | rw1    | 0x00       |
| P0DIR                                    | 0x4014  | 0x14   | 1    | rw     | 0x41       |
| P0IF                                     | 0x4018  | 0x18   | 1    | rw1    | 0x00       |
| P0IES                                    | 0x401C  | 0x1C   | 1    | rw     | 0x00       |
| P0IE                                     | 0x4020  | 0x20   | 1    | rw     | 0x00       |
| P0SEL                                    | 0x4024  | 0x24   | 1    | rw     | 0x4E       |
| P0REN                                    | 0x4028  | 0x28   | 1    | rw     | 0x80       |
| P0AFS                                    | 0x402C  | 0x2C   | 4    | rw     | 0x00000000 |
| P0TASK                                   | 0x4030  | 0x30   | 1    | rw     | 0x00       |
| <b>GPIO1</b> base 0x4100 (shared window) |         |        |      |        |            |
| P1IN                                     | 0x4100  | 0x00   | 1    | r      | 0x00       |
| P1OUT                                    | 0x4104  | 0x04   | 1    | rw     | 0x00       |
| P1OUTS                                   | 0x4108  | 0x08   | 1    | rw1    | 0x00       |
| P1OUTC                                   | 0x410C  | 0x0C   | 1    | rw1    | 0x00       |
| P1OUTT                                   | 0x4110  | 0x10   | 1    | rw1    | 0x00       |
| P1DIR                                    | 0x4114  | 0x14   | 1    | rw     | 0x10       |
| P1IF                                     | 0x4118  | 0x18   | 1    | rw1    | 0x00       |
| P1IES                                    | 0x411C  | 0x1C   | 1    | rw     | 0x00       |
| P1IE                                     | 0x4120  | 0x20   | 1    | rw     | 0x00       |
| P1SEL                                    | 0x4124  | 0x24   | 1    | rw     | 0x30       |
| P1REN                                    | 0x4128  | 0x28   | 1    | rw     | 0x00       |
| P1AFS                                    | 0x412C  | 0x2C   | 4    | rw     | 0x00000000 |
| P1TASK                                   | 0x4130  | 0x30   | 1    | rw     | 0x00       |
| <b>SPI0</b> base 0x4200 (shared window)  |         |        |      |        |            |
| SPI0CR                                   | 0x4200  | 0x00   | 4    | rw     | 0x00000000 |
| SPI0SR                                   | 0x4204  | 0x04   | 1    | r/rw1  | 0x00       |
| SPI0TX                                   | 0x4208  | 0x08   | 4    | rw     | 0x00000000 |
| SPI0RX                                   | 0x420C  | 0x0C   | 4    | r      | 0x00000000 |
| SPI0FOS                                  | 0x4210  | 0x10   | 4    | rw     | 0x00000000 |
| <b>SPI1</b> base 0x4300 (shared window)  |         |        |      |        |            |
| SPI1CR                                   | 0x4300  | 0x00   | 4    | rw     | 0x00000000 |
| SPI1SR                                   | 0x4304  | 0x04   | 1    | r/rw1  | 0x00       |
| SPI1TX                                   | 0x4308  | 0x08   | 4    | rw     | 0x00000000 |
| SPI1RX                                   | 0x430C  | 0x0C   | 4    | r      | 0x00000000 |
| SPI1FOS                                  | 0x4310  | 0x10   | 4    | rw     | 0x00000000 |
| <b>UART0</b> base 0x4400 (shared window) |         |        |      |        |            |
| UART0CR                                  | 0x4400  | 0x00   | 1    | rw     | 0x00       |
| UART0SR                                  | 0x4404  | 0x04   | 1    | r/rw1  | 0x00       |

| Register                                  | Address     | Offset    | Size | Access | Reset      |
|-------------------------------------------|-------------|-----------|------|--------|------------|
| UART0BR                                   | 0x4408      | 0x08      | 2    | rw     | 0x0000     |
| UART0RX                                   | 0x440C      | 0x0C      | 1    | r      | 0x00       |
| UART0TX                                   | 0x4410      | 0x10      | 1    | rw     | 0x00       |
| <b>UART1</b> base 0x4500 (shared window)  |             |           |      |        |            |
| UART1CR                                   | 0x4500      | 0x00      | 1    | rw     | 0x00       |
| UART1SR                                   | 0x4504      | 0x04      | 1    | r/rw1  | 0x00       |
| UART1BR                                   | 0x4508      | 0x08      | 2    | rw     | 0x0000     |
| UART1RX                                   | 0x450C      | 0x0C      | 1    | r      | 0x00       |
| UART1TX                                   | 0x4510      | 0x10      | 1    | rw     | 0x00       |
| <b>TIMER0</b> base 0x4600 (shared window) |             |           |      |        |            |
| TIM0CR                                    | 0x4600      | 0x00      | 4    | rw     | 0x00000000 |
| TIM0SR                                    | 0x4604      | 0x04      | 1    | r/rw1  | 0x00       |
| TIM0VAL                                   | 0x4608      | 0x08      | 4    | rw     | 0x00000000 |
| TIM0CMP0                                  | 0x460C      | 0x0C      | 4    | rw     | 0x00000000 |
| TIM0CMP1                                  | 0x4610      | 0x10      | 4    | rw     | 0x00000000 |
| TIM0CMP2                                  | 0x4614      | 0x14      | 4    | rw     | 0x00000000 |
| TIM0CAPn (n = 0..1)                       | 0x4618 + 4n | 0x18 + 4n | 4    | r      | 0x00000000 |
| <b>TIMER1</b> base 0x4700 (shared window) |             |           |      |        |            |
| TIM1CR                                    | 0x4700      | 0x00      | 4    | rw     | 0x00000000 |
| TIM1SR                                    | 0x4704      | 0x04      | 1    | r/rw1  | 0x00       |
| TIM1VAL                                   | 0x4708      | 0x08      | 4    | rw     | 0x00000000 |
| TIM1CMP0                                  | 0x470C      | 0x0C      | 4    | rw     | 0x00000000 |
| TIM1CMP1                                  | 0x4710      | 0x10      | 4    | rw     | 0x00000000 |
| TIM1CMP2                                  | 0x4714      | 0x14      | 4    | rw     | 0x00000000 |
| TIM1CAPn (n = 0..1)                       | 0x4718 + 4n | 0x18 + 4n | 4    | r      | 0x00000000 |
| <b>GPIO2</b> base 0x4800 (shared window)  |             |           |      |        |            |
| P2IN                                      | 0x4800      | 0x00      | 1    | r      | 0x00       |
| P2OUT                                     | 0x4804      | 0x04      | 1    | rw     | 0x00       |
| P2OUTS                                    | 0x4808      | 0x08      | 1    | rw1    | 0x00       |
| P2OUTC                                    | 0x480C      | 0x0C      | 1    | rw1    | 0x00       |
| P2OUTT                                    | 0x4810      | 0x10      | 1    | rw1    | 0x00       |
| P2DIR                                     | 0x4814      | 0x14      | 1    | rw     | 0x00       |
| P2IF                                      | 0x4818      | 0x18      | 1    | rw1    | 0x00       |
| P2IES                                     | 0x481C      | 0x1C      | 1    | rw     | 0x00       |
| P2IE                                      | 0x4820      | 0x20      | 1    | rw     | 0x00       |
| P2SEL                                     | 0x4824      | 0x24      | 1    | rw     | 0x00       |
| P2REN                                     | 0x4828      | 0x28      | 1    | rw     | 0x00       |
| P2AFS                                     | 0x482C      | 0x2C      | 4    | rw     | 0x00000000 |
| P2TASK                                    | 0x4830      | 0x30      | 1    | rw     | 0x00       |
| <b>SYSTEM</b> base 0x4900 (shared window) |             |           |      |        |            |
| SYSCLKCR                                  | 0x4900      | 0x00      | 2    | rw     | 0x0000     |
| CLKDIVCR                                  | 0x4904      | 0x04      | 1    | rw     | 0x00       |
| BLOCKPWR                                  | 0x4908      | 0x08      | 1    | rw     | 0x00       |
| CRCDATA                                   | 0x490C      | 0x0C      | 1    | rw     | 0x00       |
| CRCSTATE                                  | 0x4910      | 0x10      | 2    | rw     | 0x0000     |
| WDTPASS                                   | 0x4930      | 0x30      | 4    | w      | 0x00000000 |
| WDTCR                                     | 0x4934      | 0x34      | 1    | rw     | 0x00       |
| WDTSR                                     | 0x4938      | 0x38      | 1    | rw1    | 0x00       |
| WDTVAL                                    | 0x493C      | 0x3C      | 4    | r      | 0x00000000 |
| DConBIAS (n = 0..1)                       | 0x4940 + 4n | 0x40 + 4n | 2    | rw     | 0x0800     |
| <b>NPU</b> base 0x4A00 (shared window)    |             |           |      |        |            |
| NPUCR                                     | 0x4A00      | 0x00      | 4    | rw     | 0x00000000 |
| NPUIVSAR                                  | 0x4A04      | 0x04      | 4    | rw     | 0x00000000 |

| Register                                   | Address     | Offset    | Size | Access | Reset      |
|--------------------------------------------|-------------|-----------|------|--------|------------|
| NPUWVSAR                                   | 0x4A08      | 0x08      | 4    | rw     | 0x00000000 |
| NPUOVSAR                                   | 0x4A0C      | 0x0C      | 4    | rw     | 0x00000000 |
| NPUSR                                      | 0x4A10      | 0x10      | 4    | rw1    | 0x00000000 |
| NPUCFG1                                    | 0x4A14      | 0x14      | 4    | rw     | 0x00000000 |
| NPUCFG2                                    | 0x4A18      | 0x18      | 4    | rw     | 0x00000000 |
| <b>PWRCTRL</b> base 0x4B00 (shared window) |             |           |      |        |            |
| PWRCR                                      | 0x4B00      | 0x00      | 4    | rw/r   | 0x00000000 |
| PWRSR                                      | 0x4B04      | 0x04      | 4    | r      | 0x00000000 |
| PWRWAKE                                    | 0x4B14      | 0x14      | 4    | rw     | 0x00000000 |
| PWRSTS                                     | 0x4B18      | 0x18      | 4    | r      | 0x00000000 |
| TASKWKM                                    | 0x4B1C      | 0x1C      | 4    | rw     | 0x00000000 |
| <b>GPIO3</b> base 0x4D00 (shared window)   |             |           |      |        |            |
| P3IN                                       | 0x4D00      | 0x00      | 1    | r      | 0x00       |
| P3OUT                                      | 0x4D04      | 0x04      | 1    | rw     | 0x00       |
| P3OUTS                                     | 0x4D08      | 0x08      | 1    | rw1    | 0x00       |
| P3OUTC                                     | 0x4D0C      | 0x0C      | 1    | rw1    | 0x00       |
| P3OUTT                                     | 0x4D10      | 0x10      | 1    | rw1    | 0x00       |
| P3DIR                                      | 0x4D14      | 0x14      | 1    | rw     | 0x00       |
| P3IF                                       | 0x4D18      | 0x18      | 1    | rw1    | 0x00       |
| P3IES                                      | 0x4D1C      | 0x1C      | 1    | rw     | 0x00       |
| P3IE                                       | 0x4D20      | 0x20      | 1    | rw     | 0x00       |
| P3SEL                                      | 0x4D24      | 0x24      | 1    | rw     | 0x00       |
| P3REN                                      | 0x4D28      | 0x28      | 1    | rw     | 0x00       |
| P3AFS                                      | 0x4D2C      | 0x2C      | 4    | rw     | 0x00000000 |
| P3TASK                                     | 0x4D30      | 0x30      | 1    | rw     | 0x00       |
| <b>I2C0</b> base 0x4E00 (shared window)    |             |           |      |        |            |
| I2C0CR                                     | 0x4E00      | 0x00      | 4    | rw     | 0x00000000 |
| I2C0FCR                                    | 0x4E04      | 0x04      | 1    | w1     | 0x00       |
| I2C0SR                                     | 0x4E08      | 0x08      | 2    | r/rw1  | 0x0000     |
| I2C0MTX                                    | 0x4E0C      | 0x0C      | 1    | rw     | 0x00       |
| I2C0MRX                                    | 0x4E10      | 0x10      | 1    | r      | 0x00       |
| I2C0STX                                    | 0x4E14      | 0x14      | 1    | rw     | 0x00       |
| I2C0SRX                                    | 0x4E18      | 0x18      | 1    | r      | 0x00       |
| I2C0AR                                     | 0x4E1C      | 0x1C      | 1    | rw     | 0x79       |
| I2C0AMR                                    | 0x4E20      | 0x20      | 1    | rw     | 0x00       |
| <b>I2C1</b> base 0x4F00 (shared window)    |             |           |      |        |            |
| I2C1CR                                     | 0x4F00      | 0x00      | 4    | rw     | 0x00000000 |
| I2C1FCR                                    | 0x4F04      | 0x04      | 1    | w1     | 0x00       |
| I2C1SR                                     | 0x4F08      | 0x08      | 2    | r/rw1  | 0x0000     |
| I2C1MTX                                    | 0x4F0C      | 0x0C      | 1    | rw     | 0x00       |
| I2C1MRX                                    | 0x4F10      | 0x10      | 1    | r      | 0x00       |
| I2C1STX                                    | 0x4F14      | 0x14      | 1    | rw     | 0x00       |
| I2C1SRX                                    | 0x4F18      | 0x18      | 1    | r      | 0x00       |
| I2C1AR                                     | 0x4F1C      | 0x1C      | 1    | rw     | 0x23       |
| I2C1AMR                                    | 0x4F20      | 0x20      | 1    | rw     | 0x00       |
| <b>CLINT</b> base 0x5000 (shared window)   |             |           |      |        |            |
| MSIPn (n = 0..4)                           | 0x5000 + 4n | 0x00 + 4n | 4    | rw     | 0x00000000 |
| MTIMEL                                     | 0x5020      | 0x20      | 4    | rw     | 0x00000000 |
| MTIMEH                                     | 0x5024      | 0x24      | 4    | rw     | 0x00000000 |
| MTIMECMPnL (n = 0..4)                      | 0x5030 + 8n | 0x30 + 8n | 4    | rw     | 0xFFFFFFFF |
| MTIMECMPnH (n = 0..4)                      | 0x5034 + 8n | 0x34 + 8n | 4    | rw     | 0xFFFFFFFF |
| <b>MUTEX</b> base 0x6000 (shared window)   |             |           |      |        |            |

| Register                                     | Address      | Offset     | Size | Access | Reset      |
|----------------------------------------------|--------------|------------|------|--------|------------|
| MUTEXn (n = 0..15)                           | 0x6000 + 4n  | 0x00 + 4n  | 4    | rw     | 0x00000000 |
| <b>NFC0</b> base 0x6200 (shared window)      |              |            |      |        |            |
| NFC0CR                                       | 0x6200       | 0x00       | 4    | rw/w1  | 0x00001000 |
| NFC0SR                                       | 0x6204       | 0x04       | 4    | r/rw1  | 0x00000000 |
| NFC0UID                                      | 0x6208       | 0x08       | 4    | rw     | 0x00000000 |
| NFC0CFG                                      | 0x620C       | 0x0C       | 4    | rw     | 0x00000044 |
| NFC0TIM                                      | 0x6210       | 0x10       | 4    | rw     | 0x088004D4 |
| NFC0RXST                                     | 0x6214       | 0x14       | 4    | r      | 0x00000000 |
| NFC0IDX                                      | 0x6218       | 0x18       | 4    | rw     | 0x00000000 |
| NFC0DATA                                     | 0x621C       | 0x1C       | 4    | rw     | 0x00000000 |
| NFC0TXCTL                                    | 0x6220       | 0x20       | 4    | rw     | 0x00000000 |
| NFC0DBG                                      | 0x6224       | 0x24       | 4    | r      | 0x00000000 |
| <b>GPIO4</b> base 0x6300 (shared window)     |              |            |      |        |            |
| P4IN                                         | 0x6300       | 0x00       | 1    | r      | 0x00       |
| P4OUT                                        | 0x6304       | 0x04       | 1    | rw     | 0x00       |
| P4OUTS                                       | 0x6308       | 0x08       | 1    | rw1    | 0x00       |
| P4OUTC                                       | 0x630C       | 0x0C       | 1    | rw1    | 0x00       |
| P4OUTT                                       | 0x6310       | 0x10       | 1    | rw1    | 0x00       |
| P4DIR                                        | 0x6314       | 0x14       | 1    | rw     | 0x00       |
| P4IF                                         | 0x6318       | 0x18       | 1    | rw1    | 0x00       |
| P4IES                                        | 0x631C       | 0x1C       | 1    | rw     | 0x00       |
| P4IE                                         | 0x6320       | 0x20       | 1    | rw     | 0x00       |
| P4SEL                                        | 0x6324       | 0x24       | 1    | rw     | 0x00       |
| P4REN                                        | 0x6328       | 0x28       | 1    | rw     | 0x00       |
| P4AFS                                        | 0x632C       | 0x2C       | 4    | rw     | 0x00000000 |
| P4TASK                                       | 0x6330       | 0x30       | 1    | rw     | 0x00       |
| <b>GPIO5</b> base 0x6400 (shared window)     |              |            |      |        |            |
| P5IN                                         | 0x6400       | 0x00       | 1    | r      | 0x00       |
| P5OUT                                        | 0x6404       | 0x04       | 1    | rw     | 0x00       |
| P5OUTS                                       | 0x6408       | 0x08       | 1    | rw1    | 0x00       |
| P5OUTC                                       | 0x640C       | 0x0C       | 1    | rw1    | 0x00       |
| P5OUTT                                       | 0x6410       | 0x10       | 1    | rw1    | 0x00       |
| P5DIR                                        | 0x6414       | 0x14       | 1    | rw     | 0x00       |
| P5IF                                         | 0x6418       | 0x18       | 1    | rw1    | 0x00       |
| P5IES                                        | 0x641C       | 0x1C       | 1    | rw     | 0x00       |
| P5IE                                         | 0x6420       | 0x20       | 1    | rw     | 0x00       |
| P5SEL                                        | 0x6424       | 0x24       | 1    | rw     | 0x00       |
| P5REN                                        | 0x6428       | 0x28       | 1    | rw     | 0xC0       |
| P5AFS                                        | 0x642C       | 0x2C       | 4    | rw     | 0x00000001 |
| P5TASK                                       | 0x6430       | 0x30       | 1    | rw     | 0x00       |
| <b>IRQROUTER</b> base 0x7000 (shared window) |              |            |      |        |            |
| HnENL (n = 0..4)                             | 0x7000 + 16n | 0x00 + 16n | 4    | rw     | 0x00000000 |
| HnENM (n = 0..4)                             | 0x7004 + 16n | 0x04 + 16n | 4    | rw     | 0x00000000 |
| HnENU (n = 0..4)                             | 0x7008 + 16n | 0x08 + 16n | 4    | rw     | 0x00000000 |
| HnENX (n = 0..4)                             | 0x700C + 16n | 0x0C + 16n | 4    | rw     | 0x00000000 |
| CLAIM                                        | 0x7800       | 0x800      | 4    | rw     | 0x00000000 |
| PENDL                                        | 0x7810       | 0x810      | 4    | r      | 0x00000000 |
| PENDM                                        | 0x7814       | 0x814      | 4    | r      | 0x00000000 |
| PENDU                                        | 0x7818       | 0x818      | 4    | r      | 0x00000000 |
| PENDX                                        | 0x781C       | 0x81C      | 4    | r      | 0x00000000 |
| INSVCL                                       | 0x7820       | 0x820      | 4    | r      | 0x00000000 |
| INSVCM                                       | 0x7824       | 0x824      | 4    | r      | 0x00000000 |
| INSVCU                                       | 0x7828       | 0x828      | 4    | r      | 0x00000000 |



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| Register | Address | Offset | Size | Access | Reset      |
|----------|---------|--------|------|--------|------------|
| INSVCX   | 0x782C  | 0x82C  | 4    | r      | 0x00000000 |

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## C Errata

This appendix lists known deviations of the hardware from this manual. Each entry gives the affected silicon revision, a reference number, a description of the fault, the circumstances in which it occurs, its effect, and the recommended software workaround. There are no known errata for the Castalia MCU at the time of this revision.

Table C.1: Errata.

| Ref. | Affects | Description | Conditions | Effect | Workaround |
|------|---------|-------------|------------|--------|------------|
| none | all     | None known. |            |        |            |

## D Revision history

Table D.1: Revision history of this manual.

| Revision | Date       | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 2026-09-12 | Restructured into parts and chapters with a front matter chapter (conventions, access types, register anatomy, glossary). Register descriptions reformatted with summary tables, per-register reset values and field tables. Per-peripheral instance, signal and interrupt tables generated from the chip configuration. Reset, clocks and power consolidated into one chapter; the boot image format corrected from the ROM source; the hand-written instruction listing replaced by the ISA and CSR summary; software material moved to Appendix A. |